# Circuit Codex

User manual

What each tool does, what its terms mean, and where its numbers come from. Tap a tool to go to its section.

# Index

## PASSIVE COMPONENTS

| | |
|---|---|
| Resistor color code | Resistors |
| SMD resistor code | Resistors |
| Resistor power | Resistors |
| E-series values | Resistors |
| Series/parallel (R) | Resistors |
| Voltage divider | Resistors |
| Current divider | Resistors |
| Wheatstone bridge | Resistors |
| Delta-Y transform | Resistors |
| NTC/PTC thermistor | Resistors |
| Ceramic cap code | Capacitors |
| Film capacitor code | Capacitors |
| SMD cap code | Capacitors |
| Series/parallel (C) | Capacitors |
| RC time constant | Capacitors |
| Capacitor energy | Capacitors |
| Inductor color code | Inductors |
| Inductor SMD code | Inductors |
| Series/parallel (L) | Inductors |
| RC filter | Passive filters |
| RL filter | Passive filters |

SMD package sizes Reference

## ACTIVE & SEMICONDUCTOR DEVICES

| | |
|---|---|
| Diode forward voltage | Diodes |
| LED series resistor | Diodes |
| BJT biasing | Transistors (BJT) |
| BJT as a switch | Transistors (BJT) |
| BJT currents | Transistors (BJT) |
| MOSFET biasing | MOSFET / IGBT |
| MOSFET switching | MOSFET / IGBT |
| Half-wave rectifier | Rectifiers |
| Bridge rectifier | Rectifiers |
| Center-tap rectifier | Rectifiers |
| Rectifier ripple | Rectifiers |
| AC phase control | Thyristors (SCR, TRIAC) & MOSFET |
| Inverting | Op-amps |
| Non-inverting | Op-amps |
| Buffer | Op-amps |
| Comparators | Op-amps |
| Integrator | Op-amps |
| Differentiator | Op-amps |
| Summing | Op-amps |
| Differential | Op-amps |
| Photocell / LDR | Optoelectronics |
| Optocoupler | Optoelectronics |

| | |
|---|---|
| Wavelength ↔ color | Optoelectronics |

## DIGITAL & LOGIC CIRCUITS

| | |
|---|---|
| Logic gates | Logic gates |
| Karnaugh map | Logic gates |
| Flip-flops | Sequential logic |
| Multivibrators | Sequential logic |
| 555 timer | Sequential logic |
| Number bases | Number systems |
| ASCII table | Number systems |
| PWM (duty cycle) | Timing & interfaces |
| Switch debounce | Timing & interfaces |
| I2C pull-up resistor | Timing & interfaces |
| UART baud rate | Timing & interfaces |
| Crystal load | Timing & interfaces |
| Oscillator stability | Timing & interfaces |
| PLL settings | Timing & interfaces |
| ADC resolution | Data conversion |
| DAC resolution | Data conversion |
| SNR estimation | Data conversion |

## RF & MICROWAVE

| | |
|---|---|
| Path loss | Antennas & propagation |
| Radio wavelength | Antennas & propagation |
| Beamwidth & gain | Antennas & propagation |
| Fresnel zone | Antennas & propagation |
| Antenna length | Antennas & propagation |
| VSWR / Return loss | Transmission lines |
| Velocity factor | Transmission lines |
| Coax impedance | Transmission lines |
| Smith chart matching | Transmission lines |
| Skin depth | Transmission lines |
| Coax cable standards | Transmission lines |
| RF cascade | Power & gain |
| Intercept point | Power & gain |
| Intermodulation | Power & gain |
| Attenuator | Power & gain |
| Power divider | Power & gain |

## POWER ELECTRONICS

| | |
|---|---|
| Linear regulator | Supply & regulation |
| Zener regulation | Supply & regulation |
| DC/DC buck converter | Supply & regulation |
| DC/DC boost converter | Supply & regulation |
| Zener + transistor | Supply & regulation |
| Constant current | Supply & regulation |
| Inrush limiter (NTC) | Supply & regulation |
| SMPS flyback basics | Supply & regulation |
| Battery runtime | Battery |
| Battery C-rate | Battery |
| Internal resistance | Battery |
| Battery types & sizes | Battery |
| Button cell sizes | Battery |
| Heat sink sizing | Thermal management |

## PCB & INTERCONNECT

| | |
|---|---|
| Trace width | Trace current & resistance |
| Via current capacity | Trace current & resistance |
| Microstrip | Impedance |
| Embedded microstrip | Impedance |
| Edge-coupled microstrip | Impedance |
| Edge-coupled embedded | Impedance |
| Stripline | Impedance |
| Asymmetric stripline | Impedance |
| Edge-coupled stripline | Impedance |
| Edge-coupled asym stripline | Impedance |
| Broadside-coupled stripline | Impedance |
| Broadside-coupled, no plane | Impedance |
| Coplanar waveguide | Impedance |
| Edge-coupled coplanar | Impedance |
| Propagation delay | Impedance |
| Crosstalk | Impedance |
| Laminate materials | Impedance |
| Thermal vias | Vias & layout |
| Decoupling capacitor | Vias & layout |
| Wire sizing | Cabling |
| Wire & cable colors | Cabling |

| | |
|---|---|
| IP ratings | Electrical safety |

## TOOLS

| | |
|---|---|
| Calculator | Calculators |
| Ohm's law | Calculators |
| AC Ohm's law | Calculators |
| Kirchhoff's laws | Calculators |
| SI prefix converter | Calculators |
| Tolerance & error | Calculators |
| dB ratio | Calculators |
| dB levels | Calculators |
| RMS calculator | Calculators |
| Unit converter | Converters |
| PPM converter | Converters |
| Formula search | Reference |
| Physical constants | Reference |
| SI units | Reference |

# Resistor color code




## What it computes

The resistance, tolerance and, on 6-band parts, temperature coefficient that a resistor's colour bands stand for — and the other way round: type a value and it shows the bands. It also says whether the value is a standard E-series value and, if not, the nearest one.

## Source

**IEC 60062:2016**, *Marking codes for resistors and capacitors*. Its Table 1 gives each colour a digit, a multiplier, a tolerance and a temperature coefficient; clause 3 says how the bands are laid out. Both were read in the official preview of the standard, which runs to clause 4.2.2 — all this tool needs.

## What the bands mean

A resistor is too small to print a number on, so the value is painted as rings of colour, one digit per ring. Each colour stands for a digit from 0 to 9: black 0, brown 1, red 2, orange 3, yellow 4, green 5, blue 6, violet 7, grey 8, white 9.

- **Digit bands** — the first two (4-band parts) or three (5- and 6-band parts) give the significant figures.
- **Multiplier** — the next band says how many zeros follow, as a power of ten: red is ×100, orange ×1000. Gold (×0.1), silver (×0.01) and pink (×0.001, new in the 2016 edition) make values below 10 Ω: yellow–violet–pink is 47 mΩ.
- **Tolerance** — how far the actual resistance may be from the marked value: gold ±5% means a 1 kΩ part measures somewhere from 950 Ω to 1050 Ω. Leaving the band off means ±20%.
- **Temperature coefficient** (6-band only) — how much the resistance drifts per degree of temperature change, in parts per million per kelvin: brown is 100 ppm/K, so 0.01% per degree.

Brown–black–red–gold is therefore 1, 0, ×100, ±5%: 1 kΩ ±5%.

**Which end to start from.** The tolerance band stands apart from the others, on the right, and the standard makes it at least 1.5 times as wide as the others; the digits are bunched on the left, starting as near the end as possible. Gold and silver are never digits, so a part with gold at one end is read from the other. On 6-band parts the temperature band comes last but is not the wide one: the 2016 edition changed that, because a wide sixth band was mistaken for the tolerance.

**Standard values.** Resistors are made only in preferred values, the E-series, spaced so that neighbouring values just overlap within their tolerance. The tolerance band picks the one series the value is checked against — gold, ±5%, means E24; brown, ±1%, means E96 — and when the value is not in it, the tool gives the nearest one that is. A 1% 4.7 kΩ part therefore reads "Not in E96", though makers do sell it (see the E-series tool).

## How the value is worked out

```
4 bands:  Value = (10 × D1 + D2) × Multiplier
5, 6:     Value = (100 × D1 + 10 × D2 + D3) × Multiplier
```

Typing a value runs it backwards: the value is split into as many significant digits as the band count allows, rounded, and the remaining power of ten picks the multiplier colour. The smallest value a code can show is 10 mΩ on 4 bands (brown–black–pink) and 0.1 Ω on 5 or 6; the largest is 99 GΩ and 999 GΩ. Outside that there is no band for it, and the tool says so.

## Assumptions and limits

- **The 2016 tolerance colours.** IEC 60062:2016 made orange ±0.05%, yellow ±0.02% and grey ±0.01%. Before that edition some manufacturers used grey for a non-standard ±0.05%, so a grey tolerance band on an old part may mean that.
- **Colours are only as good as the light.** Red, orange and brown are easily confused on a real part, and so are violet and grey. When in doubt, measure.
- **Zero-ohm links** are marked with a single black band and are not a colour code as such.

## What it deliberately does not do

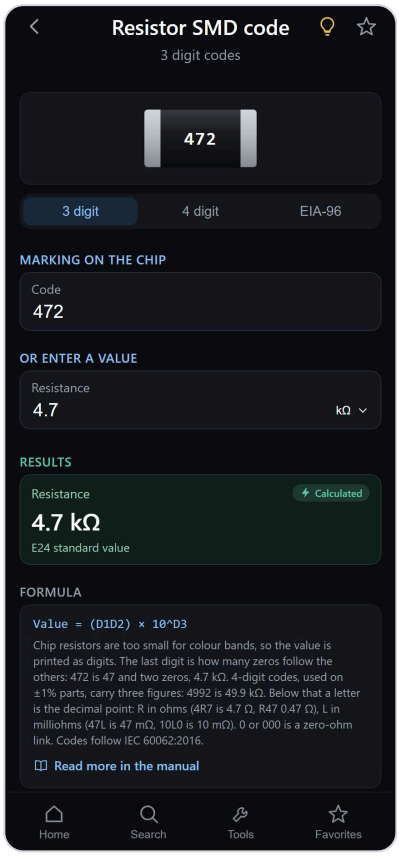
- **Capacitor and inductor colour codes** have their own tools, since the same colours mean different units and tolerances there.
- **SMD resistor codes** (the three- and four-digit and EIA-96 markings) are the SMD code tool.

## How it was checked

Against the IEC 60062:2016 table: digits, multipliers, tolerances and temperature coefficients for every colour. This check found the app had grey as ±0.05%, the pre-2016 non-standard value, and no tolerance for orange or yellow; all three now follow the standard. A second reading of the official text found two more gaps, now fixed: the pink ×0.001 multiplier was missing, and the drawing made the tolerance band no wider than the others.

Brown–black–red–gold reads 1 kΩ ±5%, 950 Ω to 1.05 kΩ, an E24 value. 47 mΩ gives yellow–violet–pink. At grey's ±0.01% the range reads 999.9 Ω to 1.0001 kΩ — at four figures it had printed as "1 kΩ", which the range now shows to enough figures to see. Typing 4.7 k gives yellow–violet–red; 0.22 Ω gives red–red–silver.

# SMD resistor code




## What it computes

The resistance printed on a surface-mount (chip) resistor, from its marking — 3-digit, 4-digit or EIA-96 — and the other way round: the marking a value would carry. It says whether the value is a standard E-series value.

## Source

**IEC 60062:2016**, *Marking codes for resistors and capacitors*, which defines the three-character code (clause 4.2.2), the four-character code (4.2.3), R as the decimal point, and in its Annex A a special three-character code for values with three figures — what the industry calls EIA-96.

What was read in the standard itself, and what was not: the official 2016 preview runs to clause 4.2.2, far enough to confirm the three-character code and its limit — it "is applicable to values from an E series up to E24", since it carries only two figures. The three- and four-character tables were read in **GOST IEC 60062-2014**, the official Russian edition of IEC 60062:2004, declared identical to it (IDT) and published free by Standartinform; its tables 3 and 4 match the official IEC 2004 preview word for word, and the 2016 versions of both tables in ROHM's application note "List of Nominal Resistance Values" (2024), which reproduces them from IEC 60062:2016. Annex A (EIA-96) is new in 2016, lies beyond the preview and is not in the ROHM note; it was checked against published examples only.

## What the marking means

A chip resistor is far too small for colour bands, so the value is printed as characters, three or four of them.

- **3-digit code** — usual on ±5% parts. The first two digits are the value's figures, the last is how many zeros follow them. 472 is 47 followed by two zeros: 4700 Ω, 4.7 kΩ. 334 is 330 kΩ; 100 is 10 Ω, not 100 Ω.
- **4-digit code** — usual on ±1% parts, which need a third figure. The first three are the figures, the last the number of zeros. 1001 is 1.00 kΩ, 4992 is 49.9 kΩ.
- **R and L** — below 10 Ω (3-digit) or 100 Ω (4-digit) there is no zero count small enough, so a letter stands where the decimal point is: R in ohms, down to 0.1 Ω, then L in milliohms. 4R7 is 4.7 Ω, R47 0.47 Ω, 47R0 47.0 Ω; 47L is 47 mΩ, 1L0 1 mΩ, 10L0 10 mΩ, 12L7 12.7 mΩ — the milliohm values are current-sense resistors. Some makers print R010 for 10 mΩ instead of 10L0; it reads the same, and the tool reads it.
- **0, 000 or 0000** — a zero-ohm link, used as a jumper that a pick-and-place machine can fit.
- **EIA-96** — on small 1% parts (0603 and below) even four characters do not fit. Two digits and a letter do: the digits are not the value but its position among the 96 values of the E96 series, 01 for 100 up to

96 for 976, and the letter is the multiplier. 01C is the first E96 value, 100, times 100: 10 kΩ. 68X is 499 × 0.1 = 49.9 Ω.

## How the value is worked out

```
3-digit:  Value = D1D2 × 10^D3
4-digit:  Value = D1D2D3 × 10^D4
EIA-96:   Value = E96[code] × multiplier
```

EIA-96 multiplier letters: Z ×0.001, Y or R ×0.01, X or S ×0.1, A ×1, B or H ×10, C ×100, D ×1000, E $\times 10^4$, F $\times 10^5$. Where two letters share a multiplier the tool reads both and prints the first.

Typing a value runs it backwards. A marking holds only two or three figures, so a value with more is rounded to what the part could carry — 12 345 Ω becomes 1232, 12.3 kΩ — and the tool says so. In EIA-96 only E96 values exist, so a value is moved to the nearest one: 4.7 kΩ becomes 66B, 4.75 kΩ.

## Assumptions and limits

- **The marking does not say the tolerance.** A 3-digit code is usually ±5% and a 4-digit or EIA-96 code ±1%, but only the datasheet or the reel label says.
- **Standard value.** The value is checked against the series of that usual tolerance: E24 for a 3-digit code, E96 for 4-digit and EIA-96. So 472 is an E24 value, and 4701 reads "Not in E96 — nearest is 4.75 kΩ": E96 has no 4.7. A 3-digit code carries two figures, enough for every E6 to E24 value; the three figures of E48 to E192 need 4 digits or EIA-96.
- **Which scheme a part uses** shows in the marking: a letter other than R or L at the end means EIA-96, and digits only are the 3- or 4-digit code, told apart by how many there are. Pick the pill that matches.
- **Small parts may not be marked at all.** 0402 and smaller usually carry no marking; the value is only on the reel.
- Some makers use their own schemes for current-sense resistors (such as a lowercase m for milliohms); those are not covered.

## What it deliberately does not do

- **Colour bands** are the Color code tool.
- **SMD capacitor and inductor markings** have their own tools, since the same characters mean picofarads or microhenries there.

## How it was checked

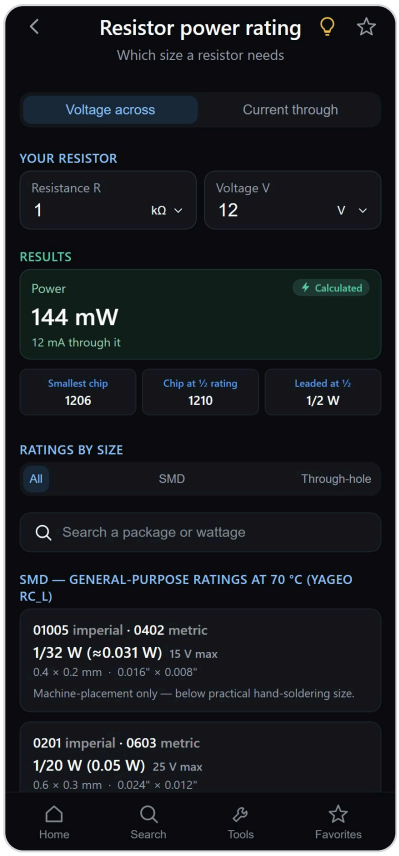
Against published examples of the codes: 334 is 330 kΩ, 222 is 2.2 kΩ, 1001 is 1.00 kΩ, 4992 is 49.9 kΩ, R300 is 0.30 Ω, 01C is 10 kΩ; also 68X is 49.9 Ω and 66B 4.75 kΩ.

The review found four faults, now fixed. 47 mΩ on the 3-digit pill printed "R047", four characters; 10 mΩ on the 4-digit pill printed "R0100", five characters, and typing R010 showed R0100 on the chip. A marking of just "0" for a zero-ohm link was rejected. And 12 345 Ω showed the code 1232 beside a resistance of 12.35 kΩ, with nothing to say the code means 12.3 kΩ.

Reading the standard's tables (in ROHM's note) then showed that the first fix was itself wrong: milliohms take the letter L, not R. The tool had printed 47 mΩ as R05, rounding it to 50 mΩ, and 12.7 mΩ as R013; the standard writes 47L and 12L7, exactly. It now writes the L forms — 47L, 1L0, L12, 10L0, 12L7 — and still reads R010.

# Resistor power

(formerly Resistor power rating)




## What it computes

Which size a resistor needs. Enter its resistance and the voltage across it, or the current through it, and the tool gives the power it dissipates, the smallest chip size that takes it, the smallest at half its rating — the usual design margin — and the through-hole wattage class at half its rating. A chip must also be rated for the voltage across it.

Below that is the reference table: how much power each common resistor size can dissipate, with its dimensions. Surface-mount parts are listed by package code, with their maximum working voltage; through-hole parts by their wattage class, with a typical body size. It can be filtered to SMD or through-hole and searched by package or wattage.

## Source

- **SMD** — Yageo's RC_L general-purpose thick-film chip resistor datasheet (V.11): rated power at 70 °C, maximum working voltage and operating temperature range for each size from 01005 to 2512. These are the conventional figures most tables repeat.
- **Vishay D/CRCW e3** (revision April 2026) as the counter-example: the same sizes rated higher, 0.125 W for 0603 and 0.25 W for 0805.
- **Through-hole** body sizes are typical ranges from distributor listings, not a standard.

## What it means

When current flows through a resistor, the resistor turns electrical power into heat. The power rating is the most heat it can get rid of, continuously, without being damaged or drifting out of tolerance. Exceed it and the part runs hot, its value shifts, and eventually it fails.

The power a resistor actually has to handle comes from what is across it or through it:

$$P = V^2 / R = I^2 \times R$$

A 1 kΩ resistor with 5 V across it dissipates 25 / 1000 = 25 mW, well inside an 0603's 100 mW. The same resistor at 12 V dissipates 144 mW: more than an 0805's 125 mW, so a 1206 at least, and a 1210 with the half-rating rule below.

**Temperature.** The rating holds only while the air around the part is at 70 °C or cooler. Above that the part can shed less heat, and the allowed power falls in a straight line to nothing at 155 °C, the maximum film

temperature — 125 °C for 0201 and 01005. A common rule is to use no more than half the rating, which leaves room for a warm enclosure and for the board around the part heating it too.

**Voltage.** Each size also has a maximum working voltage, set by the gap between its terminals. For high resistance values this limit is reached before the power one:

```
V = √(P × R), or the maximum working voltage, whichever is lower
```

An 0603 (1/10 W, 75 V) at 1 MΩ would reach its power rating at 316 V, but may only see 75 V, which is $75^2$ / 1 MΩ = 5.6 mW. For high-voltage dividers the voltage rating, not the wattage, sets the size — or several resistors in series share the voltage.

## Assumptions and limits

- **Ratings depend on the series, not only the size.** The table gives the conventional general-purpose figures; anti-surge, high-power and some newer thick-film series rate the same package two or four times higher. The part's own datasheet has the last word.
- **The board matters.** An SMD rating assumes the copper pads and board carry heat away as in the maker's test. Small pads, thin copper or neighbouring hot parts lower it.
- **Through-hole sizes vary by maker and technology**: a metal-film 1/4 W part and a carbon-film one are not the same size. The ranges are for recognising a part, not for a footprint.
- Pulses are a separate question: a short pulse can exceed the continuous rating, within limits the datasheet gives as a pulse curve.
- **The size check uses the table's figures**: the Yageo ratings for chips, each with its maximum working voltage, and the wattage class for leaded parts, whose voltage rating is not in the table and depends on the part. "Half its rating" is the usual margin for a warm enclosure and hot neighbours; the full rating holds only up to 70 °C around the part, on the maker's pad layout.
- Above 200 V no chip in the table will do; several in series share the voltage, each taking its part.

## What it deliberately does not do

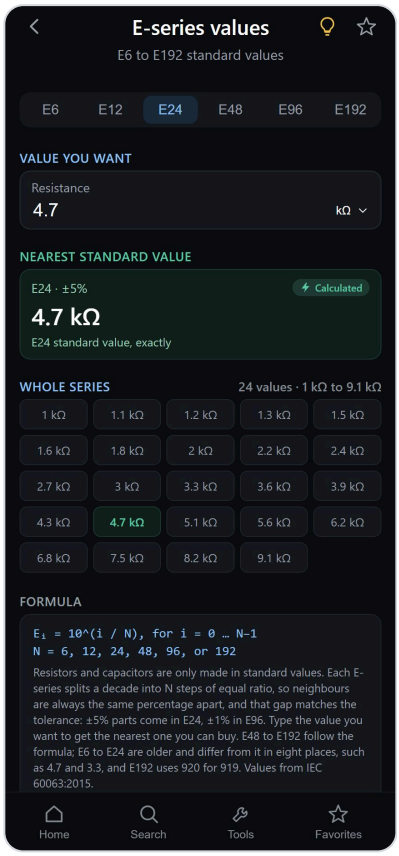
- **Package dimensions in detail** (pad sizes, heights) are the SMD package sizes tool.
- **Temperature rise** from a given power and board is a thermal calculation, not a rating lookup.

## How it was checked

Every SMD power rating, working voltage and the 125 °C / 155 °C limits against the Yageo RC_L datasheet. The review corrected the 1210's size in inches, which read 0.12″ × 0.125″, and the 0805's width to 1.25 mm; the SMD package sizes review later aligned all the sizes to makers' datasheets (1210: 0.126″ × 0.098″); it also withdrew the claim that SMD ratings are standard across manufacturers, which the Vishay datasheet contradicts. The maximum working voltages were added then.

The size check was added in the re-walk. Its reference cases: 1 kΩ at 12 V is 144 mW, a 1206 at full rating, a 1210 or a 1/2 W leaded part at half; a 0.1 Ω shunt at 2 A is 0.4 W, a 1210 or at half a 2512; 1 MΩ at 100 V is only 10 mW but needs an 0805, the first chip rated over 75 V. The re-walk also corrected this section's own example, which had said 144 mW fits "an 0805 at least" when an 0805 is rated 125 mW.

# E-series values




## What it computes

The standard value nearest to the one you want, in the series you choose (E6 to E192), how far off it is in percent, and the whole series across the decade of your value, with the nearest one highlighted.

## Source

**IEC 60063:2015**, *Preferred number series for resistors and capacitors*. It lists every series value by value, and its tolerances: E6 ±20%, E12 ±10%, E24 ±5%, E48 ±2%, E96 ±1%, E192 ±0.5% and better.

## What it means

Resistors and capacitors are not made in every value. It would be impossible to stock them all, and pointless: a ±5% 1000 Ω resistor can be anything from 950 to 1050 Ω, so a separate 1010 Ω part would overlap it completely. Instead, manufacturers make a fixed set of **preferred values**, the E-series.

Each series splits a decade — 1 to 10, 10 to 100, and so on — into a fixed number of steps: 6 for E6, 24 for E24, 96 for E96. The steps are not equal in ohms but equal in **ratio**, so each value is the same percentage above the previous one, whatever the decade. And the number of steps is chosen so that the gap between neighbours matches the tolerance: in E24 each value is about 10% above the last, so ±5% parts either side just meet. That is why the tolerance tells you the series — ±5% parts come in E24 values, ±1% in E96.

In practice: pick the tolerance you need, look up the nearest value in its series, and check whether the percent difference matters for your circuit. If it does, a tighter series has a closer value, or two parts in series or parallel can make up the difference.

## How the values are worked out

$$E_i = 10^{(i / N)},\quad i = 0 \ldots N-1,\quad \text{rounded to 2 figures (E6–E24) or 3 (E48–E192)}$$

N is the number of values in the series. E48, E96 and E192 follow this formula, with one exception: the standard lists 920 in E192 where the formula gives 919. E6, E12 and E24 are older than the formula and were chosen by hand; eight E24 values differ from it — 2.7, 3.0, 3.3, 3.6, 3.9, 4.3, 4.7 and 8.2 — which is why 4.7 kΩ, not 4.6 kΩ, is the value everyone knows. The tool uses the standard's lists, not the formula, for those.

The nearest value is the one with the smallest percentage difference from what you typed. The top of a decade counts as the start of the next one, so 97 kΩ in E24 gives 100 kΩ, not 91 kΩ.

## Assumptions and limits

- **E3** (±40%: 1.0, 2.2, 4.7) exists in the standard but is rarely used for new parts and is not offered.
- **Exactly between two values** the lower one is given; both are equally far off.
- **One tolerance, one series.** Throughout the app a tolerance picks exactly one series: ±1% means E96, and the sliders and hints step through E96 only. Manufacturers also sell the E24 values at ±1% — Yageo's RC_L datasheet lists "1% (E24/E96)" — so a 1% 4.7 kΩ part is easy to buy even though the app suggests 4.75 kΩ. The simpler rule is kept on purpose.
- **Capacitors** follow the same series, but in practice ceramic capacitors are mostly stocked in E6 or E12 values, and electrolytics in E3 or E6.
- The table shows the decade of the nearest value, so the value found is always highlighted in it: for 97 kΩ in E24 that is the 100 kΩ decade.

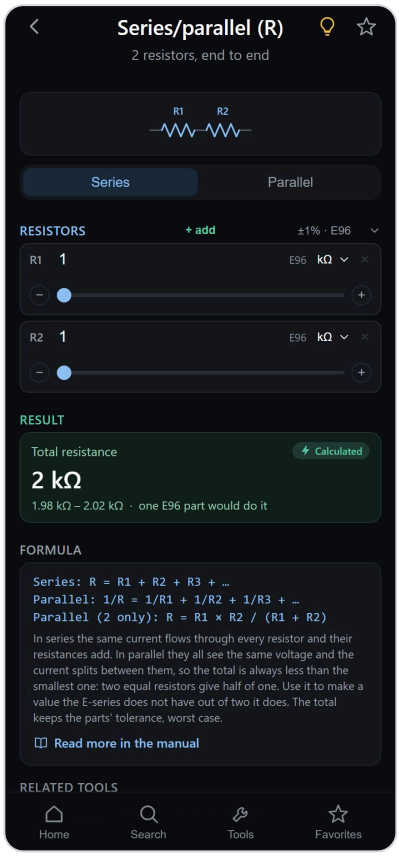
## What it deliberately does not do

- **Combining two standard values** to hit an exact one is a series/parallel question, the Series/parallel tool's job.

## How it was checked

Every E48 and E96 value against the IEC 60063 list, and E192 against the same list. The review found the E192 exception missing: the tool showed 919 where the standard has 920, so 9.2 kΩ was reported as not standard; it now uses 920. The E6, E12 and E24 lists, including the eight values that differ from the formula, match the standard. The table also showed the decade of the typed value, so a nearest value in the next decade (97 kΩ → 100 kΩ) was not highlighted; it now shows the nearest value's decade.

# Series/parallel (R)




## What it computes

The total resistance of two to four resistors connected in series or in parallel, the range it can take given the parts' tolerance, and the nearest single standard part to that total. Each resistor has a slider that steps through standard values.

## Source

Ohm's law and Kirchhoff's laws; no standard is needed beyond them. The standard values come from IEC 60063, as in the E-series tool.

## What it means

**In series** — resistors connected end to end, so the same current has to pass through each of them in turn. Each one drops a voltage, and the drops add up, so the resistances add:

```
R = R1 + R2 + R3 + …
```

The total is always more than the largest one.

**In parallel** — resistors connected side by side between the same two points, so each sees the same voltage and the current divides between them. Each gives the current another path, so the total is always less than the smallest one. It is the conductances, 1/R, that add:

```
1/R = 1/R1 + 1/R2 + 1/R3 + …
R = R1 × R2 / (R1 + R2)       for two
```

Two equal resistors in parallel give half of one; ten in parallel, a tenth. A resistor much larger than another barely changes it: 1 kΩ in parallel with 1 MΩ is 999 Ω.

**Why combine them.** The common reason is to get a value the E-series does not stock from two it does — 1 kΩ in parallel with 4.7 kΩ is 824.6 Ω, within a tenth of a percent of the E96 value 825 Ω. Others: to share the power or the voltage between several parts, each within its own rating.

## Tolerance of the total

If every resistor is off by the same fraction in the same direction, the series sum and the parallel combination are both off by exactly that fraction. So in the worst case the total has the same tolerance as the parts, no better and no worse, and that is the range the tool shows. In practice the errors of separate parts are

independent and partly cancel, so the total is usually closer than that — but a worst-case design should not count on it.

## Assumptions and limits

- **Up to four resistors**, all in series or all in parallel. Mixed networks (two in series, in parallel with a third) are worked out in steps: combine one group, then use its total as a single resistor.
- **A zero-ohm resistor in parallel** shorts the group, and the total is 0.
- Wire and contact resistance are ignored, which matters only for milliohm values.

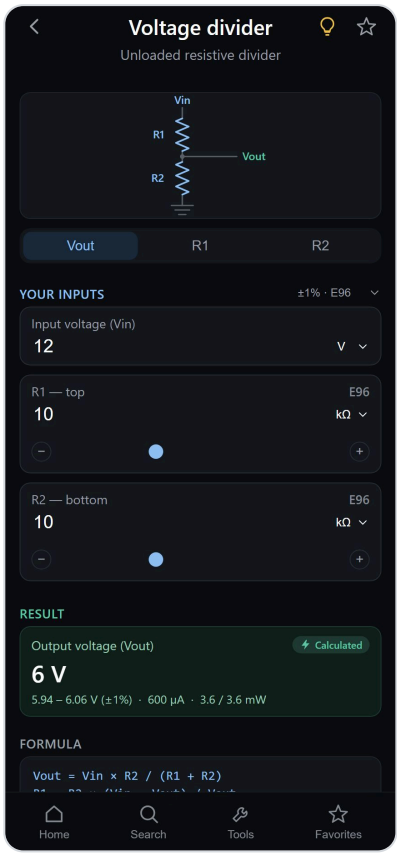
## What it deliberately does not do

- **Capacitors and inductors** combine the other way round and have their own tools.
- **Finding the best pair** for a target value automatically is not done; the sliders make trying pairs quick.

## How it was checked

By hand: 1 k + 2.2 k = 3.2 kΩ; 1 k ‖ 1 k = 500 Ω; 1 k ‖ 4.7 k = 824.6 Ω, nearest E96 825 Ω; a third 1 kΩ in parallel gives 451.9 Ω; 1 k ‖ 0 = 0 Ω. The tolerance ranges are the total × (1 ± t).

# Voltage divider




## What it computes

For two resistors in series across a voltage, any one of the output voltage, the top resistor R1 or the bottom resistor R2, from the other three. It also gives the output's worst-case spread from the resistors' tolerance, the current through the pair and the power in each resistor, and when a resistor is solved for, the nearest standard value and the output that value would give.

## Source

Ohm's law applied to two resistors in series; standard values from IEC 60063.

## What it means

A voltage divider makes a smaller voltage out of a larger one. Two resistors are connected in series from the input voltage to ground. The same current flows through both, so each drops a share of the voltage in proportion to its resistance, and the point between them — the output — sits at a fixed fraction of the input:

```
Vout = Vin × R2 / (R1 + R2)
```

Equal resistors give half. A 10 kΩ over a 4.7 kΩ gives 4.7 / 14.7, about a third. Typical uses: bringing a 12 V signal down to what a 3.3 V ADC can read, setting a reference voltage, or biasing a transistor.

Solved for either resistor:

```
R1 = R2 × (Vin − Vout) / Vout
R2 = R1 × Vout / (Vin − Vout)
```

**Choosing the values.** Only the ratio sets Vout, so 1 kΩ/1 kΩ and 1 MΩ/1 MΩ both halve the voltage. What the size decides is the current: small values waste power (12 V across 2 kΩ is 6 mA, 72 mW), large ones make the output weak, easily pulled down by whatever it feeds and more sensitive to noise. Tens of kΩ is a common middle ground for signals.

## Tolerance

The output is furthest off when the two resistors miss in opposite directions: R2 high and R1 low pushes Vout up, the reverse pulls it down.

```
Vout max = Vin × R2(1+t) / (R1(1−t) + R2(1+t))
Vout min = Vin × R2(1−t) / (R1(1+t) + R2(1−t))
```

So a divider built from ±1% parts is not a ±1% divider: 10 kΩ over 4.7 kΩ at ±1% spreads by ±1.37%. The spread is widest when the output is a small fraction of the input, and approaches ±2t.

## Loading

The formula assumes nothing draws current from the output. Anything that does — an ADC input, a transistor base, a meter — is a resistor in parallel with R2, and lowers the output. Seen from the output, the divider behaves like a source with an internal resistance of R1 ‖ R2, so a load RL lowers Vout by roughly

```
(R1 ‖ R2) / RL
```

1% for a load a hundred times R1 ‖ R2, 10% for one ten times. The fix is lower divider resistances or a buffer (the op-amp voltage follower).

## Assumptions and limits

- **Unloaded**, as above.
- **Positive voltages.** For a negative supply, enter both voltages as positive values; the ratio and the resistors are the same.
- Vout must be between 0 and Vin when solving for a resistor; a divider can only divide.
- The standard value offered is from the one series the chosen tolerance implies (±1% → E96).

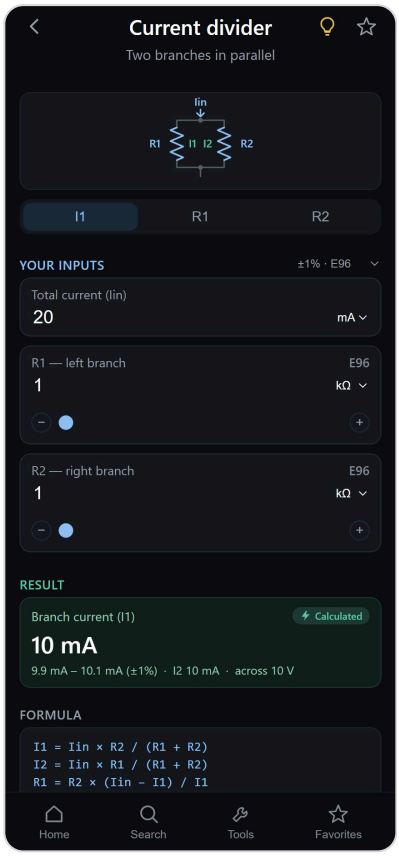
## What it deliberately does not do

- **A load resistor** is not an input; the loading rule above covers it.
- **Adjustable dividers** (a potentiometer with fixed end resistors) are not modelled.

## How it was checked

By hand: 12 V over 10 k/10 k gives 6 V, 5.94–6.06 V at ±1%, 600 µA, 3.6 mW each. 10 k over 4.7 k gives 3.837 V, 3.78–3.89 V (±1.37%). R1 for 12 V to 3.3 V with R2 = 10 kΩ is 26.36 kΩ, nearest E96 26.1 kΩ giving 3.32 V. R2 for 5 V to 3.3 V with R1 = 10 kΩ is 19.41 kΩ, E96 19.6 kΩ giving 3.31 V. Vout equal to Vin, or below zero, is refused with a reason.

# Current divider




## What it computes

For a current flowing into two resistors in parallel, any one of the current in the first branch I1, R1 or R2, from the other three. It also gives the current in the second branch, the voltage across both, the spread of I1 from the resistors' tolerance, and when a resistor is solved for, the nearest standard value and the current it would give.

## Source

Ohm's law and Kirchhoff's current law; standard values from IEC 60063.

## What it means

When a current reaches two resistors connected in parallel, it splits: part goes through each branch and the two parts add back up to the whole (Kirchhoff's current law). Both branches have the same voltage across them, so by Ohm's law each takes a current inversely proportional to its resistance — the easier path takes more:

```
I1 = Iin × R2 / (R1 + R2)
I2 = Iin × R1 / (R1 + R2)
```

This is the voltage divider turned inside out, and the place people trip: the current in branch 1 is set by the *other* resistor, R2. With 20 mA into 1 kΩ in parallel with 2 kΩ, the 1 kΩ branch takes two thirds, 13.3 mA, and the 2 kΩ branch one third, 6.7 mA.

Solved for a resistor:

```
R1 = R2 × (Iin − I1) / I1
R2 = R1 × I1 / (Iin − I1)
```

**Where it is used.** Shunts that extend a meter's range (a small resistor in parallel takes most of the current, the meter a known fraction), sharing current between parallel parts, and understanding why a low-value resistor in parallel with a circuit steals most of its current.

## Tolerance

As with the voltage divider, I1 is furthest off when the two resistors miss in opposite directions, so the spread of I1 can exceed the parts' own tolerance. When I1 is the larger share, it is less sensitive: 1 k ‖ 2 k at ±1% moves I1 by only ±0.66%, because most of the error lands in the smaller branch current.

## Assumptions and limits

- **The total current is taken as given**, as if from a current source. In a real circuit fed from a voltage, Iin itself depends on R1 ‖ R2.
- **Two branches.** For more, combine all but one into a single resistance first with the Series/parallel tool.
- I1 must be between 0 and Iin when solving for a resistor.
- The standard value offered is from the one series the chosen tolerance implies.

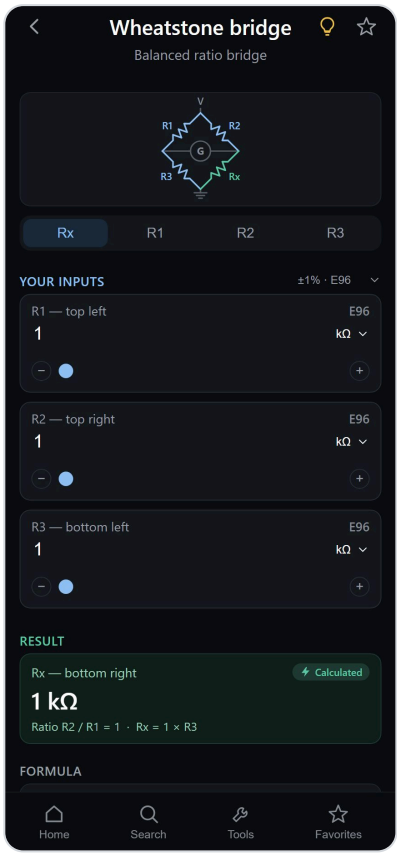
## What it deliberately does not do

- **Power in each branch** is not shown; the voltage across the branches is, and $P = V^2 / R$ from there.

## How it was checked

By hand: 20 mA into 1 k ‖ 1 k is 10 mA each at 10 V; into 1 k ‖ 2 k, 13.33 mA and 6.67 mA at 13.3 V, I1 between 13.2 and 13.4 mA at ±1%. R1 for 15 mA of 20 mA with R2 = 1 kΩ is 333.3 Ω, E96 332 Ω. R2 for 5 mA of 20 mA with R1 = 1 kΩ is 333.3 Ω. I1 above Iin is refused. The review labelled the voltage in the result line, which had been a bare "10 V" with nothing to say what it was.

# Wheatstone bridge




## What it computes

For a balanced Wheatstone bridge, any one of its four resistors from the other three. Solving for the unknown Rx, it also gives the ratio R2 / R1 that multiplies R3 into Rx; solving for one of the fixed arms, the nearest standard value.

## Source

The balance condition follows from Ohm's law applied to the two dividers that make up the bridge. The circuit is Samuel Hunter Christie's (1833), popularised by Charles Wheatstone.

## What it means

A Wheatstone bridge measures an unknown resistance by comparing it with known ones, instead of measuring a current and a voltage.

Four resistors form a diamond. A supply V is connected across the top and bottom corners, so the left side (R1 over R3) and the right side (R2 over Rx) are two voltage dividers on the same supply. A sensitive meter — a galvanometer, G — joins the midpoints of the two sides.

If the two midpoints are at the same voltage, no current flows through the meter and it reads zero. The bridge is then **balanced**, and that happens when both dividers have the same ratio:

```
R1 / R2 = R3 / Rx      so      Rx = R2 × R3 / R1
```

The supply voltage and the meter's own characteristics drop out entirely: the only thing that matters is whether the meter reads zero, which even a crude meter can tell very precisely. That is what made the bridge the standard way to measure resistance accurately.

**In use.** R1 and R2 are fixed and set the range: with R2 / R1 = 10, Rx is ten times R3. R3 is a calibrated adjustable resistor. Turn R3 until the meter reads zero, and Rx = 10 × R3.

The same circuit, left unbalanced, is how strain gauges, pressure sensors and many thermometers are read: a small change in one arm produces a small voltage across the meter's position, measured by an amplifier.

## Assumptions and limits

- **Balance only.** The tool gives the resistance that balances the bridge; it does not compute the voltage or current through the meter when the bridge is off balance.

- **The precision is that of the known arms.** Rx can be no more accurate than R1, R2 and R3 combined; with ±0.1% parts, roughly ±0.3% worst case.
- The arm used as a divisor cannot be zero: R1 when solving Rx, Rx when solving R1, R3 for R2 and R2 for R3.

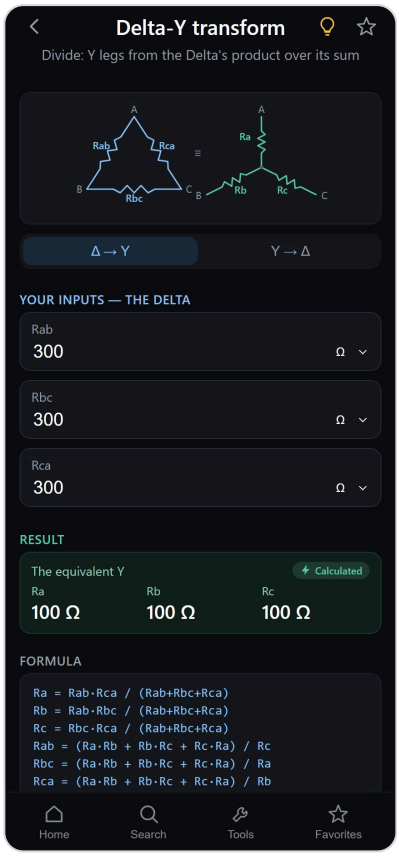
## What it deliberately does not do

- **Unbalanced output** for sensor bridges (the voltage across G as a function of a changing arm) is a different calculation, not done here.
- AC bridges for capacitance and inductance (Maxwell, Schering, Wien) are not covered.

## How it was checked

By hand: R1 = 1 k, R2 = 2 k, R3 = 3 k balance with Rx = 6 kΩ, ratio 2; each of the other three arms solved from the remaining ones gives back 1 k, 2 k and 3 k. R1 = 0 is refused. The review replaced the "nearest standard value" once shown for Rx, which is the resistance being measured rather than a part to buy, with the bridge ratio; dropped "(unknown)" from Rx's label, which stayed even when Rx was an input; and made the supply's V label neutral, which had taken R1's colour.

# Delta-Y transform




## What it computes

Converts three resistors connected as a triangle (Delta, Δ) into the three of an equivalent star (Y), or the reverse, so that the two behave identically between their three terminals.

## Source

Arthur Edwin Kennelly's transform (1899), derived from requiring the resistance between each pair of terminals to be the same in both networks.

## What it means

Three resistors can be connected to three terminals A, B and C in two ways:

- **Delta (Δ)** — a triangle. One resistor between each pair of terminals: Rab, Rbc, Rca.
- **Y (star, also called T)** — three resistors meeting at a common centre point, one leg to each terminal: Ra, Rb, Rc.

Seen only from the three terminals, a Delta and a suitably chosen Y cannot be told apart: the resistance between any two terminals is the same, whatever else is connected. So one can always be replaced by the other.

**Why that is useful.** Many networks are neither series nor parallel anywhere — the classic case is a bridge, five resistors in a diamond with one across the middle. Replacing one of its triangles by the equivalent star turns it into ordinary series and parallel combinations that can be simplified step by step. The same transform turns up in three-phase power, where loads are wired in delta or in star.

## The formulas

**Delta to Y** — each leg of the star is the product of the two Delta sides that touch its terminal, divided by the sum of all three:

```
Ra = Rab × Rca / (Rab + Rbc + Rca)
Rb = Rab × Rbc / (Rab + Rbc + Rca)
Rc = Rbc × Rca / (Rab + Rbc + Rca)
```

**Y to Delta** — each side of the triangle is the sum of the products of the legs taken in pairs, divided by the leg *opposite* that side:

```
Rab = (Ra·Rb + Rb·Rc + Rc·Ra) / Rc
Rbc = (Ra·Rb + Rb·Rc + Rc·Ra) / Ra
Rca = (Ra·Rb + Rb·Rc + Rc·Ra) / Rb
```

A way to remember them: Delta to Y divides, so the star's legs are smaller; Y to Delta multiplies, so the triangle's sides are larger. With three equal parts the Y is a third of the Delta: a 300 Ω triangle is a 100 Ω star.

## Assumptions and limits

- **All three must be greater than zero.** A zero-ohm side or leg joins two terminals directly, and the network is no longer a true three-terminal one.
- **Resistance only.** The transform holds just as well for impedances (capacitors and inductors, in AC circuits), but this tool works in ohms.

## What it deliberately does not do

- **Solving a whole network** (such as the full bridge) is not automated; the tool does the one step that makes the rest ordinary.

## How it was checked

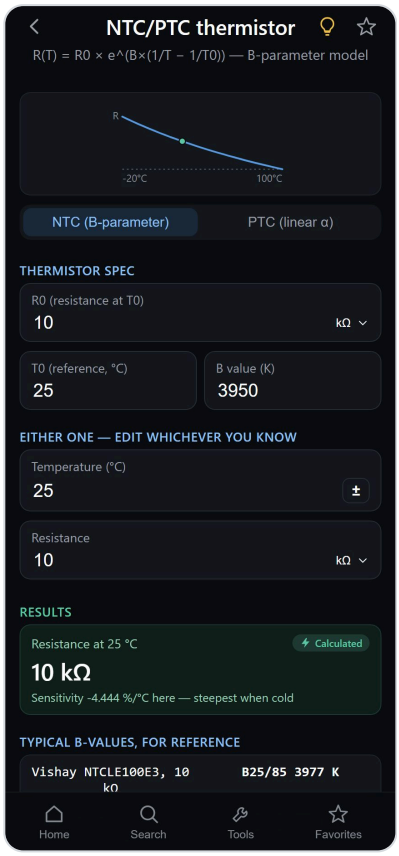
By hand: a 300 Ω Delta gives a 100 Ω Y, and back. Rab = 10, Rbc = 20, Rca = 30 Ω give Ra = 5, Rb = 3.333, Rc = 10 Ω, and those three converted back give 10, 20 and 30 Ω.

# NTC/PTC thermistor

## What it computes

For a temperature-sensing resistor, its resistance at a given temperature or the temperature for a measured resistance, from the datasheet's reference resistance and coefficient; and its sensitivity, in percent per degree, at that point. A curve from −20 to 100 °C shows where the point sits.

- **NTC** — the B-parameter model, used for ordinary NTC thermistors.
- **PTC** — a linear model, for silicon (KTY) sensors and, approximately, platinum RTDs.




## Source

- The B-parameter equation, as NTC datasheets define it. The Vishay NTCLE100E3 datasheet defines B as ln(R1/R2) / (1/T1 − 1/T2) and gives B25/85 = 3977 K for its 10 kΩ part, with a full resistance-temperature table, which is what the tool was checked against.
- The linear PTC model is the definition of a temperature coefficient α.

## What it means

A **thermistor** is a resistor whose resistance changes strongly with temperature, made to be used as a temperature sensor. Put it in a voltage divider, read the voltage with an ADC, work out the resistance, and from that the temperature.

- **NTC** (negative temperature coefficient): resistance falls as temperature rises, steeply — about 4% per °C near room temperature for a typical part. A 10 kΩ NTC is 10 kΩ at 25 °C, about 33 kΩ at 0 °C and about 1 kΩ at 85 °C.
- **PTC** (positive temperature coefficient): resistance rises with temperature. The sensors modelled here rise steadily; a KTY silicon sensor by about 0.7% per °C, a Pt100 platinum RTD by 0.385% per °C.

The datasheet gives two numbers that define the part:

- **R0 at T0** — its resistance at a reference temperature, almost always 25 °C. "10 kΩ NTC" means 10 kΩ at 25 °C.
- **B** (for an NTC), in kelvin — how steep the curve is. A higher B means a larger change per degree. Or **α** (for a linear PTC) in percent per degree.

## The formulas

```
NTC:  R = R0 × e^(B × (1/T − 1/T0))
      T = 1 / (1/T0 + ln(R/R0) / B)
PTC:  R = R0 × (1 + α × (T − T0))
```

T and T0 are in kelvin (°C + 273.15) in the NTC equations. The sensitivity is the slope of the curve as a percentage: $-B / T^2$ for the NTC, which grows steeper the colder it is, and simply α for the linear PTC.

## How good the B model is

B is measured between two temperatures, written as a suffix: **B25/85** is worked out from the resistance at 25 °C and at 85 °C, B25/50 from 25 and 50 °C. The model then goes exactly through those two points and bends away from the real curve outside them. Checked against the Vishay 10 kΩ part's own table, with B25/85 = 3977 K:

| Temperature | Datasheet | B model | Difference |
|---|---|---|---|
| −40 °C | 332.1 kΩ | 412 kΩ | +24%, about 3.5 °C |
| −20 °C | 96.4 kΩ | 107 kΩ | +11%, about 2 °C |
| 0 °C | 32.6 kΩ | 33.9 kΩ | +4%, under 1 °C |
| 85 °C | 1.07 kΩ | 1.07 kΩ | exact |
| 100 °C | 677 Ω | 685 Ω | +1% |

So: use the B value whose temperature pair covers the range you measure, and for accuracy well below 0 °C use the datasheet's table or the Steinhart-Hart equation, which fits three points instead of two.

## Assumptions and limits

- **Self-heating is ignored.** The current used to measure a thermistor also warms it. Datasheets give a dissipation factor (a few mW/°C); keep the measuring current small enough that it adds a fraction of a degree.
- **The part's own tolerance** — typically ±1 to ±5% on R25 and ±0.5 to ±3% on B — adds to the model error.
- **Switching PTCs** (resettable polymer fuses, ceramic PTC heaters and motor protectors) jump sharply by orders of magnitude at a trip temperature. No single smooth formula describes them, and they are not modelled.
- A platinum RTD is not exactly linear; the linear model is within a few tenths of a degree over 0–100 °C, and the full Callendar-Van Dusen equation is needed beyond that.

## What it deliberately does not do

- **Steinhart-Hart** three-coefficient fitting is not offered.

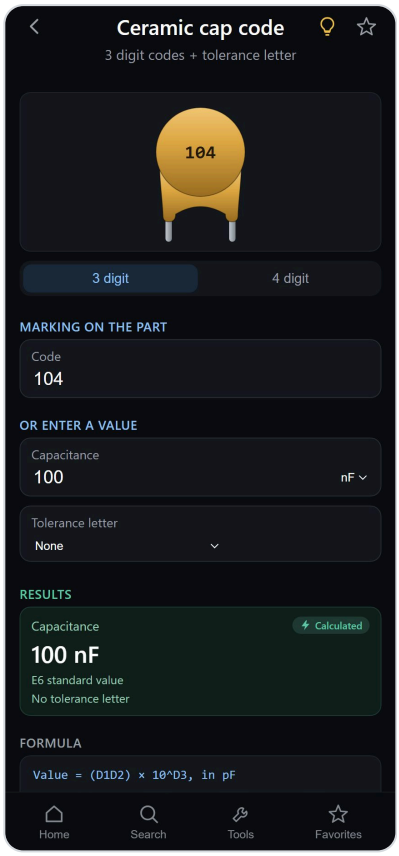
- **The divider around the sensor** (choosing the fixed resistor for best resolution) is the Voltage divider tool's arithmetic.

## How it was checked

Against the Vishay NTCLE100E3 datasheet table as above. At 85 °C the model gives 1.07 kΩ, as the datasheet does, and typing 1.07 kΩ back gives 85.0 °C; the sensitivity at 25 °C is −4.47 %/°C. The linear PTC with R0 = 1 kΩ, α = 0.7 %/°C gives 1.525 kΩ at 100 °C.

The review made the temperature field accept negative values on an iPhone, whose number keypad has no minus key; limited the resistance units to Ω, kΩ and MΩ; and replaced the table of "typical" B values, which had no source, with the Vishay figures.

# Ceramic cap code




## What it computes

The capacitance and tolerance printed on a ceramic capacitor as a three- or four-digit code with an optional letter, and the other way round: the marking for a value. It says whether the value is a standard E-series value.

## Source

- **EIA-198** (RS-198), the code for ceramic capacitors: digits in picofarads, with 8 and 9 as the multipliers ×0.01 and ×0.1 for values under 10 pF.
- **IEC 60062:2016**, for R as the decimal point, its three-character code for capacitors (clause 4.3.2) and the tolerance letters (clause 5: relative tolerances, asymmetrical ones such as Z, and absolute ones in pF).

What was read: IEC 60062's capacitor letter code (clause 4.3) and all its tolerance letters (clause 5, tables 6 to 8) in **GOST IEC 60062-2014**, the official Russian edition of IEC 60062:2004, declared identical to it (IDT) and published free by Standartinform; its tables 3 and 4 match the official IEC 2004 preview word for word. The digit code (104 = 100 nF) is given "according to IEC 60062" in Vishay's ceramic capacitor catalogue, which also writes values below 10 pF as 1p0 to 9p1. EIA-198 is not freely available: the 479 form, with 8 and 9 as ×0.01 and ×0.1, is found on parts and in reference guides but was not read in any standard.

## What the marking means

Ceramic disc and small film capacitors have no room for "100 nF", so the value is printed as a code, always counted in **picofarads** (pF). 1 nF is 1000 pF, and 1 µF is 1 000 000 pF.

- **Three digits** — the first two are the value's figures, the last is how many zeros follow them. 104 is 10 followed by four zeros: 100 000 pF, which is 100 nF or 0.1 µF — the most common capacitor there is. 472 is 4700 pF, 4.7 nF. 220 is 22 pF, not 220 pF.
- **Small values** — below 10 pF the standard puts p where the decimal point would be: 4p7 is 4.7 pF, 1p0 is 1 pF. Parts are also seen with R in its place (4R7), or with a last digit 9 meaning ×0.1 and 8 meaning ×0.01, the EIA practice: 479 is 47 × 0.1 = 4.7 pF. The tool reads all three and writes the standard's 4p7.
- **Four digits** — three figures and a multiplier, for closer values: 1002 is 10 000 pF, 10 nF.
- **A letter after the digits** is the tolerance:

| Letter | Below 10 pF | 10 pF and above |
|---|---|---|
| B | ±0.1 pF | ±0.1% |
| C | ±0.25 pF | ±0.25% |
| D | ±0.5 pF | ±0.5% |
| F | ±1 pF | ±1% |
| G | ±2 pF | ±2% |
| J | | ±5% |
| K | | ±10% |
| M | | ±20% |
| Q | | −10% / +30% |
| T | | −10% / +50% |
| S | | −20% / +50% |
| Z | | −20% / +80% |

Small capacitors use a tolerance in picofarads because a percentage of a few picofarads would be a fraction of the stray capacitance of the leads. Q, T, S and Z are uneven tolerances, for parts whose value is mostly guaranteed not to be too low; Z is typical of cheap high-capacitance ceramics. Some makers use letters of their own outside the standard — Vishay prints Y for −20/+50% and P for −0/+100% — which the tool does not read.

So **473K** is 47 nF ±10%, **104M** 100 nF ±20%, **220J** 22 pF ±5%.

**Other markings on the part.** A code such as **X7R**, **X5R**, **C0G** or **NP0** is the dielectric, which says how the capacitance changes with temperature and voltage — C0G/NP0 barely at all, X7R by ±15% over −55 to +125 °C, Y5V by much more. A number followed by V, or a code such as 1H or 2A, is the voltage rating. Neither is part of the value code.

## Assumptions and limits

- **The value is nominal.** Class 2 ceramics (X7R, X5R, Y5V) lose capacitance with DC voltage applied: a 10 µF X5R part can be well under half of that at its rated voltage. The code says nothing about it; the datasheet's DC-bias curve does.
- **SMD ceramic chips are almost never marked**: they are too small, and the value is only on the reel. The code is mostly seen on through-hole discs and on film capacitors.
- A value typed in is encoded with p below 10 pF (4p7), as the standard writes it; 4R7 and 479 mean the same and are read as well.
- **Standard value.** The tolerance letter picks the one series the value is checked against: J (±5%) E24, K (±10%) E12, M (±20%) E6, as for resistors. With no letter, or an uneven one (Q, T, S, Z), it is E6, the

series most ceramics are stocked in. Below 10 pF, B to G give a tolerance in picofarads, which implies no series, and the tool says so.

## What it deliberately does not do

- **Film capacitor markings** such as 4n7 and the voltage codes have the Film code tool; **SMD capacitor** two-character codes have their own tool.

## How it was checked

104 → 100 nF, 473K → 47 nF ±10%, 220J → 22 pF ±5%, 106M → 10 µF ±20%, 4R7 → 4.7 pF, 8R2C → 8.2 pF ±0.25 pF.

The review found three faults, now fixed. Codes ending in 8 or 9 were read as powers of ten, so 479 came out as 47 mF instead of 4.7 pF; they now follow EIA-198. B, C and D above 10 pF were read as picofarads, so 101D showed ±0.5 pF instead of ±0.5%. And the units offered ran to millifarads and farads, which no ceramic capacitor reaches; they are now pF, nF and µF. The drawing now shows the marking as typed (479 stays 479 rather than becoming 4R7).

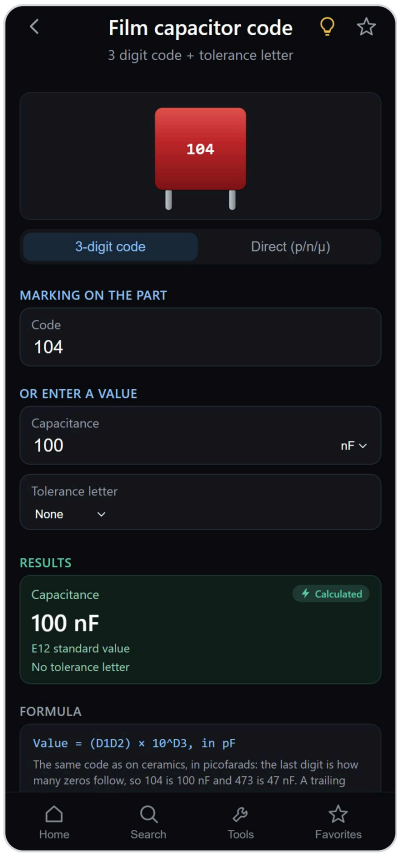
The re-walk checked the value against the tolerance's own series rather than the coarsest one it happened to be in: 182 is "Not in E6 — nearest is 1.5 nF", 182K an E12 value, 479C "no E-series implied".

A reading of the standard's text (GOST IEC 60062-2014, identical to the 2004 edition) and Vishay's catalogue then made four changes: the picofarad letters apply below 10 pF, not at 10 pF itself (100C is ±0.25%, 8p2C ±0.25 pF); the uneven letters Q, T and S were added; values below 10 pF are written 4p7, the standard's form; and 4p7 is read.

# Film capacitor code

## What it computes

The capacitance and tolerance printed on a film capacitor, in either of the two ways film parts are marked — the 3-digit picofarad code or the letter code (4n7) — and the other way round: the marking for a value.




## Source

- **EIA-198** for the 3-digit code, the same as on ceramic capacitors.
- **IEC 60062:2016** for the letter code (the "RKM" code: p, n, µ in place of the decimal point, clause 4.3.1) and the tolerance letters (clause 5).

The letter code (clause 4.3) and the tolerance letters (clause 5, table 6) were read in **GOST IEC 60062-2014**, the official Russian edition of IEC 60062:2004, declared identical to it (IDT) and published free by Standartinform; its tables 3 and 4 match the official IEC 2004 preview word for word. TDK's film capacitor marking guide says its parts use "the coded forms specified in IEC 60062:2004". EIA-198 is not freely available and was not read.

## What the marking means

Film capacitors — polyester (PET), polypropylene (PP) and similar, usually a small rectangular box with two leads — carry their value in one of two ways.

**The 3-digit code**, in picofarads. The first two digits are the figures, the last how many zeros follow: 104 is 100 000 pF = 100 nF, 473 is 47 nF, 102 is 1 nF. As on ceramics, a last digit of 9 or 8 means ×0.1 or ×0.01 for values under 10 pF, though film parts that small are rare.

**The letter code.** The unit letter stands where the decimal point would be:

- p = picofarad, n = nanofarad, µ (often written u) = microfarad
- 4n7 = 4.7 nF, n33 = 0.33 nF (330 pF), µ1 or u1 = 0.1 µF (100 nF), 2µ2 = 2.2 µF, 100p = 100 pF

It is the same idea as 4k7 for a 4.7 kΩ resistor: no decimal point to rub off or be misread.

**The tolerance letter** that often follows either form:

| Letter | Tolerance |
|---|---|
| F | ±1% |
| G | ±2% |
| H | ±3% |
| J | ±5% |
| K | ±10% |
| M | ±20% |

**H is where the sources disagree.** The standard's text gives H = ±3% (IEC 60062:2004, table 6, in its official identical GOST edition). WIMA, TDK and KEMET use H for ±2.5% in their part numbers and marking guides — TDK while stating it follows IEC 60062:2004 — and WIMA also uses F for ±1.5% and E for ±1%. The tool follows the standard's text, ±3%; on a WIMA, TDK or KEMET part, check the datasheet.

**The voltage code.** Many film capacitors add a two-character voltage code in front of the value, per the same EIA scheme: 1H = 50 V, 2A = 100 V, 2E = 250 V, 2G = 400 V, 2J = 630 V. So **2A104J** is a 100 V, 100 nF, ±5% part. Others print the voltage plainly (63 V, 400 V=).

## Assumptions and limits

- **Plain decimals** such as ".1" or "0.1" on older and larger parts mean microfarads (0.1 µF); the tool reads only the two coded forms.
- **The voltage code is not read** from the marking; type the part of the marking that follows it.
- A value typed in is encoded with the unit that gives the shortest marking; n33 and 330p mean the same.
- **Standard value.** The tolerance letter picks the one series the value is checked against: J (±5%) E24, K (±10%) E12, M (±20%) E6, G and H E48, F E96. With no letter it is E12, since film parts are mostly ±10% and stocked in E12 values.

## What it deliberately does not do

- **Ceramic codes with the pF-based tolerance letters** (B, C, D) are the Ceramic code tool; film capacitors are not made small enough to need them.
- **Electrolytic** capacitors print their value and voltage in plain text.

## How it was checked

104J → 100 nF ±5%, 473K → 47 nF ±10%, 102H → 1 nF ±3%, 4n7 → 4.7 nF, 4N7J → 4.7 nF ±5%, n33 → 330 pF, u1 and µ1 → 100 nF, 2u2K → 2.2 µF ±10%, 100p → 100 pF.

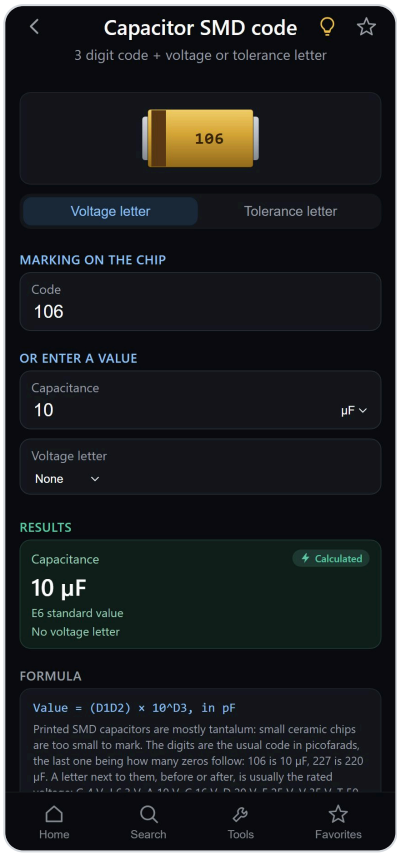
The review found the tolerance table borrowed from the inductor tool, which had no H, a common film tolerance; added it, first as ±2.5% from WIMA's datasheets, then as ±3% once the standard's own table was read. It made the unit letter case-blind (4N7 was rejected), applied the 8 and 9 multipliers and the R-code

length fix from the ceramic tool, limited the units to pF, nF and µF, and made the drawing show the marking as typed.

The re-walk checked the value against the tolerance's own series: 562 is an E12 value, 912J an E24 one, 562M "Not in E6 — nearest is 4.7 nF". A value read from a marking now shows in its own unit, n33 as 330 pF.

# SMD cap code

(formerly Capacitor SMD code)




## What it computes

The capacitance printed on a surface-mount capacitor, usually a tantalum part, with the letter printed next to it read either as the rated voltage or as the tolerance; and the other way round, the marking for a value.

## Source

- The capacitance digits are the EIA-198 picofarad code, as on ceramic parts.
- The marking layout and voltage letters from the **Kyocera-AVX TAJ** tantalum datasheet: "227 A" is 220 µF at 10 V (A = 10 V) on its larger cases; on its small cases the letter sits above the digits, "J" over "106" for 10 µF at 6.3 V. The rest of the voltage letters are the EIA tantalum code as the makers print it; the EIA document itself was not read.
- The **KEMET T491** datasheet lists what its marking holds: polarity band, picofarad code, rated voltage and a date code.

## What the marking means

Most small ceramic chip capacitors carry no marking at all — they are too small to print on, and the value is only on the reel. The SMD capacitors that are marked are mostly **tantalum** (and polymer) capacitors, which are larger and come in moulded cases.

**The digits** are the capacitance in picofarads, as on a ceramic disc: the last digit is how many zeros follow the first two. 106 is 10 000 000 pF = 10 µF; 227 is 220 µF; 475 is 4.7 µF.

**The letter** is, on most tantalum parts, the **rated voltage** — the most it may see continuously:

| Letter | Rated voltage |
|---|---|
| G | 4 V |
| J | 6.3 V |
| A | 10 V |
| C | 16 V |
| D | 20 V |
| E | 25 V |
| V | 35 V |
| T | 50 V |

It may be printed after the digits (227A) or before them (J106, often on a line of its own above); the tool reads both. Some parts instead use a **tolerance** letter (K ±10%, M ±20%). The two tables share letters — G and J mean 4 V and 6.3 V in one, ±2% and ±5% in the other — so the marking alone cannot say which is meant; the datasheet can, and the pill chooses.

**Polarity.** A tantalum capacitor is polarised: the band or bevel at one end marks the **positive** terminal — the opposite of an aluminium electrolytic, whose stripe marks the negative. Fitted backwards, a tantalum can fail short and burn.

## Assumptions and limits

- **Derate the voltage.** Tantalum ($MnO_2$) capacitors are usually run at no more than half their rated voltage, to survive turn-on surges; the letter gives the rating, not the voltage to design for.
- **Makers' own marks** — date codes, logos, ID codes — share the same face and are not read.
- A value typed in is encoded with R below 10 pF; tantalum parts never go that low.
- **Standard value.** A tolerance letter picks the one series the value is checked against (K, ±10%, E12; M, ±20%, E6). A voltage letter says nothing about tolerance, so the value is checked against E6, the series tantalum and polymer capacitors are stocked in.

## What it deliberately does not do

- **The two-character SMD capacitor code** (a letter for the figures and a digit for the multiplier, IEC 60062 Annex B) is not read.
- **Ceramic discs and film parts** have their own code tools.

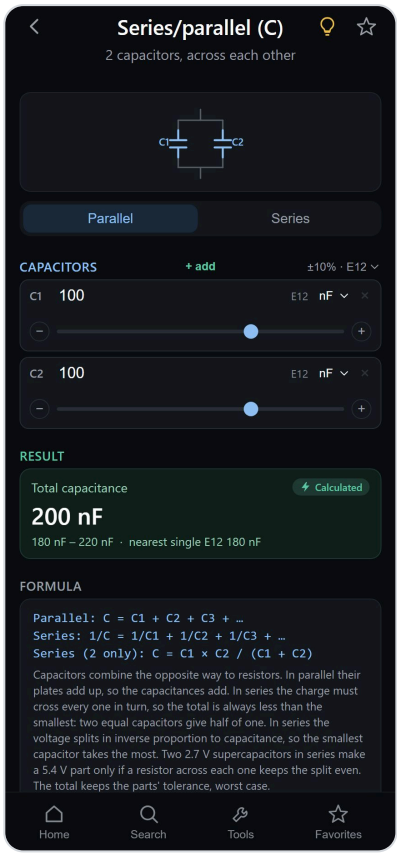
## How it was checked

Against the AVX datasheet's own examples: 227A reads 220 µF at 10 V, J106 and J 106 read 10 µF at 6.3 V. Also 475V → 4.7 µF at 35 V, 106K → 10 µF ±10%, 226M → 22 µF ±20%.

The review made the tool read a voltage letter printed before the digits (J106 was rejected), applied the 8 and 9 multipliers and R-code length fix from the ceramic tool, limited the units to pF, nF and µF, and made the drawing show the marking as typed.

The re-walk checked the value against the tolerance's own series (156A is an E6 value, 825K an E12 one), showed a value read from a code in its own unit (471 as 470 pF) and put a space in "10 V".

# Series/parallel (C)




## What it computes

The total capacitance of two to four capacitors in parallel or in series, the range it can take given their tolerance, and the nearest single standard value. Each capacitor has a slider that steps through standard values.

## Source

The charge relation Q = C × V applied to parts sharing a voltage (parallel) or a charge (series); standard values from IEC 60063.

## What it means

Capacitors combine the **opposite way to resistors**.

**In parallel** — side by side between the same two points. Each has the same voltage across it and stores its own charge, so the charges add, and so do the capacitances — it is as if the plates were one larger plate:

```
C = C1 + C2 + C3 + …
```

A 100 nF and a 10 µF in parallel make 10.1 µF. This is how a board gets both a large capacitance for slow supply dips and a small one for fast noise: in parallel they add, and each does the part of the job it is good at.

**In series** — end to end. The same charge has to appear on every capacitor in turn, and their voltages add up to the total, so it is the reciprocals that add:

```
1/C = 1/C1 + 1/C2 + 1/C3 + …
C = C1 × C2 / (C1 + C2)       for two
```

The total is always less than the smallest one. Two equal capacitors give half of one; 1 µF in series with 10 µF gives 909 nF, a little less than the 1 µF.

## Voltage in series

The reason to put capacitors in series is usually voltage: two 2.7 V supercapacitors in series make a part that can be charged to 5.4 V. But the voltage does not split evenly on its own. With the same charge on each,

```
V1 / V2 = C2 / C1
```

so the **smaller** capacitor takes the **larger** share of the voltage. Real capacitors differ by their tolerance and by their leakage, so one of them can be pushed past its rating. Series capacitors that are meant to share a voltage — supercapacitors, electrolytics in a high-voltage supply — get a resistor across each one, or an active balancing circuit, to hold the split even.

## Tolerance of the total

As for resistors, if every part is off by the same fraction in the same direction the total is off by exactly that fraction, so the worst-case range is the total × (1 ± t). Many capacitors have wide tolerances — ±10% or ±20% for ceramics and electrolytics, and some ceramics +80/−20% — so the range is often the more important number.

## Assumptions and limits

- **Up to four capacitors**, all in series or all in parallel.
- **Nominal values.** Class 2 ceramics lose capacitance with applied DC voltage and with temperature; electrolytics lose it as they age. The total is only as good as the values typed in.
- Leakage, ESR and inductance are ignored.

## What it deliberately does not do

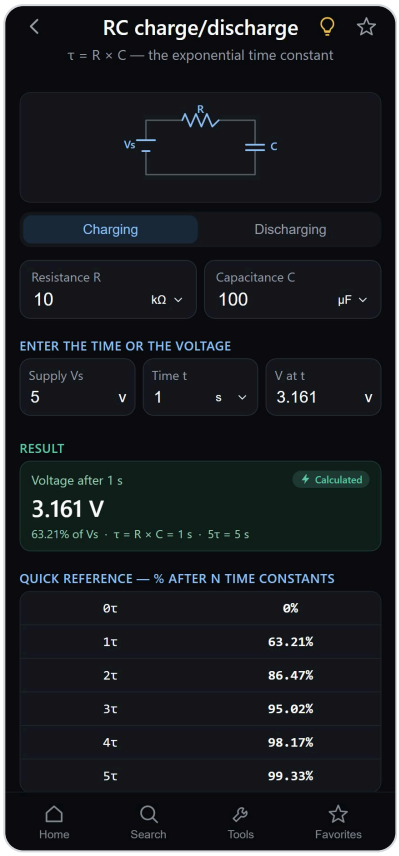
- **Balancing resistor values** for series capacitors are not calculated.
- **Resistors** have their own series/parallel tool.

## How it was checked

By hand: 100 nF + 100 nF in parallel is 200 nF, in series 50 nF; 1 µF and 10 µF in series give 909.1 nF; two 2.7 F supercapacitors in series give 1.35 F. The ranges are the total × (1 ± t).

# RC time constant

(formerly RC charge/discharge)




## What it computes

For a capacitor charging from a supply through a resistor, or discharging through one: the time constant τ, the voltage after a given time, or the time it takes to reach a given voltage. A table gives the percentage reached after one to five time constants.

## Source

The solution of the circuit's differential equation, from Ohm's law and the capacitor law I = C × dV/dt. No standard is involved.

## What it means

A capacitor stores charge, and its voltage rises as the charge builds up. When it charges through a resistor, the current is largest at the start, when the capacitor is empty and the whole supply voltage is across the resistor. As the capacitor fills, less voltage is left across the resistor, the current falls, and the charging slows. So the voltage does not climb in a straight line but bends over, approaching the supply ever more slowly.

The pace is set by one number, the **time constant**:

$$\tau = R \times C$$

10 kΩ and 100 µF make τ = 1 second. In each time constant the capacitor covers 63% of the distance that is still left:

| After | Charging, % of Vs | Discharging, % of V0 left |
|---|---|---|
| 1 τ | 63.2% | 36.8% |
| 2 τ | 86.5% | 13.5% |
| 3 τ | 95.0% | 5.0% |
| 4 τ | 98.2% | 1.8% |
| 5 τ | 99.3% | 0.7% |

It never quite arrives: mathematically the curve only approaches Vs. In practice **5τ** counts as fully charged or discharged.

**Discharging** is the same curve upside down: a charged capacitor connected across a resistor, with no supply, loses 63% of what is left in each τ.

## The formulas

```
Charging:     V(t) = Vs × (1 − e^(−t/τ))      t = −τ × ln(1 − V/Vs)
Discharging:  V(t) = V0 × e^(−t/τ)            t = −τ × ln(V/V0)
```

The second form of each answers the practical question — how long until the capacitor reaches a threshold. For example, how long a power-on reset holds, or when a slowly rising input crosses a logic level: charging to 2/3 of Vs takes τ × ln 3 = 1.1 τ, which is where the 555 timer's 1.1 R C comes from.

## Assumptions and limits

- **An ideal source and capacitor.** The supply is taken as having no internal resistance, and the capacitor as having no leakage. A leaky capacitor, such as an electrolytic, levels off below Vs.
- **Nominal R and C.** With a ±20% capacitor the times are ±20% too.
- The target voltage must lie between 0 and Vs: a charging capacitor never reaches Vs, so asking for exactly Vs has no finite answer.

## What it deliberately does not do

- **Switch debouncing and RC timing for logic inputs** have the Debounce / RC timing tool, which adds input thresholds.
- **Energy** stored in the capacitor is the Stored energy tool.

## How it was checked

By hand: 10 kΩ and 100 µF give τ = 1 s; at 1 s a 5 V charge reaches 3.161 V (63.21%); 4.5 V is reached at 2.303 s (ln 10 × τ); discharging from 5 V, 1.839 V is left at 1 s, and 0.5 V is reached at 2.303 s.

The review fixed the layout, which put the voltage result 24 px under the tab bar on the target phone; R and C now share a line, the voltage and time fields share the next, and the results sit in one row. The schematic no longer shows a supply when discharging, and the resistance units are Ω, kΩ and MΩ.

# Capacitor energy




## What it computes

The energy stored in a charged capacitor from its capacitance and voltage, or either of those from the other two; and the charge it holds.

## Source

The energy of a capacitor charged from zero: the work done moving charge onto it against its own rising voltage. A standard result; no standard document is involved.

## What it means

A capacitor stores energy in the electric field between its plates. Charging it takes work, because each bit of charge has to be pushed on against the voltage already there; that work is held, and comes back out when the capacitor discharges.

```
E = ½ × C × V²      in joules, with C in farads and V i
Q = C × V            the charge, in coulombs
```

**The square matters.** Twice the voltage stores four times the energy. A 1000 µF capacitor at 12 V holds 72 mJ — a small spark; the same capacitor at 400 V would hold 80 J. This is why the capacitors in mains power supplies, camera flashes and microwave ovens are dangerous long after the power is removed, and why they are discharged through a resistor before being touched.

**How much is a joule.** One joule is one watt for one second. A 1 F supercapacitor at 2.7 V holds 3.6 J: enough to keep a 10 mW real-time clock circuit running for six minutes.

Rearranged:

```
V = √(2E / C)       the voltage needed to store E in C
C = 2E / V²         the capacitance needed to store E at V
```

## Usable energy

A circuit seldom uses all of it. A device powered from a capacitor stops working when the voltage falls below its minimum, so the energy it can draw is the difference between two voltages:

```
E usable = ½ × C × (V start² − V min²)
```

A supercapacitor charged to 5 V feeding a circuit that needs at least 3 V delivers only $1 - (3/5)^2 = 64\%$ of what it holds. Use the tool twice, at the two voltages, and subtract.

## Assumptions and limits

- **An ideal capacitor.** Leakage slowly drains the stored energy; ESR turns some of it to heat when it is drawn fast.
- **Nominal capacitance.** Class 2 ceramics hold less than their marked value at high voltage, and electrolytics lose capacitance with age.

## What it deliberately does not do

- **Discharge time** through a load is the RC time constant tool's job.
- **Hold-up time** from the usable energy is not calculated directly; it is the usable energy divided by the power drawn.

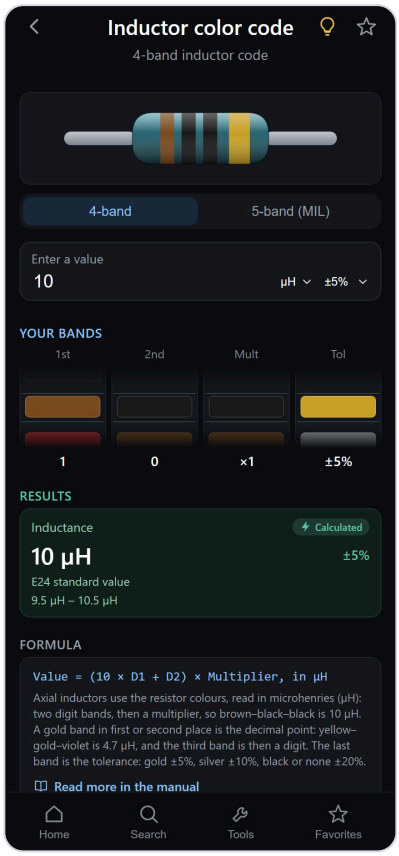
## How it was checked

By hand: 1000 µF at 12 V holds 72 mJ and 12 mC; 1 F at 2.7 V holds 3.645 J; storing 1 J in 100 µF needs 141.4 V; storing 10 J at 400 V needs 125 µF.

# Inductor color code

## What it computes

The inductance and tolerance that the colour bands on an axial (leaded) inductor stand for, and the other way round: the bands for a value. A second pill handles the 5-band form with the wide silver identifier band.




## Source

- **Inductors Inc., colour band guide**: the colours carry the same digits and multipliers as the resistor code, read in microhenries; gold among the digit bands is the decimal point; the tolerance colours are gold ±5%, silver ±10% and black ±20%; the wide silver band is a "military identifier" that does not imply military qualification.
- **Vishay IM series** (MIL-PRF-15305 molded inductors): inductance tolerances of ±1, ±3, ±5, ±10 and ±20%, of which only the last three have a colour.
- IEC 60062, the standard for resistor and capacitor codes, does not cover inductors; the inductor code borrows its colours.

## What the bands mean

Small axial inductors look like fat resistors, and use the same colours — but the value is in **microhenries (µH)**, not ohms.

- **Band 1 and 2** — the two digits (black 0, brown 1, red 2 … white 9).
- **Band 3** — the multiplier, how many zeros follow: black ×1, brown ×10, red ×100; gold ×0.1 and silver ×0.01 for small values.
- **Band 4** — the tolerance: gold ±5%, silver ±10%, black or no band ±20%.

So brown–black–black–gold is 10 × 1 = 10 µH ±5%, and red–red–brown–silver is 22 × 10 = 220 µH ±10%.

**Gold as the decimal point.** A gold band in first or second place is not a digit but the decimal point, and the band after it is then a digit too: yellow–gold–violet is 4.7 µH, gold–yellow–violet is 0.47 µH. The same 4.7 µH can also be written yellow–violet–gold, with gold as a ×0.1 multiplier; both forms are read.

**The 5-band form.** Some parts start with a wide silver band. It is an identifier in the military style and carries no digit; the four bands after it read as above. It does not by itself mean the part is military-qualified.

## Assumptions and limits

- **Tolerance colours are only three.** Precision inductors (±1%, ±2%, ±3%) are not marked with a tolerance colour; their tolerance is on the datasheet or in the part number.

- **Makers vary.** Colour-coded inductors are old enough, and made by enough small makers, that some use their own scheme. When the result looks implausible for the part's size, measure it.
- Inductance alone does not say what current the part can carry; that is on its datasheet.
- **Standard value.** The tolerance band picks the one series the value is checked against: gold (±5%) E24, silver (±10%) E12, black or none (±20%) E6.
- **Units.** The value is shown in nH, µH or mH, whichever reads best: 1000 µH is shown as 1 mH. No colour-coded inductor reaches a henry, so H is not offered.

## What it deliberately does not do

- **SMD inductor markings** are the SMD code tool.
- **Resistor bands** are the resistor Color code tool, which uses the full IEC 60062 tolerance colours.

## How it was checked

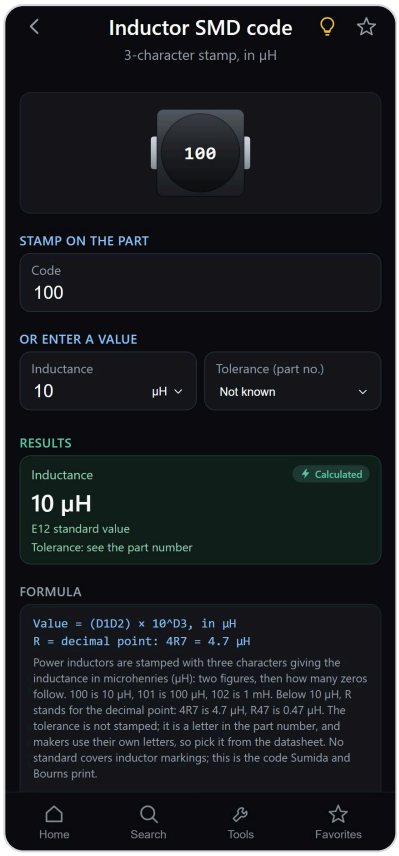
Brown–black–black–gold reads 10 µH ±5%; red–red–brown–silver reads 220 µH ±10%; yellow–gold–violet 4.7 µH; gold–yellow–violet 0.47 µH; typing 4.7 µH gives yellow–violet–gold.

The review found the tolerance list offering the resistor's precision colours, brown ±1% down to grey ±0.01%, which no colour-coded inductor uses; it is now gold, silver, black and none. It added gold as a decimal point in the digit bands, which the tool could not read, and corrected the note's claim that the silver band marks a part as military-qualified.

The re-walk checked the value against the tolerance's own series (4.7 µH with silver is an E12 value), showed a value from the bands in its natural unit (brown–black–red reads 1 mH, not 1000 µH) and dropped H from the units.

PASSIVE COMPONENTS › INDUCTORS

# Inductor SMD code




## What it computes

The inductance that the three-character stamp on a surface-mount power inductor stands for — and the other way round, the stamp for a value. With the tolerance taken from the part number, it also says whether the value is a standard one.

## Source

There is no standard for it. IEC 60062, the marking standard, covers resistors and capacitors only; inductor makers borrow its principle and apply it in microhenries. What the tool follows is what their datasheets show:

- **Sumida CDRH74** lists a "Stamp" for every part: 100 for 10 µH, 101 for 100 µH, 102 for 1.0 mH. The tolerance (M, ±20%) is in the part name, CDRH74NP-100MC, not in the stamp.
- **Bourns SRR1260** shows "100" on the body in its drawing; its tolerances, in the part number, are Y ±30%, M ±20% and K ±10%. It is "available in E12 values".
- **Bourns SRN4018**, the part drawn, shows "4R7" on its core and is 4.0 × 4.0 mm, the core at most 3.6 mm across, 4.55 mm across the terminals.

## What it means

An inductor stores energy in a magnetic field; its inductance, in henries (H), says how strongly it opposes a change of current. Power inductors in DC/DC converters are a few hundred nanohenries to a few millihenries, so the stamp counts in **microhenries** (µH): 1 µH is a millionth of a henry, 1000 nH, and 1 mH is 1000 µH.

- **Three digits** — the first two are the figures, the last is how many zeros follow. 100 is 10 and no zeros: 10 µH. 101 is 100 µH, 102 is 1000 µH, 1 mH. 470 is 47 µH, not 470.
- **R** — below 10 µH, R stands where the decimal point is: 4R7 is 4.7 µH, 1R0 1 µH, R47 0.47 µH (470 nH).
- **The tolerance is not stamped.** It is a letter in the part number, and makers do not agree on the letters — Bourns uses Y for ±30%, a letter the resistor and capacitor standard does not have — so the tool asks for the percentage.

## How the value is worked out

```
Value = D1D2 × 10^D3, in µH        (100 → 10 × 10⁰ = 10 µH)
xRy   = x.y µH                      (4R7 → 4.7 µH)
```

A stamp holds two figures, so a typed value with more is rounded to what the part could carry, and the tool says so. The shortest stamp is R10, 0.1 µH.

## Assumptions and limits

- **Not every maker uses this stamp.** Many small chip inductors carry no marking at all, and some makers print a code of their own or a colour dot. When the reading makes no sense for the part, trust its datasheet.
- **Standard value.** The tolerance picks the one series the value is checked against, as elsewhere in the app: ±5% E24, ±10% E12, ±20% and ±30% E6. With the tolerance not known it is E12, the series Bourns quotes. Power inductors are often sold in E12 values at ±20%, so an E12 value at ±20% can read "Not in E6" and still be a stock part.
- Inductance alone does not say what current the part can carry before its core saturates; that is in the datasheet, as the saturation current.

## What it deliberately does not do

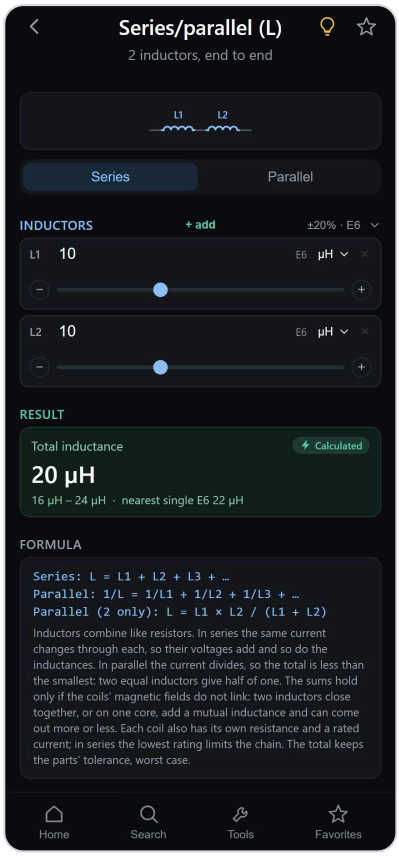
- **Colour-banded inductors** are the Inductor color code tool.
- **A 4-digit code** is not offered: no inductor datasheet read shows one.

## How it was checked

Against the stamps in the Sumida CDRH74 datasheet (100 → 10 µH, 101 → 100 µH, 102 → 1 mH) and the Bourns SRR1260 and SRN4018 drawings (100, 4R7). R47 reads 470 nH, 4.7 µH is stamped 4R7, and a typed "10M" is refused with a note that the tolerance letter belongs to the part number.

The review corrected three things. The tool offered a 4-digit mode, which no standard or datasheet shows. It read a tolerance letter as part of the stamp, where datasheets put it only in the part number, and it had no ±30%. And its note said that "100" means 10 nH on some parts, a claim no source supported; it was removed.

# Series/parallel (L)




## What it computes

The total inductance of two to six inductors, in series or in parallel, by pill.

- **Inputs:** each inductor's value, in nH, µH, mH or H, by typing or with its slider, which steps through the standard values. Rows are added and removed.
- **Tolerance:** ±5, 10 or 20 %, which sets the E-series offered: E24, E12 or E6. Each row names the nearest standard value.
- **Output:** the total, large, with its worst-case spread from the tolerance and the nearest single standard part.
- **Drawing:** the coils end to end, or side by side between two rails.
- **Remembered:** the connection and the tolerance.

Examples: two 10 µH in series, two 22 µH in parallel, 15 µH made from 10 and 4.7 µH, and 100 µH beside 47 µH.

## Source

**Tony R. Kuphaldt, Lessons in Electric Circuits, Volume I, DC, 5th edition (2021), chapter 15**, read in review: - **§15.4:** - in series "the total inductance is the sum of the individual inductors' inductances", Ltotal = L1 + L2 + … Ln; - in parallel it is "less than any one of the parallel inductors' inductances", with the formula "the same form as for calculating parallel resistances". - **§15.5, practical considerations:** - a rated current; - the series resistance of the coil's wire; - mutual inductance when the magnetic fields of two inductors "link" with each other's turns.

## Why the formulas are these

```
series:   L = L1 + L2 + … + Ln
parallel: 1/L = 1/L1 + 1/L2 + … + 1/Ln
two in parallel: L = L1 × L2 / (L1 + L2)
```

Inductance is the voltage across a coil for a given rate of change of its current. In series one current changes through every coil, so their voltages add and so do the inductances. In parallel the current divides between them, so the same change in the total makes less voltage: the total is below the smallest. Inductors combine like resistors and unlike capacitors.

## Assumptions and limits

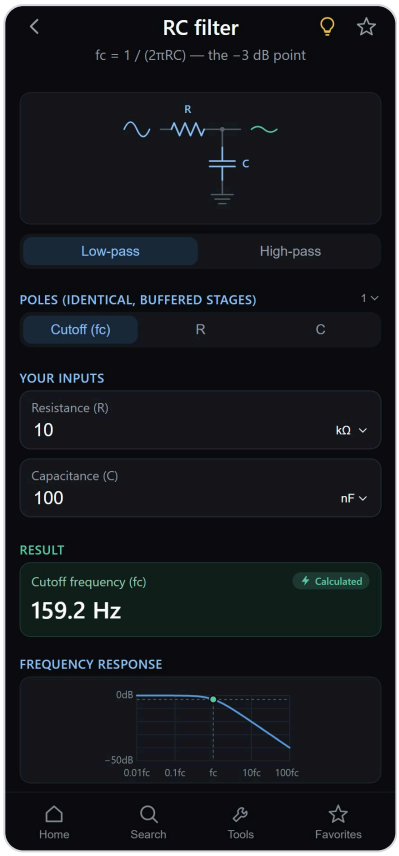
- **No coupling:** the sums hold only if the coils' fields do not link. Two inductors close together, or wound on one core, add a mutual inductance M: in series the total can be more or less than the sum,

depending on how they are wound. That case is not computed here.
- **Ideal coils:** each real one also has a resistance and a rated current. In series the lowest rating limits the chain.
- **The spread** takes every part at the same tolerance, worst case.

## How it was checked

8 cases: - 10 µH + 10 µH in series is 20 µH, ±20 % spanning 16 to 24 µH; - two 22 µH in parallel are 11 µH; - 10 µH + 4.7 µH make 14.7 µH; - 100 µH beside 47 µH is 31.97 µH; - a third 10 µH makes 30 µH; - 10 µH + 10 mH is 10.01 mH; - a negative value is refused; - the remembered connection and tolerance.

# RC filter




## What it computes

The cutoff frequency of a filter made of one resistor and one capacitor — or the resistor or capacitor for a cutoff you want — as a low-pass or a high-pass, with up to six identical stages. It draws the frequency response, and gives the attenuation and phase shift at any frequency you pick.

## Source

No standard is needed: the results follow from Ohm's law and the impedance of a capacitor. The derivation is below.

## What it means

A **filter** passes some frequencies and weakens others. A **low-pass** (resistor in series, capacitor to ground) lets slow signals through and weakens fast ones: it smooths PWM into a steady voltage, or removes noise above the band you care about. A **high-pass** (capacitor in series, resistor to ground) does the reverse: it blocks DC and passes the signal riding on it, as a coupling capacitor does in audio.

- **Cutoff frequency, fc** — where the output has fallen to 70.7% of the input (1/√2). Below it a low-pass passes nearly everything; above it, less and less.
- **Decibels (dB)** — a way of comparing two levels on a scale that suits signals: 20 × $\log_{10}$(output ÷ input). −3 dB is 70.7%, −20 dB a tenth, −40 dB a hundredth.
- **Pole** — one RC stage. Past fc a single pole weakens the signal by a further 20 dB for every tenfold increase in frequency (20 dB per decade); two poles 40 dB, and so on.
- **Phase shift** — how far the output's waveform is behind (lag) or ahead of (lead) the input's, in degrees of a cycle. At fc it is 45°.

## Why the formulas are these

A capacitor's impedance falls with frequency: Z = 1 / (2πfC). The low-pass is a voltage divider of R on top and C below, so its output is the part of the input across C:

```
Vout / Vin = 1 / (1 + j·f/fc),     fc = 1 / (2πRC)
```

At f = fc the real and imaginary parts are equal, so the magnitude is 1/√2 (−3.01 dB) and the phase −45°. Well above fc the magnitude is fc/f: ten times the frequency, a tenth of the output, −20 dB. The high-pass swaps R and C, and its response is the same with f/fc turned upside down.

Solving fc = 1 / (2πRC) for R or C gives the other two forms the tool uses.

With N identical stages each **buffered** (an op-amp between them, so none loads the next), the responses multiply: the dB figures and the phase are N times one stage's. At each stage's fc the total is already −3N dB, so the system's own −3 dB point moves in:

```
f(−3 dB) = fc × √(2^(1/N) − 1)          (low-pass; high-pass divides instead)
```

For two stages that is 0.644 × fc.

## Assumptions and limits

- **Nothing loads the output.** A load in parallel with the output changes the divider: keep it well above R (ten times R or more costs under 10%), or buffer it.
- **The source drives it with little resistance of its own.** A source resistance adds to R and lowers fc.
- **Several poles here means buffered stages.** Unbuffered RC stages in a chain load each other and give a softer knee than this.
- Parts have tolerances: a ±5% R and a ±10% C move fc by up to about 15%.

## What it deliberately does not do

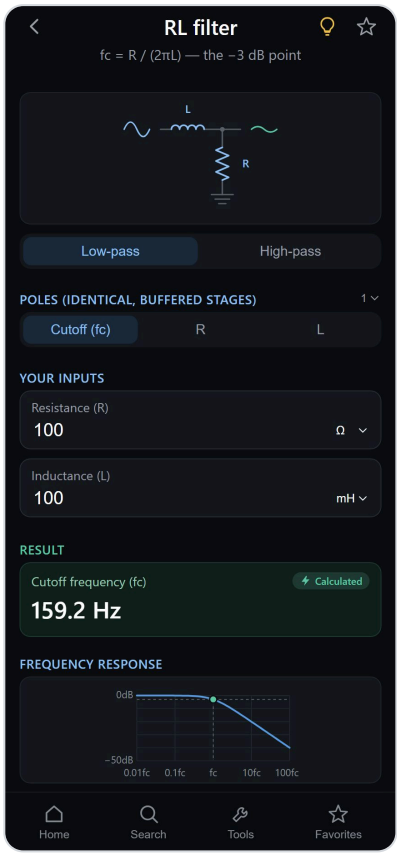
- **RL filters** have their own tool; **op-amp active filters** with gain and sharper shapes (Butterworth, Sallen-Key) are not covered.
- **The time response** — how the output settles after a step — is the RC charge/discharge tool.

## How it was checked

10 kΩ with 100 nF gives 159.2 Hz; at fc the attenuation is −3.01 dB and the phase −45°. Two buffered poles put the system's −3 dB point at 102.4 Hz, 0.644 × fc. 10 kΩ with 1 µF gives 15.92 Hz, and −35.97 dB at 1 kHz. A high-pass of the same parts is 2.13 dB down at 20 Hz. fc = 20 kHz with 1 kΩ needs 7.958 nF.

The review found the frequency to explore opening at 159.2 Hz, a rounded fc, which showed −45.01°; it now starts at fc exactly. The units offered were cut to those RC filters use: Ω to MΩ, pF to µF, Hz to MHz.

# RL filter




## What it computes

The cutoff frequency of a filter made of one resistor and one inductor — or the resistor or inductor for a cutoff you want — as a low-pass or a high-pass, with up to six identical stages; the frequency response, and the attenuation and phase shift at any frequency you pick.

## Source

No standard is needed: the results follow from Ohm's law and the impedance of an inductor. The derivation is below.

## What it means

It is the RC filter's counterpart with an inductor in place of the capacitor (see that section for what a filter, fc, a decibel, a pole and a phase shift are). An inductor's impedance **rises** with frequency, where a capacitor's falls, so the two parts swap places:

- **Low-pass** — inductor in series, resistor to ground. The inductor blocks the fast signals. Often the "resistor" is the load itself: a loudspeaker in a crossover, a circuit fed through a choke.
- **High-pass** — resistor in series, inductor to ground. The inductor shorts the slow signals to ground and lets the fast ones through, as in RF circuits where the series resistance is the 50 Ω of the system.

## Why the formulas are these

An inductor's impedance is Z = 2πfL. The low-pass is a divider of L on top and R below, so its output is the part across R:

```
Vout / Vin = R / (R + j·2πfL) = 1 / (1 + j·f/fc),     fc = R / (2πL)
```

This is the same shape as the RC low-pass, so everything that follows from it is the same: −3.01 dB and −45° at fc, 20 dB per decade beyond, and for N buffered stages a system −3 dB point at fc × √(2^(1/N) − 1). The high-pass swaps R and L and turns f/fc upside down. Solving fc = R / (2πL) for R or L gives the other two forms.

## Assumptions and limits

- **A real inductor has resistance.** Its winding's DC resistance (DCR, in the datasheet) adds to R in a low-pass and flattens the high-pass at low frequencies. For small R — an 8 Ω speaker — it matters.

- **Cores saturate.** Past its rated current an inductor with a ferrite or iron core loses much of its inductance, and fc moves up.
- **Every inductor resonates** with its own winding capacitance at its self-resonant frequency (SRF, in the datasheet); well below it the model holds, near it the part stops behaving as an inductor.
- **Nothing loads the output**, and several poles mean buffered stages, as for the RC filter.

## What it deliberately does not do

- **RC filters** have their own tool; **LC filters** (both an inductor and a capacitor, 40 dB per decade from one pair) are not covered.
- **Loudspeaker crossover design** beyond one coil — the speaker's impedance is not a pure 8 Ω across the band — is not covered.

## How it was checked

100 Ω with 100 mH gives 159.2 Hz; at fc −3.01 dB and −45°. An 8 Ω woofer crossed over at 2.5 kHz needs 509.3 µH. A 10 µH choke into 10 Ω is a 159.2 kHz low-pass, −16.07 dB at 1 MHz. 100 nH to ground in a 50 Ω system is a high-pass at 79.58 MHz.

The review made the same corrections as on the RC filter: the frequency to explore opens at fc exactly, Low-/High-pass take the mode pills, and the resistor units are ohms to megohms. Inductors keep nH to H and frequencies up to GHz, which RF inductors and supply chokes really reach.

PASSIVE COMPONENTS › REFERENCE

# SMD package sizes




## What it computes

A reference table of the common surface-mount chip sizes: each one's imperial and metric name, its length and width in inches and millimetres, and what it is typically used for. It can be searched by either name or by a dimension.

## Source

The body sizes are those manufacturers' datasheets give:

- **Panasonic ERJ** thick-film chip resistors — 01005 to 2512, e.g. 01005 0.40 × 0.20 mm, 0805 2.00 × 1.25 mm, 1210 3.20 × 2.50 mm, 2512 6.40 × 3.20 mm.
- **Vishay D/CRCW e3** (April 2026) — the same sizes, and 1218, 3.2 × 4.6 mm; 2512 is 6.3 × 3.15 mm there.
- **KEMET C1002 X7R** (September 2026) ceramic capacitors — every size in millimetres and in inches (1210: 3.20 mm (0.126″) × 2.50 mm (0.098″)), and 1808, 4.70 × 2.00 mm.
- **Murata LQM2HP** inductors — 1008 (metric 2520), 2.5 × 2.0 mm.

Where makers differ by a few hundredths of a millimetre, the metric name's nominal size is given.

## What it means

A surface-mount chip — resistor, capacitor, inductor — is a small block with metal ends, soldered flat onto pads on the board. Its **size code** names its length and width:

- **Imperial** (the usual one in the US and in most catalogues): four digits, two for the length and two for the width, in hundredths of an inch. 0805 is about 0.08″ × 0.05″.
- **Metric**: the same idea in tenths of a millimetre. The 0805 body is metric 2012, about 2.0 × 1.2 mm.

The two collide: **imperial 0402** is 1.0 × 0.5 mm, but **metric 0402** is 0.4 × 0.2 mm, the part imperial calls 01005 — a body six times smaller in area. A bare "0402" in a drawing or a bill of materials is ambiguous until you know which system it is in, which is why every row shows both.

**The names are rounded; the table gives the real size.** 0805's metric name says 1.2 mm wide, but the parts are 1.25 mm. 1210's name says 0.12″ × 0.10″; the parts are 3.2 × 2.5 mm, 0.126″ × 0.098″. The inches here are the millimetres converted, as KEMET prints them.

## Assumptions and limits

- **Sizes are nominal.** Each has a tolerance of ±0.05 mm on the smallest to about ±0.2 mm on the larger ones; the height depends on the part and its value, and is not listed.
- **Some parts use a size but not its exact body**: power inductors and electrolytic capacitors have their own case codes, not these.
- The notes on hand-soldering are practical guidance, not a specification.

## What it deliberately does not do

- **Pad (footprint) dimensions** depend on the soldering process and the maker; take them from the part's datasheet or IPC-7351.
- **Power ratings** by size are in the Resistor power tool.

## How it was checked

Every size against the datasheets above. The review found three faults. 1210 was listed as 0.12″ × 0.125″, where the parts are 0.126″ × 0.098″; the code claimed 1210, 1812 and 2512 were 0.125″ wide by exception, which the datasheets do not support for 1210. 0805 was listed as 1.2 mm wide, the metric name, where the parts are 1.25 mm. And 1806 (4516), found only in online size charts and in no maker's datasheet read, was replaced by 1808, from KEMET's; 1218, a common power-resistor size, was added from Vishay's. The inches in the Resistor power table were aligned to the same figures.

# Diode forward voltage




## What it computes

Nothing is calculated: it is a reference. It shows which way round a diode conducts, and lists the forward voltage that common real diodes drop, each at the current its datasheet quotes it for.

## Source

- **Vishay 1N4148** (Rev. 1.6, 2024): VF ≤ 1 V at 10 mA; its curves show VF at 25, 75 and 150 °C.
- **Vishay 1N4001–1N4007**: VF ≤ 1.1 V at 1.0 A.
- **Vishay 1N5817–1N5819**: VF ≤ 0.450 / 0.550 / 0.600 V at 1.0 A, and 0.750 / 0.875 / 0.900 V at 3.1 A.
- **Diodes Inc. BAT54**: VF ≤ 240 mV at 0.1 mA, 320 mV at 1 mA, 400 mV at 10 mA, 500 mV at 30 mA, 800 mV at 100 mA.
- **Taitron 1N34A** (germanium): VF ≤ 1.0 V at 5 mA.

These are guaranteed maximums; a typical part drops somewhat less.

## What it means

A **diode** lets current through one way only, like a valve. Its two leads are the **anode (A)** and the **cathode (K)**, marked by the band on the body; in the symbol the triangle points from A to K, the way current flows.

- **Forward bias** — the anode more positive than the cathode. Once the voltage across the diode reaches about its **forward voltage, Vf**, it conducts. From then on its voltage hardly changes: a little more voltage means a lot more current. So a diode must never be put straight across a supply; a resistor in series sets the current, taking the rest of the supply's voltage.
- **Reverse bias** — the cathode more positive. The diode blocks, apart from a tiny leakage current, up to its reverse rating (VRRM). Past it the diode breaks down and conducts, which destroys an ordinary diode. A Zener diode is made to work in that region, as a voltage reference.

**Vf is not one number.** It rises with the current — the BAT54 goes from 0.24 V at 0.1 mA to 0.8 V at 100 mA — and falls as the diode warms, by about 2 mV per °C for silicon. The familiar "0.6–0.7 V" is a rule of thumb for a silicon diode at a few milliamps; a 1N4007 carrying its rated 1 A may drop 1.1 V. That is why each figure here comes with its current.

**Which kind to choose.** A Schottky drops less (useful where every tenth of a volt counts, as in a supply's reverse-polarity protection) but leaks more when blocking. A silicon rectifier such as the 1N4007 blocks up to 1000 V. A small-signal diode such as the 1N4148 switches fast, at small currents.

## Assumptions and limits

- The figures are the datasheets' maximums at 25 °C, at the stated current. Between the listed currents, read the datasheet's VF–IF curve.
- Other makers' versions of the same part numbers have slightly different limits.

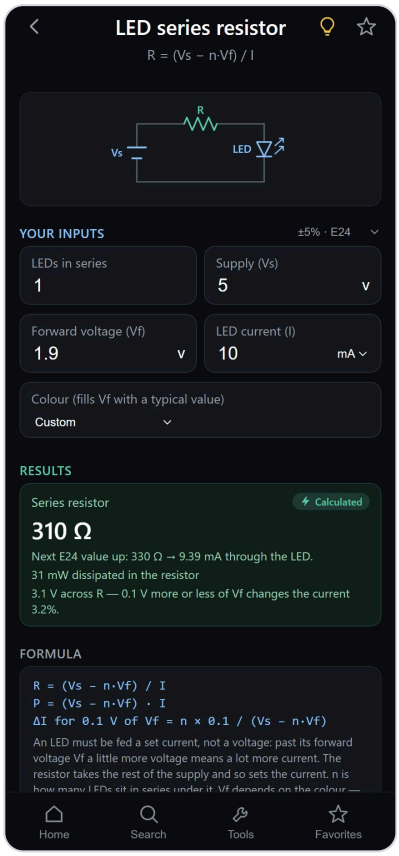
## What it deliberately does not do

- **Working out the series resistor** is the LED series resistor tool, which handles any diode's Vf.
- **Zener regulation** will have its own tool.

## How it was checked

Every figure against the datasheet named. The review found the table giving bare ranges with no current ("silicon ≈0.6–0.7 V"), which a 1N4007 exceeds at its rated current, and "germanium ≈0.2–0.3 V", which its datasheet does not state; it now gives each part's figure with its current. It also found the drawing putting a diode straight across a battery with no resistor — the circuit that burns a diode out — and the note saying Vf stays "pinned regardless of current"; both were corrected.

# LED series resistor




## What it computes

The resistor that sets an LED's current from a given supply — for one LED or several in series — the power it dissipates, the next standard value up and the current that value gives, and how much the current moves if the LED's forward voltage is 0.1 V off.

**Standard values.** The resistors' tolerance, with its E series, is picked in the "Your inputs" heading above the inputs, as in the Passive components tools: ±0.1 % to ±20 %, ±5 % being E24 and ±1 % E96; it sets the series the next value up comes from (E24 by default).

## Source

- The formula is Ohm's law, applied to the voltage the LEDs leave.
- The colour presets are typical forward voltages from **Kingbright's WP7113** 5 mm LED datasheets (typical / maximum): red 1.9 / 2.3 V and yellow 1.95 / 2.4 V at 10 mA, standard (GaP) green 2.0 / 2.4 V at 10 mA; bright (InGaN) green 3.3 / 4.1 V, blue 3.3 / 4.0 V and white 3.3 / 4.0 V at 20 mA.
- Infrared from **Vishay's TSAL6200**: 1.35 V typical, 1.6 V maximum, at 100 mA; its forward voltage falls 1.8 mV per kelvin.

## What it means

An LED is a diode that gives light (see Diode forward voltage). Like any diode, once past its **forward voltage, Vf**, its current rises steeply with the voltage, so it cannot be fed from a voltage directly: it would take as much current as the supply can give and burn out. A **series resistor** takes up the rest of the supply's voltage and so sets the current:

$$R = (Vs - n \cdot Vf) / I$$

with **n** LEDs in series, each dropping Vf, and **I** the current you want — typically 10–20 mA for an indicator, and the datasheet's maximum is the limit. The resistor turns $P = (Vs - n \cdot Vf) \times I$ into heat; choose its power rating from that (see Resistor power).

**The colour sets Vf.** Red and yellow LEDs are about 1.9–2 V; blue, white and the bright green ones about 3.3 V. The old, dimmer green is about 2 V — the tool offers both greens because they differ by more than a volt.

**Headroom.** The voltage the resistor is left with, $Vs - n \cdot Vf$, is what fixes the current. Vf varies from one LED to the next (the datasheets give a typical and a maximum a few tenths of a volt apart) and with temperature. With volts to spare, that hardly matters; with a few tenths left, it moves the current a lot. The tool shows how

much a 0.1 V change would move it, and flags it in red above 20%. A white LED on 3.3 V has no headroom at all.

**Rounding up.** The standard value offered is the next one up in the chosen series, never the nearest one down, so the LED gets at most the current asked for.

## Assumptions and limits

- Vf is taken as fixed at the current asked. It really rises slightly with current, so the true current differs a little from the figure.
- **Several LEDs in parallel on one resistor** do not share the current evenly: the one with the lowest Vf takes most of it and runs hottest. Give each its own resistor.
- High-power LEDs (hundreds of milliamps) are usually driven by a constant-current regulator rather than a resistor, which would waste too much power.

## What it deliberately does not do

- **Brightness** (luminous intensity) is in the LED's datasheet, not calculated.
- **Constant-current drivers** are not covered.

## How it was checked

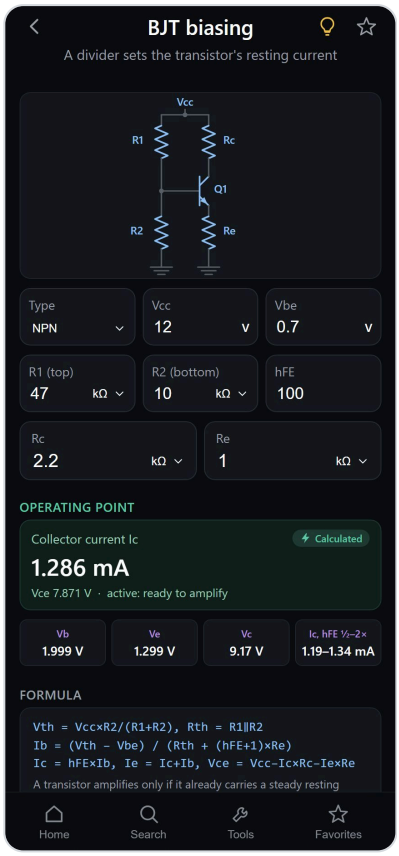
A red LED (1.9 V) on 5 V at 10 mA needs 310 Ω; the next E24 value up, 330 Ω, gives 9.39 mA. 2.0 V at 20 mA from 5 V is exactly 150 Ω, an E24 value. 345 Ω goes up to 360 Ω, not down to 330. Three red LEDs on 12 V at 20 mA need 315 Ω, 330 Ω in E24.

The review found the drawing's LED not joined to the wires either side, the standard value rounded to the nearest (so the current could exceed the one asked), and a green preset of 2.2 V that fits only the old GaP LEDs — a bright green LED is 3.3 V. The presets now follow the datasheets, the value rounds up, and the drawing is laid out like the diode screen's.

ACTIVE & SEMICONDUCTOR DEVICES › TRANSISTORS (BJT)

# BJT biasing

(formerly BJT biasing (voltage divider))




## What it computes

For an NPN or PNP bipolar transistor biased by a voltage divider with an emitter resistor, it gives: - the collector current Ic and the voltage across the transistor (Vce, or Vec for a PNP), which together are the operating point; - whether the transistor is active, saturated or off; - the voltages at the base, emitter and collector; - Ic if the current gain hFE were half or twice the value entered; - a warning when the divider is too weak.

## Source

The analysis is the standard Thevenin treatment of the divider, found in every electronics textbook. Transistor data read from the datasheets themselves: - **Diodes Inc. MMBT3904** (DS30036 rev. 27, 2025), the SOT-23 version of the 2N3904: - hFE at least 40 at 0.1 mA and 70 at 1 mA, 100–300 at 10 mA, at least 60 at 50 mA; - VCE(sat) 0.2 V maximum at 10 mA (with IB = 1 mA); - VBE(sat) 0.65–0.85 V at 10 mA. - **ST 2N3906 PNP** (2003): hFE at least 80 at 1 mA, 100–300 at 10 mA; VCE(sat) 0.25 V maximum at 10 mA.

The tool uses 0.2 V for Vce in saturation.

## What it means

A transistor amplifies only if it already carries a steady resting current. That current, and the voltage across the transistor with no signal, are its **operating point** (or Q-point).

- **The divider** — R1 and R2 divide the supply and hold the base near a fixed voltage, Vth.
- **Vbe** — the emitter sits one base-emitter voltage away from the base: about 0.6–0.7 V for silicon. So the emitter voltage is fixed, Re fixes the emitter current, and Ic is almost the same.
- **hFE** (or β) — the current gain, Ic / Ib. It varies a lot between transistors of one type: 100 to 300 for a 2N3904 at 10 mA. Here it only enters through the small base current. That is why this circuit is used so widely: the "hFE ½–2×" cell shows how little Ic moves when hFE halves or doubles.
- **Regions:**
- **active** — the transistor can amplify;
- **saturated** — fully on, with Vce about 0.2 V;
- **off** (cut off) — no current flows.

Saturated and off are both useless for an amplifier. - **Vce near half the supply** — leaves the most room for the signal to swing both ways. - **Stiff divider** — the current in the divider should be at least 10 times the base current, or the base voltage sags with hFE. This is a rule of thumb, and the tool warns below it. - **PNP** —

the same circuit upside down: the emitter and Re near the supply, the collector and Rc near ground. Switching the type swaps R1 and R2, so the base lands near the supply as a PNP needs.

## Why the formulas are these

The divider seen from the base is a source Vth = Vcc × R2 / (R1 + R2) with a resistance Rth = R1 ∥ R2. Around the loop through the base and emitter,

```
Vth = Ib × Rth + Vbe + (hFE + 1) × Ib × Re
```

so

```
Ib = (Vth − Vbe) / (Rth + (hFE + 1) × Re),   Ic = hFE × Ib
```

For a PNP, Vth is replaced by Vcc − Vth. Then Vce = Vcc − Ic × Rc − Ie × Re.

If that would leave less than the saturation voltage, the transistor is saturated instead. Ic is then (Vcc − 0.2 V) / (Rc + Re). If Vth does not exceed Vbe, it is off.

When (hFE + 1) × Re is much larger than Rth, Ib × hFE ≈ (Vth − Vbe) / Re: hFE drops out. That is the design aim.

## Assumptions and limits

- **Fixed Vbe.** Vbe also falls as the transistor warms, about 2 mV per °C (a textbook figure, not from these datasheets). Re keeps the effect small, but it is not calculated.
- **Constant hFE.** hFE is taken as constant; really it changes with Ic and temperature.
- **DC only.** Signal gain, coupling capacitors and the emitter bypass capacitor are not covered.
- **Simple saturation.** The saturation current neglects the base current's share of the emitter current.

## What it deliberately does not do

- It does not design the divider for a target current; enter values and adjust.
- Small-signal gain and input impedance are not calculated.

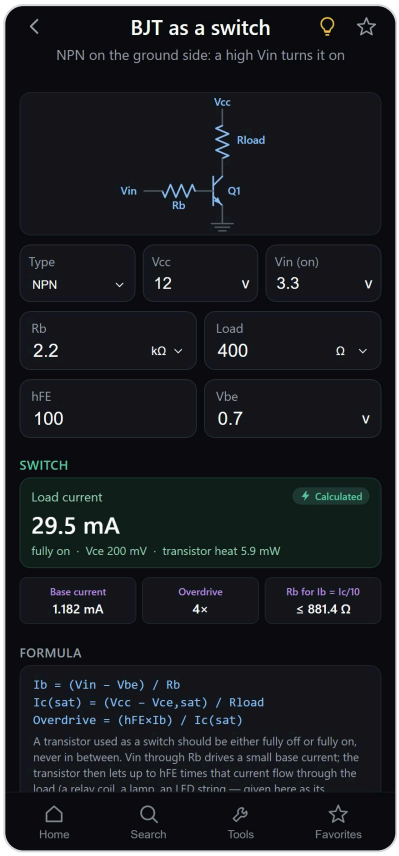
## How it was checked

All on 12 V, with Vbe 0.7 V and hFE 100: - **NPN, 47 k / 10 k divider, Rc 2.2 k, Re 1 k:** - Ic 1.286 mA, Vce 7.871 V, Vb 1.999 V, Ve 1.299 V, Vc 9.17 V; - 1.19–1.34 mA for hFE 50–200. - **Rc 3.3 kΩ:** Vce 6.456 V. - **PNP with the divider swapped (10 k / 47 k):** the same 1.286 mA. - **Rc 10 kΩ:** saturated, Ic (12 − 0.2) / 11 kΩ = 1.073 mA. - **470 k / 100 k divider:** only about 2× the base current, so the tool warns.

The review made these changes: - It put Ic first and large, with Vce and the region in words. - It added the hFE ½–2× robustness cell and the stiff-divider check. - It limited the units to real ones and remembers them. - It replaced the NPN/PNP pills with a Type choice in the fields, so the result stays above the tab bar on a

smaller phone. - It rewrote the note in plain words and added examples. - The tool was renamed BJT biasing, on Pierre's choice, to fit beside the bulb.

# BJT as a switch




## What it computes

For a bipolar transistor switching a load, it gives: - the load current and whether the transistor is fully on, partly on or off; - the voltage left across the transistor and the heat it produces; - the base current; - the overdrive, which is how much spare base drive there is; - the largest base resistor that drives the transistor the way datasheets test it.

The transistor is an NPN on the ground side of the load, or a PNP on the supply side.

## Source

Saturation data read from the datasheets themselves: - **Diodes Inc. MMBT3904** (DS30036 rev. 27, 2025): - VCE(sat) at most 0.2 V at IC = 10 mA with IB = 1 mA, and 0.3 V at 50 mA with 5 mA; - hFE 100–300 at 10 mA; - collector current at most 200 mA. - **ST 2N3906 PNP** (2003): VCE(sat) at most 0.25 V at 10 mA with 1 mA.

The saturation figures are given with the base current at a **tenth** of the collector current. The tool's "Rb for Ib = Ic/10" applies that same condition, and it uses 0.2 V for Vce when the transistor is fully on.

## What it means

A transistor used as a switch should be either fully off or fully on, never in between.

- **Base current** — Vin, through Rb, drives a small current into the base: (Vin − Vbe) / Rb, with Vbe about 0.6–0.7 V.
- **Gain (hFE)** — the transistor then allows up to hFE times that current through the load. The load can be a relay coil, a lamp or an LED string, entered as its resistance.
- **Fully on (saturated)** — the load would accept less current than that, so the transistor drops only about 0.2 V and stays cool. The load current is then set by the load alone: (Vcc − 0.2 V) / R.
- **Partly on** — there is not enough base current. The transistor drops a large voltage and turns the difference into heat. This is the fault to avoid.
- **Overdrive** — hFE × Ib divided by the current the load needs.
- hFE varies about 3:1 between transistors of one type, and falls when cold, so the design needs spare base drive.
- The tool flags overdrive below 2×. That threshold is a rule of thumb.
- The surest design follows the datasheet test condition: base current at a tenth of the collector current, whatever the gain. The "Rb for Ib = Ic/10" cell gives the largest Rb that achieves it.
- **Coils** — a relay or motor coil needs a diode across it. Otherwise the voltage spike when the current stops can destroy the transistor.
- **NPN or PNP:**

- **NPN** — switches the ground side of the load, and turns on with a high Vin.
- **PNP** — switches the supply side, and turns on when its base is pulled low. To turn it off, the base must come back up to Vcc. A 3.3 V or 5 V logic pin cannot do that on a higher supply; drive the PNP's base with an NPN instead.

## Why the formulas are these

```
Ib = (Vin − Vbe) / Rb                (NPN)
Ib = (Vcc − Vin − Vbe) / Rb          (PNP)
Ic(sat) = (Vcc − 0.2 V) / Rload
```

- **Fully on** — if hFE × Ib ≥ Ic(sat), the transistor is saturated and Ic = Ic(sat).
- **Partly on** — otherwise Ic = hFE × Ib, and Vce = Vcc − Ic × Rload.
- **Heat** — the transistor dissipates Vce × Ic.
- **Largest Rb** — Rb = (Vin − Vbe) / (Ic(sat) / 10), and the same with Vcc − Vin − Vbe for a PNP.

## Assumptions and limits

- **A resistive load.** For a coil, use its DC resistance, and fit the diode.
- **Steady state.** Switching speed is not calculated, and neither is the storage time a saturated transistor takes to turn off.
- **Ratings.** The transistor's own limits are not checked: a 2N3904 or MMBT3904 is rated 200 mA. Heavier loads need a bigger transistor or a MOSFET.

## What it deliberately does not do

- MOSFET switches have their own tool.
- Darlington pairs are not covered.

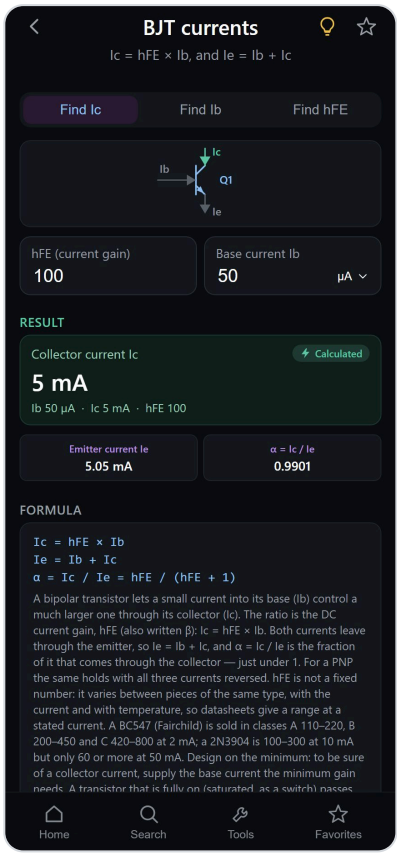
## How it was checked

All with Vbe 0.7 V and hFE 100: - **12 V, 400 Ω load, 3.3 V through 2.2 kΩ:** - 29.5 mA, fully on, 5.9 mW of heat; - Ib 1.182 mA, 4× overdrive, Rb ≤ 881.4 Ω for Ic/10. - **120 Ω on 12 V from 5 V through 1 kΩ:** 98.33 mA, 4.4× overdrive. - **47 kΩ base resistor:** partly on, 5.532 mA, with 9.787 V across the transistor. - **PNP on 5 V, 100 Ω, base pulled to 0 V through 1 kΩ:** 48 mA, with the turn-off note. - **5.6 kΩ base resistor:** on, but with under 2× overdrive, so the tool flags it.

The review made these changes: - It put the load current first and large, with the state, Vce and the transistor's heat in words. - It added the "Rb for Ib = Ic/10" design value, from the datasheets' own test condition. - It added the partly-on, low-overdrive and PNP turn-off notes, and the coil diode in the note. - It replaced the NPN/PNP pills with a Type choice in the fields. - It limited the units to real ones and remembers them. - It rewrote the note in plain words and added examples.

# BJT currents

(formerly BJT currents (hFE, IC))




## What it computes

For a bipolar transistor, enter two of base current Ib, collector current Ic and current gain hFE, and it gives the third. It also gives the emitter current Ie and α, the share of the emitter current that comes through the collector.

## Source

The relations are the definitions of the DC current gain and Kirchhoff's current law. The gain figures are read from the datasheets themselves: - **Fairchild BC546–BC550** (rev. A2, 2002): - hFE 110–800 at VCE 5 V, IC 2 mA, sold in classes A 110–220, B 200–450 and C 420–800; - collector current at most 100 mA. - **Diodes Inc. MMBT3904** (the SOT-23 2N3904, DS30036 rev. 27, 2025): hFE at least 40 at 0.1 mA, 70 at 1 mA, 100–300 at 10 mA, at least 60 at 50 mA and 30 at 100 mA.

## What it means

A bipolar transistor lets a small current into its base control a much larger one through its collector.

- **hFE** (also written β) — the DC current gain: Ic = hFE × Ib.
- **Ie** — both currents leave through the emitter of an NPN, so Ie = Ib + Ic. For a PNP, all three currents flow the other way.
- **α** — Ic / Ie, the fraction of the emitter current that comes through the collector. It is just under 1: 0.990 for hFE 100.
- **hFE is not a fixed number.** It differs between pieces of the same type, changes with the current and with temperature. That is why datasheets give a range at a stated current, and why the BC547 is sorted and sold in gain classes.
- **Design on the minimum.** To be sure of a collector current, supply the base current the *minimum* gain needs, at the current you will use. A 2N3904 is 100–300 at 10 mA but only 60 or more at 50 mA.
- **Saturated transistors.** A transistor used as a fully-on switch passes less than hFE × Ib, because the load sets the current; see BJT as a switch. Circuits that must not depend on hFE use the divider bias of BJT biasing.

The current being solved for is drawn green in the diagram.

## Why the formulas are these

```
hFE = Ic / Ib      (the definition)
Ie = Ib + Ic       (the current in equals the current out)
α = Ic / Ie = hFE / (hFE + 1)
```

## Assumptions and limits

- **Active region.** The transistor is assumed to work in the active region, where Ic = hFE × Ib holds.
- **DC gain only.** The DC gain hFE is used. The small-signal gain hfe is similar but not identical.
- **Leakage ignored.** Leakage currents (nanoamps for these parts) are neglected.

## What it deliberately does not do

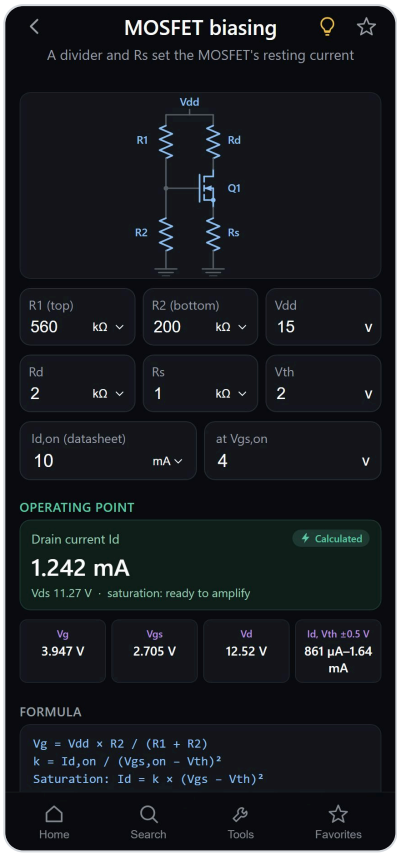
- Bias networks and switching are covered by the other two BJT tools.

## How it was checked

- **hFE 100, Ib 50 µA:** Ic 5 mA, Ie 5.05 mA, α 0.9901.
- **hFE 200 (BC547B minimum), Ib 10 µA:** Ic 2 mA, α 0.995.
- **50 mA at hFE 60 (2N3904 minimum at 50 mA):** Ib 833.3 µA.
- **20 µA in, 5.2 mA out:** hFE 260, α 0.9962.

The review made these changes: - It replaced an unchecked 2N2222 figure with gain data read from the BC547 and 2N3904 datasheets. - It added α and colours the solved current green in the diagram. - It limited the units to µA and mA for Ib, and µA, mA and A for Ic, and remembers them. - It labels the choice as "Find Ic / Ib / hFE". - It rewrote the note in plain words and added examples. - The tool was renamed BJT currents, on Pierre's choice, to fit beside the bulb.

# MOSFET biasing




## What it computes

For an N-channel enhancement MOSFET biased by a voltage divider on its gate, with a drain resistor Rd and a source resistor Rs, it gives: - the drain current Id and the voltage across the MOSFET, Vds, which together are the operating point; - the region: saturation (amplifying), triode (fully on) or off; - the gate, gate-source and drain voltages; - Id if the threshold voltage were 0.5 V lower or higher.

## Source

The device model is the textbook square law, solved for the circuit as drawn. Device data read from the datasheet itself: - **Zetex 2N7000** (issue 2, November 1994): - gate threshold Vgs(th) 0.8 V minimum, 3.0 V maximum, at Id = 1 mA; - on-state drain current Id(on) at least 75 mA at Vgs 4.5 V, Vds 10 V; - Rds(on) at most 5 Ω at Vgs 10 V, 500 mA.

The 3.7:1 threshold range is why the source resistor matters.

## What it means

A MOSFET is controlled by the voltage on its gate, which draws no current.

- **Threshold Vth** — below it, no drain current flows. Above it, the current grows with the square of how far the gate is above it.
- **The divider** — R1 and R2 divide the supply to set the gate voltage Vg. No gate current loads them, unlike a BJT's base.
- **The source resistor Rs** — pulls back: more current raises the source, which lowers Vgs, so the current settles at the operating point.
- This feedback reduces how much a different threshold moves the current. The "Id, Vth ±0.5 V" cell shows it.
- It works better with a higher gate voltage and a larger Rs.
- **k** — the device's square-law constant. It comes from one datasheet point: a drain current Id,on at a gate voltage Vgs,on.
- **The names are a trap:**
- **saturation** is where a MOSFET *amplifies*: its drain current is set by the gate, not by Vds. That is the opposite of a BJT's saturation;
- **triode** (or ohmic, or linear) is where it is fully on and acts as a resistor. A switch wants this; an amplifier does not;
- **off** — the gate is below the threshold.

## Why the formulas are these

```
Vg = Vdd × R2 / (R1 + R2)
k = Id,on / (Vgs,on − Vth)²
Saturation (Vds ≥ Vgs − Vth):   Id = k × (Vgs − Vth)²
Triode     (Vds < Vgs − Vth):   Id = k × (2 (Vgs − Vth) Vds − Vds²)
Vgs = Vg − Id × Rs,   Vds = Vdd − Id × (Rd + Rs)
```

The two device equations meet at the boundary Vds = Vgs − Vth. As Id rises, Vgs and Vds both fall, and the device current falls with them, so there is exactly one crossing. The tool finds it by bisection, in whichever region it lies.

## Assumptions and limits

- **The square law is an approximation.** Real MOSFETs follow it only roughly, and power MOSFETs poorly. Use it for understanding and first values, then measure.
- **k from one datasheet point.** Datasheet Id(on) figures are often minimums, measured in pulses.
- **Ignored effects.** Channel-length modulation (the slight rise of Id with Vds) and temperature are not modelled.
- **N-channel only.** A P-channel circuit is the mirror image.

## What it deliberately does not do

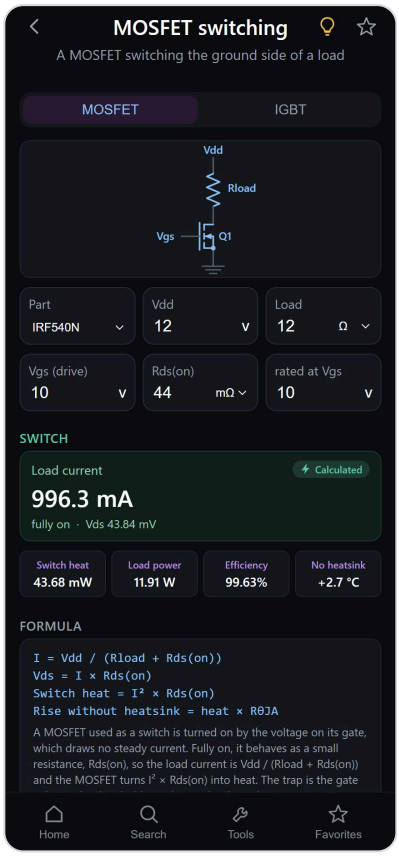
- Small-signal gain is not calculated.
- MOSFET switches have their own tool.

## How it was checked

All on 15 V with a 560 k / 200 k divider (Vg 3.947 V), Vth 2 V and 10 mA at Vgs,on 4 V (k = 2.5 mA/V²). The figures were also computed in a separate script. - **Rd 2 kΩ, Rs 1 kΩ:** - Id 1.242 mA, Vds 11.27 V, saturation; - 861 µA–1.64 mA for Vth ±0.5 V. - **Rd 1 kΩ, no Rs:** - Id 9.481 mA; - 5.24–13.4 mA for Vth ±0.5 V (the top end limited by the triode region). - **Rd 12 kΩ, Rs 1 kΩ:** triode, Id 1.127 mA, Vds 349 mV. The old version applied the saturation formula there. - **R2 50 kΩ (Vg 1.23 V):** off.

The review made these changes: - It fixed the triode region. The current and voltages were computed with the saturation equation even there; now each region uses its own. - It explains the region names in plain words. - It put Id first and large, and added the Vth ±0.5 V cell. - It limited the units to real ones and remembers them. - It rewrote the note and added examples. - A global layout fix made three-to-a-row fields slightly tighter, so a three-digit value beside a unit is no longer cut on a 375 px phone.

# MOSFET switching




## What it computes

For a MOSFET or an IGBT switching the ground side of a resistive load, fully on, it gives: - the load current and the voltage left across the switch; - the heat in the switch, the power in the load and the efficiency; - for parts whose datasheet gives it, how much the switch warms without a heatsink; - whether the gate drive is enough for the on-state figures to hold.

## Source

Device data read from the datasheets themselves:

| Part | Rds(on) max | at Vgs | Vgs(th) | RθJA (no heatsink) |
|---|---|---|---|---|
| 2N7000 (Zetex, issue 2, 1994) | 5 Ω | 10 V | 0.8–3.0 V at 1 mA | not used |
| IRLZ44 (Vishay 91328 rev. D, 2021) | 28 mΩ | 5 V | 1.0–2.0 V at 250 µA | 62 °C/W |
| IRF540N (Infineon rev. 2.1) | 44 mΩ | 10 V | 2.0–4.0 V at 250 µA | 62 °C/W |

The IRLZ44 is also specified at 39 mΩ at Vgs 4 V. The IGBT model, a knee voltage V0 plus a slope resistance Rce, is the usual straight-line fit of an IGBT datasheet's Vce–Ic curve. Its default values are illustrative, not from a particular part.

## What it means

A MOSFET used as a switch is turned on by the voltage on its gate, which draws no steady current.

- **Fully on** — it behaves as a small resistance, Rds(on). The load current is set almost entirely by the load, and the MOSFET turns $I^2$ × Rds(on) into heat.
- **The gate-voltage trap:**
- the **threshold** Vgs(th) is only where the MOSFET *starts* to conduct: 250 µA for the IRF540N, 1 mA for the 2N7000;
- **Rds(on) is guaranteed only at a stated, higher gate voltage** — 10 V for a standard part, 4.5 or 5 V for a logic-level one;
- a 3.3 V microcontroller pin cannot properly drive an IRF540N. Its threshold alone may be up to 4 V, so some pieces would not turn on at all. Use a logic-level part rated at your drive voltage, or a gate

driver.

- **The three drive verdicts:**
- **fully on** — the drive reaches the voltage the on-resistance is rated at;
- **on, but not fully** — the drive is above the highest threshold but below that voltage, so the real resistance and heat are higher than shown;
- **may not turn on** — the drive does not exceed the highest threshold the part may have.
- **Heat without a heatsink** — the heat × RθJA: about 62 °C per watt for these TO-220 parts. The tool warns above 100 °C of rise, a rule of thumb that leaves room for warm surroundings. The IRF540N's datasheet allows 175 °C at the junction; check each part's own limit.
- **IGBT** — like a MOSFET, driven by its gate voltage, and suited to high voltages. Fully on, it drops a knee voltage plus a small resistance. That figure is specified at a stated gate voltage, usually 15 V.
- **Coils** — a relay or motor coil needs a diode across it, as with any switch.

Choosing a part fills in its datasheet figures. Typing your own switches it to Custom.

## Why the formulas are these

```
MOSFET:  I = Vdd / (Rload + Rds(on)),   Vds = I × Rds(on),
         heat = I² × Rds(on)
IGBT:    I = (Vdd − V0) / (Rload + Rce), Vce = V0 + I × Rce,
         heat = Vce × I
Rise without heatsink = heat × RθJA
```

## Assumptions and limits

- **Fully-on state only.** Switching losses, the energy lost during each turn-on and turn-off, are not calculated. They matter at high switching frequencies (PWM).
- **Rds(on) at 25 °C.** It rises as the MOSFET warms (each datasheet plots it against temperature), so a hot switch dissipates more than shown.
- **A resistive load** on the ground side (low-side switch).

## What it deliberately does not do

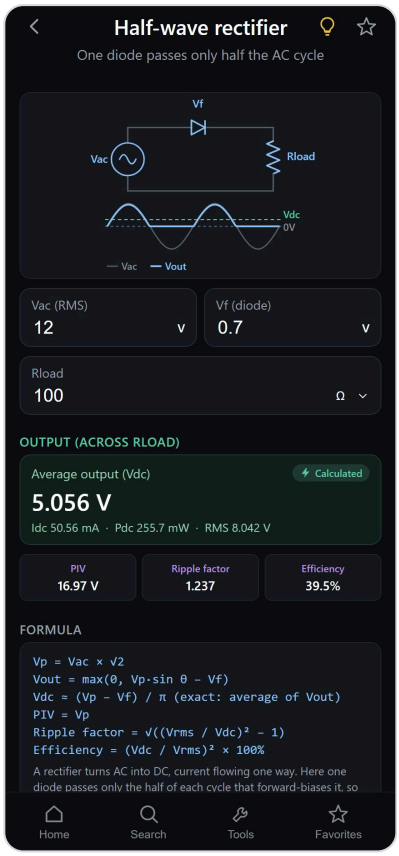
- Gate-driver design, gate charge and switching speed are not covered.
- High-side switching (P-channel, or N-channel with a driver) is not covered.

## How it was checked

These figures were also computed in a separate calculation. - **IRF540N, 12 V, 12 Ω load, 10 V drive:** 996.3 mA, 43.68 mW in the MOSFET, +2.7 °C, 99.63 % efficient. - **The same from a 3.3 V pin:** may not turn on. - **6 V of drive:** on, but not fully. - **IRLZ44, 12 V, 2 Ω, 5 V drive:** 5.917 A, 980.4 mW, +61 °C. - **IRF540N with a 1 Ω load:** about 5.8 W, so the tool says a heatsink is needed. - **IGBT, 12 V, 12 Ω, V0 0.8 V, Rce 150 mΩ, 15 V drive:** 921.8 mA, Vce 938.3 mV.

The review made these changes: - The on-state check called the switch fine once the drive was 2 V above the threshold. It now compares the drive with the gate voltage the datasheet guarantees Rds(on) or Vce(sat) at, and with the part's highest threshold. - It added datasheet parts, and the temperature rise without a heatsink. - It put the load current first and large. - It gives the IGBT its own 15 V drive. - It limited the units to real ones and remembers them. - It rewrote the note in plain words and added examples.

# Half-wave rectifier




## What it computes

For one diode feeding a resistive load from an AC source: the average (DC) output voltage and current, the DC power, the RMS output, the peak reverse voltage the diode must withstand, the ripple factor and the efficiency. It draws the input and output waveforms.

## Source

No standard is needed: the results are averages of the output waveform over one cycle, derived below. With a zero diode drop they reduce to the textbook figures for a half-wave rectifier — Vdc = Vp/π, ripple factor 1.21, efficiency $4/\pi^2$ = 40.5%.

## What it means

A **rectifier** turns alternating current into current that flows one way only. The **half-wave** rectifier is the simplest: a single diode in series with the load. During the half of each cycle that makes the diode's anode positive, it conducts and the load sees the input, less the diode's forward voltage **Vf**; during the other half it blocks and the load sees nothing. The output is one hump per cycle.

- **Vac** is the source's RMS voltage, the figure on a transformer's label; its **peak**, Vp = Vac × √2, is 1.41 times higher.
- **Vdc** is the average of the output, what a DC voltmeter reads.
- **PIV**, the peak inverse voltage, is the reverse voltage the diode must block while it is off: the whole input peak, since no current flows and the load drops nothing. The diode's rated reverse voltage (VRRM) must exceed it.
- **Ripple factor** — how much AC is left riding on the DC, as the ratio of the AC part's RMS to the DC value. 1.21 means the output is mostly ripple.
- **Efficiency** — the share of the power in the load that is DC, at most 40.5% here.

A bare half-wave rectifier is used where smoothness does not matter, or with a capacitor across the load to fill the gaps (the Rectifier ripple tool).

## Why the formulas are these

With a constant drop Vf, the output is Vout = max(0, Vp·sin θ − Vf): the diode conducts from θ1 = asin(Vf / Vp) to π − θ1, a little less than half a cycle. Averaging over the cycle:

```
Vdc   = [Vp·cos θ1 − Vf·(π/2 − θ1)] / π
Vrms² = [Vp²·((π − 2θ1)/2 + sin 2θ1 / 2) − 4·Vp·Vf·cos θ1 + Vf²·(π − 2θ1)] / 2π
```

With Vf = 0 these are Vp/π and Vp/2. The common shortcut Vdc ≈ (Vp − Vf)/π treats the output as a whole half-sine of peak Vp − Vf, and reads 2.4% high at 12 V AC with a 0.7 V diode.

Then Idc = Vdc / R, Pdc = Vdc × Idc, ripple factor = √((Vrms/Vdc)² − 1) and efficiency = (Vdc/Vrms)², the DC power over the total power in the load.

## Assumptions and limits

- The diode is an ideal switch with a constant drop; a real one's drop rises with current (see Diode forward voltage).
- The source has no resistance of its own: a transformer's winding resistance lowers the output under load.
- The load is a resistor. With a capacitor across it the output, the PIV (up to twice the peak) and the diode's current all change; that is the Rectifier ripple tool.

## What it deliberately does not do

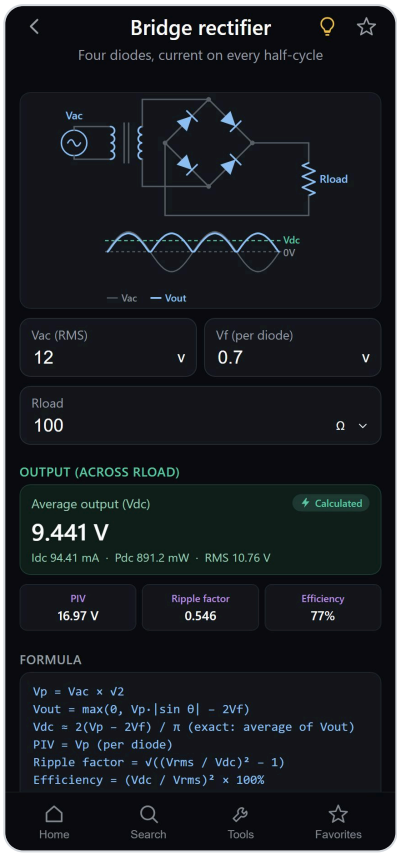
- **Full-wave rectifiers** (bridge, centre-tapped) have their own tools.
- **Smoothing and ripple voltage** with a capacitor is the Rectifier ripple tool.

## How it was checked

12 V AC with a 0.7 V diode into 100 Ω: Vp 16.97 V, Vdc 5.056 V, Idc 50.56 mA, ripple factor 1.237, efficiency 39.5%; with Vf = 0, 5.402 V (Vp/π), 1.211 and 40.5%. Both closed forms were checked against a numerical average over the cycle. 3 V AC gives 1.156 V with a 0.4 V Schottky and 1.019 V with 0.7 V silicon; 24 V AC has a 33.94 V PIV.

The review replaced the (Vp − Vf)/π shortcut with the exact averages, which the drawn waveform already followed; put Vdc first and large; limited the units to volts and ohms to megohms; and added plain definitions.

# Bridge rectifier




## What it computes

For a bridge of four diodes feeding a resistive load from an AC source: the average (DC) output voltage and current, the DC power, the RMS output, the peak reverse voltage each diode must withstand, the ripple factor and the efficiency, with the input and output waveforms.

## Source

No standard is needed: the results are averages of the output waveform, derived below. With zero diode drop they reduce to the textbook figures for a full-wave rectifier — Vdc = 2Vp/π, ripple factor 0.483, efficiency $8/\pi^2$ = 81.1%.

## What it means

A **bridge rectifier** is four diodes in a diamond. The AC source (usually a transformer's secondary) connects to two opposite corners, the load to the other two. On one half-cycle two of the diodes conduct, on the other half the other two, and in both cases the current passes through the load **the same way**. So where a half-wave rectifier (see that section) gives one pulse per cycle, the bridge gives two, and uses the whole of the input.

The price is two diodes in the current's path at every moment: each pulse is **2 × Vf** lower than the input's peak. From a low voltage that loss is a large part of the output; Schottky diodes, with a smaller Vf, reduce it.

The other figures — **Vdc**, **PIV**, **ripple factor**, **efficiency** — mean the same as for the half-wave rectifier. Each diode, while off, blocks the input's peak Vp, as in the half-wave case; a bridge does not double it.

## Why the formulas are these

The output is Vout = max(0, Vp·|sin θ| − 2Vf). Each half-cycle conducts from θ1 = asin(2Vf / Vp) to π − θ1, so the averages are twice the half-wave ones with 2Vf as the drop:

$$V_{dc} = 2\cdot[V_p\cdot\cos\theta_1 - 2V_f\cdot(\pi/2 - \theta_1)] / \pi$$

$$V_{rms}^2 = 2\cdot[V_p^2\cdot((\pi - 2\theta_1)/2 + \sin 2\theta_1 / 2) - 8\cdot V_p\cdot V_f\cdot\cos\theta_1 + 4V_f^2\cdot(\pi - 2\theta_1)] / 2\pi$$

With Vf = 0 these are 2Vp/π and Vp/√2. The shortcut 2(Vp − 2Vf)/π treats each pulse as a whole half-sine and reads 5% high at 12 V AC with 0.7 V diodes. Ripple factor and efficiency follow as for the half-wave rectifier.

## Assumptions and limits

- Each diode is an ideal switch with a constant drop.
- The source has no resistance; a transformer's winding resistance lowers the output under load.
- The load is a resistor. A smoothing capacitor changes the diodes' current into short peaks and the output into DC with a small ripple — the Rectifier ripple tool.
- Strictly, a blocking diode sees Vp − Vf, the conducting diode's drop taken off; Vp is the figure to rate it by.

## What it deliberately does not do

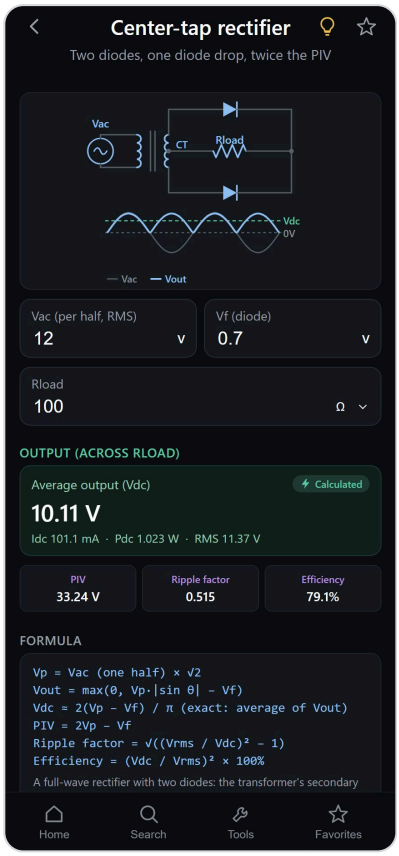
- **The centre-tapped full-wave rectifier**, two diodes and a transformer with a centre tap, has its own tool.
- **Smoothing** with a capacitor is the Rectifier ripple tool.

## How it was checked

12 V AC with 0.7 V diodes into 100 Ω: Vp 16.97 V, Vdc 9.441 V, Idc 94.41 mA, ripple factor 0.546, efficiency 77.0%; with ideal diodes 10.80 V (2Vp/π), 0.483 and 81.1%. Both closed forms were checked against a numerical average. 5 V AC gives 3.190 V with 0.7 V diodes and 3.638 V with 0.45 V Schottky ones.

The review replaced the 2(Vp − 2Vf)/π shortcut with the exact averages, which the drawn waveform already followed; put Vdc first and large; limited the units; and added plain definitions and examples.

# Center-tap rectifier




## What it computes

For a full-wave rectifier made of two diodes and a centre-tapped transformer, feeding a resistive load: the average (DC) output voltage and current, the DC power, the RMS output, the peak reverse voltage each diode must withstand, the ripple factor and the efficiency, with the waveforms.

## Source

No standard is needed: the results are averages of the output waveform, as for the bridge rectifier (see that section), with one diode drop instead of two. The peak inverse voltage follows from the circuit, below.

## What it means

The transformer's secondary winding has a connection at its middle, the **centre tap**, which becomes the output's negative side. Each end of the winding feeds the load through its own diode. On one half-cycle the top half drives the load through the top diode, on the other half the bottom half through the bottom diode — current flows through the load the same way both times, so the output is full-wave, two pulses per cycle.

- **Vac** is the RMS voltage of **one half** of the secondary. A transformer sold as "24 V CT" (centre-tapped) is two 12 V halves: enter 12.
- **One diode drop.** Only one diode conducts at a time, so each pulse loses Vf, where a bridge loses 2 × Vf — better at low voltage.
- **The catch: PIV.** While one diode conducts, the other has one end at the peak of its own half and the other end at the peak of the conducting half, of opposite sign: it blocks nearly **twice the peak**, 2Vp − Vf. Choose diodes rated for that.
- **Vdc**, **ripple factor** and **efficiency** mean the same as for the half-wave rectifier.

**Bridge or centre tap?** The centre tap saves two diodes and one drop but needs a winding twice as long, each half working only half the time, and diodes rated for twice the voltage. The bridge uses the whole winding all the time and is the usual choice today.

## Why the formulas are these

The output is Vout = max(0, Vp·|sin θ| − Vf), Vp being one half's peak. Each half-cycle conducts from θ1 = asin(Vf / Vp) to π − θ1, so:

```
Vdc   = 2·[Vp·cos θ1 − Vf·(π/2 − θ1)] / π
Vrms² = 2·[Vp²·((π − 2θ1)/2 + sin 2θ1 / 2) − 4·Vp·Vf·cos θ1 + Vf²·(π − 2θ1)] / 2π
PIV   = 2Vp − Vf
```

With Vf = 0, Vdc = 2Vp/π, ripple factor 0.483 and efficiency 81.1%, as for the bridge. The shortcut 2(Vp − Vf)/π reads 2.4% high at 12 V a half.

## Assumptions and limits

- Each diode is an ideal switch with a constant drop.
- The transformer has no resistance or leakage, and both halves are equal.
- The load is a resistor; a smoothing capacitor is the Rectifier ripple tool.

## What it deliberately does not do

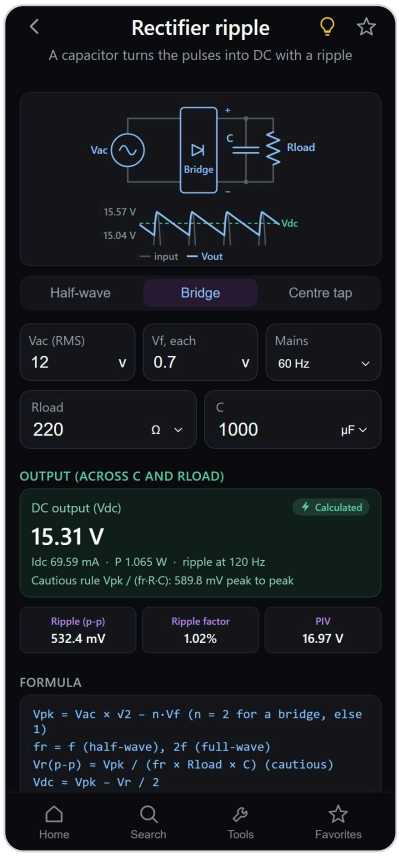
- **The bridge rectifier** has its own tool.
- **Transformer selection** (VA rating, regulation) is not covered.

## How it was checked

12 V a half, 0.7 V diodes, 100 Ω: Vp 16.97 V, Vdc 10.11 V, Idc 101.1 mA, PIV 33.24 V, ripple factor 0.515, efficiency 79.1%; with ideal diodes 10.80 V and 0.483. 5 V a half gives 3.824 V; 24 V a half needs 67.18 V PIV. Both closed forms were checked against a numerical average.

The review replaced the 2(Vp − Vf)/π shortcut with the exact averages, put Vdc first and large, limited the units, and added plain definitions and examples. The PIV, 2Vp − Vf, was already right.

# Rectifier ripple




## What it computes

For a half-wave, bridge or centre-tap rectifier with a smoothing capacitor across a resistive load: the DC output voltage, current and power, the ripple's peak-to-peak voltage and frequency, the ripple factor, the diodes' peak inverse voltage, and the classic design estimate of the ripple for comparison, with the output waveform.

## Source

No standard is needed. The figures come from simulating the circuit drawn over a steady cycle; the design estimate is the usual textbook rule. Both are explained below.

## What it means

A rectifier alone gives pulses (see the Half-wave and Bridge rectifier sections). A **smoothing capacitor** across the load charges to the peak at each pulse and, between pulses, feeds the load on its own, its voltage sagging as it discharges until the next peak tops it up. The output is then DC with a small sawtooth on top: the **ripple**.

- **Ripple frequency, fr** — how often the capacitor is topped up: the mains frequency for a half-wave, twice it for a bridge or centre tap. That is why full-wave rectification halves the ripple for the same capacitor.
- **Ripple (p-p)** — the height of the sawtooth, from peak to trough.
- **Vdc** — the average output, a little below the peak.
- **Ripple factor** — the ripple's RMS value as a percentage of Vdc.
- **PIV** — with the capacitor holding the output near the peak, a half-wave or centre-tap diode sees nearly twice the peak when off (2Vp − Vf); a bridge diode sees Vp.

More capacitance, or less load current (a larger Rload), gives less ripple.

## Why the formulas are these

**The simulation.** Over each short time step the capacitor either follows the rectified input, while that is higher (the diodes conduct), or decays through the load, v → v·e^(−dt/RC). After a few cycles to settle, one cycle gives the maximum, the minimum, the average (Vdc) and the RMS of the ripple. It models a perfect transformer — an ideal source, one that keeps its voltage whatever current is drawn — and diodes with a constant drop, so it is exact for the circuit drawn.

**The design rule.** Assuming the capacitor discharges at the full load current for the whole period 1/fr gives

```
Vr(p-p) ≈ Vpk / (fr × R × C),   Vdc ≈ Vpk − Vr / 2
```

with Vpk = Vac × √2 less one diode drop (two for a bridge). The capacitor is really recharged before the period ends, so the rule overstates the ripple — by little at light load, by a lot at heavy load: at 1 A from 2200 µF it gives 4.9 V where the circuit gives 3.4 V. That makes it a safe bound for choosing a capacitor.

## Assumptions and limits

- **An ideal source.** A real transformer's winding resistance and leakage inductance stop the capacitor charging fully to the peak and widen the diodes' conduction: Vdc comes out lower and the ripple somewhat larger than simulated. Keep margin, or use the design rule.
- **Diode current.** The diodes conduct in short pulses much larger than the average current; their peak and surge ratings matter, and are not calculated.
- The load is a resistor; a regulator after the capacitor draws a roughly constant current instead, for which the design rule is Vr ≈ I / (fr·C).

## What it deliberately does not do

- **Choosing C** for a target ripple is done by trying values; the rule above solves for it directly: C ≈ I / (fr × Vr).
- **Transformer and diode surge ratings** are not covered.

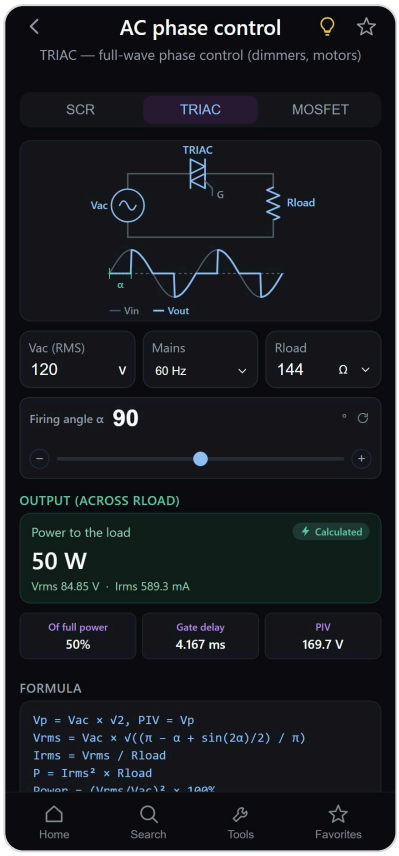
## How it was checked

Bridge, 12 V AC, 220 Ω, 1000 µF, 60 Hz: Vdc 15.31 V, ripple 532 mV p-p, design rule 590 mV. 12 Ω and 2200 µF: 13.95 V, 3.40 V p-p against the rule's 4.91 V; on a half-wave, 6.73 V and a PIV of 33.24 V. At 50 Hz with 4700 µF, 2.13 V. The simulation was checked in a separate script with finer steps.

The review found the tool half-wave only, though the name did not say so and the bridge is the common case; its numbers from the design rule while the drawn waveform came from a simulation, so the two disagreed; and a badge warning that the rule was strained, which the simulation makes unneeded. It now offers the three rectifiers, takes its figures from the simulation with the rule beside them, and is renamed Rectifier ripple.

# AC phase control

(formerly Thyristor firing angle)




## What it computes

For phase control of a resistive load from the mains, with an SCR, a TRIAC or a pair of MOSFETs, at a chosen angle: - the power (TRIAC, MOSFET) or the DC output (SCR); - the RMS voltage and current; - the share of full power; - the delay from each zero crossing at which the device is switched; - the peak voltage the device must block; - for the MOSFETs, the power they lose; - the waveforms.

## Source

No standard is needed. The results are averages of the chopped sine, derived below, and match the formulas of power-electronics textbooks for phase-controlled circuits with a resistive load.

## What it means

A **thyristor** is a semiconductor switch for AC. It stays off until a short pulse on its **gate** (G) turns it on. It then stays on by itself until the current through it falls to zero, which on AC happens at the end of every half-cycle. So it can be switched on at any point of each half-cycle, but it switches itself off at the next zero crossing.

- **SCR** (silicon-controlled rectifier) — conducts one way only, like a diode with a gate. It passes one half of each cycle, from the moment it is fired, and gives DC.
- **TRIAC** — conducts both ways and is fired in each half-cycle. It passes both halves, so the load gets AC. Lamp dimmers, heater controls and simple motor speed controls use it.
- **Firing angle, α** — how far into each half-cycle the gate is fired, from 0° (at once, full power) to 180° (never, no power). Firing later cuts off the start of each half-wave: this is **leading-edge** control.
- **Gate delay** — the same angle as a time after the zero crossing: α / 360° of a mains period, so 4.17 ms for 90° at 60 Hz. A microcontroller dimmer waits that long after it detects the zero crossing, then pulses the gate.
- **MOSFET** — a transistor switch that turns on and off whenever its gate says, not only at the zero crossing. Each MOSFET has a built-in diode (the body diode) that conducts backwards. One MOSFET alone would therefore still pass one half-wave, so an AC switch uses two back to back with their sources joined: one blocks each direction, and one gate signal drives both.
- **Cut-off angle, β** — with MOSFETs the switch is turned on at each zero crossing and off at β. This cuts the end of each half-wave instead of its start: **trailing-edge** control. β runs from 0° (off at once, no power) to 180° (never off, full power). It suits LED lamps and electronic transformers, and it is quieter, because the current starts gently from zero instead of jumping up.
- **Turns off at** — β as a time after the zero crossing.

- **Rds(on)** — the small resistance of a MOSFET that is switched on, from its datasheet (tens to hundreds of milliohms for mains parts). The load current flows through both MOSFETs, so they dissipate $I_{rms}^2 \times 2 \times R_{ds(on)}$. That power becomes heat in the MOSFETs, which is what the heatsink must carry away.
- **Of full power** — the power at this angle as a share of the power with the switch always on. Power changes slowly near 0° and 180° and fastest around 90°, where it is exactly half.
- **PIV / Vds rating** — the peak voltage the device blocks while it is off: the mains peak. Choose a part rated above it, with margin.

## Why the formulas are these

With a TRIAC the load sees $V_p \cdot \sin\theta$ from $\alpha$ to $\pi$ in each half-cycle. Its power is $V_{rms}^2 / R$, and the RMS of that chopped sine over a half-cycle is

$$V_{rms} = V_{ac} \times \sqrt{(\pi - \alpha + \sin 2\alpha / 2) / \pi} \qquad \text{(TRIAC)}$$

This is Vac at $\alpha = 0$ and 0 at $\alpha = \pi$. The share of full power is the square of Vrms / Vac, which is the bracket under the root.

An SCR conducts only one half of each cycle. So its RMS is that of a half-wave, $(V_p / 2)\cdot\sqrt{(\ldots)}$, and its average, the DC, is

$$V_{dc} = (V_p / 2\pi) \times (1 + \cos\alpha)$$

This is $V_p/\pi$ at $\alpha = 0$, the plain half-wave rectifier.

With MOSFETs the load sees $V_p \cdot \sin\theta$ from 0 to $\beta$ instead. Because the sine is symmetric about 90°, the part from 0 to $\beta$ carries the same energy as the part from $180° - \beta$ to 180°. So

$$k = (\beta - \sin 2\beta / 2) / \pi, \qquad I_{rms} = V_{ac} \times \sqrt{k} / (R_{load} + 2 \times R_{ds(on)})$$

For a resistive load, trailing-edge control at $\beta$ gives exactly the power of leading-edge control at $\alpha = 180° - \beta$. The two differ in the shape of the current, not in the power. The MOSFETs' on-resistance is in series with the load, so it is included in the current.

The gate delay (thyristors) and the turn-off time (MOSFETs) are the angle / (360° × f).

## Assumptions and limits

- **A resistive load** — lamp, heater. With a motor or a transformer the current lags the voltage, a thyristor does not turn off at the zero crossing of the voltage, and a MOSFET switching off an inductive current must absorb its energy. These formulas do not hold for such loads.
- **Switches.**
- A real thyristor drops 1–2 V while on, small against mains voltages, and needs a minimum holding current to stay on.
- The MOSFET loss is the conduction loss only. Switching losses at each turn-off, and the rise of Rds(on) as the part heats, add to it.

- Phase control chops the current sharply, which causes electrical noise and harmonics, and dimmers need filtering to meet emission limits. This is not covered.

## What it deliberately does not do

- **Gate drive** is not covered: pulse current, opto-isolation, snubbers, and the floating supply a MOSFET pair's gate needs.
- **Inductive loads** need a different analysis, as above.

## How it was checked

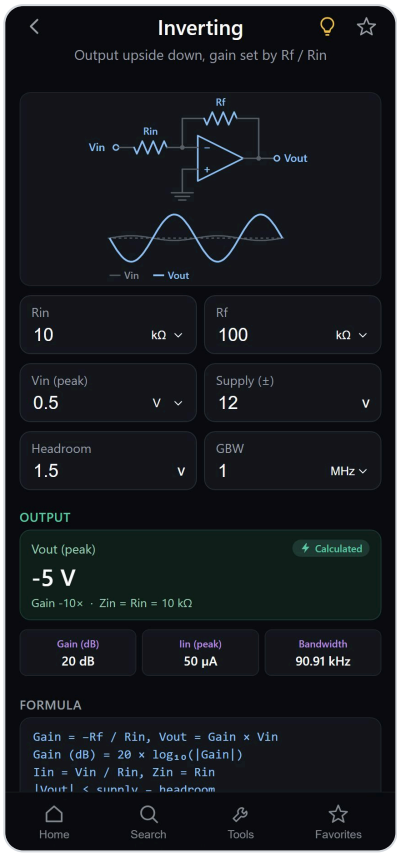
- TRIAC, 120 V, 144 Ω (a 100 W lamp), α = 90°: 84.85 V RMS and 50 W, exactly half of the 100 W at α = 0. Gate delay 4.167 ms at 60 Hz, PIV 169.7 V.
- SCR at 90°: Vdc 27.01 V (Vp / 2π), 60 V RMS, 25 W.
- A 1 kW, 230 V heater at 60°: 80.4% of full power, 3.333 ms at 50 Hz.
- SCR, 24 V AC at 45°: 9.222 V DC.
- MOSFETs, same lamp, β = 90°:
- with Rds(on) = 0: 50 W;
- with 2 × 100 mΩ: 49.86 W, and 69.25 mW lost in the MOSFETs.
- β = 120° gives 80.4%, the TRIAC's figure at 60°.

The review put the power (TRIAC) or the DC (SCR) first and large, showed the gate delay the tool already worked out, and limited the units to real ones (volts, 50 or 60 Hz, ohms to megohms). It also moved the waveform's legend off the waves and added plain definitions and examples.

On Pierre's request it then added the MOSFET pair as a third switch, with trailing-edge control, its own waveform and its conduction loss. The tool was renamed AC phase control, and the old name stays findable in search.

# Inverting

(formerly Inverting amplifier)




## What it computes

For the inverting op-amp amplifier: - the output voltage for a given input, turned upside down; - the gain, as a ratio and in decibels; - the input current and the input impedance; - where the output clips, from the supply and the op-amp's headroom; - the bandwidth, from the op-amp's gain-bandwidth product.

It draws the input and output waves.

## Source

No standard applies. The gain comes from the ideal op-amp model found in every analog textbook. The two real limits, headroom and bandwidth, were checked against manufacturers' datasheets, read from the documents themselves: - **TI TL072** (SLOS080W, July 2025): - maximum peak output ±13.5 V typical, ±12 V minimum, on ±15 V into 10 kΩ, so about 1.5 V short of each rail; - gain-bandwidth product 5.25 MHz (3 MHz for some versions). - **TI LM358** (SLOS068AB, October 2024): - the output stops 2 V typical, 3 V maximum, below the positive rail at 30 V, and 5–20 mV above the negative one; - gain-bandwidth product 0.7 MHz (1.2 MHz for the LM358B). - **Microchip MCP6002** (DS20001733L), a rail-to-rail part: - the output gets within 25 mV of each rail; - gain-bandwidth product 1 MHz.

## What it means

An **op-amp** (operational amplifier) is a chip with two inputs, "−" and "+", and one output. It amplifies the difference between its inputs by a huge amount, 100 000 times or more. It is almost always used with **feedback**: part of the output is returned to the "−" input, and the resistors then set the gain.

- **Virtual ground** — the "+" input is at 0 V here. With feedback, the op-amp drives its output until the "−" input sits at the same 0 V. So the "−" input is at ground potential without being connected to ground.
- **Rin** — the input resistor. With one end at Vin and the other at 0 V, it carries Iin = Vin / Rin.
- **Rf** — the feedback resistor, from the output back to the "−" input. The op-amp's input takes no current, so all of Iin continues through Rf.
- **Gain** — Vout / Vin = −Rf / Rin. The minus sign means the output is turned upside down: a positive input gives a negative output. For a sine wave this is a 180° phase shift.
- **Gain in dB** — 20 × $\log_{10}$ of the size of the gain. A gain of 10 is 20 dB, and a gain of 100 is 40 dB.
- **Zin** — the input impedance, the load the signal source sees. It is simply Rin, because the other end of Rin is held at 0 V.

- **Headroom** — how close the output can get to the supply. Enter the value from the op-amp's datasheet:
- a classic op-amp (TL072, LM358) stops about 1.5–2 V short of each rail;
- a rail-to-rail op-amp gets within a few tens of millivolts, so enter 0 for it.
- **Clipping** — when the gain asks for more than the output can give, the output stops at its limit and the tops of the wave are cut flat. That is distortion.
- **GBW** (gain-bandwidth product) — a figure from the datasheet: the gain the op-amp can give, multiplied by the frequency, is at most this. A 1 MHz op-amp can give a gain of 10 up to about 100 kHz.
- **Bandwidth** — the frequency where the stage's gain has fallen by 3 dB, to 71% of its low-frequency value.

## Why the formulas are these

The "−" input is at 0 V, so the current through Rin is Vin / Rin. None of it enters the op-amp, so the same current flows through Rf, from the "−" node towards the output. The output must therefore sit Iin × Rf below 0 V:

```
Vout = −Vin × Rf / Rin
```

The output can only swing to the supply less the headroom, so

```
|Vout| ≤ supply − headroom
```

beyond which it clips.

The bandwidth depends on the **noise gain**, 1 + Rf / Rin, rather than on the signal gain Rf / Rin. The noise gain is how much the op-amp's own feedback loop divides down its output, and GBW is shared out by it:

```
Bandwidth = GBW / (1 + Rf / Rin)
```

That is why a unity-gain inverter (Rf = Rin) has only half the GBW as bandwidth, whereas a unity-gain buffer has all of it.

## Assumptions and limits

- **An ideal op-amp for the gain.** Its open-loop gain is taken as infinite. A real one's is large but finite, which lowers the gain a little: well under 0.1% for ordinary gains.
- **Symmetric supply.** The supply is ± the value entered, and the headroom is taken as the same on both sides. A single-supply circuit needs its input biased to mid-supply, which is not covered.
- **Slew rate.** A large, fast output can also be limited by how fast the output can move (V/µs on the datasheet). This is not calculated.
- **Offset and bias current** add a small DC error at the output. They are not included.
- The output must also drive its load. A load below about 2 kΩ lowers the swing of most classic op-amps; see the datasheet.

## What it deliberately does not do

- It does not choose the resistors. Pick Rin for the input impedance you need (often 10 kΩ), then Rf = gain × Rin from the E-series.
- Frequency response curves and phase are not drawn.

## How it was checked

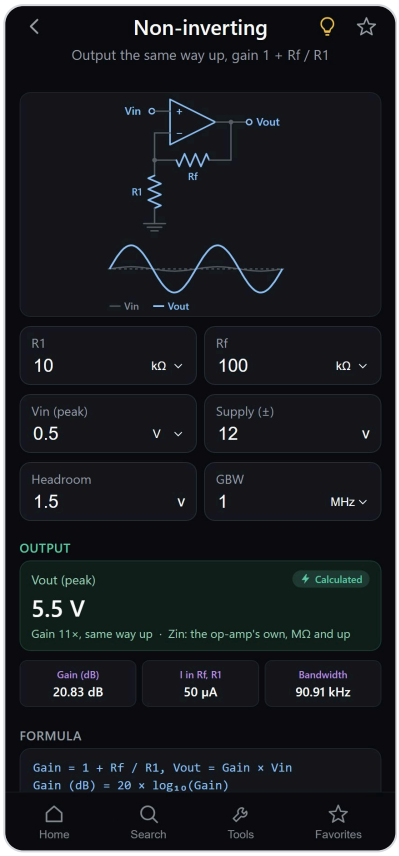
- 10 kΩ / 100 kΩ, 0.5 V peak, ±12 V, 1.5 V headroom, 1 MHz GBW:
- gain −10 (20 dB), Vout −5 V, Iin 50 µA;
- bandwidth 1 MHz / 11 = 90.91 kHz.
- Unity inverter: −1 V out, 500 kHz bandwidth.
- Mic preamp, 1 kΩ / 47 kΩ, 20 mV: −940 mV out, 20.83 kHz bandwidth.
- Gain −100, 0.2 V: the ideal −20 V clips at −10.5 V, or at −12 V with no headroom.

The review found three problems: - the output clipped at the full supply, which only rail-to-rail op-amps reach; - the gain was given without the bandwidth that limits it; - the units included mΩ, GΩ and kV.

It now has the headroom and GBW fields, shows Vout first and large, and remembers the units, and the op-amp's figures for all the op-amp tools. It also adds plain definitions, examples and an 11 px legend under the waves.

# Non-inverting

(formerly Non-inverting amplifier)




## What it computes

For the non-inverting op-amp amplifier, it gives: - the output voltage for a given input, the same way up; - the gain, as a ratio and in decibels; - the current through the feedback divider; - where the output clips, from the supply and the op-amp's headroom; - the bandwidth, from the op-amp's gain-bandwidth product.

It also draws the input and output waves.

## Source

No standard applies: the gain comes from the ideal op-amp model. The headroom and gain-bandwidth figures were read from the TL072, LM358 and MCP6002 datasheets, as listed in the Inverting section.

## What it means

The signal goes straight into the op-amp's **+ input**. Rf and R1 form a **voltage divider** from the output to ground, and its middle point feeds the − input. The op-amp drives its output until the divided-down voltage equals Vin.

- **Gain** — Vout / Vin = 1 + Rf / R1. The output is the same way up as the input (in phase), and the gain is never below 1.
- With Rf = 0, or with no R1, the gain is exactly 1: a buffer.
- **Gain in dB** — 20 × $\log_{10}$ of the gain.
- **Zin** — the signal source sees only the op-amp's own input, megohms to teraohms, and no resistor. This is the main reason to choose this circuit over the inverting amplifier, whose input impedance is only Rin.
- **Current in Rf, R1** — the divider current, supplied by the op-amp's output. Keep it to a few milliamps: resistors in the kΩ range.
- **Headroom** — how close the output can get to the supply. It is about 1.5–2 V for a TL072 or LM358; enter 0 for a rail-to-rail op-amp. **Clipping** is when the gain asks for more than that and the tops of the wave are cut flat.
- **GBW** — the op-amp's gain-bandwidth product, from its datasheet. **Bandwidth** — the frequency where the gain has fallen by 3 dB.

The headroom and GBW entered here are remembered for all the op-amp tools, since they describe the op-amp rather than the circuit.

## Why the formulas are these

The op-amp holds its − input at the same voltage as its + input, which is Vin. The divider puts Vout × R1 / (R1 + Rf) on the − input, so

```
Vout × R1 / (R1 + Rf) = Vin   →   Vout = Vin × (1 + Rf / R1)
```

The divider current is Vout / (Rf + R1). The output can only reach the supply less the headroom, so |Vout| ≤ supply − headroom.

Here the **noise gain**, which sets the bandwidth, equals the signal gain, so

```
Bandwidth = GBW / Gain
```

A gain of 101 on a 1 MHz op-amp leaves only 9.9 kHz. For audio, use a faster op-amp or split the gain over two stages.

## Assumptions and limits

The limits are the same as for the inverting amplifier: - an ideal op-amp for the gain; - a symmetric supply, with the same headroom on both sides; - slew rate, offset and bias current are not calculated; - the load and the divider together must not demand more current than the output can give.

The input voltage must also stay within the op-amp's **common-mode range** (on the datasheet), because here the inputs follow the signal. Many classic op-amps do not accept inputs close to their positive supply.

## What it deliberately does not do

- It does not choose the resistors. Pick R1 in the kΩ range, then Rf = (gain − 1) × R1 from the E-series.
- Frequency response curves and phase are not drawn.

## How it was checked

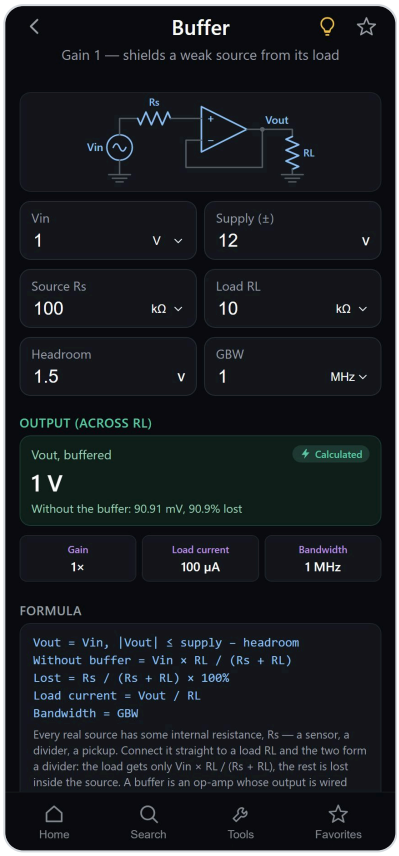
- 10 kΩ / 100 kΩ, 0.5 V peak, ±12 V, 1.5 V headroom, 1 MHz GBW:
- gain 11 (20.83 dB), 5.5 V out, 50 µA in the divider;
- bandwidth 90.91 kHz.
- Rf = R1: gain 2 (6.02 dB), 500 kHz.
- Gain 101, 20 mV: 2.02 V, 9.901 kHz.
- Gain 11, 1.5 V: the ideal 16.5 V clips at 10.5 V.
- Rf = 0: gain 1 and the whole 1 MHz.

The review found the same problems as on the inverting amplifier: - clipping at the full supply; - no bandwidth; - units such as mΩ, GΩ and kV; - a 9 px legend and textbook jargon ("series feedback", "virtual short").

It now shares the headroom and GBW fields and the waveform with the inverting amplifier. It shows Vout first and large, and adds plain definitions and examples.

# Buffer

(formerly Buffer (voltage follower))




## What it computes

For an op-amp buffer between a source and a load, it gives: - the voltage the load receives through the buffer; - what the load would get without the buffer, and how much is lost; - the load current; - where the output clips, from the supply and the op-amp's headroom; - the bandwidth.

## Source

No standard applies. The figures come from the ideal op-amp model and from Ohm's law for the source and load. - **Headroom and gain-bandwidth figures** — read from the TL072, LM358 and MCP6002 datasheets, as listed in the Inverting section. - **The 10 mA warning** — follows the TI LM358 datasheet (SLOS068AB, October 2024). At a 15 V supply it gives an output current of 20 mA minimum (30 typical) when sourcing and 10 mA minimum (20 typical) when sinking.

## What it means

Every real signal source has some resistance of its own, **Rs**: a sensor, a voltage divider, a guitar pickup, a reference. When it drives a **load**, RL, current flows, and part of the voltage is lost across Rs before it reaches the load. The two resistances form a divider. This is called **loading**.

- **Without the buffer** — the load gets Vin × RL / (Rs + RL). A 100 kΩ sensor into a 10 kΩ load delivers only 91 mV of its 1 V.
- **Buffer** — an op-amp whose output is wired straight back to its − input. It copies its + input to its output: a gain of 1, the same way up.
- Its input draws almost no current, so no voltage is lost across Rs.
- Its output drives the load from the op-amp's own supply, with almost no resistance of its own.
- **Load current** — Vout / RL. The op-amp's output has to supply it. Above about 10 mA many op-amps lose swing or limit; check the output current on the datasheet.
- **Headroom** — how close the output can get to the supply. It is about 1.5–2 V for a TL072 or LM358, and 0 for a rail-to-rail op-amp. A larger input clips: a buffer has no gain to reduce, so the input itself is too large for the supply.
- **Bandwidth** — with a gain of 1, the bandwidth is the op-amp's whole gain-bandwidth product (GBW).

The headroom and GBW entered here are remembered for all the op-amp tools.

## Why the formulas are these

Without the buffer, the same current flows through Rs and RL, so RL gets the share RL / (Rs + RL) of Vin, and the share Rs / (Rs + RL) is lost.

With the buffer, the op-amp holds its − input, which is its output, at the voltage of its + input. Almost no current flows into the + input, so nothing is dropped across Rs, and Vout = Vin, up to supply − headroom. The noise gain is 1, so the bandwidth equals GBW.

## Assumptions and limits

- An ideal op-amp, apart from the headroom.
- **Input range.** In a buffer the inputs follow the signal, so Vin must stay within the op-amp's input common-mode range on the datasheet. Many classic op-amps do not accept inputs near their positive supply.
- **Output current.** The 10 mA figure is a rule of thumb from the LM358; other parts deliver more or less.
- **Capacitive loads** (long cables) can make a buffer oscillate. This is not covered.

## What it deliberately does not do

- It does not model the source's own frequency response or noise.
- Gain other than 1 belongs to the Non-inverting tool.

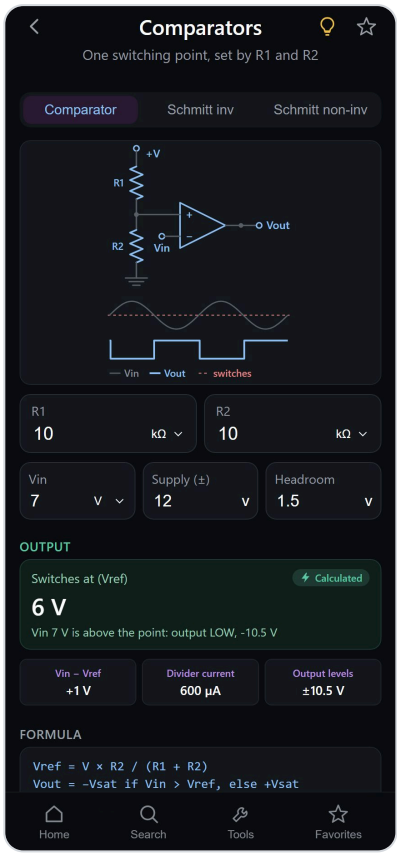
## How it was checked

- 1 V from 100 kΩ into 10 kΩ:
- buffered: 1 V, 100 µA, a 1 MHz bandwidth on a 1 MHz op-amp;
- without the buffer: 90.91 mV, 90.9% lost.
- A 10 k / 10 k divider (2.5 V from 5 kΩ) into 1 kΩ: 416.7 mV without the buffer, 2.5 V with it, 2.5 mA.
- 50 Ω into 10 kΩ: 0.498% lost.
- 11 V on ±12 V with 1.5 V headroom: clips at 10.5 V.
- 5 V into 100 Ω: 50 mA, with the warning.

The review kept the drawing from Pierre's sheet and added the source with its Rs and the load RL around it. It added the headroom, the bandwidth and the output-current warning, and put the buffered voltage first and large, with the unbuffered value beside it. It also limited the units to real ones, remembered the choices, and wrote the note in plain words with examples.

# Comparators

(formerly Comparator (± hysteresis / Schmitt trigger))




## What it computes

For an op-amp used as a comparator, in three forms, it gives: - the switching point or points: one for a plain comparator, two for a Schmitt trigger; - the hysteresis between the two points, and their centre; - whether the output is high or low for a given input, and the two output levels; - for the plain comparator, the reference divider's current.

The waveform shows a sine wave swept across the points and the output it produces.

## Source

No standard applies. The points come from the resistor network at the + input; the derivation is below. The output levels use the op-amp's headroom, read from the TL072, LM358 and MCP6002 datasheets as listed in the Inverting section.

## What it means

A **comparator** answers one question: which input is higher? It has no feedback to hold its output in between, so the output sits at one of two levels, **high** or **low**. Those levels are the supply less the op-amp's headroom, called **Vsat**. They are about 1.5–2 V short of the rails for a TL072 or LM358, and at the rails for a rail-to-rail op-amp.

- **Comparator** — R1 and R2 divide the supply into a reference, Vref, on the + input. Vin goes to the − input. Vin above Vref sends the output low, and below it high: it is inverting. The divider draws its current all the time.
- **The problem with a plain comparator** — when the input is slow or noisy near Vref, the noise carries it back and forth across the point, and the output flips many times. This is called **chatter**.
- **Schmitt trigger** — the cure is **hysteresis**: two switching points instead of one. A rising input switches at the upper point, **VT+**, and a falling one at the lower point, **VT−**. Noise smaller than the gap between them cannot flip the output back.
- **Inverting** — Rf feeds part of the output back to the divider's middle point, so the reference moves with the output. While the output is high, the point is VT+; while it is low, VT−. When Vin is between the points, the output keeps its last state.
- **Non-inverting** — Vin comes to the + input through Rin, with Rf from the output to the same input, and the − input at 0 V. The points sit symmetrically around 0 V, and the output follows the input.
- **Centre** — the middle of the two points.

The headroom entered here is remembered for all the op-amp tools.

## Why the formulas are these

**Comparator.** Vref = V × R2 / (R1 + R2), and the divider current is V / (R1 + R2).

**Inverting Schmitt.** Three resistors meet at the + input: R1 from the supply V, R2 to ground, and Rf from the output (±Vsat). With no current into the op-amp, the currents into that node sum to zero, so its voltage is

```
V+ = (V / R1 + Vout / Rf) / G,   G = 1/R1 + 1/R2 + 1/Rf
```

Putting Vout = +Vsat gives VT+, and −Vsat gives VT−. The gap is 2 × Vsat / (Rf × G). A smaller Rf makes it wider.

**Non-inverting Schmitt.** The output switches when the + input crosses 0 V. There Vin / Rin = −Vout / Rf, so

```
VT± = ± Vsat × Rin / Rf
```

and the gap is 2 × Vsat × Rin / Rf.

The output level Vsat appears in both Schmitt formulas. That is why the headroom changes the switching points, not only the output.

## Assumptions and limits

- **An op-amp used as a comparator.**
- An op-amp is not made for this. It switches slowly (its slew rate), and some op-amps misbehave with their inputs far apart.
- A dedicated comparator chip (LM393, LM339) is faster. Its output is usually open-collector: it only pulls low, a pull-up resistor makes the high level, and the high level is the pull-up's voltage. That case is not modelled.
- **Symmetric supply** of ± the value entered, with the same headroom on both sides.
- **Input range** — both inputs must stay within the op-amp's input common-mode range from the datasheet.
- The source driving Vin is taken as having no resistance of its own. In the non-inverting form, any source resistance adds to Rin.

## What it deliberately does not do

- It does not choose the resistors for given switching points. Try values instead: a smaller Rf widens the gap, and R1 and R2 move the centre.
- Switching speed and propagation delay are not calculated.

## How it was checked

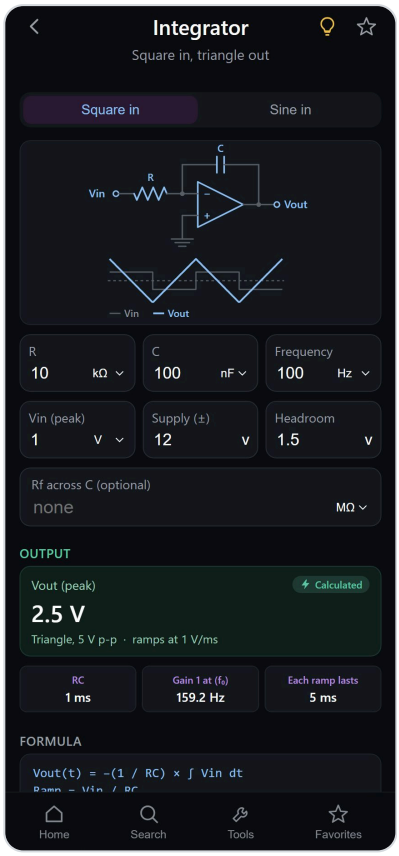
All on ±12 V with 1.5 V headroom, so the output levels are ±10.5 V: - **Comparator**, 10 k / 10 k: switches at 6 V. Vin 7 V gives a low output, −10.5 V, and the divider draws 600 µA. - **Inverting Schmitt**, 10 k / 10 k: - Rf 100 k: VT+ 6.214 V, VT− 5.214 V, a 1 V gap; - Rf 22 k: 6.833 V and 2.944 V, 3.889 V apart; - Rf 100 k on a rail-

to-rail op-amp (headroom 0): 6.286 V and 5.143 V. - **Non-inverting Schmitt**, 10 k / 100 k: ±1.05 V. A 2 V input gives a high output.

The review made several changes: - The output levels were the full supply, so the switching points were wrong for most op-amps. The shared headroom field fixes both. - The switching points come first and large. - The Schmitt mode's Rf no longer squeezes three fields into one row. - The waveform legend is now 11 px, below the plot. - Units are limited to real ones and remembered, the note is in plain words, and there are examples. - The tool was renamed Comparators to fit on one line on the phone.

# Integrator

(formerly Op-amp integrator)




## What it computes

For the op-amp integrator, with a square or a sine wave at its input: - the output's peak; - for a square, the triangle's peak-to-peak, its ramp rate and how long each ramp lasts; - for a sine, the gain, as a ratio and in decibels; - the time constant RC and the frequency $f_0$ where the gain is 1; - with the optional drift resistor Rf across C, its corner frequency fL, and the output it then gives; - where the output clips, from the supply and the op-amp's headroom.

It draws the input and output waves.

## Source

No standard applies. The output is the integral of the input, from the ideal op-amp model. The headroom figures come from the TL072, LM358 and MCP6002 datasheets, as listed in the Inverting section.

## What it means

To **integrate** is to keep adding up: the output is the running total of the input over time.

- **How the circuit does it** — the op-amp holds its − input at 0 V, so the current through R is Vin / R, and none of it enters the op-amp. All of it charges C. A capacitor charged by a steady current changes its voltage at a steady rate, so the output moves at Vin / RC volts per second. It moves downwards for a positive input: the circuit inverts.
- **RC** — the time constant, in seconds. A larger RC makes a slower ramp and a smaller output.
- **Square in, triangle out** — a square wave is a steady +Vin, then a steady −Vin. The output ramps down, then up, which makes a triangle. Its height is the ramp rate times half a period.
- **Sine in** — a sine comes out as a sine moved a quarter cycle (90°) earlier, its size multiplied by the **gain**, 1 / (2πfRC). The gain halves each time the frequency doubles (−6 dB per octave).
- **$f_0$** — the frequency where the gain is exactly 1. Below it the output is larger than the input, and above it smaller.
- **Headroom** — how close the output can get to the supply. It is about 1.5–2 V for a TL072 or LM358, and 0 for a rail-to-rail op-amp. At low frequencies the integrator asks for large outputs and clips.
- **Rf across C** (optional) — a real integrator also adds up the op-amp's own small offset voltage, so its output slowly drifts into a rail. A large resistor across C, around 100 × R, stops that. Leave the field empty for the ideal circuit.
- **fL, the Rf corner** — 1 / (2π × Rf × C). Above fL the circuit integrates; well below it, it is just an inverting amplifier of gain Rf / R.
- Under 10 × fL the tool warns that the integration is poor.

The headroom is remembered for all the op-amp tools.

## Why the formulas are these

The current into C is Vin / R. A capacitor's voltage changes at I / C per second, and the output is the far side of C from the 0 V input:

```
Vout(t) = −(1 / RC) × ∫ Vin dt
```

- **Square wave** — ±Vin holds for half a period, 1 / (2f), at the rate Vin / RC. The swing is

  Vout p-p = Vin / (2 × f × RC)

and the peak is half of it, with the triangle centred on 0 V. - **Sine** — integrating Vp × sin(2πft) gives −Vp × cos(2πft) / (2πf). With the minus sign of the circuit, that becomes +Vp × cos / (2πfRC): the same shape, a quarter cycle ahead, scaled by 1 / (2πfRC) = $f_0$ / f. - **With Rf** — Rf in parallel with C makes a first-order low-pass of DC gain Rf / R and corner fL. - A sine sees a gain of (Rf / R) / √(1 + (f / fL)$^2$) and leads by 180° − atan(f / fL). Well above fL this is the ideal $f_0$ / f and 90°. - A square settles to exponential edges. Each half period starts at +A and heads towards −(Rf / R) × Vin with time constant Rf × C. Requiring it to end at −A gives

```
    A = (Rf / R) × Vin × tanh(1 / (4 × f × Rf × C))

As Rf grows this becomes the ideal Vin / (4 × f × RC). The slope
through 0 V is always Vin / RC.
```

## Assumptions and limits

- **Drift.** Without Rf, the tool shows the ideal integrator. A real one drifts into a rail; fit Rf to model the practical circuit. The offset itself is not calculated.
- **Steady state.** The output is shown as it settles, centred on 0 V. Without Rf, its level in a real circuit depends on where it started.
- **Op-amp speed.** At high frequencies the op-amp's gain-bandwidth and slew rate limit the output. This is not calculated.
- The input is assumed to have no DC part. A DC input makes a ramp that never stops.

## What it deliberately does not do

- The output's DC error from the offset voltage, multiplied by Rf / R, is not calculated.
- Other waveshapes (pulses, ramps) are not offered.

## How it was checked

All on ±12 V with 1.5 V headroom: - Square, 10 kΩ, 100 nF, 1 V, 100 Hz: RC 1 ms, a 5 V p-p triangle (2.5 V peak), ramping 1 V/ms, each ramp 5 ms. $f_0$ 159.2 Hz. - 10 nF at 1 kHz: RC 100 µs, 5 V p-p, 10 V/ms. - Sine at 100 Hz with 10 kΩ and 100 nF: gain 1.592 (4.04 dB), 1.592 V peak. - Square at 10 Hz: the ideal 25 V peak clips

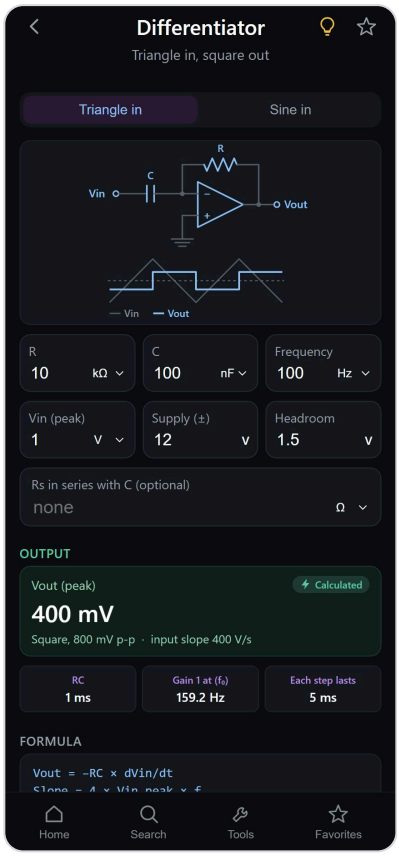
at 10.5 V. - With Rf = 1 MΩ: - fL is 1.592 Hz; - the 100 Hz square gives 2.499 V peak, against 2.5 V ideal; - a 100 Hz sine has a gain of 1.591 and leads by 90.9°; - at 10 Hz, below 10 × fL (15.92 Hz), the tool warns.

These were checked in a separate script.

The review made several changes: - It moved clipping from the full supply to the supply less the headroom. - It put the output peak first and large, with the triangle's size and ramp beside it. - It shows the ramp in V/ms or V/µs instead of kV/s. - It replaced a row with three fields squeezed into two places by two rows of three. - It moved the 11 px legend below the waves. - It limited the units to real ones and remembers them. - It rewrote the note in plain words and added examples. - On Pierre's request, it added the optional Rf across C, drawn as a second rung above C.

# Differentiator

(formerly Op-amp differentiator)




## What it computes

For the op-amp differentiator, with a triangle or a sine wave at its input, it gives: - the output's peak; - for a triangle, the square's peak-to-peak, the input's slope and how long each step lasts; - for a sine, the gain, as a ratio and in decibels, and the phase; - the time constant RC and the frequency $f_0$ where the gain is 1; - with the optional resistor Rs in series with C, its corner frequency fH and the output it then gives; - where the output clips, from the supply and the op-amp's headroom.

It draws the input and output waves.

## Source

No standard applies. The output is the derivative of the input, from the ideal op-amp model. The headroom figures come from the TL072, LM358 and MCP6002 datasheets, as listed in the Inverting section.

## What it means

To **differentiate** is to measure how fast something changes: the output follows the input's slope, not its level. It is the mirror image of the integrator.

- **How the circuit does it** — a capacitor passes current only while its voltage changes, so the current through C is C × (rate of change of Vin). The op-amp holds its − input at 0 V and sends that current through R, so Vout = −RC × slope. A rising input gives a negative output: the circuit inverts.
- **Triangle in, square out** — a triangle rises at one steady rate, then falls at the same rate. The output is therefore a steady negative level, then a steady positive one: a square. A triangle of peak Vp at frequency f rises at 4 × Vp × f.
- **Sine in** — a sine comes out as a sine a quarter cycle (90°) **behind** the input, multiplied by the gain 2πfRC. The derivative of a sine is a cosine, which is ahead, but the circuit's inversion turns it round. The gain doubles each time the frequency doubles (+6 dB per octave).
- **$f_0$** — the frequency where the gain is exactly 1.
- **Headroom** — how close the output can get to the supply: about 1.5–2 V for a TL072 or LM358, 0 for a rail-to-rail op-amp. At high frequencies the differentiator asks for large outputs and clips.
- **Rs in series with C** (optional) — because the gain keeps rising with frequency, a bare differentiator amplifies high-frequency noise enormously and can oscillate. A small resistor in series with C stops the rise. Leave the field empty for the ideal circuit.
- **fH, the Rs corner** — 1 / (2π × Rs × C). Above it the gain stops rising, at R / Rs.
- The circuit differentiates well only below about fH / 10, and the tool warns above that.

The headroom is remembered for all the op-amp tools.

## Why the formulas are these

The current into the − node is C × dVin/dt, and it all flows through R:

```
Vout = −RC × dVin/dt
```

- **Triangle** — the slope is ±4 × Vp × f, so the output is a square of ±RC × 4 × Vp × f. Each level lasts half a period.
- **Sine** — the derivative of Vp × sin(2πft) is 2πf × Vp × cos(2πft). With the minus sign it becomes −2πfRC × Vp × cos: the same shape, a quarter cycle behind, scaled by 2πfRC = f / $f_0$.
- **With Rs** — Rs in series with C makes a first-order high-pass of top gain R / Rs and corner fH.
- A sine sees a gain of 2πfRC / √(1 + (f / fH)$^2$) and lags by 90° + atan(f / fH). Well below fH this is the ideal f / $f_0$ and 90°.

- A triangle's slope drives the input current towards C × slope with time constant Rs × C, so the square gets exponential edges. Its peak is

  ```
  RC × slope × tanh(1 / (4 × f × Rs × C))
  ```

  which becomes the ideal RC × slope as Rs shrinks.

## Assumptions and limits

- **Ideal circuit without Rs.** It is shown as drawn on the reference sheet, but a real one needs Rs; fit it to model the practical circuit.
- **Input source.** The source driving Vin is taken as having no resistance of its own. A real source's resistance acts like part of Rs.
- **Op-amp speed.** The op-amp's gain-bandwidth and slew rate also limit the output at high frequencies. This is not calculated.
- **Sharp triangle corners** are taken as perfect, so the square switches instantly.

## What it deliberately does not do

- The small capacitor sometimes added across R, for more noise filtering, is not modelled.
- Other waveshapes are not offered.

## How it was checked

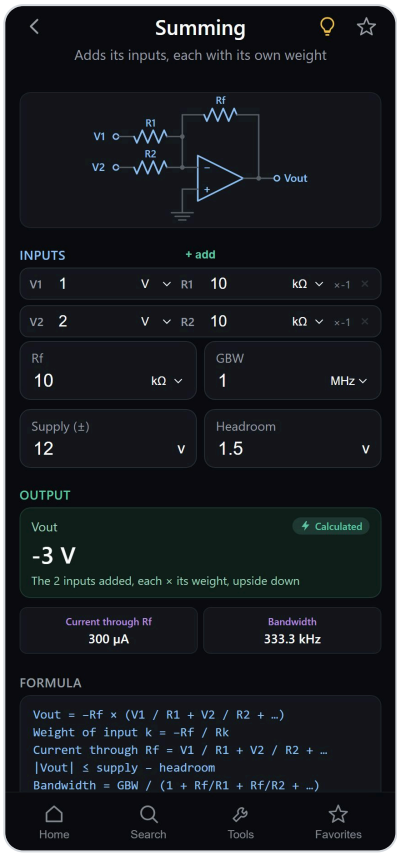
All with 10 kΩ, on ±12 V with 1.5 V headroom: - **Triangle**, 100 nF, 1 V, 100 Hz: - RC 1 ms, slope 400 V/s; - a ±400 mV square (800 mV p-p), each step 5 ms. - **Triangle**, 10 nF at 1 kHz: 4 V/ms, ±400 mV. - **Sine**, 100 nF at 1 kHz: gain 6.283 (15.96 dB), lags 90°. - **Sine** at 10 kHz: the ideal 62.83 V clips at 10.5 V. - **With Rs = 100 Ω**: - fH 15.92 kHz; - at 1 kHz the gain is 6.271 and the lag 93.6°. - **With Rs = 1 kΩ** at 1 kHz: above fH / 10 (159.2 Hz), so the tool warns that the gain tends to R / Rs = 10.

The review made these changes: - It fixed the subtitle, which said a sine comes out a quarter cycle early; the output lags by 90°. - It moved clipping from the full supply to the supply less the headroom, and put the

output peak first and large. - It shows slopes in V/s, V/ms or V/µs. - It replaced a row with three fields squeezed into two places by two rows of three. - It put an 11 px legend below the waves. - It limited the units to real ones and remembers them, rewrote the note in plain words and added examples. - On Pierre's request, it added the optional Rs, drawn in series ahead of C.

# Summing

(formerly Summing amplifier)




## What it computes

For the inverting summing amplifier, with two to four inputs (+ add and × on each input's line), it gives: - the output voltage; - each input's weight, shown on its line; - the current through Rf; - where the output clips, from the supply and the op-amp's headroom; - the bandwidth, from the op-amp's gain-bandwidth product.

## Source

No standard applies. The output comes from the ideal op-amp model. The headroom and gain-bandwidth figures come from the TL072, LM358 and MCP6002 datasheets, as listed in the Inverting section.

## What it means

- **Virtual ground** — the op-amp holds its − input at 0 V. So each input drives a current V / R through its own resistor, and is not affected by the others.
- **Adding** — those currents meet at the − input, add up, and all flow on through Rf. Rf turns the total into the output voltage, upside down (inverted).
- **Weight** — each input's weight is −Rf divided by its own resistor:
- equal resistors give a plain sum;
- unequal ones mix channels at different levels, as in an audio mixer;
- Rf equal to half the input resistors gives the average;
- binary-weighted resistors (R, 2R, 4R…) make a simple digital-to-analog converter.
- **Headroom** — how close the output can get to the supply. It is about 1.5–2 V for a TL072 or LM358, and 0 for a rail-to-rail op-amp. Past that, the output clips.
- **Bandwidth** — the op-amp's gain-bandwidth product (GBW) is shared out by the noise gain, 1 + Rf/R1 + Rf/R2 + …. Every input resistor counts, so the bandwidth is lower than any one channel's gain would suggest.

The headroom and GBW entered here are remembered for all the op-amp tools.

## Why the formulas are these

The − input is at 0 V, so input k drives Vk / Rk into it. No current enters the op-amp, so the sum flows through Rf, from the − node towards the output:

```
Vout = −Rf × (V1 / R1 + V2 / R2 + …)
```

The feedback network divides the output by 1 + Rf / (R1 ‖ R2 ‖ …) on its way back to the − input, which equals 1 + Rf/R1 + Rf/R2 + …. That noise gain sets the bandwidth: GBW / (1 + Rf/R1 + Rf/R2 + …). Each input added lowers it.

## Assumptions and limits

- An ideal op-amp for the sum.
- A symmetric supply, with the same headroom on both sides.
- Each source is taken as having no resistance of its own. A source's resistance adds to its input resistor and lowers its weight.
- Up to four inputs are offered, which is what still fits above the tab bar. More work the same way, each adding its own V / R term.

## What it deliberately does not do

- Non-inverting summing, with the inputs on the + side, is not covered.
- Subtracting belongs to the Differential tool.

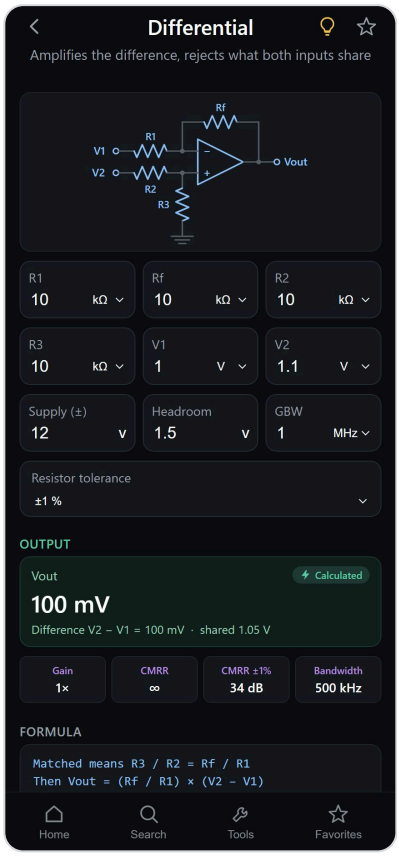
## How it was checked

All on ±12 V with 1.5 V headroom and a 1 MHz GBW: - 10 kΩ all round, 1 V + 2 V: - −3 V, weights −1 and −1, 300 µA through Rf; - bandwidth 333.3 kHz (1 MHz / 3). - **+ add**, with a third input of 1 V through 10 kΩ: −4 V, 250 kHz. - Three-channel mixer, R3 = 20 kΩ, 0.5 V on each input: −1.25 V, weight −0.5 on V3, 285.7 kHz. - Averager, Rf = 5 kΩ: −1.5 V. - 3-bit DAC, 10 / 20 / 40 kΩ, code 101 (5 V, 0 V, 5 V): −6.25 V. - 6 V + 6 V: the ideal −12 V clips at −10.5 V.

The review made these changes: - It moved clipping from the full supply to the supply less the headroom, and added the bandwidth. - It put Vout first and large, replacing a row of six small cells. - It limited the units to real ones and remembers them. - It rewrote the subtitle and note in plain words, and added examples. - The tool was renamed Summing, on Pierre's request. - Also on Pierre's request, inputs can be added, up to four. Each is one compact line with its weight, and the drawing grows one arm per input.

# Differential

(formerly Differential amplifier)




## What it computes

For the one-op-amp difference amplifier, it gives: - the output for two input voltages; - the gain; - the common-mode rejection (CMRR), both as drawn and at worst with real resistors of a chosen tolerance; - the bandwidth; - where the output clips.

When the two arms are not balanced, it also gives how much of the output is leaked common-mode voltage.

## Source

No standard applies. The output follows from the ideal op-amp model by superposition. The worst-case CMRR with resistor tolerance is derived below from the same equation. The headroom and gain-bandwidth figures come from the TL072, LM358 and MCP6002 datasheets, as listed in the Inverting section.

## What it means

A **difference amplifier** outputs the difference between its inputs, V2 − V1, times a gain, and ignores the voltage both inputs share.

- **Common-mode voltage** — the shared part, (V1 + V2) / 2. It is often much larger than the difference:
- a sensor bridge sitting at 2.5 V with millivolts between its sides;
- a current shunt at 12 V;
- hum picked up equally on both wires.
- **Balanced arms** — the circuit rejects the shared part only when R3 / R2 equals Rf / R1. The gain is then Rf / R1.
- **CMRR** (common-mode rejection ratio) — how much more the difference is amplified than the shared voltage, in dB. With exactly matched resistors it is infinite.
- **CMRR with tolerance** — real resistors are off by their tolerance. At worst, four ±1 % resistors at gain 1 leave only about 34 dB: a shared 12 V would still put about 240 mV on the output. That is why precision resistors, or a ready-made difference-amplifier chip with laser-trimmed resistors, are used.
- **Input loading** — V1 sees R1 to the virtual input, and V2 sees R2 + R3 to ground. Both draw current from their sources, unlike an instrumentation amplifier.
- **Headroom** and **bandwidth** — as on the other op-amp tools. The bandwidth is GBW / (1 + Rf / R1).

The headroom and GBW are remembered for all the op-amp tools.

## Why the formulas are these

With k = Rf / R1 and β = R3 / (R2 + R3), the + input sits at β × V2, and the op-amp holds the − input there too. So

```
Vout = β(1 + k) × V2 − k × V1
```

Writing V1 and V2 as the common part Vcm plus or minus half the difference Vd gives

```
Vout = [β(1 + k) + k] / 2 × Vd + [β(1 + k) − k] × Vcm
     =        Ad        × Vd +       Acm        × Vcm
```

Acm is zero exactly when β(1 + k) = k, which means R3 / R2 = Rf / R1. Then Ad = k.

**Worst case with tolerance.** Let the ratios Rf / R1 and R3 / R2 be off by fractions a and b. To first order, Acm = k(b − a) / (1 + k). Each ratio of two ±t resistors can be off by 2t, in opposite directions, so at worst |Acm| = 4tk / (1 + k). With Ad ≈ k:

```
CMRR (worst) = (1 + Rf / R1) / 4t
```

A higher gain helps: the same ±1 % resistors give about 49 dB at gain 10.

## Assumptions and limits

- **An ideal op-amp.** The op-amp's own CMRR, usually 70–100 dB, is assumed far better than the resistors'. With 0.1 % resistors at high gain, the two become comparable.
- **Ideal sources.** Each source is taken as having no resistance of its own. A source's resistance adds to R1 or R2 and unbalances the arms.
- **Inputs within range.** The op-amp's inputs sit at β × V2, which must stay within its input range even when V1 and V2 are higher. In the 12 V shunt example they sit at about 6 V.
- **Symmetric supply**, with the same headroom on both sides.

## What it deliberately does not do

- The three-op-amp instrumentation amplifier, with high input impedance, is not covered.
- Frequency-dependent CMRR is not calculated.

## How it was checked

- **Four 10 kΩ, 1 V and 1.1 V:** 100 mV out, CMRR ∞ as drawn, 34 dB at ±1 %, 500 kHz on a 1 MHz op-amp.
- **Bridge at 2.5 V, 10 k / 100 k:** 2.49 V and 2.51 V give 200 mV, gain 10, 48.8 dB at ±1 %, 90.91 kHz.
- **R3 = 110 kΩ instead of 100 kΩ:**
- 409.2 mV out;
- CMRR 41.6 dB;

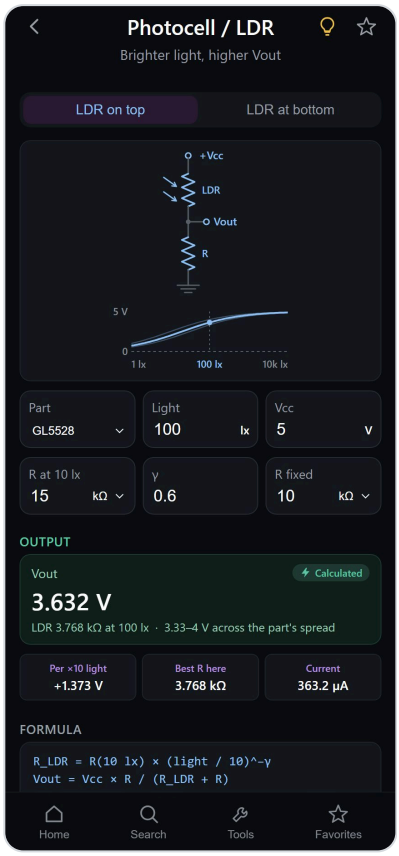
- 50.9 % of the output is leaked common-mode voltage.
- **Shunt at 12 V (11.9 V and 12 V):** 100 mV out, common-mode voltage 11.95 V.
- **±0.1 % at gain 1:** 54 dB.

These figures were checked in a separate calculation.

The review made these changes: - It moved clipping from the full supply to the supply less the headroom, and added the bandwidth. - It added the worst-case CMRR for a chosen resistor tolerance. - It put Vout first and large, replacing a row of six small cells. - It removed the waveform. That waveform showed an invented slow interference unrelated to the entered inputs. - It limited the units to real ones and remembers them. - It rewrote the note in plain words and added examples. - The tool was renamed Differential, on Pierre's request.

ACTIVE & SEMICONDUCTOR DEVICES › OPTOELECTRONICS

# Photocell / LDR




## What it computes

For a light-dependent resistor in a voltage divider with a fixed resistor, it gives: - the output voltage at a given light level; - the LDR's resistance at that light; - the range of output the part's 2:1 manufacturing spread can give; - how much the output changes for each ×10 of light; - the fixed resistor that would make the divider most sensitive; - the current drawn.

It plots the output across four decades of light.

## Source

GL55-series data from the series' datasheet, "GL55 Series CdS Photoresistor Manual" (5 pages; the copy read names no manufacturer). - **Test conditions** — light resistance is measured at 10 lx under standard light source A (2856 K), after two hours at 400–600 lx. - **γ** — defined as lg(R10 / R100), the logarithm of the ratio of the resistances at 10 and 100 lx.

| Part | R at 10 lx | Dark resistance (min) | γ | Rise / fall |
|---|---|---|---|---|
| GL5516 | 5–10 kΩ | 0.5 MΩ | 0.5 | 30 / 30 ms |
| GL5528 | 10–20 kΩ | 1 MΩ | 0.6 | 20 / 30 ms |
| GL5537-1 | 20–30 kΩ | 2 MΩ | 0.6 | 20 / 30 ms |
| GL5537-2 | 30–50 kΩ | 3 MΩ | 0.7 | 20 / 30 ms |
| GL5539 | 50–100 kΩ | 5 MΩ | 0.8 | 20 / 30 ms |
| GL5549 | 100–200 kΩ | 10 MΩ | 0.9 | 20 / 30 ms |

Other vendors' sheets for the same names differ a little. One gives 8–20 kΩ for the GL5528, for example, so the spread is real.

## What it means

An **LDR** (light-dependent resistor, photoresistor or photocell) is a resistor whose value falls as the light rises. It is thousands of ohms in a dim room, a few hundred in daylight, and megohms in the dark. The common ones are made of cadmium sulphide (CdS).

- **Lux (lx)** — the unit of illuminance, how much light falls on a surface. Rough values:
- about 1 lx at deep twilight;
- 50–100 lx in a dim room;
- 300–500 lx in an office;
- 10 000 lx and more in daylight.
- **R at 10 lx** — the datasheet's reference resistance.
- **γ** — how steeply the resistance falls: each ×10 of light divides it by 10^γ. With γ = 0.6, 100 lx gives about a quarter of the 10 lx value.
- **The divider** — the LDR and a fixed resistor R between the supply and ground. The midpoint, Vout, follows the light:
- with the LDR **on top**, brighter light gives a higher Vout;
- with it **at the bottom**, darker gives a higher Vout, as for a night light that switches on at dusk.
- **Sensitivity** — the change of Vout for each ×10 of light. It is largest when R equals the LDR's resistance at the light level you care about, which puts Vout at half the supply. "Best R here" gives that value.
- **Spread** — pieces of the same part differ by up to 2:1 at 10 lx. With a part chosen, the tool shows the range of Vout this gives, and draws it as two faint curves.

Choosing a part fills in its middle R at 10 lx and its γ. Typing your own values switches the choice to Custom.

## Why the formulas are these

The datasheet's γ = lg(R10 / R100) describes a straight line of slope −γ on log-log axes, so

```
R_LDR = R(10 lx) × (light / 10)^−γ
```

The divider gives Vout = Vcc × R / (R_LDR + R) with the LDR on top, or Vcc × R_LDR / (R_LDR + R) with it at the bottom.

Differentiating with respect to $\log_{10}$ of the light gives

```
± ln 10 × γ × Vcc × R × R_LDR / (R_LDR + R)²
```

per ×10 of light. Setting its derivative with respect to R to zero gives R = R_LDR.

## Assumptions and limits

- **The power law** is a straight-line fit through the datasheet's two defining points, 10 lx and 100 lx. It is least reliable far from them. Below about 1 lx the cell heads towards its dark resistance instead, and the tool warns. The dark resistance is measured 10 s after the light is switched off, so the cell also recovers slowly.

- **Slow.** CdS cells take tens of milliseconds to respond (20–30 ms here), so they suit day/night switching, not fast signals.
- **Temperature and history.** The resistance also drifts with temperature and with recent exposure. An LDR is not a precision light meter.
- **Light source.** The values are for an incandescent-like source (2856 K). LEDs and daylight have different spectra, and the cell peaks at 540 nm (green).
- **No loading** — the circuit reading Vout is assumed to draw no current.

## What it deliberately does not do

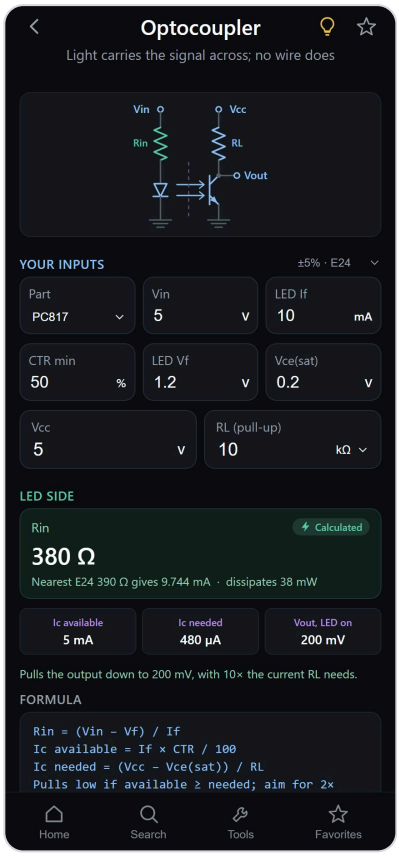
- Photodiodes and phototransistors, which are faster and linear, are not covered.
- Switching thresholds with hysteresis belong to the Comparators tool.

## How it was checked

All at 5 V, with the GL5528's middle value, 15 kΩ at 10 lx, γ 0.6: - **On top, 100 lx, R = 10 kΩ:** - LDR 3.768 kΩ, Vout 3.632 V; - 3.33–4 V across the 10–20 kΩ spread; - +1.373 V per ×10 of light, 363.2 µA. - **At the bottom, 10 lx, R = 15 kΩ:** exactly 2.5 V. - **300 lx:** 1.949 kΩ, and 2.532 V with R = 2 kΩ. - **GL5516 at 10 000 lx:** 237.2 Ω, and 4.884 V with R = 10 kΩ.

The review made these changes: - It added the GL55 parts from their datasheet, with the spread shown in the result and on the curve. - It added the choice of LDR on top or at the bottom. - It put Vout first and large. - It enlarged the curve's 8 px labels to 11 px. - It limited the units to real ones and remembers them. - It rewrote the note in plain words, with typical light levels, and added the warning below 1 lx and examples.

# Optocoupler




## What it computes

For a phototransistor optocoupler driven through a resistor, with a pull-up on the output side: - the input resistor Rin for the LED current wanted, the nearest E24 value, and the power in it; - the collector current the minimum CTR makes available; - the current the pull-up needs to pull the output low; - the output voltage with the LED on; - whether the output really pulls low, and with how much margin.

**Standard values.** The resistors' tolerance, with its E series, is picked in the "Your inputs" heading above the inputs, as in the Passive components tools: ±0.1 % to ±20 %, ±5 % being E24 and ±1 % E96; it sets the series Rin is rounded to (E24 by default).

## Source

Data read from the manufacturers' datasheets themselves.

**Sharp PC817X series** (sheet D2-A03101EN, 2003): - **CTR by rank** — measured at IF = 5 mA, VCE = 5 V, from the collector-current bins:

| Rank | CTR |
|---|---|
| A | 80–160 % |
| B | 130–260 % |
| C | 200–400 % |
| D | 300–600 % |
| Unranked | 50–600 % |

- **VF** — 1.2 V typical, 1.4 V maximum, at 20 mA.
- **VCE(sat)** — 0.1 V typical, 0.2 V maximum, at IF = 20 mA, IC = 1 mA.
- **Rise / fall time** — 4 / 3 µs typical (VCE 2 V, IC 2 mA, RL 100 Ω).

**Vishay 4N35** (document 81181, rev. 1.2, 2010): - **CTR** — at least 100 % at IF = 10 mA, VCE = 10 V. Over −55 to +100 °C it is at least 40 %, 50 % typical. - **VF** — 1.3 V typical, 1.5 V maximum, at 10 mA. - **Switching time** — 10 µs typical (VCC 10 V, IC 2 mA, RL 100 Ω). - **VCE(sat)** — no limit is given. Its curves of saturated CTR are drawn at VCE = 0.4 V, which the tool uses.

## What it means

An **optocoupler** is an LED and a light-sensitive transistor sealed in one package, facing each other across an insulating gap. The signal crosses as light, so the two circuits share no wire. They can sit at very different voltages, a microcontroller and mains equipment for example; the PC817 and 4N35 are both rated for 5 kV rms between their sides.

- **Rin** — sets the LED current If, as for any LED: Rin = (Vin − Vf) / If.
- **CTR** (current transfer ratio) — the collector current the transistor can pass for a given LED current, in %. A CTR of 50 % turns 10 mA of LED current into 5 mA available on the output side.
- **Pull-up RL** — from Vcc to the transistor's collector:
- with the LED off, the output reads high;
- with it on, the transistor pulls the output low, provided it can pass the current RL needs: (Vcc − Vce(sat)) / RL.
- **Vce(sat)** — the voltage left across the transistor when it is fully on: 0.2 V for the PC817.
- **Margin** — CTR varies widely between pieces, which is why the PC817 is sold in ranks. It is also lower when the transistor is saturated, and it falls as the LED ages and with temperature.
- Design on the **minimum** CTR, and aim for about **2×** the current RL needs. That 2× is a rule of thumb, not a datasheet figure.
- Below 1× the output stops part way down, between logic levels.

Choosing a part fills in its minimum CTR, Vf and Vce(sat). Typing your own values switches it to Custom.

## Why the formulas are these

```
Rin = (Vin − Vf) / If
Ic available = If × CTR / 100
Ic needed = (Vcc − Vce(sat)) / RL
```

If Ic available ≥ Ic needed, the transistor saturates and Vout = Vce(sat). Otherwise it passes only Ic available, and Vout = Vcc − Ic available × RL.

## Assumptions and limits

- **CTR at the test current.** CTR changes with If; both datasheets plot it against If. The ranks are specified at 5 mA (PC817) and 10 mA (4N35), so far from those currents read the curve.
- **Speed.** Phototransistor optocouplers are slow, a few microseconds, and a larger RL makes them slower. Fast signals need a logic-output part.
- **The input source** is taken as ideal, and the LED is driven with steady DC.
- **No load** on Vout other than RL.

## What it deliberately does not do

- Switching speed is not calculated.
- Triac-output and logic-output optocouplers are not covered.

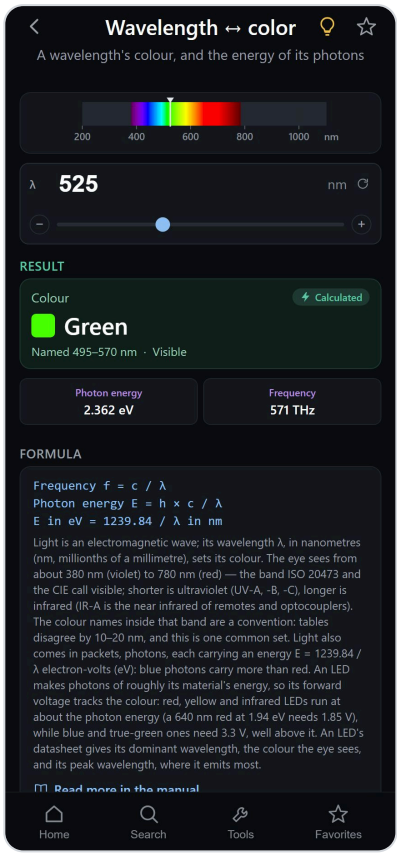
## How it was checked

- **PC817, any rank, 5 V, 10 mA, 10 kΩ to 5 V:**
- Rin 380 Ω, nearest E24 390 Ω giving 9.744 mA, 38 mW;
- 5 mA available, 480 µA needed: 10× margin, Vout 200 mV.
- **3.3 V drive at 5 mA:** 420 Ω, 5.2× margin.
- **1 kΩ to 12 V with 2 mA:** needs 11.8 mA but only 1 mA is available, so the output stops at 11 V.
- **4N35 at 10 mA from 5 V:** 370 Ω, nearest E24 360 Ω, Vout 400 mV.
- **1 mA at 50 %:** only 1× over, so the tool flags it.

The review made these changes: - The output check used a fixed 0.3 V saturation. It now takes the part's datasheet Vce(sat) and minimum CTR, from a choice of PC817 ranks or the 4N35. - It added the 2× margin advice. - It put Rin first and large, and draws it green as the calculated part. - It added the nearest E24 value. - It limited the units to real ones, rewrote the note in plain words and added examples.

# Wavelength ↔ color

(formerly Light wavelength ↔ color)




## What it computes

For a wavelength from 200 nm to 1100 nm, it gives: - the colour name, with a swatch; - the named region it falls in (ultraviolet, visible or infrared, and which part of them); - the energy of one photon, in electron-volts; - the frequency.

It marks the wavelength on a drawn spectrum.

## Source

- **Region limits** — the UV divisions follow ISO 20473:2007, for optics and photonics: UV-A 315–380 nm, UV-B 280–315 nm, UV-C 100–280 nm. The visible band is 380–780 nm, and IR-A is 780–1400 nm and IR-B 1400–3000 nm, as the CIE defines them.
- These limits were read from Opsytec's published table of spectral ranges by standard (2024), not from the standards themselves.
- The CIE notes that visible light has no precise limits: the lower one lies between 360 and 400 nm, the upper one between 760 and 830 nm.
- **Colour names** — no standard names the colours inside the visible band. Published tables disagree by 10–20 nm, and some add cyan or indigo. The tool uses one common set: violet from 380 nm, blue 450, green 495, yellow 570, orange 590, red 620 to 780.
- **The examples' LEDs** — taken from their datasheets: Kingbright WP7113SRC/DU, WP7113ZGC and WP7113QBC/D, and Vishay TSAL6200.
- **The swatch** — drawn with Dan Bruton's wavelength-to-RGB approximation, a way to put a colour on screen. It is not colorimetry.

## What it means

- **Wavelength** — light is an electromagnetic wave, and its wavelength λ, in nanometres (millionths of a millimetre), sets its colour. The eye sees about 380 nm (violet) to 780 nm (red).
- Shorter than that is **ultraviolet**: UV-A, then UV-B, then UV-C.
- Longer is **infrared**: IR-A is the near infrared that remote controls and optocouplers use.
- **Photon energy** — light comes in packets, photons, each carrying an energy E = 1239.84 / λ electron-volts (eV). Blue photons carry more than red.
- **LEDs** — an LED makes photons of roughly its material's energy, so its forward voltage tracks its colour:
- red, yellow and infrared LEDs run at about the photon energy: a 640 nm red, at 1.94 eV, needs 1.85 V;
- blue and true-green LEDs need about 3.3 V, well above it.
- **Dominant and peak wavelength** — an LED datasheet gives both. The dominant wavelength is the colour the eye sees; the peak is where the LED emits most. Names follow the dominant one.

- **Naming varies** — Kingbright sells its WP7113ID, at 617 nm, as "High Efficiency Red", yet 617 nm is orange in the set used here.

## Why the formulas are these

```
f = c / λ,   E = h × f = h × c / λ
```

With h × c = 1239.84 eV·nm, E in eV is 1239.84 divided by λ in nm. With c = 299 792 458 m/s, 525 nm is 571 THz.

## Assumptions and limits

- **Single wavelengths.** Real sources emit a band. LEDs are 20–45 nm wide (half-width, from the same datasheets), and white light mixes many wavelengths.
- **Approximate swatch.** Screens cannot show pure spectral colours. The swatch is an approximation, and blank outside 380–780 nm.

## What it deliberately does not do

- Colour mixing and colour spaces (RGB, CIE xy) are not covered.
- The eye's sensitivity (lumens) is not calculated.

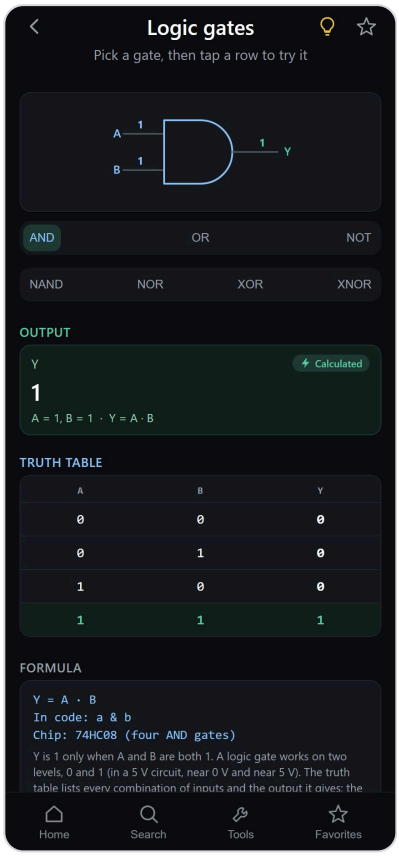
## How it was checked

- **525 nm:** green, visible, 2.362 eV, 571 THz.
- **Red LED at 640 nm:** red, 1.937 eV.
- **617 nm:** orange in this set, 2.009 eV.
- **IR LED at 940 nm:** infrared, IR-A, 1.319 eV.
- **760 nm:** still red, since visible light runs to 780 nm.
- **365 nm:** ultraviolet, UV-A.

The review made these changes: - Infrared started at 750 nm; it now starts at 780 nm, the limit ISO 20473 and the CIE use. - It added the named UV and IR regions. - It removed a source attribution ("Britannica") that had never been checked. - It corrected the claim that photon energy is the floor of an LED's forward voltage: a red LED runs slightly below it. - It put the colour first and large, with its swatch. - It enlarged the strip's 9 px labels to 11 px. - It added LED examples from datasheets and rewrote the note in plain words. - It shortened the name to Wavelength ↔ color, which fits on the phone.

# Logic gates

(formerly Basic gates (AND, OR, NOT, NAND, NOR, XOR, XNOR))




## What it computes

For each of the seven basic gates (AND, OR, NOT, NAND, NOR, XOR and XNOR), it shows: - the gate's symbol; - its truth table; - its Boolean expression; - the same operation written in code; - a common chip that contains it.

Tap a row of the truth table to set the inputs. The symbol then shows the levels on its wires, and the output Y appears large.

## Source

- **Symbols** — these are the "distinctive shape" symbols: a D for AND, a shield for OR, a triangle for NOT. Wikipedia's "Logic gate" article says ANSI/IEEE Std 91-1984 (and its supplement 91a-1991) defines both these and the rectangular IEC-style symbols, and that the distinctive shapes come from MIL-STD-806. The standard itself was not read. The app uses the distinctive shapes throughout, as it uses zigzag resistors.
- **Chips** — Texas Instruments' product pages confirm the SN74HC00 (four NAND gates), SN74HC04 (six inverters), SN74HC08 (four AND) and SN74HC32 (four OR), all for 2–6 V. Distributor listings of TI parts confirm the 74HC02 (four NOR) and 74HC86 (four XOR).
- The 74HC7266 (four XNOR) is in a Philips/Nexperia datasheet listed by a distributor. TI's own version is obsolete.

## What it means

A logic gate works on two levels, 0 and 1. In a 5 V circuit these are a voltage near 0 V and one near 5 V. The truth table lists every combination of inputs and the output each gives. A bar over an expression means "inverted".

| Gate | Output is 1 when… | Expression | In C / Python |
|---|---|---|---|
| AND | A and B are both 1 | $Y = A \cdot B$ | a & b |
| OR | A or B, or both, is 1 | $Y = A + B$ | a \| b |
| NOT | A is 0 (it inverts) | $Y = \bar{A}$ | !a / not a |
| NAND | not both are 1 | $Y = \overline{(A \cdot B)}$ | !(a & b) / not (a & b) |
| NOR | both are 0 | $Y = \overline{(A + B)}$ | !(a \| b) / not (a \| b) |
| XOR | A and B differ | $Y = A \oplus B$ | a ^ b |
| XNOR | A and B are the same | $Y = \overline{(A \oplus B)}$ | !(a ^ b) / not (a ^ b) |

- **NAND and NOR are universal** — any logic, however complex, can be built from either one alone. A NAND with its inputs tied together is a NOT.
- **XOR and AND add two bits** — XOR gives the sum and AND the carry, which makes a half adder.
- **Unused inputs** — on CMOS chips such as the 74HC series, an input left floating can make the gate switch at random and draw extra current. Tie every unused input to 0 V or to the supply.

## Why the formulas are these

Each truth table is computed from the gate's function, so a row cannot disagree with the symbol. The code forms use the bitwise operators on the values 0 and 1. In Python, `not` returns True or False rather than 1 or 0.

## Assumptions and limits

- **Ideal logic levels.** Real gates have input thresholds, output drive limits and a propagation delay, given on each chip's datasheet. They are not modelled.
- **Two-input versions.** Chips with three or more inputs per gate exist (for example triple 3-input AND); the logic extends the same way.

## What it deliberately does not do

- Simplifying a larger expression belongs to Karnaugh map simplification.
- Flip-flops, which remember, have their own tool.

## How it was checked

- **AND:** with A = 1, B = 1, Y = 1; tapping the 0 1 row gives 0.
- **XOR:** 1 0 gives 1, and 1 1 gives 0, the sum bit of a half adder.
- **NOT:** A = 1 gives Y = 0.
- **NAND:** with both inputs at 1, Y = 0 (the NOT example).
- **Remembered:** the chosen gate comes back when the tool reopens.

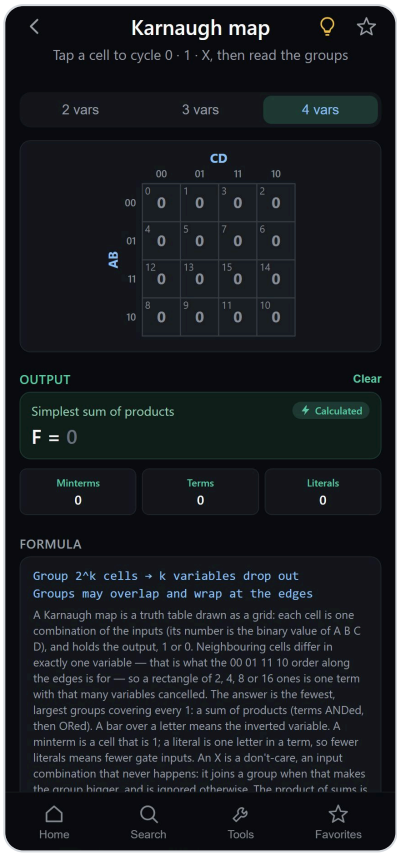
The review made these changes: - It corrected the standard cited ("ANSI/IEEE 315") to IEEE Std 91. - It made the truth table tappable, with the levels shown on the symbol and Y large. - For each gate, it added the

description in plain words, the code form and a common 74HC chip, with the CMOS unused-input advice. - It added examples and remembers the chosen gate. - The tool was renamed Logic gates, on Pierre's choice, to fit on one line.

DIGITAL & LOGIC CIRCUITS › LOGIC GATES

# Karnaugh map

(formerly Karnaugh map simplification)




## What it computes

Takes a truth table for 2, 3 or 4 variables, where each cell is 0, 1 or a don't-care. It returns: - the minimum sum-of-products expression, with each product term drawn as a group on the map in its own colour; - the minimum product of sums, under the expression; - the exclusive-OR form, when the function is a parity.

Tap a cell to cycle it through 0, 1 and X. The number of variables is remembered, and four examples in the bulb load ready-made maps.

**Plain words:** - **Cell** — one combination of the inputs. Its small number is the binary value of A B C D, and it holds the output. - **Minterm** — a cell that is 1. - **Literal** — one letter in a term, plain or barred. A bar means the inverted variable. - **Sum of products** — terms ANDed, then ORed together. - **Product of sums** — ORs ANDed together. - **Don't-care (X)** — an input combination that never happens, so its output may be whatever helps.

## Source

The map is Karnaugh's (1953), itself a reworking of the Veitch diagram. The minimisation is Quine–McCluskey, which is an exact algorithm rather than a heuristic; the covering step is solved by exhaustive search, which is affordable at four variables and would not be at twenty.

## Why the formulas are these

The whole method rests on the ordering along the edges: **00, 01, 11, 10**, not 00, 01, 10, 11. That is Gray code, and it means any two side-by-side cells differ in exactly one variable. When two adjacent cells are both 1, that variable appears once true and once complemented, so it cancels:

$$A \cdot \bar{B} \cdot C + A \cdot B \cdot C = A \cdot C \cdot (\bar{B} + B) = A \cdot C$$

Extend it and a rectangle of 2^k cells cancels k variables. A pair drops one, a block of four drops two, eight drops three. That is the entire trick — the map is just an arrangement that puts algebraically adjacent terms physically next to each other.

**The edges wrap.** The first and last columns also differ in one variable (00 and 10 differ only in the middle bit), and so do the first and last rows. The map is really a torus. That is why the four corners of a four-variable map form one group of four, and why a group can run off one edge and continue on the other. A torus

rectangle has no single box on a flat sheet, so the tool draws wrapping groups as two or four pieces in the same colour.

**Quine–McCluskey.** Terms are combined pairwise wherever they differ in one bit, the shared bit becoming a dash; repeat until nothing more combines. What never combined is a **prime implicant** — a group that cannot be made larger. Then a cover must be chosen: a minterm covered by exactly one prime implicant forces that implicant in (it is *essential*), and the rest is a set-cover problem solved here by branch and bound.

The cover is ordered **fewest terms first, then fewest literals**. The second key matters: several covers can tie on term count while differing in how many variables survive, and the answer a textbook prints is the one with fewer letters.

**Don't-cares** are included when forming groups but never required to be covered. That is where most of the saving usually comes from: an X adjacent to a group lets the group double in size, dropping another variable for free.

**Parity is what the map cannot see.** A checkerboard of ones has no two adjacent cells, so every group is a single cell and the sum of products comes out at full length — twelve of twelve literals on a three-variable map, nothing saved at all. That function is exclusive-OR:

```
A ⊕ B ⊕ C = 1 whenever an odd number of A, B, C are 1
```

Two XOR gates implement it against four ANDs and an OR for the sum of products. The map cannot find this because XOR is not a sum of products in any small form — its minimal SOP genuinely is the long one. The tool therefore tests, separately from the map, whether the function is the parity of any subset of its variables (or the complement of one) and prints that form when it is.

## Assumptions and limits

- Two to four variables. Five and six variable maps exist, drawn as stacked pairs, but they stop being readable and the whole point of the map is that you can see it.
- The result is a minimum, not *the* minimum: several different expressions can tie on both term and literal count, and the tool shows one of them.
- Literal count is a proxy for cost, not a gate count. It ignores that inverters may be free if a signal is already available complemented, and that fan-in is limited in real logic families.
- No account of hazards. A minimal cover can contain a static hazard — a glitch when two inputs change at once — which is removed by adding a redundant group that the minimiser will never choose because it costs a term.

## What it deliberately does not do

- **More than four variables**, as above.
- **Hazard-free covers.** Adding the redundant terms is a different objective from minimisation and would make the answers disagree with what a textbook shows.

## How it was checked

A second minimiser was written independently for the purpose: enumerate every possible cube, discard those covering a zero, and find the smallest cover by iterative deepening. The two were compared over 3500 random functions at 2, 3 and 4 variables, and 12000 more were checked only for the expression reproducing the truth table at every care position.

That comparison earned its keep. Term counts matched from the start, but 34 of the first 3400 came back with more literals than the reference — the cover search was returning whichever minimum-size cover it reached first, with no tie-break on literals. A correct minimum by term count, and still not the printed answer.

The parity detector was checked exhaustively over all 256 three-variable functions: exactly eight are parity forms (three pairs and one triple, each with its complement), and each one's exclusive-OR form reproduces its truth table.

**The product of sums** (added in the 2026 review) is the same exact minimisation run on the zeros, keeping the same don't-cares. De Morgan's law then turns each group of zeros into a sum of inverted letters:

$$\bar{F} = \overline{AB} + \ldots \quad \rightarrow \quad F = (A + B)(\ldots)$$

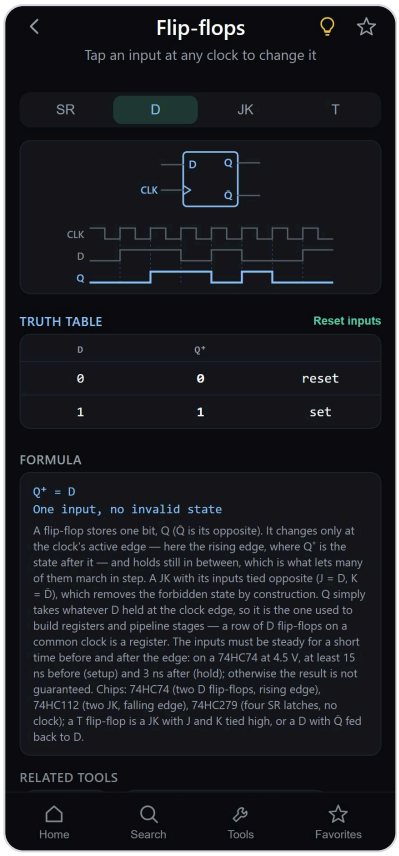
It was checked in the browser over 6000 random functions of 2 to 4 variables, with don't-cares. At all 47,615 positions that are not don't-cares, the product of sums reproduced the truth table.

**The examples:** - **Majority of three** — AB + AC + BC, with the product of sums (A + B)(A + C)(B + C). - **BCD digit 5 or more** — with codes 10–15 as don't-cares, A + BC + BD. - **4-input parity** — eight terms of four letters, and A ⊕ B ⊕ C ⊕ D. - **The four corners** — they wrap around into the single term $\bar{B}\bar{D}$.

The review made these changes: - It added the product of sums, which an earlier version left out as not worth a second screen. It needs only one line. - It put the answer first and large. - It moved the cells' numbers above the group outlines, at 11 px. - It added plain definitions, examples, and remembers the number of variables. - The tool was renamed Karnaugh map, on Pierre's choice, to fit beside the bulb.

# Flip-flops

(formerly Flip-flops (SR, D, JK, T))




## What it computes

For the four flip-flop types (SR, D, JK and T), it shows: - the symbol; - the truth table: what the next clock edge does in each input case; - the characteristic equation; - an eight-clock timing diagram, with the output Q computed by running the inputs through the flip-flop.

Tap an input at any clock in the timing diagram to flip it, and Q is recomputed. "Reset inputs" brings back the demonstration sequence. The chosen type is remembered.

## Source

The behaviour of each type is textbook logic. The chips and timing figures were read from manufacturers' documents: - **Nexperia 74HC74** (rev. 9, 2 April 2024): - a dual D flip-flop, triggered on the rising clock edge, with asynchronous active-low set and reset; - at 4.5 V and 25 °C, D must be steady at least 15 ns before the edge (setup) and 3 ns after it (hold); - the clock runs up to at least 24 MHz. - **Nexperia 74HC112** — a dual JK flip-flop, triggered on the *falling* edge, with set and reset. - **ST M74HC279** — four SR latches, with no clock.

## What it means

A **flip-flop** stores one bit, Q; $\bar{Q}$ is its opposite. It changes only at the clock's active edge, here the rising edge, and holds still in between. That is what lets many of them, such as the bits of a register or a counter, change in step. $\mathbf{Q^+}$ is the state after the next edge.

- **SR** (set / reset):
- S = 1 sets Q to 1, R = 1 resets it to 0, and both at 0 holds it;
- S = R = 1 is forbidden: both outputs are driven the same way, and when the inputs are released together the state is unpredictable. The timing diagram draws that unknown stretch dashed with a "?", until the next set or reset defines Q again;
- on its own, without a clock, the circuit is an SR *latch*.
- **D** (data) — Q takes the value D had at the edge and ignores D in between. One D flip-flop is one bit of a register.
- **JK** — like SR, but J = K = 1 toggles Q instead of being forbidden. It is the general-purpose type: D and T are both a JK with its inputs tied.
- **T** (toggle) — T = 1 flips Q at every edge, so Q is a square wave at half the clock frequency, with a 50 % duty cycle whatever the clock's.
- Chained, n stages divide by $2^n$: a ripple counter.
- It is built from a JK with J and K tied high, or a D with $\bar{Q}$ fed back to D.

- **Setup and hold** — the inputs must be steady for a short time before and after the edge, or the result is not guaranteed. A datasheet gives both, as above for the 74HC74.

## Why the formulas are these

The characteristic equations follow from the truth tables:

```
SR:  Q⁺ = S + R̄ · Q        (with S · R = 0)
D:   Q⁺ = D
JK:  Q⁺ = J · Q̄ + K̄ · Q
T:   Q⁺ = T ⊕ Q
```

In the timing diagram, each input holds across a whole clock cycle, and Q updates on the rising edge that ends it. Q starts at 0. After SR's forbidden input, Q is unknown. A hold or toggle keeps an unknown unknown, and only a set or reset ends it.

## Assumptions and limits

- **One edge for all four.** All four types are drawn edge-triggered on the rising edge. Some chips act on the falling edge (74HC112), and latches are level-sensitive instead.
- **Ideal timing.** Propagation delay, setup and hold violations and metastability are not simulated; the inputs are taken as meeting the setup and hold times.
- **No set or reset pins.** The asynchronous set and reset inputs of real chips are not drawn.

## What it deliberately does not do

- Counters and shift registers built from several flip-flops are not simulated.

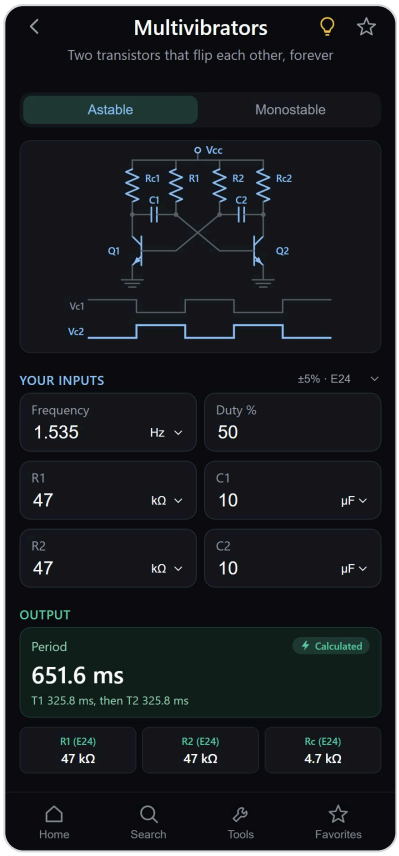
## How it was checked

- **D, demonstration inputs:** Q runs 0 0 1 1 0 1 0 0.
- **Tapping D high at the first clock** makes it 0 1 1 1 0 1 0 0, and "Reset inputs" restores it.
- **T held at 1:** 0 1 0 1 0 1 0 1, half the clock frequency.
- **SR with set, hold, S = R = 1, two holds, reset, hold:** 0 1 1 ? ? ? 0 0. Q is unknown from the forbidden input until the reset.

The review made these changes: - It fixed the forbidden SR input. The old diagram quietly kept Q unchanged, as if the state were known; it now shows it unknown until the next set or reset. - It made the inputs tappable, with a reset. - It enlarged the 10 px labels to 11 px. - It added setup and hold times and the chips from Nexperia and ST documents. - It added plain-word explanations and examples, and remembers the type. - The tool was renamed Flip-flops, on Pierre's choice, to fit beside the bulb.

# Multivibrators

(formerly Multivibrators (astable/monostable))




## What it computes

For the classic two-transistor multivibrator, the tool works in two modes.

**Astable (free-running):** - the frequency and duty cycle from the two resistor–capacitor pairs, or the resistors from a frequency and duty you type; - the period and its two halves; - the same circuit in standard E-series values, with a suggested collector resistor.

**Monostable (one-shot):** - the pulse width from R and C, or R from the width you type; - the capacitor's recovery time; - the shortest time between triggers and the highest trigger rate; - the E-series value of R.

Every field works in both directions, and the timing diagrams are drawn to the computed times.

**Standard values.** The resistors' tolerance, with its E series, is picked in the "Your inputs" heading above the inputs, as in the Passive components tools: ±0.1 % to ±20 %, ±5 % being E24 and ±1 % E96; it sets the series the suggested resistors are rounded to and the tolerance their spread uses.

## Source

- **The timing** is the standard textbook analysis of the circuit. Each half-cycle is one RC time constant times ln 2, about 0.693 RC, with Vbe and Vce(sat) neglected as the references do.
- **The emitter–base limit** is read from the datasheets: VEBO is 6 V for a 2N3904 (Diodes Inc. MMBT3904, DS30036) and a BC547 (Fairchild, rev. A2), and 5 V for the BC548–BC550 (same Fairchild sheet).

## What it means

A **multivibrator** is two transistors that switch each other on and off through capacitors.

- **Astable** — it has no stable state, so it oscillates by itself: an LED flasher, a tone.
- **Monostable** — it has one stable state. A trigger flips it for one timed pulse, and it returns by itself: a one-shot.
- **T1 and T2** — the two halves of the astable's cycle, set by R1·C1 and R2·C2.
- **Duty** — the share of the period spent in the first half, T1.
- **How a half-cycle works** — when one transistor switches on, its collector drops and pulls the other transistor's base, through the coupling capacitor, to about −Vcc. That base then climbs back through its resistor toward +Vcc and switches on as it passes zero, after ln 2 × R × C. The supply voltage cancels out, so the frequency does not depend on it.

- **The catch with high supplies** — the reversed base reaches about −Vcc. A small transistor's emitter–base junction withstands only 5–6 V in reverse (VEBO). On a higher supply it breaks down, and the circuit runs faster than the formula says.
- **The collector resistor Rc** — squeezed from both sides:
- at least R / β (β, or hFE, is the current gain), or the transistor stops switching fully on;
- at most about R / 10, or the collector's rising edge softens.

The tool suggests R / 10 in your E-series. - **Monostable recovery** — after each pulse the capacitor must recharge through Rc, about 5 × Rc × C, before the next trigger does anything useful. Triggering faster gives short pulses, not more of them. The large result is that shortest interval. - **Rb** (monostable) only holds Q1's base low while Q2 is on. It sets no timing, so it is not calculated.

## Why the formulas are these

```
T1 = ln 2 × R1 × C1,   T2 = ln 2 × R2 × C2
Frequency = 1 / (T1 + T2),   Duty = T1 / (T1 + T2)
Pulse width = ln 2 × R × C,   Recovery ≈ 5 × Rc × C
Shortest time between triggers = Pulse width + Recovery
```

- **ln 2** — a capacitor charging toward +Vcc from −Vcc crosses zero when half the 2·Vcc swing is done. That takes RC × ln 2.
- **Typing a frequency** — scales both resistors and keeps the duty and the capacitors.
- **Typing a duty** — shares the period between them and keeps the frequency.
- **The capacitors are never rewritten**, since they are the part you have in the drawer.

## Assumptions and limits

- **Ideal switches.** Vbe and Vce(sat) are neglected. With them, the base starts at about Vbe − (Vcc − Vce(sat)) and switches on at Vbe rather than 0 V, so each half takes a little longer: about 0.72 RC instead of 0.693 RC on 9 V, and 0.74 RC on 5 V. The real frequency is therefore a few percent lower than calculated, more so on low supplies.
- **The 5 RC recovery** is a rule of thumb (99 % recharged).
- **Capacitor tolerance**, often ±20 % for electrolytics, matters more than the resistors' tolerance for the timing.

## What it deliberately does not do

- The 555 timer versions have their own tool.
- Bistable multivibrators, which are flip-flops, have their own tool.

## How it was checked

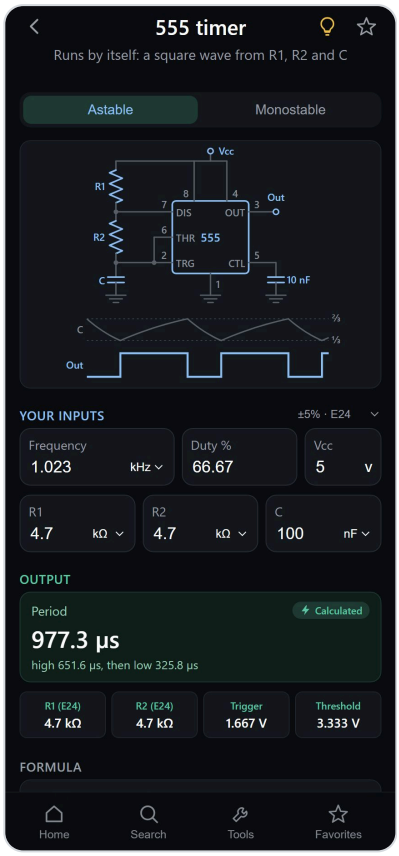
- **47 k / 10 µF on both sides:** period 651.6 ms (T1 = T2 = 325.8 ms), 1.535 Hz, Rc 4.7 kΩ suggested.
- **68 k / 10 µF:** 1.061 Hz.
- **10 k and 100 k with 10 µF:** T1 69.31 ms, T2 693.1 ms, duty 9.091 %.

- **10 k / 68 nF:** 1.061 kHz.
- **Monostable, 68 k / 10 µF with Rc 6.8 k:**
- pulse 471.3 ms;
- recovery 340 ms;
- shortest interval 811.3 ms, up to 1.233 Hz.
- **Typing 2 Hz:** both 10 µF resistors become 36.067 kΩ.

The review made these changes: - It put the period, or the shortest trigger interval, first and large. - It limited the units to real ones (Ω–MΩ, pF–µF, µs–s). - It enlarged the timing labels to 11 px. - It added plain definitions, with the VEBO figures from datasheets. - It added examples, and remembers the mode and the tolerance. - The tool was renamed Multivibrators, on Pierre's choice.

# 555 timer

(formerly 555 Timer (astable/monostable))




## What it computes

The tool covers the 555 timer chip in its two classic circuits.

**Astable (free-running):** - the frequency and duty cycle from R1, R2 and C, or the resistors from a frequency and duty you type; - the period, with its high and low times; - the two resistors in standard E-series values; - the trigger and threshold voltages, ⅓ and ⅔ of Vcc.

**Monostable (one-shot):** - the pulse width from R and C, or R from the width you type; - R in standard E-series values; - the trigger and threshold voltages.

Every field works in both directions. The timing diagram, with the capacitor's voltage above the output, is drawn to the computed times. The mode and the tolerance are remembered.

**Standard values.** The resistors' tolerance, with its E series, is picked in the "Your inputs" heading above the inputs, as in the Passive components tools: ±0.1 % to ±20 %, ±5 % being E24 and ±1 % E96; it sets the series the suggested resistors are rounded to and the tolerance their spread uses.

## Source

Texas Instruments, **NA555 / NE555 / SA555 / SE555 datasheet, SLFS022K** (revised March 2026), was read for: - **the formulas**: - astable: tH ≈ 0.693 (RA + RB) C, tL ≈ 0.693 RB C, f ≈ 1.44 / ((RA + 2 RB) C), duty (RA + RB) / (RA + 2 RB); - monostable: tw ≈ 1.1 RA C.

RA and RB are R1 and R2 here. - **the levels**: trigger about ⅓ Vcc and threshold about ⅔ Vcc. At 5 V the trigger level is 1.1–2.2 V (1.67 V typical). - **the supply**: 4.5 to 16 V for the NE555. - **the frequency**: TI advises 100 kHz or below, to reduce distortion, and the CMOS TLC555 above that. - **the pulse**: the shortest monostable pulse is about 10 µs. - Trig must be back high at least 10 µs before the pulse ends, or the output stays high. - Retriggering cannot happen during the timing cycle. - **the largest resistors**: the threshold pin draws up to 250 nA (30 nA typical). TI gives the resulting limit as RA + RB ≈ 3.4 MΩ at 5 V and ≈ 10 MΩ at 15 V. - **the smallest resistors**: TI's timing figures and test circuits start at RA = 1 kΩ. No explicit minimum is stated. The discharge switch is specified at 0.4 V maximum while sinking 8 mA, at 5 V. - **the pins**: RESET is tied to Vcc when it is not used. The CONT pin may be decoupled with a capacitor, 0.01 µF in TI's figure.

## What it means

A **555** is a timer chip, with two comparators, a latch and a discharge switch inside. The pins in the drawing are: - **TRG**, pin 2, the trigger: a voltage below ⅓ Vcc starts a high output; - **THR**, pin 6, the threshold: a

voltage above ⅔ Vcc ends it; - **DIS**, pin 7, the discharge: shorted to ground while the output is low, which empties C; - **CTL**, pin 5, the control voltage: it sets the two levels, and 10 nF to ground keeps them steady; - **pin 4**, reset, tied to Vcc so it never resets.

The modes work like this: - **Astable** — C charges through R1 + R2 up to ⅔ Vcc. The output then goes low, and pin 7 drains C through R2 down to ⅓ Vcc, when the output goes high again. - **Duty** — the share of each period the output spends high. Since C charges through R1 + R2 but drains through R2 alone, the duty is always above 50 %. - Ask for 50 % or less, and a diode appears across R2 to take it out of the charge path. - Its forward drop is ignored, as datasheets ignore it, so the real duty is a little off. - **Monostable** — the output rests low, with C held empty. A trigger sends the output high, and C charges through R until it reaches ⅔ Vcc, which ends the pulse. - **Vcc cancels out** of every time, because both levels are fractions of it. It only sets the voltages you would probe and the resistor limits.

## Why the formulas are these

```
Astable:    t high = ln 2 × (R1 + R2) × C    (with the diode: ln 2 × R1 × C)
            t low  = ln 2 × R2 × C
            Frequency = 1 / (t high + t low)
Monostable: Pulse width = ln 3 × R × C ≈ 1.1 × R × C
```

- **ln 2** — charging from ⅓ toward Vcc, C reaches ⅔ when half the remaining gap is closed, which takes RC × ln 2. Draining from ⅔ toward 0, it reaches ⅓ at half, again RC × ln 2.
- **ln 3** — the monostable starts from an empty C and must reach ⅔ of the way, leaving a third: RC × ln 3 = 1.0986 RC, TI's 1.1 RC.
- **The timing diagram** uses the same curves, so they need no RC. Each stretch is drawn as 1 − ⅔ × 2^−s rising, ⅔ × 2^−s falling, and 1 − 3^−s for the monostable, where s runs from 0 to 1 across it.
- **Typing a frequency** scales both resistors. **Typing a duty** redistributes them. C is never rewritten, since it is the part you have in the drawer.

## Assumptions and limits

The tool cautions, without refusing, when: - **R1 (or R) is under 1 kΩ**, where TI's figures start. Pin 7 then sinks Vcc / R1 during every low time, or the whole time a monostable waits; the caution shows that current. - **The charging resistance is too large.** That is R1 + R2, R1 alone with the diode, or the monostable's R, against TI's limit: 3.4 MΩ at 5 V and 10 MΩ at 15 V, joined by a straight line. - The line keeps the charging current at the threshold around 500 nA, twice the pin's 250 nA maximum. - Applying it to the monostable's R is this tool's extension: TI states it for the astable, but it is the same pin sensing the same capacitor. - **The frequency is over 100 kHz**, or **the pulse is under 10 µs**. - **Vcc is outside 4.5–16 V.**

Beyond those: - **The levels are typical.** The real trigger and threshold levels spread, as above, so the real times do too. - **Capacitor tolerance**, often ±20 % for electrolytics, usually matters more than the resistors'. - **The bipolar NE555 is modelled.** CMOS versions such as the TLC555 use the same formulas, but have their own limits.

## What it deliberately does not do

- Two-transistor multivibrators have their own tool.
- Switch debouncing has its own tool, Debounce / RC timing.

## How it was checked

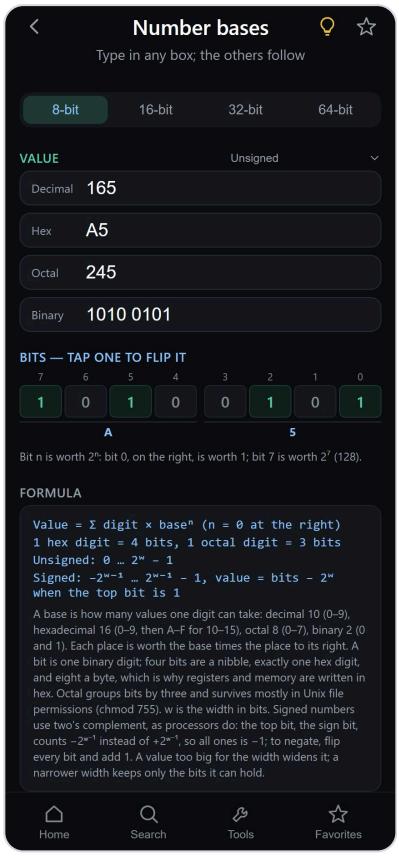
- **4.7 k, 4.7 k, 100 nF:** period 977.3 µs (1.023 kHz), duty 66.67 %, levels 1.667 V and 3.333 V at 5 V.
- **10 k, 680 k, 1 µF:** 949.6 ms (1.053 Hz), high 478.3 ms, duty 50.36 %.
- **1 kHz at 50 % with 10 nF:** the diode switches in, and R1 = R2 = 72.135 kΩ.
- **Monostable, 91 k and 100 µF:** 9.997 s.
- **Monostable, 1 s with 100 nF:** R = 9.1024 MΩ, cautioned at 5 V (over 3.4 MΩ), not at 15 V (under 10 MΩ).
- **R1 = 470 Ω at 5 V:** the caution shows 10.6 mA into pin 7.

The review made these changes: - It put the period, or the pulse width, first and large. - It added the capacitor's voltage to the timing diagram. - It replaced the fixed 10 MΩ caution with TI's supply-dependent limit. The old wording claimed a 1 kΩ minimum "set by datasheets", which TI does not state, and the 1 kΩ caution now says what it rests on. - It added TI's 100 kHz, 10 µs and 4.5–16 V cautions. - It limited the units to real ones (Ω–MΩ, pF–µF, Hz–kHz, µs–s, Vcc in V). - It enlarged the 9–10 px labels to 11 px, and wrote the 10 nF out. - It added plain definitions of every pin, and examples. - It remembers the mode and the tolerance. - The tool was renamed 555 timer, on Pierre's choice.

DIGITAL & LOGIC CIRCUITS › NUMBER SYSTEMS

# Number bases

(formerly DEC / HEX / OCT / BIN conversion)




## What it computes

One value, shown in decimal, hexadecimal, octal and binary at once. - **Four live boxes.** Type in any of them, and the other three follow. - **A register width:** 8, 16, 32 or 64 bits. - **Unsigned or signed.** Signed numbers are read as two's complement, the way processors store negative numbers: FFF6 in 16 bits is −10. - **One button per bit.** Each is numbered from bit 0 on the right, and tapping one flips it. The hex digit of each group of four bits sits under it.

The width and the signed choice are remembered.

## Source

The arithmetic is standard: positional notation and two's complement. No datasheet is involved. The four widths are those of the usual fixed-size integer types (uint8 to uint64, int8 to int64).

## What it means

- **A base** is how many values one digit can take:
- decimal, 10 (0–9);
- hexadecimal, 16 (0–9, then A–F for 10–15);
- octal, 8 (0–7);
- binary, 2 (0 and 1).

Each place is worth the base times the place to its right. - **A bit** is one binary digit. Four bits are a **nibble**, which is exactly one hex digit, and eight bits are a **byte**: two hex digits. That is why registers and memory are written in hex. - **Octal** groups bits by three. It survives mostly in Unix file permissions (chmod 755), rarely in electronics. - **Bit 0**, on the right, is the least significant bit (LSB), worth 1. The top bit is the most significant (MSB). - **Two's complement** — in a signed number the top bit, the **sign bit**, counts $-2^{w-1}$ instead of $+2^{w-1}$, where w is the width. - All ones is therefore −1. - To negate a number, flip every bit and add 1. - Adding and subtracting work the same for signed and unsigned values, which is why processors use it.

## Why the formulas are these

```
Value    = Σ digit × baseⁿ          (n = 0 at the right)
Unsigned:  0 … 2ʷ − 1
Signed:    −2ʷ⁻¹ … 2ʷ⁻¹ − 1,   value = bits − 2ʷ when the top bit is 1
```

- **One hex digit is four bits** because 16 = $2^4$, and one octal digit is three bits because 8 = $2^3$. So hex and octal are read straight off the binary, four or three bits at a time.
- **A negative decimal** is stored as $2^w$ plus the value: −10 in 16 bits is 65 536 − 10 = 65 526 = FFF6.

## Assumptions and limits

- **Whole numbers only, up to 64 bits.** Every value is an exact big integer, not a JavaScript number, so all 64 bits are exact. Typing more than 64 bits is refused, and the box goes back to the last value.
- **Widening.** A value too big for the chosen width moves up to the next width that holds it (8 → 16 → 32 → 64). Choosing a wider width keeps the value, and a negative one is sign-extended (FFF6 → FFFFFFF6).
- **Narrowing.** Choosing a narrower width keeps only the bits it can hold, as the register would, and the tool says so.
- **No padding.** The boxes show no leading zeros, so 165 in 16 bits reads A5, not 00A5. The bit row shows every bit of the width.

## What it deliberately does not do

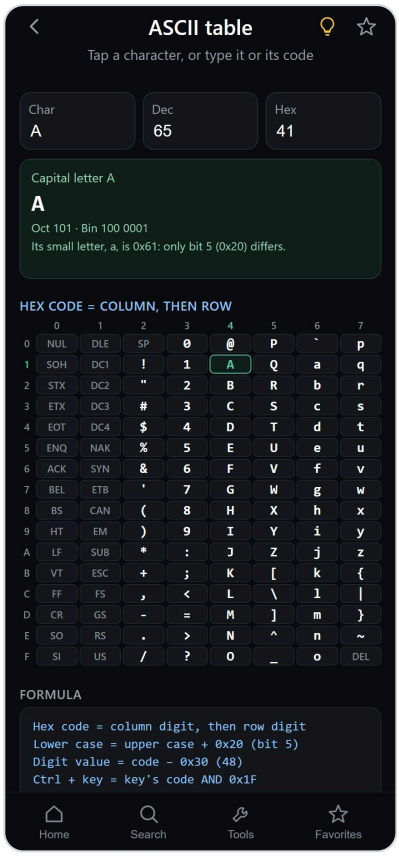
- Characters and their codes have their own tool, the ASCII table.
- Fractions, floating point (IEEE 754), BCD and Gray code are not converted.

## How it was checked

- **165:** A5, 245, 1010 0101.
- **FFFFFFFFFFFFFFF:** 18446744073709551615 and 17777777777777777777.
- Before the review this came back as 10000000000000000 in hex, and as 18446744073709552000 in decimal.
- JavaScript numbers hold integers exactly only up to $2^{53}$.
- **9007199254740993** ($2^{53}$ + 1): 20000000000001.
- **300** typed at 8 bits widens to 16 bits: 12C, 0001 0010 1100.
- **Signed 8-bit FF:** −1.
- **Signed 32-bit −1:** FFFFFFFF, 37777777777.
- **Signed −10 at 8 bits, widened to 16:** FFF6, still −10.
- **1A5 narrowed to 8 bits:** A5, with a note that the upper bits were dropped.
- **Examples:**
- 0x88 = 136 (bits 7 and 3);
- FFF6 as int16 = −10;
- FFFFFFFF = 4,294,967,295. As a 32-bit millisecond counter, that is 49.7 days before it wraps.

The review made these changes: - It fixed the conversion of values above $2^{53}$, which were silently wrong. Values are now exact BigInts. - It replaced the base pills, the input and the four separate results with four live boxes. - It added the register width and signed (two's complement) reading, with a ± key for typing negatives. - It added the tappable bit row, with the hex digit under each nibble and the sign bit marked. - It added plain definitions, examples and remembered settings. - The tool was renamed Number bases, on Pierre's choice.

# ASCII table




## What it computes

The 128 ASCII characters on one screen, as the classic chart of 8 columns by 16 rows. Tap a character, or type it in one of the three live boxes: - **Char** takes a character, or a control character's name (LF, ESC, XOFF, TAB, SP…); - **Dec** takes the decimal code, 0–127; - **Hex** takes the hex code, 00–7F.

The card shows, for the chosen character: - its name, with the standard mnemonic for a control character; - its codes in octal and in binary, the binary grouped as column and row; - the C escape (\n, \t, \0…) and the Ctrl key that sends it, where there is one; - in plain words, what a control character does, how a letter relates to its other case, or a digit to its value.

## Source

**RFC 20, ASCII format for Network Interchange** (October 1969), was read for this. It is based on USAS X3.4-1968, the US standard ASCII, and gives: - **the mnemonics and full names** of the 33 control characters, NUL to US and DEL. Code 9 is HT, horizontal tabulation, so the tool no longer calls it TAB (it still finds it by that name); - **the definitions** behind each plain-word meaning; - **the chart:** bits b7–b5 give the column (0–7), and b4–b1 the row (0–15).

The modern uses are common knowledge, not from RFC 20: - C's escapes; - XON/XOFF serial flow control on DC1/DC3; - Ctrl+C sending ETX and Ctrl+D sending EOT in a terminal; - CR LF line endings; - ESC starting a terminal control sequence.

## What it means

ASCII (American Standard Code for Information Interchange) gives 128 characters a 7-bit code. - **Codes 0–31 and 127 are control characters:** commands for a terminal, printer or modem rather than symbols. RFC 20 groups some of them: - communication controls frame and acknowledge messages: SOH, STX, ETX, EOT, ENQ, ACK, NAK, SYN, ETB; - format effectors move the print position: BS, HT, LF, VT, FF, CR; - information separators split data, from the largest unit to the smallest: FS, GS, RS, US. - **Codes 32–126 are printed characters.** 32, the space, is a printed character that prints nothing. - **Codes above 127 are not ASCII.** Different 8-bit sets give them different characters (Latin-1, Windows-1252…). In UTF-8 they only appear inside multi-byte characters.

Uses worth knowing in electronics: - **CR LF** ends a line on Windows and on many serial devices. Linux and macOS end a line with LF alone. - **XOFF (DC3) and XON (DC1)** pause and resume a serial sender: software flow control, Ctrl+S and Ctrl+Q. - **ESC [** starts a terminal control sequence, which sets colours and moves the cursor. - **SOH, EOT, ACK, NAK and CAN** run XMODEM, a simple serial file-transfer protocol.

## Why the formulas are these

```
Hex code      = column digit, then row digit
Lower case    = upper case + 0x20        (bit 5)
Digit value   = code − 0x30              (48)
Ctrl + key    = key's code AND 0x1F      (bits 5 and 6 cleared)
```

- **The chart's columns are the top three bits** and its rows the low four. A column is one hex digit, 0–7, and a row is the other, 0–F. So A, column 4 and row 1, is 0x41.
- **Letters were placed so the two cases differ in one bit.** A to Z run 0x41–0x5A, and a to z run 0x61–0x7A: setting bit 5 lowers the case.
- **Digits were placed so the low four bits are the value.** 0 to 9 run 0x30–0x39, so subtracting 0x30 (or masking with 0x0F) gives the number.
- **Ctrl clears bits 5 and 6.** Ctrl+C is C (0x43) AND 0x1F, which is 3, ETX; Ctrl+[ is ESC.

## Assumptions and limits

- **US ASCII only.** The national variants of ISO 646 replace a few printed characters, such as # or [, and are not shown.
- **Uses, not rules.** The modern uses in the card say what a code usually does today. A particular device may assign its own meanings.

## What it deliberately does not do

- Numbers in other bases have their own tool, Number bases.
- Turning text into bytes, for example "AT\r\n" into 41 54 0D 0A, was left out on Pierre's choice. It could become a tool of its own.

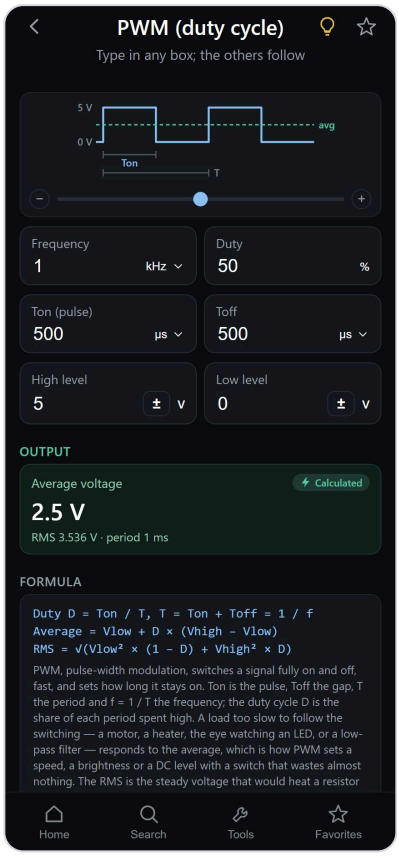
## How it was checked

- **A:** 65, 0x41, octal 101, binary 100 0001; column 4, row 1.
- **Code 9:** HT, Horizontal tab (RFC 20's name).
- **LF:** 10, 0x0A, octal 012, binary 000 1010, \n, Ctrl+J.
- **Hex 7E:** 126, the tilde.
- **XOFF typed by name:** 19, DC3.
- **ESC:** octal 033, binary 001 1011, Ctrl+[.
- **127:** DEL.
- **Out of range:** 200 and é are reported as not ASCII.
- **<:** shown as a character, and named less-than sign.
- **Space:** shown as SP.
- **7:** 55; value = 55 − 48 = 7.
- **The chart's height:** constant for all 128 characters. At 375 × 812 it ends 21 px above the tab bar.

The review made these changes: - It replaced the searchable list with RFC 20's chart, and the search with three live boxes. - It named code 9 HT, after the standard, instead of TAB. - It added names and plain meanings for the 33 control codes, C escapes, Ctrl keys, and the case and digit relations. - It escaped <, >

and &, which were inserted as raw HTML. - It added examples: CR LF, XOFF/XON, a digit to its number, upper and lower case.

# PWM (duty cycle)




## What it computes

For a pulse-width-modulated signal, the tool links six live boxes: - **Frequency** and **Duty**, the usual way to specify a PWM signal; - **Ton** (the pulse) and **Toff** (the gap), the way it is read off a scope; - **High level** and **Low level**, in volts.

From them it computes: - **the average voltage**, shown large; - **the RMS voltage**; - **the period**.

The waveform shows two periods, with the average as a dashed line. The slider under it sets the duty. The high and low levels are remembered.

## Source

The formulas follow from the definitions. Two examples were checked against documents: - **The Arduino reference for analogWrite()** (reference-en, read on GitHub), which gives: - the PWM frequency: 490 Hz on most Uno, Nano and Mini pins, and 980 Hz on pins 5 and 6; - the value range: 0 (always off) to 255 (always on). - **An ebm-papst DC fan catalogue page** (speed setting, p. 175). It states a four-wire interface "in accordance with Intel specifications for 12 VDC fans, PWM frequency 25 kHz". Intel's own 4-wire PWM fan specification could not be opened (the copies found refused access), so it was not read.

The hobby-servo figures are the common convention, not a standard: - a pulse every 20 ms (50 Hz); - 1 ms to 2 ms wide, with 1.5 ms at the centre.

## What it means

**PWM**, pulse-width modulation, switches a signal fully on and off, fast, and sets how long it stays on. - **Ton** — the pulse, how long the signal stays high. - **Toff** — the gap, how long it stays low. - **T** — the period, Ton + Toff. **f = 1 / T** is the frequency. - **The duty cycle D** — the share of each period spent high, from 0 % (always off) to 100 % (always on). - **The average** — a load too slow to follow the switching responds to the average. That load can be a motor, a heater, the eye watching an LED, or a low-pass filter. This is how PWM sets a speed, a brightness or a DC level with a switch that wastes almost nothing: it is either fully on or fully off. - **The RMS** — the steady voltage that would heat a resistor the same. With a low level of 0 V, the power into a resistor is D times the full power. - **A negative low level** — an H-bridge switching between +V and −V. The average then goes from −V to +V as the duty goes from 0 to 100 %.

The boxes work like this: - **Typing a frequency** rescales the pulse and keeps the duty. - **Typing a duty** redistributes it and keeps the frequency. - **Ton and Toff** are independent, so two readings off a scope can be typed one after the other.

## Why the formulas are these

```
D = Ton / T,   T = Ton + Toff = 1 / f
Average = Vlow + D × (Vhigh − Vlow)
RMS     = √(Vlow² × (1 − D) + Vhigh² × D)
```

- **The average** is the time-weighted mean of the two levels: Vhigh for a share D of the time and Vlow for the rest.
- **The RMS** is the square root of the time-weighted mean of the squares, by definition.

## Assumptions and limits

- **Ideal edges.** The rise and fall times are taken as zero. With real edges the average shifts a little, most at very short pulses.
- **Ideal averaging.** The average is what a perfect filter, or an infinitely slow load, would see. How much ripple remains depends on the filter or the load: see the RC filter tool.
- **Frequency is a design choice the tool does not make.** It must be high enough for the load to average: no visible flicker, no audible whine (20 kHz is the usual upper limit of hearing). It must also be low enough for the switching losses to stay small.

## What it deliberately does not do

- Timer register values (prescaler, top, compare) are not computed: they differ from one microcontroller family to the next.
- Filtering the PWM into a DC level is the RC filter tool's job.

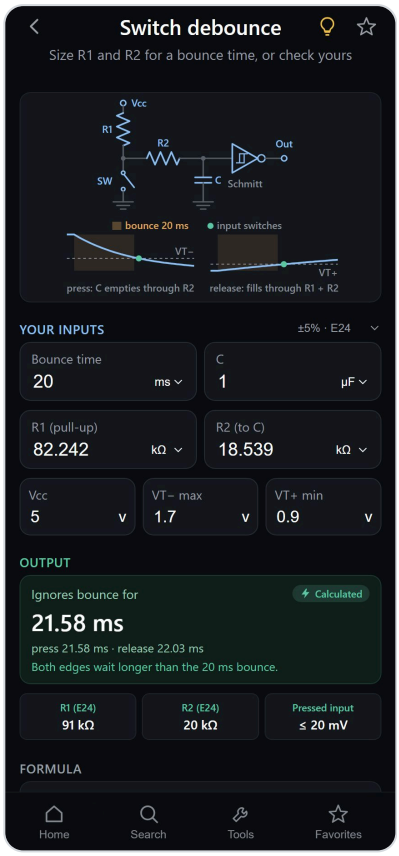
## How it was checked

- **1 kHz, 50 %, 5 V / 0 V:** Ton 500 µs, average 2.5 V, RMS 3.536 V, period 1 ms.
- **25 % at 1 kHz:** 250 µs on, 750 µs off, average 1.25 V.
- **2 kHz** with the duty kept at 50 %: 250 µs on.
- **Ton 200 µs, then Toff 800 µs:** 1 kHz and 20 %.
- **150 % typed:** held at 100 %, Toff 0, average 5 V.
- **+12 V / −12 V at 75 %:** average 6 V, RMS 12 V.
- **Examples:**
- servo: 1.5 ms on, 18.5 ms off, 7.5 %;
- analogWrite(128): 490 Hz, 50.2 %, 2.51 V;
- PC fan at 40 %: 16 µs at 25 kHz;
- 12 V motor at 75 %: 9 V.

The review made these changes: - It replaced the two mode pills with six live boxes. - It put the average voltage first and large. - It redrew the waveform: - the pulse in the part blue, not the domain colour; - two periods, with the average dashed in green; - 11 px labels instead of 9 px; - the duty slider under it. - It limited the units to real ones (Hz–MHz, ns–s), and the levels to V with a ± key. - It added plain definitions, examples, and remembered levels.

# Switch debounce

(formerly Debounce / RC timing)




## What it computes

It sizes the RC debouncer from Jack Ganssle's guide: Vcc through R1 to the switch, R2 from the switch to C, and C feeding a Schmitt-trigger input. - **Sizing.** Type the bounce time, C, Vcc and the input's worst-case thresholds (VT− at its highest, VT+ at its lowest), and R1 and R2 are sized so that both the press and the release take the bounce time to change the input. - **Checking.** Type R1 and R2 to check a circuit you have.

The output shows: - **the delays of the circuit built from standard values,** rounded up to the next value of the chosen series. The shorter delay is shown large, with the press and release delays beneath; - **the two resistors** in standard values; - **the pressed input level** caused by input leakage through R2.

Two small graphs show C's voltage after a press, falling through R2, and after a release, rising through R1 + R2. Each is drawn against the bounce window, with a dot where it crosses the threshold: green after the bounce, amber inside it. Vcc, the thresholds and the series are remembered.

**Standard values.** The resistors' tolerance, with its E series, is picked in the "Your inputs" heading above the inputs, as in the Passive components tools: ±0.1 % to ±20 %, ±5 % being E24 and ±1 % E96; it sets the series the sized resistors are rounded to and the tolerance their spread uses.

## Source

- **Jack Ganssle, *A Guide to Debouncing*, part 2** (ganssle.com) was read for:
- the circuit;
- the method: R2 from the press, R1 + R2 from the release, each against the worst-case threshold;
- his worked example: 20 ms, 74AHCT14 at 5 V; with 0.1 µF R2 = 185 kΩ, and with 1 µF R2 = 18 kΩ, R1 + R2 = 101 kΩ, R1 = 82 kΩ;
- the leakage concern;
- his measured bounce times: most switches well under 10 ms, so he designs for 20 ms.

His equations are images on the page and could not be read as text. They were rebuilt from the RC charge law, and reproduce all three of his resistor values. - **TI SN74AHCT14 datasheet (SCLS246T, October 2023)** was read for the thresholds: - VT+ (rising) is 0.9–1.9 V at 4.5 V, and 1.0–2.1 V at 5.5 V; - VT− (falling) is 0.5–1.5 V at 4.5 V, and 0.6–1.7 V at 5.5 V; - Ganssle's 0.9 V and 1.7 V are the extremes; - the input current is ±1 µA. - **Nexperia 74HC_HCT14 datasheet (rev. 10, 29 February 2024)**, table 10, was read for the 74HC14 thresholds: - at 4.5 V, VT+ is 1.7–3.15 V and VT− 0.9–2.0 V; - the input leakage is ±1.0 µA over temperature.

## What it means

- **Contact bounce** — a switch's contacts bounce for a few milliseconds when it is pressed or released. A fast input would see every bounce as another press.
- **The RC delay** — C slows the input down. Pressing empties C through R2; releasing fills it through R1 + R2. The input changes only once C crosses its threshold, after the bouncing is over.
- **R1** — the pull-up. It also limits the current through the switch while it is held.
- **R2** — sets the press delay, and limits the current the contacts take from C.
- Without R2 (C straight across the switch), a press registers on the very first contact, and the contacts take C's full discharge.
- The bounce after that first contact is still filtered: an opening is too short for C to recharge past VT+ through R1.
- **A Schmitt-trigger input** — it has two thresholds, VT− for a falling input and VT+ for a rising one, with a gap between them (hysteresis). So a slow, noisy edge switches it only once.
- Ganssle warns never to feed C straight to a microcontroller input, as few have any hysteresis. An input without it does unpredictable things on a slow edge.
- Some microcontrollers do have Schmitt-trigger inputs: check the datasheet, and use its thresholds.
- **Worst case** — thresholds vary between parts and with the supply. The highest VT− and the lowest VT+ give the shortest delays, so those are the values to design with.
- **Pressed input** — the input's leakage (up to 1 µA) flows through R2 while the switch is held, lifting the input up to 1 µA × R2 above ground.
- Ganssle rejected 180 kΩ (0.18 V, against a lowest threshold of 0.5 V) and moved to 1 µF to bring R2 down to 18 kΩ.
- The tool cautions from 0.1 V.

## Why the formulas are these

```
Press delay   = R2 × C × ln(Vcc / VT−)
Release delay = (R1 + R2) × C × −ln(1 − VT+ / Vcc)
Sizing:  R2 = t / (C × ln(Vcc / VT−)),   R1 + R2 = t / (C × −ln(1 − VT+ / Vcc))
```

- **Press:** C falls from Vcc toward 0 through R2 alone, following V = Vcc × e^(−t/R2C). It reaches VT− after R2C × ln(Vcc / VT−).
- **Release:** C rises from 0 toward Vcc through R1 + R2, following V = Vcc × (1 − e^(−t/(R1+R2)C)). It reaches VT+ after (R1 + R2)C × −ln(1 − VT+ / Vcc).
- **Bounce makes the real delays longer.** Each opening during a press lets C climb back a little, so these delays are the shortest the circuit can give, which is what the design needs.
- **Typing the bounce time, C or the thresholds** sizes R1 and R2. **Typing R1 or R2** checks them, as typed.
- **The standard values are rounded up,** so the built circuit never waits less than the bounce time. Ganssle rounded to the nearest values (18 k, 82 k): that gives 19.4 ms and 19.8 ms, a little under his 20 ms, which already doubles the bounce he measured.

## Assumptions and limits

- **Full charge and discharge.** Each delay starts from C fully charged or fully empty. A press released before C has emptied gives a shorter release delay.
- **Leakage only through R2 is shown.** The released input also sits up to 1 µA × (R1 + R2) below Vcc.
- **Bounce time is the switch's.** Measure it, or use Ganssle's 20 ms. Some switches bounce far longer.

## What it deliberately does not do

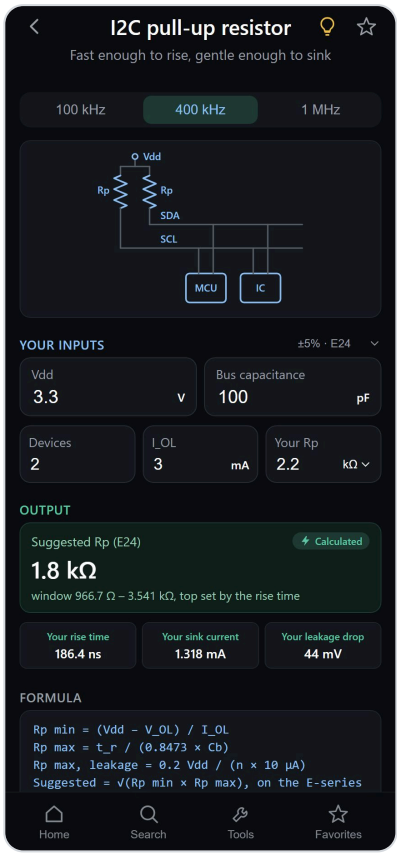
- Firmware debouncing, which Ganssle's guide also covers, is not modelled.
- The RC charge itself, with any threshold, has its own tool: RC charge/discharge.

## How it was checked

- **20 ms, 1 µF, 74AHCT14 at 5 V (1.7 V, 0.9 V):**
- exact R2 18.539 kΩ and R1 82.242 kΩ, so R1 + R2 = 100.78 kΩ (Ganssle's 101 kΩ);
- E24, rounded up: 20 kΩ and 91 kΩ;
- delays: press 21.58 ms, release 22.03 ms;
- pressed input ≤ 20 mV.
- **0.1 µF:** R2 = 185.39 kΩ (Ganssle's 185 kΩ). With 200 kΩ (E24) the pressed input sits at 200 mV, and the tool cautions.
- **50 ms:** R2 46.347 kΩ, R1 205.6 kΩ.
- **10 k and 10 k typed with 1 µF:** press 10.79 ms, release 3.969 ms, shorter than 20 ms.
- **74HC14 at 4.5 V (2.0 V, 1.7 V), 1 µF, 20 ms:** 27 kΩ and 18 kΩ (E24), shorter delay 21.35 ms.
- **R2 = 0:** the press delay is 0 s, and the tool says why.

The review made these changes: - It replaced the single-resistor circuit, C straight across the switch, with Ganssle's two-resistor circuit and his worst-case method. - It made Vcc count. It was an input the result never used, because the threshold was a percentage. - It replaced the guessed 50 % threshold with VT− and VT+ in volts, from the datasheet. - It replaced the unsourced bounce-time table with Ganssle's measurements. - It added the two graphs and the leakage check. - It limited the units to real ones (Ω–MΩ, pF–µF, µs–s, volts). - It put the result first and large, added examples, and remembers the input's thresholds. - The tool was renamed Switch debounce, on Pierre's choice.

# I2C pull-up resistor




## What it computes

For an I²C bus, given the speed mode, the supply, the bus capacitance and the number of devices, it computes: - **the window** of pull-up resistances that work; - **a suggested value** inside it, snapped to a standard E-series and shown large. The result says which limit sets the top of the window: the rise time, or the devices' leakage; - **for the Rp you have:** its rise time, the current it asks a device to sink, and how far leakage pulls the high level down through it.

The mode, the series and Vdd are remembered.

**Standard values.** The resistors' tolerance, with its E series, is picked in the "Your inputs" heading above the inputs, as in the Passive components tools: ±0.1 % to ±20 %, ±5 % being E24 and ±1 % E96; it sets the series the suggested Rp is rounded to and the tolerance its spread uses.

## Source

**NXP UM10204, I²C-bus specification and user manual, rev. 7.0 (1 October 2021)**, was read for every limit: - **Section 7.1, pull-up resistor sizing:** - the rise time is measured between 0.3 and 0.7 Vdd, which gives T = 0.8473 × RC; - Rp(max) = t_r / (0.8473 × Cb); - Rp(min) is set by the minimum sink current: 3 mA for Standard and Fast mode, 20 mA for Fast-mode Plus; - portable, low-power designs "can use a value toward the higher end of the range". - **Table 10, the I/O stages:** - V_OL1 is 0.4 V at 3 mA when Vdd > 2 V; - V_OL2 is 0.2 Vdd at 2 mA when Vdd ≤ 2 V, for Fast mode and Fast-mode Plus only; - I_OL at 0.4 V is 3, 3 and 20 mA; - each I/O pin is at most 10 pF. - **Table 11, the bus lines:** - the rise time t_r is at most 1000, 300 and 120 ns; - the bus capacitance Cb is at most 400, 400 and 550 pF; - the high-level noise margin is 0.2 Vdd. - **Section 7.4, input leakage:** each connection may draw up to 10 µA at the high level, and "due to the required noise margin of 0.2 VDD for the HIGH level, this input current limits the maximum value of Rp."

## What it means

- **Open-drain.** I²C lines are open-drain: a device can only pull a line low. Nothing drives it high; the pull-up resistor Rp brings it back. That single fact produces the bounds, pulling in opposite directions.
- **I_OL** — the current a device is rated to sink while holding the line at the output-low level V_OL. Too small an Rp asks for more.
- **The rise time t_r** — the time the line takes to climb from 0.3 to 0.7 Vdd. Too large an Rp makes it too slow for the mode.
- **The bus capacitance Cb** — every pin on the bus, up to 10 pF each, plus the wiring. It is the input people are most wrong about, and the whole upper bound scales with it.

- **Leakage** — every device may leak up to 10 µA into the line. Through Rp, that pulls the high level down, and it must not cost more than the 0.2 Vdd noise margin. With many devices this, and not the rise time, sets the top of the window.

## Why the formulas are these

```
Rp min          = (Vdd − V_OL) / I_OL        V_OL = 0.4 V, or 0.2 Vdd at 2 V or les
Rp max          = t_r / (0.8473 × Cb)
Rp max, leakage = 0.2 Vdd / (n × 10 µA)
Suggested       = √(Rp min × Rp max), on the E-series
```

- **The lower bound is the pull-down.** When a device asserts the line, its transistor has to hold it below V_OL while the pull-up pushes (Vdd − V_OL) / Rp into it.
- **The upper bound is the rise.** Going up, the line is just an RC: the pull-up charging the bus capacitance. An RC charge reaches 0.3 Vdd at RC × ln(1/0.7), and 0.7 Vdd at RC × ln(1/0.3). The difference is RC × ln(7/3) = 0.8473 RC, the factor the spec prints.
- **The leakage bound** is Ohm's law. n devices × 10 µA through Rp must drop no more than 0.2 Vdd.
- **The suggestion** is the geometric mean of the two bounds: the centre of the window on a logarithmic scale, which is what "equal margin on both sides" means when the ends can differ by a decade. The spec's own advice for low-power designs is the high end instead.
- **Finding Cb.** Measure the rise time with a known Rp, and the same formula gives Cb = t_r / (0.8473 × Rp).
- **When the window closes.** Rp max falls as the bus capacitance rises, while Rp min does not move. At 3.3 V in Fast mode the two cross at about 366 pF. Past that no resistor rises in time without overloading the pull-down. That is a dead end, not a poor choice. The ways out are:
- a shorter bus, or fewer devices;
- a slower mode;
- a bus buffer that splits the bus into segments (UM10204 section 7.2).

## Assumptions and limits

- **Bus capacitance is an estimate.** UM10204 gives no capacitance per centimetre of wiring, so the tool does not guess one. Measure it on a long bus, as above.
- **One lumped capacitance and an ideal resistor.** At Fast-mode Plus, with long tracks, the line no longer behaves as one RC.
- **Standard mode at 2 V or less** has no output-low level in UM10204. The tool keeps 0.4 V at 3 mA and says so.
- **I_OL follows the mode and the supply,** unless you type a value: a particular part may be rated differently.

## What it deliberately does not do

- **Current-source or switched pull-ups** (UM10204 section 7.2.4): a different circuit, with different arithmetic.

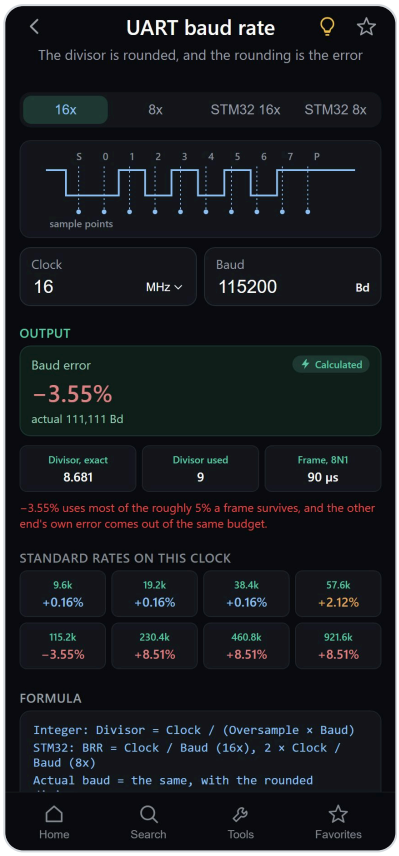
- **Bus buffers and repeaters:** they split a bus into segments, each with its own capacitance and resistor. Run the tool once per segment.
- **Level shifters:** a MOSFET level shifter between two voltage domains puts a pull-up on each side, and the two interact.

## How it was checked

- **400 kHz, 3.3 V, 100 pF, 2 devices:**
- window 966.7 Ω – 3.541 kΩ, 1.8 kΩ suggested;
- with 2.2 kΩ: rise time 186.4 ns, sink current 1.318 mA, leakage drop 44 mV.
- **100 kHz, same bus:** up to 11.8 kΩ, 3.3 kΩ suggested.
- **100 kHz with 10 devices:** leakage caps Rp at 6.6 kΩ, and 2.4 kΩ is suggested.
- **1.8 V at 400 kHz:** I_OL drops to 2 mA, Rp min is 720 Ω, and 1.6 kΩ is suggested.
- **1 MHz, 3.3 V, 200 pF:** 145–708.1 Ω, 330 Ω suggested.
- **370 pF at 400 kHz and 3.3 V:** no value works.

The review made these changes: - It added the output-low level below 2 V, 0.2 Vdd at 2 mA. The tool used 0.4 V at 3 mA at any supply, which put Rp min too low on a 1.8 V bus (467 Ω instead of 720 Ω). - It added the input-leakage limit (UM10204 section 7.4), with a Devices field. - It replaced the unsourced "a few pF per centimetre of track" with the spec's 10 pF per pin and the measure-the-rise-time method. - It put the suggestion first and large, and named which limit sets the window. - It added examples, remembered the mode, series and Vdd, enlarged the 10 px labels to 11 px, and added plain definitions.

# UART baud rate




## What it computes

From a peripheral clock and a wanted baud rate, it computes: - **the divisor** the UART needs, and the divisor actually written once it is rounded; - **the rate that divisor gives,** and the resulting **error**, shown large; - **the frame time** for 8N1.

There are two kinds of divider, each at 16x or 8x oversampling: - **16x / 8x:** the integer divider of the AVR, the 16550 family and PIC. - **STM32 16x / 8x:** the finer divider of STM32's USART.

A table gives the error of every standard rate on the same clock; tap one to try it. The drawing shows the sample points walking off the bits as the error grows. The mode and the clock are remembered.

## Source

There is no single standard for baud generation: it is per-device. Three sources were read: - **ST's STM32G0 HAL driver** (stm32g0xx_hal_uart.h and .c, on GitHub): - UART_DIV_SAMPLING16 is round(PCLK / baud); - UART_DIV_SAMPLING8 is round(2 × PCLK / baud); - the authorized BRR is UART_BRR_MIN 0x10 (16) to UART_BRR_MAX 0xFFFF. - **The Arduino AVR core** (HardwareSerial.cpp, ArduinoCore-avr on GitHub): - begin() tries U2X, the 8x mode, first; - it falls back to 16x only for 57600 at 16 MHz (for bootloader compatibility), or when the divisor would exceed 4095. - **The error budget** is derived below from the sampling behaviour, not quoted.

## What it means

- **A UART** sends bytes one bit at a time, with no clock wire. Both ends agree on a rate, the baud, in bits per second.
- **Its own oscillator** — it has none. It divides whatever clock it is given, and the divisor is a whole number, so the rate comes out a little off.
- **Oversampling** (16x or 8x) — how many clock periods of the divided clock the receiver counts per bit, to find the middle of each bit.
- **8N1** — one start bit, eight data bits, no parity, one stop bit: ten bits per frame.
- **Register names:**
- an AVR writes UBRR = divisor − 1;
- an STM32 writes BRR, which at 16x is the divisor shown. At 8x the HAL computes USARTDIV, shown here, and repacks it into BRR.

## Why the formulas are these

```
Integer:  Divisor = Clock / (Oversample × Baud),  Actual = Clock / (Oversample × rc
STM32:    BRR = round(Clock / Baud) at 16x,  round(2 × Clock / Baud) at 8x
          Actual = Clock / BRR,  or 2 × Clock / USARTDIV
Error  = (Actual − Baud) / Baud
```

**Integer dividers.** At 16 MHz, 115200 baud and 16x, the divisor comes to 8.681. Written as 9, the line runs at 111,111 baud, 3.55 % slow. - That single fact is why 14.7456 MHz crystals exist: 14,745,600 / 16 = 921,600, which divides exactly into every standard rate. 11.0592 MHz and 18.432 MHz are the same trick at other speeds. - **An Arduino Uno does better than that figure.** Its core uses 8x mode first: 16 MHz / (8 × 17) = 117,647 baud, 2.12 % fast.

**STM32.** STM32's USART divides in whole clock cycles per bit, sixteen times finer than an integer 16x divider. The same 16 MHz at 115200 writes BRR = 139: 115,108 baud, −0.08 %.

**How much error a link survives.** There is no clock shared between the two ends. The receiver finds the falling edge of the start bit, starts its own counter, and samples each bit at what it believes is the middle. Taking the receiver as the reference, it samples bit *n* at

```
(n + 0.5) × T_rx
```

while the transmitter's bit *n* occupies

```
[ n × T_tx , (n+1) × T_tx ]
```

With r = T_tx / T_rx, the sample lands in the right bit only while

```
n × r  <  n + 0.5  <  (n+1) × r
```

Nothing resynchronises inside a frame, so the worst case is the last bit. For 8N1 that is the stop bit, n = 9:

```
9 r < 9.5      →  r < 1.0556
9.5 < 10 r     →  r > 0.95
```

So the two clocks may differ by about −5.0 % to +5.56 %, usually quoted as ±5 %. - It is slightly asymmetric because the sample point is fixed at mid-bit, while the bit it must land in is what stretches. - That budget covers **both** ends together. Hence 2 % each as the practical rule: it leaves room for the other end, edge jitter and finite rise time. - **The tool's colours:** blue up to 2 %, amber to 3 %, red beyond. The caution says "fails outright" only past 5 %.

The error does **not** accumulate across a message. The receiver re-finds the start edge on every frame, so the drift resets ten bits at a time.

**The drawing shows exactly this.** The waveform is drawn at the transmitter's real bit width, stretched or squeezed by the error, while the sample marks stay evenly spaced where the receiver puts them. Raise the error, and the marks that fall outside their own bit turn red.

## Assumptions and limits

- **8N1 is assumed.** Parity or a second stop bit makes the frame longer and the budget tighter.
- **One end's error.** The receiver is taken as exact. In reality both ends have error and the two add, which is why the caution starts at 2 % rather than 5 %.
- **STM32 as ST's HAL writes it,** read from the G0 driver only; the other STM32 families' drivers were not read. Other vendors' fractional dividers (SAM, nRF and others) have their own resolution, and are not modelled.
- **Oversampling is a divider stage, not noise immunity.** 8x doubles the top speed from a given clock, and halves the margin around each sample.

## What it deliberately does not do

- **Auto-baud detection**, where the receiver measures a known character.
- **Anything above the physical layer:** framing errors, break detection, flow control.

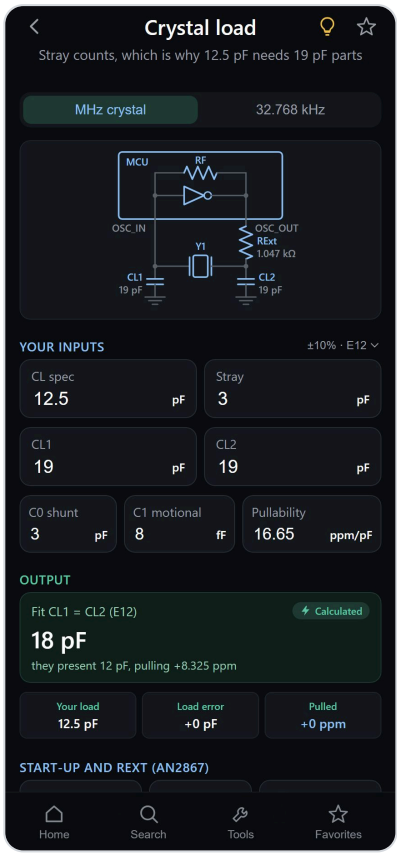
## How it was checked

- **16 MHz, 115200, integer 16x:** divisor 8.681, 9 used, 111,111 Bd, −3.55 %.
- **16 MHz at 9600:** 104, +0.16 %.
- **Arduino Uno (8x), 16 MHz at 115200:** 17, 117,647 Bd, +2.12 %.
- **STM32 16x, 16 MHz at 115200:** BRR 139, 115,108 Bd, −0.08 %. At 8x, USARTDIV 278 gives the same.
- **14.7456 MHz at 115200:** exactly 8, 0.00 %.
- **8 MHz at 115200, integer 16x:** 4 used, +8.51 %, fails.
- **STM32 at 1 MHz and 115200:** BRR would be 9, under the HAL's minimum of 16, so the tool refuses it.

The review made these changes: - It added STM32's divider from ST's HAL. The tool modelled only integer dividers, which overstated an STM32's error about fortyfold at 16 MHz and 115200. - It put the error first and large. - It reworded the caution: a single end at 3.55 % does not fail, it uses most of the shared budget; only past 5 % does a frame fail. - It added the Arduino detail (8x first), checked against the core's source, and examples. - It remembers the mode and the clock, and enlarged the 9 px labels to 11 px.

# Crystal load

(formerly Crystal load capacitance)




## What it computes

For a Pierce oscillator, it computes: - **the two capacitors it needs,** so the crystal sees the load it was cut for. The pair to buy, in the chosen E-series, is shown large, with the load it really presents and how far that pulls the frequency; - **for the pair in the fields:** its load, its error and its pull; - **start-up, after AN2867:** - from the frequency, the crystal's ESR and the MCU's gm, the gain margin, with and without REXT; - REXT's starting value; - from the swing you measure on OSC_IN, the crystal's drive level against its rating, and so whether REXT is needed at all.

The drawing is the oscillator as it is really built: - the MCU, with its inverting amplifier and feedback resistor RF inside; - the OSC_IN and OSC_OUT pins (OSC32_IN and OSC32_OUT for a 32.768 kHz crystal); - the crystal Y1, REXT, and CL1 and CL2, with their values.

The crystal type and the series are remembered.

**Standard values.** The load capacitors' tolerance, with its E series, is picked in the "Your inputs" heading above the inputs, as in the Passive components tools (the series the CL1 and CL2 suggestions come from).

## Source

- **ST AN2867, Oscillator design guide, rev. 10 (DocID15287),** was read in a copy hosted by Raltron (st.com reset the connection):
- **figure 5, "Pierce-oscillator circuitry":** Inv, the internal inverter working as the amplifier; RF, the internal feedback resistor; Q, the crystal; REXT, an external resistor "to limit the inverter output current"; CL1 and CL2, the load capacitors; Cs, the stray, which is the pin capacitance plus the PCB;
- **section 3.3:** $CL = CL1 \times CL2 / (CL1 + CL2) + Cs$, and the worked example: a 15 pF crystal, assuming Cs = 5 pF, needs CL1 = CL2 = 20 pF;
- **section 3.7:** pullability from Cm, C0 and CL;
- **section 3.2, typical RF:** 10–25 MΩ at 32.768 kHz, down to 0.47–5 MΩ at 20 MHz;
- **section 3.4, the gain margin:**
  - $gmcrit = 4 \times ESR \times (2\pi F)^2 \times (C0 + CL)^2$, and gm / gmcrit should be at least 5;
  - the worked example: an 8 MHz Fox crystal (C0 7 pF, CL 10 pF, ESR 80 Ω) on an STM32F1 (gm 25 mA/V) gives gmcrit 0.23 mA/V and a margin of 107;
  - its LSE table gives gm from 5 to 25 µA/V across the drive modes of the first family listed;
- **section 3.5.3, REXT:**
  - it limits the drive level, and with CL2 keeps the oscillator off overtones;
  - it is mandatory only if the crystal would dissipate more than its rated drive level, and otherwise "not recommended", 0 Ω;

    - its initial value is the reactance of CL2, 1 / (2πF × CL2): 1326 Ω at 8 MHz with 15 pF;
    - it is then trimmed with a potentiometer, and added to ESR when the margin is rechecked;
- **section 3.5.2, the drive level from a measurement:**
    - the crystal current is the current through CL1, so the swing Vpp measured on CL1 gives DL = ESR × (π F Ctot)² × Vpp² / 2, with Ctot = CL1 + Cs/2 + Cprobe;
    - the probe must add no more than 1 pF;
    - section 3.5.1 gives the other way, a current probe, with I_pp,max = 2√2 × √(DLmax / ESR).
- **The drive-level ratings are Epson's:** 0.5 µW max for the FC-135, and 200 µW max (10 µW recommended) for the FA-238, an MHz crystal.
- **Epson FC-13F / FC-135 specification sheet,** read for the 32.768 kHz values:
- C1 3.4 fF and C0 1.0 pF typical;
- CL 7, 9 or 12.5 pF;
- ±20 ppm at 25 °C.
- **The pulling expression** comes from the crystal's equivalent circuit, the Butterworth–Van Dyke model: a motional arm of L1, C1 and R1 in series, all in parallel with the shunt capacitance C0 of the electrodes and holder.

## Why the formulas are these

**The two capacitors are in series, not in parallel.** This is the step people get backwards. Each one runs from one crystal terminal to ground, so the loop the crystal drives is: out through CL1, along the ground, back through CL2. Two capacitors in series:

```
CL1 × CL2 / (CL1 + CL2)
```

which for an equal pair is just half of one of them. Add the stray capacitance of the two tracks and the two oscillator pins — that sits directly across the same nodes, so it adds on:

```
Load = CL1 × CL2 / (CL1 + CL2) + Stray
```

Turn it round for a symmetric pair and the capacitors come out at

```
CL1 = CL2 = 2 × (CL spec − Stray)
```

A crystal marked 12.5 pF on a board with 3 pF of stray therefore wants **19 pF** parts, not 25 pF and not 12.5 pF. Both of those are common mistakes, and the second is the worse one.

**Why the load changes the frequency at all.** A crystal has two resonances. At series resonance the motional arm is purely resistive; a little above it the arm looks inductive and resonates with everything capacitive across it — C0 and whatever load the circuit adds. The parallel-resonant frequency is

```
fp = fs × [ 1 + C1 / (2 × (C0 + CL)) ]
```

A crystal is cut and trimmed so that this lands on the marked frequency **for one particular CL**. Present a different load and it lands somewhere else, which is the whole reason the number is on the datasheet.

Differentiating that with respect to CL gives the sensitivity, which is what the tool reports as pullability:

$$\frac{d(\Delta f/f)}{dCL} = -\frac{C1}{2 \times (C0 + CL)^2}$$

The sign is negative: **more load, lower frequency**. For a typical MHz part with C1 = 8 fF and C0 = 3 pF at CL = 12.5 pF that comes to about 16.6 ppm/pF, so fitting 22 pF where 19 pF was wanted — 1.5 pF too much load — pulls it roughly 25 ppm slow. Epson's FC-135 watch crystal (C1 3.4 fF, C0 1.0 pF) pulls 9.3 ppm/pF at 12.5 pF. In its 7 pF version it pulls 26.6 ppm/pF: the lower the load, the steeper the slope, so a low-load crystal wants the tightest stray.

## Where the inputs come from, and what they are called

Three of the inputs are easy to look for in the wrong place.

- **C0** — on the crystal datasheet, as `C0`, "shunt capacitance" or "static capacitance". Often given only as a maximum; that is usable.
- **C1** — the motional capacitance, `C1` on datasheets that give it, after the Butterworth–Van Dyke model. **Many do not.** When it is missing, look for a pullability figure in ppm/pF, or a "frequency versus load capacitance" curve, and type that into Pullability instead: the tool works C1 back from it. C1 is the one that stays fixed when the load changes — it is the crystal — while pullability is its slope at whatever load is chosen, so changing CL spec recomputes pullability, not C1.
- **Stray** — **never** on the crystal datasheet, because it is not the crystal. It is the capacitance of the two oscillator pins, sometimes listed in the MCU datasheet, plus the two tracks. ST's AN2867 assumes 5 pF in its worked example; take that as an order of magnitude, not a value to copy.

The load capacitors are called **CL1 and CL2** rather than C1 and C2, which is how MCU application notes label them and which keeps C1 free for its datasheet meaning. Every capacitance here only ever comes in one unit — pF for load, stray and shunt, fF for motional — so the fields show the unit fixed rather than offering a picker.

**When CL says "Series".** Some datasheets offer, or list, a series-resonant cut: the crystal is trimmed to its marked frequency with no load at all. A Pierce oscillator cannot present no load, so the part runs above its marking by the whole parallel-resonance offset,

$$\frac{\Delta f}{f} = \frac{C1}{2 \times (C0 + Load)}$$

which is not a trim but a few hundred ppm: 8 fF against 3 pF + 12.5 pF is +258 ppm. Type 0 in CL spec to see it. The fix is to order the part with a load capacitance, or to use it in a circuit that runs it at series resonance.

## Assumptions and limits

- **Stray is an estimate, and it dominates the mistakes.** It is a few picofarads for the pins plus short tracks (AN2867's example takes 5 pF), but a long or guarded layout can be well outside that. If the frequency matters, measure.

- **The presets are starting points.**
- The 32.768 kHz preset is the Epson FC-135's typical values.
- The MHz preset is illustrative: no manufacturer table was found to quote for C0 and C1.
- Use your crystal's datasheet: a wrong C1 scales the pullability in proportion.
- The pulling figure is a **slope taken at the specified load**, so it is exact for small errors and increasingly optimistic for large ones, since the true curve flattens as CL grows.
- Sizing assumes a **symmetric pair**. Asymmetric capacitors are legitimate and the tool will report the load they present, but it will not propose them.

## What it deliberately does not do

- **RF is drawn, not sized.** It is inside the MCU on the parts AN2867 covers. Its section 3.2 gives typical values, from 10–25 MΩ at 32.768 kHz to 0.47–5 MΩ at 20 MHz.
- **The drive level comes from your measurement,** not from a model. Type the peak-to-peak swing you read on OSC_IN.
- The Vpp defaults (1 V, 0.5 V) only illustrate a reading.
- A probe heavier than 1 pF both loads the oscillator and inflates the figure: a 10 pF probe reads twice the drive of a 1 pF one at 8 MHz here.
- **Temperature and ageing.** The load error here is a fixed offset; drift over temperature is the crystal's cut and belongs with the ppm tooling.

## How it was checked

Recomputed by hand: 12.5 pF specified with 3 pF stray gives 19 pF capacitors and a presented load of exactly 12.5 pF, 0 ppm. Fitting 22 pF instead presents 14 pF, 1.5 pF over, and pulls −24.97 ppm against 16.65 × 1.5 = 24.98 by hand. Fitting 33 pF presents 19.5 pF and pulls −116.5 ppm against 16.65 × 7 = 116.6. The watch preset (FC-135) returns 9.328 ppm/pF, against 3.4 fF / (2 × (1.0 + 12.5 pF)$^2$) = 9.33 by hand.

Further checks: - **AN2867's example** (15 pF, 5 pF stray) gives 20 pF each, the same as the note. - **The E12 pair for 12.5 pF with 3 pF stray** is 18 pF, which presents 12 pF and pulls +8.325 ppm. - **The FC-135's 7 pF version** needs 8 pF parts and pulls 26.56 ppm/pF. - **A Series crystal with 18 pF parts** runs +266.7 ppm fast. - **Start-up reproduces AN2867:** - C0 7 pF, CL 10 pF, 8 MHz, ESR 80 Ω and gm 25 mA/V give a margin of 107; - REXT with CL2 = 15 pF is 1326 Ω. - **At the defaults** (8 MHz, C0 3 pF, CL 12.5 pF, 19 pF parts): - REXT 1.047 kΩ; - margin 129 without REXT, 9.135 with it. - **Drive level at 8 MHz** (19 pF parts, 3 pF stray, 1 pF probe, ESR 80 Ω): - 1 Vpp gives 11.68 µW, 5.84 % of the FA-238's 200 µW, so REXT is 0 Ω; - 5 Vpp gives 292 µW, over the rating, so REXT is needed; - a 10 pF probe reads 23.5 µW for the same 1 Vpp. - **The FC-135 at 0.5 Vpp:** 42.86 nW, against its 0.5 µW rating. - **The FC-135 at 15 µA/V:** - margin 6.934; - with REXT (255.6 kΩ) it would fall to 1.491, so the tool says to fit 0 Ω; - at the lowest drive, 5 µA/V, the margin is 2.311, too low.

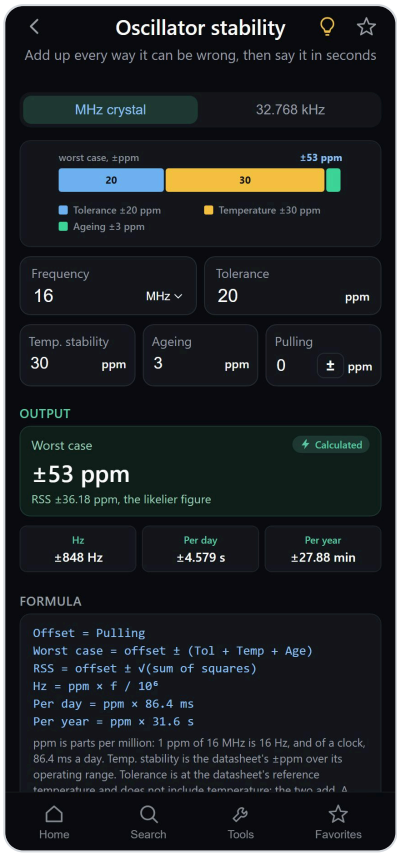
The pullability was wrong by twelve orders of magnitude on first write: C1 / (2(C0+CL)$^2$) is *per farad*, not dimensionless, because the numerator is first order in capacitance and the denominator is second. It needs multiplying by one picofarad to become per-pF. The hand check is what caught it.

The review made these changes: - It redrew the schematic as the real Pierce oscillator of AN2867 figure 5, on Pierre's request. - It replaced the 32.768 kHz preset (C0 1.5 pF, C1 2.5 fF, from no source) with the Epson FC-135's typical values, and corrected the claim that a watch crystal has "a tenth the motional capacitance" and pulls "around 6 ppm/pF". - It replaced the "2–5 pF" stray range, which AN2867 does not state in what was

read, with its worked example's 5 pF. - It put the pair to buy first and large, with what it pulls. - It added the start-up check and RExt's starting value from AN2867, after Pierre asked why RExt was drawn but not calculated. - It added the drive-level check from a scope reading, so the tool says whether RExt is needed at all. - It added examples, remembers the crystal type and the series, and was renamed Crystal load, on Pierre's choice.

# Oscillator stability

(formerly Oscillator stability (ppm → Hz))




## What it computes

The total frequency error budget of a crystal oscillator — tolerance, temperature, ageing and load pulling combined — as a worst case and as a root-sum-square estimate, then expressed as hertz and as time gained or lost per day and per year. The worst case is shown large, the RSS beneath it. For a 32.768 kHz crystal, the parabola's coefficient B and turnover T0 are fields, taken from the crystal's datasheet. The crystal type is remembered.

For a single ppm figure applied to a single value, the **PPM converter** under Tools does that; this tool is the budget that produces the figure.

## Source

The contributions and their names are the ones on crystal datasheets: frequency tolerance at a reference temperature, frequency stability over the operating temperature range, and ageing, usually per first year. The reference is 20 °C on some datasheets and 25 °C on others — one Petermann-Technik page quotes both for the same part — so the field is labelled Tolerance without a temperature.

The 32.768 kHz temperature curve is a parabola, whose two numbers come from the crystal's datasheet. **Epson's FC-135 specification sheet,** read for this review, gives: - B −0.04 ppm/°C$^2$ max, and a turnover of 25 ± 5 °C; - a tolerance of ±20 ppm at 25 °C; - ageing of ±3 ppm in the first year.

These are the tool's defaults. The −0.034 ppm/°C$^2$ often quoted as typical (±0.006) has −0.04 as its worst end.

## Why the formulas are these

**Two kinds of error, kept apart.** Tolerance, temperature stability and ageing are *limits*: the part is somewhere within ± each of them, and which way it sits is not known. The tuning fork's temperature drift and the load pulling are *offsets*: they have a known sign and they move the whole window. Mixing the two kinds is how budgets end up either too pessimistic or wrong-signed, so the tool computes

```
Offset     = Drift + Pulling
Worst case = Offset ± (Tolerance + Temperature + Ageing)
```

and reports the extreme of that window with its sign. A window of −69 to −23 ppm is a clock that can only lose time, and writing it as ±69 ppm would throw that away.

**Worst case and RSS.** The worst case adds every limit at full size in the same direction. It is what datasheets call *overall stability*, and it is the right figure to design against when failure is not an option. If the contributions are independent, all of them sitting at their extremes together is unlikely, and the root-sum-square

```
RSS = Offset ± √(Tolerance² + Temperature² + Ageing²)
```

is the likelier spread. Offsets are added directly, never squared: they are not random.

**The tuning-fork parabola.** A 32.768 kHz crystal is a quartz tuning fork, and its frequency falls off either side of a turnover temperature:

```
Δf/f = B × (T − T0)²        (FC-135: B −0.04 ppm/°C² max, T0 25 °C)
```

It only ever goes down. - **With the FC-135's worst-case B:** −25 ppm at 0 °C (2.2 seconds a day), and −81 ppm at −20 °C (7 seconds a day). - **With the typical −0.034:** −21 ppm and −69 ppm. That is why a watch or a data logger left outdoors in winter runs slow, and why the tool asks for the temperature the fork will actually see rather than a spec. MHz crystals are AT-cut plates, whose curve is a cubic set by the cut angle; their datasheets fold it into one ±ppm figure over the range, which is what the MHz pill asks for.

**Into time.** A day is 86 400 s, so 1 ppm is 86.4 ms a day, and a year of 31 557 600 s makes 1 ppm about 31.6 s a year. Twenty ppm is 1.73 s a day or 10.5 minutes a year.

## Assumptions and limits

- **Ageing is not linear.** It is fastest in the first months and slows roughly logarithmically, so ten years is much less than ten times the first-year figure. Enter the total you expect over the period, not a rate.
- **The parabola uses one B and one T0.** The datasheet gives B as a maximum and T0 with a tolerance (±5 °C on the FC-135). Typing the worst B, and the turnover furthest from your temperature, gives the worst case. Near the turnover it barely matters; at the extremes it does.
- The MHz temperature figure is taken as a symmetric ± limit, which is how it is specified, even though the real AT-cut curve is not symmetric.
- Pulling is carried in as a signed figure from the crystal load tool and is treated as exact.
- **Tolerance never includes temperature on a crystal datasheet.** The temperature stability is a separate line, often a separate ordering table, and the two add: ±20 ppm tolerance with ±30 ppm over −40/+85 °C is up to ±50 ppm before ageing. A packaged **XO or TCXO** is different: its "frequency stability" is often one all-inclusive figure covering tolerance, temperature, supply and load, sometimes ageing too. Read the footnote; if it is all-inclusive, enter it as Tolerance and leave the rest at zero.

## What it deliberately does not do

- **Compensated oscillators.** A TCXO or an RTC with digital trimming cancels most of the temperature term; its datasheet gives a residual stability, which can be entered as Temp. stability on the MHz pill.
- **Short-term stability and jitter.** Phase noise and Allan deviation are a different question from where the average frequency sits.

- **Protocol limits.** Whether a budget meets USB, CAN or Ethernet timing needs those standards' own tolerances and belongs with the tools for each.

## How it was checked

By hand, MHz defaults: 20 + 30 + 3 = 53 ppm worst case; √(400 + 900 + 9) = 36.18 ppm RSS; 53 ppm of 16 MHz is 848 Hz; 53 × 86.4 ms = 4.579 s a day; 53 × 31.56 s = 27.88 minutes a year. With −25 ppm of pulling the window becomes −78 to +28 ppm and the worst case reads −78, signed.

32.768 kHz at 0 °C, with the FC-135's B of −0.04: - drift $-0.04 \times 25^2 = -25$ ppm; - with ±23 ppm of limits the window is −48 to −2 ppm, worst −48 ppm.

At −10 °C it is −49 ppm of drift, −72 ppm worst, 6.221 s a day slow. A typed B of −0.034 at −10 °C gives −41.65 ppm of drift, −64.65 ppm worst. At 25 °C only the limits remain: ±23 ppm, 12.1 minutes a year. Switching pills resets pulling, which on first build carried over from one family to the other.

The review made these changes: - It made B and the turnover fields, defaulting to the FC-135 sheet. The tool hard-coded −0.034 ppm/°C$^2$ about 25 °C, which is a typical value and not the datasheet's worst case. - It put the worst case first and large. - It enlarged the 9–10 px chart labels to 11 px. - It added a plain definition of ppm, examples, and the remembered type. - It was renamed Oscillator stability, on Pierre's choice.

# PLL settings

(formerly PLL multiplication factor)




## What it computes

Every combination of input divider M, feedback multiplier N and output divider that a chip's PLL allows, walked exhaustively, with the ones that land closest to a target frequency listed first — each with its error in ppm and the frequencies at the phase detector and the VCO, so the limits can be seen being respected. The chosen setting is shown large, with its error and its internal frequencies; tap another row to choose it. Presets cover the STM32F1, F4, G0 and G4 and the RP2040 and RP2350; Generic takes any integer-N PLL's limits from its manual. The chip is remembered.

## Source

The limits are taken from the vendors' datasheets and code rather than from summaries, which disagree with each other:

- **STM32F4** — ST's HAL driver (`stm32f4xx_hal_rcc.h`, `stm32f4xx_hal_rcc_ex.h`): PLLM 2–63, PLLN 50–432, PLLP 2, 4, 6 or 8, PLLQ 2–15, and the VCO input between 1 and 2 MHz. The VCO output window of 100–432 MHz and the 168 MHz system clock of the F405/407 are from RM0090.
- **STM32F1** (the F103 of the Blue Pill) — DS5319 rev 18, table 27: PLL input 1–25 MHz, PLL output 16–72 MHz. The F1 has no output divider: its PLL output is the system clock. From the F1 HAL: PLLMUL 2–16, the crystal divided by 1 or 2 on its way in (PLLXTPRE), and USB clocked by the PLL output divided by 1 or 1.5 — so USB works only with the PLL at 48 or 72 MHz.
- **STM32G4** (G474) — DS12288 rev 2, table 49: PLL input 2.66–16 MHz, VCO 96–344 MHz, and the R output that clocks the core up to 150 MHz, or 170 MHz with the regulator in Range 1 boost mode (section 3 of the same datasheet). Dividers from the G4 HAL: PLLM 1–16, PLLN 8–127, PLLR 2, 4, 6 or 8.
- **STM32G0** (G071) — DS12232 rev 4, table 43: PLL input 2.66–16 MHz, VCO 96–344 MHz, R output 12–64 MHz. Dividers from the G0 HAL: PLLM 1–8, PLLN 8–86, PLLR 2–8.
- **RP2040** — the Pico SDK's own calculator, `vcocalc.py`: reference divider 1–63 with at least 5 MHz after it, FBDIV 16–320, VCO 750–1600 MHz, and two post-dividers each 1–7.
- **The RP2040's rated clock** is from the Pico SDK 2.1.1 release notes: it is now "certified to run at a system clock of 200Mhz when using a regulator voltage of at least 1.15 volts". The SDK raises the voltage itself when needed, and keeps 125 MHz as the default. Before, 133 MHz was the most that was supported.
- **RP2350** — its datasheet, section 8.6, lists the RP2040's limits again (at least 5 MHz after REFDIV, VCO 750–1600 MHz, FBDIV 16–320, post-dividers 1–7) and gives 150 MHz as the most the RP2350 supports.

The STM32 datasheets were read in the copies distributors host (RS and DigiKey), because st.com refused the download; later revisions exist and were not compared. ST's own code disagrees with these tables: its LL utils files check the VCO from 64 MHz on both families and the G0's PLL input at 4–8 MHz, and the G4's reference manual (RM0440 rev 3) gave a 64–344 MHz VCO and a 2.66–8 MHz input. Asked about it on ST's community forum in April 2020, an ST technical moderator answered that the datasheet's PLL characteristics are the correct values and that RM0440 would drop its numbers, so the tables are used here.

## Why the formulas are these

An integer-N PLL has one loop and three dividers:

```
PFD = Input / M
VCO = PFD × N
Out = VCO / P
```

The input is divided down to the **phase-frequency detector** (PFD) rate. The VCO runs at N times that, because the loop compares the VCO divided by N against the PFD reference and steers the VCO until they match. The output is the VCO divided down again. Each stage has a window it must stay inside: the PFD because the detector and its filter are designed for a range, the VCO because the oscillator only tunes over a range. That is what makes this a search rather than a division — the obvious M and N often put one of the stages outside its window.

The STM32F1 has no output divider, so there Out = PFD × N, and its window is the PLL output's. The output divider is P on the F4, R on the G0 and G4, and two dividers in series on the RP2040 and RP2350.

**Ranking.** Rows are ordered by error first. Among equally good rows:

- on the STM32F1 and F4, a row that also gives USB exactly 48 MHz comes first, since USB will not work on anything else;
- then the **higher PFD**. ST's manual recommends a 2 MHz VCO input on the F4 to limit jitter, and in general a faster comparison lets the loop correct the VCO more often and filter less, which is quieter;
- then the higher VCO, quieter for the same reason. It costs current: the G0 and G4 datasheets give their PLL 200 µA typical at a 96 MHz VCO and 520 µA at 344 MHz, so pick a lower-VCO row from the table when power matters more.

That is why 8 MHz to 168 MHz on the STM32F4 comes out as M 4, N 168, P 2, Q 7 (PFD 2 MHz, VCO 336 MHz, USB 48 MHz) ahead of the equally exact M 8, N 336, which runs the detector at 1 MHz.

**The RP2040's and RP2350's two post-dividers** multiply, so the tool offers each distinct product once, as the pair with the larger first divider — the SDK prefers that split for power.

**Notes above a plain rating.** The RP2040 above 133 MHz needs its core regulator at 1.15 V or more, and the G4 above 150 MHz needs Range 1 boost mode (`PWR_REGULATOR_VOLTAGE_SCALE1_BOOST` in the HAL); the tool says so under the result. Above the rating itself it says the part is not rated for it.

## Assumptions and limits

- **Integer-N only.** Many modern PLLs add a fractional part to N, which can hit almost any target but trades it for spurs and depends on each vendor's modulator. Not modelled.
- The preset limits are for the named parts. Other members of a family can differ — the F401, for one, needs its VCO in 192–432 MHz, and the F105/F107 multiply by 4–9 or 6.5 after a 1–16 prescaler — and the G0 and G4 presets come from the G071 and G474 datasheets, so check another member's own table, or use Generic with the right reference manual when in doubt.
- The G0 and G4 windows are Range 1's. Range 2, the low-power range, narrows the VCO to 96–128 MHz and the system clock to 26 MHz on the G4 and 16 MHz on the G0; not modelled.
- Lock time, loop-filter design and phase noise are not addressed; the tool answers only which settings are legal and how close they land.
- On Generic, the search stops at three million combinations to stay responsive and says so; narrow the ranges for a complete answer.

## What it deliberately does not do

- **Vendor clock trees beyond the PLL** — AHB and APB prescalers, peripheral clock muxes, flash wait states for the chosen frequency. Those are the next steps after this one, and every family arranges them differently.
- **The G0's and G4's other PLL outputs.** P and Q feed peripherals; the tool sets R, the system clock. The G4 can clock USB from its separate 48 MHz oscillator, HSI48, instead of the PLL.
- **Other presets.** The Generic pill covers any integer-N PLL given its manual; a preset is only worth adding for a part used often enough to save the reading.

## How it was checked

Against the configurations the vendors ship: 8 MHz to 168 MHz on the STM32F4 gives M 4, N 168, P 2, Q 7 with USB at exactly 48 MHz, and from a 25 MHz crystal M 25, N 336, P 2, Q 7 — the standard setting for that crystal. 180 MHz is flagged as above the part's 168 MHz rating and as leaving USB at 45 MHz.

On the STM32F1, 8 MHz to 72 MHz gives M 1, N 9 with USB at 72 / 1.5 = 48 MHz, the setting of ST's own F1 USB examples; at 64 MHz the tool warns that USB cannot get its 48 MHz. On the STM32G4, 16 MHz (HSI16) to 170 MHz gives M 4, N 85, R 2 with the boost-mode note — the setting of ST's NUCLEO-G474RE examples, which switch boost mode on — and 150 MHz gives M 4, N 75, R 2 without it. On the STM32G0, 16 MHz to 64 MHz gives M 1, N 20, R 5 (VCO 320 MHz).

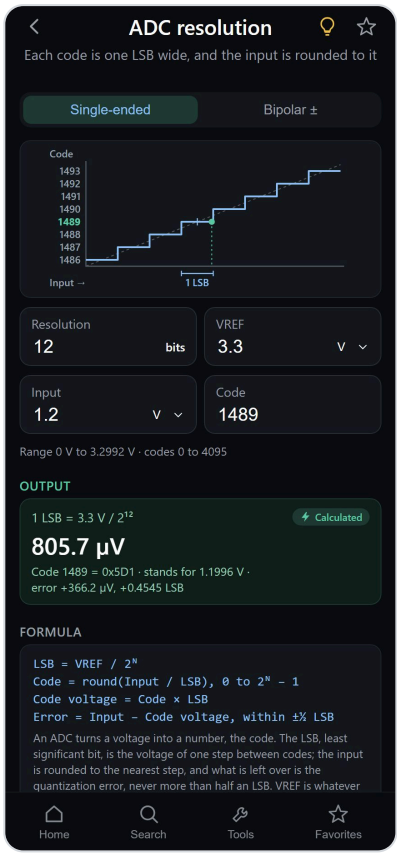
On the RP2040, 12 MHz to 125 MHz gives VCO 1500 MHz with post-dividers 6 and 2, and 133 MHz gives VCO 1596 MHz with 6 and 2 — both exactly the Pico SDK's own default settings. 200 MHz gives VCO 1200 MHz with 6 and 1, and the note about the 1.15 V core voltage; 250 MHz is flagged as above the certified 200 MHz. On the RP2350, 12 MHz to 150 MHz gives VCO 1500 MHz with 5 and 2, the SDK's default in `clocks.h`; 200 MHz is flagged as above its 150 MHz. The full RP2040 search takes about 3 ms.

The review made these changes: - It raised the RP2040's rating from 133 MHz to the 200 MHz certified in Pico SDK 2.1.1, with the voltage condition shown above 133 MHz. - It put the chosen setting first and large. - It enlarged the 9–10 px diagram labels to 11 px, and moved the VCO frequency clear of the USB branch. - It added plain definitions of the PFD, the VCO and the dividers, examples, and the remembered chip. - It was renamed PLL settings, on Pierre's choice. - It added the STM32F1, G0 and G4 and the RP2350, on Pierre's

choice, with limits read from their datasheets, the chips split over two rows of pills. - It fixed a chip switch that carried the RP2040's 1.15 V note over to the next chip. - Without a USB branch, the diagram drops its empty top band, which keeps Generic's result above the tab bar with the second row of pills.

# ADC resolution

(formerly ADC resolution / quantization)




## What it computes

For an ideal N-bit converter and its reference: the size of one step (the LSB), shown large since it is the resolution, then the code a given input produces, the voltage that code stands for, and the quantization error between the two, in volts and in LSB. It works both ways: type a voltage to get the code, or type a code — say, one read off a register — to get the voltage. The range of inputs and codes is shown under the fields. The reference and the input can be typed in mV or V; the mode and both units are remembered.

Two pills cover the two ways converters are built:

- **Single-ended** — 0 to VREF, codes 0 to $2^N - 1$. Every MCU ADC (AVR, STM32, RP2040, ESP32) works this way.
- **Bipolar ±** — −FSR to +FSR, codes $-2^{N-1}$ to $2^{N-1} - 1$ in two's complement. Differential converters such as the ADS1115 and the HX711 work this way.

Examples: an Arduino Uno on its 5 V reference, an LM35 temperature sensor on the Uno's 1.1 V reference, an STM32 at 12 bits on 3.3 V, the ADS1115 at its default ±2.048 V, and the HX711 load-cell converter at 24 bits.

## Source

- The ideal transfer function is the one in IEEE Std 1241 and the one MCU datasheets measure their errors against. The ATmega328P datasheet (Atmel-42735B, section 28.6) defines offset error as the deviation of the first transition from its ideal place "at 0.5 LSB", gain error against a last transition 1.5 LSB below full scale, the quantization error as always ±0.5 LSB, and ±0.5 LSB as the ideal absolute accuracy. The same chip is specified at ±2 LSB absolute accuracy, and its internal reference at 1.0 V minimum, 1.1 V typical, 1.2 V maximum.
- Arduino's analogReference() page gives the Uno's DEFAULT reference as 5 V and INTERNAL as 1.1 V on the ATmega328P.
- The bipolar pill follows the TI ADS1115 datasheet (SBAS444E): the full-scale range is set by the PGA to ±6.144, ±4.096, ±2.048 (the power-up default), ±1.024, ±0.512 or ±0.256 V; Table 7-1 gives the LSB as 125 µV at ±4.096 V and 62.5 µV at ±2.048 V; output is binary two's complement, clipping at 7FFFh and 8000h.
- The HX711 datasheet (Avia Semiconductor) gives its full-scale input as ±0.5(AVDD/GAIN), ±20 mV at gain 128 with a 5 V AVDD, 24-bit two's complement output from 800000h to 7FFFFFh, and 50 nV rms input noise at gain 128 and 10 samples a second.
- The LM35 is 10 mV/°C (TI SNIS159H).

## Why the formulas are these

**The step.** N bits give $2^N$ codes. Spread over the reference:

```
Single-ended:  LSB = VREF / 2^N
Bipolar:       LSB = 2 × FSR / 2^N
```

The bipolar span is twice FSR because it runs from −FSR to +FSR. The ADS1115 datasheet writes its Equation 4 as LSB = FSR / $2^{16}$, but its own table (125 µV at ±4.096 V) only works with FSR as the whole 8.192 V span — the tool asks for the ± figure, as the datasheet's register table quotes it, and doubles it.

**The code.** Code k stands for k × LSB and covers half an LSB either side, so the ideal converter rounds:

```
Code = round(Input / LSB)
```

The first transition, 0 to 1, is at ½ LSB. That is why the quantization error is ±½ LSB rather than 0 to −1 LSB: the steps are centred on the ideal straight line, and the error is the sawtooth between the line and the staircase. The drawing shows eight codes around the input at scale, with the dashed ideal line through the step centres.

**The top of the range.** The highest code is $2^N - 1$, not $2^N$, so the largest voltage a code stands for is VREF − 1 LSB: 3.2992 V for 12 bits on 3.3 V. An input at VREF itself reads the top code, a whole LSB short.

**Two's complement.** A bipolar converter's register holds negative codes as $2^N$ + code, so −1 reads FFFFh and −FS reads 8000h. The hex in the result shows the register as it would be read, the Code field the signed value it means.

**Dividing by $2^N - 1$.** Arduino's ReadAnalogVoltage example computes code × (5.0 / 1023.0), and ST's LL macro `__LL_ADC_CALC_DATA_TO_VOLTAGE` divides by `__LL_ADC_DIGITAL_SCALE`, 4095 at 12 bits. That treats the top code as exactly VREF, stretching the scale: the reading is k / ($2^N$ − 1) instead of k / $2^N$ of VREF, high by up to one LSB near the top. Small beside a real ADC's errors, but a systematic one, and it is why a 3.3 V input "reads 3.3 V" on those examples and 3.2992 V here.

**Resolution is not accuracy.** A finer LSB only helps while it stays above the noise and the reference's tolerance. The HX711's 2.33 nV step sits under 50 nV rms of noise, about 21 LSB, so its last four bits or so are noise; SNR estimation turns noise into effective bits.

## Assumptions and limits

- **An ideal converter.** Offset, gain error, INL, DNL and noise are not modelled; on a real MCU ADC they add up to several LSB, far more than the quantization shown here (the ATmega328P's ±2 LSB, for one). Their datasheet figures are in LSB, which is what this tool gives the size of.
- **VREF is taken as exact.** On most MCUs it is the supply, so the supply's own tolerance scales every reading. A 1% regulator is 41 LSB at 12 bits. An internal reference is no cure by itself: the ATmega328P's 1.1 V is 1.0 to 1.2 V from part to part, ±9 %, so measure it once and use that figure.
- **Inputs past the range** read the end code; the tool says so and the error then is the overshoot, not quantization. On the ADS1115 the ±4.096 and ±6.144 V settings are scaling only — the pins

themselves never go past VDD + 0.3 V or below GND − 0.3 V.
- **Single-ended on a bipolar part.** The ADS1115 measuring one input against ground uses only codes 0000h to 7FFFh, so it is a 15-bit converter in that mode. Model it on the bipolar pill with 16 bits, and read only the positive half.

## What it deliberately does not do

- **SNR, ENOB, oversampling.** The noise that quantization adds, and what averaging buys back, is the next tool, SNR estimation.
- **DAC.** The reverse conversion has its own tool.
- **Offset-binary and sign-magnitude codes.** Some bipolar converters output these instead of two's complement; they are rare enough on current parts to leave out.

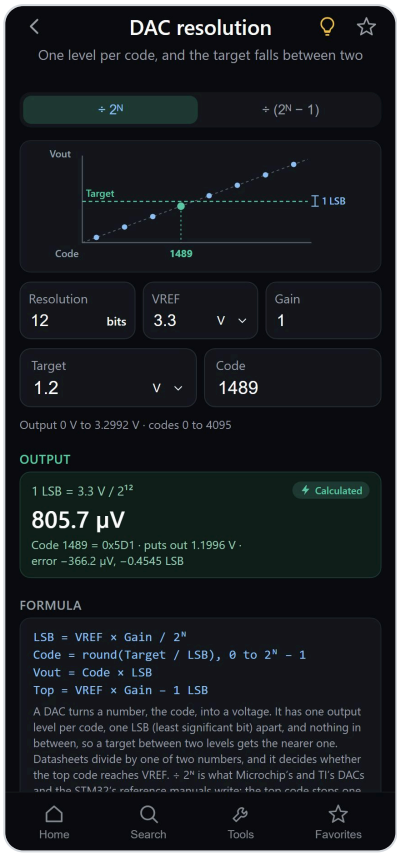
## How it was checked

By hand, single-ended 12 bits on 3.3 V: LSB = 3.3 / 4096 = 805.66 µV; 1.2 V / 805.66 µV = 1489.45, code 1489 = 0x5D1, code voltage 1.19963 V, error +366.2 µV = +0.4545 LSB. Code 4095 gives 3.29919 V; 4 V reads 4095 and is flagged as over range.

The examples: 3.3 V on the Uno's 5 V reference reads 676 (4.883 mV steps, −0.16 LSB); an LM35 at 25 °C, 250 mV, reads 233 on the 1.1 V reference, in steps of 1.074 mV, 0.11 °C, against 0.49 °C on 5 V; −1 V on the ADS1115 at ±2.048 V reads −16000, 0xC180; 5 mV on the HX711 at ±19.53 mV reads 2147484, 0x20C49C, in steps of 2.328 nV.

The review made these changes: - It was renamed ADC resolution, on Pierre's choice; the old name still finds it in search. - The LSB became the large result, with the code, its voltage and the error on the line under it. - The reference and input fields take mV or V, remembered with the mode. - The drawing's labels went from 9–10 px to 11 px, with a left margin that grows with the code's digits. - It added the five examples, checked against their datasheets. - In formula search, the cards of renamed tools had lost their link to the tool; they now find a tool by its old name too.

# DAC resolution




## What it computes

For an N-bit DAC, its reference and its output gain: the step between neighbouring output levels (the LSB), shown large since it is the resolution, then the code to write for a target voltage, the voltage that code really puts out, and how far that is from the target, in volts and in LSB. It works both ways — type the voltage you want, or type a code to see what it outputs. The output range is shown under the fields. The target and VREF can each be typed in mV or V, each with its own picker; the divisor pill and both units are remembered.

Examples: an MCP4725 on 5 V, an MCP4822 at ×2, an STM32 DAC set to 0.7 V, the ESP32's 8-bit DAC and the SAMD21's 10-bit DAC (the Arduino Zero's).

## Source

Divided by $2^N$, the top code one LSB under VREF:

- **Microchip MCP4725** (DS22039D): Equation 4-1 gives the ideal LSB as VREF / $2^n$, which equals (full-scale − zero-scale) / ($2^n$ − 1); VREF is VDD on this part.
- **Microchip MCP4802/4812/4822** (DS20002249B): VOUT = VREF × D / 4096 with the GA bit at 1 (×1), and 2 × VREF × D / 4096 with GA at 0 (×2), from the internal 2.048 V reference; its table gives 0.5 mV per step at ×1 and 1 mV at ×2.
- **TI DAC7311** (SBAS442D): Equation 1, VOUT = AVDD × D / $2^n$, read in TI's HTML datasheet viewer.
- **STM32.** On ST's community forum, a user quotes RM0444 (STM32G0) as dividing by 4096, and in a 2017 thread on the STM32F373, whose reference manual also writes 4096, ST engineer K.Oku explains why: the DAC "is kind of R-2R implementation", and its top code "has one resistor (corresponds 1LSB) under the VREF". RM0444 itself was not read; st.com refused the download. The STM32G071 datasheet (DS12232 rev 4, table 59) gives the output as 0 to VREF+ with the buffer off and 0.2 V to VREF+ − 0.2 V with it on.

Divided by $2^N$ − 1, the top code at VREF:

- **SAMD21** (Atmel-42181, the Arduino Zero's microcontroller): VDAC = DATA / 0x3FF × VREF.
- **ESP32** (technical reference manual, section 31.4.3): DACn_OUT = VDD3P3_RTC × PDACn_DAC / 255, with VDD3P3_RTC typically 3.3 V.

And ST's software, against its own manuals: the STM32G0 HAL's comment writes DAC_OUTx = VREF+ × DOR / 4095, with the example (3.3 × 868) / 4095 = 0.7 V, and its LL driver's `DAC_DIGITAL_SCALE_12BITS` is 4095, which `__LL_DAC_CALC_VOLTAGE_TO_DATA` multiplies by.

## Why the formulas are these

**One level per code.** A DAC puts out one voltage for each code and nothing in between:

```
Vout = Code × LSB
```

so a target that is not exactly a level gets the nearer one, at most half an LSB away. That is why the picture is points, not a staircase: eight levels around the target at scale, the target a line between two of them, the chosen one in green.

**Two divisors, and why it matters.** How big the step is depends on what the datasheet divides by:

```
÷ 2^N:        LSB = VREF × Gain / 2^N          top code = VREF × Gain − 1 LSB
÷ (2^N − 1):  LSB = VREF × Gain / (2^N − 1)    top code = VREF × Gain exactly
```

A resistor string or R-2R ladder makes $2^N$ equal steps between ground and VREF, of which the top code uses $2^N - 1$, so the output stops one LSB short: that is the ÷ $2^N$ datasheets, the STM32 among them. The SAMD21 and ESP32 documents divide by $2^N - 1$ instead. The difference is under one LSB, but it is systematic, largest near the top of the range. Pick the pill your part's datasheet uses.

**ST's 4095.** ST's HAL and LL code divide by 4095, so a code they compute is up to one LSB low near the top on a DAC that, by ST's manuals, divides by 4096. At 0.7 V on 3.3 V, the HAL comment's 868 puts out 0.69932 V; 869, the nearer level on the 4096 scale, puts out 0.70012 V.

An earlier version of this tool took the STM32 to be ÷ 4095, on a forum user's bench readings from a NUCLEO-G071RB. Those readings cannot decide it: on the 2.5 V reference used there, the two divisors give the same code, or codes one LSB (0.61 mV) apart, at the 400, 800 and 1200 mV targets, yet the readings differ by about 4 mV, six or seven LSB, so something else changed between the two runs.

**Gain.** Many DACs follow the ladder with an amplifier. On the MCP4822 the GA bit picks ×1 or ×2 from its internal 2.048 V reference, which is how a 2.048 V reference makes a 4.095 V range at 1 mV a step.

## Assumptions and limits

- **The output cannot reach the rails.** The MCP4822's amplifier swings from 10 mV to VDD − 40 mV, and its accuracy holds only between those. The STM32 DAC with its output buffer on spans 0.2 V to VREF+ − 0.2 V. The top and bottom few codes on the tool's range are therefore not really available on those parts, and with ×2 gain the range can exceed VDD, which the output will not follow either.
- **An ideal DAC.** Offset, gain error, INL and DNL are not modelled; they are specified in LSB, which is what this tool gives the size of.
- **VREF is taken as exact.** When it is VDD, as on the MCP4725, the supply's tolerance scales every output.

## What it deliberately does not do

- **Bipolar DACs** (±VREF output with offset-binary or two's-complement codes) are rare on current general-purpose parts; the bipolar pill on the ADC tool covers the arithmetic if needed.

- **PWM as a DAC.** Resolution there is set by the timer and the filter, not a ladder; it is a different calculation.
- **SNR and ENOB** are the next tool.

## How it was checked

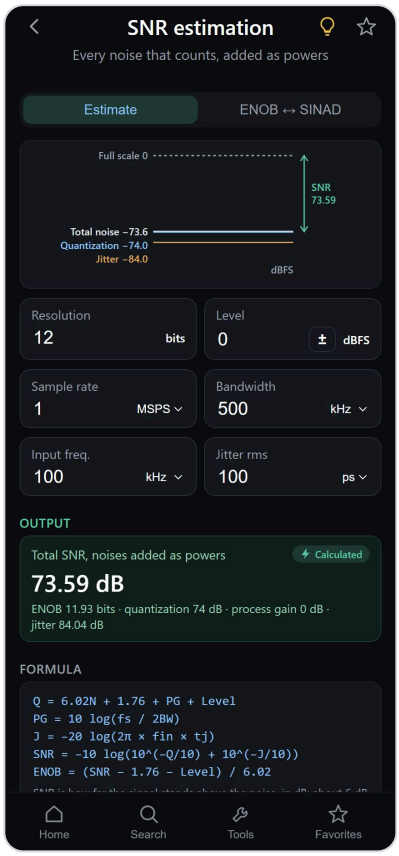
÷ $2^N$, 12 bits, 3.3 V: LSB 805.66 µV; 1.2 V gives code 1489 (0x5D1), output 1.19963 V, error −366.2 µV = −0.4545 LSB. Code 4095 outputs 3.29919 V, one LSB short of VREF.

÷ ($2^N$ − 1): LSB 3.3 / 4095 = 805.86 µV; code 4095 outputs exactly 3.3 V; 1.2 V gives code 1489, output 1.19993 V.

The examples: the MCP4725 on 5 V puts 3.3 V at code 2703, 0xA8F, 0.44 mV low; the MCP4822 at ×2 steps exactly 1 mV, so 3 V is code 3000; the STM32 at 0.7 V gets 869, +122 µV; the ESP32's 8 bits ÷ 255 step 12.94 mV, so 1.5 V is code 116 at 1.501 V; the SAMD21's 10 bits ÷ 1023 put code 310 at exactly 1 V.

The review made these changes: - The STM32 moved from the ÷ ($2^N$ − 1) side to ÷ $2^N$, after reading ST's engineer's answer and its manuals' 4096; the second pill now rests on the SAMD21 and ESP32 documents, and the forum bench test is no longer cited as evidence. Analog Devices is no longer named: none of its datasheets was read. - The LSB became the large result, with the code, its output and the error on the line under it. - The target field takes mV or V, remembered with the pill. - The drawing's labels went from 9–10 px to 11 px. - It added the five examples, the first cases (13), and in every tool's examples a value now stays on one line with its unit.

# SNR estimation




## What it computes

Two things, one per pill.

- **Estimate** — the signal-to-noise ratio an ADC can reach from the noises that can be calculated: quantization, the process gain of oversampling, and sampling-clock jitter. Each is shown on its own, then added as noise powers into one SNR and expressed as an effective number of bits (ENOB).
- **ENOB ↔ SINAD** — the conversion between the two figures ADC datasheets quote, both ways, with how many of the converter's bits noise and distortion take away.

The total SNR is the large result on Estimate, with the ENOB it is worth and its three terms on the line under it; on ENOB ↔ SINAD the effective bits are large, with the bits lost and the noise floor. The pill and the units of the sample rate, bandwidth, input frequency and jitter are remembered.

The picture is a level diagram in dBFS: full scale at the top, the signal, and each noise floor as a line. The gap from the signal down to the total floor is the SNR, and the highest noise line is the one worth fixing.

Examples: 16-bit audio at 44.1 kHz, ×16 oversampling, 1 ps of jitter at 100 MHz, a signal at −20 dBFS, and the ADCs of the RP2040 and the STM32G0.

## Source

Walt Kester's Analog Devices tutorials:

- **MT-001**, "Taking the Mystery out of the Infamous Formula, SNR = 6.02N + 1.76 dB": the ideal SNR of an N-bit converter for a full-scale sine measured over dc to fs/2, and the correction 10 log(fs / 2BW) it calls process gain.
- **MT-003**, "Understand SINAD, ENOB, SNR, THD, THD + N, and SFDR": ENOB from SINAD, and its Equation 2, which adds the level below full scale so ENOB is normalised to full scale.
- **MT-007**, "Aperture Time, Aperture Jitter, Aperture Delay Time": the SNR limit set by jitter, and that the total jitter is the root-sum-square of the sampling clock's and the ADC's own aperture jitter.

analog.com refused a re-read of these during the review, so the formulas were checked again elsewhere:

- **TI's SLYT379**, part 1 of an Analog Applications Journal series on clock jitter: its Equation 2 is SNR = −20 log(2π × fIN × tJitter), its Equation 1 adds the quantization, thermal and jitter SNRs as noise powers, and its Equation 3 combines the clock's and the ADC's aperture jitter as a root-sum-square.
- Wikipedia's article on ENOB quotes MT-003's ENOB = (SINAD − 1.76) / 6.02.

- National Semiconductor's note on aperture time and jitter (TI SNAA018) agrees that doubling the sample rate and filtering gains 3 dB. Its jitter example does not agree with the formula: it gives 73 dB for 10 ps at 10 MHz, where −20 log(2π × 10 MHz × 10 ps) is 64 dB, because it divides by the full-scale range where the sine's rms value belongs (20 log 2√2 = 9 dB).

Datasheet figures:

- **RP2040**, section 4.9.3, table 566, typical, measured at 250 ksps with the input spanning codes 1 to 4094: SNR 61.5 dB, THD −55 dB, SFDR 59.9 dB, SINAD 54.0 dB, ENOB 8.7. These are the ENOB pill's defaults.
- **STM32G071**, DS12232 rev 4, table 58: SINAD 63 dB typical (62.5 dB minimum) and ENOB 10.2 (10.1) at VDDA = VREF+ = 3 V and 25 °C.

## Why the formulas are these

**Quantization.** Rounding to the nearest code leaves an error spread evenly over ±½ LSB, whose rms value is q / √12. A full-scale sine has an rms value of $2^N$ q / (2√2). Their ratio, in dB, is

```
SNR = 6.02N + 1.76 dB
```

— 6.02 dB per bit, which is 20 log 2, plus a constant from the shape of a sine. 12 bits: 74.0 dB. 16 bits: 98.1 dB.

**Level.** The quantization noise stays where it is when the signal gets smaller, so the SNR falls one dB for every dB below full scale. A 12-bit ADC reading a signal at −20 dBFS has 54 dB of SNR, the same as an ideal 8.7-bit one. That is the most common way resolution is wasted.

**Process gain.** Quantization noise is spread evenly from dc to fs/2. Filter the result digitally down to a bandwidth BW and only the share of the noise inside BW remains:

```
PG = 10 log(fs / 2BW)
```

Four times the bandwidth needed is 6.02 dB, one extra bit. MT-001's example: a 65 MSPS ADC with channels 30 kHz wide gains 30.3 dB, 65 dB becoming 95.3 dB.

**Jitter.** If the instant of sampling wanders by tj rms, the error it causes is the signal's slope times that wander. For a full-scale sine at fin the result is independent of amplitude:

```
SNR(jitter) = −20 log(2π × fin × tj)
```

It depends on the input frequency, not the sample rate. 100 ps is harmless at 1 kHz (124 dB) and ruinous at 100 MHz (24 dB). MT-007 notes that 14-bit performance at 100 MHz needs under 0.1 ps; the tool gives 84 dB at exactly 0.1 ps, just under the 86 dB of 14 bits.

**Adding them.** Uncorrelated noises add as powers:

```
SNR = −10 log(10^(−Q/10) + 10^(−J/10))
```

so the total is always a little worse than the worse of the two, and a term 10 dB better than the other barely counts. When jitter is the lower one, the tool says so — more bits will not help.

**What ENOB means.** ENOB, the *effective number of bits*, answers the question "how many bits is this converter really worth?". The resolution on the datasheet, 12 bits say, only counts the codes the ADC can output. A real ADC also adds its own noise and distortion, so the last few bits are buried in them: they change from reading to reading without telling you anything about the signal. ENOB is the resolution of a perfect converter that would be exactly as noisy as the real one. A 12-bit ADC with an ENOB of 8.7 gives, on a changing signal, the same quality as a flawless 8.7-bit ADC; the bottom 3.3 bits are noise. ENOB is not a whole number, because it is worked back from a measured ratio, not counted.

**What SINAD means.** SINAD, the *signal-to-noise-and-distortion ratio*, is that measured ratio: a pure sine is fed in, and the power of the sine is compared with the power of everything else in the output — noise and the harmonics that distortion adds. SNR is the same measurement leaving the harmonics out, which is why a datasheet's SNR is always the higher of the two. Because a perfect N-bit converter has an SINAD of 6.02N + 1.76 dB, the two figures are the same fact in different units, and datasheets quote either.

**ENOB from the ratio.** The SNR equation solved for N, with the level term of MT-003:

```
ENOB = (SNR − 1.76 − Level) / 6.02
```

On the ENOB pill SINAD takes the place of SNR, as the datasheets do. The RP2040's 54.0 dB gives 8.68 bits, and its datasheet says 8.7: of its 12 bits, 3.3 are lost to noise and distortion. Its noise alone, an SNR of 61.5 dB, would leave 9.9; its THD of −55 dB — its INL and DNL errors (the DNL spikes are erratum RP2040-E11) — costs more than the noise. The STM32G071's 63 dB gives 10.17 bits, its datasheet's 10.2.

## Assumptions and limits

- **Oversampling needs noise.** Process gain assumes the quantization error is random. A clean, slow signal on a quiet ADC can give the same code every time, and averaging a constant gives nothing. MT-001 warns about this correlation; the cure is noise or dither of about an LSB at the input.
- **Thermal noise and distortion are not estimated.** They are the ADC's own and only its datasheet knows them. The Estimate pill gives the ceiling the architecture allows; the ENOB pill gives what the part actually does.
- **No noise shaping.** Delta-sigma converters push quantization noise out of band and gain far more than 10 log per octave of oversampling. Their datasheets give SNR directly.
- **SINAD at other levels.** MT-003's level correction assumes the noise does not change with level. Distortion usually drops at lower levels, so ENOB normalised from a −6 dBFS measurement can read better than at full scale.

## What it deliberately does not do

- **THD, SFDR, THD + N.** Read from the datasheet, not calculated.

- **Noise-free resolution and effective resolution** from rms input noise, the measure for dc and slow signals. MT-003 warns not to confuse them with ENOB; they are a different calculation from different data.

## How it was checked

Default Estimate: 12 bits, 1 MSPS, 500 kHz, 0 dBFS, 100 kHz, 100 ps. Q = 6.02 × 12 + 1.76 = 74.0 dB; J = −20 log($2\pi \times 10^{5} \times 10^{-10}$) = 84.04 dB; total 73.59 dB; ENOB 11.93 bits.

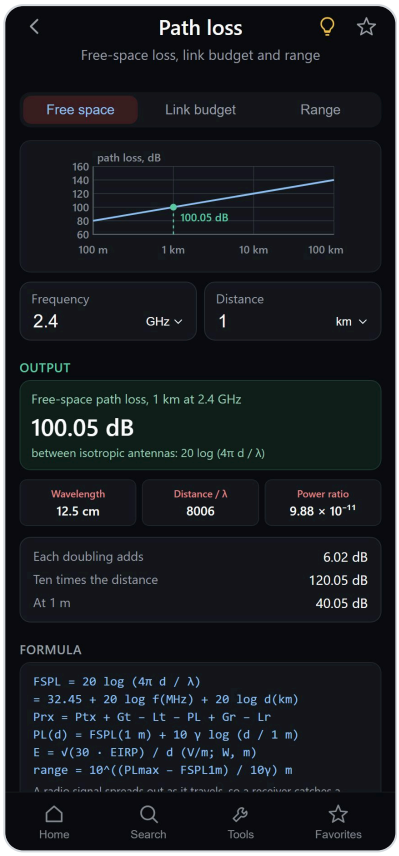
MT-001's process-gain example: 65 MSPS and 30 kHz give 30.35 dB. MT-007's jitter example: 14 bits, 100 MHz, 0.1 ps give J = 84.04 dB, total 81.91 dB, with the jitter warning. 40 MHz of bandwidth at 65 MSPS is refused as past fs/2. With no jitter at −1 dBFS, 14 bits give 85.04 dB and an ENOB of 14.0, normalised.

ENOB pill: SINAD 54.0 dB at 0 dBFS gives 8.678 bits — the RP2040's 8.7 — with 3.32 bits lost; ENOB 10 gives SINAD 61.96 dB.

The examples: 16 bits over the whole band with no jitter, 98.08 dB; 12 bits keeping 31.25 kHz of a 1 MSPS band, process gain 12.04 dB, total 86.04 dB, 14 bits; 14 bits at 250 MSPS with 1 ps at 100 MHz, jitter 64.04 dB, total 64.01 dB, 10.34 bits, with the warning; 12 bits at −20 dBFS, 54 dB and an ENOB still of 12; the STM32G0's 63 dB, 10.17 bits.

The review made these changes: - The total SNR and the ENOB became the large results, with their terms on the line under them. - Rates, frequencies and jitter are kept in hertz and seconds, so a remembered unit only changes how a field shows them; the pill and units are remembered. - The Estimate fields went from three to a row to two: at 375 px the sample rate's unit was cut to "MS…". - The diagram's labels went from 8–9 px to 11 px, with values to one decimal. - It added the six examples, the first cases (12) and a formula-search card.

# Path loss




## What it computes

How much of a radio signal gets from one antenna to another, three ways (pills):

- **Free space:** the frequency and the distance give the free-space path loss in dB, between isotropic antennas. In three cells: the wavelength, the distance in wavelengths and the power ratio (the share of the power that arrives). Under them: what each doubling adds (6.02 dB), the loss at ten times the distance, the loss at 1 m, and a warning when the distance is under 5 wavelengths. The drawing is the loss against distance on a logarithmic axis, your distance marked.
- **Link budget:** the transmitter's power, the transmitting antenna's gain and line loss, the receiving antenna's gain and line loss, the receiver's sensitivity, the frequency, the distance and the exponent γ. The big result is the received power in dBm, with the margin over the sensitivity in the line under it ("the link closes" or "no link"). In three cells: the EIRP, the path loss and the margin. Under them: the EIRP in watts, the received power in watts (pW, fW) and the field strength, in free space, for that EIRP at that distance (V/m and dBµV/m). The drawing is the received level against distance, the sensitivity dashed, your distance marked with its margin.
- **Range:** the same equipment, and the margin you want instead of the distance. The big result is the distance at which the signal falls to the sensitivity plus that margin, with that level and the other system's unit (miles for metres and kilometres, the other way round for miles) in the line under it. In three cells: the path loss allowed, the EIRP and γ. Under them: the range in free space (γ 2) and with γ 3 and γ 4. The drawing is the level against distance with the sensitivity (grey) and the level needed (green) dashed, the range marked.
- **Units:** frequency kHz, MHz or GHz; distance m, km, mi, nmi or ft; the transmit power dBm, mW or W; the antenna gains dBi or dBd (the two boxes change together).
- **Remembered:** the pill and the units.

Examples: a Wi-Fi link across a field, a geostationary satellite, how far a LoRa node reaches, and what each doubling of distance costs.

## Terms

- **Path loss:** how much weaker a signal is at the receiving antenna than at the transmitting one, in dB. In free space, the **free-space path loss** (the ITU's "free-space basic transmission loss") is between **isotropic antennas**, the reference ones that radiate and receive equally in every direction.
- **dBi, dBd:** an antenna's gain in dB over an isotropic antenna, or over a half-wave dipole; dBd = dBi − 2.15, the dipole's own 2.15 dBi.

- **EIRP:** the effective isotropic radiated power: the transmitter's power less its line loss, plus the antenna's gain; the power an isotropic antenna would need to radiate the same in the antenna's best direction.
- **Line loss:** what the cable and connectors between the radio and its antenna take, in dB (the Coax cable standards tool gives it for a cable).
- **Sensitivity:** the weakest signal at the receiver's input that it can use, in dBm; it depends on the bandwidth and the data rate.
- **Link margin (fade margin):** how many dB the received power is above the sensitivity: the cushion against fading, rain, bodies and misalignment.
- **Path-loss exponent γ:** how fast the loss grows with distance, 10 γ dB for each tenfold distance: 2 in free space, larger where walls, trees and buildings are in the way.
- **Far field, Fresnel zone:** the region well beyond the antennas (the distance much more than the wavelength) where the formula holds; and the ellipsoid round the straight path in which obstacles take signal away.

## Sources

- **Recommendation ITU-R P.525-3, "Calculation of free-space attenuation" (09/2016; read as PDF text):** the field of an e.i.r.p. p (watts) at a distance d (metres), $e = \sqrt{(30\,p)}/d$ V/m (eq. 1); the free-space basic transmission loss between isotropic antennas, $L_{bf} = 20 \log (4\pi d/\lambda)$ dB with d and λ in the same unit (eq. 3), and the same for f in MHz and d in km, $32.4 + 20 \log f + 20 \log d$ (eq. 4). The constant is 32.4478, so the rounded 32.4 reads up to 0.05 dB low (2.4 GHz at 1 km: 100.004 dB by eq. 4, 100.052 by eq. 3): the tool uses eq. 3 with c = 299 792 458 m/s. Later versions of the recommendation were not read (the address tried for version 4 did not exist).
- **Wikipedia, "Free-space path loss" (last edited 20 July 2026; read through a text-extraction tool):** $FSPL = (4\pi d/\lambda)^2$ between isotropic antennas, the definition credited to IEEE Std 145; that the term is "a little misleading", the frequency dependence belonging to the receiving antenna's aperture and not to the path; valid with the antennas in each other's far field ($d \gg \lambda$) and the path clear of obstructions out to 0.6 of the first Fresnel zone; the dB forms $20 \log d + 20 \log f - 147.55$ (metres, hertz) and $20 \log d + 20 \log f + 92.45$ (km, GHz), which the tool's constant agrees with.
- **Wikipedia, "Link budget" (11 October 2026; read):** $P_{rx} = P_{tx} + G_{tx} - L_{tx} - L_{fs} - L_m + G_{rx} - L_{rx}$, and the link margin as "a safety margin between the received power and minimum power required by the receiver"; it cites Pozar, Microwave Engineering, 4th ed. (not read). The tool's margin is the cushion left over, so it has no separate $L_m$.
- **Wikipedia, "Log-distance path loss model" (15 July 2026; read):** $L = L_0 + 10\,\gamma \log (d/d_0) + X_g$, $L_0$ the loss at the reference distance $d_0$ from close-in measurements or the Friis model, $d_0$ typically 1 to 10 m for a microcell, $X_g$ a zero-mean Gaussian term (the shadowing). Its table of measured exponents γ (and standard deviations σ), credited to Rappaport, Wireless Communications: Principles and Practice (Prentice-Hall, 2002; not read): vacuum 2.0; grocery store at 914 MHz 1.8 (σ 5.2 dB); retail store at 914 MHz 2.2 (8.7); office with soft partitions 2.4 at 900 MHz (9.6) and 2.6 at 1.9 GHz (14.1); with hard partitions 3.0 at 1.5 GHz (7); textile or chemical works 2.0 at 1.3 GHz (3.0) and 2.1 at 4 GHz; offices 2.2 and commercial buildings 1.7 at 60 GHz (3.92 and 7.9). The tool takes $d_0 = 1$ m and $L_0$ the free-space loss there, so γ = 2 is exactly free space, and does not model $X_g$: its σ, 3 to 14 dB in that table, is why a margin is wanted.
- **Not read:** Friis' "A Note on a Simple Transmission Formula" (Proc. IRE, 1946), Pozar, Rappaport's book, IEEE Std 145, ITU-R P.341 and P.526. The 2.15 dB between dBi and dBd is the textbook directivity of a

half-wave dipole (10 log 1.64); no source was read for it. The 5 wavelengths at which the tool warns is its own rule of thumb: the sources say only "far field, d ≫ λ".

## Why the formulas are these

An isotropic transmitter of power P spreads it evenly over a sphere, so at a distance d the power density is $P/(4\pi d^2)$. An isotropic receiving antenna collects it over the aperture $\lambda^2/(4\pi)$, so what arrives is $P\,(\lambda/(4\pi d))^2$ and the loss is $(4\pi d/\lambda)^2$, $20 \log (4\pi d/\lambda)$ in dB: 6.02 dB for each doubling of the distance, and for each doubling of the frequency, which comes from the aperture, not from the path. With f in MHz and d in km the constant is $20 \log (4\pi \times 10^9/c) = 32.448$; with d in metres and f in MHz, −27.55; for hertz and metres, −147.55.

The field of an EIRP p at d follows from the power density: $S = p/(4\pi d^2) = e^2/(120\,\pi)$ (the 120 π Ω of free space), so $e = \sqrt{(30\,p)}/d$, ITU-R P.525's eq. 1.

The link budget adds what the equipment does in decibels: the power the antenna is given is Ptx − Ltx, its gain adds Gtx (EIRP = Ptx − Ltx + Gtx), the path takes PL, and the receiver gets EIRP − PL + Grx − Lrx; the margin is that less the sensitivity. For a path that loses more than free space the loss is PL(d) = FSPL(1 m) + 10 γ log (d / 1 m), the log-distance model with d0 = 1 m: for γ = 2 it is eq. 3 itself, since $20 \log (4\pi\, d\, f/c) = 20 \log (4\pi f/c) + 20 \log d$. Solving it for the distance at which the margin is a given M:

```
d = 10^((EIRP + Grx − Lrx − sensitivity − M − FSPL(1 m)) / (10 γ))   metres
```

## Assumptions and limits

- **Free space or a fitted exponent.** The tool has no terrain, ground reflection (the two-ray model), diffraction, atmospheric absorption, rain or foliage; γ is yours to choose, and a real path varies round it by the shadowing σ.
- **Far field, clear path.** Closer than about 5 wavelengths the formula is not reliable, and the tool says so (its own rule of thumb); for a large antenna the far field starts much farther out, at about $2D^2/\lambda$ (D the antenna's largest dimension: 13.9 m for a 0.6 m dish at 5.8 GHz), which the Beamwidth & gain tool gives. The path must be clear of obstructions out to 0.6 of the first Fresnel zone for the free-space figures to apply: the Fresnel zone tool is where to check it.
- **Antennas aligned and the polarization matched.** The gains are in the direction of the link; a misalignment or a polarization mismatch is margin you must keep.
- **The sensitivity is yours.** It depends on the bandwidth, the data rate and the noise figure; the receiver's data sheet gives it for each setting.
- **The field strength** is the free-space far-field figure for the EIRP, whatever γ you give.

## How it was checked

26 cases. The expected values come from a Python implementation of the same formulas: - Free space: 1 km at 2.4 GHz, 100.05 dB (λ 12.5 cm, 8006 wavelengths, a power ratio of $9.88 \times 10^{-11}$, 40.05 dB at 1 m); a geostationary satellite, 12 GHz over 35 786 km, 205.11 dB; 100 m 80.05 dB and 200 m 86.07 dB; a mile, 104.18 dB; 868 MHz over 1 km, 91.22 dB; 0.5 m flagged as under 5 wavelengths; a missing distance or frequency refused; - Link budget: 2.4 GHz over 1 km with 20 dBm, 2 dBi at each end, 0.5 dB lines and −90

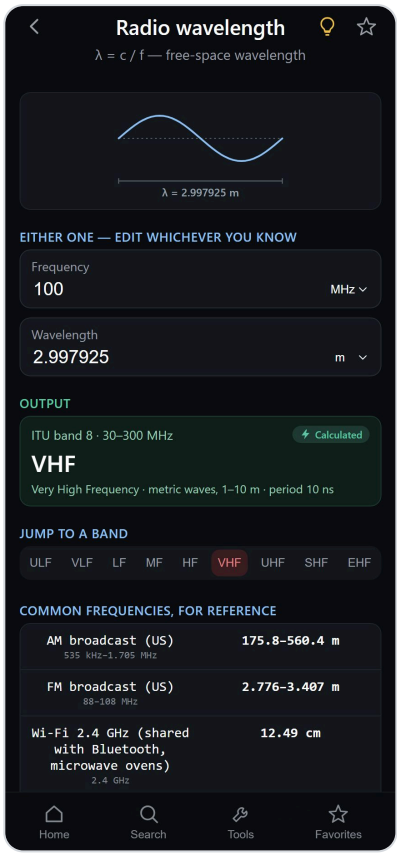
dBm: EIRP 21.5 dBm (141 mW), −77.05 dBm (19.7 pW), +12.95 dB, 2.06 mV/m; a −70 dBm receiver (7.05 dB short); the power in watts; the antenna in dBd (−76.90 dBm); γ 3 (−107.05 dBm, 17.05 dB short); a negative line loss and a zero exponent refused; - Range: 103 dB allowed, 1.4 km (0.872 mi), with γ 3 and 4 125 m and 37.5 m (411 ft); a LoRa node at 868 MHz, 14 dBm, −130 dBm, 10 dB margin, 137 dB allowed: 195 km in free space, 3.36 km at γ 3, 441 m at γ 4; no margin, 4.44 km; in miles; a missing margin refused; - the four examples.

The Python twin also showed that the log-distance form with γ = 2 equals eq. 3 to $3 \times 10^{-14}$ dB over 110 combinations of frequency and distance (30 cm to 384 000 km, 100 kHz to 120 GHz), that solving the range and then running the link budget at it returns the margin asked for to $10^{-14}$ dB, and that $e^2/(120\,\pi)$ equals $p/(4\pi\,d^2)$.

RF & MICROWAVE › ANTENNAS & PROPAGATION

# Radio wavelength

(formerly Wavelength (RF), then Wavelength ↔ frequency)




## What it computes

A frequency and its free-space wavelength, each from the other — type either — with the period of one cycle and the ITU band the frequency falls in: its symbol and number, its full name, and the name ITU gives its wavelengths (metric waves for VHF, decimetric for UHF). The band is the large result, since the wavelength already sits in its field, and the drawing carries the wavelength's value. A row of buttons jumps to the middle of each named band, and a table gives the wavelengths of common allocations.

The units are remembered. When typing one field leaves the other unreadable in its unit — 100 Hz is 2 997 925 000 mm — that field moves to the unit that reads best, without changing the remembered choice.

Examples: FM radio at 100 MHz, AM radio at 1 MHz, Wi-Fi at 2.4 GHz, GPS L1, a 1 m wavelength typed as such, and exactly 30 MHz, on a band edge.

## Source

- **c = 299 792 458 m/s**, exact: the metre has been defined from the speed of light since 1983.
- **The bands** are ITU-R Recommendation V.431-8 (08/2015), table 1, which follows No. 2.1 of the Radio Regulations: band N runs from 0.3 × 10ᴺ to 3 × 10ᴺ Hz, "lower limit exclusive, upper limit inclusive", with the symbols ULF (band 3) to EHF (band 11) and a metric name for each band's wavelengths, from hectokilometric to decimillimetric waves. Band 12, 300 GHz to 3 THz, has no symbol in the table.
- **The reference rows** are US allocations from the FCC: the AM broadcast band, 535 to 1705 kHz (47 CFR 73.14); the FM broadcast band, 88 to 108 MHz (73.310); and Wi-Fi's 6 GHz band, 5.925 to 7.125 GHz (15.407), all read through Cornell's copy of the CFR. GPS L1, 1575.42 MHz, is 154 times the 10.23 MHz reference that IS-GPS-200 section 3.3.1.1 specifies: the change notice read here gives the 10.23 MHz, and the multiplier comes from secondary sources.

## Why the formulas are these

```
λ = c / f
f = c / λ
T = 1 / f
```

A wave travels at c and repeats every period T, so one cycle covers c × T, which is the wavelength. In free space only: in a cable or on a PCB the wave is slower, by the velocity factor, and so its wavelength is shorter.

**Bands by decade.** ITU's bands are decades of frequency, and since λ = c/f each is a decade of wavelength too: band N spans 10^(8−N) to 10^(9−N) m, which is where the metric names come from — VHF, 30 to 300 MHz, is 1 to 10 m, the metric waves. The decades are nominal: c is not quite 3 × $10^8$ m/s, so 30 MHz is 9.993 m rather than 10.

**Edges.** Each band includes its upper limit and excludes its lower one, so exactly 30 MHz is HF, band 7, and anything above it is VHF.

## Assumptions and limits

- **Free space.** Air is close enough to it for this purpose; coax, twisted pair and PCB traces are not, and need their velocity factor.
- **ITU bands 3 to 12 only.** Below 300 Hz and above 3 THz the tool shows no band. V.431 also lists URSI's extension to lower bands and bands 13 to 15 up to 3000 THz, which reach the infrared; light has its own tool, Wavelength ↔ color.
- **The reference table is the US's.** Other regions allocate these bands differently.

## What it deliberately does not do

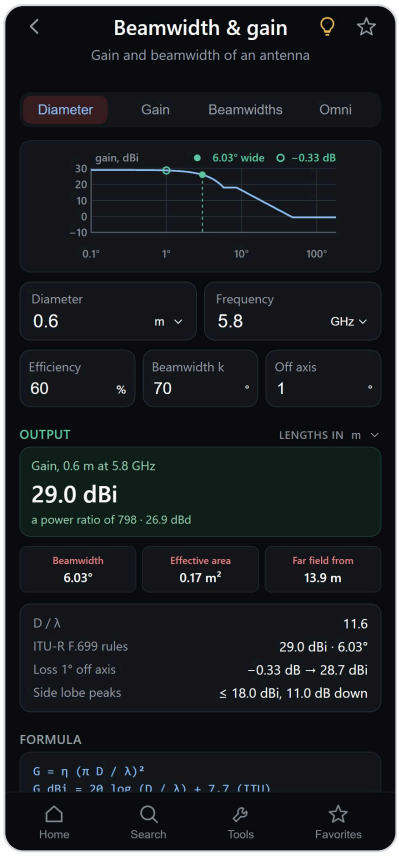
- **Antenna lengths.** A quarter-wave or half-wave antenna is a fraction of this wavelength, shortened further by end effects: that is the Antenna length tool's job.
- **Velocity factor.** Its own future tool.
- **Band plans.** Amateur, cellular and microwave allocations are reference tools of their own.

## How it was checked

100 MHz gives 2.997925 m: VHF, ITU band 8, metric waves 1–10 m, period 10 ns. Exactly 30 MHz gives HF and 9.993082 m, and 30.001 MHz gives VHF. A 1 m wavelength gives 299.7925 MHz. GPS L1 keeps its decimals: 1575.42 MHz, 19.02937 cm. The EHF button lands on the band's geometric middle, 94.86833 GHz. 1 kHz is ULF, hectokilometric waves, 100–1000 km; 500 GHz is band 12; 100 Hz is outside the bands, its wavelength shown as 2997.925 km.

The review made these changes: - It was renamed Radio wavelength, on Pierre's choice: with the examples bulb, "Wavelength ↔ frequency" wrapped. - Band edges now follow ITU, the upper limit in the band: the tool had put exactly 30 MHz in VHF. - ULF (band 3) and band 12 were added, with each band's number and ITU wavelength name. - The AM row read 525 kHz, which is not the US band's edge; it is now the FCC's 535–1705 kHz, and the rows say US. - The band and the period became one result; the wave on the drawing went from the domain colour to part blue and carries the wavelength's value. - Frequency and wavelength are kept in hertz and metres with remembered units, shown to seven figures so GPS's .42 survives. - It added six examples and the first cases (11).

# Beamwidth & gain




## What it computes

How strongly an antenna beams its power and how wide the beam is, from whichever of the antenna's figures you have (pills):

- **Diameter:** a dish's diameter and the frequency, with its aperture efficiency, a beamwidth constant k and an aiming error, give the gain. The big result is the gain in dBi; under it the same as a power ratio and in dBd. In three cells: the half-power beamwidth, the effective area and the distance from which the far field starts. Under them: D/λ; the ITU-R F.699 rules of thumb for the same dish (its gain and its beamwidth); the loss from the aiming error and the gain left; and the side lobe peaks (the level they stay under, and how far below the main beam that is). A dish under one wavelength across is flagged. The drawing is the antenna's gain against the angle off its axis, the angle on a logarithmic scale: the main lobe, then the reference pattern of ITU-R F.699 for the side lobe peaks. The −3 dB point is a green dot with a dotted line down to its angle, the aiming error a ring; both are named, with their values, in the row above the plot.
- **Gain:** the gain as a data sheet gives it (dBi or dBd) and the frequency. The big result is the beamwidth; in three cells the equivalent dish's diameter, its effective area and the far-field distance; under them the same rows. It is the circular dish that would give that gain at the efficiency you assume.
- **Beamwidths:** a sector antenna's horizontal (azimuth) and vertical (elevation) half-power beamwidths; the frequency is optional. The big result is its gain by the sectoral rule of ITU-R F.1336, 31 000 / (θH × θV). In three cells: the directivity by F.1336's own formula (the gain with no losses in the antenna), the gain in dBd and the equivalent D/λ (the dish that would give that gain, by F.699's rule). Under them: the equivalent dish (needs the frequency), how much of the sphere the beam covers, the loss from the aiming error in each plane, and a flag when the horizontal beamwidth is over about 120° or the vertical one over about 45°, where the rule is outside its range or rougher. The drawing is the two beams as sectors, each as wide as its beamwidth: the horizontal one seen from above, the vertical one from the side.
- **Omni:** an omnidirectional antenna, one that radiates alike in every direction round the horizon and squeezes its beam in elevation, like a doughnut: a whip, or a collinear array of stacked dipoles. Its vertical beamwidth and its gain are linked: type either and the other is calculated (the box that is calculated carries the word "result"). The efficiency is the antenna's radiation efficiency (100 % is the ITU's assumption), the side-lobe k is the parameter of the ITU's pattern (0.7 for typical antennas below 3 GHz, 0 for ones with improved side lobes and for 3 to 70 GHz), and the elevation is the angle off the horizon. The big result is the gain (when the beamwidth is typed) or the vertical beamwidth (when the gain is). In three cells: the directivity, the gain in dBd and how much of the sphere the beam covers. Under them: the simplified rule of F.1336 for the same antenna (its gain, 10 log (107.6 / θ), or its beamwidth, 107.6 × 10^(−0.1 G)), the loss at the elevation angle and the gain left, how far down the

far side lobes stay (12 − 10 log k dB; "falling away" for k = 0), and a flag when the gain is outside the 8 to 13 dBi the ITU states its pattern for. The drawing is the gain against the angle off the horizon, to 90°, the angle on a logarithmic scale: the main lobe, then the ITU's side lobe peaks, marked as in the Diameter pill.
- **Units:** frequency kHz, MHz or GHz; the diameter in m, cm, ft or in; the gain in dBi or dBd (in the Gain and Omni pills). The "lengths in" picker at the Output heading is the diameter's own: the diameter box, the diameter result, the effective area (in that unit squared) and the far-field distance (metres and kilometres, or feet and miles) follow it, and choosing a unit there changes the box.
- **Remembered:** the pill and the units.

Examples: a 2 ft dish at 13 GHz (CommScope's data sheet), a 2° miss on a 1.2 m dish, how big a 30 dBi antenna is, a 90° by 30° sector panel, and the ITU's measured 5.8 GHz omnidirectional antenna.

## Terms

- **Gain (dBi, dBd):** how much more power an antenna sends, or picks up, in its best direction than an isotropic antenna, which treats every direction alike. In dB: dBi against the isotropic antenna, dBd against a half-wave dipole (dBd = dBi − 2.15, the dipole's own gain).
- **Directivity:** the same figure for an antenna with no losses; the gain is the directivity times the radiation efficiency, the share of the power given to the antenna that it radiates.
- **Half-power beamwidth (beamwidth, −3 dB beamwidth):** the angle between the two directions, one each side of the best, where the power has fallen to half, 3 dB, of the peak.
- **Aperture, effective area:** the area through which an antenna collects a wave. A dish's physical area is $\pi D^2/4$; its effective area, $A_e = G\,\lambda^2/4\pi$, is the area that really counts.
- **Aperture efficiency (η):** the effective area over the physical area: what uneven illumination, spill-over past the edge, the feed's shadow and surface errors take away. 55 to 70 % is usual.
- **Beamwidth constant (k):** the number in $\theta = k\,\lambda/D$, in degrees: about 70 for a typical dish, 58.96 for an evenly lit aperture.
- **Sectoral antenna:** one that covers a sector of the horizon, up to about 120° wide in the ITU's rules, with a narrow beam in elevation: a base-station panel.
- **Omnidirectional antenna:** one with the same gain in every direction of azimuth; only its elevation pattern is shaped. Its vertical (elevation) beamwidth, θ3 in ITU-R F.1336, is the angle between the directions above and below the horizon where the power is half the peak.
- **Side-lobe k (F.1336):** how high the side lobes of an omnidirectional antenna stand above what an improved design would give: 0.7 for typical antennas below 3 GHz, 0 for improved ones and for 3 to 70 GHz.
- **Main lobe, side lobes:** the beam, and the weaker lobes beside it. The side lobe peaks are what interference studies use.
- **Far field:** the region far enough out that the pattern no longer changes with the distance, beyond about $2D^2/\lambda$; nearer, the formulas for a distant receiver do not hold.
- **Off axis:** the angle between the direction the antenna points and the one in question; the aiming error (the angle off the horizon, for an omni).
- **Beam solid angle (Ω):** the beam's size as an angle in space, in steradians; $\Omega = 4\pi/D$, a share 1/D of the whole sphere (what the tool calls "sphere covered").

## Sources

- **Recommendation ITU-R F.699-9, "Reference radiation patterns for fixed wireless system antennas … 100 MHz to 174.8 GHz" (02/2026; read as PDF text, and the pages with the equations rendered and read as printed):** recommends 3 gives the maximum gain from the diameter, G_max ≈ 20 log (D/λ) + 7.7 dBi (about 48 dBi for D/λ = 100) and the reverse; recommends 4 gives D/λ ≈ 70/θ and G_max ≈ 44.5 − 20 log θ for the −3 dB beamwidth θ in degrees. Recommends 2 is the reference pattern of the side lobe peaks: the main lobe G_max − $2.5 \times 10^{-3}$ $(D\ \varphi/\lambda)^2$ up to φ_m = (20 λ/D) √(G_max − $G_1$), with $G_1$ = 2 + 15 log (D/λ); then $G_1$ up to φ_r = 15.85 (D/λ)^−0.6 (D/λ over 100) or to 100 λ/D (100 or less); then 32 − 25 log φ or 52 − 10 log (D/λ) − 25 log φ; then a floor, −10 dBi from 48° (1 to 70 GHz; 10 − 10 log (D/λ) for D/λ up to 100), or −20 dBi and −10 log (D/λ) from 120° (70 to 174.8 GHz), and for 100 MHz to 1 GHz (D/λ over 0.63) −2 − 5 log (D/λ) from φ_s = 144.5 (D/λ)^−0.2. Its notes say the pattern is for rotationally symmetric antennas and can be used for square or polygonal reflectors and flat panels through an equivalent D/λ found from the maximum gain (Note 5), that a real antenna may be worse than it over some angles and that it should not be read as a limit (Notes 2, 3), and that it models some practical antennas poorly from 5° to 70° (Note 9); Annex 1 says measurements confirm it to D/λ of about 180.
- **Recommendation ITU-R F.1336-5, "Reference radiation patterns of omnidirectional, sectoral and other antennas … 400 MHz to about 70 GHz" (01/2019; read as PDF text, and pages 4, 9, 12 and 29 rendered and read as printed):** recommends 2.1 gives the elevation pattern of the side lobe peaks of an omnidirectional antenna: $G_0$ − 12 $(\theta/\theta 3)^2$ up to θ4 = θ3 √(1 − log (k + 1) / 1.2), $G_0$ − 12 + 10 log (k + 1) up to θ3, then $G_0$ − 12 + 10 log ((θ/θ3)^−1.5
- k) to 90°, with θ3 = 107.6 × 10^(−0.1 $G_0$) ($G_0$ the gain in dBi, θ3 the beamwidth in degrees); 2.3 and 2.4 set k to 0.7 for typical antennas from 400 MHz to 3 GHz and to 0 for antennas with improved side lobes and all those from 3 to 70 GHz. Annex 1 builds the pattern on a collinear array of dipoles spaced 3λ/4, states it for 8 to 13 dBi (further work being needed above), gives the directivity from the beamwidth, D = 10 log [191 √(0.818 + 1/θ3) − 172.4] dBi (5a), says k = 0.7 stands for side lobes about 13.5 dB below the peak and k = 0.5 for about 15 dB, and compares a measured 9.4 dBi antenna at 5.8 GHz, θ3 = 8.7°, with it. Annex 2 derives the directivity from an exponential elevation pattern: D = 107.64 e^($\theta 3^2$/36400) / θ3 for an omnidirectional antenna (23a, θ3 in degrees; Table 2 compares it with the directivity of a cos^2N pattern, to within 0.062 dB), the same with the exponential set to 1 (23b, under 6 % error below 45°), and for a sector D = k e^($\theta 3^2$/36400) / (φs θ3) with k = 36 400 up to φs = 120° and 38 750 above (34, 35; the transition said to need study). Its measured sectoral antennas (25 to 29.5 GHz: 90° by 2.5°, directivity 22.1 dBi, gain 20.5 to 21.4 dBi) lead to the provisional gain $G_0$ = 10 log (31 000 / (φs θ3)) of recommends 3.3 (28a), for a sector of about 120° or less and an elevation beamwidth under about 45°.
- **Condon and Ransom, "Essential Radio Astronomy", chapter 3 (NRAO; the first 100 000 of its 174 658 characters, read through a text-extraction tool):** the beam solid angle Ω_A = 4π/G (eq. 3.34), the average effective area $\lambda^2/4\pi$ (3.41) and A_e = $\lambda^2$ G/4π (3.46), a half-power beamwidth of 0.89 λ/D for a uniform one-dimensional aperture (3.82) and about 1.2 λ/D for a cosine taper, "typical of most radio telescopes" (3.96), the aperture efficiency as the effective over the geometric area (3.111) and its being 70 % or less for most radio telescopes, and the far-field distance R ≈ $2D^2/\lambda$, defined by a path error of less than λ/16 at the edge (3.64). The coefficient for a uniform circular aperture is not on that page.
- **Wikipedia, "Parabolic antenna" (read through a text-extraction tool):** G = e_A $(\pi\ d/\lambda)^2$, an aperture efficiency of 0.55 to 0.70 for typical dishes (0.7 to 0.8 for offset feeds), and θ = k λ/d with k

about 70 for a typical dish; it gives 57.3 for an ideal uniform one, which the exact value below, 58.96, corrects. **"Near and far field":** $d_F = 2D^2/\lambda$, with $d_F$ much greater than D and than λ.

- **Köppen, "Notes and Tools about Paraboloidal Antennas" (Kiel; read through a text-extraction tool):** the half-power beamwidth of a uniformly illuminated circular dish as 58.95701° × λ/D. It cites Hansen (1976) and Baars (2007) for how the constant grows with the edge taper; neither was read.
- **CommScope, "VHLP2-13-2WH/A" product specifications (revised 23 October 2015; read from a distributor's copy of the PDF):** a 0.6 m (2 ft) dish, 12.700 to 13.250 GHz; gain 35.8, 36.0 and 36.2 dBi at the low, middle and top of the band; beamwidth 2.7° both ways; front-to-back ratio 61 dB.
- **Ubiquiti, RD-5G30-LW tech specs (the maker's page):** 5.1 to 5.9 GHz, 30 dBi, 5.8° (3 dB) in both polarisations, 650 × 650 × 386 mm.
- **Not read:** Kraus, Balanis and IEEE Std 145 (the definitions of gain and of the far field); ITU-R F.1245 and the satellite patterns S.580 and S.465; F.1336's Annexes 3 to 5 (the pattern at azimuth and elevation together, and downtilt) and its recommends 4 (low-gain antennas); the CommScope 80 GHz sheets (the maker's server refused them). The 41253 mentioned below is derived, not quoted, and dBd = dBi − 2.15 is the textbook figure, as in Path loss.

## Why the formulas are these

**Gain from the size.** An isotropic antenna receives from a wave over an area $\lambda^2/4\pi$; the gain of an antenna with effective area $A_e$ is the ratio, $G = 4\pi A_e/\lambda^2$ (NRAO 3.41, 3.46). A circular dish of diameter D and efficiency η has $A_e = \eta \pi D^2/4$, so

```
G = η (π D / λ)²
```

and in dB, $G = 20 \log (D/\lambda) + 20 \log \pi + 10 \log \eta = 20 \log (D/\lambda) + 9.94 + 10 \log \eta$. The ITU's 7.7 is that with $10 \log \eta = -2.24$ dB, an efficiency of 59.7 %: its rule $G = 20 \log (D/\lambda) + 7.7$ is within 0.025 dB of η = 60 % for any D/λ from 1 to 1000.

**The beamwidth.** The far field of a circular aperture is the transform of its illumination. For an evenly lit one it is $2 J_1(u)/u$ with $u = \pi D \sin \varphi/\lambda$; the −3 dB point is at u = 1.6163, a full width of 1.0290 λ/D radians, 58.96° × λ/D (Köppen's 58.95701°); the first null is at $\sin \varphi = 1.2197\ \lambda/D$ and the first side lobe is 17.57 dB down. A taper that lights the middle more than the edge lowers the side lobes and widens the beam, to about 70° × λ/D for a typical dish (the ITU's and Wikipedia's figure; NRAO's 1.2 λ/D is 68.8°). The tool's default k is 70.

**The far field.** From a point at distance R the path to the edge of an aperture of diameter D is longer than to its centre by about $D^2/(8R)$. Asking for no more than λ/16 gives $R \geq 2D^2/\lambda$ (NRAO 3.64).

**Aiming.** Near the peak the pattern is a parabola in dB, $G(\varphi) = G_{max} - 12 (\varphi/\theta)^2$, which is 3 dB down at $\varphi = \theta/2$. The ITU's main lobe is the same parabola for $\theta = 69.28\ \lambda/D$, since $12 / 69.28^2 = 2.5 \times 10^{-3}$: at 1° for D/λ = 11.6 both give 0.33686 dB. The ITU's omnidirectional main lobe is the same parabola again, with θ3 the full beamwidth. The tool writes it with the beamwidth you use, so k changes it, and continues with the ITU's pattern beyond $\varphi_m$.

**Omnidirectional antennas.** F.1336 takes the elevation pattern as $e^{(-a\theta^2)}$, half the peak at θ3/2, which fixes the constant, $a = 4 \ln 2 / \theta 3^2 = 2.7726 / \theta 3^2$ (θ3 in radians). The directivity is 4π over the integral of the pattern over the sphere; round the horizon the integral is 2π, and in elevation it is the integral of $e^{(-a\theta^2)} \cos \theta$, which the ITU evaluates as the infinite integral $\sqrt{(\pi/a)}\ e^{(-1/(4a))}$. That gives D = 107.64

e^(θ3²/36400) / θ3 with θ3 in degrees. The tool's numerical integration of the same pattern agrees with it to 0.0005 dB up to a 60° beam, 0.0055 dB at 78° and 0.021 dB at 90°. The exponential is 1 for narrow beams and 1.06 at 45°; leaving it out gives the simple 107.6 / θ3, and θ3 = 107.6 × 10^(−0.1 G). A half-wave dipole's 78° gives 2.12 dBi, against its own 2.15. The gain is the directivity times the radiation efficiency, G = η D; the ITU takes η = 100 %, but its measured 9.4 dBi antenna, 8.7° wide, has a directivity of 10.93 dBi, so its efficiency is 70 %: the tool's efficiency box is for such an antenna. To go from the gain to the beamwidth the tool solves the equation for θ3 (D falls steadily with θ3 up to 90°, so there is one answer; a directivity under 1.74 dBi, a 90° beam, has none).

**Sector antennas.** The same integral with a flat azimuth pattern of width φs gives D = 38 750 e^(θ3²/36400) / (φs θ3) (the 107.64 above is 38 750/360, the omnidirectional case), and an exponential azimuth pattern gives 36 400, within 0.3 dB of it; F.1336 takes 36 400 up to φs = 120° and 38 750 above. Numerical integration of the flat case agrees with its formula to 0.001 dB. The practical rule, 31 000 / (φs θ3), comes from measurements: with the gain 0.7 dB under the directivity and the exponential dropped, the ITU fits 31 000 to its first set of antennas and 34 000 to the second. For a 90° by 30° sector that is 10.6 dBi against a directivity of 11.4 dBi. The familiar 41253 / (θH θV), the sphere's 41 253 square degrees ($4\pi (180/\pi)^2$) over a rectangle, is the same idea without the shape of the beam or any losses: 11.8 dBi for that sector.

## Assumptions and limits

- **A dish-like aperture.** The gain, beamwidth and pattern rules of the first two pills are for a circular aperture lit smoothly from the middle; flat panels and horns follow only through an equivalent dish (F.699 Note 5).
- **The efficiency is yours.** 55 to 70 % is typical for a dish, 60 % matches the ITU's rules, and it falls where the dish surface is rough compared with the wavelength (the Ruze loss, which the tool does not model). The data sheets below imply 56 to 60 %.
- **The sector rule is provisional.** F.1336 gives 31 000 / (φs θ3) for sectors of about 120° or less and elevation beamwidths under about 45°, on two sets of measurements at 25 to 29.5 GHz, and says the transition at 120° needs study; the tool flags beamwidths beyond those limits.
- **The omnidirectional rules** are for a collinear array of dipoles and an elevation beam up to 90° (a directivity from 1.74 dBi); the ITU's pattern is stated for 8 to 13 dBi, from 400 MHz to 70 GHz, and further work is said to be needed above that, particularly above 3 GHz; the tool says so outside 8 to 13 dBi. The efficiency of a real antenna is below the ITU's 100 %: its beam is narrower for its gain.
- **The pattern is the ITU's reference.** It is a line through the side lobe peaks, for interference studies, not a measured pattern; real antennas can be worse over some angles (F.699 Notes 2, 3 and 9). The dish pattern is defined from 100 MHz to 174.8 GHz; its printed ranges follow one another only above D/λ of 100/48 (1 to 70 GHz), 100/120 (70 GHz and up) and 0.63 (under 1 GHz, the limit F.699 states), so the tool draws the main lobe only below those, and outside the frequencies. Its pieces meet to 0.031 dB (the printed constants are rounded).
- **The aiming loss** is the parabola inside the main lobe and the pattern beyond; for the Beamwidths pill, per plane, the parabola only, to an angle of the beamwidth.
- **The far field** is for the largest dimension of the antenna, and holds when the distance is also much larger than D and λ.
- **Under one wavelength across** the aperture formulas do not apply (the tool's own flag), and a beam wider than 180° is not a beam.

## How it was checked

69 cases. The expected values come from a Python implementation of the same formulas: - Diameter: 0.6 m at 5.8 GHz, 29.0 dBi (a ratio of 798, 26.9 dBd), 6.03°, 0.17 $m^2$, far field from 13.9 m, the ITU rules 29.0 dBi and 6.03°, 0.33 dB lost 1° off axis, side lobes up to 18.0 dBi, 11.0 dB down; the CommScope dish at 12.7, 12.975 and 13.25 GHz (35.8, 36.0 and 36.2 dBi; 2.75°, 2.7°, 2.64°); a 1.2 m dish 2° off axis (35.0 dBi, 3.02°, 5.28 dB); 11.7 GHz (35.1 dBi, 2.99°); 70 % efficiency; k = 58.96; the diameter in cm and in ft; a 3 m dish at 11 GHz (D/λ over 100, 16.56 dB lost 1° off axis); 80 GHz; 500 MHz; far off axis; frequencies and sizes outside the pattern; a flag under one wavelength; the missing figures refused; - Gain: 30 dBi at 5.8 GHz (0.672 m, 5.39°, 0.213 $m^2$, the ITU's 0.674 m and 5.31°), 36 dBi at 12.975 GHz, the gain in dBd, 70 % efficiency, feet, a low gain flagged, a missing gain refused; - Beamwidths: 90° by 30° (10.6 dBi, 8.4 dBd, directivity 11.4 dBi, D/λ 1.4, 7.24 % of the sphere), 60° by 60°, 6° by 6°, 120° by 90° and 150° by 30° flagged, no frequency, 50° off axis, feet, missing and over-large beamwidths refused; - Omni: a 14° beam (8.9 dBi, 6.7 dBd, 12.9 % of the sphere, the ITU rule 8.9 dBi, 1.53 dB lost 5° off the horizon, side lobes 13.5 dB down); a 9 dBi gain (a 13.6° beam, the other box following, and the "result" tags); the ITU's measured antenna (9.4 dBi at 70 % efficiency and k 0: 8.67°); the gain in dBd; k = 0; 30° and 0° off the horizon; beams of 3°, 40° and 70 % efficiency, which fall outside 8 to 13 dBi; the 90° and 78° beams; the missing, over-large and too-low figures refused; - the length unit linked between the diameter box and the Output heading, the pills and the units remembered, the five examples, and the formula-search card.

Against data sheets, with η 60 % and k 70:

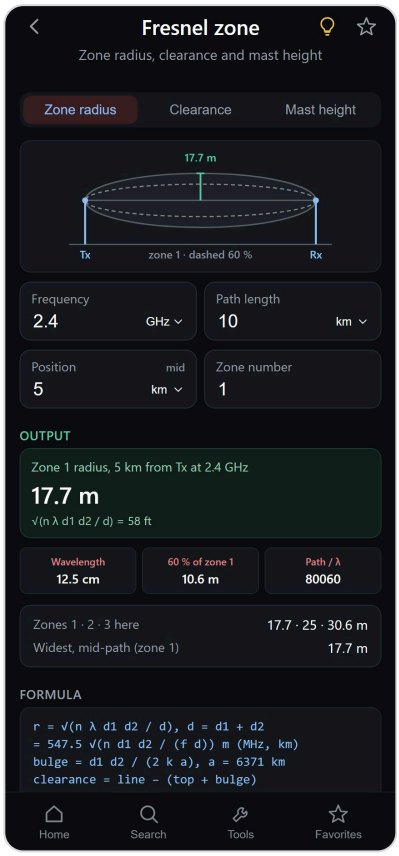
| Antenna | Data sheet | This tool |
|---|---|---|
| CommScope VHLP2-13-2WH/A, 0.6 m, 12.7 GHz | 35.8 dBi | 35.8 dBi |
| the same, 12.975 GHz | 36.0 dBi, 2.7° | 36.0 dBi, 2.7° |
| the same, 13.25 GHz | 36.2 dBi | 36.2 dBi |
| Ubiquiti RD-5G30-LW, 650 mm square, 5.5 GHz | 30 dBi, 5.8° | for 30 dBi: a 0.708 m dish, 5.4° |

The CommScope sheet implies an efficiency of 59.6, 59.8 and 60.1 % at its three frequencies, and the ITU's rule gives 35.99 dBi at mid-band. The Ubiquiti dish is square: its sheet's 30 dBi at 650 mm implies 56 % efficiency, the equal-area circle is 0.733 m across (the tool's equivalent dish 0.708 m), and 70 λ over the 650 mm side is 5.87°, the sheet's 5.8°.

Against the ITU's own numbers (F.1336-5): - Fifteen of the rows of its Table 2 (2N from 2 to 74) are reproduced: the beamwidths 2 arccos (0.5^(1/2N)) and the directivities (2N + 1)!! / (2N)!! of the cos^2N pattern to the printed digits, and eq. 23a to 0.0001 dB; - its sector example, 90° by 2.5°, gives a directivity of 22.09 dBi by eq. 34 (the text says 22.1 dBi) and 21.39 dBi by the gain rule (the measured gain is 20.5 to 21.4 dBi); - its measured omnidirectional antenna (9.4 dBi, 8.7°) has a directivity of 10.93 dBi by eq. 23a, an efficiency of 70.2 %; a 9.4 dBi gain at 70 % gives 8.67°; the lossless rule would say 12.4°; - the pattern's three regions meet to $10^{-8}$ dB at θ4 and θ3 for k of 0, 0.3, 0.7 and 1, and solving eq. 23a for the beamwidth and then computing the directivity again returns the beamwidth to rounding; - for a half-wave dipole, 78°, eq. 23a gives 2.12 dBi (2.15 dBi); 14° gives 8.88 dBi, and a 9 dBi omnidirectional antenna 13.6° by eq. 23a and 13.5° by the simplified rule.

The Python implementation also showed that the dish pattern's pieces meet at their boundaries to 0.031 dB at most (the main lobe meets $G_1$ exactly), that $44.5 - 20 \log \theta$ is within 0.10 dB of $20 \log (70/\theta) + 7.7$, and that the $2.5 \times 10^{-3}$ of the ITU's main lobe is $12 (\varphi/\theta)^2$ with $\theta = 69.28 \lambda/D$.

# Fresnel zone




## What it computes

How much room a radio path needs round the straight line between two antennas, and what an obstacle in that room costs, three ways (pills):

- **Zone radius:** the frequency, the path length, a point on the path (its distance from the transmitting antenna) and the zone number give the radius of that Fresnel zone at that point. The point starts at the middle of the path and follows it when the path changes, until you type one; the "mid" beside the box says so, and becomes a button that takes you back. In three cells: the wavelength, 60 % of the first zone's radius there (the clearance to keep) and the path length in wavelengths. Under them: the radii of zones 1, 2 and 3 there, and the widest radius of the zone you chose, at mid-path. The drawing is the path seen from the side: the zone as an ellipse, the 60 % one dashed (zone 1 dashed instead when you chose a higher zone), the radius marked in green.
- **Clearance:** the two antennas' heights, the top of an obstacle and its distance from the transmitting antenna, the Earth radius factor k and the share of the first zone you want kept clear. The big result is the loss in dB the obstacle causes, taken as a knife-edge; the line under it says how much of the first zone is clear and how that compares with what you want, or "blocked" when the top is above the line of sight. In three cells: the clearance, its share of the first zone's radius and that radius there. Under them: the height of the line of sight at the obstacle, the obstacle's height with the Earth's bulge added, and the clearance from which the loss is nil (55.2 % of the radius, with its length). The drawing is the same path with a hill whose top is where the clearance puts it, the clearance in green.
- **Mast height:** the same problem solved the other way: for the transmitting antenna you give, the height the receiving antenna needs so that the line clears the obstacle by the share you want. The big result is that height. In three cells: the height if both ends are alike, the clearance wanted in length, and the Earth's bulge. Under them: the line's height at the obstacle as a sum (the obstacle, the bulge, the clearance) and the first zone's radius there; and a note when the receiving antenna comes out below the datum, which means the transmitting antenna alone clears the obstacle.
- **Units:** frequency kHz, MHz or GHz; distances m, km, mi, nmi or ft; heights m or ft. A distance unit and the height unit go together: miles or feet for the distances put the heights in feet, metres or kilometres put them in metres, and the other way round (nautical miles leave the heights as they are). Both distance boxes change together, and so do the three height boxes. The zone radius is given in the height unit, with the other unit in the line under it.
- **Remembered:** the pill and the units.

Examples: a 5.8 GHz link across 5 km, the masts needed over a 20 m hill, the Earth's bulge on a 50 km hop, and a hill just inside the zone.

## Terms

- **Fresnel ellipsoid, Fresnel zone:** the radio energy from one antenna to another does not follow a line but fills a region round it. Take the points M whose path through them, AM + MB, is longer than the straight one AB by n half-wavelengths: they form an ellipsoid, the n-th Fresnel ellipsoid, with the antennas at its foci. The zone of order n is the part between ellipsoids n − 1 and n; the first zone is inside the first ellipsoid.
- **Zone radius (r):** the radius of the n-th ellipsoid at a point: the distance from the line of sight to its edge there, widest at mid-path and nil at the antennas. "F1" or "r1" is the first one.
- **Clearance:** how far the line of sight passes above the top of an obstacle there; negative when the top is above it (blocked).
- **The 60 % rule:** keep the path clear of obstacles to 60 % of the first zone's radius, and the path behaves as if it were free space.
- **Datum:** the level the heights are measured from: sea level, or the level ground both antennas stand on.
- **Earth bulge:** how far the Earth's surface rises between two points on it, above the straight line joining them: d1 d2 / (2 a), the product of the distances to the two ends over twice the Earth's radius.
- **k, the effective Earth radius factor:** the atmosphere bends radio rays slightly toward the ground, so a ray can be treated as straight if the Earth is made larger by the factor k: the effective radius is k × 6371 km. 4/3 is the usual value; 1 means no bending.
- **Knife-edge, ν, J:** an idealised obstacle with a sharp top and no thickness. ν (nu) packs its geometry into one number, √2 × (how far its top is above the line of sight) / r1; J(ν) is the loss in dB the obstacle causes, by diffraction.

## Sources

- **Recommendation ITU-R P.526-15, "Propagation by diffraction" (10/2019; read as PDF text; the extraction dropped the Greek letters and the root signs, so each equation was read with its dimensions checked, and the numbers below were checked against it):** § 2.1 defines the ellipsoids by AM + MB − AB = nλ/2 (eq. 1) and gives the radius of the n-th at a point d1 and d2 from the ends as an approximation, $R_n = \sqrt{(n \lambda d1\, d2/(d1 + d2))}$ (eq. 2), and in metres for f in MHz and d in km as $550 \sqrt{(n\, d1\, d2/(f\,(d1 + d2)))}$ (eq. 3); it calls the part between ellipsoids n − 1 and n the zone of order n, and takes a path as line-of-sight, with negligible diffraction, when nothing lies inside the first ellipsoid. § 2.3 starts the diffraction zone where the clearance falls to 60 % of the first zone's radius, and § 2.5 lists among the conditions for an isolated obstacle a clearance of at least 0.6 of it on both sides. § 4.1 gives the knife-edge parameter, $\nu = h \sqrt{(2\,(d1 + d2)/(\lambda\, d1\, d2))}$ (eq. 26, h the top's height above the straight line, negative below it; eq. 33 is the same in practical units, with 0.0316 for h and λ in metres and d in km), the loss J(ν) as Fresnel integrals (eq. 30) and, for ν above −0.78, the approximation $J = 6.9 + 20 \log (\sqrt{((\nu - 0.1)^2 + 1)} + \nu - 0.1)$ dB (eq. 31), which the tool uses. It assumes a diffraction angle under about 0.2 rad (12°) and says the data are mainly for VHF and shorter waves (above 30 MHz). § 3.2 gives the clearance for zero diffraction loss as $h_{req} = 0.552 \sqrt{(\lambda\, d1\, d2/d)}$ (eq. 23). Attachment 2 gives the ray's height at a point of the path, $(h_u\, d_{iv} + h_v\, d_{ui})/d_{uv}$ (eq. 123), the terrain's, $h_i + d_{ui}\, d_{iv}/(2\, a_e)$ (eq. 124), the clearance $h_z$ as their difference (eq. 121), $F1 = \sqrt{(\lambda\, d_{ui}\, d_{iv}/d_{uv})}$ (eq. 122), the normalised clearance $C_F = h_z/F1$ (eq. 120) and $\nu = -\sqrt{2}\, C_F$ (eq. 125). The text before eq. 44 gives $a_e = k \times 6371$ km, and § 1 suggests an equivalent Earth radius of 8 500 km when nothing better is known (k = 1.334, 0.06 % above 4/3).

- **Recommendation ITU-R P.526-14 (01/2018; read as PDF text):** the same eq. 2 and 3 and the same 60 % statements.
- **Not read:** later versions of P.526; ITU-R P.530 (the design of terrestrial line-of-sight links, where clearance criteria with several values of k are given), P.310 (the definition of k) and P.834 (refraction); any textbook derivation of the Fresnel zones or of the knife-edge loss. The 5 wavelengths at which the tool warns is its own rule of thumb (see Assumptions and limits).

## Why the formulas are these

**The zone radius.** A point M at a distance r off the line, d1 from one antenna and d2 from the other along it (r much smaller than d1 and d2), is $\sqrt{(d1^2 + r^2)} \approx d1 + r^2/(2\,d1)$ from the first antenna and $d2 + r^2/(2\,d2)$ from the second. The path through it is longer than the straight one, d1 + d2, by $r^2/2 \times (1/d1 + 1/d2) = r^2\,d/(2\,d1\,d2)$. Setting that to n λ/2 gives

```
r = √(n λ d1 d2 / d)      d = d1 + d2
```

(eq. 2). It is widest at mid-path, $r = \sqrt{(n\,\lambda\,d)/2}$, nil at the antennas, symmetric about the middle, and the n-th is √n times the first. With f in MHz λ = 299.792 458/f, and with d1 and d2 in km the factor 1000 turns km into m: $r = \sqrt{(299.792\,458 \times 1000)}\,\sqrt{(n\,d1\,d2/(f\,d))} = 547.53\,\sqrt{(n\,d1\,d2\,/(f\,d))}$. P.526's eq. 3 rounds the 547.53 to 550, which reads 0.45 % high; the tool does not round it.

**The clearance and the loss.** The ray passes at a height (h1 d2 + h2 d1)/d over the datum at a point d1 from the first antenna, a linear mix of the two antennas' heights (eq. 123). The Earth's surface, seen from a straight line between two points on it, lies d1 d2/(2 a_e) above that line (the chord theorem for a circle of radius a_e, a small sagitta, eq. 124), so the top of an obstacle at height h_o sits at h_o + d1 d2/(2 k a) over the straight line; the clearance is the difference. In the knife-edge formula $\nu = h\,\sqrt{(2\,(d1 + d2)/(\lambda\,d1\,d2))} = \sqrt{2}\,h / r1$, where h is the top's height above the line: h = −clearance gives $\nu = -\sqrt{2} \times \text{clearance}/r1$ (eq. 125). J(ν) comes from the Fresnel integrals (eq. 30); eq. 31 is a simple fit to it.

Eq. 31 is exactly 0 dB at ν = −0.7806 ($\sqrt{(u^2 + 1)} + u = 10^{(-6.9/20)}$ with u = ν − 0.1 gives u = −0.8806), and −0.7806/(−√2) = 0.552: the clearance of 0.552 r1 in P.526's eq. 23 is the point where the knife-edge loss starts. The tool takes no loss at or above it. The 60 % rule has some margin over that: at 60 % the exact knife-edge field is 0.37 dB above free space.

**The mast.** The line's height at the obstacle must be h_o + bulge + (share wanted) × r1 = T. With the transmitting antenna at h1 and the receiving one at h2 it is (h1 d2 + h2 d1)/d, which equals T for

```
h2 = h1 + (T − h1) × d / d1
```

and with the same height at both ends the line is level and the height is T.

## Assumptions and limits

- **One thin obstacle.** The tool looks at the one obstacle you describe and treats its top as a knife-edge. A real hill is rounded, which loses more (P.526 § 4.2 adds a term for it), and a path with several obstacles, or trees and buildings, needs the methods of its later sections (Bullington, Epstein-Peterson,

Deygout, ...), none of which the tool has. The figures are an estimate to within a few dB for a single sharp ridge, and a minimum for anything else.

- **No ground reflection and no fading.** A path clear to 60 % of the first zone can still fade from a reflection off the ground or water; the 60 % rule is about diffraction by obstacles only.
- **The approximation of eq. 2.** It holds when the point is many wavelengths from either antenna. The exact ellipsoid of eq. 1 is a little wider: +0.0002 % at mid-path of 10 km at 2.4 GHz, +0.25 % at 50 wavelengths from an antenna, +2.5 % at 5 wavelengths (+7.2 % for zone 3), +12 % at one wavelength. The tool flags a point under 5 wavelengths from an antenna; that limit is its own, and the Fresnel picture itself, which assumes the paraxial limit, stops meaning much there.
- **J below its zero.** Under 55.2 % of clearance the tool says 0 dB. The exact knife-edge curve there ripples round 0 dB: −1.4 dB (a gain) at ν = −1.22 (a clearance of 86 % of r1) and +1.1 dB at ν = −1.87 (132 %). A real path has a ground and other things that move such ripples; the tool does not model them.
- **Eq. 31 against the integrals.** The approximation is within 0.12 dB of eq. 30 over ν from −0.78 to 20 (the worst at ν = 4.6).
- **k is yours.** The real value changes with the weather, and a link is usually checked at a smaller k as well; the tool takes the one you give. P.530, which has the criteria for that, was not read.
- **Heights share a datum.** If the antennas are on ground of different levels (or the obstacle is), add each one's elevation to its height first; the Earth bulge is measured from the straight line between the antennas' bases.
- **Small angles and VHF and up.** P.526's knife-edge formulas are for diffraction angles under about 12° and frequencies above 30 MHz; the tool does not stop you outside them.
- **The distances are along the ground.** The tool uses d1 and d2 as the distances to the ends and does not correct for a steep path.

## How it was checked

51 cases. The expected values come from a Python implementation of the same formulas (with the Fresnel integrals of scipy for the check of eq. 31): - Zone radius: 2.4 GHz over 10 km, mid-path, 17.7 m (58 ft; λ 12.5 cm, 80 060 wavelengths, 60 % 10.6 m; zones 1, 2 and 3: 17.7, 25 and 30.6 m); 2 km from the transmitter 14.1 m; zone 2 and 3 at mid-path 25 m and 30.6 m; 5.8 GHz over 5 km 8.04 m; 145 MHz over 30 km 125 m; a 5-mile path in feet and a 600 m one in metres; nautical miles; a point 0.5 m from an antenna flagged; a missing frequency or path, a position off the path and a zone number that is not whole refused; - The position: it follows the middle when the path changes (10 km on a 20 km path), stays where it was typed, and goes back with the "mid" button; - Clearance: 2.4 GHz over 10 km, 30 m antennas, a 20 m hill mid-path, k 4/3: bulge 1.47 m, clearance +8.53 m, 48.3 % of the zone, 0.66 dB; a 5 m hill (133 %, no loss); a 35 m hill (blocked by 6.47 m, 10.43 dB); antennas of 10 and 50 m with the hill 2 km from the lower one (the line 18 m there); k = 1 (bulge 1.96 m); a 50 km hop at 6 GHz (bulge 36.8 m, 9.34 dB); 100 % and 0 % wanted; the same path in miles and feet; the missing heights, k = 0, 200 % wanted refused; a flag near an antenna; - Mast height: 34.1 m at the receiving end (32.1 m at both) for the same hill; 54.1 m with the transmitting antenna at 10 m; 100 % wanted; a 100 m transmitting antenna, which leaves the receiver below the datum; the 50 km hop (104 m, 81.8 m); a hill 9 km along; k = 1; - the unit links (miles put heights in feet, feet put distances in miles, metres put them in kilometres, nautical miles leave the heights), the pill and the units remembered, the four examples, and the formula-search card.

The Python implementation also showed that: - eq. 31 is 0 dB at ν = −0.78062, a clearance of 0.55198 r1 (P.526 eq. 23: 0.552); - eq. 31 stays within 0.123 dB of eq. 30 for ν from −0.78 to 20 (the worst at ν = 4.57), for

example 6.03 against 6.02 dB at ν = 0 and 13.93 against 13.86 dB at ν = 1; - running the mast height and then the clearance at the height it gives returns the share wanted to $2 \times 10^{-13}$ over 2000 random paths (10 MHz to 100 GHz, 100 m to 316 km, k from 2/3 to 2), and the same with a level line at the same height at both ends; - the loss against the clearance, from eq. 31 (the exact value in brackets):

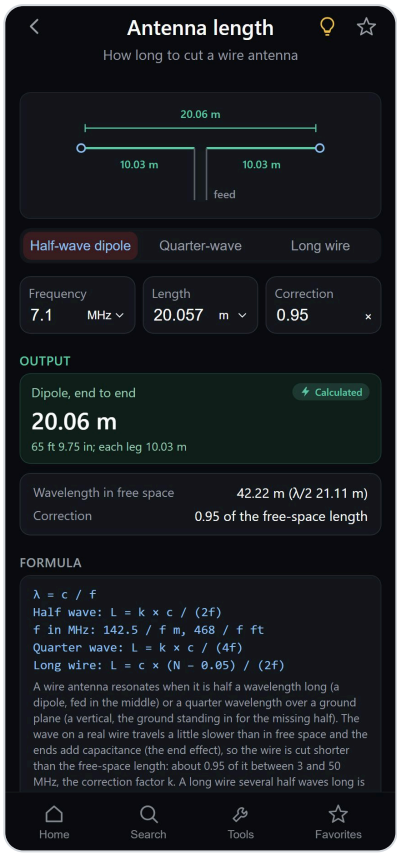
| **Clearance (share of r1)** | **Loss, dB** |
|---|---|
| 60 % | 0 (−0.37) |
| 50 % | 0.49 (0.42) |
| 40 % | 1.48 (1.37) |
| 20 % | 3.65 (3.60) |
| 0 (grazing) | 6.03 (6.02) |
| blocked by 25 % | 9.08 (9.04) |
| blocked by 100 % | 16.34 (16.33) |
| blocked by 200 % | 21.92 (22.02) |

- the Earth's bulge at mid-path, k = 4/3 (k = 1): 1.47 m (1.96 m) at 10 km, 5.89 m (7.85 m) at 20 km, 36.8 m (49.1 m) at 50 km.

RF & MICROWAVE › ANTENNAS & PROPAGATION

# Antenna length

(formerly Antenna length (dipole/monopole))




## What it computes

How long to cut a wire antenna for a frequency, or the frequency a wire of a given length is cut for:

- **Three kinds** (pills):
- a **half-wave dipole**, fed in the middle;
- a **quarter-wave** vertical over a ground plane;
- a **long wire** of N half waves, fed at one end.
- **Two live boxes:** the frequency and the length. Type either and the other follows.
- **The correction factor** k: 0.95 by default. Outside 3 to 50 MHz it is flagged, because that is the only range FM 24-18 gives it for. The long wire uses N − 0.05 half waves instead, and its box takes a whole number.
- **The big result:** the length, in metres, centimetres or feet. Below it:
- the same length in feet and inches (to the quarter inch), or in metres when feet are shown;
- each leg of a dipole;
- the free-space wavelength, with its half or quarter.
- **Drawing:** the antenna (the calculated part, green) with its length; insulators and the ground plane are blue, the feed is grey.
- **Remembered:** the kind of antenna and both units.

Examples:

- a 40 m band dipole at 7.1 MHz;
- a CB quarter-wave whip at 27 MHz;
- FM 24-18's 468 / f rule at 10 MHz;
- FM 24-18's worked long wire, 3 half waves at 7 MHz.

## Terms

- **Wavelength (λ):** the distance a radio wave travels during one cycle, c / f. At 7 MHz it is 42.8 m.
- **Half-wave dipole (Hertz antenna):** a wire half a wavelength long, cut in the middle where the feed line connects. It resonates there: it looks like a plain resistance, about 73 Ω in free space, to the transmitter.
- **Quarter-wave (Marconi) antenna:** half a dipole, standing over ground or a ground plane (a metal sheet, or radials). The ground acts as a mirror and supplies the missing quarter wave, so it resonates at the same frequency as the full dipole.
- **End effect:** a real wire seems electrically longer than it is, because of its thickness and the capacitance at its ends and supports. So a wire resonates shorter than the free-space half wave.

- **Correction factor (k):** how much shorter to cut it, as a share of the free-space length; 0.95 means 5 % shorter.
- **Long wire:** a wire several half waves long, used on harmonics; the end effect matters only at its ends, so FM 24-18 takes 0.05 of a half wave off the total rather than 5 %.

## Source

- **FM 24-18, Tactical Single-Channel Radio Communications Techniques, Headquarters, Department of the Army, 30 September 1987, §3-8 "Antenna Length"** (read, the EverySpec copy):
- an antenna's physical and electrical lengths are "never the same": the reduced velocity of the wave on the wire and the end effect make it seem longer electrically. The diameter-to-length ratio and the insulators and clamps contribute;
- use a correction of 0.95 between 3.0 and 50.0 MHz;
- a half-wave antenna is 150 × 0.95 / f = 142.5 / f metres, or 492 × 0.95 / f = 468 / f feet, f in MHz;
- a long wire (one wavelength or longer, for harmonic operation) is 150 (N − 0.05) / f metres, or 492 (N − 0.05) / f feet, N being the number of half waves. Its example is 3 half waves at 7 MHz: 150 × 2.95 / 7 = 63.2 m;
- §3-8 later gives 468 / f feet again for the half-wave doublet.
- **NEETS Module 10, Introduction to Wave Propagation, Transmission Lines, and Antennas, NAVEDTRA 14182, September 1998, chapter 4** (read):
- Hertz antennas are half a wavelength long and insulated from ground;
- Marconi antennas are a quarter wavelength, grounded at one end or worked against a counterpoise, which "provides the equivalent of an additional 1/4 wavelength";
- a grounded quarter-wave antenna resonates at the same frequency as the ungrounded half-wave, because ground acts as an electrical mirror image;
- a half-wave antenna in free space has a radiation resistance of 73 Ω; over a ground plane, anything from 0 to about 100 Ω.
- **c = 299 792 458 m/s**, exact. FM 24-18 rounds c / 2 to 150 m·MHz and 492 ft·MHz; the tool uses c itself. That is why it gives 46.72 ft where 468 / 10 gives 46.8 ft, and 63.17 m where the manual's example gives 63.2 m.

The quarter-wave applies FM 24-18's 0.95 to half the dipole. FM 24-18 states the factor for half-wave and long-wire antennas only, and NEETS gives the quarter-wave as half the dipole; the factor is not separately sourced for it.

## Why the formulas are these

```
λ = c / f
Half wave:     L = k × c / (2f)       (k = 0.95: 142.5 / f m, 468 / f ft)
Quarter wave:  L = k × c / (4f)
Long wire:     L = c × (N − 0.05) / (2f)
```

A wire resonates when a wave goes to its end and back in step with the source: half a wavelength, end to end. The end effect makes the wire seem longer, so it is cut short by k. A quarter wave over ground resonates like the full dipole, because its mirror image in the ground supplies the other half. A long wire is N

half waves end to end, but the end effect acts only at its two ends: it loses a fixed 0.05 of a half wave, not 5 % of the whole.

The length box works the formula backwards: f = (the factor in front) / L.

## Assumptions and limits

- **Thin wire in the clear.** The real factor depends on the ratio of the wire's diameter to its length and on the insulators and clamps (FM 24-18), and also on its height above ground and what is near it. 0.95 is FM 24-18's figure for HF wire antennas.
- **3 to 50 MHz for k = 0.95.** Outside it the tool flags the factor and still computes. Type another k if you have a better figure for your antenna, from its maker or from measuring it.
- **Cut long, then trim.** Cut the wire a little long, then trim it on the air with an SWR meter or an antenna analyser.
- **The feed point.** The tool gives the resonant length only. It does not match the feed-line impedance: a dipole in free space is 73 Ω (NEETS), less or more near the ground; a quarter-wave is fed at its low-impedance grounded end; an end-fed long wire is high and needs a tuner.
- **Ground plane.** A quarter-wave needs a good ground or radials (each about a quarter wave long). Their length is not computed here.

## What it deliberately does not do

- **The wavelength alone:** Radio wavelength has the bands and the period.
- **Loaded or shortened antennas** (coils, capacitance hats), Yagis and loops: each would be its own tool.
- **Coax stubs and velocity factor:** a cable's length per wavelength is a different correction, for a future Velocity factor tool.

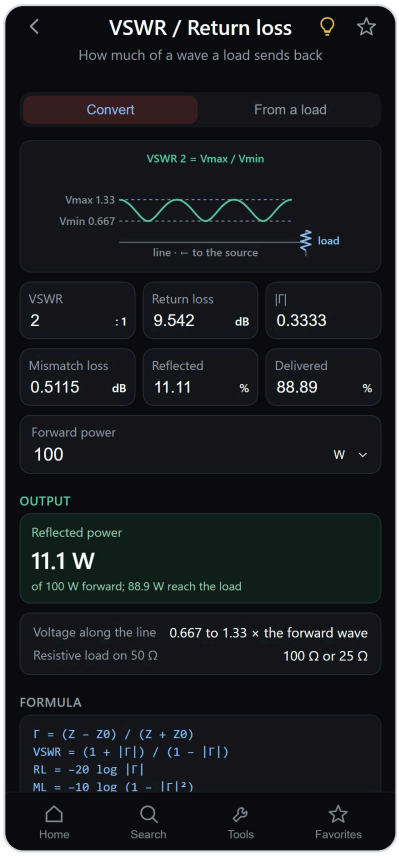
## How it was checked

8 cases:

- 7.1 MHz half wave: 20.06 m, 65 ft 9.75 in, legs of 10.03 m, λ 42.22 m.
- 27 MHz quarter wave: 2.637 m, 8 ft 7.75 in; λ/4 2.776 m.
- FM 24-18's long wire, 3 half waves at 7 MHz: 63.17 m (the manual's rounded 150 gives 63.2 m).
- 20 m typed in the length box: 7.1201 MHz.
- 144 MHz flags the 0.95.
- A correction of 0 and 2.5 half waves are refused.
- 468 / f at 10 MHz: 46.72 ft, 14.24 m, legs of 23.36 ft.

RF & MICROWAVE › TRANSMISSION LINES

# VSWR / Return loss




## What it computes

How much of a wave a load sends back, and what that costs:

- **Two pills:** Convert (the six usual measures of a mismatch, linked) and From a load (a line's Z0 and a load R + jX).
- **Convert:** six boxes, VSWR, return loss, |Γ|, mismatch loss, the reflected and the delivered share: type in any one, the five others follow; the box typed in keeps its text until you leave it. With the forward power, the big result is the reflected power in watts; below it the voltage range along the line, and the two resistive loads that give this VSWR on 50 Ω (50 × VSWR and 50 / VSWR).
- **From a load:** Γ = (Z − Z0)/(Z + Z0) for Z = R + jX; the big result is the VSWR, then |Γ| and its angle, return loss, Γ as re + j im, mismatch loss, reflected and delivered power.
- **Units:** mW, W or kW for the forward power; the rest fixed (dB, %, Ω, : 1).
- **Drawing:** the standing wave along the line, |1 + Γ e^(−j2βd)| over a wavelength and a half back from the load (a zigzag, US style), Vmax and Vmin dashed: their ratio is the VSWR. From a load, Γ's angle sets where the peaks fall (a short puts a null at the load, an open a peak).
- **Remembered:** the pill and the power unit.

Examples: a 2:1 VSWR, 20 dB return loss, a 75 Ω load on 50 Ω coax, a short antenna of 25 − j50 Ω.

## Terms

- **Reflection coefficient Γ:** the reflected wave over the incident one at the load, complex: 0 matched, magnitude 1 for an open, a short or a pure reactance.
- **VSWR (voltage standing wave ratio, SWR):** the incident and reflected waves add into a standing wave; VSWR is its maximum over its minimum, (1 + |Γ|)/(1 − |Γ|): 1 matched, infinite for total reflection.
- **Return loss:** how many dB the reflected wave lies below the incident one, −20 log |Γ|; positive, larger is better.
- **Mismatch loss:** the power that does not reach the load because of the reflection, in dB, −10 log (1 − $|\Gamma|^2$), for a source matched to the line.
- **Forward power:** the power travelling toward the load, as a directional wattmeter reads it.

## Sources

- **Keysight AN 1449-3, "Fundamentals of RF and Microwave Power Measurements (Part 3)", literature number 5988-9215EN** (read): Γ = b/a (eq. 2-1), Γ = (Z − Zo)/(Z + Zo) (2-5), $|a|^2$ the incident power, $|b|^2$ the reflected (2-2, 2-3); SWR = (|a| + |b|)/(|a| − |b|) = (1 + ρ)/(1 − ρ), "VSWR is shortened to

SWR"; the "Zo mismatch loss", the ratio of its equations 2-22 to 2-21 in dB: with a Zo generator (Γg = 0), 10 log(1/(1 − |Γ|$^2$)) (its printed equations 2-25 and 2-26 did not survive the text extraction; the tool derives them from 2-21 and 2-22).

- **Keysight 5965-7707EN, "Understanding the Fundamental Principles of Vector Network Analysis", 10 August 2022** (read): "Return loss = −20 log(ρ)", "the number of decibels that the reflected signal is below the incident signal", "always expressed as a positive number", infinite at the characteristic impedance, 0 dB for an open or short; VSWR = Emax / Emin = (1 + ρ)/(1 − ρ).

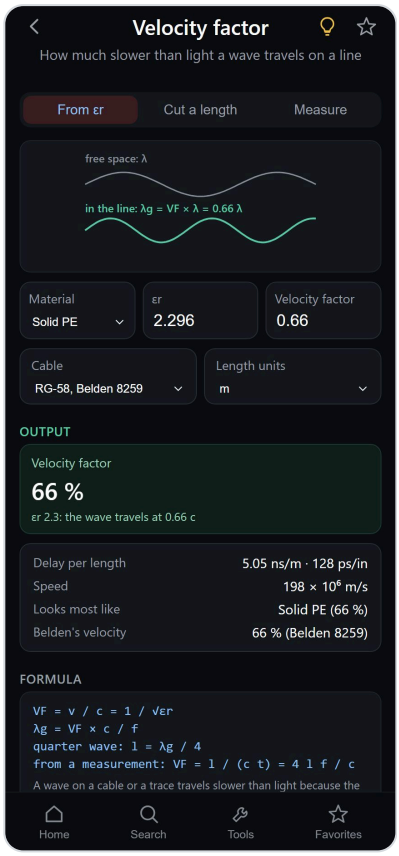
## Assumptions and limits

- **One frequency, lossless line:** a lossy line attenuates the reflected wave on its way back, so the VSWR read at the transmitter is better than at the load.
- **Mismatch loss with a matched source:** with a mismatched source the power transferred depends on both reflections (AN 1449-3's mismatch uncertainty).
- **Resistive loads on 50 Ω:** the two pure resistances with this VSWR; a complex load has the same VSWR on a circle of impedances.

## How it was checked

7 cases, the expected values from the formulas above: - VSWR 2: |Γ| 0.3333, return loss 9.542 dB, mismatch loss 0.5115 dB, 11.11 % reflected, 11.1 W of 100 W, loads 100 Ω or 25 Ω; - return loss 20 dB: VSWR 1.222, mismatch loss 0.04365 dB, 1 W; - a VSWR under 1 is refused; - 75 Ω on 50 Ω: Γ 0.2, VSWR 1.5, return loss 14 dB, 0.177 dB, 4 W; - 25 − j50 Ω on 50 Ω: Γ = (625 − j5000)/8125 = 0.0769 − j0.615, VSWR 4.27, return loss 4.15 dB, 38.5 W; - a short: |Γ| 1 at 180°, VSWR ∞, return loss 0 dB, 100 W back; - kW re-expresses the forward power.

# Velocity factor




## What it computes

How much slower than light a wave travels on a cable or trace, and the lengths that follow:

- **Three pills:**
- **From εr:** the Material picker (air, PTFE, solid and foam PE at Belden's 66 % and 82 %), εr and the velocity factor as linked boxes (typing one sets the other), a Cable picker (Coax impedance's seven Belden cables, which set εr; typing a value returns it to Custom);
- **Cut a length:** the same line plus a frequency: the quarter-wave length λg/4 as the big result, then the half and whole guided wavelength, the free-space wavelength and the quarter wave's delay;
- **Measure:** a cable's length, its one-way delay and the frequency of its first resonance, linked (resonance = 1/(4 × delay)); type the delay or the resonance and the velocity factor of the cable follows.
- **The big result:** VF in % (or the quarter-wave length); below it the delay per length (ns/m and ps/in), the speed, and the dielectric it looks most like (solid PE, foam PE, PTFE).
- **Units:** mm, cm, m, in or ft for lengths (the order of metric and imperial in the results follows the one picked); kHz, MHz or GHz.
- **Drawing:** a wave in free space (grey) over the same wave in the line (green), λg = VF × λ, with λg/4 marked in Cut a length.
- **Remembered:** the pill and the units.

Examples: RG-58 at 66 %, a quarter wave of it at 100 MHz, Wadell's PTFE resonator, a 10 m cable measured by its resonance.

## Terms

- **Velocity factor (velocity of propagation):** the wave's speed on the line as a share of the speed of light: 1/√εr, εr the dielectric's (εeff for a microstrip).
- **Guided wavelength λg:** the wavelength in the line, λ × VF.
- **Quarter-wave section or stub:** a line λg/4 long; it turns a short into an open and the reverse, and is the usual length of a balun or a matching section.
- **Resonance:** with the far end shorted, the input impedance of a lossless line is j Z0 tan(2π l/λg), which is infinite where l = λg/4: the first peak, which a network analyser or a grid-dip meter finds.
- **TDR (time-domain reflectometer):** an instrument that sends a pulse down a cable and times its echo; half the round trip is the delay.

## Sources

- **Brian C. Wadell, Transmission Line Design Handbook, 1991** (Pierre's scan, read): §2.8.3 (p32–35): Zsc = j Z0 tan(2πl/λ) (2.8.3.2), infinite at l = λ/4 (2.8.3.7), λg = c/(f √εr) (its example); the example 2.8.3.1: PTFE (εr 2.10) at 402 MHz, λg/4 = 0.1287 m = 5.07 in with c = 3 × $10^8$ m/s.
- **Belden technical data sheets** (see Coax impedance): "velocity of propagation" 66 % (PE), 70 % (PTFE), 82 % (foam PE).
- **c = 299 792 458 m/s** (SI), so the tool's 402 MHz example reads 0.12865 m = 5.065 in, rounding to Wadell's 5.07 in.

## Assumptions and limits

- **Lossless, no dispersion:** one εr; a cable's VF varies a little with frequency and with its braid and jacket.
- **Open ends and connectors** add a little capacitance and length, so a measured quarter-wave resonance reads slightly low; trim on the air.
- **Shortening of antennas** (the 0.95 end-effect factor of Antenna length) is a different effect, for a wire in air.

## How it was checked

8 cases, the expected values from the formulas above and a Python check: - RG-58: 66 %, 5.05 ns/m, 128 ps/in (128.37), 198 × $10^6$ m/s, solid PE; - typing VF 0.82: εr 1.487, foam PE, 4.07 ns/m, 103 ps/in, cable picker Custom; - a quarter wave of RG-58 at 100 MHz: 49.5 cm, half 98.9 cm, whole 1.98 m, free space 3 m; - Wadell's PTFE resonator: 5.07 in (12.9 cm), VF 69 %; - a 10 m cable resonating at 4.95 MHz: 66 %; a 50 ns delay: 66.7 % and 5 MHz; a delay faster than light is refused; - inches put the imperial figure first.

# Coax impedance




## What it computes

The impedance and the other line constants of a round coaxial cable, from its dimensions, or the dimensions for an impedance:

- **Inputs:** the inner conductor's diameter d, the dielectric's diameter D (the inside of the shield), the dielectric (Material picker, εr or the velocity factor, linked: typing either sets the other), a listed cable, and a frequency for the loss.
- **Impedance is a live box:** type an impedance and D is set to give it (d and εr kept), D drawn green; leave the box and it shows Z0 again.
- **Cable picker:** seven Belden cables; picking one sets d, D and εr ($1/VF^2$, the maker's velocity of propagation), and the results add its construction and Belden's own impedance, velocity and capacitance, and at 100 MHz or 1 GHz its attenuation. Typing a dimension or εr returns the picker to Custom.
- **The big result:** Z0 (or D for the impedance typed); below it D/d and εr.
- **Below it:** C and L per metre, the velocity factor and the delay per metre, the frequency where TE11 starts, the loss of smooth copper (per 100 m, or per 100 ft in inches or beside Belden's figure).
- **Units:** mm or in for the diameters, linked; MHz or GHz.
- **Drawing:** the cable cut across: shield, dielectric, inner conductor, drawn to the D/d entered (kept between 1.5 and 8), with D and d in mm and in.
- **Remembered:** the units.

| Cable (Belden) | d, D (in) | Belden Z0, VF, pF/m | from the sizes | dB/100 ft at 100 MHz, 1 GHz |
|---|---|---|---|---|
| RG-58, 8259 | 0.035 (19×33), 0.117 PE | 50, 66 %, 101 | 47.8 Ω, 106 pF/m | 5.4, 22.8 |
| RG-58, 9201 | 0.033 solid, 0.116 PE | 52, 66 %, 93.5 | 49.7 Ω, 102 pF/m | 3.8, 14.5 |
| RG-174, 8216 | 0.019 (7×34), 0.060 PE | 50, 66 %, 101 | 45.5 Ω, 111 pF/m | 8.4, 34.0 |
| RG-316, 83284 | 0.020 (7×34), 0.060 PTFE | 50, 70 %, 95 | 46.1 Ω, 103 pF/m | 8.3, 29.0 |
| RG-213, 8267 | 0.089 (7×21), 0.285 PE | 50, 66 %, 101 | 46.1 Ω, 110 pF/m | 1.9, 8.0 |
| RG-59, 8241 | 0.023 solid, 0.146 PE | 75, 66 %, 66.9 | 73.1 Ω, 69.1 pF/m | (sheet's table empty) |
| RG-6, 1694A | 0.040 solid, 0.180 foam PE | 75, 82 %, 53.1 | 73.9 Ω, 55.0 pF/m | 1.95, 6.3 |

The formula, exact for a smooth line, reads 1 to 9 % low from the nominal sizes, most with stranded conductors: a stranded conductor's outline is larger than its effective diameter, and a braid's inside diameter is not the insulation's nominal one. Belden's figures are the measured truth for its cables; the formula is the one to design with.

## Terms

- **d, D:** the inner conductor's outside diameter, and the dielectric's diameter, which is the inside diameter of the shield.
- **Characteristic impedance Z0:** the ratio of voltage to current of a wave on the line; set by D/d and εr only.
- **Velocity factor VF:** the wave's speed as a share of light's, 1/√εr; makers give it as "velocity of propagation", 66 % for solid PE.
- **TE11:** the first wave pattern other than the normal (TEM) one; above its cutoff a cable can carry two kinds of wave at different speeds, which spoils it.
- **Skin effect:** at high frequency current flows in a thin skin at the conductors' surface, so their resistance, and the loss, rise with √f.
- **Solid, foam PE, PTFE:** the usual dielectrics: polyethylene solid or foamed with gas (lower εr, faster), and PTFE for heat and high frequency.

## Sources

- **Brian C. Wadell, Transmission Line Design Handbook, Artech House, 1991** (Pierre's scanned copy, read):

- §3.2.1 (p47–49): $Z0 = \eta0/(2.0\ \pi\ \sqrt{\varepsilon r})\ \ln(D/d)$ (3.2.1.1), "exact for the case of an infinitely long line"; the TE11 cutoff $\lambda c \approx \pi\ (D + d)/2.0$ (3.2.1.12), "generally accurate to 3% for 50 Ω lines"; the example (§3.2.1.1): 50 Ω around a 0.040 in pogo pin needs D = 0.092 in in air and 0.134 in in PTFE (εr 2.1);
- §2.7.1 (p25): $Rs = 1/(\sigma\delta) = \sqrt{(\pi\ f\ \mu/\sigma)}$ (2.7.1.3) and $\alpha \approx R/(2.0\ Z0)$ (2.7.6); its printed $\delta = 1/\sqrt{(\omega\mu\sigma)}$ (2.7.1.5) is √2 short of the δ its own Rs implies, $1/\sqrt{(\pi\ f\ \mu\ \sigma)}$; the tool works from Rs. Its coax loss shortcuts (3.2.1.2 and 3.2.1.3) do not state their units, so the tool does not use them.
- **Copper:** IACS, 1/58 Ω·mm²/m at 20 °C, as in Wire sizing (NBS Handbook 100).
- **The cables:** Belden technical data sheets 8259, 9201, 8216, 83284, 8267, 8241 and 1694A (catalog.belden.com, read, revisions dated February to May 2026): nominal conductor and insulation diameters, materials, impedance, velocity of propagation, capacitance, attenuation in dB/100 ft. 8241's attenuation table is printed empty. Times Microwave's LMR-400 sheet could not be downloaded, so LMR cables are not listed.
- **Solid and foam PE's εr:** $1/VF^2$ from Belden's 66 % and 82 %: 2.30 and 1.49; PTFE 2.1 as Wadell's example.

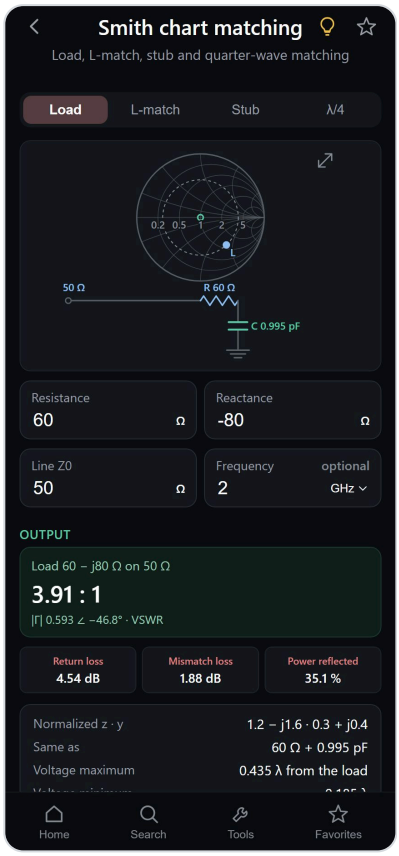
## Assumptions and limits

- **Smooth, round, concentric, uniform:** the formula; eccentric or partly filled lines (Wadell 3.2.2, 3.2.3) differ.
- **Loss:** solid smooth copper only, no dielectric loss, no braid, plating or stranding: a lower bound, often half a real cable's.
- **TE11:** the simple estimate, within about 3 % (Wadell).

## How it was checked

9 cases, the expected values from the formulas above (a Python check of the same formulas): - RG-58, Belden 8259: 47.8 Ω, 106 pF/m, 241 nH/m, VF 0.66, 5.05 ns/m, TE11 from 32.6 GHz, smooth copper 3.36 dB/100 ft at 100 MHz against Belden's 5.4; - Wadell's pin: 49.9 Ω in air, 50.0 Ω in PTFE; 50 Ω typed gives D = 0.0921 in (his 0.092); - RG-213 at 1 GHz: 46.1 Ω, TE11 from 13.3 GHz, 4.38 against 8.0 dB/100 ft; - RG-6: foam PE, 73.9 Ω, 4.07 ns/m; - inches re-express both diameters; - at 50 MHz the maker's loss row goes and the copper loss reads 7.8 dB/100 m; - D smaller than d is refused.

# Smith chart matching




## What it computes

A load sits at the end of a transmission line. When its impedance is not the line's, part of the wave is sent back. The tool shows how bad that is and gives three ways of curing it (pills). You give the load as a resistance R above zero and a reactance X (negative for a capacitor), both in ohms, and the line's impedance Z0.

The drawing is the same for the four pills: the Smith chart with the load on it and whatever the tool found, and under it the circuit that was calculated, the line coming in on the left and the load, ZL, on the right. The parts you gave are blue, the ones the tool calculated green, the wires grey. Tap the drawing and the chart fills the box while the circuit goes; tap again and it comes back.

- **Load:** how well the load matches the line. The big result is the VSWR (3.91 : 1) with |Γ| and its angle under it. In three cells: the return loss, the mismatch loss and the share of the power reflected. Under them: the normalized impedance z and admittance y; "Same as", the load as its resistance in series with an inductor or a capacitor at the frequency (the frequency is optional here, and only this row waits for it); the distance from the load to the first voltage maximum, in wavelengths, and to the first minimum. The chart has the load, L, and the dashed circle of constant |Γ| through it; the circuit is the line ending in that resistance and reactance.
- **L-match:** a series part and a shunt part, ordinary inductors or capacitors at the frequency, that match the load. The big result is the two values ("92.7 fF · 6.08 nH"); under it which part is which ("shunt C at the load, then series L") and which solution of how many. In three cells: the loaded Q, the band where the return loss stays at 15 dB or better, and the load as its resistance with an inductor or capacitor. Under them, every solution on a row, sorted by loaded Q (up to four); tap a row to pick it: the chart draws its path, from the load through "1", the point after the first part, to the centre, and the circuit shows its parts numbered from the line (L1, C1; C1 and C2 when both are capacitors). A load whose resistance equals Z0 needs only a series part, and that one is listed once.
- **Stub:** a length of line from the load, then a stub, a short length of line with its far end shorted or open, across the line (shunt) or in series with it. The Output heading holds the choices: shunt or series, short or open, and the unit for lengths. The big result is the distance from the load to the stub in wavelengths, with the same in length units and "solution 1 of 2" under it. In three cells: the length of a short stub, of an open stub, and the lumped part that does the same job ("shunt L 2.7 nH"). Under them: the two solutions on rows to tap, the stub lengths in the length unit, and the 15 dB band for the short stub and for the open one. The chart draws both solutions, the one you picked bright. The circuit shows the line, the stub you chose and the load: a shunt stub hangs from the line and ends in ground (short) or in the air (open); a series stub breaks the line and runs out and back, closed at its end (short) or not (open). Its label carries the stub's length and, with a frequency, the same in the length unit.
- **λ/4:** a quarter-wave section of another impedance, placed where the line shows a real impedance. The big result is the section's impedance; under it its length, where it goes (at the voltage minimum or

maximum) and the impedance the line shows there. In three cells: the section's impedance, the length of line before it in wavelengths and the 15 dB band. Under them: the two places (sorted by distance from the load) on rows to tap, and the line's length before the section in the length unit. The circuit is line, section, the line of that length, the load; with the load already at the voltage maximum (a real load above Z0) there is no line before the section and the drawing leaves it out.
- **Units:** frequency kHz, MHz or GHz; lengths mm, cm, m, in or ft, set in the Output heading. The velocity factor (Stub and λ/4) turns wavelengths into lengths: λ in the line is c × VF / f.
- **Remembered:** the pill, the frequency unit, the length unit, shunt or series and short or open. The enlarged chart is not.

Examples: a 60 − j80 Ω load on 50 Ω, an L-match from 200 − j100 Ω to 100 Ω, a shunt stub and a series stub, and a quarter-wave match for 100 Ω.

## Terms

- **Impedance Z = R + jX:** what a load presents to a signal at one frequency, in ohms. R, the resistance, dissipates power; X, the reactance, stores and returns it, positive for an inductor and negative for a capacitor. The j marks a quantity a quarter of a cycle out of step with the other.
- **Z0, the line's impedance:** the impedance a wave meets travelling along the line, set by its construction (50 Ω and 75 Ω are the usual ones). A load of exactly Z0 sends nothing back.
- **Normalized z = Z / Z0 = r + jx:** the load in units of the line's impedance. The chart is drawn for it, so one chart serves every Z0.
- **Admittance Y = 1 / Z = G + jB, y = Y / Y0 = g + jb:** the same load seen as how easily current flows, in siemens: G the conductance, B the susceptance, positive for a capacitor and negative for an inductor. Parts in parallel add their admittances, parts in series add their impedances, which is why a matching job switches between the two.
- **Reflection coefficient Γ:** the share of the voltage wave sent back by the load, with its phase: Γ = (Z − Z0) / (Z + Z0). |Γ| runs from 0 (a match) to 1 (everything reflected, as from a short, an open or a pure reactance).
- **VSWR (voltage standing wave ratio):** incoming and reflected waves add up to a standing wave on the line; the VSWR is its highest voltage over its lowest, (1 + |Γ|) / (1 − |Γ|), and 1 when matched.
- **Return loss, mismatch loss, power reflected:** the same |Γ| in other words. Return loss, −20 log |Γ| dB, says how far down the reflected wave is; mismatch loss, −10 log (1 − $|\Gamma|^2$) dB, the power the mismatch costs the load; power reflected is $|\Gamma|^2$.
- **Smith chart:** the disk of every Γ with |Γ| ≤ 1, which holds every load with R ≥ 0. Its grid shows, at each Γ, the load's r and x: circles of constant resistance all touching at the right-hand edge, and arcs of constant reactance (the top half inductive, the bottom half capacitive). The centre is the match, Γ = 0.
- **Moving along the line:** towards the generator, Γ keeps its size and turns clockwise round the centre, a full turn for every half wavelength. A point therefore stays on its dashed circle of constant |Γ|.
- **Voltage maximum, minimum:** places on the line where the standing wave peaks or dips. There the impedance is real, Z0 × VSWR at a maximum and Z0 / VSWR at a minimum, a quarter wavelength apart.
- **Series part, shunt part:** a series part sits in the line, a shunt part joins the line to ground. Adding series reactance moves the load's point along its circle of constant resistance; adding shunt susceptance moves it along its circle of constant conductance.
- **L-match:** two lumped parts, one series and one shunt, drawn like an L or a turned L. **Loaded Q:** how sharply it is tuned, the reactance over the resistance at its middle point; the lower, the wider the band.

- **Stub:** a length l of line, shorted or open at the far end, used as a reactance. Across the line it adds susceptance $-\cot \beta l$ (short) or $\tan \beta l$ (open); in series, reactance $\tan \beta l$ (short) or $-\cot \beta l$ (open), in units of 1 / Z0 and Z0, with $\beta l = 2\pi\, l / \lambda$. A shorted stub a quarter wave long is an open circuit; an open one is a short.
- **Quarter-wave section, λ/4:** a line a quarter wavelength long, of impedance ZT, turns a real impedance R into $ZT^2 / R$. With $ZT = \sqrt{(Z0\ R)}$ it makes R look like Z0.
- **Velocity factor (VF):** how fast the wave travels in the line, as a fraction of the speed of light; a wavelength in the line is $c \times VF / f$.
- **15 dB band:** the frequencies around the design frequency over which the return loss stays at 15 dB or better, $|\Gamma| \le 0.178$, about 3 % of the power reflected. 15 dB is this tool's own yardstick, not a standard.

## Sources

- **Wikipedia, "Smith chart" (read through a fetch tool):** the chart as a plot of Γ with $z = (1 + \Gamma)/(1 - \Gamma)$ and the circles of constant resistance and reactance; the admittance chart as the same chart turned half a turn; movement along the line as rotation on the outer wavelength scale, the angle changing by 2π over λ/2; and a worked example the tool reproduces: a 50 Ω line at 800 MHz and a load of 17.5 Ω in series with 6.5 nH, $z = 0.35 + j0.65$, matched either by 0.079 λ (29.6 mm) of line and a 2.6 pF series capacitor, or by 0.329 λ (123 mm) of line and a 6.5 nH shunt inductor. The article does not say in which direction the rotation goes; that comes from the line equations below.
- **Michael Steer, Fundamentals of Microwave and RF Design, § 10.4 "The L matching network" (LibreTexts page, read through the fetch tool):** the L-match between two real resistances, with $Q = \sqrt{(R_{high} / R_{low} - 1)}$, and two examples: 100 Ω to 1700 Ω at 900 MHz (Q 4, 70.7 nH, 0.416 pF) and 50 Ω to 75 Ω (Q 0.7071, a series reactance of 35.35 Ω and a shunt one of 106.1 Ω). That page has no method for a complex load.
- **Lecture slides "Impedance Matching and Tuning", Çankaya University (ece358.cankaya.edu.tr; PDF text extracted and read):** L-sections: the load 200 − j100 Ω to 100 Ω at 500 MHz, read off the chart as 0.92 pF with 38.8 nH or 46.1 nH with 2.61 pF, and $z = 0.5 + j0.3$, with a series $x = 0.2$ and a shunt $b = 1.0$, or $x = -0.8$ and $b = -1.0$; single-stub tuning: a short shunt stub for 60 − j80 Ω on 50 Ω at 2 GHz (0.110 λ and 0.095 λ, or 0.260 λ and 0.405 λ) and an open series stub for 100 + j80 Ω (0.120 λ and 0.397 λ, or 0.463 λ and 0.103 λ); the loads modelled as 200 Ω + 3.18 pF, 60 Ω + 0.995 pF and 100 Ω + 6.37 nH to plot |Γ| against frequency; short stubs preferred on coaxial cable (a short radiates less) and open stubs on a printed board (no via to ground). Their title is also that of the matching chapter of Pozar's Microwave Engineering (from memory; the book was not read) and they do not cite it. Their quarter-wave formula is an image the extraction lost; it is derived below.
- **University of Tennessee, ECE341 lecture notes 9, "Impedance matching: lumped elements and single stub" (Fall 2021; PDF text extracted and read):** $y_L = 0.4 + j0.8$: a line of 0.178 − 0.115 = 0.063 λ where $y = 1 + j1.58$, a short stub of 0.09 λ, and a second place, the mirror image, at 0.322 − 0.115 = 0.207 λ where $y = 1 - j1.58$ (the extracted text lost that minus sign).
- **Not read:** Pozar's book itself, Smith's papers, and anything on double-stub tuning (the tool does not do it) or on stubs in microstrip, where junction and open-end effects are modelled.

## Why the formulas are these

**The chart.** With $\Gamma = u + jv$, $z = (1 + \Gamma)/(1 - \Gamma)$ gives $r = (1 - u^2 - v^2) / ((1 - u)^2 + v^2)$ and $x = 2v / ((1 - u)^2 + v^2)$. Setting r constant and completing the square gives $(u - r/(1 + r))^2 + v^2 = 1/(1 + r)^2$, a circle of centre $(r/(1 + r), 0)$ and radius $1/(1 + r)$; setting x constant gives $(u - 1)^2 + (v - 1/x)^2 = 1/x^2$, a circle of centre $(1, 1/x)$ and

radius 1/|x|, of which only the part inside the unit circle is drawn. The chart shows r and x at 0.2, 0.5, 1, 2 and 5. Seen as y = 1/z the same chart has its points turned half a turn: Γ(y) = −Γ(z).

**Along the line.** A wave on a lossless line has Γ(d) = Γ_L e^(−j2βd) at a distance d from the load toward the generator, β = 2π/λ. The size stays, the angle falls by 4π d/λ: clockwise, a full turn at d = λ/2. At a voltage maximum Γ(d) is real and positive, so the first one is where the angle has fallen to 0, d = (arg Γ_L mod 360°) / 720° wavelengths, and a minimum is a quarter wavelength from it. There z = S at a maximum and 1/S at a minimum, S the VSWR.

**The L-match.** Series first (at the load): adding jx to z = R + jX gives R + j(X + x). Its admittance must have conductance 1/Z0: R / (R² + (X + x)²) = 1/Z0, so (X + x)² = R (Z0 − R), which has solutions only for R ≤ Z0:

```
x = −X ± √(R (Z0 − R))
```

What is left is the susceptance −(X + x) / (R Z0), which a shunt part cancels:

```
b = (X + x) / (R Z0)        Q = |X + x| / R
```

Shunt first: the load's admittance is G + jB_L with G = R / (R² + X²) and B_L = −X / (R² + X²); adding jb gives G + j(B_L + b), whose impedance must have resistance Z0: G / (G² + (B_L + b)²) = Z0, so (B_L + b)² = G (Y0 − G), possible for G ≤ Y0 = 1/Z0:

```
b = −B_L ± √(G (Y0 − G))    x = Z0 (B_L + b) / G    Q = |B_L + b| / G
```

Up to four solutions, two per order. A series x is an inductor L = x/ω if positive, a capacitor C = −1/(ωx) if negative; a shunt b is a capacitor C = b/ω if positive, an inductor L = −1/(ωb) if negative, ω = 2π f. For a real load the shunt-first Q is √(Y0/G − 1) = √(R/Z0 − 1), Steer's formula. When R = Z0 exactly, the series-first root is single (x = −X, b = 0) and the shunt-first one ends in the same lone series part, which is listed once.

**The stub.** With a = y_L (shunt stub) or z_L (series stub) = g + jb, a line of electrical length βd changes it to a(d) = (a + jt) / (1 + j a t), t = tan βd: the same law for an admittance as for an impedance. Its real part is g (1 + t²) / ((1 − bt)² + g² t²); setting it to 1 gives

```
(b² + g² − g) t² − 2 b t + (1 − g) = 0
t = (b ± √(g ((1 − g)² + b²))) / (b² + g² − g)
```

and d/λ = atan(t) / 2π, taken in [0, ½). When b² + g² − g is 0 (a on the circle through 0 and 1) one root is t = ∞, d = λ/4, and the other is t = (1 − g) / (2b). Moving along the line from a gives a(d) = 1 + j b_d; the stub must add s = −b_d. A shunt stub, shorted, has −cot βl = s, so l/λ = atan2(1, −s) / 2π; open, tan βl = s, so l/λ = (atan s mod π) / 2π. A series stub has the two swapped. The lumped part with the same s: shunt, C = s / (ω Z0) if s > 0 and L = −Z0 / (ω s) if s < 0; series, L = s Z0 / ω if s > 0 and C = −1 / (ω s Z0) if s < 0. The distance d is of the line's wavelength, so d × c × VF / f is its length.

**The quarter-wave section.** A line of impedance Zt and electrical length θ turns Z_L into Zt (Z_L + j Zt tan θ) / (Zt + j Z_L tan θ), which at θ = 90° is Zt² / Z_L. For the line to see Z0 the section needs Zt² = Z0 R, with R the

real impedance at that point: Z0 × S at a voltage maximum, Z0 / S at a minimum, so Zt = Z0 √S or Z0 / √S. With $Zt^2$ = Z0 Z_L the reflection coefficient at any θ is

```
Γ = (Z_L − Z0) / (Z_L + Z0 + j 2 Zt tan θ)
```

so its size, $(Z_L - Z0)^2$ / ($(Z_L + Z0)^2$ + 4 Z0 Z_L $\tan^2$ θ) under a root, is smallest at θ = 90°.

**The band.** For the design frequency f0 the lines and stubs keep their length in wavelengths at f0, so their electrical length is proportional to f/f0; a lumped part keeps ωL or −1/(ωC); the load is its resistance in series with X f/f0 (an inductor, X > 0) or X f0/f (a capacitor), as the slides model theirs. |Γ| is computed at steps of 0.2 % of f0 from f0 downward and upward until it passes 10^(−15/20) = 0.178, and each crossing is refined by halving 50 times. A band that has not closed within ±80 % is given as "wider than ±80 %".

**The drawing.** A series part x moves z to z + jx along the circle of constant resistance; a shunt part b moves y to y + jb along the circle of constant conductance; a length of line d moves Γ clockwise round the dashed circle by 720° per wavelength. The tool samples those paths and draws them. A dotted green circle that appears is the circle of conductance 1, g = 1 (the mirror image of the r = 1 circle): a shunt stub's line, and an L-match that starts with its series part, must bring the load onto it, from where one shunt move reaches the centre.

## Assumptions and limits

- **Ideal parts.** Lines lose nothing; inductors and capacitors have no resistance, no stray capacitance and no self-resonance. At gigahertz that is often untrue: the default L-match at 2 GHz asks for a 92.7 fF capacitor, less than the stray capacitance of a solder pad or a via. The tool solves the circuit; it does not know what can be built.
- **Stubs and junctions.** The stub and the line have the same Z0 and velocity factor. A T-junction adds a little discontinuity and an open end behaves as if the line went a little further (its fringing field), so a real open stub is cut slightly shorter; neither is modelled.
- **One frequency.** Every design is exact at the design frequency and narrow-band. The 15 dB band shows how narrow, for the load model above; a real antenna or device has an impedance that varies in its own way.
- **A match to the line.** The tool brings the load to Γ = 0 on a line of Z0, the usual aim. It does not conjugate-match a source with a reactance of its own.
- **R above zero.** A negative resistance (an active load) is outside the chart. A load of exactly Z0 is "already matched".
- **Numbers, not pictures.** Everything is calculated, not read off the chart; chart readings in the sources differ from the exact values by up to about 2.5 % (2.6 against 2.55 pF, 6.5 against 6.38 nH, 38.8 against 39.0 nH).
- **L-match choices.** Up to four networks match one load; the first listed has the lowest loaded Q and so the widest band. The tool does not say which is a low-pass or a high-pass, nor which of them can be built with parts that exist.
- **The λ/4 section** matches only a real impedance, so for a load with reactance the tool first moves along the line to the voltage maximum or minimum, and the line before the section counts in the band.
- **The circuit is a diagram.** Boxes and parts are not drawn to scale; the numbers beside them are the sizes.

- **The 15 dB** and the ±80 % limit are this tool's choices; three digits are shown.

## How it was checked

74 cases and the formula-search card. The expected values come from a Python implementation of the same formulas, written apart from the JavaScript, and cover every figure and every label of the circuit: - Load: 60 − j80 Ω on 50 Ω (3.91 : 1, |Γ| 0.593 ∠ −46.8°, 4.54 dB, 35.1 %, z 1.2 − j1.6, y 0.3 + j0.4, 60 Ω + 0.995 pF, voltage maximum at 0.435 λ), 100 Ω and 25 Ω (VSWR 2, the maximum at the load and at 0.250 λ), a match, a near short and a short, the load as 6.5 nH, 3.18 nH, 1.99 nF and 99.5 fF, with and without a frequency, a 75 Ω line; - L-match: the slides' two examples, Steer's two, a load with four solutions, a lone series part, 10 Ω to 50 Ω at 14 MHz, a 10 + j5 Ω load at 500 kHz, picking each row, and the numbering of the parts on the circuit; - Stub: shunt and series, short and open, both solutions, velocity factor 0.66, lengths in cm and in inches, no frequency, a stub right at the load, and a load on the circle through 0 and 1 (one stub exactly a quarter wave away); - λ/4: 60 − j80, 100 Ω at 1 GHz in a 0.66 line, 25 Ω, 300 Ω on 75 Ω, both rows, no frequency; - the refusals (Z0, R, X, frequency, velocity factor, a matched load), the units and the remembered choices, the five examples, and the tap that enlarges the chart.

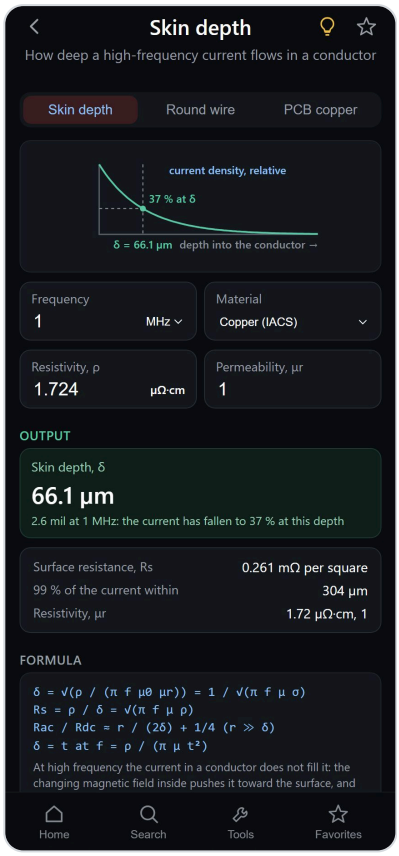
About 1 200 more cases, on 240 random loads (R 0.5 Ω to 1.5 kΩ, X ±600 Ω, Z0 10 to 400 Ω, 100 kHz to 20 GHz) through all four pills, compared every figure and label with the Python implementation; they were run and not kept.

Published answers reproduced (the tool gives the exact value, the sources read a chart):

| Example | Source | Source reads | Tool gives |
|---|---|---|---|
| 200 − j100 Ω on 100 Ω, 500 MHz | Çankaya | 0.92 pF + 38.8 nH; 46.1 nH + 2.61 pF | 0.923 pF + 39.0 nH; 46.1 nH + 2.60 pF |
| z = 0.5 + j0.3 | Çankaya | x 0.2, b 1.0; x −0.8, b −1.0 | 10 Ω and 0.02 S on 50 Ω: the same |
| Shunt stub, 60 − j80 Ω on 50 Ω | Çankaya | d 0.110 and 0.260 λ; short 0.095 and 0.405 λ | 0.110 and 0.259; 0.095 and 0.405 |
| Series stub, 100 + j80 Ω on 50 Ω | Çankaya | d 0.120 and 0.463 λ; open 0.397 and 0.103 λ | 0.120 and 0.463; 0.398 and 0.102 |
| y_L = 0.4 + j0.8 | Tennessee | d 0.063 and 0.207 λ; short stub 0.09 λ; y = 1 ± j1.58 | 0.063 and 0.207; 0.090 |
| 17.5 Ω + 6.5 nH, 800 MHz, 50 Ω | Wikipedia | 0.079 λ, 2.6 pF in series; 0.329 λ, 6.5 nH in shunt | 0.079 λ (29.6 mm), 2.55 pF; 0.329 λ (123 mm), 6.38 nH |
| 100 Ω to 1700 Ω, 900 MHz | Steer | Q 4, 70.7 nH, 0.416 pF | Q 4, 70.7 nH, 0.416 pF |
| 50 Ω to 75 Ω | Steer | Q 0.7071, 35.35 Ω, 106.1 Ω | Q 0.707; 56.3 nH and 15.0 pF at 100 MHz, which are 35.35 Ω and 106.1 Ω |

The Python implementation also showed that: - simulating every network at its design frequency, over 5 000 random loads (R 0.3 Ω to 1.6 kΩ, X ±800 Ω, Z0 10 to 400 Ω, 100 kHz to 20 GHz), leaves |Γ| at most $4 \times 10^{-13}$ for the L-matches and $3 \times 10^{-11}$ for the stubs, short and open, and for the quarter-wave sections; - the band of a quarter-wave section at the load equals the closed form above: 0.65846 to 1.34154 f0 for 100 Ω on 50 Ω and for 25 Ω on 50 Ω, 0.84509 to 1.15491 f0 for 300 Ω on 75 Ω, the same to five digits; - the slides' two L-matches, which they say differ little in bandwidth, have 15 dB bands of 401.2 to 606.1 MHz (41 % of 500 MHz) and 428.1 to 662.9 MHz (47 %); - the slides' loads, 60 − j80 Ω at 2 GHz and 100 + j80 Ω, are 60 Ω + 0.9947 pF and 100 Ω + 6.366 nH, as they write them.

# Skin depth




## What it computes

How deep a high-frequency current flows in a conductor, and what that does to a wire's or a foil's resistance:

- **Three pills:**
- **Skin depth:** δ at a frequency, with the surface resistance Rs (ohms per square) and the depth within which 99 % of the current flows (δ ln 100 = 4.6 δ);
- **Round wire:** a wire's diameter gives the AC resistance as a multiple of its DC resistance, the two in mΩ/m, the frequency where δ equals the radius (where the skin effect begins) and the share of the wire's area that carries the current;
- **PCB copper:** a foil's thickness (oz, µm or mil) gives the frequency where δ equals it; and at the frequency typed, δ, t/δ, Rs and the foil's DC resistance per square.
- **Inputs:** the frequency (Hz to GHz), the material, its resistivity ρ in µΩ·cm and relative permeability µr (typing ρ sets the picker to Custom).
- **Materials (ρ at 20 °C):** copper as wire is sold (IACS) 1.7241; pure copper 1.678; silver 1.587; gold 2.214; aluminum 2.650.
- **Units:** the wire's diameter in mm or mil, the foil's thickness in oz, µm or mil, linked as in the impedance tools.
- **Drawing:** the current density e^(−x/δ) against depth with 37 % at δ marked; the wire's section with its green ring δ thick; the foil seen on edge with δ from each face.
- **Remembered:** the pill and the units.

Examples: copper at 1 MHz, silver at 1 GHz, a 1 mm wire at 1 MHz, 1 oz copper on a PCB.

## Terms

- **Skin effect:** at high frequency the changing magnetic field inside a conductor pushes the current toward its surface; the resistance rises.
- **Skin depth δ:** the depth at which the current density has fallen to 1/e (37 %) of its surface value.
- **Surface resistance Rs:** ρ/δ, in ohms per square: the resistance of a square of the skin, whatever its size.
- **Rac/Rdc:** the wire's AC resistance over its DC resistance.
- **µr:** the relative permeability; 1 for copper, silver, gold and aluminum, hundreds for iron, which has a much shallower skin.
- **Proximity effect:** current crowding caused by a neighbouring conductor (turns of a coil); not included.

## Sources

- **Brian C. Wadell, Transmission Line Design Handbook, 1991** (Pierre's scan, read): §2.7.1 (p25): Rs = 1/(σδ) = √(π f μ/σ), δ = "the depth at which the current density is reduced to 1/e"; its printed δ = 1/√(ωμσ) (2.7.1.5) is √2 short of the δ its own Rs implies, 1/√(π f μ σ); the tool uses the latter (copper 66.1 µm at 1 MHz).
- **David Kuphaldt, Lessons in Electric Circuits, Vol. I, ch. 12, "Specific resistance at 20 °C"** (ibiblio copy, read): copper 1.678, silver 1.587, gold 2.214, aluminum 2.650 µΩ·cm. Other sources list gold at 2.44; copper as wire is sold is the IACS value 1.7241 (1/58 Ω·mm²/m, NBS Handbook 100 / Copper Development Association), as in Wire sizing: 1.3 % deeper than pure copper.
- **The round wire:** the textbook solution of Maxwell's equations for an isolated straight round wire, Z/Rdc = (z/2) J0(z)/J1(z), z = (1 − j) r/δ, real part Rac/Rdc. It is not from a page read; the tool computes it by power series up to r/δ = 12 and by r/(2δ) + 1/4 + 3δ/(32r) above, and both agree with scipy's Bessel functions to 6 figures (r/δ = 7.566: 4.04524; 23.93: 12.21687).

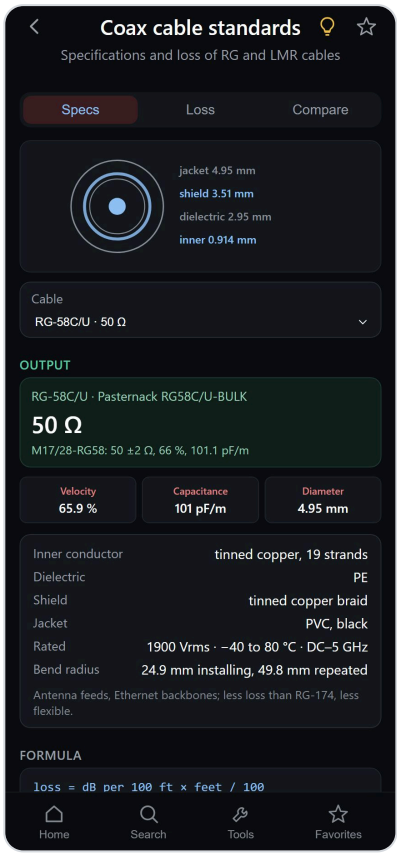
## Assumptions and limits

- **A smooth, uniform, isolated conductor at 20 °C:** resistivity rises 0.4 % per °C; surface roughness (PCB copper) adds loss.
- **Foils and planes:** the foil's DC resistance and Rs are per square for current in the plane; a trace's current runs on one face as well as the edges, so its loss needs a field solver.
- **Magnetic metals:** μr is taken as constant, and iron's falls with frequency.

## How it was checked

12 cases, the expected values from the formula and a Python check (scipy for the Bessel ratios): - copper at 1 MHz: 66.1 µm (2.6 mil), 0.261 mΩ per square, 99 % within 304 µm; silver at 1 GHz 2.005 µm, 7.92 mΩ per square; gold 74.9 µm and aluminum 81.9 µm at 1 MHz; - μr 100 with ρ typed: the picker goes to Custom; - a 1 mm copper wire: at 1 MHz × 4.05, 22 and 88.8 mΩ/m, skin effect from 17.5 kHz, 24.7 % of the area; at 100 kHz × 1.45; at 10 MHz × 12.2 (the large-wire form), 268 mΩ/m; - 1 oz foil: 3.61 MHz; t/δ 0.527 at 1 MHz and 5.27 at 100 MHz; - a zero frequency and a μr under 1 are refused.

# Coax cable standards




## What it computes

The common RG coaxial cables and the Times Microwave LMR family: what each one is, and how much signal it loses. 23 cables, each from a maker's data sheet: 14 RG types that have a row in the M17 table (RG-174A, 316, 58C, 142B, 223, 400, 402, 213, 214, 393 on 50 Ω; RG-59B, 6A, 11A on 75 Ω; RG-62A on 93 Ω), RG-8A/U and RG-8X (no M17 row), and LMR-100A, 195, 200, 240, 400, 600 and the 75 Ω LMR-400. Three pills:

- **Specs:** the cable cut across, to scale (jacket, shield, dielectric, inner conductor, with their diameters). The big result is the impedance, with the M17 standard's own line for that type under it (slash sheet, impedance and tolerance, velocity, capacitance). In three cells: the velocity, the capacitance and the diameter. Under them: the inner conductor, the dielectric, the shield, the jacket, what the sheet rates it for (voltage, temperature, top frequency) and its bend radius. A note says what the cable is for, where a source says (L-com's guide, Times' sheet for LMR-400), and where the sheet differs from the standard.
- **Loss:** the cable, the frequency (MHz or GHz), the length (m, ft, cm or in) and the power in (mW or W). The drawing is the cable's loss against frequency on logarithmic axes, the sheet's figures as dots, the frequency of the run marked. The big result is the loss of the run in dB, with how it was read from the sheet. In three cells: dB per 100 ft, dB per 100 m and the power out. Under them: the sheet's two nearest figures, the output level in dBm, the largest CW power the sheet allows at that frequency, the delay of the run, and a warning when the frequency is over the sheet's top frequency.
- **Compare:** every cable of one impedance (50, 75 or 93 Ω) for the frequency and length you give, best first, with a bar for each. A "~" marks a loss beyond the cable's sheet (an estimate); the cable you picked last is in white.
- **Remembered:** the pill, the cable, the impedance and the three units.

Examples: 30 m of RG-58 at 100 MHz, LMR-400 against RG-213 at 144 MHz, a GPS antenna lead of RG-316, and what RG-6 is.

## Terms

- **RG:** the prefix of the old American military coax series. Today a loose name for a cable of that size and kind, not a specification (see Sources).
- **/U, A, B, C:** "universal", for general use; the letters are revisions of the type (RG-58, RG-58A, RG-58B, RG-58C).
- **MIL-DTL-17 (formerly MIL-C-17), M17 slash sheet:** the military specification for flexible and semi-rigid coax; each cable it defines has a slash sheet, M17/28-RG58 for RG-58, giving its construction and limits.

- **Impedance Z0:** the ratio of voltage to current of a wave on the line; 50 Ω for radio and test gear, 75 Ω for video and cable TV, 93 Ω for some data links.
- **Velocity of propagation (VP, velocity factor):** the share of light's speed at which a wave travels on the cable, 1/√εr; the dielectric sets it.
- **Capacitance:** per metre; with Z0 it fixes the velocity, C = 1/(Z0 VP c).
- **Loss (attenuation):** how much weaker the signal is at the far end of a matched cable, in dB, for 100 ft or 100 m, rising with frequency.
- **Typical, maximum:** a sheet's typical loss is what a sample usually does; a maximum (Pasternack's RG-59B) is a limit it should stay under.
- **CW power:** the largest continuous (not peak) power the cable can carry at that frequency without overheating; it falls as the loss rises.
- **Dielectric withstanding voltage:** the voltage the insulation survives in a test (Times' sheets); not an operating voltage.
- **Bend radius:** how tightly the cable may be bent, once when installing and repeatedly.

## Sources

- **The standard: the M17 table (a distributor's reprint, "MIL-C-17", from Sycor; no date or author; read):** each slash sheet's M17 number and QPL number, conductor, dielectric, shield, jacket, impedance with its tolerance, velocity, capacitance in pF/ft and pF/m, maximum operating voltage, temperature range and test frequency; 14 of its rows are used. It is a transcription, not the specification: it has misprints (RG-58's capacitance reads 38.8 pF/ft beside 101.1 pF/m, RG-11's 20.8 beside 67.6; only 30.8 and 20.6 fit), so the pF/m column is used, and every row's capacitance was checked against 1/(Z0 Vp c): all within 0.5 %. The abbreviations (BC, TC, SC, CCS, SCCS) are not defined in it; they are read in the usual sense (bare, tinned, silver-plated copper; copper-clad steel, silver-plated). The slash sheets themselves, and the DLA's qualified products list, were not read.
- **What "RG" is:** Wikipedia, "Coaxial cable" (read through a text-extraction tool, the first 100 000 characters; no edit date seen): RG was "originally a U.S. military designation from World War II", the numbers listed in MIL-HDBK-216 (1962), "/U" for universal and A, B, C for revisions, superseded by the M17 designation, and no standard guarantees a cable sold as an "RG-# type". Military & Aerospace Electronics, "MIL-DTL-17 Requirements for Hi-Rel/MIL-SPEC Coaxial Cable Assemblies and a Note on RG Coax" (4 March 2019, no byline shown; read): MIL-DTL-17 was MIL-C-17; RG is "a generalized industry term, and not a standardized set of cable specifications"; a cable meets MIL-DTL-17 only if it meets every category, and a MIL-DTL-17 supplier should be on the qualified products list, which readily available RG coax is not.
- **What each is for:** L-com, "RG Coax Cable Applications" (a one-page technical brief, no date; read): typical use, best feature and trade-off of RG174, 188A, 316, 58C, 142B, 59A/B, 6, 223, 213, 179B and 187; used, paraphrased, for the cables it covers (not RG-188A, 179B, 187, 59A, which are not listed here). For LMR-400, Times' own sheet ("Replacement for Air Dielectric type RG8 cable").
- **The cables, Pasternack:** the data sheets of RG174A/U-BULK, RG316/U-BULK, RG58C/U-BULK, RG142B/U-BULK, RG223/U-BULK, RG400/U-BULK, RG402/U-BULK, RG393/U-BULK, RG213/U-BULK, RG214/U-BULK, RG59B/U-BULK, RG6A/U-BULK, RG11A/U-BULK, RG62A/U-BULK, RG8A/U-BULK and RG8X-BULK (PDFs, read as text; revision 1.0 and later as printed): impedance, velocity, capacitance, the typical loss at 3 to 10 frequencies (a maximum for RG-59B), the largest CW power where given, construction, ratings.
- **The cables, Times Microwave:** the LMR-100A (as Fairview Microwave's LMR-100A-PVC, with its -FR and -FR-FT sheets), LMR-195, LMR-200, LMR-240, LMR-400 (REV 1.3, 2020), LMR-600 and LMR-400-75

sheets as Fairview Microwave prints them (read as PDF text): loss and largest CW power at ten frequencies from 50 MHz, and the construction. Times' own site refused automated downloads, so its sheets were not read.
- **Not read:** the MIL-DTL-17 slash sheets; Belden's sheets again (the Coax impedance tool lists seven of them); the MIL-HDBK-216 table.

## Slips found in the sheets, and what was done

- Pasternack RG-142B: capacitance "96.45 [316.44] pF/ft [pF/m]"; the 96.45 is the pF/m figure (29.4 pF/ft). Used as pF/m.
- RG-316 and RG-400: 32 pF/ft (105 pF/m), where their 50 Ω and 69 to 70 % imply 96.7 and 95.3 pF/m. RG-62A: 83 % with 47.57 pF/m, where 83 % implies 43.2 (the standard has 81 % and 44.3). Shown as the sheets print them, with the differences from the standard pointed out in the note.
- RG-6A and RG-62A at 10 MHz: 0.08 and 2.7 dB per 100 ft. Both break the √f law and their own other figures (it gives about 0.86 and 0.82); the two lines are left out and the loss below the first remaining figure goes as √f.
- LMR-100A-PVC's first line (50 MHz: 3.9 dB, 230 W) is LMR-400's 900 MHz line; the -FR and -FR-FT sheets say 5.1 dB and 180 W, which are used.
- RG-8X: no velocity on the sheet; taken from 1/(Z0 C c) = 78.2 % and starred.
- LMR sheets give no shield diameter (except LMR-400): the shield ring is drawn at a default thickness there.
- Pasternack's RG-316 loss at 1 GHz (38 dB) sits about 10 % above the √f and f law through its other figures, and RG-8A/U's 400 MHz figure about 9 % under it: left as printed.

## How the loss is read

Loss in a coax has two parts: the conductors, from the current crowded into their skin, which grows as √f, and the dielectric, which grows as f. Times prints it as K1 √f + K2 f dB per 100 ft, f in MHz (their 75 Ω LMR-400: 0.115570 √f + 0.000260 f). The tool does not fit that law: it reads the maker's own figures, exactly at each, and between two figures it follows a straight line on logarithmic axes (a power law):

```
a(f) = a1 × (a2/a1)^(ln(f/f1) / ln(f2/f1))
```

so the number at a sheet's frequency is the sheet's. Below the first figure the loss goes as √f from it (the conductor term dominates there); above the last it continues along the last segment's slope, held between 0.5 and 1 (the two terms' limits). Those two ranges are estimates and are flagged ("~", and a line saying so). The loss of the run is the loss per 100 ft times the length in feet over 100; the power out is Pin × 10^(−loss/10); the delay is the length over VP × c. The largest CW power is read the same way between the sheet's powers (a single figure is shown as one), and not outside them.

Between the figures the reading differs from the K1 √f + K2 f law through the same figures by 4.1 % or less for 18 of the 23 cables; the other five scatter more in their own figures (RG-316/U 10.2 %, RG-8A/U 8.7 %, LMR-600 6.7 %, LMR-100A 6.0 %, RG-8X 5.3 %). Against Times' own formula for the 75 Ω LMR-400 it stays within 3.6 % (the sheet rounds to 0.1 dB, and the formula and the sheet's lines differ by up to 3.8 % at their own points).

## Assumptions and limits

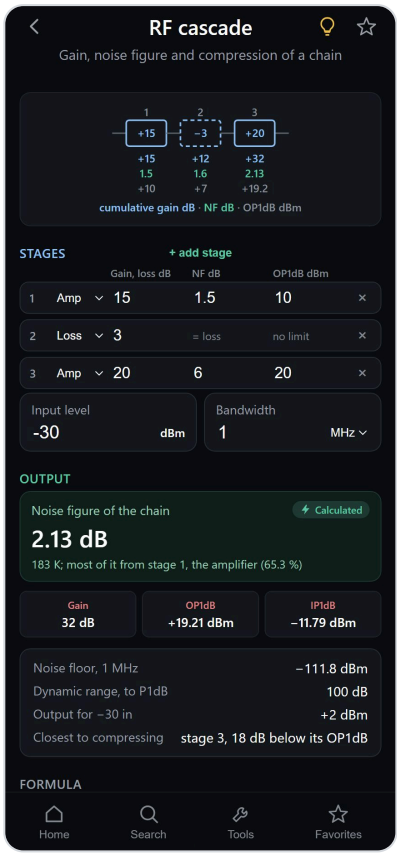
- **Typical figures, not guarantees.** A cable, even from the same maker, differs between samples and batches; the standard sets maxima.
- **The same name is not the same cable.** RG-58's loss at 100 MHz is 4.9 dB per 100 ft on Pasternack's sheet, 5.4 on Belden's 8259 and 3.8 on its 9201 (the Coax impedance tool lists them): the plain name fixes no loss. The M17 row is the only thing that does.
- **A matched line, no connectors.** A mismatch (VSWR) adds loss and connectors add some; the VSWR / Return loss tool counts the first.
- **Power ratings** are the sheet's, for a matched load and ordinary conditions; they are not derated for heat, altitude or mismatch.
- **Frequency ranges** are the sheets': beyond a sheet's top frequency the loss is an estimate and the cable may not be usable (higher modes).
- **23 cables** only; semi-rigid RG-402 is the one tube; no connectors, twin-ax, hard-line or fibre.

## How it was checked

31 cases. The expected values come from a Python twin of the loss maths and the sheets' own figures: - Specs: RG-58C/U (50 Ω, 65.9 %, 101 pF/m, 4.95 mm; M17/28-RG58's line; construction, ratings, bend radius, use), LMR-400 (no M17 row, 85 %, 78.4 pF/m, a withstand voltage), RG-8X (velocity from the capacitance, starred), RG-316/U and RG-62A/U (the differences from the standard named), RG-6A/U (agrees), RG-402/U (a tube); - Loss: RG-58C/U, 10 m at 100 MHz: 4.9 dB/100 ft, 1.61 dB, 691 mW, 28.4 dBm, 50.6 ns; 100 ft: 4.9 dB; 100 mW in; zero length; 1 MHz (√f, 0.443 dB/100 ft, flagged) and 6 GHz (68 dB/100 ft, 22.3 dB, flagged); LMR-400 at a sheet frequency (2.7 dB, 830 W) and at 300 MHz between two (2.21 dB/100 ft, 2.18 dB over 30 m, 1.02 kW); RG-59B's maximum; RG-213/U at 2 GHz over its sheet; RG-6A/U at 10 MHz without the bad line; RG-316/U at 1.575 GHz over 3 m (45.3 dB/100 ft, 4.45 dB); LMR-100A at 50 MHz (5.1 dB, 180 W); RG-8X (3.66 dB, 252 W, 130 ns); no frequency refused; - Compare: 50 Ω at 100 MHz over 10 m (LMR-600 0.254 dB first, RG-316/U 3.61 dB last, three estimates marked), 75 Ω, 93 Ω, 144 MHz over 30 m (LMR-400 1.45 dB, RG-213/U 2.66 dB) and 1.575 GHz over 3 m; - the four examples.

Every cable's figures were also checked before use: its loss figures against the K1 √f + K2 f law (19 of the 23 within 4.5 %; LMR-100A 6.4 %, LMR-600 7.2 % because its 50 MHz line is a rounded 0.5 dB, RG-8A/U 9.0 %, RG-316/U 9.3 %; after the slips above), its capacitance against 1/(Z0 VP c), and, for the M17 rows, the impedance recomputed from the conductor and dielectric diameters and the velocity (1 to 9 % under the stated impedance, as stranding and air spaces make an effective diameter differ: no diameter is grossly misprinted).

# RF cascade




## What it computes

The gain, noise figure and compression of a chain of amplifiers and losses, seen from its input (it takes the place of the planned "Cascade gain, noise figure, P1dB"):

- **Stages:** up to five, each a row: Amp (gain in dB, noise figure in dB, OP1dB in dBm, left blank when unknown) or Loss (an attenuator, a filter or a cable: its loss in dB; its noise figure is its loss and it does not compress). "+ add stage" adds one, × removes one. A mixer is an Amp with a negative gain (its conversion loss) and its noise figure.
- **System:** the input level (dBm) and the bandwidth (Hz to GHz).
- **The big result:** the chain's noise figure, with its noise temperature and the stage that adds most of the noise and its share.
- **Three cells:** the chain's gain, its OP1dB (output-referred 1 dB compression point) and its IP1dB (input-referred).
- **Below them:** the noise floor at the input in that bandwidth, the dynamic range from the floor to the input P1dB, the output level for the input level, and the stage closest to compressing with its margin, or a warning when one is over its OP1dB.
- **Drawing:** the chain as boxes, each stage's own gain inside (dashed for a loss), and under each, the cumulative gain (blue), noise figure (green) and OP1dB (grey) up to and including that stage.
- **Remembered:** the bandwidth unit.

Examples: Rohde & Schwarz's two amplifiers, a 3 dB cable before and after a low-noise amplifier, and a receiver front end.

## Terms

- **Noise factor F, noise figure NF:** F is how much a stage worsens the signal-to-noise ratio, as a ratio; NF = 10 log F, in dB. A noiseless stage has NF 0 dB.
- **Gain G:** output over input power; in a chain, gains multiply (their dB add). A loss L dB is a gain of −L dB.
- **Friis equation:** how the stages' noise factors combine: each is divided by the gain ahead of it, so the first stage dominates.
- **Noise temperature Te:** the same noise as a temperature, Te = 290 K × (F − 1); unlike F it adds across stages.
- **Noise floor:** the thermal noise k T B in the bandwidth, raised by the noise figure, referred to the input (−174 dBm/Hz + 10 log B + NF).
- **P1dB:** the power at which gain has dropped by 1 dB. OP1dB is measured at the output, IP1dB at the input: OP1dB = IP1dB + G − 1 dB.
- **Dynamic range:** here, from the noise floor up to the input P1dB.

## Sources

- **Rohde & Schwarz, application note 1SL378, "The Cold Source Technique for Noise Figure Measurements", Mike Leffel and Christopher Stumpf, version 0e, December 2021** (read): the Friis cascade equation (39), F = F1 + (F2 − 1)/G1 + … + (FN − 1)/(G1 … GN−1), in linear noise factor and gain; the noise-temperature form Te = T1 + T2/G1 + … (40) and Te = T0 (F − 1) (9) with T0 = 290 K; NF = 10 log F (3); the noise density −174 dBm/Hz (21); "a 3 dB attenuator has −3 dB of gain, or +3 dB of loss" (8); and its example 3.7.2, 20 dB at 2 dB then 20 dB at 11 dB, 2.31 dB, which the tool reproduces. It also notes that equations (38) and (39) hold only when the input noise is the standard N0 = k T0 B.
- **A loss's noise figure:** the note says a passive device's noise temperature is its physical temperature (so at 290 K, Te = T0 (L − 1), and F = L): the equality NF = loss is derived from it, not quoted.
- **The P1dB cascade:** RF Essentials, "What is Cascaded P1dB?" (a web glossary page, no author or date shown; read): the reciprocal sum of the stages' output compression points, each divided by the gains after it, "≈", for "worst-case in-phase compression", the last stages dominating, and OP1dB = IP1dB + Gain − 1 dB; RF Cafe, "Cascaded P1dB" (undated; read, its formula is an image): apply the cascaded intercept-point equation to each device's P1dB as an estimate, the same conversion, and "a well-known rule-of-thumb is to subtract 10 to 15 dB to the IP3 value to estimate the P1dB value". These are website pages, not standards or application notes: the formula is the standard cascaded-intercept sum.
- **Not read:** Friis's original paper (1944), and Analog Devices' "System Noise-Figure Analysis for Modern Radio Receivers" (gated; only its summary, which warns that the cascade equation changes with mixers).

## Why the compression formula is this one

Taking the chain stage by stage from the input, with Gk the linear gain of stage k and OPk its OP1dB in mW:

```
1/OP(1) = 1/OP1
1/OP(k) = 1/OPk + 1/(OP(k−1) × Gk)
```

so an earlier stage's limit counts for less when the stages after it have gain (it is referred to the output, 1/(OP(k−1) × Gk)), and a stage with no limit (a loss, or a blank) adds nothing but still scales the earlier ones.

## Assumptions and limits

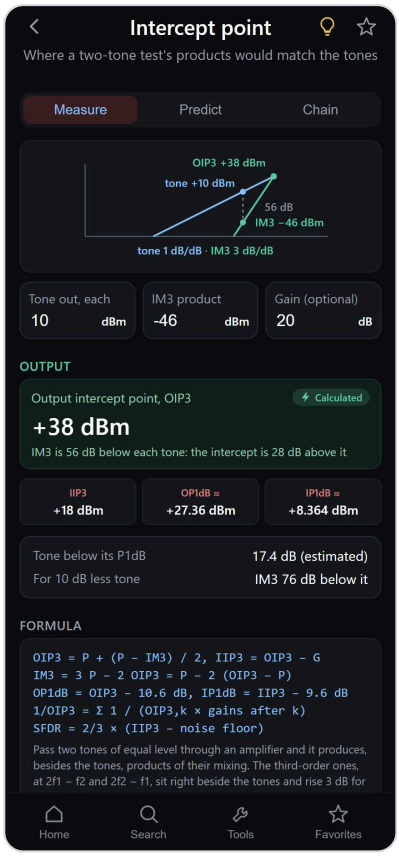
- **Matched, linear in the small-signal sense:** gains and noise figures are the available gains and noise figures into 50 Ω with the source at 290 K; mismatch between stages changes them.
- **A loss is at 290 K.** A warmer pad adds more noise.
- **The noise figure of a mixer** depends on whether the image is filtered (single-sideband) or not (double-sideband, 3 dB lower): enter the one that matches your receiver.
- **The P1dB cascade is an estimate,** pessimistic by assuming every stage compresses in phase; it holds only when none of the stages operates far into compression. Check each stage's own margin (the last row).
- **Dynamic range** here is to P1dB; spur-free dynamic range uses the third-order intercept (see Intercept point).

## How it was checked

12 cases, the expected values from a Python implementation of the same formulas, and Rohde & Schwarz's example: - the default chain, an LNA (15 dB, 1.5 dB, +10 dBm), a 3 dB loss and an amplifier (20 dB, 6 dB, +20 dBm): NF 2.13 dB (2.1275), 183 K, 32 dB, OP1dB +19.21 dBm, IP1dB −11.79 dBm, floor −111.8 dBm in 1 MHz, 100 dB range, +2 dBm out for −30 in, stage 3 18 dB below its OP1dB; - Rohde & Schwarz's two amplifiers: 2.31 dB, 40 dB; - a 3 dB cable before a 1 dB LNA, 4.0 dB; after it, 1.03 dB (1.0342); - LNA, 3 dB pad, mixer (−6 dB, 6 dB, +5 dBm), IF amplifier (18 dB, 4 dB, +15 dBm): NF 3.04 dB, 24 dB, OP1dB +13.08 dBm and IP1dB −9.92 dBm. RF Essentials' table for the compression of the same chain reads +10.00, +7.00, −0.55 and +13.15 dBm per stage where the sum gives +10.00, +7.00, −0.455 and +13.08: a rounding difference of under 0.1 dB; - a first-stage noise figure of 3 dB: 3.45 dB; 0 dBm in flags stage 3, 12 dB over its OP1dB; adding then removing a stage; a blank gain, a noise figure under 0 dB and a zero bandwidth refused; kHz re-expresses the bandwidth.

RF & MICROWAVE › POWER & GAIN

# Intercept point




## What it computes

How linear an amplifier, or a whole chain of stages, is, summed up by its third-order intercept point (IP3), three ways (pills):

- **Measure** (you have run a two-tone test): the level of each tone at the output and the level of one third-order product (IM3), both in dBm, and the gain in dB (optional). The big result is the output intercept point (OIP3), with how far the product sits below each tone. Under it: the input intercept point (IIP3, needs the gain), the estimated 1 dB compression points OP1dB and IP1dB, how far the tone is below that compression, and how far down the product would be for 10 dB less tone.
- **Predict** (you know the IP3, say from a data sheet): IIP3, Gain and OIP3 are linked boxes (type any two and the third follows; the intercept you typed last is the one held when the gain changes), the level of each input tone and the IM3 rejection you want, in dB. The big result is the IM3 product at the output, with its rejection. Under it: the output tone, the product as it appears at the input, OP1dB; then the largest tone that still gives the rejection you asked for (at the output and at the input), and how far the tone is below its compression.
- **Chain** (several stages): up to five rows, each Amp (gain in dB, OIP3 in dBm, left blank when unknown) or Loss (an attenuator, a filter or a cable: its loss in dB, with no distortion of its own); "+ add stage" adds a row, × removes one. Then the input tone level and the noise floor at the input, in dBm. The big result is the chain's OIP3, with the stage that adds most of the distortion and its share. Under it: the gain, the IIP3 and the spur-free dynamic range (SFDR); the IM3 product for that input level (at the output and in dBc) and the tone at the output. Each row shows the chain's OIP3 up to and including that stage.
- **Drawing:** Measure and Predict: the tone line (blue, 1 dB per dB) and the IM3 line (green, 3 dB per dB) meeting at the intercept, with the tone and product levels marked and the gap between them, the rejection; not to scale. Chain: the stages as boxes (dashed for a loss), each stage's own gain inside, and under each the cumulative gain (blue) and OIP3 (green).
- **Remembered:** the pill.

Examples: Keysight's amplifier (−10 dBm in, 20 dB gain, OIP3 +38 dBm), the same figures worked backwards from a two-tone test, a receiver's input IP3 with two −40 dBm interferers, and an LNA, a pad and an amplifier in a chain.

## Terms

- **Two-tone test:** two equal sine waves, close in frequency (f1 and f2), applied to the device. The output shows the two tones and, beside them, their intermodulation products.

- **IM3:** the third-order products, at 2f1 − f2 and 2f2 − f1. Their level rises 3 dB for every 1 dB the tones rise (the tones rise 1 dB).
- **dBc:** dB relative to the carrier, here to each tone: an IM3 of −56 dBc is 56 dB below the tone.
- **Third-order intercept point (IP3, TOI):** the level at which the tone line and the IM3 line, each extended along its small-signal slope, would meet. It is never reached, because the device compresses first: a figure of merit, the higher the more linear.
- **OIP3, IIP3:** the intercept read on the output power axis or on the input one. They differ by the gain: IIP3 = OIP3 − G.
- **P1dB:** the power at which the gain has fallen by 1 dB. OP1dB is read at the output, IP1dB at the input: OP1dB = IP1dB + G − 1 dB.
- **Spur-free dynamic range (SFDR):** the range of input tone levels from the noise floor up to where the IM3 product reaches that floor.
- **Noise floor:** the noise power in the bandwidth of interest, referred to the input (the RF cascade tool computes it).

## Sources

- **Agilent (now Keysight) application note 1408-9, "Amplifier and CW Swept Intermodulation-Distortion Measurements" (publication 5988-9474EN; read):** the products at n f1 + m f2; the third-order ones as the concern because they sit beside the tones and rise three times as fast; and, for two equal tones, IP3 (dBm) = the output power of a tone + ½ × (that power − the larger third-order product), in dB. It requires the measurement to be made in the amplifier's linear region. Its measured examples (a −1.258 dBm tone with a −42.03 dBm product: 19.13 dBm; −0.6 and −39.3: printed as 19, this gives 18.75) are reproduced. Its example amplifier has a 1 dB compression point of +3 dBm and an intercept of +14 dBm, 11 dB apart; the model here puts them 10.6 dB apart.
- **Keysight application note 1408-17, "Making Accurate Intermodulation Distortion Measurements with the PNA-X Network Analyzer" (5989-7265EN; read), appendix B:** an amplifier with an OIP3 of +38 dBm and 20 dB of gain, driven at −10 dBm per tone: +10 dBm out, the product at −46 dBm, −56 dBc. The tool reproduces it (the first example).
- **The compression point:** Wikipedia, "Third-order intercept point" (an encyclopedia page, not a standard; read): for a weakly nonlinear odd-order power-series model, the 1 dB compression point lies about 9.6 dB below the intercept (a factor 0.10875, −9.636 dB), and "approximately 10 dB" below it as a rule of thumb for many amplifiers; the page cites no source for either. The 9.636 dB is re-derived below and used as is. RF Cafe, "Cascaded P1dB" (read, see RF cascade) gives the rule of thumb as 10 to 15 dB below the IP3.
- **The cascade:** RF Essentials, "What is Cascaded IP3?" (a web glossary page, no author or date shown; read): 1/OIP3 = 1/OIP3,n + 1/(OIP3,n−1 × Gn) + 1/(OIP3,n−2 × Gn × Gn−1) + …, in linear power; products adding in phase degrade the linearity fastest, and design tools assume that worst case. Its worked table for four stages (an LNA of 16 dB and +22 dBm, a filter of −1.5 dB, a mixer of −6 dB and +18 dBm, an IF amplifier of 12 dB and +25 dBm) lists the cumulative OIP3 as +22.00, +20.50, +12.65 and +10.90 dBm. The first two follow its formula; the third is 0.25 dB under the formula's +12.90 and the last 11 dB under its +21.94. The tool follows the formula and treats the table's last entry as an error. The page was read through a text-extraction tool, not as raw HTML, so a transcription slip cannot be ruled out.
- **Derived here, not quoted:** the relations in the next section, the share of each stage, and the SFDR = ⅔ (IIP3 − noise floor), a form in common use for which no source was read.

- **Not read:** Analog Devices' tutorial 749, "Use Selectivity to Improve Receiver Intercept Point" (the fetch failed), and any standard that defines the intercept point.

## Why the formulas are these

For a weakly nonlinear stage, output = a1 x + a3 $x^3$ with a3 < 0 (it compresses). Two equal tones of amplitude A give a fundamental a1 A and an IM3 product of amplitude ¾ |a3| $A^3$, so product over tone = ¾ (|a3|/a1) $A^2$, which grows with the square of the tone: 2 dB per dB of amplitude, 3 dB per dB of power against the tone's 1. It reaches 1 at $A^2$ = 4 a1 / (3 |a3|), the intercept. In dB, with P the tone level and IP3 the intercept on the same axis, the product sits 2 (IP3 − P) below the tone:

```
IM3 = 3 P − 2 IP3
IP3 = P + (P − IM3) / 2        (Keysight's formula)
```

A single tone is compressed by the same term: its gain falls by the factor 1 − $A^2$/$A_{IP}^2$. For 1 dB, 1 − $A^2$/$A_{IP}^2$ = 10^(−1/20), so $A^2$/$A_{IP}^2$ = 0.10875, which is −9.636 dB. Hence IP1dB = IIP3 − 9.636 dB and, with the 1 dB of gain lost, OP1dB = OIP3 − 10.636 dB.

For a chain, stage k with input intercept IIP3,k and the gain Gprev ahead of it puts a distortion voltage at the chain's output proportional to Gprev / IIP3,k, which is 1/(OIP3,k × the gain after k) once referred to the output. Taken in phase (the worst case), the voltages add, and so do these terms:

```
1/OIP3 = Σ 1 / (OIP3,k × G(k+1) × … × Gn)
```

A stage with no limit (a loss, or a blank OIP3) adds no term, but its gain still weighs the stages before it. Taken from the input, stage by stage:

```
1/OIP3(k) = 1/OIP3,k + 1/(OIP3(k−1) × Gk)
```

which gives the cumulative figure on each row. A stage's share is its term over the sum.

The product referred to the input is 3 P − 2 IIP3. It reaches the noise floor N when P = (2 IIP3 + N)/3, so the range from N up to there is (2 IIP3 + N)/3 − N = ⅔ (IIP3 − N).

## Assumptions and limits

- **Two equal tones, close together,** in a device without memory (the intercept does not depend on their spacing).
- **The 3:1 slope holds only well below compression.** Near the P1dB the product stops following the line. The tool flags a tone within 10 dB of its estimated P1dB, and one over it, in amber.
- **The compression point is an estimate.** The cubic model puts it 10.6 dB under the OIP3; real amplifiers are 10 to 15 dB under, Keysight's example 11 dB. If the data sheet gives the P1dB, trust that.
- **Third order only.** Fifth-order terms bend the product's line near compression and are not modelled.
- **The chain sum is the in-phase worst case.** If the stages' products had random phases they would add less: for two equal contributions the root-sum-square would give 1.5 dB more OIP3 than the in-

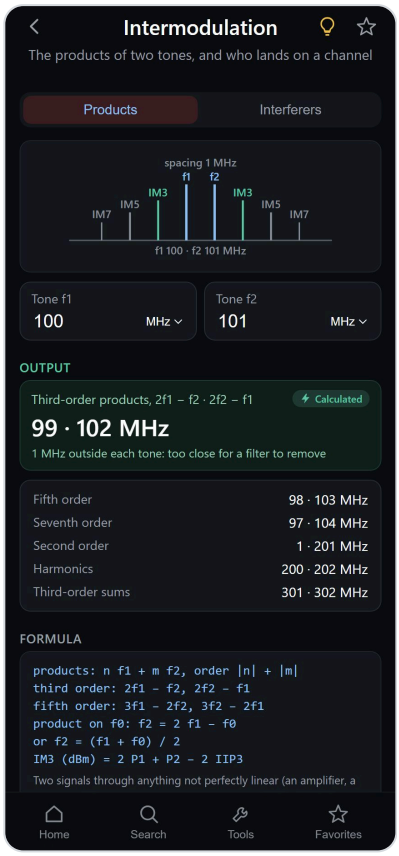
phase sum.

- **A loss has no limit here.** A passive part is far more linear than an amplifier (RF Essentials' example gives its filter +100 dBm) and is entered as no limit. A mixer is an Amp with its conversion loss as a negative gain and its own OIP3.
- **The noise floor is yours to choose,** and the SFDR follows it: take it for the bandwidth you need (the RF cascade tool gives it).

## How it was checked

14 cases; the expected values come from a Python implementation of the same formulas and from the published examples: - Measure, +10 dBm tones with a −46 dBm product and 20 dB of gain: OIP3 +38 dBm, 56 dB below the tone, IIP3 +18 dBm, OP1dB +27.36 dBm, IP1dB +8.364 dBm, the tone 17.4 dB under its P1dB, the product 76 dB down for 10 dB less tone; the same without the gain (the input figures show "—"); a +25 dBm tone with a +20 dBm product flagged 8.14 dB over its estimated P1dB; a product as strong as the tone refused; - Keysight AN 1408-9's example, −1.258 dBm and −42.03 dBm: +19.13 dBm; - Predict, Keysight's amplifier: −46 dBm product, 56 dB down, −66 dBm at the input, the largest tone for 50 dB +13 dBm out and −7 dBm in; typing an IIP3 of 18 with 20 dB gives an OIP3 of 38, and a gain of 10 then keeps the IIP3 and moves the OIP3 to 28; a receiver with an IIP3 of −10 dBm and −40 dBm tones: −100 dBm, 60 dB down; a blank gain refused; - Chain, an LNA (15 dB, +25 dBm), a 3 dB loss and an amplifier (20 dB, +30 dBm): 32 dB, OIP3 +29.73 dBm, IIP3 −2.266 dBm, SFDR 71.8 dB for a −110 dBm floor, 94.1 % of the distortion from stage 3, a −83.47 dBm product (−75.5 dBc) for −40 dBm in; RF Essentials' four stages: +21.94 dBm by the formula (the table says +10.90), +20.5 and +12.9 dBm after stages 2 and 3, IIP3 +1.438 dBm; no OIP3 entered shows "—" and "nothing limits the chain"; a negative loss refused; adding then removing a stage.

# Intermodulation




## What it computes

What two signals leave behind in anything not perfectly linear, and which pair of signals lands on a channel. Two pills:

- **Products:** the frequencies of two tones f1 and f2 (in either order; one unit picker, Hz to GHz, sets both). The big result is the third-order products, 2f1 − f2 and 2f2 − f1, with how far outside each tone they sit (one tone spacing). Under it: the fifth order (3f1 − 2f2, 3f2 − 2f1), the seventh order (4f1 − 3f2, 4f2 − 3f1), the second order (f2 − f1, f1 + f2), the harmonics (2f1, 2f2) and the third-order sums (2f1 + f2, f1 + 2f2). A product that falls below 0 Hz is shown at its mirror image, marked *.
- **Interferers:** the channel you listen on (f0) and one interferer (f1). The big result is the frequency of a second interferer that puts a third-order product on the channel: 2f1 − f2 = f0 gives f2 = 2f1 − f0 (A, as far beyond f1 as f1 is from the channel) and 2f2 − f1 = f0 gives f2 = (f1 + f0)/2 (B, halfway between the channel and f1); a value at or below 0 Hz shows "—". Under it: the fifth-order partners, (3f1 − f0)/2 and (f0 + 2f1)/3. With the receiver's IIP3, the level of each interferer and the noise floor (all in dBm, at the receiver's input): the product's level at the input (3 × level − 2 × IIP3), how far it sits above or below the floor, and how far the floor rises when the product adds to it ("unchanged" under 0.05 dB).
- **Drawing:** Products: the spectrum as lines, the tones (blue), IM3 (green), IM5 and IM7 (grey). Interferers: the channel (green), the interferer (blue), and the partners A and B (dashed blue).
- **Remembered:** the pill and the frequency unit.

Examples: two close tones at 100 and 101 MHz, Keysight's PNA-X tones (5.00001 and 5.00011 GHz), and an interferer at 100.2 MHz near a 100 MHz channel.

## Terms

- **Intermodulation:** the mixing of two or more signals in a nonlinear device (an amplifier, a mixer, a diode, a corroded joint): the output holds signals at the sums and differences of their multiples.
- **Product, order:** n f1 + m f2 for whole numbers n and m, of order |n| + |m|. IM3, IM5 and IM7 are the odd orders with the tones' difference (2f1 − f2, 3f1 − 2f2, 4f1 − 3f2 and their mirror images).
- **Harmonic:** a multiple of one tone alone (2f1, 2f2).
- **Interferer:** an unwanted signal near the channel; two of them can put a product on it.
- **Mirror image (*):** a product at −f is the same tone as one at +f, so it is shown at f with a star.
- **IIP3:** the receiver's input intercept (see Intercept point).

## Sources

- **Agilent (now Keysight) application notes 1408-9 and 1408-17 (see Intercept point; read):** the output holds signals at n f1 + m f2 for whole n and m; the third-order products 2f2 − f1 and 2f1 − f2 are the concern because they sit close to the tones and so cannot be filtered out. Their example, tones at 5.00001 and 5.00011 GHz, gives products at 4.99991 and 5.00021 GHz, which the tool reproduces. For tones of different level, note 1408-17 gives the high-side product as P(f1) − 2 × (IP3 − P(f2)) and the low-side one as P(f2) − 2 × (IP3 − P(f1)), at the receiver: the tool's level for equal interferers is the same formula.
- **Derived here, not quoted:** the definition of the order as |n| + |m| (the notes name the third-order products and give n f1 + m f2 but do not define the order), the fifth- and seventh-order products, the second-order products and sums, and the partner frequencies of the Interferers pill (the third-order equation solved for the second tone).
- **Not read:** a standard that defines intermodulation orders.

## Why the frequencies are these

A nonlinearity of order N makes terms of the form (A cos ω1t + B cos ω2t)^N. Each product of cosines splits into cosines at the sum and the difference of their frequencies, so the output holds n f1 + m f2 with |n| + |m| up to N. With Δ = f2 − f1, the odd-order products with opposite signs land at

```
2f1 − f2 = f1 − Δ        2f2 − f1 = f2 + Δ
3f1 − 2f2 = f1 − 2Δ      3f2 − 2f1 = f2 + 2Δ
4f1 − 3f2 = f1 − 3Δ      4f2 − 3f1 = f2 + 3Δ
```

equally spaced by Δ either side of the pair: the picture the tool draws. That is why no filter that passes the tones can stop them.

Backwards, a second interferer f2 puts IM3 on the channel f0 when 2f1 − f2 = f0, so f2 = 2f1 − f0, or when 2f2 − f1 = f0, so f2 = (f1 + f0)/2. The same for the fifth order: 3f1 − 2f2 = f0 gives f2 = (3f1 − f0)/2, and 3f2 − 2f1 = f0 gives f2 = (f0 + 2f1)/3.

The product at the receiver's input, for two interferers of level P each and an input intercept IIP3, is 3 P − 2 IIP3 (see Intercept point). The floor is the noise N plus that product in power: the rise is 10 log(1 + 10^((IM3 − N)/10)), which is 3 dB when the product equals the noise and 10.4 dB when it is 10 dB above it.

## Assumptions and limits

- **Equal levels:** both interferers are taken at the same level; for unequal ones the product is 2 P1 + P2 − 2 IIP3 for 2f1 − f2 (two of the tone whose frequency is doubled).
- **The 3:1 rule** holds below compression only; with strong interferers the real product can differ.
- **The intercept is the receiver's, at its input.** If a filter before the first nonlinear stage removes an interferer, it makes no product; if the gain ahead of that stage is high, use the chain's IIP3 (Intercept point).
- **Orders above the third** are listed by frequency only; their levels are not computed. Second-order products (f2 − f1, f1 + f2) and harmonics need the second-order intercept, which the app does not

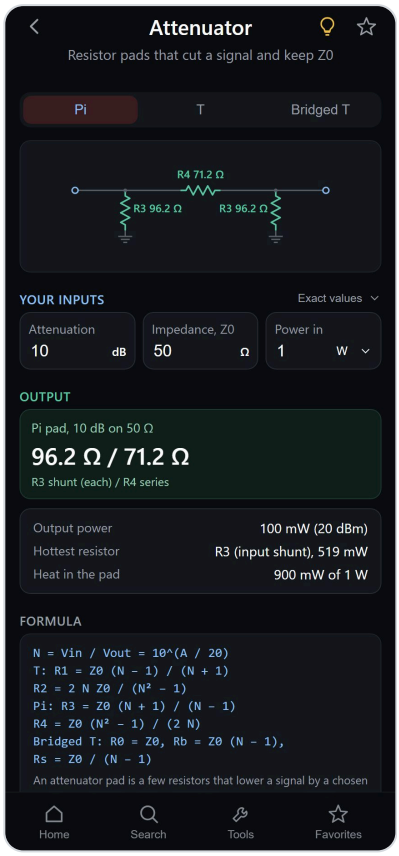
compute.

- **The tool does not know your receiver's passband,** so it does not say whether a product falls in it.

## How it was checked

10 cases; the expected values come from a Python implementation of the same equations and from Keysight's example: - 100 and 101 MHz: IM3 at 99 and 102 MHz, IM5 at 98 and 103, IM7 at 97 and 104, second order 1 and 201, harmonics 200 and 202, third-order sums 301 and 302; the tones typed in the other order give the same; - Keysight AN 1408-17's tones, 5.00001 and 5.00011 GHz: 4.99991 and 5.00021 GHz (and IM5 4.99981 · 5.00031, IM7 4.99971 · 5.00041); - 10 and 100 MHz: an IM3 below 0 Hz shown as 80* and the other at 190 MHz; two equal tones refused; - a 100 MHz channel with an interferer at 100.2 MHz: 100.4 or 100.1 MHz, fifth order 100.3 and 100.1333, with −40 dBm interferers and an IIP3 of −10 dBm an IM3 of −100 dBm, 10 dB over a −110 dBm floor, which rises 10.4 dB; at −60 dBm the product is −160 dBm, 50 dB under the floor, which does not move; an interferer at 30 MHz has no partner below the channel, only 65 MHz; one on the channel refused; kHz re-expresses the frequencies.

# Attenuator




## What it computes

The resistors of a pad that cuts a signal by a set number of dB while looking like the line's impedance Z0 from both ends:

- **Three pills:** Pi (a series resistor and a shunt resistor to ground at each end), T (a series resistor in each arm and one to ground) and Bridged T (two arms of Z0, a bridge across them and a shunt).
- **Inputs:** the attenuation (dB), the impedance Z0 (50, 75 or 600 Ω typically) and the input power (mW or W). The resistors' tolerance is picked in the "Your inputs" heading, as in the Passive components tools: Exact values, or ±0.1 % to ±10 %, each with its E series (±1 % is E96, ±5 % E24, ±10 % E12).
- **The big result:** the two values that define the pad (T: R1 and R2, Pi: R3 and R4, Bridged T: Rb and Rs, whose two arms are Z0).
- **With a tolerance picked:** the big result is the nearest values of its E series; below it come the ideal values, then the pad's real loss and its input return loss with the standard parts (the analysis of the actual circuit); the bridged T's arms of Z0 are rounded too.
- **Below it:** the output power in W and dBm, the hottest resistor and its dissipation (to pick the resistors' rating), and the heat in the whole pad; over 20 dB, a warning to split the pad in two.
- **Drawing:** the pad as a schematic, US zigzag resistors in green with their values (the standard ones when picked), ports in blue.
- **Remembered:** the pill, the series and the power unit.

Examples: a 10 dB Pi pad on 50 Ω, the T pad of the same loss (Kuphaldt's), the Pi pad in E96 values, a 20 dB bridged T on 75 Ω.

| Pad (Z0) | T: R1, R2 | Pi: R3, R4 | Bridged T: Rb, Rs |
|---|---|---|---|
| 3 dB (50 Ω) | 8.55, 141.9 Ω | 292.4, 17.61 Ω | 20.63, 121.2 Ω |
| 10 dB (50 Ω) | 25.97, 35.14 Ω | 96.25, 71.15 Ω | 108.1, 23.12 Ω |
| 20 dB (50 Ω) | 40.91, 10.10 Ω | 61.11, 247.5 Ω | 450, 5.556 Ω |
| 6 dB (75 Ω) | 24.92, 100.4 Ω | 225.7, 56.03 Ω | 74.64, 75.36 Ω |

## Terms

- **Attenuation A (dB):** the power ratio's 10 log; N = 10^(A/20) is the voltage ratio Vin/Vout (3.16 at 10 dB, 10 at 20 dB).
- **Constant-impedance pad:** a pad whose input sees Z0 when its output has a Z0 load, and the reverse; T and Pi pads are symmetric, so they work either way round.

- **Return loss:** how many dB the reflection at the pad's input is below the incident wave (see VSWR / Return loss); infinite for an exact pad.
- **Shunt, series:** a resistor to ground, and one in the signal's path.
- **Heat:** the power that does not reach the load is dissipated in the resistors, mostly the one nearest the input.

## Sources

- **Ian Poole, "Resistive Attenuator Pads: Pi & T Circuit Design", Electronics Notes** (read): N = Vin/Vout; T: R1 = R0 (N − 1)/(N + 1) and R2 = R0 · 2N/($N^2$ − 1); Pi: R3 = R0 (N + 1)/(N − 1) and R4 = R0 ($N^2$ − 1)/(2N); the advice not to attempt more than about 20 dB in one section, and that bridged T pads suit PIN-diode variable attenuators.
- **Kuphaldt, Lessons in Electric Circuits Vol. III ch. 1, "Attenuators"** (ibiblio copy, read; its formulas are images, so the text gives only the values): the pads from the voltage ratio K, the 50 Ω, 10 dB T pad of R1 = 25.97 Ω and R2 = 35.14 Ω, and Zin = R1 + (R2 || (R1 + Z)) = 25.97 + 24.03 = 50 Ω, which the tool reproduces; and "multiply all R values by the ratio (600/50) to correct for 600 Ω matching".
- **Bridged T:** the arms R0 = Z0, the bridge Rb = Z0 (N − 1) and the shunt Rs = Z0/(N − 1) were not found in readable text; they are derived here and checked by nodal analysis of each pad at 1, 3, 6, 10, 20 and 30 dB: the input impedance is Z0 and the loss A to every figure shown. The same analysis gives the power in each resistor and the results with standard values.

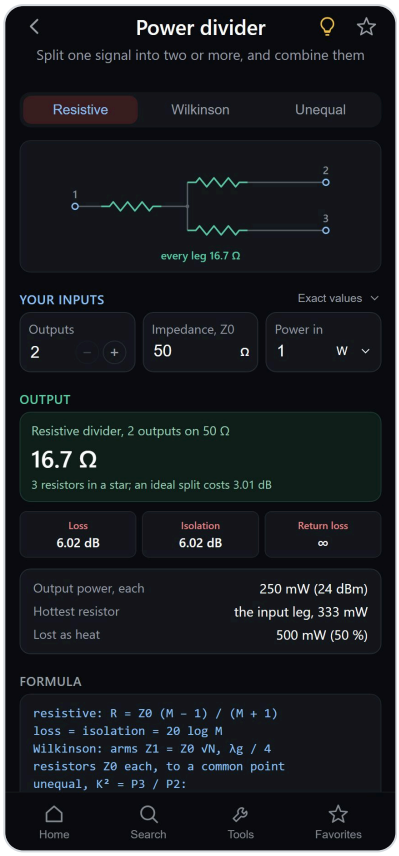
## Assumptions and limits

- **Equal source and load impedance Z0, resistive:** no minimum-loss pad between different impedances.
- **Ideal resistors:** at high frequency a resistor's stray inductance and capacitance, the board's layout and the shielding between sections set the real loss; use small non-inductive surface-mount parts.
- **Power ratings:** the hottest resistor's dissipation at the input power entered; derate for temperature and for peaks.
- **The T pad's heat sits mostly in R1 (52 % of the input at 10 dB), the Pi pad's in the input shunt, R3.**

## How it was checked

11 cases, the expected values from the formulas above and a Python nodal analysis (input impedance and loss at 10, 20 and 30 dB): - the 10 dB Pi pad on 50 Ω at 1 W: 96.2 and 71.2 Ω, 100 mW (20 dBm) out, 519 mW in the input shunt, 900 mW of heat; - Kuphaldt's T pad: 26 and 35.1 Ω; - the Pi pad in E96 (95.3 and 71.5 Ω): 10.05 dB and 53.8 dB return loss; the T pad in E24 (27 and 36 Ω): 10.2 dB and 36.4 dB; - the 20 dB bridged T on 75 Ω at 100 mW: 75, 675 and 8.33 Ω, 1 mW out, 81 mW in the input arm; - 600 Ω: 312 and 422 Ω; 1 dB T: 2.88 and 433 Ω; 30 dB in one Pi pad: 53.3 and 790 Ω, flagged; - mW re-expresses the power; 0 dB and over 100 dB are refused.

# Power divider




## What it computes

How to split one signal into two or more shares, and combine them back, with every port matched to Z0. Three pills:

- **Resistive:** a star of resistors: the input and each of 2 to 16 outputs are joined to one centre node through a resistor R. The big result is R, the same in every leg. In three cells: the loss from the input to each output, the isolation between two outputs and the worst return loss. Under them: the power out of each output, the hottest resistor (the input leg) and the power lost as heat, for the input power you give.
- **Wilkinson:** 2 to 8 equal outputs on quarter-wave arms. The big result is the arm impedance, with the quarter-wave length for your frequency and line (millimetres and inches, or centimetres, metres and feet for long lines). The cells: the isolation resistor (one resistor for two outputs, a star of N resistors of Z0 for more), the loss (the split itself, 10 log N dB) and the 20 dB band as ± a percentage of the centre frequency. Under them: the return loss and isolation at the centre frequency (∞ with exact values), the band as a frequency range, and the heat in the resistor if an output opens or shorts.
- **Unequal:** a two-way Wilkinson with the split you ask for, P3/P2 from 0 to 20 dB (port 3 gets the larger share). The big result is the two arm impedances (to port 2 and to port 3); the cells: the isolation resistor and the two output lines Z04 and Z05 that bring each output back to Z0; under them: the loss to each port, the match and isolation, the 20 dB band and the heat as above.
- **Inputs:** the number of outputs (type it, or step it with the − and + in its box; hold to repeat; a Wilkinson takes 8 at most, so a larger count comes down to 8 when you change pill) or the split, Z0, the power in (mW or W) and, for the two quarter-wave pills, the frequency (kHz to GHz) and the line's εeff and velocity factor (linked boxes: VF = 1/√εeff). To combine signals instead of splitting one, the outputs are the inputs. The tolerance at the top of the inputs picks the E series for the resistors: "Exact values" shows the ideal ones, any series shows the loss, the match and the isolation the standard resistor really gives.
- **Drawing:** the circuit, ports numbered, the resistors and quarter-wave lines in green. Up to four outputs are drawn; more are drawn as two, dots and the last.
- **Remembered:** the pill, the tolerance and both units.

Examples: a 2-way resistive divider on 50 Ω, the same in E24 resistors, a Wilkinson for 1 GHz on 50 Ω, a 3-way Wilkinson on 75 Ω at 100 MHz and a 2:1 unequal split.

## Terms

- **Divider, splitter, combiner:** one network. Run backwards, the outputs are inputs and the input is the output.

- **Port:** a connection to the outside, each matched to Z0. Port 1 is the input, ports 2 and up the outputs.
- **Loss (insertion loss):** how much weaker the signal is at an output than at the input, in dB. It includes the unavoidable split: 3.01 dB for two equal outputs, 10 log N for N.
- **Isolation:** how much weaker a signal is at one output when it is applied to another. High is good.
- **Return loss:** how much weaker the reflected wave is than the wave that entered a port, in dB. ∞ is a perfect match.
- **Quarter-wave line:** a line a quarter of a wavelength long at the centre frequency (90°). It turns a load R into $Z^2/R$, Z being its own impedance.
- **Isolation resistor (Wilkinson):** the resistor across the outputs. It carries no current while the outputs are equal and in step, and absorbs the difference when they are not.
- **εeff, velocity factor:** the effective dielectric constant a line sees, and the share of light's speed a wave travels at: VF = 1/√εeff.
- **K:** √(P3/P2), the unequal split's amplitude ratio ($K^2$ = P3/P2).
- **20 dB band:** the frequencies round the centre where every return loss and every output-to-output isolation is at least 20 dB.

## Sources


- **Marki Microwave, "6 Ways to Make an N-Way Power Splitter" (application note, posted 24 June 2015, modified 13 July 2023; read):** the direct N-way wye resistive splitter, R = Z0 (N − 1)/(N + 1), N the number of outputs; the branched resistive splitters lose 6 dB per split and the direct wye as much; and the Wilkinson dissipates nothing in its isolation resistors in splitting use. It gives no isolation figure for the resistive types and no N-way Wilkinson impedances.
- **University of Utah, lecture notes L23, "ECE 6130 Wilkinson Power Divider", text section 7.3 (a PDF with no author name in its metadata, created 11 October 2006; read):** the equal Wilkinson is matched at all ports, with complete isolation between the outputs, lossless when the outputs are matched and lossy when they are not (only reflected power is lost); by even and odd modes S11 = S22 = S33 = 0, S21 = S31 = −j/√2, S23 = 0, and all the odd-mode power is lost in the resistor. It leaves the unequal split as an exercise.
- **Benha University, ECE 344 Microwave Fundamentals, Lecture 08, "Power Dividers and Couplers, Part 1" (Dr. Sherif Hekal, 30 April 2018; read, but the equations in its text layer are garbled, so only its statements and examples were used):** the resistive three-port divider has all ports matched, but half of the input power is dissipated in the resistors and the outputs are not isolated; no lossless divider can match all three ports; the Wilkinson is matched at all ports, isolates the outputs and loses no power between port 1 and ports 2 and 3 (50 Ω, 70.7 Ω arms, 100 Ω).
- **Maloratsky and London, "Quarter-wavelength N-way power dividers/combiners: historical aspects and new modifications", Microwave Journal, 1 September 2003 (read; its general equations are images and did not come through as text):** for N outputs, line impedance Z1 = Z0 √N; the resistors go to a floating common node; theoretically infinite isolation between any two ports at the centre frequency; the bandwidth narrows as N grows; planar layouts are difficult for N > 2.
- **Wikipedia, "Wilkinson power divider" (last edited 20 August 2025; read):** the unequal split with $K^2$ = P3/P2 and R = Z0 (K + 1/K); its arm equations came out garbled, and it cites Pozar, Microwave Engineering, 3rd ed. (2005) for them, and Wilkinson's paper, IRE Trans. MTT, vol. 8, pp. 116–118, January 1960.
- **Huettner et al. (Nuvotronics), "How Hot Do Those Isolation Resistors Get?", Microwave Journal, 13 May 2025 (read):** with the amplifier pairs in phase no power goes to the isolation resistor; with a phase difference all the mismatched power goes to one resistor. It gives no formula.

- **RF Essentials, "Wilkinson Divider" (a web glossary page, no author or date; read through a text-extraction tool):** 70.7 Ω and 100 Ω for two ways, 10 log N for the split, and a one-section bandwidth of about 25 to 30 % (criterion not given; the 36 % here is for 20 dB). Its unequal-split equations ($K^2$ = P2/P3, arms Z0 √(K (1 + $K^2$)) and Z0 √((1 + $K^2$)/K)) were simulated as given, resistor across the arm ends, and leave a 15 dB match at $K^2$ = 2, so they are not used.
- **Derived here and checked by simulation:** every equation in the next section, the heat in the resistors, and the bands.
- **Not read:** Pozar, Microwave Engineering, §7.2 and §7.3 (the book); Wilkinson's 1960 paper; Microwaves101's Wilkinson page (it returned "not found").

## Why the formulas are these

**Resistive.** With every leg R and the M outputs ended in Z0, the input sees R + (R + Z0)/M. Setting that to Z0 gives R (M + 1) = Z0 (M − 1), so R = Z0 (M − 1)/(M + 1). Every port sees the same star, so the transmission between any two ports, input to output or output to output, is 1/M: the loss and the isolation are both 20 log M dB (6.02 dB for two outputs, 9.54 for three, 12.0 for four). The input leg carries the input current and dissipates (R/Z0) of the input power, (M − 1)/(M + 1) of it; each output leg (R/Z0)/$M^2$ of it; together (M − 1)/M of it is heat.

**Wilkinson, N equal outputs.** Each arm, a quarter wave of impedance Z1 loaded with Z0, looks like $Z1^2$/Z0 from the input; N in parallel look like $Z1^2$/(N Z0), which is Z0 when Z1 = Z0 √N. For the isolation, send signals of opposite signs to the outputs, adding up to zero: the arms' input ends are then a virtual ground, a quarter-wave line shorted at one end looks open at the other, and each output sees only its star resistor: matched when that resistor is Z0. (Between any two outputs that is 2 Z0, the 100 Ω of the two-way, so for two outputs one 2 Z0 resistor does.) Signals in step leave the common point at the outputs' voltage: no current in the resistors.

**Unequal, $K^2$ = P3/P2.** Load the arm ends with R2 = K Z0 and R3 = Z0/K (any pair with the ratio $K^2$ would do; this one has R2 R3 = $Z0^2$ and keeps the output lines near Z0), brought back to Z0 by quarter-wave lines of Z04 = √(Z0 R2) = Z0 √K and Z05 = Z0/√K. Driven from port 1, the arm ends must be at the same voltage (no current in R), which gives R2/Z02 = R3/Z03 and, for the power split ($R3/Z03^2$)/($R2/Z02^2$) = $K^2$, R2/R3 = $K^2$. Matching the input, $Z02^2$/R2 || $Z03^2$/R3 = Z0, then gives Z02 = Z0 √(K (1 + $K^2$)) and Z03 = Z0 √((1 + $K^2$)/$K^3$). For the isolation, drive port 2 and ask for no voltage at the end of arm 3. The voltage V2 at the end of arm 2 sends a current V2/R through R into the end of arm 3; that end, held at zero, is a short, which a quarter-wave line passes on as a current V1/Z03 (V1 the voltage at the input node); and with port 1 ended in Z0, V1 = Z0 V2/Z02. Equating the two currents gives R = Z02 Z03/Z0 = Z0 (K + 1/K) = R2 + R3. R goes between the arm ends, ahead of the transformers. For K = 1 these are 70.7 Ω, 100 Ω and no transformers.

**Heat with an output open or shorted.** The output k that fails reflects all the power it was given, P_k. Of that, the fraction $|S1k|^2$ goes back to the input, none to the other outputs (they are isolated), and the rest to the resistors: P_k (1 − $|S1k|^2$). For N equal outputs that is (P/N)(1 − 1/N) = P (N − 1)/$N^2$ of the input power P; for the unequal split, either port gives $K^2$/(1 + $K^2$)$^2$ P. In the star of N resistors the failed output's resistor takes (N − 1)/N of it and each other 1/(N (N − 1)).

**The band.** The arms' length is 90° × f/f0. The solver computes the S-parameters round the centre, and the band is the span where every return loss and every output-to-output isolation is at least 20 dB. For two equal outputs it is 0.819 to 1.18 f0; it narrows with the outputs (±11.1 % for three, ±8.56 % for four, ±5.18 % for eight) and with the split (about ±15 % for 3 dB, ±9.4 % for 6 dB, ±2.3 % for 20 dB).

## Assumptions and limits

- **Ideal lossless lines.** The loss shown is the split only. A real line adds its own loss, and the resistors and junctions add parasitics that matter at GHz. No discontinuity is modelled.
- **One section.** Wider bands need several sections; they are not computed.
- **The same εeff for every line,** though lines of different impedance have different widths and so slightly different εeff: use the figure for the trace you mean (the microstrip tool gives it).
- **N > 2 needs a floating common point.** On a PCB that takes crossovers or vias (Maloratsky and London); the tool computes the ideal circuit.
- **The heat is the worst case:** total reflection at one output, the others matched, at the centre frequency. For a combiner, two inputs of unequal amplitude or phase give the resistor $|a1 - a2|^2/2$ (the odd-mode power).
- **Power in** is the power a matched source can deliver.
- **Standard values:** one value for every leg or resistor, the nearest in the series to the ideal one.
- **Unequal:** port 3 gets the larger share; swap the ports for the other way round. Splits over 20 dB are refused (the transformers get extreme).
- **Not covered:** hybrid and directional couplers and the two-resistor "leveling" splitter of test equipment.

## How it was checked

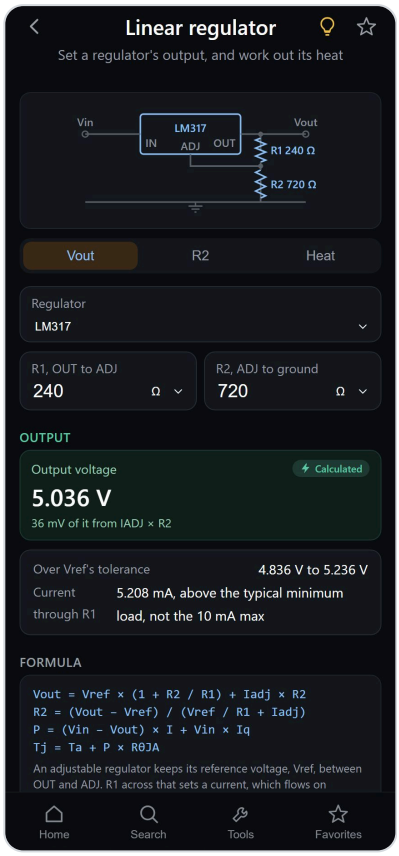
32 cases. The expected values come from a Python S-parameter solver (numpy, nodal analysis with quarter-wave line matrices) and the app's own solver agreed with it to the digits shown: - Resistive: 2 outputs on 50 Ω, 16.7 Ω, 6.02 dB (loss and isolation), matched, 250 mW out of 1 W, 333 mW in the input leg, 500 mW lost; in E24, 16 Ω: 5.93 dB, a 39.9 dB match, 490 mW lost; 4 outputs, 30 Ω, 12 dB, 750 mW lost; 3 outputs on 75 Ω, 37.5 Ω, 9.54 dB; 8 outputs in E96, 39.2 Ω, 18.1 dB, 49.1 dB; 100 mW in; one and 17 outputs refused; - Wilkinson: 2 outputs on 50 Ω at 1 GHz, VF 66 %: 70.7 Ω, 49.5 mm, 100 Ω, 3.01 dB, ±18.1 %, 250 mW; 3 outputs on 75 Ω at 100 MHz: 130 Ω, 49.5 cm, 4.77 dB, ±11.1 %, 222 mW (148 mW in one resistor); 4 outputs (100 Ω, ±8.56 %, 187 mW) and 8 (141 Ω, ±5.18 %, 109 mW); E12 star resistors, 47 Ω: a 33.7 dB match and 39.7 dB isolation; 10 MHz, 4.95 m; VF 1, 74.9 mm; VF 0.5 and εeff 4 linked; nine outputs and no frequency refused; - Unequal: 3 dB: 103 and 51.5 Ω, 106 Ω, 59.4 and 42.1 Ω, 4.76 and 1.76 dB, 0.852 to 1.15 GHz, 222 mW; 4:1: 158 and 39.5 Ω, 125 Ω, 70.7 and 35.4 Ω, 160 mW; 0 dB: the equal Wilkinson; 20 dB; 6 dB in E24 (120 Ω for 125 Ω: a 36.1 dB match, 42.1 dB isolation); 25 dB refused; - the outputs' − and + (to 4, stopping at 2 and at 16 and, for a Wilkinson, at 8, and a count of 12 coming down to 8 on changing pill); - three of the examples.

The solver also confirmed, to numerical zero at the centre frequency, the equal and N-way Wilkinson (every port matched, every pair of outputs isolated), the unequal one for $K^2$ from 1 to 100 with the exact split, and that the resistor placed after the output transformers, or the transformers left out, or RF Essentials' arms, do not give a matched and isolated divider.

POWER ELECTRONICS › SUPPLY & REGULATION

# Linear regulator

(formerly Linear regulator (LDO))




## What it computes

A linear regulator, picked from the common ones, in two groups: - **Variable** (adjustable with two resistors): LM317, LM1117-ADJ, AMS1117-ADJ, LM1086-ADJ, the negative LM337, or a custom part with your own Vref and dropout. - **Fixed:** 7805, 7812, 7815, AMS1117-3.3, AP2112-3.3, MCP1700-3.3.

Three ways, by pill:

| Pill | Inputs | Output |
|---|---|---|
| Vout | R1 (OUT to ADJ) and R2 (ADJ to ground) | the output voltage |
| R2 | the wanted output and R1 | R2, and the nearest 1 % value with the output it gives |
| Heat | Vin, Vout, the load current, the package or RθJA, the ambient | the heat in it, the junction temperature |

- **Vout** also gives the spread over Vref's tolerance and the share of the adjust-pin current.
- **Vout and R2** say whether the current through R1 is above the datasheet's minimum load.
- **Heat** also gives:
- the efficiency;
- whether there is enough voltage across the regulator (the LM317's 3 V, the LM337's 2.5 V, the 78xx's input for regulation, the others' dropout);
- the most current before the junction reaches 125 °C, or the part's rating if that is lower;
- a warning past the rated current, the input voltage or the voltage it takes across it.
- **A fixed part** has no divider to set, so Vout and R2 are greyed out and Heat takes its output from the part. It also shows the output's tolerance and its own current to ground, Iq, which adds Vin × Iq to the heat: 8 mA in a 7805, 1.6 µA in an MCP1700, which matters on a battery.
- **The LM337** works with negative voltages; they are shown with their minus, and can be typed with or without it.
- **Drawing:** the regulator with its IN, OUT and ADJ pins and the two resistors, R2 green when it is the answer; a fixed part with its GND pin to ground and its output voltage.
- **Remembered:** the regulator, the pill and the units.

Examples: an LM317 at 5 V, R2 for 12 V, an LM1117 from 5 V to 3.3 V at 500 mA, an LM317 from 12 V to 5 V at 1 A, an LM1117 then an MCP1700 from a 3.7 V cell, a 7805 from 12 V at 500 mA, and an LM337 at −5 V.

**Standard values.** The resistors' tolerance, with its E series, is picked in the "Your inputs" heading above the inputs, as in the Passive components tools: ±0.1 % to ±20 %, ±5 % being E24 and ±1 % E96; on the R2 pill, it sets the series R2 is rounded to (E96 by default).

## Source

Read in review, each from the maker's datasheet: - **TI LM317 (SLVS044Z, April 2025):** - VO = VREF (1 + R2 / R1) + (IADJ × R2), equation 2, "IADJ is typically 50 µA and negligible in most applications"; the typical application uses R1 = 240 Ω; - VREF 1.25 V, from 1.2 to 1.3 V; - adjustment pin current 50 µA, 100 µA max; - minimum load current 3.5 mA, 10 mA max; - recommended: VO 1.25 to 37 V, VI − VO 3 to 40 V, IO 0.01 to 1.5 A, TJ 0 to 125 °C; - RθJA 66.8 °C/W (SOT-223), 23.5 (TO-220, KCS), 38.0 (TO-263). - **TI LM1117 (SNOS412Q, January 2023):** - VREF 1.25 V, from 1.238 to 1.262 V at 25 °C; - dropout 1.1 V at 100 mA, 1.15 V at 500 mA, 1.2 V at 800 mA (typical, 25 °C); - adjust pin current 60 µA, 120 µA max; - minimum load current 1.7 mA, 5 mA max; - 800 mA, input to 15 V, TJ 0 to 125 °C; - RθJA 61.6 °C/W (SOT-223), 45.1 (TO-252), 23.8 (TO-220), 41.3 (TO-263); - "PD = (TJ(max) – TA) / RθJA". - **Advanced Monolithic Systems AMS1117 (ds1117):** - reference 1.250 V, 1.232 to 1.268 V at 25 °C; the -3.3 gives 3.251 to 3.349 V; - adjust pin current 55 µA, 120 µA max; minimum load 5 mA, 10 mA max; - dropout 1.1 V typical, 1.3 V max, at 0.8 A, "decreasing at lower load currents"; - quiescent current of the fixed versions 5 mA, 11 mA max; - up to 1 A, input to 15 V, junction to 125 °C; - θJA 90 °C/W (SOT-223, from 46 to over 90 with the copper area), 80 (TO-252), 160 (SO-8). - **TI LM1086 (SNVS039J, April 2015):** - VREF 1.25 V, 1.238 to 1.262 V at 25 °C; - adjust pin current 55 µA, 120 µA max; minimum load 5 mA, 10 mA max; - dropout 1.3 V typical, 1.5 V max, at 1.5 A; - 1.5 A, 29 V maximum between input and output (ADJ); - RθJA 23.0 °C/W (TO-220), 40.8 (TO-263), 35.9 (WSON-8). - **TI LM337 (SLVS047L, January 2015):** - reference −1.25 V, from −1.2 to −1.3 V over load and temperature; - adjustment current 65 µA, 100 µA max; - minimum output current 2.5 mA, 10 mA max (up to 40 V across it); - recommended VI − VO −2.5 to −37 V, 1.5 A, TJ 0 to 125 °C; - RθJA 24.8 °C/W (TO-220, KCS), 25.3 (TO-263), 30.3 (TO-252). - **TI LM340 / LM7805 family (SNOSBT0L, September 2016):** - output 4.8 to 5.2 V, 11.5 to 12.5 V, 14.4 to 15.6 V at 25 °C; - "input voltage required to maintain line regulation" 7.5 V, 14.6 V, 17.7 V; dropout 2 V at 1 A; - quiescent current 8 mA (the only figure, a maximum); - 1.5 A, DC input to 35 V; - RθJA 23.9 °C/W (TO-220), 44.8 (TO-263), 62.1 (SOT-223), from its thermal information table. A footnote in the same datasheet gives older figures (54 °C/W TO-220, 174 SOT-223); the table is used. - **Diodes AP2112 (DS39724 Rev. 2-2, June 2017):** - the -3.3 gives 3.251 to 3.350 V (± 1.5 %); - dropout 5 mV at 10 mA, 125 mV at 300 mA, 250 mV at 600 mA (typical); - quiescent current 55 µA, 80 µA max; - 600 mA, input 2.5 to 6.0 V; - θJA 184 °C/W (SOT-23-5, the AP2112K), 120 (SOT-89-5), 114 (SO-8). - **Microchip MCP1700 (DS20001826E):** - output within ± 2 % at 25 °C (± 3 % over temperature); - dropout 178 mV typical, 350 mV max, at 250 mA; - quiescent current 1.6 µA, 4 µA max; - 250 mA, input 2.3 to 6.0 V, junction to 125 °C; - θJA 212 °C/W (SOT-23), 104 (SOT-89), 92 (TO-92), 91 (2×2 DFN).

## Why the formulas are these

```
Vout = Vref (1 + R2/R1) + Iadj × R2
R2 = (Vout − Vref) / (Vref / R1 + Iadj)
P = (Vin − Vout) × I + Vin × Iq,   Tj = Ta + P × RθJA
Imax = ((125 °C − Ta) / RθJA − Vin × Iq) / (Vin − Vout)
```

The regulator holds Vref across R1, so R1 sets a current of Vref / R1. That current and the adjust current flow on through R2, whose voltage adds to Vref to make the output. A fixed regulator has the same divider inside, and its own working current flows from the input to ground. The regulator drops the rest of the input as heat, and the junction runs RθJA degrees per watt above the air.

## Assumptions and limits

- **RθJA** is the datasheet's, measured on the board it names. A real board, smaller copper or still air can make it worse; a heatsink makes it better. Type your own figure if you have one.
- **The dropout** is the typical figure at 25 °C. Between the LM1117's and the AP2112's listed currents it is interpolated. For the AMS1117, LM1086 and MCP1700 only the full-load figure is given, so it is used at every current: their real dropout is lower at light loads. It is a little higher over temperature.
- **An adjustable part's own current** goes to the output through R1 and R2, so it is not added to the heat; the minimum-load line covers it.
- **125 °C** is the junction limit used for every part: the recommended maximum of most of them. Some allow 150 °C.
- **Other fixed voltages** (1.8 V, 2.5 V, 5 V versions…) work with the Custom part in Heat, without the Iq term.

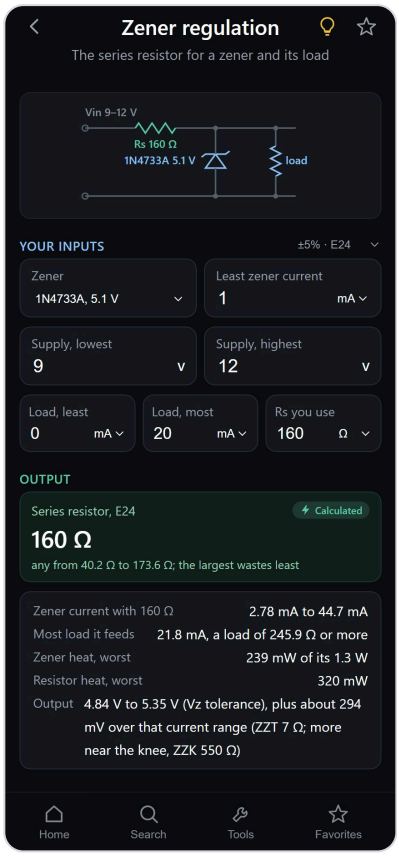
## How it was checked

17 cases: - the LM317 with 240 and 720 Ω gives 5.036 V, 36 mV from the adjust current, and 4.836 to 5.236 V over Vref's tolerance; - 240 Ω carries 5.208 mA, above the typical minimum load only; - R2 for 12 V is 2.044 kΩ, and 2.05 kΩ gives 12.03 V; - the LM1117 from 5 to 3.3 V at 500 mA: 850 mW, 77 °C; - the LM317 from 12 to 5 V at 1 A: 190 °C, over the limit; - the LM1117 with 0.4 V across it is under its dropout, and its current is capped at its 800 mA rating; - the MCP1700 with the same 0.4 V is above its 178 mV dropout: 40 mW, 33 °C, 1.6 µA to ground (5.92 µW); - a 7805 from 12 V at 500 mA: 3.5 W plus 96 mW from its 8 mA, 111 °C in a TO-220, 584 mA before 125 °C; - the LM337 with 240 and 720 Ω gives −5.047 V, and its −12 V input shows with its minus; - the AP2112 at 300 mA needs 125 mV, so 100 mV will not regulate; - the LM1086 with 30 V across it is over its 29 V; - a fixed part keeps only Heat; - a custom 0.8 V reference gives 3.2 V; - an input below the output is refused; - picking a package fills in its RθJA; - the remembered regulator and pill.

POWER ELECTRONICS › SUPPLY & REGULATION

# Zener regulation

(also found as Zener voltage stabilization)




## What it computes

The series resistor for a zener shunt regulator, on one screen:

- **Inputs:** the zener (a 1N4728A–1N4761A, 1.3 W, or a BZX55, 500 mW, each with its datasheet figures, or a custom Vz and power rating), the least zener current, the supply's lowest and highest voltage, the load's least and most current, and the resistor you use.
- **The big result:** the largest E24 value inside the window that works, with the window itself: under (Vin min − Vz max) / (IL max + Iz min) so the zener still gets its least current at the lowest supply and the heaviest load, and over (Vin max − Vz min) / (Iz max + IL min) so it stays within its current at the highest supply and the lightest load. The largest wastes the least. When the window is empty it says so, and why.
- **The resistor you use** follows that value until you type your own. For it, the tool gives:
- the zener current at the two extremes, flagged under the least current (the zener drops out) or over Iz max;
- the most load it can feed while keeping the least zener current, as a current and as the smallest load resistance;
- the zener's and the resistor's heat at the worst case;
- the output: the zener's tolerance band, plus ZZT times the current range.
- **The least zener current** starts at the part's knee test current, IZT2 (1, 0.5 or 0.25 mA on the 1N47xxA, 1 or 0.5 mA on the BZX55); it can be raised for better regulation.
- **Units:** currents in µA, mA or A (one unit for all three), the resistor in Ω, kΩ or MΩ.
- **Drawing:** the supply, Rs, the reverse-biased zener and the load.
- **Remembered:** the part and the units.

Examples: Kuphaldt's 12.6 V from 45 V through 1 kΩ, the same with a 500 Ω load, 5.1 V at 20 mA from 9–12 V with a 1N4733A, and a BZX55C5V1 for 5 mA.

**Standard values.** The resistors' tolerance, with its E series, is picked in the "Your inputs" heading above the inputs, as in the Passive components tools: ±0.1 % to ±20 %, ±5 % being E24 and ±1 % E96; it sets the series the series resistor is chosen from (E24 by default).

## Source

- **Tony R. Kuphaldt, Lessons in Electric Circuits, vol. III (Semiconductors), ch. 3, "Zener diodes"** (read, ibiblio HTML): the zener is reverse-biased; "zener diodes regulate voltage by acting as complementary loads"; worked example 12.6 V, 45 V, 1 kΩ: 32.4 mA, the zener 408 mW, the resistor

1.05 W; with a 500 Ω load "this leaves 7.2 mA of current through the zener diode"; with 100 kΩ the regulator only works down to a 38.889 kΩ load; Vz "may be anywhere from 5 percent to 10 percent in error".

- **Vishay 1N4728A to 1N4761A, document 85816 rev. 2.7, 7 Nov 2024** (read): Vz 3.3 to 75 V, ± 5 %; Ptot 1300 mW "provided that leads at a distance of 4 mm from case are kept at ambient temperature"; per part IZT1, IZT2, ZZT at IZT1, ZZK at IZT2 and the regulator current IZM (e.g. 1N4733A: 5.1 V, 49 mA, 1 mA, 7 Ω, 550 Ω, 178 mA). IZM × Vz comes to about 0.9 W, under the 1.3 W Ptot: the tool checks the current against IZM and the heat against Ptot, so a design can pass one and not the other, as the sheet's own figures do.
- **Vishay BZX55, document 85604 rev. 2.0, 7 Nov 2024** (read): Vz 2.4 to 75 V with min / nom / max per part, IZT1 5 mA (2.5 mA from 39 V), IZT2 1 mA (0.5 mA), ZZ, ZZK, Ptot 500 mW (4 mm leads at 25 °C); Iz max is taken as Ptot / Vz.

## Why the formulas are these

```
Rs ≤ (Vin min − Vz max) / (IL max + Iz min)
Rs ≥ (Vin max − Vz min) / (Iz max + IL min)
Iz = (Vin − Vz) / Rs − IL,   Pz = Vz × Iz,   PR = (Vin − Vz)² / Rs
```

The resistor carries the load's current and the zener's together, and drops what the supply has over Vz. At the lowest supply and the heaviest load the least current reaches the zener: the resistor must be small enough that it is still above the knee. At the highest supply and the lightest load the most reaches it: the resistor must be large enough to keep it within its rating. Taking Vz at the edge of its tolerance that makes each case worst keeps the window honest.

## Assumptions and limits

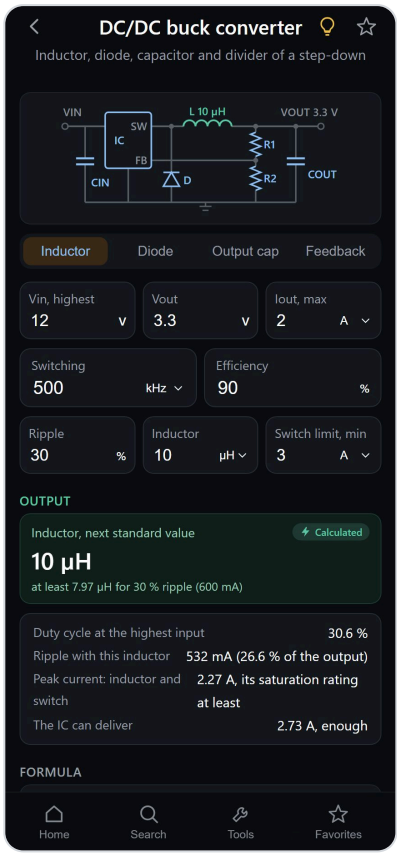
- **ZZT** is the zener's dynamic resistance at its test current; near the knee it is ZZK, much higher, so the output moves more there than the tool's figure.
- **The power ratings** assume the leads held at ambient 4 mm from the body; on a board, a zener runs hotter and must be derated (Vishay's figure 1).
- **The supply's ripple** is part of its range: enter the lowest valley and the highest peak.
- **For more current,** a zener feeding a transistor (Zener + transistor stabilization, planned) or a linear regulator is the better circuit.

## How it was checked

15 cases: - 1N4733A, 9–12 V, 0–20 mA: 40.2 to 173.6 Ω, E24 160 Ω; 2.78 to 44.7 mA, 239 mW in the zener, 320 mW in the resistor; - its output: 4.84 to 5.35 V, plus about 294 mV over that range; - Kuphaldt's 1 kΩ: 32.4 mA, 408 mW, 1.05 W, 12.6 V; - Kuphaldt's 500 Ω load: 7.2 mA left for the zener; - 1 kΩ from 45 V feeds 31.4 mA keeping 1 mA: a load of 401.3 Ω or more; - Kuphaldt's 100 kΩ: 324 µA, too little for a 1 mA knee; - BZX55C5V1 from 11–13 V at 0–5 mA: 910 Ω, 48.7 mW; - too large a resistor starves the zener; too small passes IZM (while under Ptot); a BZX55 at 22 Ω goes over 500 mW; - a supply range too wide leaves no resistor; - a supply under the zener voltage is refused; - picking a part brings its knee current (1N4740A: 0.25 mA); - the remembered part.

POWER ELECTRONICS › SUPPLY & REGULATION

# DC/DC buck converter




## What it computes

The power stage of a buck (step-down) converter in continuous conduction, the way TI's application note lays it out, on one screen:

- **Shared inputs:** the highest input voltage, the output voltage, the most output current, the switching frequency and the efficiency η (90 % by default).
- **Four pills:**
- **Inductor:** the ripple you allow (a share of the output current) gives the smallest inductor, and the box proposes the next standard value up (E6: 1, 1.5, 2.2, 3.3, 4.7, 6.8). It follows the inputs until you type your own. Then the duty cycle, the ripple with that inductor, the peak current (the inductor's saturation rating and the switch's), and, with the switch's minimum current limit, the most the IC can deliver, flagged when under the output current. A limit of 0 skips that row.
- **Diode:** its average current, the power at its forward voltage VF, the reverse voltage and peak current it must stand. A synchronous converter has a second switch instead, and no diode loss.
- **Output cap:** the least capacitance for the output ripple allowed, the ripple its ESR adds, and the capacitance a load step needs to stay within an overshoot. The larger of the two is the result.
- **Feedback:** from the IC's feedback voltage VFB and R2, R1 exact and as the nearest E96 resistor, the output it gives, and the divider current against 100 × the FB pin's bias current IFB.
- **Units:** the currents in mA or A (one picker moves them all), the frequency in kHz or MHz, the inductor in nH or µH, IFB in nA or µA, R2 in Ω or kΩ; ripple and overshoot in mV, ESR in mΩ.
- **Drawing:** the power stage as TI draws it, with the feedback divider: the IC with its SW and FB pins, the input capacitor, the diode, the inductor, R1 and R2 with their middle wired back to the FB pin (the wire crosses the diode's lead above the diode, without a junction), and the output capacitor last, by the output. The part the pill works out is green.
- **Remembered:** the pill and the units.

Examples: TI's TPS54331 design (its inductor, its feedback divider, its output capacitor), a 12 V to 5 V Schottky diode, and a 2 MHz 5 V to 1.2 V.

**Standard values.** The resistors' tolerance, with its E series, is picked in the "Your inputs" heading above the inputs, as in the Passive components tools: ±0.1 % to ±20 %, ±5 % being E24 and ±1 % E96; on the Feedback pill, it sets the series R1 is rounded to (E96 by default).

## Terms

- **Buck converter:** a switching regulator that steps a voltage down. A switch connects the inductor to the input for part of each cycle; the rest of the time a diode (or a second switch) lets the inductor's

current keep flowing to the output.

- **Duty cycle (D):** the fraction of each cycle the switch is on. Ideally Vout / Vin; the losses make it a little more.
- **Efficiency (η):** output power over input power.
- **Inductor ripple (ΔIL):** how far the inductor's current rises and falls each cycle, peak to peak.
- **Continuous conduction:** the inductor's current never falls to zero; the formulas assume it.
- **Saturation current:** above it the inductor's core saturates and its inductance collapses; it must exceed the peak current.
- **ESR:** the capacitor's equivalent series resistance.
- **VFB, IFB:** the voltage the IC regulates its FB pin to, and the small current that pin draws.
- **Synchronous converter:** one with a second switch in place of the diode.

## Source

- **TI SLVA477C, "Basic Calculation of a Buck Converter's Power Stage", Brigitte Hauke, December 2011, revised September 2026** (read, all 9 pages): D = Vout / (Vin(max) × η) (eq. 1; η "e.g., estimated 90%" in section 2, 85 % in its appendix); ΔIL = (Vin(max) − Vout) × D / (fs × L) (eq. 2); IMAXOUT = ILIM(min) − ΔIL / 2 (eq. 3); ISW(max) = ΔIL / 2 + IOUT(max), "the peak current, the inductor, the integrated switch(es), and the external diode have to withstand" (eq. 4); L = Vout (Vin − Vout) / (ΔIL fs Vin) (eq. 5) with ΔIL "20% to 40% of the output current" (eq. 6); IF = IOUT(max) (1 − D), PD = IF × VF (eqs. 7, 8); the divider current at least 100 × IFB, R2 = VFB / IR1/2, R1 = R2 (VOUT / VFB − 1) (eqs. 9 to 11); COUT(min) = ΔIL / (8 fs ΔVOUT) (eq. 12); ΔVOUT(ESR) = ESR × ΔIL (eq. 13); COUT(min),OS = ΔIOUT² L / (2 VOUT VOS) (eq. 14). It has no worked numbers.
- **TI TPS54331 datasheet, SLVS839H, October 2023, section 8.2** (read): its design is 7 to 28 V in, 3.3 V, 3 A, 570 kHz, 30 mV of output ripple; LMIN = VOUT (VIN(max) − VOUT) / (VIN(max) × KIND × IOUT × FSW) (eq. 8), KIND = 0.3, "calculated as 5.7 µH", and "a large value was selected: 6.8 µH"; the divider R5 = 10.2 kΩ over R6 = 3.24 kΩ on its 0.8 V reference, "resulting in a 3.31-V output voltage"; two 47 µF ceramics at the output.

## Why the formulas are these

```
D = Vout / (Vin × η)
L = Vout (Vin − Vout) / (ΔIL × fs × Vin)
ΔIL = (Vin − Vout) × D / (fs × L),   Ipeak = Iout + ΔIL / 2
IF = Iout (1 − D),   PD = IF × VF
Cout = ΔIL / (8 fs ΔV),   Cout = ΔI² L / (2 Vout Vos)
R1 = R2 (Vout / VFB − 1)
```

While the switch is on, the inductor sees Vin − Vout and its current rises at (Vin − Vout) / L for D / fs seconds: that rise is the ripple. The inductor's current averages the output current, so its peak is half the ripple above it. While the switch is off the diode carries it, a fraction 1 − D of the time. The ripple current flows into the output capacitor, whose charge swings by ΔIL / (8 fs) each cycle, hence the capacitance for a ripple voltage. When the load jumps, the energy ½ L ΔI² the inductor must gain comes from the capacitor first, ½ C ((Vout + Vos)² − Vout²) ≈ C Vout Vos, hence the load-step capacitance.

**Which input voltage.** SLVA477 puts the typical input in the inductor formula; the TPS54331 datasheet puts the highest, where the ripple is the largest, and so does this tool, so the ripple stays within what you allow over the whole input range. With 28 V in, it gives the datasheet's 5.7 µH.

**The datasheet's ripple.** TPS54331's ripple formula (its eq. 9) divides by a further 0.8 and leaves out η: it gives 0.939 A of ripple and 3.47 A of peak with 6.8 µH, where SLVA477's formulas, which this tool uses, give 0.835 A and 3.42 A.

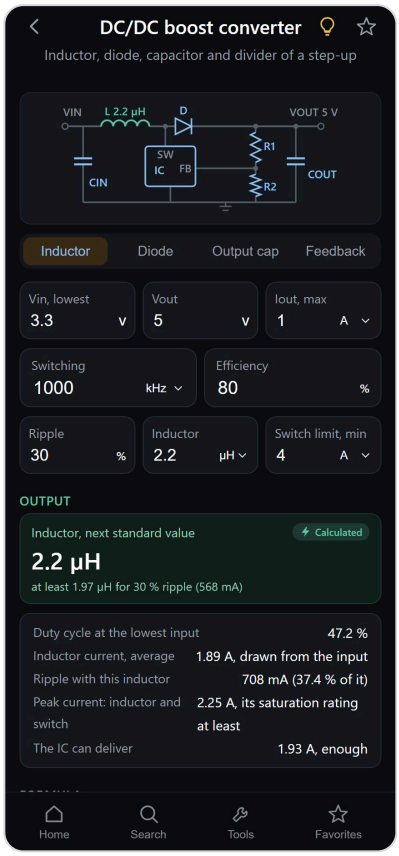
## Assumptions and limits

- **Continuous conduction only:** at light load the inductor's current reaches zero and the formulas no longer hold.
- **The IC decides much:** use the inductor and capacitor ranges its datasheet gives, above all when its compensation is internal (SLVA477 section 7); the switching frequency, the current limit, VFB and IFB come from it too.
- **Ceramic capacitors lose capacitance** with DC bias and temperature: X5R or better (SLVA477), and check the value at the working voltage.
- **The input capacitor** is the datasheet's minimum, raised when the input is noisy (SLVA477 section 6); it is not computed here.
- **No losses beyond η and the diode:** the switch's and the inductor's losses, and the inductor's RMS current, are not computed.

## How it was checked

11 cases: - TPS54331's design, 28 V to 3.3 V at 3 A, 570 kHz, 30 %: 5.67 µH at least (its 5.7 µH), the next E6 value 6.8 µH (its choice); - the same at 90 %: D 13.1 %, 835 mA of ripple, 3.42 A peak, a 3.5 A limit delivering 3.08 A; - its divider: R1 for 3.3 V over 3.24 kΩ on 0.8 V is 10.13 kΩ, E96 10.2 kΩ (its R5) gives 3.319 V (its 3.31 V); - its output capacitor: 6.1 µF for 30 mV of ripple, 10.3 µF for a 1 A step within 100 mV; - 12 V to 5 V at 2 A: the diode carries 1.07 A and burns 537 mW at 0.5 V; - 5 V to 1.2 V at 2 MHz: 760 nH at least, so 1 µH; - the proposed inductor follows the output current (1 A: 22 µH) until one is typed, and a typed one stays while its ripple follows; - a switch limit too low is flagged; - the current pickers move together; - a step-up is refused.

# DC/DC boost converter




## What it computes

The power stage of a boost (step-up) converter in continuous conduction, as TI's application note lays it out, on one screen, the twin of DC/DC buck converter:

- **Shared inputs:** the lowest input voltage (the worst case for a boost), the output voltage, the most output current, the switching frequency and the efficiency η (80 % by default).
- **Four pills:**
- **Inductor:** the ripple you allow, as a share of the average inductor current, gives the smallest inductor, and the box proposes the next standard value up (E6). It follows the inputs until you type your own. Then the duty cycle, the average inductor current (what the input draws), the ripple with that inductor, the peak current (the inductor's saturation rating and the switch's), and, with the switch's minimum current limit, the most the IC can deliver, flagged when under the output current. A limit of 0 skips that row.
- **Diode:** it carries the whole output current; its power at VF, the reverse voltage (the output) and the peak current it must stand.
- **Output cap:** the least capacitance for the output ripple allowed, the ripple its ESR adds, the total, and the peak current into it.
- **Feedback:** as in the buck: R1 exact and as the nearest E96 resistor, the output it gives, and the divider current against 100 × IFB.
- **Units:** the currents in mA or A (one picker moves them all), the frequency in kHz or MHz, the inductor in nH or µH, IFB in nA or µA, R2 in Ω or kΩ; ripple in mV, ESR in mΩ.
- **Drawing:** the power stage as TI draws it: the inductor from the input to the switch node, the IC's switch below it, the diode on to the output, R1 and R2 with their middle wired to the FB pin, the output capacitor last. The part the pill works out is green.
- **Remembered:** the pill and the units.

Examples: TI's TPS61030 design (its inductor and its output capacitor), TPS61023's feedback divider, and a 5 V to 12 V Schottky diode.

**Standard values.** The resistors' tolerance, with its E series, is picked in the "Your inputs" heading above the inputs, as in the Passive components tools: ±0.1 % to ±20 %, ±5 % being E24 and ±1 % E96; on the Feedback pill, it sets the series R1 is rounded to (E96 by default).

## Terms

- **Boost converter:** a switching regulator that steps a voltage up. The switch charges the inductor from the input; when it opens, the inductor pushes its current through the diode into the output.
- **Duty cycle (D):** the fraction of each cycle the switch is on; ideally 1 − Vin / Vout.

- **Inductor current (IL):** in a boost it is the input current, larger than the output current by Vout / Vin (and the losses).
- The other terms (efficiency, ripple, continuous conduction, saturation current, ESR, VFB, IFB, synchronous) are as in DC/DC buck converter.

## Source

- **TI SLVA372D, "Basic Calculation of a Boost Converter's Power Stage", Brigitte Hauke, November 2009, revised November 2022** (read, all 10 pages): D = 1 − VIN(min) × η / VOUT (eq. 1; η "e.g. estimated 80%" in section 2, 85 % in section 8); ΔIL = VIN(min) × D / (fS × L) (eq. 2); IMAXOUT = (ILIM(min) − ΔIL / 2) × (1 − D) (eq. 3); ISW(max) = ΔIL / 2 + IOUT(max) / (1 − D) (eq. 4); L = VIN (VOUT − VIN) / (ΔIL fS VOUT) (eq. 5) with ΔIL = (0.2 to 0.4) × IOUT(max) × VOUT / VIN (eq. 6); IF = IOUT(max), PD = IF × VF (eqs. 7, 8); the divider as in SLVA477 (eqs. 9 to 11); COUT(min) = IOUT(max) × D / (fS × ΔVOUT) (eq. 12); ΔVOUT(ESR) = ESR × (IOUT(max) / (1 − D) + ΔIL / 2) (eq. 13). It has no load-step formula and no worked numbers.
- **TI TPS61030 datasheet, SLUS534G, March 2015, section 11.2** (read): 1.8 to 5.5 V in, 5 V out, 600 kHz; "for an output current of 1000 mA at 5 V, at least 3500 mA of average current flows through the inductor at a minimum input voltage of 1.8 V" (its eq. 4, with 0.8 for the losses); L = VBAT (VOUT − VBAT) / (ΔIL f VOUT) with ΔIL "10% × IL": "the desired inductor has the value of 5.5 µH. In typical applications a 6.8 µH inductance is recommended"; the switch's current limit 4500 mA at 5 V; "with a chosen ripple voltage of 10 mV, a minimum capacitance of 100 µF is needed"; 80 mΩ of ESR adds 80 mV.
- **TI TPS61023 datasheet, SLVSF14B, August 2020, section 8.2** (read): 2.7 to 4.35 V to 5 V at 1.5 A; VREF 0.595 V typical, FB leakage 4 nA typical, 20 nA maximum; R1 732 kΩ, R2 100 kΩ; the ripple best under 40 % of the average inductor current; η "use 90% for most applications"; its current limit is a valley limit, not the peak limit SLVA372 assumes.

## Why the formulas are these

```
D = 1 − Vin × η / Vout,   IL = Iout / (1 − D)
L = Vin (Vout − Vin) / (ΔIL × fs × Vout)
ΔIL = Vin × D / (fs × L),   Ipeak = IL + ΔIL / 2
IF = Iout,   PD = IF × VF
Cout = Iout × D / (fs × ΔV)
R1 = R2 (Vout / VFB − 1)
```

While the switch is on, the inductor sees the input and its current rises at Vin / L for D / fs seconds: that rise is the ripple. The output gets the inductor's current only while the switch is off, a fraction 1 − D of the time, so the inductor carries Iout / (1 − D) on average, which is Iout × Vout / (Vin × η). While the switch is on, the capacitor alone feeds the load for D / fs seconds, losing Iout × D / fs of charge: hence its size. The diode passes every bit of the output current. The ESR sees the full inductor current at the instant the switch opens.

**Which input voltage.** The lowest input gives the largest duty cycle, the largest inductor current and the largest switch current, so the tool uses it everywhere, as the TPS61030 and TPS61023 datasheets do; SLVA372 puts the typical input in its inductor formula.

**The ripple's base.** SLVA372 takes the ripple as a share of Iout × Vout / Vin; both datasheets take it as a share of the average inductor current, which includes the losses, and so does this tool: that is how TPS61030's 10 % gives its 5.5 µH.

**Where TPS61030 differs.** Its capacitor formula, Iout (Vout − Vin) / (f ΔV Vout), is SLVA372's with η left out, so it asks 107 µF ("100 µF") where SLVA372's asks 119 µF. Its ESR ripple is Iout × ESR, 80 mV; SLVA372 counts the inductor's peak current, as the ESR really sees it, and gets 290 mV.

## Assumptions and limits

- **Continuous conduction only.**
- **The IC decides much:** use its datasheet's inductor and capacitor ranges, above all with internal compensation (SLVA372 section 7); its frequency, current limit, VFB and IFB come from it too.
- **Peak current limits only:** the IC row assumes the switch limit is a peak limit, as SLVA372 does; an IC with a valley limit (TPS61023) can deliver more, by its own datasheet's formula.
- **Ceramic capacitors lose capacitance** with DC bias: X5R or better, and check the value at the working voltage.
- **The input capacitor** is the datasheet's minimum; not computed here.

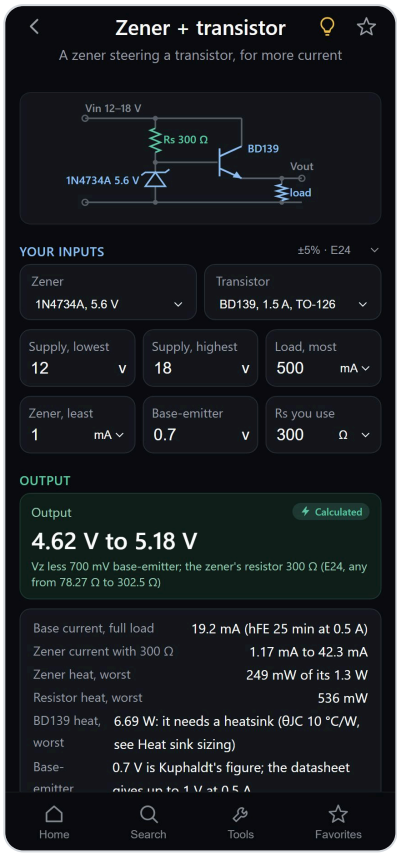
## How it was checked

8 cases: - TPS61030's design, 1.8 V to 5 V at 1 A, 80 %, 600 kHz, 10 %: 3.47 A in the inductor (its "at least 3500 mA"), 5.53 µH at least (its 5.5 µH), the next E6 value 6.8 µH (its recommendation); - the same: D 71.2 %, 314 mA of ripple, 3.63 A peak, its 4.5 A limit delivering 1.25 A; - its output capacitor: 119 µF for 10 mV, and its 80 mΩ ESR adds 290 mV; - TPS61023's divider: R1 for 5 V over 100 kΩ on 0.595 V is 740.3 kΩ, E96 732 kΩ (its R1) gives 4.95 V, 298 × its 20 nA maximum leakage; - 5 V to 12 V at 0.5 A: the diode carries 500 mA, 200 mW at 0.4 V, and blocks 12 V; - the proposed inductor follows the output current (0.5 A: 4.7 µH); - a switch limit too low is flagged; - a step-down is refused.

POWER ELECTRONICS › SUPPLY & REGULATION

# Zener + transistor

(formerly Zener + transistor stabilization)




## What it computes

A zener holding the base of an NPN emitter follower, which passes the load current: on one screen,

- **Inputs:** the zener (the same 1N4728A–1N4761A and BZX55 parts as Zener regulation, or custom), the transistor (BD139, TIP31C, 2N3055, or custom with its least hFE), the supply's lowest and highest voltage, the most load current, the zener's least current, the base-emitter drop (0.7 V by default), and the zener's resistor you use.
- **The big result:** the output, Vz less the base-emitter drop, over the zener's tolerance; under it, the zener's resistor: the largest E24 value in its window, or why none works.
- **For the resistor you use:**
- the base current at full load, IL / (hFE + 1), with the hFE used;
- the zener current at the two extremes, flagged under its least current or over its maximum;
- the zener's and the resistor's heat at worst;
- the transistor's heat at worst, (Vin max − Vout) × IL, compared with its free-air rating and its rating with the case at 25 °C, pointing to Heat sink sizing when it needs one;
- its current and VCEO ratings;
- the base-emitter drop the datasheet gives at a high current.
- **The resistor box** follows the suggestion until you type your own.
- **Units:** currents in µA, mA or A, the resistor in Ω, kΩ or MΩ.
- **Drawing:** the supply on the collector, the resistor to the base, the reverse-biased zener to ground, the emitter feeding the load.
- **Remembered:** the zener, the transistor and the units.

Examples: 5 V at 500 mA from 12–18 V with a 1N4734A and a BD139; 12 V at 0.5 A from 16–20 V with a 1N4743A and a TIP31C; a 2N3055 at 4 A, where no resistor works.

**Standard values.** The resistors' tolerance, with its E series, is picked in the "Your inputs" heading above the inputs, as in the Passive components tools: ±0.1 % to ±20 %, ±5 % being E24 and ±1 % E96; it sets the series the zener's resistor is chosen from (E24 by default).

## Source

- **Tony R. Kuphaldt, Lessons in Electric Circuits, vol. III, ch. 4** (read, ibiblio HTML), "common collector application: voltage regulator": a common-collector transistor amplifies the current "so that the zener diode circuit only has to handle the amount of current necessary to drive the base of the transistor";

"the load voltage will be approximately 0.7 volts less than the zener diode voltage"; when one transistor's gain is not enough, several can be combined (a Darlington). His ch. 9 shows the zener alone at 5.1 V and 500 mA from 12–18 V would dissipate "close to 5W".

- **ST BD135/BD139, September 2001** (read, a copy on MIT's wiki): IC 1.5 A, VCEO 80 V, Ptot 12.5 W at Tcase ≤ 25 °C and 1.25 W at Tamb ≤ 25 °C, VBE 1 V at IC 0.5 A, hFE 25 min at 0.5 A (40 at 150 mA), Rth j-case 10 °C/W.
- **Fairchild TIP31 series, rev. A, February 2000** (read): TIP31C VCEO 100 V, IC 3 A, PC 40 W at TC 25 °C and 2 W at Ta 25 °C, hFE 25 min at 1 A and 10 to 50 at 3 A, VBE(sat) 1.8 V at 3 A. θJC is taken as (150 − 25) / 40 = 3.125 °C/W from the derating, which the sheet does not print.
- **2N3055, a two-page sheet whose maker is not printed** (read): VCEO 60 V, IC 15 A, 115 W at TC 25 °C, RthJC 1.52 °C/W, hFE 20 to 70 at 4 A, VBE(on) 1.5 V at 4 A. onsemi's own sheet could not be downloaded.
- **The zeners:** Vishay's sheets, as in Zener regulation.

## Why the formulas are these

```
Vout = Vz − VBE
IB = IL / (hFE + 1)
Rs ≤ (Vin min − Vz max) / (IB + Iz min)
Rs ≥ (Vin max − Vz min) / Iz max
P transistor = (Vin max − Vout min) × IL
```

The emitter follows the base less one diode drop, so the output is set by the zener. The emitter carries the load; the base needs that divided by the gain plus one, and that is all the zener's resistor has to supply on top of the zener's own least current. With no load the base takes nothing, so the whole resistor current goes through the zener: that sets the other end of the window. The transistor drops everything between the supply and the output, at the full load current.

## Assumptions and limits

- **hFE** is the datasheet's minimum at the current it names; it falls at higher currents (the TIP31C's 25 at 1 A is 10 at 3 A). For a load near the transistor's rating, type the lower figure as a custom part.
- **VBE** is 0.7 V by default, Kuphaldt's figure at modest current; the datasheets give up to 1 to 1.8 V at amps, so the output falls a little as the load grows.
- **No current limit:** a short on the output puts the whole supply across the transistor; this circuit has no protection.
- **A base needing more than the zener can give** needs a driver transistor (a Darlington), not drawn here.

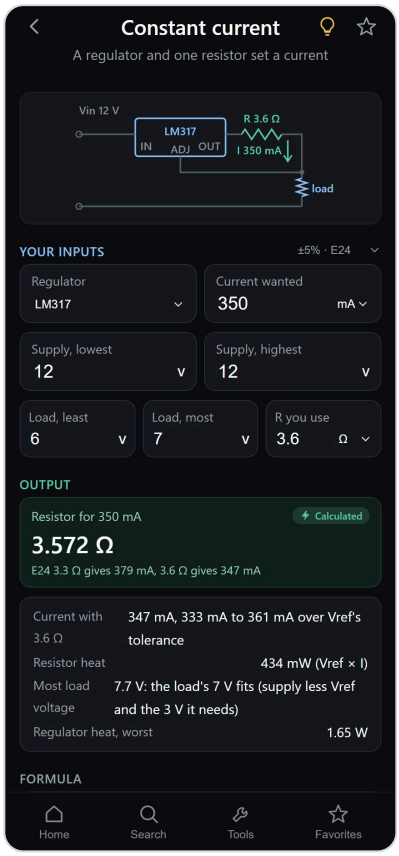
## How it was checked

9 cases: - 1N4734A and BD139, 12–18 V, 500 mA: base 19.2 mA, a 78.27–302.5 Ω window, 300 Ω, output 4.62–5.18 V; - with 300 Ω: 1.17 to 42.3 mA in the zener, 249 mW, 536 mW in the resistor; - the BD139 at 6.69 W: over free air, under its case rating; - 1N4743A and TIP31C, 16–20 V, 0.5 A: a 110.9–120.6 Ω window, 120 Ω, 353 µA left at worst, 4.17 W; - a 2N3055 at 4 A needs 190 mA of base: no resistor works; - a 50 mA load stays

within the BD139's free-air rating (669 mW); - 2 A is flagged over the BD139's 1.5 A; - a custom hFE of 100 needs 4.95 mA; - the remembered transistor and zener.

# Constant current

(formerly Constant current source/driver)




## What it computes

An adjustable regulator made into a current source with one resistor, on one screen:

- **Inputs:** the regulator (LM317, LM1117-ADJ, AMS1117-ADJ, LM1086-ADJ, with the figures read for Linear regulator, or a custom Vref and dropout), the current wanted, the supply's lowest and highest voltage, the load's least and most voltage (an LED string, a battery), and the resistor you use.
- **The big result:** the resistor, Vref / (I − Iadj), and the E24 values either side of it with the current each gives.
- **The resistor you use** follows the nearest E24 value until you type your own; a new current wanted brings it back. When the nearest would pass the part's rating, it takes the larger one instead (Pierre: an LM1086 at 1450 mA picked 820 mΩ, 1.52 A, over its 1.5 A; now 910 mΩ, 1.37 A). For it:
- the current, with its spread over the reference's tolerance;
- the resistor's heat, Vref × I;
- the most load voltage it can drive at the lowest supply (supply less Vref and the part's headroom or dropout at that current), flagged when the load needs more;
- the regulator's heat at the highest supply and the lowest load voltage;
- warnings under the part's minimum load current (typical, then guaranteed) and over its current rating, naming the resistor.
- **Units:** the current in µA, mA or A, the resistor in Ω, kΩ or MΩ.
- **Drawing:** the regulator with IN, OUT and ADJ, R from OUT, ADJ tied to the load side of R, the load to ground.
- **Remembered:** the regulator and the units.

Examples: TI's 50 mA charger (24 Ω), 350 mA for a power LED, 20 mA for an LED string, and 5 mA, too little for an LM317.

**Standard values.** The resistors' tolerance, with its E series, is picked in the "Your inputs" heading above the inputs, as in the Passive components tools: ±0.1 % to ±20 %, ±5 % being E24 and ±1 % E96; it sets the series the resistor is chosen from (E24 by default).

## Source

- **TI LM317, SLVS044Z, April 2025** (read): "By connecting a fixed resistor between the adjustment pin and output, the LM317 is also able to be used as a precision current regulator"; §8.3.3, a precision current limiter, Ilimit = 1.2 V / R1; §8.3.7, a 50 mA constant-current battery charger, "ICHG = 1.25V ÷

24 Ω. Make sure VI is greater than VBAT + 4.25V. (1.25V [VREF] + 3V [headroom])". The tool takes the reference at its worst, 1.3 V, so it asks for 4.3 V over the battery.

- **The parts' figures:** as in Linear regulator (TI LM317 and LM1117, AMS AMS1117, TI LM1086): Vref and its spread, the adjust-pin current, the minimum load current, the headroom or dropout, the current rating.

## Why the formulas are these

```
I = Vref / R + Iadj
R = Vref / (I − Iadj)
Vload max = Vin min − Vref − headroom
P resistor = Vref × (I − Iadj),   P regulator = (Vin max − Vload min − Vref) × I
```

The regulator holds Vref between OUT and ADJ. With only R between them, the current through R is Vref / R whatever the load; it flows on into the load, joined by the adjust-pin current. The regulator keeps this up as long as it has its headroom: the supply must cover the load's voltage, Vref and that headroom. Whatever the load does not use, the regulator drops as heat.

## Assumptions and limits

- **The minimum load current** is the least the regulator needs to regulate at all (LM317: 3.5 mA typical, 10 mA guaranteed); a current source below it is not assured.
- **The LM1117's dropout** is interpolated between its listed currents; the others' is their full-load figure, as in Linear regulator.
- **Vref's tolerance** moves the current by the same proportion (the LM317's 1.2 to 1.3 V is ± 4 %); a 1 % resistor adds its own 1 %.
- **The regulator's heat** needs a heatsink past a watt or two; see Heat sink sizing.

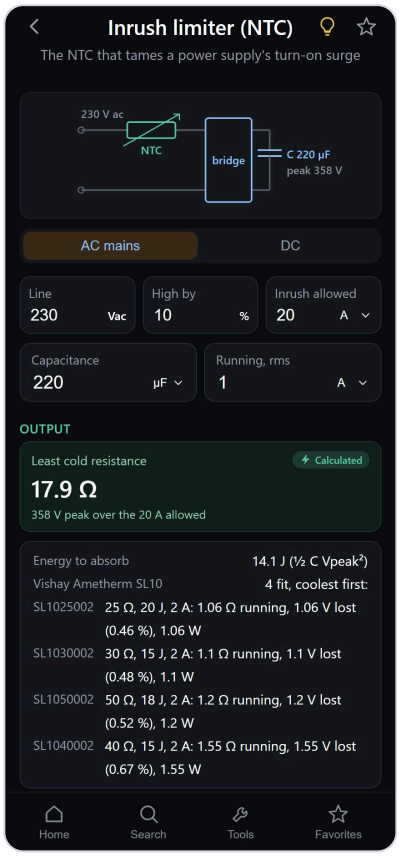
## How it was checked

13 cases: - TI's 24 Ω: 52.1 mA (50 to 54.2 mA over Vref), 65.1 mW in R; - that charger from 12 V: 7.7 V left for the battery, 248 mW in the regulator; - 350 mA: 3.572 Ω, 3.6 Ω (E24) gives 347 mA, 434 mW; - 20 mA: 62 Ω gives 20.2 mA, three LEDs at 9.6 V fit under 9.7 V; - 5 mA is under the LM317's guaranteed 10 mA, 2 mA under even its typical 3.5 mA; - a load needing more than the supply leaves is flagged; - the LM1117 needs only its 1.13 V dropout at 347 mA: 9.61 V left; - a typed resistor holds (4.7 Ω gives 266 mA); - over 1.5 A is over the LM317's rating; - an LM1086 at 1450 mA picks 910 mΩ (1.37 A), not the nearer 820 mΩ (1.52 A), which is flagged when typed; - the remembered regulator.

POWER ELECTRONICS › SUPPLY & REGULATION

# Inrush limiter (NTC)

(formerly Inrush current limiter (NTC))




## What it computes

An NTC thermistor in series with a supply, to hold down the surge that charging its capacitor draws at turn-on, on one screen:

- **AC mains or DC** (pills): the line voltage (or the DC supply), how far it can run high (%), the highest inrush current allowed, the load capacitance, and the running current.
- **The big result:** the least cold resistance, the peak voltage over the inrush allowed; the peak is √2 × the highest line voltage on AC.
- **Below it:**
- the energy the thermistor must absorb, ½ C Vpeak$^2$;
- the Vishay Ametherm SL10 parts that fit on all counts (the low end of their R25 tolerance still over the least resistance, their energy rating over the need, their current rating over the running current, 265 Vac at most), the coolest first, each with its resistance, energy and current ratings, its running resistance, the voltage it loses (and that share of the supply) and the heat;
- over 2 % of the supply lost (Pierre's threshold), a flag: a bypass relay after start-up, or an active (MOSFET) soft-start, loses less;
- when none fits, why, and that a larger series is needed.

Why an NTC rather than a resistor (Pierre asked): cold, it holds the surge down; warmed by its own current, it falls 10 to 50 times (TDK), so it loses little while running. At 230 V and 1 A an SL1025002 is about 1 Ω hot, 1 V lost (0.4 %) and 1 W; a fixed 18 Ω limiting the same surge would lose 18 V and 18 W for good. It is not free: some volts remain, and a hot NTC barely limits a quick restart. It need not be big: the energy (½ C V$^2$) and the running current set its size, and it should run at 30 to 100 % of its rating (TDK) to heat up. It suits mains supplies (about 100 to 265 Vac) and higher DC rails, where its drop is under 1 %; on a 12 V rail at 3 A the same parts lose 4.5 to 7 %, and an active soft-start is the better choice. - **Drawing:** the supply, the NTC (the calculated part, green), the bridge on AC, the capacitor, and the peak voltage. - **Units:** both currents in mA or A, the capacitance in µF or mF, each with its own picker. - **Remembered:** AC or DC, and the units.

Examples: Eaton's 110 Vac + 10 %, Ametherm's 265 Vac and 100 A, a 230 V supply with 220 µF, a 48 V DC rail, and a 12 V rail at 3 A.

## Source

- **Eaton, "Understanding the current limiting capabilities of power NTC thermistors", technical note ELX1247, July 2022** (read): the minimum zero-power resistance is the application's voltage divided by "the maximum desired peak surge current" (equation 2); the peak voltage from the maximum line voltage (equation 3): "a 110 VAC power supply with line voltage fluctuations that can

inflate that value by 10% would yield a Vmax voltage of 121 V. Then, maximum peak voltage (Vpeak) would be 171 V"; the energy of the load capacitance must be known to select the part; the running current must not exceed its rating. Its equations are images in the PDF; the text gives them in words.

- **TDK Electronics, "NTC Inrush Current Limiters, application notes", July 2019, and EPCOS application note 2013** (read): the thermistor is connected in series, never in parallel; it drops by a factor of 10 to 50 when hot; its maximum permissible capacitance is tested per IEC 60539-1; it needs one to two minutes to cool before limiting fully again; the running resistance can be fitted as $R = k \cdot I^n$ between 30 and 100 % of Imax.
- **Vishay Ametherm SL10, document 24503, revision 13 Aug 2026** (read): per part R25 and tolerance, the maximum steady current up to 65 °C, the maximum recommended energy (J), 265 Vac maximum, the maximum capacitance at 120 Vac, the resistance at 100 % and 50 % of the maximum current. Its own figures follow ½ C $V_{peak}^2$: 2000 µF at 120 Vac (170 V peak) is 28.8 J against the part's 30 J; 1300 µF is 18.7 J against 20 J. SL1050001 is left out: its table gives a lower resistance at 50 % than at 100 % current.
- **Ametherm's inrush page** (read): its example, 265 Vac (375 V peak) and 100 A, "select a thermistor that can handle 13 A of current and has a capacity of at least 126 Joules"; 1800 µF reproduces 126 J, though the page does not print the capacitance.

## Why the formulas are these

```
Vpeak = √2 × V line × (1 + high-line %)     (DC: V × (1 + %))
R cold ≥ Vpeak / I inrush
E = ½ × C × Vpeak²
P running = I² × R hot
```

At turn-on the capacitor is empty: the surge is the peak voltage over whatever resistance is in the way, so that resistance sets the surge. Charging a capacitor through a resistance costs as much energy in the resistance as ends up in the capacitor, ½ C $V^2$, and the cold NTC must take it without cracking. Once running, the current heats it until its resistance is small.

## Assumptions and limits

- **Running resistance:** the datasheet's points at 100 % and 50 % of the part's current, interpolated between; under half its current it is higher than the 50 % figure, and the tool says only that.
- **Ambient:** the SL10's currents hold up to 65 °C; above that, derate (the datasheet's curve).
- **Restarting:** a hot thermistor barely limits; it needs one to two minutes off to cool (TDK).
- **The wiring and the capacitor's ESR** also limit the surge; the tool leaves them out, which is on the safe side.
- **One series:** the SL10 (10 mm discs, up to 5 A and 30 J); bigger supplies need a larger series.

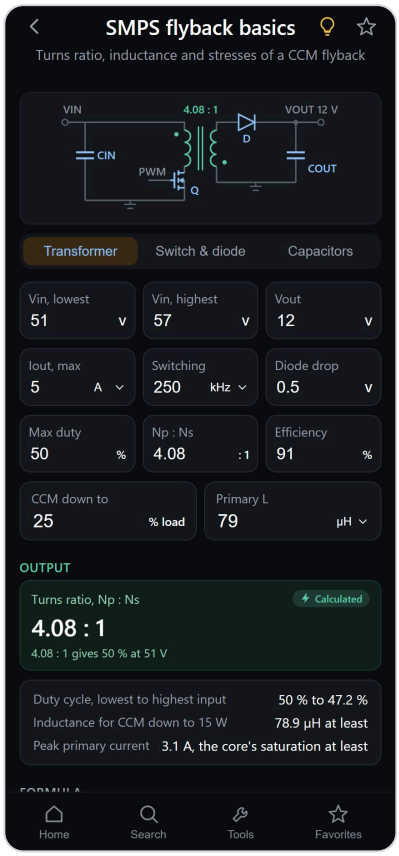
## How it was checked

12 cases: - Eaton's 110 Vac + 10 %: 171 V peak, 8.56 Ω for 20 A, 6.88 J; - six SL10 parts fit, coolest first (SL1050002, 440 mΩ, 880 mV lost, 0.8 %, 1.76 W); - at 230 V and 1 A the hot NTC loses under 1 % (SL1025002: 0.46 %); - a 12 V rail at 3 A: every part loses over 2 % (SL102R005: 4.5 %), flagged; - Ametherm's

265 Vac, 100 A: 3.75 Ω, 126 J, no SL10 fits; - 230 V + 10 %, 220 µF, 30 A: 11.9 Ω, 14.1 J, six fit; - a part under half its current is over its light-load resistance; - a 48 V DC rail: 5.04 Ω, 1.27 J, three parts carry 3 A; - over 265 Vac and over 5 A are flagged; - the remembered AC or DC; - capacitance in mF and running current in mA: 220 µF reads 0.22, 1 A reads 1000.

POWER ELECTRONICS › SUPPLY & REGULATION

# SMPS flyback basics




## What it computes

The power stage of an isolated flyback converter in continuous conduction (CCM), the way TI's John Betten designs one, on one screen:

- **Shared inputs:** the input range (lowest and highest), the output voltage and current, the switching frequency and the output diode's drop.
- **Three pills:**
- **Transformer:** from the maximum duty cycle you choose at the lowest input, the turns ratio Np : Ns, proposed until you type your own (Betten rounds his 4.08 to 4); the duty cycle over the input range with it; the primary inductance that keeps the converter in CCM down to a share of full load you set, proposed (two figures, rounded up) until you type your own, flagged when yours is lower; the peak primary current.
- **Switch & diode:** the MOSFET's flat-top drain voltage and the 25 to 50 % more to allow for ringing, the peak primary current, the diode's reverse voltage, its current while it conducts and its loss.
- **Capacitors:** the least output capacitance for the ripple allowed, its RMS current, and the input capacitance for the input ripple.
- **Units:** the current in mA or A, the frequency in kHz or MHz, the inductance in µH or mH, each ripple in mV or V with its own picker.
- **Drawing:** the input capacitor, the primary winding and the n-channel MOSFET Q (gate plate, three-dash channel, body arrow tied to the source, as in the MOSFET tools) on the primary ground; the secondary wound the other way (the dots), the diode, and the output capacitor last on the isolated secondary ground. The pill's parts are green.
- **Remembered:** the pill and the units.

Examples: TI's 60 W design (its turns ratio and inductance as calculated, then rounded to its 4 : 1 and 80 µH), its MOSFET and diode, and its capacitors.

## Terms

- **Flyback converter:** a switching supply that stores energy in a transformer's core while its switch is on, and releases it to the output through the diode while the switch is off. The two windings are wound the opposite way (the dots) and isolate the output from the input.
- **MOSFET (the switch, Q):** an n-channel enhancement MOSFET. FET (field-effect transistor) is the family it belongs to, with the JFET; Betten's article just says "FET".
- **Turns ratio (Np : Ns):** primary turns over secondary turns.
- **The dots (winding sense):** each dot marks the end of a winding that goes positive at the same moment as the other dotted end. In a flyback the dots sit at opposite ends, primary at the input end, secondary at the ground end: the windings are wound (or connected) the opposite way. While the MOSFET is on, the input drives the primary's dotted end positive, so the secondary's dotted (ground)

end is positive too and its other end, the diode's anode, is negative: the diode blocks, no current flows to the output, and the energy builds up in the core (in its magnetizing inductance, the primary L). When the MOSFET turns off, the core's field keeps the current going and every winding's voltage reverses: the diode's anode goes positive, the diode conducts and the stored energy "flies back" to the output capacitor and the load (Betten, Power Tips #98: during the on-time the diode "is reverse-biased because of T1's secondary winding polarity"; at turn-off the "secondary voltage polarity reverses"). That is why a flyback's transformer works as two coupled inductors, and why the primary inductance matters. Both windings' turns face the core in the drawing, as in any transformer symbol; only the dots show the sense.

- **Reflected voltage:** while the diode conducts, the output plus the diode drop appear across the primary multiplied by Np/Ns; the MOSFET sees the input plus that.
- **Flat-top voltage:** the MOSFET's drain voltage while off, before the ringing that the transformer's leakage inductance adds.
- **CCM / DCM:** in continuous conduction the transformer's current never falls to zero within a cycle; below a certain load it does (discontinuous conduction, DCM).
- **Primary inductance (L):** the inductance of the primary winding; it sets the load at which the converter leaves CCM.
- **RMS current:** the heating value of a current; capacitors are rated for it.

## Source

- **John Betten, "Designing a CCM flyback converter", TI technical article SSZTCY0, February 2024** (read): a 60 W, 53 V to 12 V at 5 A flyback at 250 kHz, 51 to 57 V in, 50 % maximum duty cycle at 51 V. Eq. 1, the volt-second balance; eqs. 2, 3: Nps = Vinmin × dmax / ((Vout + Vd)(1 − dmax)), 51 V × 0.5 / (12.5 V × 0.5), "4:1 step-down"; eq. 4: d = Nps (Vout + Vd) / (Vin + Nps (Vout + Vd)), 0.47 at 57 V; eq. 5: Vdsmax = Vinmax + Nps (Vout + Vd) = 107 V, "derate the FET voltage an additional 25-50%, selecting a 200V FET"; eq. 6: the diode's reverse voltage Vout + Vinmax / Nps, 26 V; eq. 7: ID1 = Ioutmax / (1 − dmax), 10 A; eq. 8: PD1 = Ioutmax × Vd, 1.7 W at the 0.33 V of the diode chosen; eq. 9: Lmin = Vinmin² dmax² η / (2 fsw Poutmin), 80 µH at 0.91 and 15 W, Poutmin "typically 20-30% of POUTMAX"; eq. 10: Ipripk = Ioutmax / ((1 − dmax) Nps) + Vinmin dmax / (2 Lpri fsw), 3.14 A; eq. 14: Coutmin = Ioutmax dmax / (fsw Vripout), 83 µF for 0.12 V; eq. 15, the output RMS current, 5 A; eq. 16: Cinmin = Ipripk dmax / (2 fsw Vinrip), 2 µF for 1.5 V. The PDF's text lost the square root in eq. 15; at 50 % both readings give 5 A, and the tool uses Iout √(d / (1 − d)).
- **John Betten, "Power Tips #98: Designing a DCM flyback converter", TI** (read): the same converter in discontinuous conduction, its equations for the turns ratio with an idle time, the inductance and the peak current. The tool does not compute DCM; this is for further reading.
- **Not used:** Unitrode's "60 Watt Flyback Regulator" (TI SLUP072, read): its primary inductance is derived in appendices missing from TI's PDF; Fairchild's AN-4137 could not be downloaded.

## Why the formulas are these

```
Np/Ns = Vin(min) × dmax / ((Vout + Vd)(1 − dmax))
d = N (Vout + Vd) / (Vin + N (Vout + Vd))
Vds = Vin(max) + N (Vout + Vd),   Vdiode = Vout + Vin(max) / N
L = Vin(min)² d² η / (2 fs Pout(min))
Ipk = Iout / ((1 − d) N) + Vin(min) d / (2 L fs)
Cout = Iout d / (fs ΔVout),   Cin = Ipk d / (2 fs ΔVin)
```

The core must end each cycle where it started: the input across the primary for the on-time d must equal the reflected output, (Vout + Vd) × N, for the off-time 1 − d. That gives the turns ratio for the duty cycle chosen, and the duty cycle at any input once the ratio is set. The output gets current only during 1 − d, so the diode carries Iout / (1 − d) while it conducts, and the primary that times 1/N, plus half the ripple: the peak. While the switch is on the output capacitor alone feeds the load, hence its size.

**Which duty cycle.** Betten works his currents and capacitors at dmax = 0.5 although his rounded 4 : 1 gives 0.495 at 51 V. The tool works everything at the duty cycle the turns ratio in use gives at the lowest input, so with 4 : 1 it shows 3.11 A of peak current where he gives 3.14 A, 77.3 µH where he gives 80 µH, 82.5 µF where he gives 83 µF and 2.05 µF where he gives 2 µF. With his exact 4.08 : 1 they agree.

## Assumptions and limits

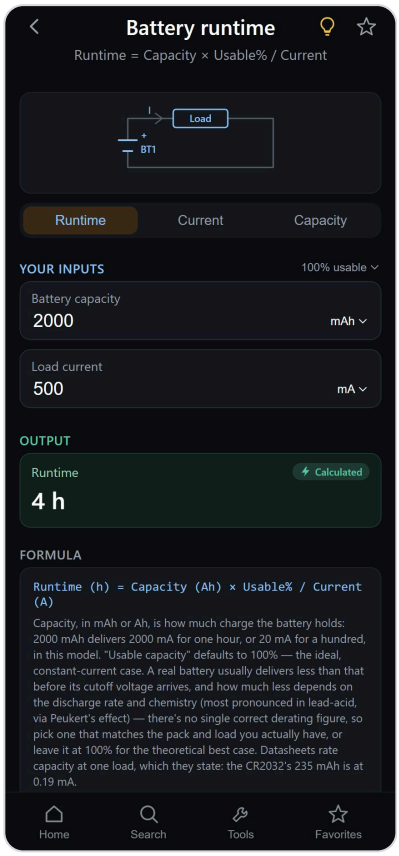
- **CCM at full load only:** below Pout(min) the converter enters DCM, the duty cycle falls and these formulas no longer hold.
- **The diode drop** is one figure for both the turns ratio and the loss; Betten assumes 0.5 V for the ratio and uses his diode's 0.33 V for the loss.
- **Ringing:** the flat-top voltages leave out the leakage-inductance spike; the 25 to 50 % allowance is Betten's, and a snubber or clamp is usually needed.
- **Not computed:** the transformer's core, turns and wire, the leakage inductance, the current-sense resistor, the MOSFET's losses, the control loop and any second output.

## How it was checked

7 cases: - TI's design: 51 V at 50 %, 12 V + 0.5 V, gives 4.08 : 1, and 78.9 µH at 91 % keeps CCM down to 15 W (its 80 µH); 50 % to 47.2 % over the input; - rounded to its 4 : 1: 49.5 % to 46.7 % (its 0.47 at 57 V), 3.11 A peak with its 80 µH; - 107 V flat-top on the MOSFET (its 107 V), 26.3 V on the diode (its 26 V), 9.9 A while it conducts, 2.5 W at 0.5 V; - 82.5 µF for 120 mV, 2.05 µF for 1.5 V, 4.95 A RMS in the output capacitor; - the proposed turns ratio follows the input (40 V: 3.2 : 1); - a typed inductance under the minimum is flagged; - a 100 % maximum duty is refused.

# Battery runtime

(formerly Runtime / capacity estimation)




## What it computes

Any one of runtime, load current and battery capacity from the other two, one pill each, with a "usable" share of the capacity (100% by default). The answer is the large result; under a runtime of more than two days the same span is given in days, and past two years in years. The pill, the usable share and the units are remembered.

Examples: a CR2032 at the load its datasheet rates it at, a sensor averaging 20 µA on one, an 18650 at 500 mA, the current that lasts a day, and the capacity a 1 A load needs for 8 hours.

## Source

- **Energizer CR2032 datasheet:** typical capacity 235 mAh to 2.0 V, rated at 15 kΩ (a drain of 0.19 mA at 2.9 V), 1245 hours to 2.0 V in that test, and about 1% a year self-discharge.
- **Panasonic NCR18650B datasheet:** rated capacity at least 3200 mAh (typical 3350 mAh), discharged to 2.5 V.
- **Energizer E91 (AA alkaline) datasheet:** it gives capacity as a graph against continuous discharge current, 25 to 500 mA, not as one number — which is why the tool leaves the usable share to you rather than inventing one figure.

## Why the formulas are these

```
Runtime = Capacity × Usable / Current
Current = Capacity × Usable / Runtime
Capacity = Current × Runtime / Usable
```

Capacity in ampere-hours is a charge: a current times a time. A battery that delivers 2000 mA for one hour holds 2000 mAh, and in this model gives 20 mA for 100 hours. The CR2032 checks it: 235 mAh at 0.19 mA is 1237 hours, and Energizer measured 1245.

## Assumptions and limits

- **Constant current.** Real loads pulse; use the average current, and know that heavy pulses cost more than their average on small cells.
- **Capacity depends on the load.** Datasheets rate capacity at one drain, which they state; at higher currents a cell generally delivers less, most markedly lead-acid (Peukert's effect). The usable share is where that goes, and no single figure fits every cell and load.

- **Cutoff voltage.** The rated capacity is to a stated end voltage (2.0 V for the CR2032, 2.5 V for the NCR18650B); a circuit that quits earlier gets less.
- **Self-discharge and temperature** are not modelled: over a year and more, the CR2032's 1% a year starts to count, and cold cuts capacity.

## What it deliberately does not do

- **Cell capacities** are in Battery types & sizes and Button cell sizes.
- **C-rate**, the current as a multiple of capacity, has its own tool.

## How it was checked

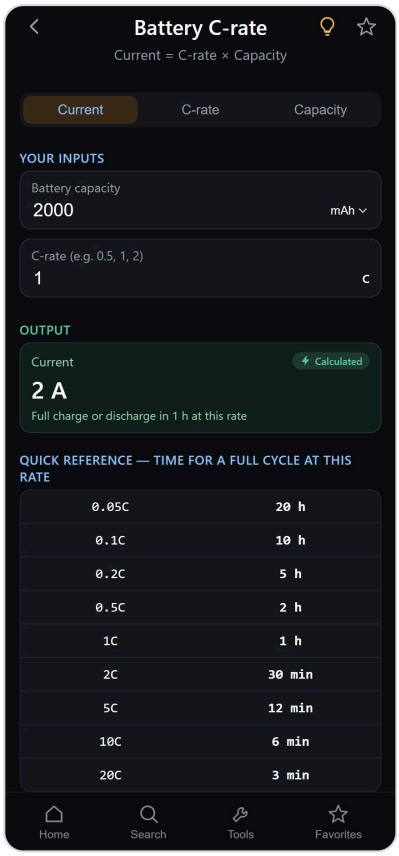
2000 mAh at 500 mA gives 4 h; at 80% usable, 3 h 12 min. 235 mAh at 0.19 mA gives 1237 h, about 51.54 days; at 20 µA, 11750 h, about 489.6 days. 3200 mAh at 500 mA gives 6 h 24 min. 2000 mAh over 24 h allows 83.3 mA. 1 A for 8 h at 85% usable needs 9.41 Ah.

The review made these changes: - The answer gained a line in days or years for long runtimes. - Values are kept in Ah, A and h, so a remembered unit only changes how a field shows them; the pill, the usable share and the units are remembered. - The drawing gained the cell's + plate, its designator BT1 and the current arrow its comment had promised. - It added five examples from the CR2032 and NCR18650B datasheets, the first cases (10) and this section.

POWER ELECTRONICS › BATTERY

# Battery C-rate

(formerly C-rate)




## What it computes

Any one of current, C-rate and capacity from the other two, one pill each, with the time a full charge or discharge takes at that rate, and a table of common C-rates and their times. The pill and the units are remembered.

Examples: a Panasonic 18650's standard charge, the same cell at 1C, a CR2032 at the load its datasheet rates it at, and the capacity a 2 A load needs to stay at 0.5C.

## Source

- **Panasonic NCR18650B datasheet:** standard charge CC-CV at 1625 mA to 4.20 V; capacity at least 3250 mAh at 25 °C, rated capacity at least 3200 mAh at 20 °C; its cycle-life and temperature curves discharge at 1C.
- **Energizer CR2032 datasheet:** 235 mAh rated at 15 kΩ, a drain of 0.19 mA.

## Why the formulas are these

```
Current = C-rate × Capacity
Time = 1 / C-rate
```

1C is the current that would move the battery's rated capacity in one hour: a 2000 mAh cell at 1C draws 2 A, at 0.5C 1 A for two hours. The C is the capacity, not the coulomb, though capacity in Ah is itself a charge (1 Ah = 3600 C). The 18650's standard charge checks it: 1625 mA into 3250 mAh is exactly 0.5C.

## Assumptions and limits

- **A notation, not a permission.** A cell's allowed charge and discharge C-rates are in its datasheet; this tool converts, it does not judge.
- **The time is nominal.** A real charge slows in its constant-voltage phase, and a real discharge stops at the cutoff voltage, both before 1 / C-rate.
- **Capacity is the rated one.** Which figure a datasheet's C refers to — the rated or the typical capacity — is its own; the NCR18650B quotes both.

## What it deliberately does not do

- **Runtime from a real load** is Battery runtime, with a usable share.
- **Peukert's effect and temperature** are not modelled.

## How it was checked

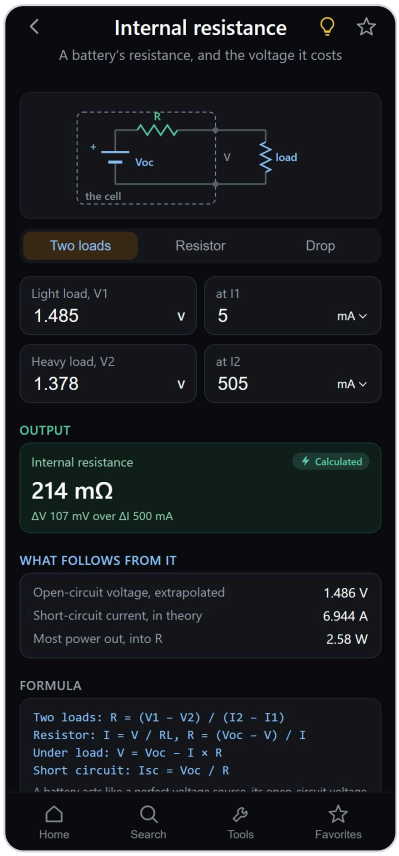
2000 mAh at 1C gives 2 A and 1 h; at 0.2C, 400 mA and 5 h. 1625 mA into 3250 mAh gives 0.5C and 2 h. 1C on 3200 mAh gives 3.2 A. 0.19 mA from 235 mAh gives 0.0008085C, 1237 h. 2 A at 0.5C needs 4 Ah.

The review made these changes: - Values are kept in Ah and A, so a remembered unit only changes how a field shows them; the pill and the units are remembered. - It added four examples from the NCR18650B and CR2032 datasheets, the first cases (9) and this section.

POWER ELECTRONICS › BATTERY

# Internal resistance

(formerly Battery internal resistance)




## What it computes

A battery's internal resistance, and the voltage it costs under load, three ways, by pill:

| Pill | Inputs | Output |
|---|---|---|
| Two loads | the voltage at a light and at a heavy current | R, and the open-circuit voltage extrapolated |
| Resistor | the voltage open, then across a known resistor | R, and the current drawn |
| Drop | the open-circuit voltage, R and a load current | the voltage at the terminals |

- **Measured modes also give:**
- the short-circuit current, Voc / R, in theory;
- the most power the cell can deliver, into a load equal to R.
- **Drop mode also gives:**
- the power lost inside the cell, $I^2R$;
- the power delivered;
- the current that would pull the cell down to a cutoff voltage you set.
- **Units:** currents in µA, mA or A, resistance in mΩ or Ω, the test resistor in Ω or kΩ.
- **Drawing:** the cell as an ideal source behind its resistance, in a dashed box, feeding the load.
- **Remembered:** the pill and the units.

Examples: Energizer's dual pulse and voltage-drop examples, a 9 V battery with a 100 Ω resistor, a lithium-ion cell at two loads, and that cell at 10 A.

## Source

**Energizer, "Battery Internal Resistance", technical bulletin, version 1.1.0, December 2005** (data.energizer.com), read in review: - **The definition:** internal resistance is "the opposition to the flow of current within the battery", electronic plus ionic. - **The dual pulse:** a low stabilising drain, then a heavier pulse of about 100 ms; the resistance is "the change in voltage by the change in current". Its example: 5 mA then 505 mA, 1.485 V to 1.378 V, 0.107 V / 500 mA = 0.214 Ω. Fresh alkaline cylindrical cells run "approximately 150 to 300 milliohms". - **Voltage under load:** "multiplying the total effective resistance by the current". Its example: 1 A on 0.1 Ω from 1.6 V leaves 1.5 V. - **Its cautions:** - a 1000 Hz impedance test reads less than this total effective resistance; - the resistance rises as the cell discharges and as the temperature drops; - results depend on the technique, and are "a general guideline and not a precise value".

## Why the formulas are these

```
two loads:  R = (V1 − V2) / (I2 − I1),  Voc = V1 + I1 × R
resistor:   I = V / RL,  R = (Voc − V) / I
under load: V = Voc − I × R,  loss = I² × R
Isc = Voc / R,  Pmax = Voc² / (4R)
```

A cell is modelled as a perfect source of its open-circuit voltage behind its resistance. The voltage falls in step with the current, so two readings give the slope, R. The most power comes out when the load equals R, and half of it then heats the cell.

## Assumptions and limits

- **One resistance:** a real cell's resistance has a fast electronic part and a slower ionic part, and changes with charge, age and temperature. The figure is for the conditions it was measured in.
- **The open-circuit voltage** in Two loads is extrapolated along the straight line; it may differ from a cell left at rest.
- **The short-circuit current** is a limit, not a test: shorting a cell, a lithium one above all, can burn it.

## How it was checked

11 cases: - Energizer's dual pulse is 214 mΩ, extrapolating to 1.486 V open; - Energizer's drop leaves 1.5 V and loses 100 mW; - a 9 V battery at 8.6 V across 100 Ω has 6.977 Ω inside, at 86 mA; - a lithium-ion cell at two loads is 50 mΩ; - that cell at 10 A keeps 3.5 V, loses 5 W, and reaches 3.0 V at 20 A; - equal currents, readings the wrong way round, and a loaded voltage above the open one are refused; - a unit change keeps the value; - the remembered choices; - the test resistor in kΩ: 15 Ω reads 0.015.

POWER ELECTRONICS › BATTERY

# Battery types & sizes




## What it computes

A searchable list of cylindrical cells — AAAA to D, 9V, the lithium AA and AAA, CR2, CR123A and the Li-ion 14500, 18650, 21700 and 26650 — each with its IEC and ANSI designations where they exist, dimensions, a capacity, and its chemistry and voltage. A filter splits alkaline/NiMH from lithium, and is remembered; the search matches the size, either designation or the text.

## Source

Energizer's datasheets, one per size, read in review:

- **Designations:** AAAA IEC LR61 (also LR8D425), ANSI 25A; AAA LR03, 24A; N LR1, 910A; AA LR6, 15A; C LR14, 14A; D LR20, 13A; 9V 6LR61 (also 6LF22), 1604A; lithium AAA (L92) FR03, 24-LF; lithium AA (L91) FR6 (FR14505), 15-LF; CR123A CR17345, 5018LC.
- **Lithium iron disulfide against alkaline:** the L91 AA weighs 15 g against the E91 alkaline AA's 23 g, the L92 AAA 7.6 g against the E92's 11.5 g; both lithium cells run from −40 to 60 °C and keep 25 years on the shelf, against 10 years for the alkaline AA; the L91 allows 2.5 A continuous.
- **CR123A:** 1500 mAh to 2.0 V, rated at 100 Ω; 1.5 A continuous.

Not from those sheets: the dimensions are the industry-standard maxima the sheets draw but whose figures could not be read as text, the CR2's sheet was not found, and the "≈" capacity ranges are typical market figures, because Energizer gives alkaline capacity only as a graph against the load.

## Why the formulas are these

There is no formula: it is a reference. Two conventions are worth knowing. The IEC designation encodes the chemistry in its letters (L alkaline, F lithium iron disulfide, C lithium manganese dioxide, R round) and the size after them; the Li-ion names encode their dimensions, the first two digits the diameter in mm and the next two the length, so a 21700 is 21 × 70 mm.

## Assumptions and limits

- **Capacity depends on the load.** A capacity is meaningful only with the drain it was measured at; the ranges here are orientation, not a spec. For a real design, read the cell's datasheet graph at your current.
- **NiMH** cells in the same sizes are 1.2 V and have their own capacities.

## What it deliberately does not do

- **Button and coin cells** are their own tool, Button cell sizes.
- **Runtime** from a capacity is Battery runtime.

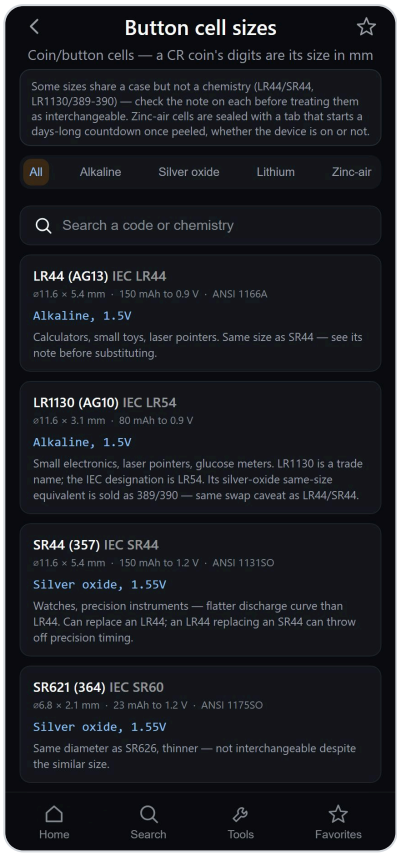
## How it was checked

Searching "fr6" finds the lithium AA, "1604" the 9V, "17345" the CR123A with 1500 mAh and ANSI 5018LC; the Lithium filter starts with the lithium AAA and is remembered.

The review made these changes: - The lithium AA and AAA showed Energizer's model numbers L91 and L92 where the IEC designation belongs; they now read FR6 and FR03. The CR123A gained its IEC CR17345, and every cell with one its ANSI code. - The lithium cells' claimed advantages — a third lighter, below freezing, long shelf life — now carry Energizer's figures; the CR123A's capacity is Energizer's 1500 mAh rather than an approximation. - The search covers the ANSI codes; the filter is remembered. - It added the first cases (5) and this section.

# Button cell sizes

(formerly Button cells)




## What it computes

A searchable list of coin and button cells — alkaline, silver oxide, lithium and zinc-air — each with its IEC and ANSI designations where known, dimensions, a capacity with the end voltage it is measured to, its chemistry and voltage, and a note on uses and substitutes. A filter by chemistry is remembered; the search matches the common name, either designation or the chemistry.

## Source

Energizer's datasheets, read in review, give each cell's typical capacity and designations:

- **Alkaline:** LR44 (Energizer A76) 150 mAh to 0.9 V at 7.5 kΩ, IEC LR44, ANSI 1166A; LR1130 (189) 80 mAh to 0.9 V, IEC **LR54**.
- **Silver oxide**, to 1.2 V: SR44 (357) 150 mAh at 6.8 kΩ, IEC SR44, ANSI 1131SO; SR621 (364) 23 mAh at 68 kΩ, IEC SR60, ANSI 1175SO; SR626 (377) 27 mAh at 45 kΩ, IEC SR66, ANSI 1176SO; SR920 (371) 35 mAh at 30 kΩ, IEC SR69.
- **Lithium**, to 2.0 V: CR1220 37 mAh (ANSI 5012LC), CR1616 60 mAh, CR1632 130 mAh, CR2016 100 mAh (5000LC), CR2025 170 mAh (5003LC), CR2032 235 mAh at 15 kΩ (5004LC), CR2450 620 mAh (5029LC).

Not from those sheets: the dimensions, which the sheets draw but could not be read as text, and the zinc-air capacities, which remain typical market ranges.

## Why the formulas are these

There is no formula: it is a reference. A lithium coin's name gives its size: the first two digits are the diameter in mm and the last two the thickness in tenths of a mm, so a CR2032 is 20 × 3.2 mm. The IEC numbers of alkaline and silver-oxide cells (LR54, SR66) are size codes, not dimensions, which is why trade names like LR1130 and SR626 stay in use.

## Assumptions and limits

- **Capacity is to an end voltage at one load,** given with each figure; a heavier load or a higher cutoff gets less.
- **Same size, different chemistry:** LR44 and SR44 share a case at 1.5 V and 1.55 V; silver oxide's flatter curve suits watches, so the substitution works one way only.
- **Zinc-air** cells start their life when the tab is peeled.

## What it deliberately does not do

- **Cylindrical cells** are Battery types & sizes.
- **Runtime** is Battery runtime.

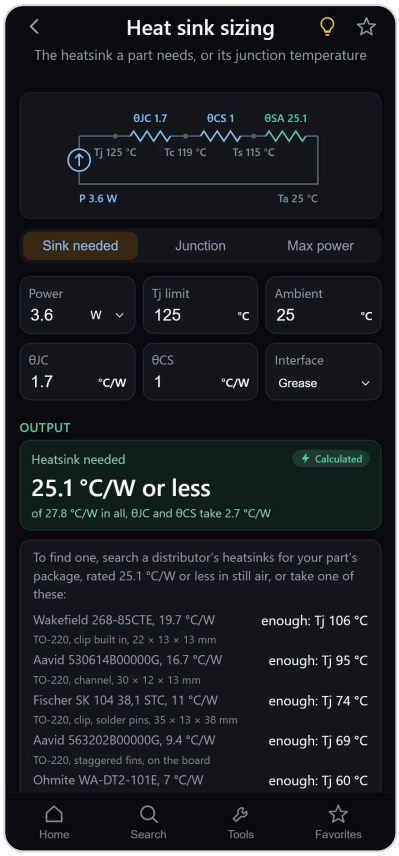
## How it was checked

Searching "lr54" finds the LR1130 with 80 mAh to 0.9 V; "cr2016" shows 100 mAh to 2.0 V and ANSI 5000LC; "sr66" finds the SR626; the Zinc-air filter is remembered.

The review made these changes: - Capacities are Energizer's typical figures with their end voltage, where the table had ranges — three of which (CR1616, CR2016, CR2025) sat below Energizer's figure. - IEC and ANSI designations were added; LR1130's IEC name is LR54, and the silver-oxide cells' are SR44, SR60, SR66 and SR69. The search covers them. - The filter is remembered; it added the first cases (4) and this section.

# Heat sink sizing

(also takes in the planned Thermal resistance (θJA / θJC))




## What it computes

The thermal chain from a part's junction to the air, three ways, by pill:

| Pill | Inputs | Output |
|---|---|---|
| Sink needed | power, junction limit, ambient, θJC, θCS | the heatsink θSA to buy, at most |
| Junction | power, ambient, θJC, θCS, θSA | the junction temperature |
| Max power | junction limit, ambient, θJC, θCS, θSA | the most power the part can dissipate |

- **The terms**, each a thermal resistance in °C per watt:
- **θJC**, junction to case: the part's datasheet, the figure through the tab or the bottom of the package (often written RθJC(bot));
- **θCS**, case to sink: the interface, grease, a pad or mica;
- **θSA**, sink to air: the heatsink's rating, usually in still air.
- **Sink needed** says how to find a heatsink: search a distributor's heatsinks for the part's package, rated at the answer or less in still air. It then checks five real heatsinks, small to large (19.7 to 7 °C/W), each with its maker and order code and what it fits (package, clip or channel, solder pins, size), and gives the junction temperature each one would reach. When θJC and the interface alone use up the whole budget, it says no heatsink is enough and gives the most power an ideal one (0 °C/W) would allow.
- **Junction** also gives the case (tab) and heatsink temperatures, which is what a finger or a thermocouple meets.
- **Max power** also gives the heatsink temperature at that power.
- **A picked heatsink** is named, with what it fits, in Junction and Max power.
- **Pickers:** the interface fills θCS with a TO-220 figure (grease, dry, mica with grease, dry mica); the heatsink fills θSA. Any value can be typed instead, and the picker then reads Custom.
- **Drawing:** the chain as a circuit, the power as a source driving into the junction, θJC, θCS and θSA in series to the ambient, with the temperature of each node; the unknown in green.
- **Units:** power in mW or W. **Remembered:** the pill and the unit.

Examples: a 7805 dissipating 3.6 W (from the Linear regulator tool's 12 V to 5 V at 500 mA) in a TO-220 with grease, the same insulated with mica in a 50 °C box, 7 W on an SK 104, the most power on an SK 104 with dry mica, and 50 W in a TO-220.

## Source

- **Fairchild (onsemi) AN-4166, "Heat Sink Mounting Guide", rev. 1.0.0, 2014** (read): the junction-to-ambient resistance "is the sum of" junction-case, case-heat sink, heat sink and heat sink-ambient resistances (figure 1); the contact resistance between case and sink depends on surface roughness and contact pressure, and grease or a pad fills the air gap. Its table 1 gives interface materials per area (°C·in$^2$/W), not per package, so it is not used for the figures.
- **Motorola AN-778, "Mounting Considerations for Power Semiconductors"** (read, a scanned copy; its successor is onsemi AN1040, which onsemi's site refused to serve), table 1, TO-220AB on a 6-32 screw at 8 in-lb: metal to metal 1.2 °C/W dry, 1.0 lubed (grease); with a 2 mil mica insulator 3.4 dry, 1.6 lubed. "Dry interface values are subject to wide variation." Its text: grease typically reduces the interface resistance 2 to 1 to 3 to 1 with an insulator.
- **θJC in the examples:** TI LM340/LM7805 family (SNOSBT0L, September 2016), thermal information table, TO-220 RθJC(bot) 1.7 °C/W.
- **Heatsinks,** each from its maker's own sheet (copies hosted by distributors), in still air; the makers rate them at different points, as noted:
- Wakefield-Vette 268-85CTE: "channel style TO-220 heat sink featuring integrated mounting clip", 21.55 × 13.21 × 13.08 mm, 19.73 (the sheet prints no unit or condition; taken as °C/W in still air, like the others);
- Aavid (Boyd) 530614B00000G: TO-220, "channel style heat sink with wide mounting surface", 29.97 × 11.68 × 12.70 mm, 16.70 °C/W "based on a 75 °C rise in natural convection";
- Fischer Elektronik SK 104 38,1 STC: TO-220 drilling, "transistor clip mounting", solder pins, 34.9 × 12.7 mm, 38.1 mm long, Rth 11 K/W (Fischer's product page, a copy downloaded from Arrow);
- Aavid (Boyd) 563202B00000G: TO-220, "space saving staggered fin heat sink", board-mounted, 9.40 °C/W at a 75 °C rise;
- Ohmite WA-DT2-101E: "heat sink for TO-247 & TO-220", integral spring clip, solderable feet, vertical through-hole, Rs-a 7 °C/W in natural convection at 10 W.

An Aavid 577202B00000G (24.4 °C/W) was in the first version; it was dropped because only distributor listings of it were read.

## Why the formulas are these

```
Tj = Ta + P × (θJC + θCS + θSA)
θSA = (Tj − Ta) / P − θJC − θCS
Pmax = (Tj − Ta) / (θJC + θCS + θSA)
Tc = Ta + P × (θCS + θSA),   Ts = Ta + P × θSA
```

Heat flowing through a thermal resistance raises the temperature across it by θ × P, as a current through a resistor raises a voltage. The three steps are in series, so their resistances add, and each node sits above the ambient by the power times everything between it and the air. The heatsink you need is what is left of the allowed total, (Tj − Ta) / P, once the part and the interface have taken theirs; a smaller θSA is a bigger heatsink.

## Assumptions and limits

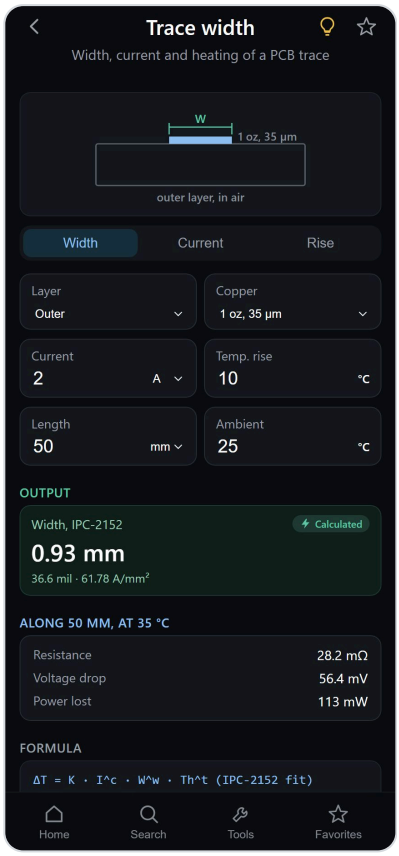
- **All the heat goes through the heatsink.** A part also loses a little through its leads and the board, so the real junction runs slightly cooler.
- **θSA is not constant.** Makers rate a heatsink at one temperature rise, in still air, mounted a given way; it improves as the sink gets hotter and with any airflow, and gets worse in a closed box or mounted flat.
- **The interface figures are for a TO-220 on a screw.** Other packages, clips, thermal pads and torques differ; type the pad maker's figure.
- **The ambient** is the air around the heatsink, which inside a box is above room temperature.
- **125 °C** is the default junction limit, the recommended maximum of most regulators; type the part's own.

## How it was checked

11 cases: - 3.6 W, 125 °C from 25 °C, θJC 1.7 and grease 1.0: 25.1 °C/W or better, out of 27.8 in all; the Wakefield gives 106 °C, the SK 104 74 °C, and the hint names 25.1 °C/W; - mica and grease in a 50 °C box: 17.5 °C/W, the Wakefield too weak at 133 °C, the Aavid 530614 at 122 °C, the SK 104 at 101 °C; - a picked heatsink is named with what it fits; - 7 W through 1.7 + 1.0 + 11 °C/W: junction 121 °C, tab 109 °C, sink 102 °C; - from 50 °C through 16.1 °C/W in all: 4.66 W at most, the sink at 101 °C; - 50 W in a TO-220: no heatsink is enough, 37 W with an ideal one; - the interface picker fills 3.4 °C/W, and a typed value reads Custom; - a unit change keeps the power (3.6 W reads 3600 mW); - a junction limit under the ambient is refused; - the remembered pill.

# Trace width

(replaces PCB trace width / resistance, Trace current capacity (ampacity) and Trace current — IPC-2152 method)




## What it computes

How hot a copper trace on a PCB runs for its current, by IPC-2152, three ways, by pill:

| Pill | Inputs | Output |
|---|---|---|
| Width | the current and the temperature rise allowed | the width needed |
| Current | the width and the temperature rise | the current it carries |
| Rise | the width and the current | how far above the ambient it runs |

- **The trace:** an outer layer (in air) or an inner one (inside the board), in 0.5, 1, 2 or 3 oz copper, the width in mm or mil.
- **Along its length:** the resistance at the temperature it runs at, the voltage drop and the power lost, for a length in mm, cm or in and an ambient temperature.
- **Also shown:** the width in the other unit and the current density, in A/mm².
- **Switching pills** keeps the answer: a width just solved is the width the Rise pill then heats.
- **Remembered:** the pill, the layer, the copper and the units (the current in mA or A).

Examples: 2 A at a 10 °C rise, Brooks and Adam's 10 A on 100 mil of 2 oz, what a 10 mil trace carries, 3 A on an inner layer, and the drop along a 5 cm power trace.

## Source

**IPC-2152 (2009), Standard for Determining Current Carrying Capacity in Printed Board Design**, through **Brooks and Adam, "Trace Currents and Temperatures Revisited", UltraCAD Design, 2015**, read in review. The standard itself is sold by IPC and was not read; the paper digitised its curves and fitted equations to them.

- **Table 3-1 and appendix 6:** $\Delta T = K \cdot I^{c} \cdot W^{w} \cdot Th^{t}$, with W the width and Th the thickness in mils and I in amperes:

| Trace | K | c | w | t |
|---|---|---|---|---|
| outer, any copper | 215.3 | 2 | −1.15 | −1.0 |
| inner, 0.5 oz | 130 to 110 (by width) | 2 | −1.10 | −1.52 |
| inner, 1 oz | 200 | 1.9 | −1.10 | −1.52 |
| inner, 2 oz | 300.3 | 2 | −1.15 | −1.52 |
| inner, 3 oz | 200 to 300 (by width) | 1.9 | −1.15 | −1.52 |

For 0.5 and 3 oz the appendix lists the constant at a few widths; the tool interpolates between them. One appendix row prints the 0.5 oz thickness exponent as +1.52; taken as −1.52 like its neighbours. - **The paper's example** checks the units: a 100 mil, 2 oz trace at 10 A rises "just under 40 °C" outside and "only 33 °C" inside. The tool gives 39.38 °C and 32.52 °C. - **Its findings:** IPC-2152 allows "approximately 25 % lower" current than the older IPC-2221 curves on outer layers, and inner traces "cool almost as well or better" than outer ones, through the board.

IPC-2221 is not used, as Pierre chose: IPC-2152 is the newer standard.

## Why the formulas are these

```
ΔT = K · I^c · W^w · Th^t                    IPC-2152, as fitted
R = ρ · L / (W · Th) · (1 + 0.00393 (T − 20 °C)),  ρ = 1/58 Ω·mm²/m
T = Tambient + ΔT,  drop = I · R,  loss = I² · R
```

A trace heats as $I^2R$ and sheds the heat into the board and the air; it settles where the two balance. The width and the thickness do not count alike (−1.15 against −1.0), so the shape of the trace matters, not only its cross-section. Copper weight is the mass per area: 1 oz/ft² is taken as 1.37 mil, 34.8 µm, the figure with which the paper's example comes out. The resistance uses the same copper as Wire sizing: 1/58 Ω·mm²/m at 20 °C, rising 0.393 % per °C, at the temperature the trace runs at.

## Assumptions and limits

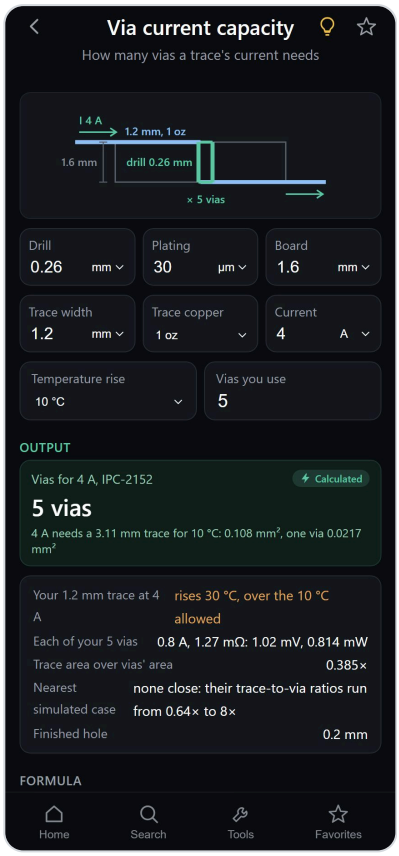
- **The fits follow IPC-2152's test conditions:** a trace alone on a board, without the planes and pours that cool real ones further. A real board usually runs cooler.
- **Copper weights:** 0.5, 1, 2 and 3 oz, the ones the paper fits.
- **Not here:** vias, which carry their own current limit, and traces in a vacuum, which the paper fits separately.

## How it was checked

11 cases: - 2 A at a 10 °C rise needs 0.93 mm, 36.6 mil; - the paper's 10 A on 100 mil of 2 oz: 39.38 °C outside, 32.52 °C inside; - 10 mil of 1 oz carries 948 mA at a 10 °C rise; - 3 A on an inner 1 oz layer at 20 °C needs 0.889 mm; - 5 cm of 0.5 mm at 2 A: 54.5 mΩ, 109 mV, 218 mW; - a width solved and heated again

gives back its 10 °C; - a unit change keeps the width; - a zero current is refused; - the remembered choices; - the current in mA: 2 A reads 2000, the width unchanged.

# Via current capacity




## What it computes

How many vias a current needs, on one screen:

- **Inputs:** the drill, the plating in the hole, the board thickness, the trace feeding the via (width and copper weight), the current, the temperature rise you allow (a picker, 5 to 60 °C), and the number of vias you use.
- **The current sets the copper:** IPC-2152, through the same Brooks & Adam fit as Trace width (outer layer), gives the trace width the current needs for that rise. The vias match the larger of that and your trace. The first version used your trace alone, so the current changed nothing: Pierre put in 100 A and still got 2 vias.
- **The big result:** that many vias, IPC-2152's area rule: together at least the copper cross-section of the trace.
- **The vias you use** follow that count until you type your own.
- **Below it:**
- how hot your trace runs at that current, flagged when over the rise allowed ("it would burn" past 1000 °C);
- each via's current, resistance, voltage drop and heat;
- the trace's area over your vias' area;
- Brooks & Adam's nearest simulated case to that ratio, quoted as they give it, or "none close" outside their 0.64× to 8×;
- the finished hole, the drill less twice the plating.
- **Units:** metric (mm, plating in µm, areas in $mm^2$) or mil for everything (areas in $mil^2$, the simulated cases' widths too); every length has a picker, and changing any one changes them all. Copper in oz, 1 oz = 34.8 µm as in Trace width.
- **Drawing:** a trace on each side of the board joined by the via, the current going through it.
- **Remembered:** the units, the copper weight and the rise. The current has its own mA/A picker.

Examples: Brooks & Adam's via beside a 1.2 mm trace at 4 A (30 °C rise), the same via with a 0.66 mm trace, 10 A on a 2.5 mm 2 oz trace with two vias, 20 A through a 30 mm 2 oz pour, and a 0.3 mm drill with 25 µm plating.

## Source

**Douglas G. Brooks and Johannes Adam, "Via Currents and Temperatures", July 2015** (read; UltraCAD paper, from a copy shared on the KiCad forum): - **IPC-2152's rule,** as they quote it from page 26 of the standard: "The cross-sectional area of a via should have at least the same cross-sectional area as the conductor or be larger than the conductor coming into it. If the via has less cross-sectional area than the conductor, then multiple vias can be used". IPC-2152 itself was not read. - **The via's area,** $\pi(r^2 - (r - th)^2)$, r the via's radius, th the plating; their via, 0.26 mm drill and 0.030 mm plating, is 0.0217 $mm^2$, "approximately equivalent to a 1 Oz. 26 mil (.66 mm) wide trace". - **Their model:** two 60 mm traces on either side of 1.6 mm FR-4, joined by that via, simulated with TRM (Thermal Risk Management). - **Their tables** (trace at the

thermocouple, via at its midpoint): - 0.4 mm, 1 oz, 2 A: trace 91.4, via 82.4 °C; - 0.66 mm, 3 A: 86.9, 80.9 °C; - 1.2 mm, 4 A: 74.1, 75.8 °C; - 1.8 mm, 5 A: 70.8, 76.8 °C; - 2.5 mm, 6 A: 68.1, 77.6 °C; - 2.5 mm, 2 oz, 10 A: 93.5, 117.9 °C; - two vias, 2.5 mm 1 oz, 7 A: 88.6, 92.4 °C; 2 oz, 10 A: 89.6, 97.3 °C. - **Their conclusion:** the rule is "extremely conservative"; "It is not the current that determines the via temperature. The trace itself has much more influence over the via temperature." A filled hole changes little with solder, more with copper. - **Resistance:** IACS copper, 1/58 Ω·mm²/m at 20 °C (NBS Handbook 100), as in Wire sizing.

## Why the formulas are these

```
A via = π (r² − (r − b)²)
A trace = w × t,   w needed: IPC-2152 for I and the rise
vias = ⌈max(A trace, A needed) / A via⌉
R via = ρ × l / A via,   per via: I / N, drop I R / N, heat (I/N)² R
```

Current flows in the via's plated wall, a copper ring; the rule asks for as much copper in the vias as in the trace, so they heat no more than it. The current decides how much copper the trace needs to stay within the rise, so it decides the vias too. Brooks and Adam show the trace and the board cool the via, so it runs near the trace's temperature well past that point; the nearest case is there to show how far.

## Assumptions and limits

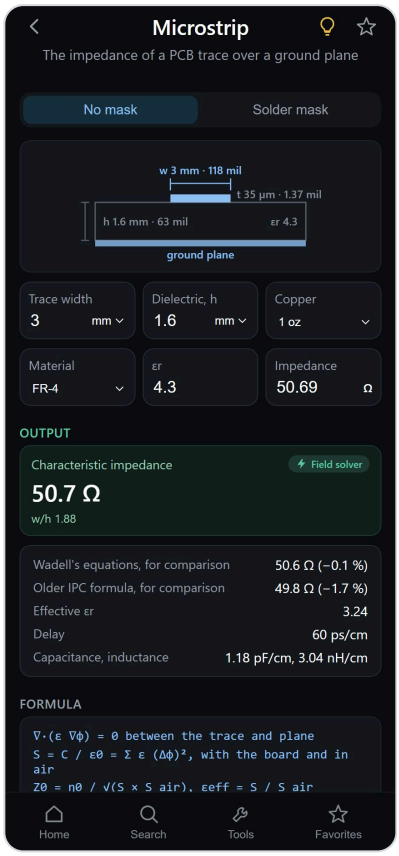
- **The width the current needs** is for an outer-layer trace, as in their via model (traces on both faces); an inner trace runs a little cooler (see Trace width).
- **The nearest case is a pointer, not a prediction.** Their cases are for a 1.6 mm board and 60 mm traces at one current each; nothing is interpolated or scaled from them.
- **The ratio** uses 34.8 µm per oz for the trace and their via's own area for their cases, so their 0.66 mm trace comes to 1.06×, not exactly 1.
- **Plating is assumed even;** they note it often is not.
- **The resistance is at 20 °C;** a hot via is a little higher (0.393 % per °C).
- **Heat from the part itself** is not included: for vias carrying heat rather than current, see Thermal vias.

## How it was checked

15 cases: - 4 A at a 10 °C rise needs a 3.11 mm, 1 oz trace: 5 of their 0.0217 mm² vias, 0.8 A each; the 1.2 mm trace rises 30 °C, over the 10 allowed; - five vias against the 1.2 mm trace (0.385×) are outside their cases; - 100 A needs an 838 mm trace for 10 °C, 1346 vias, and the 1.2 mm trace would burn; - a 30 °C rise lets 4 A use the 1.2 mm trace, so its own area sets 2; - one via at 4 A is their 1.2 mm case, 1.7 °C hotter than the trace; - the 0.66 mm trace at 3 A rises 33 °C, one via is their case, 6 °C cooler; - 10 A on a 2.5 mm 2 oz trace rises 40.1 °C (not flagged over 40); 9 vias; with two, their case is 7.7 °C over the trace; - 20 A through a 30 mm 2 oz pour rises 9 °C; 97 vias match its copper; - a 0.3 mm drill with 25 µm plating: 0.0216 mm², 1.28 mΩ; - a typed via count stays when the trace changes; - plating as thick as the hole's radius is refused; - mil turns every length and area to mil (122 mil, 168 mil²); - every length's picker switches them all; - the remembered copper weight and rise; - the current in mA: 4 A reads 4000.

# Microstrip

(formerly Microstrip impedance)




## What it computes

The characteristic impedance of a trace on an outer layer over a ground plane, or the width that gives an impedance, on one screen:

- **Inputs:** the trace width, the dielectric's thickness h (to the ground plane), the copper weight (its thickness t, ½ to 3 oz), the material (FR-4, polyimide or PTFE at AN-905's round figures, a maker's laminate from Laminate materials, or custom) and its dielectric constant εr.
- **Width and impedance are two live boxes:** type a width and the impedance box shows Z0; type the impedance you need and the width box shows the width that gives it (shown again when you leave the box).
- **The big result:** Z0 from the field solver (as in Edge-coupled embedded), or the width for the impedance typed, with the width in the other unit and w/h. For the moment the solver takes, the badge reads "Solving…" and Wadell's value stands in.
- **Below it:**
- Wadell's equations' Z0, the big result until 2026-11-07, and how far it is from the solver's;
- the older, simpler IPC-2141 formula's Z0, and how far it is; flagged when w > 2h, outside its range;
- the effective dielectric constant, from the solver;
- the delay per length, per cm in mm and per inch in mil;
- the capacitance and inductance per length.
- **Units:** mm or mil for the width, the thickness and the solder mask; any of their pickers switches them all.
- **Drawing:** the board in cross-section, the trace with its w and t, the dielectric with its h and εr, the ground plane; w, h and t in both mm and mil (Pierre), the unit in use first. The trace is green when its width is the answer.
- **Solder mask (pill):** "No mask" (the default) or "Solder mask", with the mask on the board, on the copper and its εr; see Solder mask below.
- **Remembered:** the unit, the copper weight, the mask pill and its unit.

Examples: AN-905's three slide-rule examples (25 mil on 20 mil, the same in metric, the width for 93 Ω on 30 mil), 50 Ω on a 1.6 mm two-layer board and on the 0.2 mm prepreg of a four-layer board, and the same under solder mask (the examples without one set "No mask").

## Terms

- **Microstrip:** a trace on the outer layer of a board with a ground plane under it, separated by the dielectric (the board's insulating material).

- **Characteristic impedance (Z0):** the ratio of voltage to current a fast signal sees as it travels along the trace, set by its geometry and the dielectric, not by its length. A line driven and ended in its Z0 carries a signal without reflections.
- **Dielectric constant (εr):** how much the material slows the field compared with vacuum; FR-4 about 4.3.
- **Effective dielectric constant (εeff):** part of the field runs in the air above the trace, so the signal sees less than εr.
- **Effective width (w'):** the trace's thickness widens it electrically; Wadell's equations replace w by w'.
- **Delay (Td):** the time a signal takes per length of trace.
- **C0, L0:** the trace's capacitance and inductance per length.
- **oz:** copper weight in ounces per square foot; 1 oz is 34.8 µm thick (as in Trace width).

## Source

- **Zachariah Peterson, "High Speed Design Basics", Altium, 3 May 2022** (read; Pierre's copy, slide 23 rendered, its equations being images):
- IPC-2141: Z0 = 87 / (εr + 1.41)^0.5 × ln(5.98h / (0.8w + t)), marked "Only accurate for 50 Ohm traces";
- Wadell's equations, marked "Highly accurate": Z0 = 60 / (2εr + 2)^0.5 × ln(1 + 4h/w' × [(14 + 8/εr)/11 × 4h/w' + √(((14 + 8/εr)/11)$^2$ (4h/w')$^2$
    - $\pi^2$ (1 + 1/εr)/2)]), where w' = w + ((1 + 1/εr)/2)(t/π) ln(4e / √((t/h)$^2$ + ((1/π)/(w/t + 1.1))$^2$)); and εeff = (εr + 1)/2 + (εr − 1)/2 × ((1 + 12h/w)^−0.5 + 0.04 (1 − w/h)$^2$) if w < h, (εr + 1)/2 + (εr − 1)/2 × (1 + 12h/w)^−0.5 if w > h;
- slide 25 ranks IPC-2141 lowest in accuracy, Wadell's equations moderate, field solvers high.
- **Brian C. Wadell, "Transmission Line Design Handbook", Artech House, 1991, §3.5.1, pages 93–95** (read, Pierre's scanned copy): the slide's equations, Wheeler's Z0 with Schneider's εeff as improved by Hammerstad and Bekkadal, as Bogatin recommends after comparing them with measurements. Wadell writes the factor η0 / (2√2 π √(εr + 1)), where the slide has 60 / √(2εr + 2): the same with η0 taken as 120π, 0.07 % above η0 = 376.73 Ω; the tool keeps the slide's 60. The thickness correction is Wheeler's, "the most accurate" by Owens and Potok's comparison; "Error in Z0 is less than 2% for any εr, w"; the εeff equations are "accurate to within 1%" for εr ≤ 16 and 0.05 ≤ w/h ≤ 20, for a paper-thin trace (the copper's thickness is not in them).
- **National Semiconductor AN-905, "Transmission Line RAPIDESIGNER Operation and Applications Guide", James A. Mears, May 1996, TI SNLA035** (read, appendix A rendered): the same formula as IPC-2141, written Z0 = 60 / √(0.475 εr + 0.67) × ln[4h / (0.67 (0.8w + t))] "if w ≤ 2h (maximum error ≅ 3%)"; its slide-rule examples: a 25 mil trace on 20 mil of εr 4.3 with 1 oz copper, 63 Ω; 0.127 cm on 0.102 cm, 64 Ω; 93 Ω on 30 mil of εr 3.8 with ½ oz, 18 mil (0.051 cm). Appendix D: FR-4 and G-10 4.3, polyimide 4.2, PTFE 2.4.
- **Walt Kester, ADI tutorial MT-094, "Microstrip and Stripline Design", Rev. 0, January 2009** (read, Pierre's copy): its eq. 3, quoted from IPC-2141A, is the same IPC formula, in mils; its example: a 20 mil, 1 oz (T 1.4 mil) trace on 10 mil of FR-4 at εr 4.0 is "about 50 Ω" by the W/H = 2 rule of thumb, while eq. 3 "predicts Zo to be about 46 Ω"; "The IPC microstrip equation is most accurate between 50 and 100 Ω, but is substantially less so for lower (or higher) impedances"; 75 Ω takes "about 8.3 mils". The tool gives 46.2 Ω by IPC-2141A and 48.9 Ω by Wadell for that trace, and 8.35 mil for 75 Ω by IPC-2141A (8.56 by Wadell).
- **IPC-2141A, "Design Guide for High-Speed Controlled Impedance Circuit Boards", March 2004, §4.2.2** (read, Pierre's scanned copy): its surface microstrip formulas [4-4]–[4-6] are Wadell's above, the same εeff, but with εr,eff written where Wadell has εr in the Z0 formula; read that way they come out

12 to 14 % high, so the tool keeps Wadell's. The 87/√(εr + 1.41) formula, which Peterson's slide labels IPC-2141 and MT-094 quotes from IPC-2141A, is in neither the 2004 text nor its errata; AN-905 (1996) already uses it, so it is presumably the first IPC-2141's (not read). The tool shows it as the "older IPC formula".

- **The field solver:** js/fieldsolver.js, checked against Polar's boundary-element microstrip table to 0.02 % (Edge-coupled embedded's section); tools/fd_lines.py, a separate implementation, gives the values in every case below.
- **Not read:** IPC-2141 (1996) itself, and the papers Wadell draws on (Wheeler's, Schneider's, Hammerstad and Bekkadal's).

## Why the formulas are these

The solver, the big result:

```
∇·(ε ∇ϕ) = 0 between the trace (1 V) and the plane (0 V)
S = C / ε0 = Σ ε (Δϕ)², once with the board, once in air (S_air)
Z0 = η0 / √(S × S_air),   εeff = S / S_air
Td = 33.36 √εeff ps/cm,   C0 = Td / Z0,   L0 = Z0 × Td
```

Wadell's equations, the first comparison:

```
d = √((t/h)² + (1/π / (w/t + 1.1))²)
w' = w + (1 + 1/εr)/2 × t/π × ln(4e / d)
x = 4h / w',   a = (14 + 8/εr) / 11
b = √(a² x² + π² (1 + 1/εr) / 2)
Z0 = 60 / √(2εr + 2) × ln(1 + x (a x + b))
εeff = (εr+1)/2 + (εr−1)/2 / √(1 + 12h/w)   (+ (εr−1)/2 × 0.04 (1 − w/h)² for w < h
Td = 33.36 √εeff ps/cm,   C0 = Td / Z0,   L0 = Z0 × Td
```

A wider trace, or a ground plane closer under it, adds capacitance per length and so lowers the impedance. Wadell's formula follows the field over the whole range of w/h, where IPC-2141's logarithm is a fit good near 50 Ω and for w ≤ 2h. The signal travels at the speed of light divided by √εeff (33.36 ps/cm is 1 cm at the speed of light), and Z0 = √(L0 / C0) with Td = √(L0 C0) gives C0 and L0. The width for an impedance has no closed form here: Z0 falls steadily as the width grows, so the tool halves the range of widths until it lands on it.

**How the formulas compare with the solver** (1 oz copper):

| Case | Solver | Wadell | IPC-2141 | εeff, solver (Wadell) |
|---|---|---|---|---|
| 25 mil on 20 mil, εr 4.3 | 62.27 Ω | +0.20 % | +0.70 % | 3.08 (3.16) |
| 1.27 mm on 1.02 mm, εr 4.3 | 63.20 Ω | +0.11 % | +1.31 % | 3.11 (3.16) |
| 3.07 mm on 1.6 mm, εr 4.3 | 50.00 Ω | −0.12 % | −2.00 % | 3.24 (3.26) |
| 0.36 mm on 0.2 mm, εr 4.3 | 50.00 Ω | +0.10 % | −4.54 % | 3.13 (3.25) |
| 4 mm on 1.6 mm, εr 4.3 | 42.43 Ω | −0.28 % | −6.95 % | 3.32 (3.34) |
| 20 mil on 10 mil, εr 4.0 | 48.93 Ω | −0.02 % | −5.49 % | 2.99 (3.07) |

Wadell's equations stay within 0.3 % of the solver for the impedance. The older IPC formula is within 2 % on the thick boards, but 4.5 to 5.5 % low on thin dielectric (0.2 mm, 10 mil) and 7 % low beyond w = 2h. Wadell's effective εr leaves out the copper's thickness, whose sides put more of the field in the air: it reads 0.35 to 3.7 % high with 1 oz copper, while for a paper-thin trace it is within 0.2 % of the solver's (four of the cases computed again at t = 0). The delay the tool gives therefore comes from the solver's εeff.

## Solder mask

The "Solder mask" pill coats the board and the trace with a solder mask of its own εr, as on almost every outer layer: c1 thick on the bare board, c2 over the copper's top and sides (a conformal coat; mask thins over copper, so c2 is usually less than c1), in the lengths' unit, mm or mil, their pickers moving with the others'. The solver puts the mask's εr in those cells and solves as before; a second solve without the mask gives the row "Without the mask"; the formula rows, which know no mask, say "no mask".

- **Defaults:** Sanwa Chemical's SPSR-950 C-20 LPI mask (Florida CirTech technical bulletin, revision 072511, read): "a dried thickness of 15-20 microns"; dielectric constant at 1 MHz 2.8 as received and 3.5 after 96 hours at 55 °C and 95 % RH. The tool takes 20 µm on the board, 15 µm over the copper and εr 3.5, the sheet's damp value; the sheet gives one thickness only, so enter the board maker's c1 and c2 for a real board. (Taiyo's PSR-4000 sheet, the most common mask, could not be opened.)
- **Checked:** a mask of εr 1 gives the bare trace's impedance (49.955 against 49.963 Ω, 0.36 mm on 0.2 mm, 1 oz); a mask of the board's εr, c1 = c2 + t, makes a uniform cover and gives Embedded microstrip's solver value to 0.001 % (46.865 Ω, cover to h + 64.8 µm).
- **What it does:** on 0.36 mm over 0.2 mm FR-4 the default mask lowers Z0 2.7 %, so 50 Ω needs 0.342 mm instead of 0.36; on 3 mm over 1.6 mm, 0.5 %. A pair is hit harder (Edge-coupled microstrip).

## Assumptions and limits

- **Quasi-static field solution:** dispersion, losses and copper roughness are left out; the trace is a rectangle. The board maker's stack-up calculation decides.
- **The mask's shape:** a uniform coat c1 on the board and c2 around the trace; real mask is thicker in the corners at the trace's foot.
- **εr changes with frequency and between laminates:** FR-4's 4.3 is a nominal figure; use the maker's value at your frequency.
- **No losses:** the copper's resistance and the dielectric's loss are left out.

## How it was checked

13 cases, the expected values from tools/fd_lines.py's app_microstrip: - AN-905's 25 mil on 20 mil, εr 4.3, 1 oz: 62.3 Ω (62.27); Wadell 62.4 Ω (+0.2 %), IPC-2141 62.7 Ω (+0.7 %), its slide rule 63 Ω; 149 ps/in; - the same in metric, 1.27 mm on 1.02 mm: 63.2 Ω (IPC-2141 64 Ω, its slide rule 64 Ω), εeff 3.11; - 93 Ω on 30 mil, εr 3.8, ½ oz: 17.9 mil (17.906; its slide rule's 18 mil); - 50 Ω on 1.6 mm FR-4: 3.07 mm (3.070), where IPC-2141 gives 49.0 Ω; - 50 Ω on 0.2 mm: 0.36 mm (0.3595), IPC-2141 4.5 % low; - 75 Ω at εr 3.8: 0.739 pF/cm and 4.16 nH/cm, the material becoming custom; - beyond w = 2h the IPC-2141 comparison is flagged (4 mm on 1.6 mm, 42.4 Ω); - the length pickers move together; - PTFE sets εr to 2.4; - MT-094's 20 mil on 10 mil at εr 4.0: 48.9 Ω (48.93), its IPC-2141A eq. 3 46.2 Ω (its "about 46 Ω"); - 50 Ω under the default mask on 0.2 mm, 1 oz: 0.342 mm (0.34169), 51.4 Ω without the mask (+2.8 %); - the default trace under the mask: 50.4 Ω (50.421), 50.7 Ω without it; - a mask εr under 1 is refused.

The first version used AN-905's (IPC-2141's) formula; Pierre chose Wadell's equations once his copy of Peterson's slides showed both; since 2026-11-07 the big result is the field solver's (Pierre: the most accurate value first).

# Embedded microstrip

(formerly Embedded microstrip impedance)




## What it computes

The impedance of a microstrip covered by more dielectric (solder mask over an outer-layer trace, or prepreg over an inner trace with no plane above), or the width that gives an impedance under that cover:

- **Inputs:** the trace width w, the dielectric under it h1, H2 from the ground plane to the top of the cover, the copper weight, the material and εr (the cover taken as the same εr).
- **Width and impedance are two live boxes:** type a width and the impedance box shows the covered Z0; type the impedance you need and the width box shows the width that gives it covered.
- **The big result:** the covered Z0 from the field solver (as in Edge-coupled embedded), with the uncovered one, solved too, and how much the cover takes off; or the width, with the Z0 it would have uncovered. For the moment the solver takes, the badge reads "Solving…" and IPC-2141A's value stands in.
- **Below it:**
- IPC-2141A's formula, the big result until 2026-11-07, and how far it is from the solver's;
- the cover b = H2 − h1 and how much of it is over the copper's top, flagged when it is thinner than the copper;
- the deep limit, the impedance once the cover is thick enough that all the field is in the dielectric;
- the effective εr uncovered and covered, from the solver;
- the delay per length, covered.
- **Units:** mm or mil for the lengths, moving together; oz for the copper.
- **Drawing:** the ground plane, the dielectric with the trace on h1, the cover up to H2, in mm and mil. The trace is green when its width is the answer.
- **Remembered:** the length unit and the copper weight.

Examples: solder mask over a 50 Ω trace on 0.2 mm, the width for 50 Ω under that mask, no cover (the surface microstrip), and 0.2 mm of prepreg over the trace.

## Terms

- **Embedded microstrip:** a microstrip with dielectric over the trace as well as under it, but no plane above (a plane above makes it a stripline).
- **h1, H2, b:** h1 is the dielectric between the plane and the bottom of the trace; H2 runs from the plane to the top of the cover; b = H2 − h1 is the cover measured from the bottom of the trace (IPC-2141A figure 4-3(b)).
- **Solder mask:** the thin coating over the outer layers; its thickness over the copper comes from the board maker.

- **Deep limit:** under a thick enough cover, the trace sees εr all round; Z0 falls to the uncovered value × √(εeff / εr) and goes no lower.
- **Z0, εr, εeff, oz:** as in Microstrip.

## Source

- **IPC-2141A, "Design Guide for High-Speed Controlled Impedance Circuit Boards", March 2004, §4.2.3** (read, Pierre's scanned copy): embedded microstrip Z0,embed = Z0,surf / √(e^(−2b/h) + (εr/εr,eff)(1 − e^(−2b/h))) [4-7], with b = h2 − h1 [4-8]; its figure 4-3(b) draws h1 from the plane to the trace and h2 from the plane to the top of the cover; Z0,surf and εr,eff are its surface microstrip's [4-4]–[4-6].
- **IPC-2141A errata** (read, a Wayback Machine copy): writes [4-7]'s exponent with h1, the dielectric under the trace, as the tool does.
- **Uncovered Z0 and εeff in the formula:** Wadell's equations, as in Microstrip (IPC-2141A's [4-6] εeff is the same; for its [4-4] see there).
- **The field solver:** js/fieldsolver.js, as in Microstrip, with the cover from the board up to H2 and air above it; tools/fd_lines.py's app_microstrip, a separate implementation, gives the values below.

## Why the formulas are these

The solver, the big result, solved twice, with the cover and without:

```
∇·(ε ∇ϕ) = 0, the board and the cover up to H2, air above
S = C / ε0 = Σ ε (Δϕ)², once with the dielectric, once in air (S_air)
Z0 = η0 / √(S × S_air),   εeff = S / S_air,   Td = 33.36 √εeff ps/cm
deep limit = η0 / (√εr × S_air) = Z0 √(εeff / εr)
```

The deep limit needs no third solve: with dielectric all round, the capacitance is εr times the one in air, which the covered solve already has.

IPC-2141A's formula, the comparison:

```
b = H2 − h1,   k = e^(−2b/h1)
Z0,emb = Z0 / √(k + (εr / εeff)(1 − k))
εeff,emb = εeff k + εr (1 − k),   Td = 33.36 √εeff,emb ps/cm
deep limit = Z0 √(εeff / εr)
```

Uncovered, part of the field runs in the air over the trace and the line sees εeff. A cover puts dielectric where that air was: the line sees more of εr, its capacitance rises and Z0 falls. The exponential blends the two cases, k = 1 with no cover, k → 0 under a cover a few h1 thick. With b = 0 the formula gives the surface microstrip back. The width for an impedance starts from the formula's, then the solver's answer corrects it.

## Assumptions and limits

- **One εr for the board and the cover,** as IPC-2141A's formula has; a solder mask's own εr, from its maker, may differ.
- **No plane above the cover.**
- **Uniform cover:** real solder mask is thinner over the copper's edges and thicker between traces.
- **Quasi-static field solution:** the trace a rectangle, no losses; the board maker's stack-up calculation decides.

**How IPC-2141A's formula compares with the solver** (1 oz copper, FR-4):

| Case | Solver | IPC-2141A [4-7] |
|---|---|---|
| 0.36 mm on 0.2 mm, 10 µm of mask over the copper | 47.52 Ω | −0.44 % |
| the same, 20 µm of mask | 47.17 Ω | −0.57 % |
| the same, 30 µm of mask | 46.87 Ω | −0.70 % |
| dielectric only beside the trace, up to its top | 47.96 Ω | −0.37 % |
| cover b 0.1 mm (H2 0.3 mm) | 46.05 Ω | −1.07 % |
| cover b 0.2 mm (H2 0.4 mm) | 44.70 Ω | −1.13 % |
| cover b 1 mm (H2 1.2 mm) | 42.88 Ω | +1.33 % |
| 0.15 mm on 0.1 mm, 20 µm of mask | 48.45 Ω | −0.39 % |
| 3 mm on 1.6 mm, filled to the copper's top | 50.26 Ω | +0.06 % |

The formula stays within −1.1 to +1.3 % for the impedance; its deep limit misses more, 43.45 Ω against the solver's 42.63 Ω for the 0.36 mm trace (+1.9 %) and 46.1 Ω against 44.5 Ω for 0.15 mm on 0.1 mm (+3.7 %), since Wadell's εeff, on which it rests, leaves out the copper's thickness.

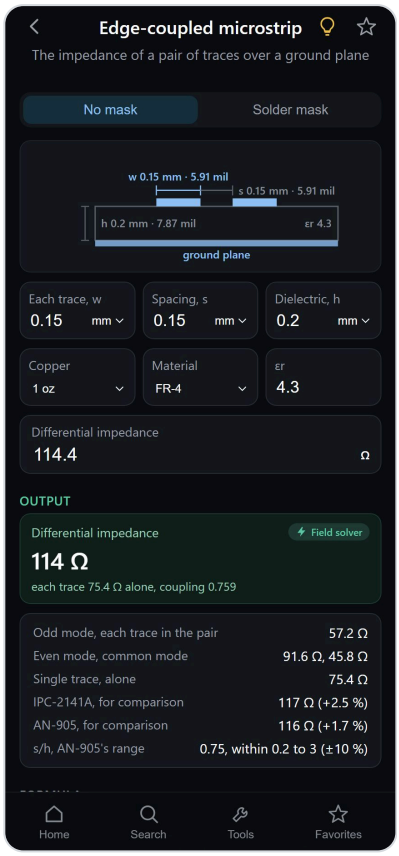
## How it was checked

5 cases, the expected values from tools/fd_lines.py's app_microstrip: - 0.36 mm on 0.2 mm of FR-4 under 20 µm of mask: 47.2 Ω (47.168; 50.0 Ω uncovered, the cover takes 5.6 %), IPC-2141A 46.9 Ω (−0.6 %), εeff 3.13 → 3.51, 62.5 ps/cm; - 50 Ω under that mask: 0.321 mm (0.32116); - no cover, 3 mm on 1.6 mm: 50.7 Ω, Microstrip's value; - 0.2 mm of prepreg over the 0.36 mm trace: 44.7 Ω (44.696), deep limit 42.6 Ω, IPC-2141A 44.2 Ω (−1.1 %); - the default: 47.2 Ω (47.162).

Until 2026-11-07 [4-7] was the big result, being within 1.6 % of fd_lines' first embedded() check; Pierre chose the field solver for every impedance tool (the most accurate value first).

# Edge-coupled microstrip

(formerly Differential microstrip impedance, then Differential microstrip)




## What it computes

The differential impedance of two edge-coupled traces on an outer layer over a ground plane, or the trace width that gives it at a spacing:

- **Inputs:** each trace's width w, the gap between them s, the dielectric h, the copper weight, the material and εr.
- **Width and differential impedance are two live boxes:** type a width and the impedance box shows Zdiff; type the Zdiff you need (90 Ω for USB, 100 Ω for Ethernet or LVDS) and the width box shows the width at your spacing.
- **The big result:** Zdiff from the field solver (as in Edge-coupled embedded), with each trace's impedance alone and the coupling (Zdiff over twice that); or the width for the Zdiff typed. For the moment the solver takes, the badge reads "Solving…" and IPC-2141A's value stands in.
- **Below it:**
- the odd-mode impedance, each trace's share of the pair, Zdiff / 2;
- the even-mode impedance, both traces driven alike, and the common-mode impedance, half of it;
- one trace alone, from the solver as in Microstrip;
- Zdiff by IPC-2141A's pair formula, the big result from 2026-11-05 to 2026-11-07, and by AN-905's simpler factor, each with its difference from the solver's;
- s/h against AN-905's range, 0.2 to 3, flagged outside.
- **Units:** mm or mil for the three lengths and the solder mask; any of their pickers switches them all.
- **Drawing:** the pair on the dielectric over the ground plane, w and s as a chain dimension, h and εr, in mm and mil. The traces are green when their width is the answer.
- **Solder mask (pill):** as in Microstrip; with it, a fourth solve, the pair without the mask, and the formula rows say "no mask".
- **Remembered:** the unit, the copper weight, the mask pill and its unit.

Examples: AN-905's two slide-rule examples, USB's 90 Ω and Ethernet's or LVDS's 100 Ω on a 0.2 mm layer.

## Terms

- **Differential pair:** two traces carrying a signal as equal and opposite voltages; the receiver reads the difference, so noise common to both cancels.
- **Differential impedance (Zdiff):** the impedance the driver sees between the two traces.
- **Odd-mode impedance:** the impedance of one trace while the other carries the opposite signal, half of Zdiff.
- **Even-mode and common-mode impedance:** the impedance of one trace while the other carries the same signal; the two in parallel, half of it, is what a common-mode signal (noise on both) sees.

- **Coupling:** brought close, each trace's field reaches the other; Zdiff falls below twice one trace's impedance.
- **Edge-coupled:** side by side on the same layer (AN-905's only case).
- **Z0, εr, oz:** as in Microstrip.

## Source

- **National Semiconductor AN-905, TI SNLA035** (read, appendix A rendered): for microstrip, Zdiff ≅ 2 Z0 (1 − 0.48 e^(−0.96 s/h)) ohms; "These functions were derived from empirical data … Various sources cite possible errors of up to ±10%. The practical ranges for Z0 and Zdiff are from 20Ω to about 150Ω"; the slide rule's range "0.20 ≤ s/h ≤ 3.0"; "Both conductors of the pair must have the same physical cross sectional dimensions"; its examples: two 75 Ω microstrips 0.10 cm apart on a 0.076 cm substrate, factor 0.86, 128 Ω; 10 mils apart on 20 mils, factor 0.702, 105 Ω.
- **Single-trace Z0 in the formulas:** Wadell's equations, as in Microstrip.
- **The field solver:** js/fieldsolver.js, as in Microstrip, the pair solved with the traces at +1 V and −1 V (odd mode) and both at +1 V (even mode); tools/fd_lines.py's app_microstrip, a separate implementation, gives the values below.
- **IPC-2141A, March 2004, §4.4.1** (read, Pierre's scanned copy): its edge-coupled surface microstrip is not AN-905's exponential: [4-24]–[4-41] give the odd-mode impedance Z0,o = Z0 √(εeff/εeff,o) / (1 − Z0/η0 √εeff Q10), with εeff,o and ten fitted terms Q1 to Q10 in u = w/h and g = s/h, citing its references 2 and 6. §4.4 warns that the balanced-line formulas "exist in various forms, some of which are not in agreement". It was the big result from 2026-11-05 (Pierre: the most accurate value first) until the field solver took over on 2026-11-07; AN-905's factor, the first version's, is shown for comparison too.

**IPC-2141A's pair formula, and how it was checked.** Its terms [4-25]–[4-41] are, in full:

```
u = w/h,  g = s/h,  Z0 and εeff one paper-thin trace alone
a0 = 0.7287 (εeff − (εr+1)/2)(1 − e^(−0.179u))
b0 = 0.747 εr / (0.15 + εr),  c0 = b0 − (b0 − 0.207) e^(−0.414u)
d0 = 0.593 + 0.694 e^(−0.562u)
εeff,o = ((εr+1)/2 + a0 − εeff) e^(−c0 g^d0) + εeff
Q1 = 0.8695 u^0.194,  Q2 = 1 + 0.7519g + 0.189g^2.31
Q3 = 0.1975 + (16.6 + (8.4/g)^6)^−0.387 + ln(g^10 / (1 + (g/3.4)^10)) / 241
Q4 = 2Q1 / (Q2 (e^−g u^Q3 + (2 − e^−g) u^−Q3))
Q5 = 1.794 + 1.14 ln(1 + 0.638 / (g + 0.517g^2.43))
Q6 = 0.2305 + ln(g^10 / (1 + (g/5.8)^10)) / 281.3 + ln(1 + 0.598g^1.154) / 5.1
Q7 = (10 + 190g²) / (1 + 82.3g³),  Q8 = e^(−6.5 − 0.95 ln g − (g/0.15)^5)
Q9 = ln(Q7) (Q8 + 1/16.5),  Q10 = (Q2Q4 − Q5 e^(ln(u) Q6 u^−Q9)) / Q2
Z0,o = Z0 √(εeff/εeff,o) / (1 − Z0/η0 √εeff Q10),  Zdiff = 2 Z0,o
```

It is printed correctly: against tools/fd_lines.py (the field solver of Edge-coupled stripline, with the board and the air above it, checked against Polar's microstrip table to 0.3 %) it is within 0.3 % for paper-thin traces (0.1 to 0.5 mm wide, 0.05 to 0.5 mm apart, on 0.2 mm of εr 4.3). Its terms carry no copper thickness, and with 1 oz copper it reads 2 to 18 % high. The tool widens each trace to Wadell's w' and narrows the gap by as much:

then it is within −0.4 to +2.8 % of the solver, where AN-905's factor is within −2.1 to +11.1 % (worst for a 0.05 mm gap).

**How the formulas compare with the solver** (1 oz copper, εr 4.3):

| Case | Solver Zdiff | IPC-2141A | AN-905 |
|---|---|---|---|
| AN-905's 75 Ω traces 1 mm apart on 0.76 mm | 131.72 Ω | +0.51 % | −1.58 % |
| 10 mil apart on 20 mil | 106.88 Ω | +1.01 % | −1.34 % |
| USB: 0.315 mm, 0.2 mm apart on 0.2 mm | 90.00 Ω | +1.69 % | −2.38 % |
| 100 Ω: 0.251 mm, 0.2 mm apart on 0.2 mm | 100.00 Ω | +1.94 % | −1.16 % |
| the default, 0.15 mm, 0.15 mm apart on 0.2 mm | 114.40 Ω | +2.54 % | +1.66 % |

For USB, IPC-2141A's 0.326 mm would give 88.5 Ω, 1.6 % under the 90 Ω wanted; the solver's width is 0.315 mm.

## Why the formulas are these

The solver, the big result, solved three times:

```
odd mode: the traces at +1 V and −1 V,  Zodd = η0 / √(S × S_air)
Zdiff = 2 Zodd
even mode: both at +1 V,  common mode = Zeven / 2
one trace alone, as in Microstrip
```

IPC-2141A's pair formula, the first comparison:

```
w' = w + (1 + 1/εr)/2 × t/π × ln(4e / √((t/h)² + (1/π / (w/t + 1.1))²))
u = w'/h,   g = (s − (w' − w))/h
the terms above, with Z0 and εeff of a paper-thin trace w' wide
Zdiff = 2 Z0,o
AN-905, for comparison: Zdiff = 2 Z0 × (1 − 0.48 e^(−0.96 s/h))
```

Far apart, the two traces are independent and the pair is two lines in series: Zdiff = 2 Z0. Brought closer, their mutual capacitance and inductance lower it. IPC's terms fit that fall for paper-thin traces, through the odd mode's own effective εr and the coupling term Q10; the copper's thickness makes each trace act wider (w', as in Microstrip) and so the gap narrower. AN-905 fits the fall with a single exponential in s/h. The width for a Zdiff starts from IPC-2141A's, the spacing kept, then the solver's answer corrects it.

## Solder mask

As in Microstrip (its section has the mask's model, defaults and source). On a pair the mask fills part of the gap between the traces, where the odd mode's field is strongest, so it lowers Zdiff more than one trace's Z0:

0.15 mm traces 0.15 mm apart on 0.2 mm FR-4 read 114.4 Ω bare and 107.0 Ω under the default mask (−6.5 %); USB's 90 Ω on 0.2 mm, 0.2 mm apart, needs 0.287 mm under the mask instead of 0.315 mm.

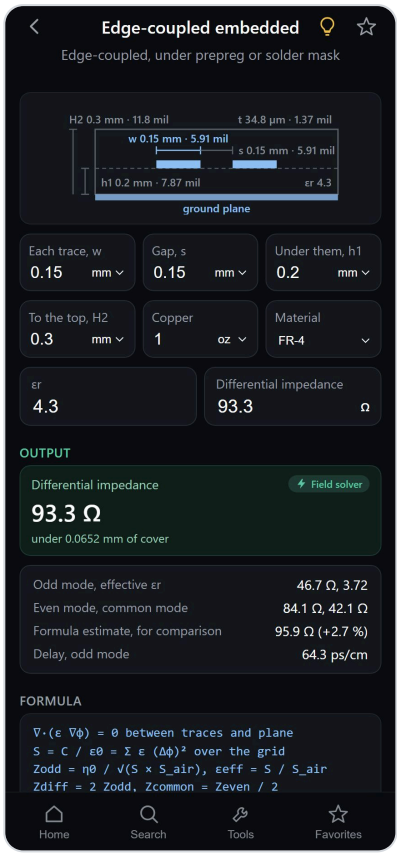
## Assumptions and limits

- **Quasi-static field solution:** the traces rectangles, no losses; the board maker's calculation decides for a controlled-impedance board.
- **Equal traces, edge-coupled;** the mask, when used, a uniform coat.
- **IPC's pair formula with copper:** the w' widening and narrower gap are the tool's, checked against the field solver, not IPC's.
- **AN-905's row:** its factor was fitted with AN-905's own Z0 (the older IPC-2141 formula); here it multiplies Wadell's.
- **Spacing to other traces:** AN-905 asks for more than 2s to the next pair, to keep crosstalk and the impedance in hand.

## How it was checked

10 cases, the expected values from tools/fd_lines.py's app_microstrip: - AN-905's 75 Ω traces (75 Ω by Wadell's equations, 74.8 Ω by the solver) 1 mm apart on 0.76 mm: 132 Ω (131.72), coupling 0.881; IPC-2141A 132 Ω (+0.5 %), AN-905 130 Ω (−1.6 %; its slide rule 128 Ω); even mode 83.3 Ω, common mode 41.6 Ω; - 10 mil apart on 20 mil: 107 Ω (106.88), 53.4 Ω odd mode; IPC-2141A 108 Ω (+1.0 %), AN-905 105 Ω (−1.3 %), s/h 0.5; - USB 90 Ω on 0.2 mm, 0.2 mm apart: 0.315 mm (0.31510), 53.7 Ω alone; - 100 Ω on the same layer: 0.251 mm (0.25055), 50 Ω odd mode; - the default pair: 114 Ω (114.40); IPC-2141A 117 Ω (+2.5 %), AN-905 116 Ω (+1.7 %); common mode 45.8 Ω; - USB 90 Ω under the default mask: 0.287 mm (0.28684), 94.1 Ω without it (+4.5 %); - the default pair under the mask: 107 Ω (106.958), 114 Ω without it (+7.0 %), IPC-2141A (no mask) 117 Ω; - a spacing beyond 3 h is flagged; - a gap the copper closes is refused; - the three length pickers move together.

# Edge-coupled embedded




## What it computes

The differential impedance of two edge-coupled traces buried in the board: on h1 of dielectric over a ground plane, covered by the same dielectric up to H2 above the plane, with no plane on top (Saturn PCB Toolkit's "Edge Cpld Embed"); or the trace width that gives a differential impedance at a gap. It is the first tool computed by the app's field solver (below):

- **Inputs:** each trace's width w, the gap s, the dielectric under the traces h1, the height of the cover's top above the plane H2, the copper's thickness t (its own picker), the material and εr.
- **Width and differential impedance are two live boxes:** type a width and the impedance box shows Zdiff; type the Zdiff you need and the width box shows the width that gives it at your gap.
- **The big result:** Zdiff from the field solver, with the cover over the copper; or the width for the Zdiff typed. For the moment the solver takes, the badge reads "Solving…" and the formula estimate stands in.
- **Below it:**
- the odd-mode impedance and its effective εr;
- the even-mode impedance and the common-mode impedance, half of it;
- the formula estimate (below), and how far it is from the solver;
- the delay per length in the odd mode.
- **Units:** mm or mil for the four lengths; oz, µm or mil for the copper. Any picker switches them all: mil anywhere makes every length mil, mm or µm makes them metric; oz, a copper weight, stays oz.
- **Drawing:** the plane, the dielectric under the traces, the cover over them up to H2, w and s as a chain dimension, h1, H2, t and εr, in mm and mil. The traces are green when their width is the answer.
- **Remembered:** the length unit and the copper unit.

Examples: 100 Ω under 0.1 mm of prepreg, the default pair under solder mask only, the same pair bare (no cover), and buried under 1 mm.

## Terms

- **Embedded (buried) pair:** a differential pair with dielectric over it as well as under it, and no plane above (a plane above makes it a stripline).
- **Odd mode:** the two traces at equal and opposite voltages, as a differential signal drives them. Zdiff = 2 Zodd.
- **Even mode:** both traces at the same voltage, as noise picked up by both drives them. The common-mode impedance, the two traces in parallel against the plane, is Zeven / 2.
- **Effective εr (εeff):** the permittivity the signal sees, between that of the air and that of the board, since part of the field runs in each; under a thick cover it reaches εr.

- **Field solver:** a program that finds the impedance from the cross- section itself, by solving for the electric field, rather than from a formula fitted to some shapes.
- **h1, H2, εr, oz:** as in Embedded microstrip.

## The field solver

The app lays the cross-section on a grid of points, finer near the copper's edges, where the field changes fastest, and coarser far away. Each trace is held at 1 V (−1 V for the second trace in the odd mode, so the plane midway between them is at 0 V), the plane at 0 V, and the potential at every other point is found so that each one balances its neighbours (Laplace's equation). The energy of that field gives the capacitance per length, once with the dielectric and once with air everywhere; the two give the impedance and the effective εr:

```
∇·(ε ∇ϕ) = 0
S = C / ε0 = Σ over the grid's edges of ε (Δϕ)²
Z0 = η0 / √(S × S_air),   εeff = S / S_air,   η0 = 376.73 Ω
```

Each case is solved on three grids, every cell halved from one to the next, and the three results are extrapolated to an infinitely fine grid (Z = (4R2 − R1)/3, R1 = 2Z1 − Z0, R2 = 2Z2 − Z1). The air above the board is closed by a grounded box 80 h1 away; nearer, at 20 h1, it read 0.1 to 0.3 % low. The equations are solved by conjugate gradients with a modified incomplete-Cholesky preconditioner, in a background thread (a Web Worker), so typing never waits: about 0.1 s per solve on an iPhone 15 Pro Max, and a width for a typed Zdiff in two or three solves.

**How the solver was checked:** - **Cohn's exact stripline** (zero thickness, centred between two planes; the single strip and the odd mode of a pair, as Burkhardt et al. quote them): within 0.005 %. Cohn's 30π takes η0 as 120π, which is 0.069 % high; with that taken out the solver agrees to the third decimal. - **Polar Instruments' boundary-element table** (Burkhardt et al., table 1: 1.5, 0.45 and 3.3 mm on 0.794 mm, εr 4.2, 35 µm copper): 50.635, 89.648 and 30.090 Ω against 50.63, 89.63 and 30.09 Ω, within 0.02 %. - **tools/fd_lines.py,** a separate implementation in Python (a direct sparse solver, scipy), with the same grids: it gives the app's values in every case below. - **tools/fs_bench.html** runs nine reference cases in the browser and times them: 1.16 s in all on Pierre's iPhone 15 Pro Max, before the solver was made three times faster.

## Source

- **No published formula** for this structure was found: IPC-2141A (March 2004) has the bare pair (§4.4.1) and one covered trace (§4.2.3); Wadell's "Transmission Line Design Handbook" (1991), §4.5.3, is a pair under a metal lid with air between; a web search found calculators that name the structure without giving their equations. Saturn PCB Toolkit lists it; its online help does not say how it computes it (read).
- **A. J. Burkhardt, C. S. Gregg and J. A. Staniforth, "Calculation of PCB Track Impedance", Polar Instruments, IPC Printed Circuits Expo 1999** (read): Cohn's exact expressions (their equations 4–8) and their table 1, computed by their boundary-element method and checked against 1500 measured boards; the numerical principle, the capacitance found from the potential.
- **The estimate:** IPC-2141A §4.4.1 [4-24]–[4-41] and §4.2.3 [4-7], as in Edge-coupled microstrip and Embedded microstrip.

## Why the formulas are these

The estimate shown at once, and kept for comparison: the bare pair's odd mode by IPC-2141A's pair formula with the copper allowed for, as Edge-coupled microstrip computes it, once in the board (Zbare) and once in air (Zair), then IPC-2141A [4-7]'s cover blend applied to the odd mode's effective εr:

```
k = e^(−2b/h1),   b = H2 − h1
εbare = (Zair / Zbare)²
Zodd ≈ Zair / √(εbare k + εr (1 − k))
```

It gives the bare pair with no cover and the right limit under a deep one; in between it reads 1 to 4 % high in the examples. No formula of it is needed for the answer, which is the solver's.

## Assumptions and limits

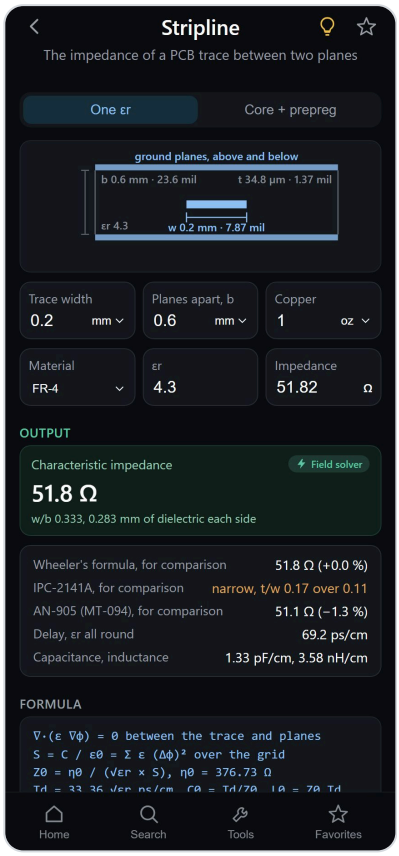
- **Quasi-static:** the field solved is the one at low frequency; the frequency dependence (dispersion), the losses and copper roughness are left out.
- **Rectangular traces of equal width;** an etched trace's trapezoid is not modelled. The cover has the board's εr.
- **One plane, nothing else nearby:** the next pair or a plane above would change the answer; the grounded box is 80 h1 away.
- **The model is exact to about 0.05 %; a real board is not:** εr's tolerance, the etching and the copper's thickness move the impedance more. The board maker's stack-up calculation decides.

## How it was checked

8 cases, the expected values from tools/fd_lines.py's app_microstrip: - the default pair, 0.15 mm traces 0.15 mm apart on 0.2 mm, H2 0.3 mm: 93.3 Ω (93.30), odd mode 46.7 Ω with εeff 3.72, even mode 84.1 Ω, common mode 42.1 Ω, estimate 95.9 Ω (+2.7 %), 64.3 ps/cm; - 100 Ω typed: 0.115 mm traces (0.1154), odd mode 50 Ω, common mode 46.3 Ω; - 100 Ω under 0.1 mm of prepreg, 0.2 mm gap: 0.146 mm (0.14596); - under 20 µm of solder mask only: 99 Ω (98.99), estimate +2.7 %; - no cover: 114 Ω (114.40), where Edge-coupled microstrip's formula reads 117 Ω; - under 1 mm: 86.8 Ω (86.76), εeff 4.3; - H2 below h1 is refused; - the four length pickers move together.

# Stripline

(formerly Stripline impedance)




## What it computes

The characteristic impedance of a trace on an inner layer, centred between two ground planes, or the width that gives an impedance, on one screen:

- **Inputs:** the trace width, the spacing between the planes b, the copper's thickness t (in oz, µm or mil, its own picker), the material (FR-4, polyimide or PTFE at AN-905's round figures, a maker's laminate from Laminate materials, or custom) and its dielectric constant εr.
- **Width and impedance are two live boxes:** type a width and the impedance box shows Z0; type the impedance you need and the width box shows the width that gives it.
- **The big result:** Z0 from the field solver (as in Edge-coupled embedded), or the width for the impedance typed, with w/b and the dielectric on each side, h = (b − t)/2. For the moment the solver takes, the badge reads "Solving…" and Wheeler's value stands in.
- **Below it:**
- Wheeler's formula, the big result until 2026-11-07, and how far it is from the solver's;
- IPC-2141A's simpler formula set: its wide-strip formula for w/b of 0.35 and over, its narrow-strip one below; the narrow one's limits (t/b ≤ 0.25, t/w ≤ 0.11) are named when the trace is outside them, as 1 oz copper on a 0.2 mm trace is;
- AN-905's formula (MT-094's eq. 7), flagged outside b − t > 2w and b > 4t; each with its difference from the solver's;
- the delay per length, the signal seeing the full εr;
- the capacitance and inductance per length.
- **Units:** mm or mil for the width and the spacing; oz, µm or mil for the copper. Any picker switches them all: mil anywhere makes every length mil, mm or µm makes them metric; oz, a copper weight, stays oz.
- **Drawing:** the two ground planes, the dielectric between them with b and εr, the trace centred with its w and t, lengths in mm and mil. The trace is green when its width is the answer.
- **Remembered:** the length unit and the copper unit.

Examples: the two stripline resonators of Hafner et al. (their table I), 50 Ω between planes 0.6 mm apart in FR-4, and AN-905's 35 Ω stripline at εr 2.4.

## Terms

- **Stripline:** a trace on an inner layer with a ground plane above and below, the dielectric filling the space between.

- **b, h:** b is the spacing between the planes; h = (b − t)/2 is the dielectric between the trace and each plane.
- **Characteristic impedance (Z0), εr, C0, L0, oz:** as in Microstrip.
- **Delay:** all the field runs in the dielectric, so a stripline is slower than a microstrip on the same material: Td depends on εr only.

## Source

- **D. Hafner, M. Dressel and M. Scheffler, "Surface-resistance measurements using superconducting stripline resonators", Rev. Sci. Instrum. 85, 014702 (2014), arXiv 1309.5331, appendix A** (read): Wheeler's approximate impedance of a stripline (H. A. Wheeler, IEEE Trans. MTT-26, 866, 1978), Zc = 30/√ε × ln[1 + ½ (16h/πw')(16h/πw' + √((16h/πw')$^2$ + 6.27))], with the equivalent zero-thickness width w' = w + (t/π) ln(e / √((t/(4h + t))$^2$ + ((1/4π)/(w/t + 1.10))^(6/(3 + t/h)))); its table I, computed with it: t 1 µm, w 155 µm, εr 10, h 430 µm, 50.06 Ω; w 45 µm, h 127 µm, 49.79 Ω. The tool reproduces both to the hundredth. The text extraction of the PDF garbled the first term of w'; the form t/(4h + t), as on Peterson's slide, is the one that reproduces the table.
- **Zachariah Peterson, "High Speed Design Basics", Altium, May 2022, slide 24** (read, Pierre's copy, rendered):
- IPC-2141: Z0 = 94.15 / (√εr (W/(B − T) + Cf/π)) "when W/B > 0.35", Cf = 2B/(B − T) ln(B/(B − T) + 1) − (T/(B − T)) ln(1/(1 − T/B)$^2$ − 1), marked "Only accurate for 50 Ohm traces";
- Wadell's equations, "Highly accurate": the same equation as Wheeler's above, but printed with 60/√εr where Hafner et al. have 30/√εr, εr where they have e, and πT/4(W + 1.1T) where they have (1/4π)/(W/T + 1.1). Their table rules out the slide's version (its 60 alone would double the impedance), so the tool follows Hafner et al.
- **National Semiconductor AN-905, TI SNLA035, appendix A** (read): stripline Z0 = 60/√εr × ln[4b / (0.67π (0.8w + t))], best "if b − t > 2w and b > 4t"; its stripline scales: delay 56 ps/cm at εr 2.8; a 35 Ω stripline at εr 2.4, 1.47 pF/cm and 1.8 nH/cm.
- **Walt Kester, ADI MT-094, "Microstrip and Stripline Design", January 2009** (read, Pierre's copy): IPC-2141A's symmetric stripline Z0 = 60/√εr × ln(1.9B / (0.8W + T)) (eq. 7), B = 2H + T, "accuracy … typically on the order of 6%" (Bogatin); this is AN-905's formula (4 / (0.67π) = 1.900).
- **IPC-2141A, March 2004, §4.2.4** (read, Pierre's scanned copy):
- formula set 1 [4-9]–[4-11] is slide 24's "Wadell" equation exactly, with its 60 and πt/4(w + 1.1t): the slide copied IPC's misprints, and Hafner et al.'s table rules them out as above;
- formula set 2: for w/b < 0.35, t/b ≤ 0.25 and t/w ≤ 0.11, Z0 = 60/√εr × ln(4b/(πD)), D = w/2 × (1 + t/(πw) (1 + ln(4πw/t)) + 0.551 (t/w)$^2$) [4-12], [4-14], the strip treated as a round wire of diameter D; for w/b ≥ 0.35, Z0 = 94.15 / ((w/b)/(1 − t/b) + θ/π) [4-15], θ = 2b/(b − t) ln((2b − t)/(b − t)) − t/(b − t) ln((2bt − t$^2$)/(b − t)$^2$) [4-16]. [4-15] is printed without √εr; slide 24's version, with it, is the one that gives ohms in a dielectric, and the tool uses it;
- the 60 ln(1.9B/(0.8W + T)) formula MT-094 attributes to IPC-2141A is not in the 2004 text; the tool labels it AN-905's.
- **Brian C. Wadell, "Transmission Line Design Handbook", Artech House, 1991, §3.6.2 "Centered Stripline, Triplate or Sandwich Line", pages 126–127** (read, Pierre's scanned copy): Wheeler's formula with the same w′ as Hafner et al.'s, "stated error … less than 1.5%", and the side walls negligible beyond 1.5 b (the solver's are 5 b away). Two things in its printing:
- its factor η0 / (2π √εr) is twice Hafner et al.'s 30/√εr = η0 / (4π √εr) with η0 = 120π: the 60 that IPC-2141A [4-9] and slide 24 carry;

- its b must be read as b − t, twice the dielectric between the trace and each plane (its eq. 3.6.2.2, "2.0 h = b"): read that way its three b terms are Hafner et al.'s exactly; read as the spacing between the planes, as its figure 3.6.2.1 draws b, Z0 comes out 3 to 4 % above the solver with 1 oz copper on 0.6 mm (+3.0 % at the default trace, +4.1 % at 35 Ω).
- **The field solver:** js/fieldsolver.js, checked against Cohn's exact zero-thickness stripline to 0.005 % (Edge-coupled embedded's section); tools/fd_lines.py's app_stripline, a separate implementation, gives the values below.
- **Not read:** Wheeler's 1978 paper itself.

## Why the formulas are these

The solver, the big result, as in Microstrip but all in one dielectric:

```
S_air = C_air / ε0, the trace at 1 V, both planes at 0 V
Z0 = η0 / (√εr × S_air)
Td = 33.36 √εr ps/cm,   C0 = Td / Z0,   L0 = Z0 × Td
```

Wheeler's formula, the first comparison:

```
h = (b − t) / 2,   m = 6 / (3 + t/h)
d = √((t/(4h + t))² + (1/4π / (w/t + 1.1))^m)
w' = w + t/π × ln(e / d)
x = 16h / (π w'),   y = x (x + √(x² + 6.27)) / 2
Z0 = 30 / √εr × ln(1 + y)
```

Wheeler's formula fits the exact solution for a zero-thickness strip between two planes, and replaces the real strip by a slightly wider one of zero thickness, w', to account for its copper. All the field is in the dielectric, so the wave travels at c / √εr. The width for an impedance starts from Wheeler's, then the solver's answer corrects it.

**How the formulas compare with the solver:**

| Case | Solver | Wheeler | IPC-2141A | AN-905 |
|---|---|---|---|---|
| Hafner et al.'s 155 µm, 1 µm thick, h 430 µm, εr 10 | 50.05 Ω | +0.00 % | −0.2 % | −2.5 % |
| 0.2 mm, 1 oz, b 0.6 mm, εr 4.3 | 51.82 Ω | +0.00 % | out of range | −1.3 % |
| 0.217 mm, 1 oz, b 0.6 mm, εr 4.3 | 50.00 Ω | −0.00 % | −0.6 % | −1.7 % |
| 0.683 mm, 1 oz, b 0.6 mm, εr 2.4 | 35.00 Ω | −0.34 % | −0.05 % | −25 %, out of range |

Wheeler's formula is within 0.01 % of the solver in the first three cases, and 0.3 % low for the strip wider than the spacing, where IPC-2141A's wide-strip formula does better. For the single stripline the solver mostly confirms Wheeler's formula; it is the same engine as for the stripline pairs, where the formulas part further.

## Core and prepreg

The "Core + prepreg" pill gives the stripline two dielectrics, as a real stack-up has: the core the trace is etched on, under it, and the prepreg pressed over it, which fills around the copper and up to the other plane (boxes "Core εr" and "Prepreg εr"; the material picker sets both). The solver takes the two layers (εr under the trace's bottom face, the prepreg's from there up) and gives the impedance and an effective εr, from which the delay; the comparison formulas, which know one εr, take the mean weighted by the dielectric's thickness on each side, as the manual used to ask the user to do by hand. With one εr (the default) the tool is as before.

- **Checked:** a paper-thin trace centred between the planes has a field symmetric about its plane, so its εeff is exactly the mean of the two: core 3 and prepreg 5 give εeff 4.00000 and the same impedance as one εr of 4 (61.617 Ω, 0.2 mm in 0.6 mm); two equal εr give the one-εr value.
- **What it does:** the default 0.2 mm trace with 1 oz copper, core 3.8 and prepreg 4.6: 52.2 Ω and εeff 4.233, where the thickness-weighted mean is 4.223: the prepreg around the copper's sides counts a little more than its thickness.

## Assumptions and limits

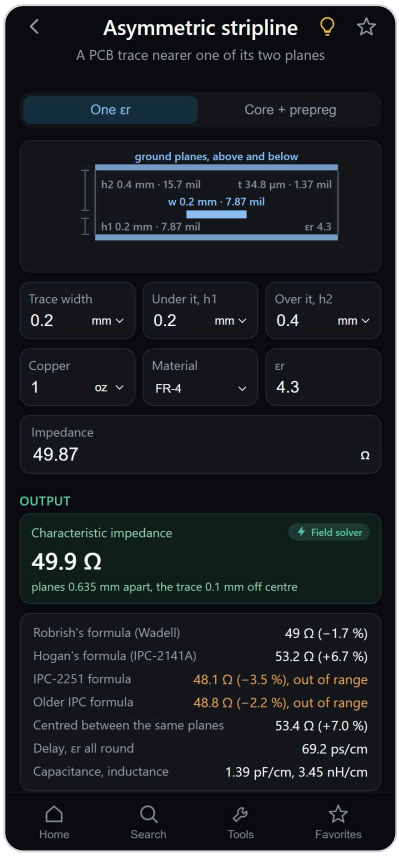
- **Symmetric stripline:** the trace centred between the planes; an off-centre trace is Asymmetric stripline's case.
- **One dielectric or core and prepreg:** with the pill, two layers split at the trace's bottom face (see Core and prepreg); more layers are averaged by the user.
- **Quasi-static field solution:** the trace a rectangle, the planes wide (the solver's side walls 5 b away), no losses; the board maker's stack-up calculation decides.

## How it was checked

9 cases, the expected values from tools/fd_lines.py's app_stripline: - Hafner et al.'s two resonators: 50.05 Ω (50.054; their table I's Wheeler value 50.06) and 49.79 Ω (49.792); IPC-2141A's narrow-strip formula 49.9 Ω for the first; - 50 Ω between planes 0.6 mm apart, FR-4, 1 oz: 0.217 mm (0.21735), IPC-2141A's wide-strip formula −0.6 %, AN-905 −1.7 %; - 35 Ω at εr 2.4: 1.48 pF/cm and 1.81 nH/cm (AN-905's 1.47 and 1.8), 51.7 ps/cm; - the same 35 Ω, a strip wider than the spacing: 0.683 mm (0.68273), Wheeler's formula 0.3 % low; - the default 0.2 mm trace between planes 0.6 mm apart: 51.8 Ω (51.819), Wheeler +0.0 %, AN-905 51.1 Ω, outside IPC-2141A's narrow-strip limit (t/w 0.17); - the copper's own picker (1 oz is 34.8 µm); - copper as thick as half the spacing is refused; - the length pickers move together.

# Asymmetric stripline

(formerly Dual stripline impedance, then Dual stripline)




## What it computes

The impedance of a trace between two ground planes but nearer one of them (the asymmetric, offset or off-centre stripline, as each trace is on a board with two signal layers between the same pair of planes), or the width that gives an impedance:

- **Inputs:** the trace width w, the dielectric under it h1 and over it h2, the copper's thickness t (its own picker), the material and εr.
- **Width and impedance are two live boxes:** type a width and the impedance box shows Z0; type the impedance you need and the width box shows the width that gives it.
- **The big result:** Z0 from the field solver (as in Edge-coupled embedded), with the spacing between the planes and how far the trace is off centre; or the width for the impedance typed. For the moment the solver takes, the badge reads "Solving…" and an estimate stands in.
- **Below it,** each with its difference from the solver:
- Robrish's formula, flagged outside its ±2 % range;
- Hogan's formula, the one IPC-2141A reprints;
- the two dual-stripline formulas, the trace seen as one of two signal layers between the planes: IPC-2251's, and the older IPC one that other calculators still use ("Older IPC formula"); each flagged outside its range (W/(H − T) < 0.35, T/H < 0.25), and absent when the far side leaves no room for a second layer;
- the same trace centred between the same planes;
- the delay per length, the signal seeing the full εr;
- the capacitance and inductance per length.
- **Units:** mm or mil for the three lengths; oz, µm or mil for the copper. Any picker switches them all: mil anywhere makes every length mil, mm or µm makes them metric; oz, a copper weight, stays oz.
- **Drawing:** the two planes, the trace drawn nearer the closer one, h1, h2, t and εr, lengths in mm and mil. The trace is green when its width is the answer.
- **Remembered:** the length unit and the copper unit.

Examples: a 0.2 mm trace 0.2 and 0.4 mm from the planes, the width for 50 Ω on those layers, a centred trace (Stripline's case), a narrow trace close to one plane, and two dual striplines: LearnEMC's case with a 2 mil trace, and a 0.43 mm trace well outside the dual formulas' range.

## Terms

- **Asymmetric (offset) stripline:** a stripline whose trace is not centred: h1 ≠ h2. A board with two signal layers between the same planes (a dual stripline) has each layer's traces offset like this; the tool leaves out the other layer's traces.

- **Name:** IPC-2141A's. The tool was first called Dual stripline; Pierre renamed it on 2026-11-05, since that name is used for the two-layer stack-up, not for one offset trace. Saturn PCB Toolkit calls it "Stripline Asym", Polar "offset stripline", Wadell "off-center stripline".
- **h1, h2, B:** the dielectric between the trace and the lower and the upper plane; B = h1 + h2 + t is the spacing between the planes.
- **Z0, εr, C0, L0, oz:** as in Microstrip and Stripline.

## Source

- **The field solver** (js/fieldsolver.js; how it works and how it was checked are in Edge-coupled embedded's section). For this case it reproduces tools/fd_lines.py, a separate implementation, in every case below.
- **Brian C. Wadell, "Transmission Line Design Handbook", Artech House, 1991, §3.6.3 "Off-Center Stripline"** (read, Pierre's scanned copy). It gives three methods:
- Robrish's, extending Cohn's (3.6.3.1–3.6.3.13): for a narrow strip, $w/(b - t) < 0.35$, a round wire of diameter $d0 = w\,(0.5008 + 1.0235x - 1.0230x^2 + 1.1564x^3 - 0.4749x^4)$, $x = t/w$, and $Z0 = \eta0/(2\pi\sqrt{\varepsilon r}) \times \cosh^{-1}(\sin(\pi c/b)\coth(\pi d0/2b))$, c the strip's centre above the lower plane; for a wide one, the parallel plates on either side and the edges' fringing (3.6.3.6–3.6.3.10). "Agreement with exact analytic solutions is better than 0.5 %", and ±2 % against numerical solutions for $0.2 < c/b < 0.8$, $w > t$, $t/b < 0.2$;
- Howe's parallel plates and fringing (3.6.3.14–3.6.3.20), which Wadell says differ substantially from the centred equations when the strip is centred; not used here;
- Hogan's (3.6.3.21–3.6.3.23, from Sean Hogan's 1988 note "Closed Form Transmission Line Equation"): $Z0 = (1/\sqrt{\varepsilon r})[Z0,ctr(1, w, b, t) - \Delta Z0,air]$, $Z0,air = 2\,Z0,ctr(1, w, h1, t)\,Z0,ctr(1, w, h2, t) /$ (their sum), $\Delta Z0,air = (0.26\pi/8)\,Z0,air^2\,(0.5 - (h1 + t/2)/(h1 + h2 + t))\text{^}2.2\,((t + w)/(h1 + h2 + t))\text{^}2.9$, Z0,ctr the centred stripline (3.6.2.1). No range or accuracy is given.
- **IPC-2141A, March 2004, §4.2.5, [4-17]–[4-19]** (read, Pierre's scanned copy): Hogan's formula, citing its reference 2, with $1/\varepsilon r$ where Wadell has $1/\sqrt{\varepsilon r}$. That $1/\varepsilon r$ is a misprint: a uniform dielectric divides the impedance in air by $\sqrt{\varepsilon r}$, and with $1/\varepsilon r$ a centred FR-4 stripline would come out at half its value.
- **The dual-stripline formulas.** For two signal layers between the planes, H is the dielectric between a trace and its nearer plane and C (D in the book below) the dielectric between the two signal layers; for the tool's trace, H = the smaller of h1 and h2, and C = the larger − t − H, the room left for the other layer.
- **IPC-2251, "Design Guide for Electronic Packaging Utilizing High-Speed Techniques", 4th working draft, February 2001**, as the LearnEMC dual stripline calculator (learnemc.com, also on the Missouri S&T EMC Laboratory's site) publishes it, in its code and its equation image (read; the IPC document itself not read): $Z0 = ½\,[60/\sqrt{\varepsilon r} \times \ln(8H / (0.67\pi\,(0.8W + T))) + 60/\sqrt{\varepsilon r} \times \ln(8(H + C) / (0.67\pi\,(0.8W + T)))]$, valid for $W/(H - T) < 0.35$, $T/H < 0.25$; the calculator's notes give it within 1 % of a 2D field solver for $0.08 < W/H < 0.35$ at εr 4, H 15 mil, C 10 mil, T 1.37 mil. It is the mean of two AN-905 striplines (Stripline's section), planes 2H and 2(H + C) apart.
- **The older IPC dual stripline formula**, as Mark I. Montrose, "Printed Circuit Board Design Techniques for EMC Compliance — A Handbook for Designers", 2nd edition, prints it (§4.2.4 "Dual or Asymmetric Stripline Topology", eq. 4.10, page 117 of Pierre's PDF; that page read, the rest of the book not): $Z0 = (80/\sqrt{\varepsilon r})\,\ln[1.9\,(2H + T) / (0.8W + T)]\,[1 - H / (4\,(H + D + T))]$, the same range, "approximate"; the page adds that both layers should be routed at right angles to each other. Other calculators still use it: it reproduces one's 32.689 Ω for W 0.4318, H 0.254, C 0.2032 mm, T 18 µm, εr 4.6. Its IPC source (IPC-D-317A, 1995, or the first IPC-2141, 1996) was not read.

- **Which is the latest:** IPC-2251 followed IPC-D-317A; IPC-2141A (2004, read) gives Hogan's formula for the offset trace instead, and no IPC-2141B was found (2026-11-07).
- **A correction:** this manual first called the full-B first term and the Z0,SS(b = h1) arguments of [4-17] and [4-19] misprints too. Wadell prints them the same way: they are Hogan's formula as published, and it is that formula which is inaccurate off centre.

## Why the formulas are these

```
∇·(ε ∇ϕ) = 0 between the trace (1 V) and the planes (0 V)
S = C / ε0 = Σ ε (Δϕ)² over the grid
Z0 = η0 / (√εr × S),   η0 = 376.73 Ω
Td = 33.36 √εr ps/cm,   C0 = Td / Z0,   L0 = Z0 × Td
```

Between two planes all the field runs in one dielectric, so one solve, in air, gives the impedance for any εr. The nearer plane takes more of the trace's field, its capacitance rises and Z0 falls below the centred value.

**How the formulas compare with the solver** (FR-4, 1 oz):

| Case | Solver | Robrish | Hogan | Centred |
|---|---|---|---|---|
| 0.2 mm, h1 0.2, h2 0.4 mm | 49.87 Ω | −1.7 % | +6.7 % | +7.0 % |
| 0.5 mm, h1 0.2, h2 0.4 mm | 30.64 Ω | −1.5 % | +7.1 % | +8.9 % |
| 0.1 mm, h1 0.1, h2 0.5 mm | 50.81 Ω | −1.8 % (c/b 0.19) | +33 % | +34 % |
| 0.2 mm centred, b 0.6 mm | 51.82 Ω | −2.8 % (wide form) | 0.0 % | 0.0 % |

**The dual-stripline formulas against the solver** (one trace, the other layer left out):

| Case | Solver | IPC-2251 | Older IPC dual |
|---|---|---|---|
| εr 4, H 15, C 10, T 1.37 mil, W 1 mil | 107.13 Ω | −1.3 % | +6.1 % |
| the same, W 2 mil | 96.63 Ω | −0.3 % | +6.5 % |
| the same, W 3 mil | 89.21 Ω | −0.1 % | +6.2 % |
| the same, W 4 mil | 83.42 Ω | −0.0 % | +5.6 % |
| εr 4.6, H 0.254, C 0.2032 mm, T 18 µm, W 0.4318 mm (W/(H − T) 1.83) | 38.27 Ω | −7.1 % | −14.6 % |

Within their range IPC-2251's mean of two striplines is within 0.3 % of the solver above W = 2 mil, the older formula 5 to 6.5 % high; outside it both read low, the older one by 15 % on the wide trace.

Robrish's stays within 3 %; Hogan's is barely below the centred value, its offset term being a fraction of an ohm, so it reads 7 to 33 % high off centre. Wadell's centred stripline (3.6.2.1), which Hogan's uses, is printed with $\eta_0/(2\pi\sqrt{\varepsilon_r})$, twice Wheeler's $\eta_0/(4\pi\sqrt{\varepsilon_r})$ (Stripline's section); the tool uses Wheeler's. The estimate shown while the solver works is the tool's earlier formula: the two halves (planes 2h1 + t and 2h2 + t apart) in parallel, minus Hogan's term; it is within 0.3 % in the first case but −6.2 % for the wide trace and +4.9 % near a plane.

## Assumptions and limits

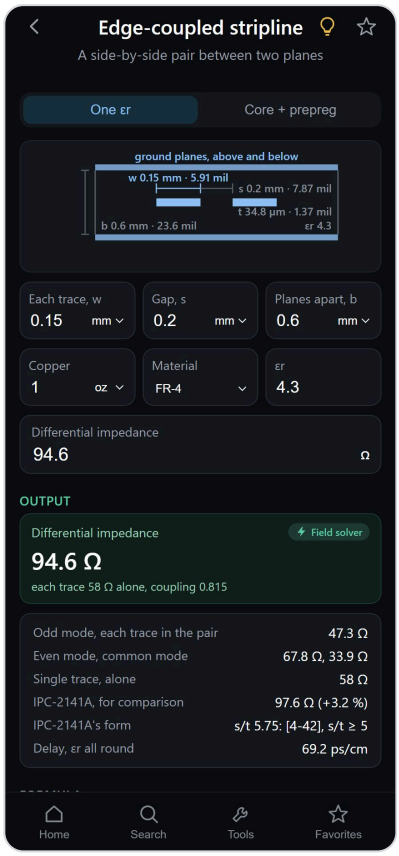
- **Quasi-static, no losses:** dispersion, losses and roughness are left out.
- **One dielectric or core and prepreg:** with the pill, the core under the trace (h1) and the prepreg from its bottom face up, as in Stripline's Core and prepreg; more layers are averaged by the user.
- **One trace:** traces on the other signal layer are left out.
- **Rectangular trace;** the board maker's stack-up calculation decides.

## How it was checked

8 cases, the expected values from tools/fd_lines.py's app_stripline: - 0.2 mm trace 0.2 and 0.4 mm from the planes, FR-4, 1 oz: 49.9 Ω (49.87); Robrish 49 Ω (−1.7 %), Hogan 53.2 Ω (+6.7 %), centred 53.4 Ω (+7.0 %); 69.2 ps/cm; 1.39 pF/cm and 3.45 nH/cm; - 50 Ω on those layers: 0.199 mm (0.19881); - centred, planes 0.6 mm apart: 51.8 Ω (51.82); Hogan's equals the centred value there, Robrish's wide form 2.8 % low; - 0.1 mm trace 0.1 and 0.5 mm from the planes: 50.8 Ω (50.81); Hogan +33.4 %; Robrish's flagged outside its range; - h1 and h2 swapped: the same impedance; - the three length pickers move together; - dual stripline, LearnEMC's case with a 2 mil trace: 96.6 Ω (96.631); IPC-2251 96.3 Ω (−0.3 %), the older formula 103 Ω (+6.5 %); - dual stripline, the 0.4318 mm trace: 38.3 Ω (38.269); IPC-2251 35.6 Ω (−7.1 %) and the older formula 32.7 Ω (−14.6 %), both flagged outside their range.

# Edge-coupled stripline

(formerly Edge-coupled stripline)




## What it computes

The differential impedance of two edge-coupled traces centred between two ground planes (a differential pair on an inner layer), or the trace width that gives it at a gap:

- **Inputs:** each trace's width w, the gap between them s, the spacing between the planes b, the copper's thickness t (its own picker), the material and εr.
- **Width and differential impedance are two live boxes:** type a width and the impedance box shows Zdiff; type the Zdiff you need (90 Ω for USB, 100 Ω for Ethernet or LVDS) and the width box shows the width at your gap.
- **The big result:** Zdiff from the field solver (as in Edge-coupled embedded), with each trace's impedance alone and the coupling (odd mode over one trace alone); or the width for the Zdiff typed. For the moment the solver takes, the badge reads "Solving…" and IPC-2141A's value stands in.
- **Below it:**
- the odd-mode impedance, each trace's share of the pair, Zdiff / 2;
- the even-mode and common-mode impedance, as in Edge-coupled microstrip;
- one trace alone, from the solver as in Stripline;
- Zdiff by IPC-2141A's form, the big result until 2026-11-07, and its difference from the solver's;
- which of IPC-2141A's two forms applies, by s/t;
- the delay per length, the signal seeing the full εr.
- **Units:** mm or mil for the three lengths; oz, µm or mil for the copper. Any picker switches them all: mil anywhere makes every length mil, mm or µm makes them metric; oz, a copper weight, stays oz.
- **Drawing:** the two planes, the pair centred between them, w and s as a chain dimension, b, t and εr, in mm and mil. The traces are green when their width is the answer.
- **Remembered:** the length unit and the copper unit.

Examples: 100 Ω and 90 Ω between planes 0.6 mm apart with a 0.2 mm gap, a close pair (IPC's narrow-gap form), and a pair far enough apart that it no longer couples.

## Terms

- **Differential pair, differential impedance (Zdiff), odd mode, coupling:** as in Edge-coupled microstrip.
- **Edge-coupled:** side by side on the same layer; a pair one above the other is broadside-coupled.
- **Fringing capacitance (Cf):** the capacitance from a strip's edge to the planes, beyond the part directly under and over it. A thicker strip has more.

- **Paper-thin strip:** a strip of zero thickness, for which Cohn found the impedance exactly; the thick strip is worked out from it.
- **b, t, εr, oz:** as in Stripline.

## Source

- **IPC-2141A, "Design Guide for High-Speed Controlled Impedance Circuit Boards", March 2004, §4.4.2, figure 4-6(b)** (read, Pierre's scanned copy), citing its references 2 and 7:
- [4-42] for s/t ≥ 5: Z0,o(w/b, t/b, s/b) = {1/Z0(w/b, t/b) − (C'f(t/b)/C'f(0)) [1/Z0,o(w/b, 0, s/b) − 1/Z0(w/b, 0)]}^−1;
- [4-43] for s/t ≤ 5: Z0,o = {1/Z0(w/b, t/b) + [1/Z0,o(w/b, 0, s/b) − 1/Z0(w/b, 0)] − (2/(377ε)) (C'f(t/b) − C'f(0) − ε/s)}^−1;
- Z0(w/b, t/b) its stripline [4-9]; Z0(w/b, 0) and Z0,o(w/b, 0, s/b) the zero-thickness impedances of its references 2 and 7;
- [4-44] C'f(t/b) = (0.0885 εr/π) {2b/(b − t) ln((2b − t)/(b − t)) − ((2b − t)/(b − t)) ln($b^2$/(b − t)$^2$ − 1)}; [4-45] C'f(0) = (0.0885 εr/π) 2 ln 2.
- **A. J. Burkhardt, C. S. Gregg and J. A. Staniforth, "Calculation of PCB Track Impedance", Polar Instruments, IPC Printed Circuits Expo 1999** (read): Cohn's exact zero-thickness stripline, Z0 = η0/(4√εr) × K(k)/K(k'), k = sech(πw/2b), k' = tanh(πw/2b), and the coupled pair's odd mode with k0 = tanh(πw/2b) coth(π(w + s)/2b) (their equations 4, 5, 7, 8); their table 1, a surface microstrip computed by their boundary-element solver: 30.09, 50.63 and 89.63 Ω for 3.3, 1.5 and 0.45 mm on 0.794 mm, εr 4.2, 35 µm copper.
- **Brian C. Wadell, "Transmission Line Design Handbook", Artech House, 1991, §4.6.5, pages 234–235** (read, Pierre's scanned copy): its C′f(t/b) has 1/(1 − t/b) − 1 = t/(b − t) as the second coefficient, and its odd mode's fringing term C′f,o = C′f(0, s/b) + εt/s (4.6.5.9): two of the three corrections below.
- **The field solver:** js/fieldsolver.js, checked against Cohn's exact zero-thickness pair to 0.005 % (Edge-coupled embedded's section); tools/fd_lines.py's app_stripline, a separate implementation, gives the values below.
- **Not read:** Cohn's papers themselves, and IPC's references 2 and 7.

## Why the formulas are these

The solver, the big result, solved three times, all in one dielectric:

```
odd mode: the traces at +1 V and −1 V,  Zodd = η0 / (√εr × S_air)
Zdiff = 2 Zodd,   even mode: both at +1 V,  common mode = Zeven / 2
one trace alone, as in Stripline
```

IPC-2141A's form, the comparison:

```
In air:   Z0, Zo,0 = 30π K(k)/K(k'), Cohn's paper-thin strip alone and in the pair
          Z(t): Wheeler's strip alone, with its copper
          ΔY = 1/Zo,0 − 1/Z0
s/t ≥ 5:  1/Zo = 1/Z(t) + (Cf(t)/Cf(0)) ΔY
s/t < 5:  1/Zo = 1/Z(t) + ΔY − (2/η0)(Cf(t) − Cf(0) − t/s)
Zdiff = 2 Zo / √εr
```

The odd-mode admittance of a paper-thin strip in a pair is known exactly; so is one strip alone. Their difference ΔY is what the neighbour adds. For a strip with copper, IPC starts from that strip alone, Z(t), and adds ΔY, scaled by how much the copper raises the edge's fringing capacitance when the gap is wide. Across a narrow gap the two copper walls face each other: for the odd mode the plane midway between them is at 0 V, adding a parallel-plate capacitance t/(s/2) per wall in units of ε. In one dielectric everything scales as 1/√εr, so the tool works in air (IPC's ε becomes ε0) and divides at the end. K is computed by the arithmetic-geometric mean, K(k) = π / (2 agm(1, k')).

**Three misprints corrected.** - [4-42] subtracts the scaled ΔY: a strip would then have a higher impedance next to its odd-mode partner than alone, which the field contradicts (13 to 50 % off). With the sign that [4-43] and the physics give, it adds it. - [4-43] ends in ε/s, which is not a capacitance per length; the copper walls' parallel-plate term is εt/s, and the field solver agrees with that within 1.4 %. - [4-44]'s second coefficient (2b − t)/(b − t) multiplies ln 0 as t goes to zero, where [4-45] says C'f(0) = 2 ln 2 (in units of ε0 εr, over π). With t/(b − t), as in IPC's own θ [4-16] $(b^2/(b - t)^2 - 1 = (2bt - t^2)/ (b - t)^2)$, it does.

**How IPC-2141A's form compares with the solver** (1 oz copper, planes 0.6 mm apart, εr 4.3):

| Case | Solver Zdiff | IPC-2141A |
|---|---|---|
| 100 Ω: 0.125 mm, 0.2 mm gap ([4-42]) | 100.00 Ω | +3.6 % |
| 90 Ω: 0.175 mm, 0.2 mm gap ([4-42]) | 90.00 Ω | +2.9 % |
| 0.15 mm, 0.2 mm gap, the default ([4-42]) | 94.60 Ω | +3.2 % |
| 0.15 mm, 0.1 mm gap ([4-43]) | 76.82 Ω | +1.0 % |
| 0.15 mm, 2 mm apart | 116.03 Ω | +0.0 % |

Corrected, the form is 1 to 3.6 % high with 1 oz copper; it was the big result until 2026-11-07, when the solver took over. At 100 Ω the difference is in the width: IPC-2141A's 0.14 mm gives 96.7 Ω, the solver's 0.125 mm gives 100 Ω.

## Core and prepreg

As in Stripline: the "Core + prepreg" pill puts the core's εr under the traces and the prepreg's from their bottom faces up; the solver takes the two layers, the formulas the thickness-weighted mean. The default pair with core 3.8 and prepreg 4.6: 95.1 Ω (95.080), 68.8 ps/cm.

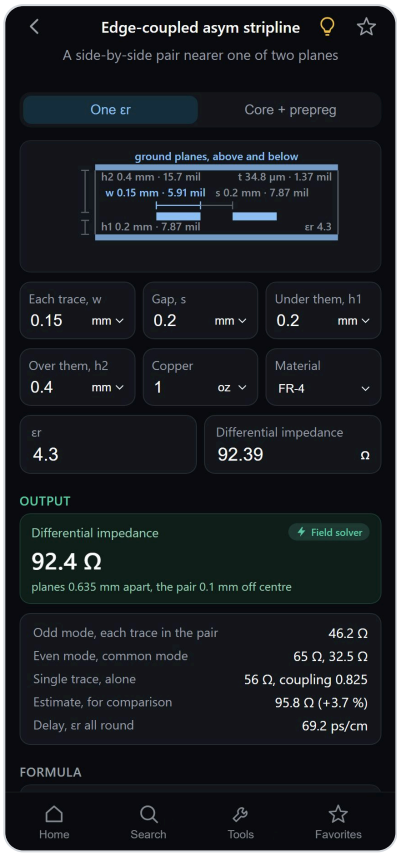
## Assumptions and limits

- **Quasi-static field solution:** the traces rectangles, no losses; the board maker's stack-up calculation decides. For the formula, §4.4 itself warns that the balanced-line formulas "exist in various forms, some of which are not in agreement".
- **Centred between the planes, equal traces, one dielectric, no losses.**
- **The jump at s/t = 5:** IPC's two forms differ there by about 2 %; the solver has no such jump.

## How it was checked

6 cases, the expected values from tools/fd_lines.py's app_stripline: - 100 Ω, planes 0.6 mm apart, FR-4, 1 oz, 0.2 mm gap: 0.125 mm traces (0.12513), 61.8 Ω alone, [4-42]; - 90 Ω, same planes and gap: 0.175 mm (0.17450); - 0.15 mm traces 0.1 mm apart: 76.8 Ω (76.815), coupling 0.662, IPC's [4-43] 77.6 Ω (+1.0 %); even mode 74.5 Ω, common mode 37.2 Ω; - 2 mm apart: 116 Ω (116.03), twice one trace's 58 Ω; - the default pair: 94.6 Ω (94.596), IPC-2141A 97.6 Ω (+3.2 %), 69.2 ps/cm; - the three length pickers move together.

# Edge-coupled asym stripline




## What it computes

The differential impedance of two edge-coupled traces between two ground planes but nearer one of them (the asymmetric, offset or off-centre edge-coupled stripline, Saturn's "Edge Cpld Int Asym"), as a pair is on a board with two signal layers between the same planes, or the trace width that gives it at a gap:

- **Inputs:** each trace's width w, the gap s, the dielectric under the traces h1 and over them h2, the copper's thickness t (its own picker), the material and εr.
- **Width and differential impedance are two live boxes,** as in Edge-coupled stripline.
- **The big result:** Zdiff from the field solver (as in Edge-coupled embedded), with the spacing between the planes, b = h1 + t + h2, and how far the pair is off centre; or the width for the Zdiff typed. For the moment the solver takes, the badge reads "Solving…" and the estimate stands in.
- **Below it:**
- the odd-mode impedance, each trace's share, Zdiff / 2;
- the even-mode and common-mode impedance, as in Edge-coupled microstrip;
- one trace alone, from the solver, and the coupling (odd mode over it);
- the estimate, and how far it is from the solver's;
- the delay per length, the signal seeing the full εr.
- **Units:** mm or mil for the four lengths; oz, µm or mil for the copper. Any picker switches them all: mil anywhere makes every length mil, mm or µm makes them metric; oz, a copper weight, stays oz.
- **Drawing:** the two planes, the pair drawn nearer the closer one, w and s as a chain dimension, h1, h2, t and εr, in mm and mil. The traces are green when their width is the answer.
- **Remembered:** the length unit and the copper unit.

Examples: 100 Ω and 90 Ω with 0.2 mm under and 0.4 mm over the pair, a pair close to one plane, and the centred pair, Edge-coupled stripline's.

## Terms

- **Differential pair, Zdiff, odd mode, even mode, common mode, coupling:** as in Edge-coupled microstrip.
- **h1, h2, b, t, εr, oz:** as in Asymmetric stripline.

## Source

- **No published formula read covers this pair.** Wadell's §4.6.7 "Asymmetric Edge-Coupled Striplines" (Transmission Line Design Handbook, 1991, pages 239–240, read, Pierre's scanned copy) is a different structure: a centred pair of unequal widths, worked from each line's capacitances (Cristal's method). IPC-2141A (March 2004, read) has the centred pair only (§4.4.2).
- **The field solver:** js/fieldsolver.js, its stripline cross-section with the pair's gap, the odd or even mode and h1; tools/fd_lines.py's app_stripline, a separate implementation with the same settings, gives the values below.
- **The estimate:** the tool's, built from IPC-2141A's centred pair as Edge-coupled stripline corrects it.

## Why the formulas are these

The solver, the big result, solved three times, all in one dielectric:

```
odd mode: the traces at +1 V and −1 V,  Zodd = η0 / (√εr × S_air)
Zdiff = 2 Zodd,   even mode: both at +1 V,  common mode = Zeven / 2
one trace alone, as in Asymmetric stripline
```

The estimate, the comparison and the width search's start:

```
Zo,a = IPC-2141A's centred pair between planes 2h1 + t apart
Zo,b = the same between planes 2h2 + t apart
1/Zo = (1/Zo,a + 1/Zo,b) / 2,   Zdiff = 2 Zo
```

The pair's field divides between the two planes: the part going down sees the pair as if it were centred between planes 2h1 + t apart, the part going up as if centred between planes 2h2 + t apart, each with half the capacitance; the two halves add, as in Asymmetric stripline's and Broadside-coupled stripline's copper term. It leaves out how the field of one half spills into the other, so it reads high:

| Case (1 oz, εr 4.3) | Solver Zdiff | Estimate |
|---|---|---|
| 0.15 mm, gap 0.2, h1 0.2, h2 0.4 mm | 92.39 Ω | +3.7 % |
| 0.1 mm, gap 0.15, h1 0.15, h2 0.3 mm | 91.80 Ω | +1.7 % |
| 0.15 mm, gap 0.2, h1 0.15, h2 0.6 mm | 87.66 Ω | +4.9 % |
| 0.1 mm, gap 0.1, h1 0.1, h2 0.4 mm | 76.41 Ω | +2.1 % |
| 0.25 mm, gap 0.3, h1 0.3, h2 0.5 mm | 92.30 Ω | +2.1 % |
| 0.2 mm, gap 0.2, centred, h1 = h2 = 0.2 mm | 77.18 Ω | +1.5 % |

Centred, the estimate is IPC-2141A's own pair, whose error it carries (up to +3.6 % in Edge-coupled stripline).

## Assumptions and limits

- **Traces on the other signal layer left out:** a trace under or over the pair changes its impedance; the tool takes the space between the planes as plain dielectric.
- **Quasi-static field solution:** the traces rectangles, one dielectric (or core and prepreg with the pill, as in Stripline), no losses; the board maker's stack-up calculation decides.

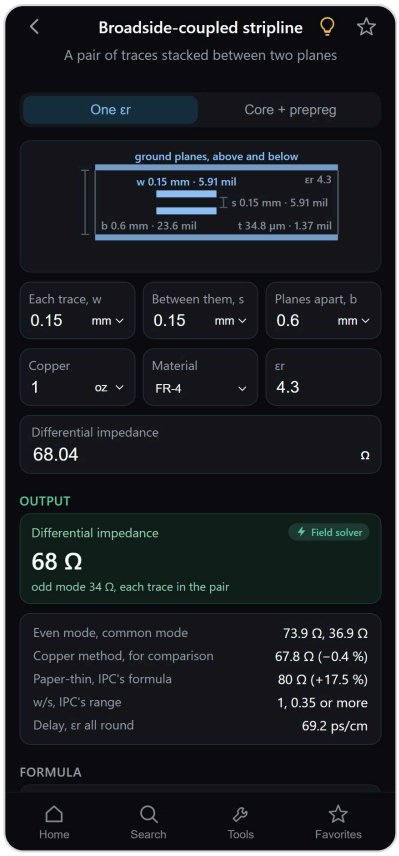
## How it was checked

7 cases, the expected values from tools/fd_lines.py's app_stripline: - the default: 0.15 mm traces, 0.2 mm gap, 0.2 mm under and 0.4 mm over, FR-4, 1 oz: 92.4 Ω (92.386), odd mode 46.2 Ω, even mode 65 Ω, common mode 32.5 Ω, 56 Ω alone, coupling 0.825, estimate 95.8 Ω (+3.7 %), 69.2 ps/cm; - 100 Ω on the same layers: 0.117 mm (0.11662); - USB 90 Ω: 0.162 mm (0.16208); - close to one plane, 0.1 mm traces 0.1 mm apart, 0.1 and 0.4 mm away: 76.4 Ω (76.413), estimate +2.1 %; - centred, h1 = h2: 94.6 Ω, Edge-coupled stripline's value, the estimate IPC-2141A's 97.6 Ω; - the four length pickers move together; - a zero gap is refused.

The pair placed nearer the upper plane (h1 and h2 swapped) gives the same 92.4 Ω, as its mirror image must.

# Broadside-coupled stripline

(formerly Broadside stripline)




## What it computes

The differential impedance of two traces one above the other on two inner layers (a broadside-coupled pair), centred between two ground planes, or the trace width that gives it:

- **Inputs:** each trace's width w, the dielectric between the two traces s, the spacing between the planes b, the copper's thickness t (its own picker), the material and εr.
- **Width and differential impedance are two live boxes,** as in Edge-coupled stripline.
- **The big result:** Zdiff from the field solver (as in Edge-coupled embedded), with the odd-mode impedance; or the width for the Zdiff typed. For the moment the solver takes, the badge reads "Solving…" and the copper method's value stands in.
- **Below it:**
- the even-mode and common-mode impedance, as in Edge-coupled microstrip;
- Zdiff by the tool's copper method, the big result until 2026-11-07, and its difference from the solver's;
- IPC-2141A's paper-thin Zdiff, and how much higher it reads;
- w/s against IPC's range, 0.35 or more, flagged below;
- the delay per length, the signal seeing the full εr.
- **Units:** mm or mil for the three lengths; oz, µm or mil for the copper. Any picker switches them all: mil anywhere makes every length mil, mm or µm makes them metric; oz, a copper weight, stays oz.
- **Drawing:** the two planes, the pair stacked between them with w and s, b, t and εr, in mm and mil. The traces are green when their width is the answer.
- **Remembered:** the length unit and the copper unit.

Examples: 100 Ω between planes 0.8 mm apart, a 0.15 mm pair 0.15 mm apart, wide traces close together, and narrow traces flagged, all with 1 oz copper.

## Terms

- **Broadside-coupled:** the two traces of a pair face each other across the dielectric between two layers, instead of side by side.
- **Paper-thin:** copper of zero thickness. Here the traces face each other with their full width, so their thickness matters more than for a side-by-side pair.
- **Odd mode, even mode, common mode, Zdiff:** as in Edge-coupled microstrip.

## Source

- **IPC-2141A, "Design Guide for High-Speed Controlled Impedance Circuit Boards", March 2004, §4.4.3, figure 4-6(c)** (read, Pierre's scanned copy), citing its references 2, 8 and 9: $Z_{0,o} = (293.9/\sqrt{\varepsilon_r})\ (s/b) / \tanh^{-1}(k)$ [4-46], k the solution of $w/b = (1/\pi)[\ln((1 + R)/(1 - R)) - (s/b)\ \ln((k + R)/(k - R))]$ [4-47], $R = \sqrt{(k(kb - s)/(b - ks))}$ [4-48]; "nearly exact for w/s ≥ 0.35, and is more accurate for substrates with large values of εr". No copper thickness.
- **The field solver:** js/fieldsolver.js, as in Edge-coupled embedded, on the lower half of the cross-section: the mid-plane between the traces at 0 V for the odd mode, a mirror for the even mode; tools/fd_lines.py's app_broadside, a separate implementation with the same settings, gives the values below.
- **Not read:** IPC's references 2, 8 and 9.

## Why the formulas are these

The solver, the big result, solved twice on the lower half:

```
odd mode: the mid-plane at 0 V,  Zodd = η0 / (√εr × S_air),  Zdiff = 2 Zodd
even mode: the mid-plane a mirror (both traces at +1 V),  common mode = Zeven / 2
```

The even mode was checked against its limit: as the traces close up, two paper-thin strips in contact are one strip, each carrying half its charge, so Zeven tends to twice Cohn's single stripline: 4.1, 2.4, 1.3, 0.8 and 0.45 % under it for gaps of 8, 4, 2, 1 and 0.5 µm (0.15 mm strips, planes 0.6 mm apart), the gap's own thickness closing as it shrinks.

IPC-2141A's paper-thin formula, a comparison:

```
R = √(k (kb − s) / (b − ks))
w/b = (ln((1+R)/(1−R)) − (s/b) ln((k+R)/(k−R))) / π,  solved for k
Zo = 295.9/√εr × (s/b) / tanh⁻¹(k),   Zdiff = 2 Zo
```

For the odd mode the plane midway between the two traces is at 0 V, so each trace is a strip between its ground plane and that mid-plane; the formula is the exact (conformal-mapping) solution for a paper-thin strip there. k has no closed form: the tool halves the range from s/b to 1 until w/b comes out. The width for a Zdiff is found the same way.

**The constant.** Against the field solver (tools/fd_lines.py, as in Edge-coupled stripline) the printed formula's ratio is 0.9934 for every geometry tried: its shape is exact, and the constant that matches is $\eta_0\pi/4 = 295.9$, not 293.9. With 295.9 the tool is within 0.1 % of the solver for paper-thin traces at w/s ≥ 0.35 (80.0 against 79.97 Ω for 0.15 mm traces 0.15 mm apart, planes 0.6 mm apart), 1.2 % low at w/s 0.25.

**The copper method.** 1 oz copper lowers Zdiff by 6 to 25 % against the solver: each trace's outer face moves 35 µm closer to its plane, and its edges add fringing. From 2026-11-05 to 2026-11-07 the big result allowed for it by the tool's own method, not IPC's, now a comparison row:

```
h1 = (b − s)/2 − t,  the copper's outer face to its plane
Zpar(t) = halves in parallel: Z(2h1 + t) ‖ Z(s + t), Wheeler's striplines, ×2 / √εr
1/Zo,cu = 1/Zo,thin(planes b − 2t apart) + 1/Zpar(t) − 1/Zpar(0)
```

The paper-thin pair with the planes 2t closer, plus the admittance the copper adds to one strip between its plane and the mid-plane, from the two half-striplines of Asymmetric stripline (without its offset term, which fails this close to a plane). Against the solver, for w 0.1 to 0.5 mm, s 0.1 to 0.2 mm, b 0.4 to 0.8 mm, εr 4.3: −1.4 to −0.1 % for ½, 1 and 2 oz.

**How the formulas compare with the solver** (1 oz copper, εr 4.3):

| Case | Solver Zdiff | Copper method | IPC paper-thin |
|---|---|---|---|
| 100 Ω: 0.0917 mm, s 0.2 mm, b 0.8 mm | 100.00 Ω | −0.32 % | +20.4 % |
| 0.15 mm, s 0.15 mm, b 0.6 mm | 68.04 Ω | −0.43 % | +17.5 % |
| 0.3 mm, s 0.1 mm, b 0.6 mm | 35.62 Ω | −0.61 % | +10.2 % |
| 0.05 mm, s 0.2 mm, b 0.6 mm (w/s 0.25) | 114.45 Ω | −1.03 % | +29.3 % |

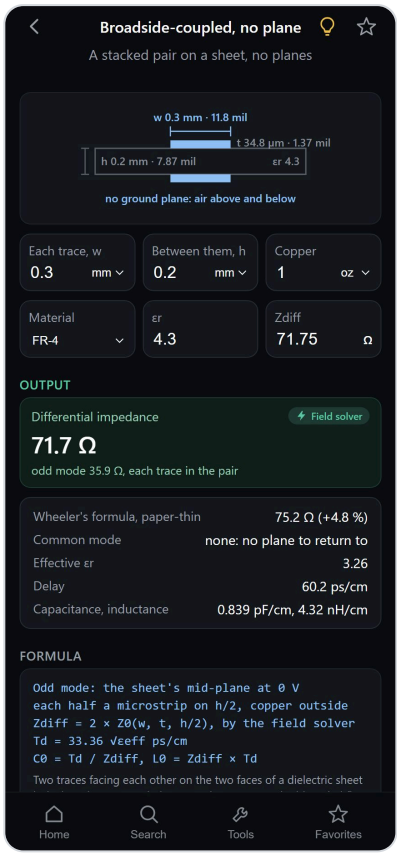
## Assumptions and limits

- **Quasi-static field solution:** the traces rectangles, no losses; the board maker's stack-up calculation decides.
- **The copper method is the tool's,** checked against the solver over the range above only.
- **Centred pair, equal traces, one dielectric:** core and prepreg are averaged by the user, or core and prepreg with the pill (the core between the traces, the prepreg outside, as in Stripline's Core and prepreg).

## How it was checked

7 cases, the expected values from tools/fd_lines.py's app_broadside: - 100 Ω with 1 oz, planes 0.8 mm apart, 0.2 mm between: 0.0917 mm traces (0.09173); paper-thin they would read 120 Ω (+20.4 %); - 0.15 mm traces 0.15 mm apart, planes 0.6 mm, 1 oz: 68 Ω (68.044), odd mode 34 Ω, even mode 73.9 Ω, common mode 36.9 Ω; copper method 67.8 Ω (−0.4 %), paper-thin 80 Ω (+17.5 %); 69.2 ps/cm; - 0.3 mm traces 0.1 mm apart: 35.6 Ω (35.615), copper method −0.6 %, paper-thin 39.2 Ω; - 0.05 mm traces 0.2 mm apart: 114 Ω (114.45), copper method −1.0 %, w/s 0.25 flagged; - no copper: 80 Ω, IPC's paper-thin formula (with 295.9) within 0.01 %; - copper reaching the planes is refused; - traces as far apart as the planes are refused.

# Broadside-coupled, no plane




## What it computes

The differential impedance of two traces facing each other on the two faces of a dielectric sheet, with no ground plane anywhere (a double-sided flex pair: the parallel-strip line, Wadell's "paired strips"), or the trace width that gives it:

- **Inputs:** each trace's width w, the sheet between them h, the copper's thickness t (its own picker), the material and εr.
- **Width and Zdiff are two live boxes:** type a width and the Zdiff box shows Zdiff; type the Zdiff you need and the width box shows the width.
- **The big result:** Zdiff from the field solver, with the odd-mode impedance; or the width for the Zdiff typed, with w/h. For the moment the solver takes, the badge reads "Solving…" and Wheeler's value stands in.
- **Below it:**
- Wheeler's paper-thin formula, and how far it is from the solver's;
- the common mode: none, with no plane to return through;
- the effective εr, from the solver;
- the delay per length;
- the capacitance and inductance per length between the two traces.
- **Units:** mm or mil for the two lengths; oz, µm or mil for the copper. Any picker switches them all: mil anywhere makes every length mil, mm or µm makes them metric; oz, a copper weight, stays oz.
- **Drawing:** the sheet with a trace on each face, w, h, t and εr, in mm and mil. The traces are green when their width is the answer.
- **Remembered:** the length unit and the copper unit.

Examples: 100 Ω on a 0.1 mm polyimide flex, wide traces on a thin sheet with 1 oz and paper-thin, and narrow paper-thin strips.

## Terms

- **Paired strips (parallel-strip line):** two flat conductors facing each other across a dielectric sheet; the field runs between them, partly in the sheet and partly in the air around it.
- **No plane, no common mode:** with no ground plane, a signal on both traces at once has nothing to return through; only the differential (odd) mode is a transmission line.
- **Differential pair, Zdiff, odd mode:** as in Edge-coupled microstrip.
- **εr, εeff, Td, C0, L0, oz:** as in Microstrip.

## Source

- **Brian C. Wadell, "Transmission Line Design Handbook", Artech House, 1991, §3.3.5 "Paired Strips", pages 71–72** (read, Pierre's scanned copy), a = w/2, b = h/2 (3.3.5.3, 3.3.5.4), paper-thin:
- wide strips, a/b > 1: Z0 = η0/√εr × {a/b + (1/π) ln 4 + (εr + 1)/(2π εr) ln[π e (a/b + 0.94)/2] + (εr − 1)/(2π $\varepsilon r^2$) ln(e $\pi^2$/16)}^−1 (3.3.5.1), "Stated error is less than 1% for wide strips";
- narrow strips, a/b < 1: Z0 = η0/(π√εr) × [ln(4b/a) + (1/8)$(a/b)^2$ − (εr − 1)/(2(εr + 1)) (ln(π/2) + ln(4/π)/εr)] (3.3.5.2), "less than 1% for narrow strips".
- **A misprint corrected:** as printed, 3.3.5.2 reads 20 % below the solver for εr 3.5 (126.0 against 156.8 Ω for 0.1 mm strips on 0.2 mm). With η0/(π√((εr + 1)/2)) it comes within 0.3 % (157.2 Ω), and that factor is Wheeler's narrow microstrip formula's, η0/(π√(2(εr + 1))), doubled with h → b: the pair is two microstrips on the sheet's mid-plane. In air (εr 1) the two factors agree.
- **The field solver:** js/fieldsolver.js's microstrip on h/2, as in Microstrip; tools/fd_lines.py's app_microstrip, a separate implementation, gives the values below.

## Why the formulas are these

```
odd mode: the sheet's mid-plane at 0 V
each half: a microstrip w wide, t thick, on h/2 of the sheet
Zdiff = 2 × Z0,ms(w, t, h/2),   εeff the half's
Td = 33.36 √εeff ps/cm,   C0 = Td / Zdiff,   L0 = Zdiff × Td
```

The two traces carry equal and opposite voltages, so by symmetry the plane midway through the sheet is at 0 V: each trace with its half of the sheet is a microstrip over a ground plane, and the two are in series between the traces. Wheeler's formulas, from conformal mapping, cover the paper-thin strips; the copper's thickness adds capacitance at the edges and lowers Zdiff:

| Case | Solver | Wheeler |
|---|---|---|
| 0.5 mm on 0.1 mm, εr 3.5, paper-thin | 32.90 Ω | +0.01 % |
| 0.1 mm on 0.2 mm, εr 3.5, paper-thin | 156.80 Ω | +0.24 % (corrected) |
| 0.3 mm on 0.2 mm, εr 4.3, paper-thin | 75.68 Ω | −0.63 % |
| 0.5 mm on 0.1 mm, εr 3.5, 1 oz | 31.87 Ω | +3.3 % |
| 0.3 mm on 0.2 mm, εr 4.3, 1 oz | 71.75 Ω | +4.8 % |
| 0.0863 mm on 0.1 mm, εr 4.2, 18 µm | 100.00 Ω | +9.3 % |

The width for a Zdiff starts from Wheeler's, then the solver's answer corrects it.

## Assumptions and limits

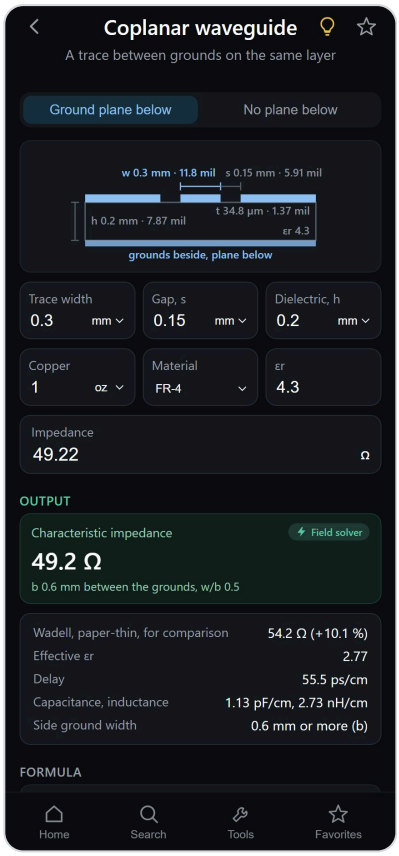
- **Air on both sides, nothing nearby:** a ground plane, a metal case or a cover layer near the pair changes Zdiff (with planes it is Broadside-coupled stripline).

- **Equal traces facing each other exactly;** a sideways offset between the layers raises Zdiff.
- **Quasi-static field solution:** the traces rectangles, no losses; the flex maker's calculation decides.

## How it was checked

7 cases, the expected values from tools/fd_lines.py's app_microstrip: - the default, 0.3 mm traces on 0.2 mm of FR-4, 1 oz: 71.7 Ω (71.748), odd mode 35.9 Ω, Wheeler 75.2 Ω (+4.8 %), εeff 3.26, 60.2 ps/cm, 0.839 pF/cm and 4.32 nH/cm; - 100 Ω on 0.1 mm polyimide, 18 µm: 0.0863 mm (0.08632); - 0.5 mm on 0.1 mm, εr 3.5, 1 oz: 31.9 Ω (31.867), Wheeler +3.3 %; - the same paper-thin: 32.9 Ω (32.901), Wheeler +0.0 %; - 0.1 mm on 0.2 mm, εr 3.5, paper-thin: 157 Ω (156.80), Wheeler's narrow form, corrected, +0.2 %; - the two length pickers move together; - a zero sheet is refused.

# Coplanar waveguide




## What it computes

The characteristic impedance of a trace with ground copper on both sides of it on the same layer, with or without a ground plane under the board, or the trace width that gives an impedance at a gap:

- **Two pills:** "Ground plane below", the grounded coplanar waveguide (GCPW), the default; "No plane below", the coplanar waveguide (CPW) on its own, air under the board.
- **Inputs:** the trace width w, the gap s between the trace and each ground, the dielectric h, the copper's thickness t (in oz, µm or mil, its own picker), the material and εr.
- **Width and impedance are two live boxes:** type a width and the impedance box shows Z0; type the impedance you need and the width box shows the width that gives it, the gap kept.
- **The big result:** Z0 from the field solver (as in Edge-coupled embedded), or the width for the impedance typed, with b = w + 2s, the distance between the grounds' edges, and w/b. For the moment the solver takes, the badge reads "Solving…" and Wadell's value stands in.
- **Below it:**
- Wadell's formula for the mode, and how far it is from the solver's (for the grounded mode it is the paper-thin one, so the row says so);
- the effective εr, from the solver;
- the delay per length, per cm in mm and per inch in mil;
- the capacitance and inductance per length;
- how wide the ground copper beside the trace should be, each side: 5 b or more with no plane (Wadell §3.4.1), b or more with a plane below (§3.4.3's notes, which also tie the two sides together).
- **Units:** mm or mil for the three lengths; oz, µm or mil for the copper. Any picker switches them all: mil anywhere makes every length mil, mm or µm makes them metric; oz, a copper weight, stays oz.
- **Drawing:** the trace between its two grounds on the substrate, the plane under it in the grounded mode, w and s as a chain dimension, h, t and εr, in mm and mil. The trace is green when its width is the answer.
- **Remembered:** the mode, the length unit and the copper unit.

Examples: Wadell's 50 Ω probe on alumina, 50 Ω with a plane below on a 0.2 mm layer and on a 1.6 mm board, and a CPW with no plane below.

## Terms

- **Coplanar waveguide (CPW):** a trace with ground copper on both sides of it on the same layer; the field runs mostly across the two gaps.
- **Grounded coplanar waveguide (GCPW):** the same with a ground plane under the board as well (also called conductor-backed, CBCPW), the usual form on RF boards; the side grounds are tied to the plane

by rows of vias.

- **s, b:** s is the gap between the trace's edge and the ground's edge; b = w + 2s is the distance between the two grounds' edges (Wadell's a and b are the trace's width and that distance).
- **Z0, εr, εeff, Td, C0, L0, oz:** as in Microstrip.
- **K(k):** the complete elliptic integral of the first kind; Wadell's formulas are conformal mappings, which turn the cross-section into a parallel-plate one whose capacitance is a ratio of two such integrals.

## Source

- **Brian C. Wadell, "Transmission Line Design Handbook", Artech House, 1991** (read, Pierre's scanned copy):
- §3.4.1, "Coplanar Waveguide", pages 73–76: Z0 = 30π/√εeff,t × K(kt')/K(kt) (3.4.1.1); εeff,t = εeff − (εeff − 1)/((b − a)/2 / 0.7t × K(k)/K'(k) + 1) (3.4.1.2); εeff = 1 + (εr − 1)/2 × K(k')K(k1)/(K(k) K(k1')) (3.4.1.3); k = a/b, kt = at/bt; k1 = sinh(πat/4h)/sinh(πbt/4h) (3.4.1.6); at = a + 1.25t/π (1 + ln(4πa/t)), bt = b − the same (3.4.1.8, 3.4.1.9); "The ground plane should extend greater than 5b on each side of the gap"; its example 3.4.1.1, a 50 Ω probe, a = 10 mil, t = 1 mil, h = 25.2 mil of 96 % alumina (εr 10): b = 0.0234 in (it prints "= 0.615 cm" for 0.0594 cm);
- §3.4.3, "Coplanar Waveguide with Ground", page 79: Z0 = 60π/√εeff / (K(k)/K(k') + K(k1)/K(k1')) (3.4.3.1), k = a/b, k1 = tanh(πa/4h)/ tanh(πb/4h) (3.4.3.5), εeff = (1 + εr r)/(1 + r), r = K(k')K(k1)/(K(k)K(k1')) (3.4.3.6), no copper thickness; page 80: to avoid microstrip-like modes, the side ground should extend more than b from the trace on each side, and jumpers can tie its two halves together.
- **The field solver:** js/fieldsolver.js, its coplanar cross-section; tools/fd_lines.py's app_coplanar, a separate implementation with the same settings, gives the values below (the two agree to the last digit shown: 49.218 and 66.165 Ω for the first cases below).
- **Not read:** the papers Wadell draws on (Ghione and Naldi, Gupta et al.).

## Why the formulas are these

The solver, the big result:

```
∇·(ε ∇ϕ) = 0: the trace at 1 V, the side grounds (and the plane) at 0 V
S = C / ε0 = Σ ε (Δϕ)², once with the board, once in air (S_air)
Z0 = η0 / √(S × S_air),   εeff = S / S_air
Td = 33.36 √εeff ps/cm,   C0 = Td / Z0,   L0 = Z0 × Td
```

The side grounds run to the solver's side wall, 20 b past the gap, and its box is 40 b above the copper and, with no plane, 40 b below the board: a paper-thin CPW in air then comes within 0.001 % of its exact impedance, η0/4 × K(k')/K(k) (120.48 Ω for w/b 0.5). A box at 10 b, the side wall 5 b past the gap, read 0.15 % low.

Wadell's formulas, the comparison, grounded and paper-thin:

```
k = a/b,   k1 = tanh(πa/4h) / tanh(πb/4h),   k' = √(1 − k²)
r = K(k') K(k1) / (K(k) K(k1')),   εeff = (1 + εr r) / (1 + r)
Z0 = 60π / √εeff / (K(k)/K(k') + K(k1)/K(k1'))
```

and with no plane, with the copper:

```
at, bt = a ± 1.25t/π (1 + ln(4πa/t)),   kt = at/bt
k1 = sinh(πa/4h) / sinh(πb/4h)
εeff = 1 + (εr − 1)/2 × K(k′) K(k1) / (K(k) K(k1′))
εeff,t = εeff − (εeff − 1) / ((b − a)/2 / (0.7t) × K(k)/K(k′) + 1)
Z0 = 30π / √εeff,t × K(kt′) / K(kt)
```

**One reading corrected.** Eq. 3.4.1.6 prints k1 with at and bt; with them Wadell's example needs b = 25.9 mil for 50 Ω. With a and b, as in the εeff equation that uses k1, it needs 23.43 mil, his 23.4 mil: the tool reads it that way. K(k)/K(k′) is computed by the arithmetic-geometric mean, and k′ from √((b − a)(b + a))/b, which keeps its precision for wide traces.

**How Wadell's formulas compare with the solver** (εr 4.3 unless noted):

| Case | Solver | Wadell |
|---|---|---|
| No plane, paper-thin, 0.3 mm, s 0.15 mm, h 1.6 mm | 74.14 Ω | +0.09 % |
| No plane, 1 oz, 0.3 mm, s 0.15 mm, h 1.6 mm | 66.17 Ω | −9.6 % |
| No plane, 1 oz, 1 mm, s 0.2 mm, h 1.6 mm | 53.97 Ω | −7.9 % |
| No plane, 1 oz, 0.5 mm, s 0.25 mm, h 0.2 mm | 77.69 Ω | −3.8 % |
| No plane, Wadell's probe, εr 10 | 52.64 Ω | −5.1 % |
| Plane below, paper-thin, 0.3 mm, s 0.15 mm, h 0.2 mm | 53.91 Ω | +0.53 % |
| Plane below, paper-thin, 0.3 mm, s 0.15 mm, h 1.6 mm | 73.42 Ω | +0.06 % |
| Plane below, paper-thin, 1 mm, s 0.2 mm, h 1.6 mm | 54.86 Ω | +0.07 % |
| Plane below, 1 oz, 0.3 mm, s 0.15 mm, h 0.2 mm | 49.22 Ω | +10.1 % |
| Plane below, 1 oz, 1 mm, s 0.2 mm, h 1.6 mm | 51.33 Ω | +7.0 % |

Paper-thin, both formulas meet the solver within 0.5 %: the conformal mappings are sound. The copper is what they handle badly. Its walls face each other across the gap and add capacitance, about t/s per wall on top of the gap's own field, so 1 oz on 0.15 mm gaps lowers Z0 by 9 to 11 % in the solver. §3.4.3 has no copper term and reads 7 to 10 % high; §3.4.1's correction overshoots and reads 4 to 10 % low. On Wadell's own probe, 52.6 Ω where he designed for 50 Ω.

## Assumptions and limits

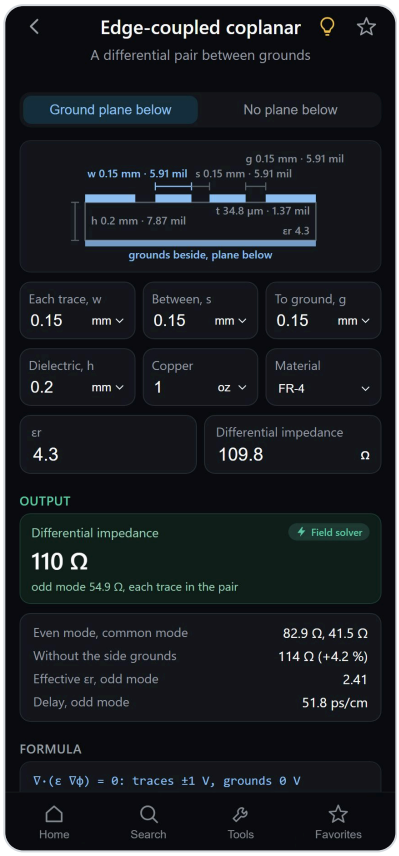
- **Wide side grounds:** the solver takes them out to 20 b; narrower ground copper raises Z0 (Wadell's 5 b, or b with a plane below, keeps it close).
- **Grounds tied together:** in the grounded mode, the side grounds and the plane are at the same potential, as stitching vias make them; a CPW without them can carry other modes (Wadell p80).

- **Quasi-static field solution:** the trace and the grounds rectangles, one dielectric, no solder mask, no losses; the board maker's stack-up calculation decides.

## How it was checked

10 cases, the expected values from tools/fd_lines.py's app_coplanar: - the default, plane below: 0.3 mm trace, 0.15 mm gaps on 0.2 mm FR-4, 1 oz: 49.2 Ω (49.219), εeff 2.77, 55.5 ps/cm, 1.13 pF/cm and 2.73 nH/cm; Wadell paper-thin 54.2 Ω (+10.1 %); - Wadell's example 3.4.1.1: 10 mil probe, b 23.4 mil, 1 mil copper, 25.2 mil of εr 10, no plane: 52.6 Ω (52.639); his formula 50.0 Ω (−5.1 %), the 50 Ω he solved for; - 50 Ω with a plane below on 0.2 mm, 0.15 mm gaps: 0.29 mm (0.29035); - 50 Ω with a plane below on 1.6 mm, 0.2 mm gaps: 1.09 mm (1.08937); - no plane, 0.3 mm on 1.6 mm, 0.15 mm gaps: 66.2 Ω (66.166), Wadell 59.8 Ω (−9.6 %), side grounds 3 mm (5 b); - the default without the plane: 70.5 Ω (70.530), Wadell 64.3 Ω (−8.8 %); - paper-thin copper: 53.9 Ω (53.909), Wadell +0.5 %; - the copper's own picker (1 oz is 34.8 µm); - the three length pickers move together; - a zero gap is refused.

# Edge-coupled coplanar




## What it computes

The differential impedance of an edge-coupled pair with ground copper beside it on the same layer (the differential or edge-coupled coplanar waveguide), with or without a ground plane under the board, or the trace width that gives it at given gaps:

- **Two pills,** as in Coplanar waveguide: "Ground plane below" (the edge-coupled CPWG, the default) and "No plane below".
- **Inputs:** each trace's width w, the gap s between the two traces, the gap g from each trace to its ground, the dielectric h, the copper's thickness t (its own picker), the material and εr.
- **Width and Zdiff are two live boxes;** a Zdiff typed sets the width, the gaps kept.
- **The big result:** Zdiff from the field solver, with the odd-mode impedance; or the width for the Zdiff typed.
- **Below it:**
- the even-mode and common-mode impedance;
- the same pair without the side grounds (moved 50 times the cross-section away), the comparison: how much the grounds lower Zdiff;
- the odd mode's effective εr and the delay per length.
- **Units:** mm or mil for the four lengths; oz, µm or mil for the copper. Any picker switches them all: mil anywhere makes every length mil, mm or µm makes them metric; oz, a copper weight, stays oz.
- **Drawing:** the pair between its two grounds on the substrate, the plane under it in the grounded mode, w, s and g, h, t and εr, in mm and mil. The traces are green when their width is the answer.
- **Remembered:** the mode, the length unit and the copper unit.

Examples: USB's 90 Ω and 100 Ω with a plane below on a 0.2 mm layer, the grounds moved far away (Edge-coupled microstrip's case), and the default pair with no plane below.

## Terms

- **Edge-coupled coplanar waveguide:** a differential pair whose outer edges each face a ground on the same layer, g away; with a plane below as well, the side grounds are tied to it by rows of vias.
- **Differential pair, Zdiff, odd mode, even mode, common mode:** as in Edge-coupled microstrip.
- **s, g:** s the gap between the two traces, g between each trace and its ground. (Wadell writes s for the first and d for the second.)
- **εr, εeff, t, oz:** as in Coplanar waveguide.

## Source

- **The field solver:** js/fieldsolver.js, its coplanar cross-section extended to a pair (half the cross-section, the pair's centre line at 0 V for the odd mode, a mirror for the even); tools/fd_lines.py's app_coplanar, a separate implementation with the same settings, gives the values below. Checked against two limits: with the grounds 5 mm away and a plane below it gives Edge-coupled microstrip's solver value to 0.003 % (114.401 against 114.397 Ω for 0.15 mm traces 0.15 mm apart on 0.2 mm); with the traces 3 mm apart the odd and even modes meet within 0.3 %. The single Coplanar waveguide is unchanged (49.219 Ω).
- **Brian C. Wadell, "Transmission Line Design Handbook", Artech House, 1991, §4.4.2 "Edge-Coupled CPW" and §4.4.3 "Edge-Coupled CPWG", pages 194–198** (read, Pierre's scanned copy). Neither is usable as a comparison: §4.4.3's odd and even impedances (4.4.3.1–4.4.3.12, after Hanna) use K(β1)/K′(β1) and K(β1k1)/K′(β1k1) for the plane below, with β1 and k1 defined nowhere on those pages; §4.4.2's (4.4.2.1–4.4.2.7) needs incomplete elliptic integrals solved for k5 and k7 and "assume[s] … that the substrate is infinitely thick". The tool shows no formula rather than guess the missing terms.
- **Not read:** Hanna's 1985 paper and the others Wadell cites.

## Why the formulas are these

The solver, three times:

```
odd mode: the traces at +1 V and −1 V, the grounds (and plane) at 0 V
Zodd = η0 / √(S × S_air),   Zdiff = 2 Zodd,   εeff = S / S_air
even mode: both traces at +1 V,  common mode = Zeven / 2
without the side grounds: the odd mode with g = 50 (w + s + h)
```

The grounds beside the pair add capacitance to each trace and lower Zdiff, and keep the field, and so the crosstalk to other traces, close to the pair. A typed Zdiff sets the width: the search starts from a rough estimate (Wadell's single CPW or CPWG, as Coplanar waveguide computes it, with s/2 and g combined harmonically; −30 to +36 %) and the solver's answers correct it.

| Case (1 oz, FR-4) | Zdiff | Without the side grounds |
|---|---|---|
| 0.15 mm, s 0.15, g 0.15 mm, h 0.2 mm, plane below | 109.82 Ω | 114.40 Ω, +4.2 % |
| the same, no plane below | 118.82 Ω | 135.40 Ω, +14.0 % |

With a plane below, the plane already holds most of the field and the side grounds take 4 %; with none, they take 14 %.

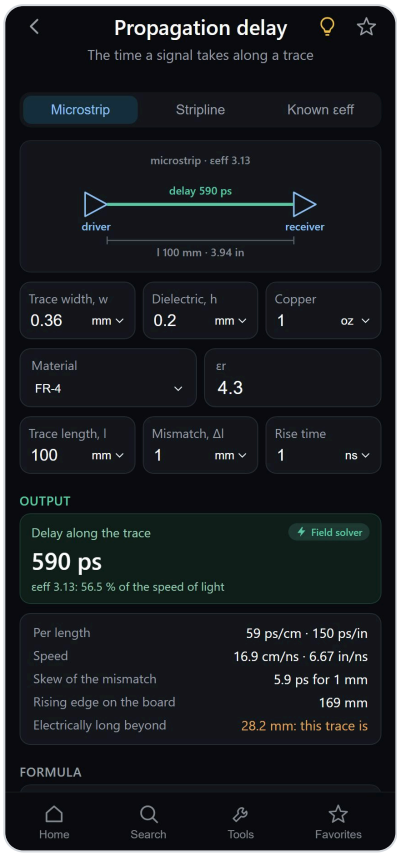
## Assumptions and limits

- **Wide side grounds,** tied to the plane in the grounded mode, as stitching vias make them.
- **Equal traces, the two gaps to ground equal.**
- **Quasi-static field solution:** the traces and grounds rectangles, one dielectric, no solder mask, no losses; the board maker's stack-up calculation decides.

## How it was checked

8 cases, the expected values from tools/fd_lines.py's app_coplanar: - the default, plane below, 0.15 mm traces 0.15 mm apart, 0.15 mm to the grounds, 0.2 mm FR-4, 1 oz: 110 Ω (109.823), odd mode 54.9 Ω, even mode 82.9 Ω, common mode 41.5 Ω, without the side grounds 114 Ω (+4.2 %), εeff 2.41, 51.8 ps/cm; - USB 90 Ω: 0.254 mm (0.25427); - 100 Ω: 0.195 mm (0.19481); - the grounds 5 mm away: 114 Ω (114.401), the same without them; - no plane below: 119 Ω (118.819), common mode 65.9 Ω, without the side grounds 135 Ω (+14.0 %); - the pill switches to no plane; - the four length pickers move together; - a zero gap to the ground is refused.

# Propagation delay




## What it computes

The time a signal takes along a trace, and what follows from it. It takes the place of the planned "Effective dielectric constant" and "Signal propagation delay" tools (Pierre, 2026-11-08), as every impedance tool already shows both for its own structure:

- **Three pills:** Microstrip (the trace on an outer layer: εeff from the field solver for its width, dielectric and copper), Stripline (between two planes in one dielectric: εeff is εr exactly, whatever the trace) and Known εeff (typed, for any other line: coplanar, embedded, a maker's figure).
- **Inputs:** the trace's width w, the dielectric h and the copper (microstrip), the material and εr (or εeff), the trace's length l, a length mismatch Δl, and the rise time.
- **The big result:** the delay along the trace; below it εeff and the speed as a share of the speed of light.
- **Below it:** the delay per length (ps/cm and ps/in), the speed (cm/ns and in/ns), the skew of the mismatch, the length of the rising edge on the board (tr × v), and the length beyond which the trace is electrically long, orange when this trace is.
- **Units:** mm or mil for the cross-section and the mismatch, oz, µm or mil for the copper, mm, cm or in for the length, linked (in switches the others to mil); ps or ns for the rise time.
- **Drawing:** the trace seen from above, driver to receiver, its delay in green, the length in mm and in.
- **Remembered:** the pill and the units.

Examples: a 50 Ω microstrip, length matching in a stripline, a known εeff in inches.

## Terms

- **Effective dielectric constant, εeff:** the εr of a uniform material that would give the line's speed; between 1 and εr when part of the field is in air (microstrip), εr when all of it is in the dielectric (stripline).
- **Propagation delay:** the time for an edge to travel the trace, l √εeff / c; per length it is √εeff / c, 33.356 ps/cm × √εeff.
- **Skew:** the difference in arrival time of two signals, here from a length mismatch Δl.
- **Rising edge on the board:** how much of the trace the edge spans while it rises, tr × v; a trace much shorter than it behaves as a lumped load.
- **Electrically long:** a trace whose delay is a large enough share of the rise time that its reflections matter, and that should be terminated.

## Sources

- **The delay:** the speed of a TEM wave, c/√εr, c = 299 792 458 m/s exactly (SI); for a microstrip's quasi-TEM wave, εeff from the app's field solver (as in Microstrip), Hammerstad's formula while it works.
- **Electrically long:** Barry Olney, "Beyond Design: High-speed Rules of Thumb", I-Connect007 (Design007 Magazine), 21 November 2024 (read): drivers need "provision for termination" when the "length (in inches) is equal to or greater than the rise and fall time (in ns)", and "the propagation speed of a signal is ~6 in/ns in FR-4"; together, a delay of a sixth of the rise time. The tool takes tr/6 with the trace's own speed: 24.1 mm (0.949 in) for a stripline in FR-4 at 1 ns, his 1 in.
- **The caution:** Howard Johnson, "Resonance in Short Transmission Line", High-Speed Digital Design Online Newsletter vol. 6 issue 06, 2003 (read): the delay-to-rise-time ratio "doesn't actually determine whether a termination is needed"; a short line with a low-impedance driver and a capacitive load can still ring.

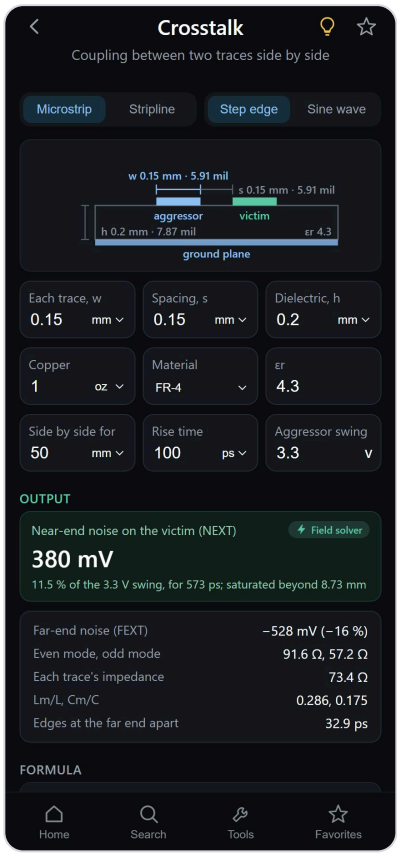
## Assumptions and limits

- **Lossless, no dispersion:** εeff at the frequencies of the edge; a real laminate's εr falls slowly with frequency, and a microstrip's εeff rises toward εr at high frequency.
- **Stripline:** one dielectric all round; with core and prepreg of different εr, use Known εeff with the Stripline tool's value.
- **The rule of thumb:** tr/6 is a guide (see Johnson above); other authors use tr/4 or tr/2.

## How it was checked

7 cases, the expected values from tools/fd_lines.py's microstrip and the formulas above: - the default microstrip, 0.36 mm on 0.2 mm FR-4, 1 oz, 100 mm: εeff 3.13 (3.13056), 59.0 ps/cm, 150 ps/in, 16.9 cm/ns, 590 ps, 5.9 ps for 1 mm, the edge 169 mm at 1 ns, long beyond 28.2 mm; - the stripline length-matching example: 346 ps for 50 mm, 3.46 ps for 0.5 mm, long beyond 12.0 mm at 0.5 ns; - a known εeff of 3, 6 in: 880 ps, 147 ps/in, long beyond 1.14 in; - Olney's rule: a stripline in FR-4 at 1 ns is long beyond 24.1 mm; - 20 mm at 1 ns is not long; - inches on the length switch the cross-section to mil; - a zero rise time is refused.

# Crosstalk




## What it computes

The crosstalk a step or a sine on one trace (the aggressor) induces on its neighbour (the victim), two equal traces side by side, as a share of the aggressor's signal:

- **Two pairs of pills, side by side:** Microstrip (the pair on an outer layer over a plane) or Stripline (the pair centred between two planes); Step edge (a logic edge with a rise time, below) or Sine wave (a sine at one frequency, see Sine wave).
- **Inputs:** each trace's width w, the spacing s between them, the dielectric h (microstrip) or the spacing between the planes b (stripline), the copper's thickness t, the material and εr, the length the traces run side by side, the aggressor's rise time, and its voltage swing on the line (3.3 V unless typed).
- **The big result:** the near-end noise on the victim (NEXT) in volts, the coefficient times the aggressor's swing; below it the coefficient in %, how long the plateau lasts (the round trip 2T), and whether the coupled length saturates it.
- **Below it:**
- the far-end noise (FEXT) on the victim in volts, with its share in %, negative in microstrip (an inverted pulse about as long as the rise time), zero in stripline, shown as "0 V: the two couplings cancel" (in one dielectric the capacitive and inductive parts are equal and opposite, Lm/L = Cm/C); flagged as an upper bound when the even and odd modes' edges arrive further apart than the rise time;
- the pair's even- and odd-mode impedances, from the solver;
- each trace's impedance with its neighbour present, ζ = √(L/C);
- the coupling ratios Lm/L and Cm/C;
- in microstrip, how far apart the even and odd edges reach the far end.
- **Units:** mm or mil for the cross-section; oz, µm or mil for the copper; mm, cm or in for the coupled length: any of them switches the others, mil or in to imperial, mm, µm or cm to metric (oz stays oz); ps or ns for the rise time; MHz or GHz for the sine's frequency.
- **Drawing:** the aggressor (blue) and the victim (green) over the plane or between the planes, w and s as a chain dimension, h or b, εr, in mm and mil.
- **Remembered:** the pills and the units.

Examples: the 3w rule (spacing three trace widths), a short run with a slow edge (below saturation), a stripline pair (no FEXT), and a 100 MHz clock (Sine wave).

## Sine wave

With "Sine wave" the rise time box becomes Frequency (MHz or GHz), and the tool gives the victim's noise for a sine on the aggressor:

- **The big result:** the near-end noise in dB, 20 log of its share of the aggressor's amplitude (−30 dB is 3.2 %), as RF crosstalk is quoted and whatever the swing (Pierre); below it the same in volts (the share times the aggressor's swing; a peak-to-peak swing gives peak-to-peak noise) and in %.
- **Below it:** the far-end noise in dB and volts ("0 V: the two couplings cancel" in stripline); the quarter-wave frequency, where the coupled run is a quarter wavelength long, with the run's length in wavelengths at the chosen frequency; the even and odd modes, ζ, Lm/L and Cm/C as for the step.
- **Drawing:** in place of the cross-section, the near-end (green) and far-end (blue) noise in dB against frequency, on a log scale over a few decades around the quarter wave and the chosen frequency, which is marked.

How: the pair splits exactly into its even mode (both traces at the same voltage) and odd mode (opposite voltages), each an ordinary line with its own impedance Z and speed v from the field solver. Each end of each trace is taken to have a resistor R = ζ, the trace's impedance; the aggressor's source is 2 × the swing behind R, so a matched line carries the swing. Each mode is then one line of electrical length θ = 2π f l / v, with half the source and R at both ends; from the standard line equations (A = D = cos θ, B = j Z sin θ, C = j sin θ / Z):

```
V_end = swing × R / (2R cos θ + j sin θ (Z + R²/Z))
V_in  = V_end × (cos θ + j (Z/R) sin θ)
near = V_in,even − V_in,odd,   far = V_end,even − V_end,odd
```

This is worked out here from those equations, not taken from a paper; it is exact for lossless lines with all four ends in R, at any frequency and any coupling, microstrip included.

- **What it shows:** the near-end noise grows with frequency, 6 dB per octave at first, peaks near the quarter wave at twice the step's plateau, falls to a null at the half wave and repeats; Shlepnev's remark that a real signal's crosstalk can reach twice the step's is this peak. The far-end noise grows with frequency in microstrip and is zero in stripline.
- **Checked:** at the quarter wave (cos θ = 0) the near end is exactly $Ze^2/(Ze^2 + R^2) - Zo^2/(Zo^2 + R^2)$; in stripline, $R^2 = Ze\,Zo$, that is (Ze − Zo)/(Ze + Zo) = Cm/C, twice the step's ¼ (Lm/L + Cm/C) = ½ Cm/C: 17.82 % against 2 × 8.915 %. At low frequency the far end is the step's FEXT times 2π f tr: 16.00 % × 2π × 100 MHz × 100 ps = 1.005 %, against 1.004 %. The scratchpad script xt_sine.py, on tools/fd_lines.py's modes, gives the values.

| **Sine, FR-4, 1 oz, 50 mm, 3.3 V** | **near end** | **far end** |
|---|---|---|
| microstrip, 0.15 mm, s 0.15, h 0.2 mm, 100 MHz | 4.12 %, −27.7 dB | 1.00 %, −40.0 dB |
| the same at 1 GHz | 22.5 %, −12.9 dB | 9.85 %, −20.1 dB |
| stripline, 0.15 mm, s 0.2, b 0.6 mm, 711.28 MHz (quarter wave) | 17.8 %, −15.0 dB | 0 |

## Terms

- **Crosstalk:** a signal one trace induces on another through their mutual capacitance Cm and mutual inductance Lm.
- **Near end, far end:** the victim's end next to the aggressor's driver, and its end next to the aggressor's receiver.
- **NEXT (near-end, backward):** a plateau at the near end that grows with the coupled length until the length's round trip, 2T, reaches the rise time, then stays: the length saturates it.
- **FEXT (far-end, forward):** a pulse at the far end, as long as the rise time, growing with the coupled length and with faster edges; the capacitive and inductive parts cancel when both modes travel at one speed, as in a stripline.
- **Even and odd modes:** the pair driven alike and driven opposite; any signal on one trace is half of each.
- **Aggressor swing:** the voltage step the aggressor's driver puts on the line: about the logic swing for a stiff CMOS driver, half the open-circuit voltage for a driver matched to the line. The victim's noise is the coefficient times it.
- **Saturation length:** the coupled length whose round trip equals the rise time, tr × v / 2.
- **3w rule:** a common rule of thumb, traces three widths apart to keep crosstalk low; the example takes the edge-to-edge spacing s = 3w.

## Source

- **Yuriy Shlepnev, "How Interconnects Work: Crosstalk Quantification", Signal Integrity Journal, 10 March 2025** (read; its formulas as an equation image on the page): the step-response coefficients of a coupled segment of length l, rise time tr, K_B = l/(2(T1 + T2)) (L21/ζ1 − ζ2 C21) min(1, (T1 + T2)/tr) at the near end, K_F = −l/(2 max(|T1 − T2|, tr)) (L21/ζ1 + ζ2 C21) at the far end, T1, T2 the traces' flight times, ζ1, ζ2 their impedances, L21 and C21 the mutual inductance and capacitance; "relatively accurate for lossless or low loss cases", "an estimate of the maximal possible step crosstalk", and the step "may underestimate the actual peak-to-peak crosstalk for short links by up to 2x". It cites Bracken's DesignCon 2016 paper "Improved Formulas for Crosstalk Coefficients" (not read).
- **The field solver:** js/fieldsolver.js, the microstrip and stripline pairs' odd and even modes, as in Edge-coupled microstrip and Edge-coupled stripline; tools/fd_lines.py gives the values below.

## Why the formulas are these

From the two modes, per length, with ve, vo = c/√εeff,e, c/√εeff,o:

```
Ce = 1/(Ze ve),  Le = Ze/ve,  Co = 1/(Zo vo),  Lo = Zo/vo
C = (Ce + Co)/2,  Cm = (Co − Ce)/2,  L = (Le + Lo)/2,  Lm = (Le − Lo)/2
ζ = √(L/C),   v = 1/√(L C),   T = l / v
```

and, for two equal traces (T1 = T2 = T, ζ1 = ζ2 = ζ, C21 = −Cm, as the capacitance matrix's off-diagonal term is negative):

```
NEXT = l/(4T) (Lm/ζ + ζ Cm) × min(1, 2T/tr) = ¼ (Lm/L + Cm/C) × min(1, 2T/tr)
FEXT = −l/(2 tr) (Lm/ζ − ζ Cm)
saturation length = tr v / 2
```

The even mode sees the pair's capacitance less its mutual part and its inductance plus its mutual part, the odd mode the reverse, so the four mode values give L, Lm, C and Cm exactly. In a stripline, one dielectric all round, Lm/L = Cm/C and FEXT cancels; in a microstrip part of the field is in air, Cm/C is smaller than Lm/L and FEXT is negative.

**Checked:** the stripline's FEXT comes out 0 (Lm/L = Cm/C = 0.1783); the saturated NEXT agrees within 2 % with the near-end voltage the two modes give a matched driver, Ze/(Ze + Z0) − Zo/(Zo + Z0), Z0 = √(Ze Zo) (11.72 against 11.52 %, 3.41 against 3.40 %).

| Case (FR-4, 1 oz, 100 ps) | NEXT | FEXT |
|---|---|---|
| microstrip, 0.15 mm, s 0.15, h 0.2 mm, 50 mm | 11.52 % | −16.00 % |
| the same, s 0.45 mm (3w) | 3.40 % | −9.19 % |
| the same as the first, 10 mm, 1 ns | 1.32 % | −0.32 % |
| stripline, 0.15 mm, s 0.2, b 0.6 mm, 50 mm | 8.92 % | 0 |

## Assumptions and limits

- **A step on matched, lossless lines:** the coefficients are the plateau and the pulse a step gives, the victim terminated in its impedance at both ends; other terminations reflect the noise and change the voltages; with losses, long lines or reflections the real waveform differs, and for short links the peak-to-peak crosstalk of a real signal can be up to twice the step's (Shlepnev).
- **FEXT's growth:** it is linear in the length; once the even and odd edges reach the far end further apart than the rise time, the pulse stops growing, and the tool flags the value as an upper bound.
- **The sine's terminations:** every end in R = ζ; a victim left open or shorted at one end, or a driver of another impedance, changes the standing waves and the voltages.
- **Two equal traces, no other neighbours;** the cross-sections and their limits as in Edge-coupled microstrip and Edge-coupled stripline (no solder mask here).

## How it was checked

10 cases, the expected values from tools/fd_lines.py's modes and the formulas above, the swing 3.3 V unless said: - the default microstrip pair, 0.15 mm traces 0.15 mm apart on 0.2 mm FR-4, 1 oz, 50 mm side by side, 100 ps: NEXT 11.5 % (11.519), FEXT −16 % (−16.003), saturated beyond 8.73 mm, even and odd modes 91.6 and 57.2 Ω, ζ 73.4 Ω, Lm/L 0.286 and Cm/C 0.175, the edges 32.9 ps apart; 380 mV at the near end for 573 ps, −528 mV at the far end; - the same with a 1.8 V aggressor: 207 mV and −288 mV; - the 3w rule, 0.45 mm apart: NEXT 3.4 % (3.397), 112 mV, FEXT −9.19 %, −303 mV; - 10 mm at 1 ns: below saturation (reached at 87.3 mm), NEXT 1.32 %, FEXT −0.32 %; - the stripline pair: NEXT 8.9 % (8.915), FEXT 0; - a zero coupled length is refused; - sine, the 100 MHz clock example: −27.7 dB (136 mV, 4.12 %) at the near end, −40.0 dB (33.1 mV) at the far end, quarter wave at 873 MHz, the run 0.0286 λ; - sine at 1 GHz: −12.9 dB (744 mV, 22.5 %), far end

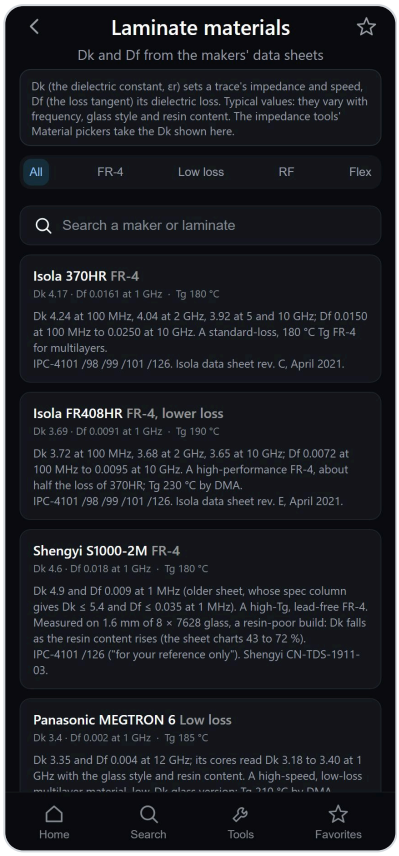
−20.1 dB (325 mV); - sine on the stripline pair at its quarter wave, 711.28 MHz: −15.0 dB (588 mV, 17.8 %), twice the step's 8.9 %, far end 0; - a zero frequency is refused.

# Laminate materials

## What it computes

A reference of real PCB laminates from their makers' data sheets, and the list behind every impedance tool's Material picker (Pierre, 2026-11-08). It takes the place of the planned "Laminate material reference (Dk/Df) — IPC-4101":

- **Cards:** maker and name, the kind (FR-4, low loss, RF, flex), Dk and Df at the frequency the sheet gives, Tg; then how Dk and Df move with frequency, a note, the slash sheet the maker states, and the sheet read.
- **Filters and search:** All, FR-4, Low loss, RF, Flex; search any maker, name or word.
- **In the impedance tools:** each Material picker lists FR-4, polyimide and PTFE at AN-905's round figures (4.3, 4.2, 2.4), then these laminates grouped by maker, then Custom; picking one sets εr to the Dk on its card. Typing εr sets the picker to the laminate with that value, or to Custom.
- **Remembered:** the filter.



| Laminate | Kind | Dk the pickers take | Df | Tg |
|---|---|---|---|---|
| Isola 370HR | FR-4 | 4.17 at 1 GHz | 0.0161 | 180 °C |
| Isola FR408HR | FR-4, lower loss | 3.69 at 1 GHz | 0.0091 | 190 °C |
| Shengyi S1000-2M | FR-4 | 4.6 at 1 GHz | 0.018 | 180 °C |
| Panasonic MEGTRON 6 | Low loss | 3.40 at 1 GHz | 0.002 | 185 °C |
| Rogers RO4003C | RF | 3.55 design (8–40 GHz) | 0.0027 at 10 GHz | > 280 °C |
| Rogers RO4350B | RF | 3.66 design (8–40 GHz) | 0.0037 at 10 GHz | > 280 °C |
| Rogers RT/duroid 5880 | RF, PTFE | 2.20 design (8–40 GHz) | 0.0009 at 10 GHz | — |
| DuPont Pyralux AP | Flex | 3.2 at 10 GHz | 0.003 | 220 °C |

## Terms

- **Dk (dielectric constant, εr, relative permittivity):** how much the material slows a signal and raises a trace's capacitance; it sets the impedance and the delay.
- **Df (dissipation factor, loss tangent, tan δ):** the share of the field's energy the material turns into heat per cycle; it sets the dielectric loss, which grows with frequency.
- **Design Dk:** Rogers' value for circuit design, "an average number from several different tested lots of material", measured on circuits from 8 to 40 GHz (differential phase length); the process Dk (clamped stripline, IPC-TM-650 2.5.5.5) is the one for quality acceptance, and Rogers warns it "may be incorrect for design purposes, especially microstrip designs".
- **Tg (glass transition temperature):** where the resin softens and its expansion rises sharply; DSC and DMA give different figures (both shown where the sheet gives both).
- **Resin content, glass style:** a laminate is woven glass in resin; glass has a higher Dk than the resin, so thin, resin-rich layers (106, 1080) read lower than thick, glass-rich ones (7628). Makers give Dk per construction; the cards give one typical figure.
- **Slash sheet:** the IPC specification sheet a material is qualified to: IPC-4101 for rigid FR-4-type laminates, IPC-4103 for high-frequency ones, IPC-4204 for flexible ones. The cards give the slash sheets the makers state; IPC-4101 itself was not read.
- **Core, prepreg:** the cured, copper-clad laminate, and the uncured sheets that bond the cores; the same resin system usually has both.

## Sources

All typical values, read from the makers' sheets: - **Isola 370HR** data sheet, revision C, last updated 15 April 2021: Tg 180 °C (DSC), Dk 4.24, 4.17, 4.04, 3.92, 3.92 and Df 0.0150, 0.0161, 0.0210, 0.0250, 0.0250 at 100 MHz, 1, 2, 5 and 10 GHz; IPC-4101 /98 /99 /101 /126. - **Isola FR408HR** data sheet, revision E, last updated 15 April 2021 (a copy hosted by Michigan State University; Isola's server refused): Tg 190 °C (DSC), 230 °C (DMA); Dk 3.72, 3.69, 3.68, 3.64, 3.65 and Df 0.0072, 0.0091, 0.0092, 0.0098, 0.0095 at the same frequencies; IPC-4101 /98 /99 /101 /126. - **Shengyi S1000-2M** typical data sheet CN-TDS-1911-03: Tg 180 °C (DSC), 185 °C (DMA), permittivity 4.6 and loss tangent 0.018 at 1 GHz (IPC-TM-650 2.5.5.9) on 1.6 mm of 8 × 7628, IPC-4101/126 "for your reference only"; an older S1000-2M/S1000-2MB sheet: 4.9 and 0.009 at 1 MHz, its spec

column ≤ 5.4 and ≤ 0.035. - **Panasonic MEGTRON 6** laminate R-5775(N), prepreg R-5670(N), data sheet No. 17061630, June 2017, low-Dk glass cloth version: Tg 185 °C (DSC), 210 °C (DMA); Dk 3.40, Df 0.002 at 1 GHz and 3.35, 0.004 at 12 GHz on a 0.75 mm core; Dk 3.18 to 3.40 at 1 GHz across its cores. - **Rogers RO4000 series** data sheet, PUB 92-004, 2022: RO4003C process Dk 3.38 ± 0.05 and RO4350B 3.48 ± 0.05 at 10 GHz, design Dk 3.55 and 3.66; Df 0.0027 and 0.0037 at 10 GHz, 0.0021 and 0.0031 at 2.5 GHz; Tg

> 280 °C; IPC-4103 /10 (RO4003C), /11 and /240 (RO4350B); the design Dk "decreases by about 0.1 as the core thickness decreases from 0.020” to 0.004”". - **Rogers RT/duroid 5870/5880** data sheet, Publication 92-101, 2022: 5880 Dk 2.20 ± 0.02 at 10 GHz, design Dk 2.20 (8 to 40 GHz), Df 0.0004 at 1 MHz and 0.0009 at 10 GHz. - **DuPont Pyralux AP** data sheet EI-10124 (2/25): AP9121 Dk 3.4 and 3.2, Df 0.002 and 0.003 at 1 MHz and 10 GHz; Tg 220 °C (TMA); IPC-4204/11. - **The round figures:** AN-905's Appendix D (see Microstrip): FR-4 and G-10 4.3, polyimide 4.2, PTFE 2.4.

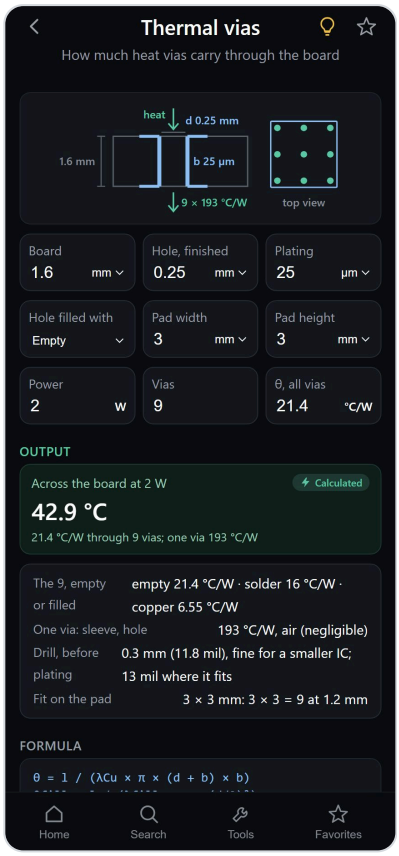
## Assumptions and limits

- **Typical, not guaranteed:** every sheet says so; a stack-up's real Dk comes from the board maker, for the glass styles and resin contents it uses.
- **One Dk per laminate:** the pickers take one figure; Dk changes with frequency (a few % per decade for FR-4, less for the RF laminates) and with the construction.
- **Not every laminate:** a short list of widely used ones; Custom takes any other value.

## How it was checked

5 cases: - RF shows the three Rogers laminates and hides the FR-4s; - searching "megtron" finds MEGTRON 6 alone; - RO4350B's card reads Dk 3.66 · Df 0.0037 at 10 GHz, Tg > 280 °C; - in Microstrip, RO4350B sets εr 3.66: 3 mm on 1.6 mm with 1 oz, 54.4 Ω (tools/fd_lines.py 54.430); - in Microstrip, typing εr 4.17 picks 370HR: 51.4 Ω (51.387).

# Thermal vias




## What it computes

The thermal resistance of plated vias carrying heat from a hot part's pad (an exposed-pad QFN, a regulator's tab, a MOSFET's drain) through the board to the copper on the other side. One screen, from the board thickness, the finished hole, the plating and the fill:

- **Vias and θ are two live boxes:** type the number of vias and the θ box shows their resistance; type the θ you need and the vias box shows the fewest that reach it (rounded up, so the real θ is a little lower, shown when you leave the box). The first version had them on two pills, and the vias box of one had nothing to do with the target of the other; Pierre asked why 9 vias became 20, and the boxes are now linked.
- **The big result** is the temperature difference across the board at the power you give, or the vias' θ when the power is 0.
- **The terms:** l is the board thickness, d the finished hole (after plating), b the copper plating in the hole; the drill is d + 2b.
- **With and without filling:** the hole can be empty (air), filled with solder, or filled with copper; whichever is picked, a row gives the same vias empty, solder-filled and copper-filled side by side.
- **Also:** one via's sleeve and hole resistances, and the temperature difference across the board at a power you give.
- **The drill** (d + 2b), in mil too, against Pierre's rule: a 13 mil drill (0.33 mm, the size AN-1520 recommends), down to 10 mil for a smaller IC; under 10 mil it is flagged with AN-1520's finding below.
- **The pad** (optional): the exposed pad's width and height. A row gives how many vias fit whole on it at AN-1520's 1.2 mm (47 mil) pitch, as columns × rows; it is flagged when you ask for more, and asks for the vias near the pad's edge when you use fewer. Past 16 vias a row says AN-1520 saw little more gain.
- **Units:** metric (board, hole and pad in mm, plating in µm) or mil for all of them; every length has a picker, the pad's two included, and changing any one changes them all.
- **Drawing:** the board in cross-section, the plated sleeve and its pads, the hole shaded when filled, the heat going through; beside it the pad from above, every place a via fits, the vias in use filled green, the outer ones first.
- **Remembered:** the fill and the units.

The result is a θ in °C/W like any other in the thermal chain: it sits between the part's pad and the copper below, as θCS does in Heat sink sizing.

Examples: Fineline's single via, nine of them under a 2 W pad, the same filled with solder, the vias needed for 10 °C/W, a 0.8 mm board, and AN-1520's 7 × 7 mm pad with 0.33 mm vias.

## Source

- **Fineline, "Temperature Management for Printed Circuit Boards", Albert Schweitzer, November 2021** (read): θ = l / (λ × A) with A = π (d + b) b; with b = 25 µm, d = 250 µm, l = 1.6 mm and λCu = 384 W/m·K, "a single thermal via has a thermal resistance of 193 K/W"; the air in the hole (λ 0.026 W/m·K) in parallel gives 192.98 K/W, "negligible"; a via array is the single via's value divided by the number of vias; counting the FR-4 between vias lowers it by "2-3 K/W per via". Its table of conductivities gives copper ≈ 400, solder (SnCuNi) ≈ 64, FR-4 ≈ 0.2 to 0.9 W/m·K.
- **TTM Technologies ANN-8804 rev B, "Power Handling Considerations for SMD Terminations, Attenuators and Resistors"** (read): the board's thermal resistance is the reciprocal sum of the dielectric, the via plating and the via fill, each θ = L / (kA); "typical SAC solder has a thermal conductivity of 57 W/m-K"; copper fill lowers it "by a large factor", most via-fill pastes barely help. Its table 3 (one 0.035" via, 1 mil plating, in 0.1" × 0.1" of board: 27.8 °C/W empty, 13.8 solder, 3.5 copper) does not print the board thickness or the conductivities, so it is not a check case; with a 0.032" board the same model lands within a few percent of it.
- **TI AN-1520, "A Guide to Board Layout for Best Thermal Resistance for Exposed Packages", SNVA183B, September 2006, revised April 2013** (read, with its figures): for the via-diameter study "the maximum number of vias is determined based on the pad size and 1.2 mm distance between vias"; its figure 13 (36-lead LLP, 7 × 7 mm pad, 0.2 mm vias, 4-layer JEDEC board) draws that maximum as a 6 × 6 grid, 36 vias, which is the count of vias whole on the pad at a 1.2 mm pitch; "make the thermal vias near the periphery of the exposed DAP if the maximum number of vias is not applicable"; "use 0.33 mm diameter of vias if possible": 0.2 mm "may increase the θJA by about 15~25%" on a pad holding 4 vias or fewer, "generally less than 10%" with 9 or more. Read off figure 13 (6.4 mm die): θJA about 26.7 °C/W with no vias, 22.5 with 4, 20.6 with 8, 19.0 with 16, 17.8 with 36; the 2.1 mm die runs about 33.6, 29.8, 28.0, 26.2, 24.7. In this revision figure 12's panels (b) and (c), the via diameter and distribution, are a copy of figure 11(b); only the text gives the diameter figures. Its "diameter" is not said to be the drill or the finished hole; the tool compares it with the drill.
- **Pierre's rule:** a 13 mil drill for thermal vias, down to 10 mil for a smaller IC.
- **Not read:** the Micron study of thermal via size and spacing Pierre mentioned (AN-1520 covers the same question); TI SPRADB7, whose via formula did not survive text extraction.

## Why the formulas are these

$$\theta_{sleeve} = l / (\lambda_{Cu} \times \pi\,(d + b)\,b)$$
$$\theta_{hole} = l / (\lambda_{fill} \times \pi\,(d/2)^2)$$
$$\theta_{via} = \theta_{sleeve} \parallel \theta_{hole}, \quad \theta_N = \theta_{via} / N, \quad \Delta T = P \times \theta_N$$

Heat conducts straight down the copper sleeve, a tube of wall b around the hole; its cross-section is the ring π (d + b) b. A filled hole is a second column of material beside it, so the two are in parallel. Vias side by side share the heat, so N of them in parallel divide θ by N.

## Assumptions and limits

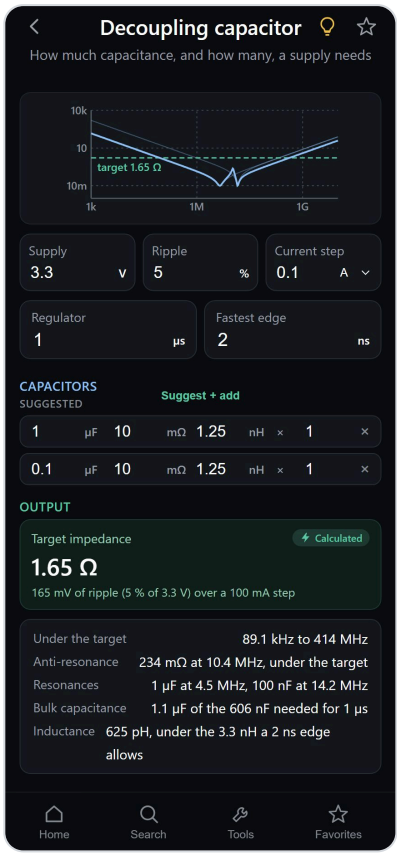
- **Straight conduction only:** the heat is assumed to reach every via evenly and to leave the far side freely. Spreading in the pads and planes, and the solder joint above, add resistance not counted here.
- **The FR-4 between vias is left out,** which slightly overstates θ (Fineline: 2-3 K/W per via).

- **Solder fill** assumes the hole is completely filled; voids and wicking during reflow are why many datasheets ask for filled or tented vias.
- **Copper at 384 W/m·K,** Fineline's figure; plated copper can be a little lower.
- **Size and spacing** follow AN-1520 (1.2 mm pitch, 0.33 mm) and Pierre's 13/10 mil rule; a fab's own minimum drill and annular ring come first.
- **The pad's vias and θJA:** the vias' θ here keeps falling as 1/N, but AN-1520's whole-package θJA does not: from 16 to 36 vias it fell only about 1.2 °C/W, because the spreading in the board and the air beyond then dominate. The vias' θ is one link in that chain.

## How it was checked

20 cases: - Fineline's via: 193 °C/W, air negligible, drill 0.3 mm (11.8 mil, fine for a smaller IC); - nine in parallel: 21.4 °C/W in the θ box, 42.9 °C across the board at 2 W; - the nine empty, solder-filled and copper-filled: 21.4, 16 and 6.55 °C/W; - solder in the hole: 572 °C/W beside the 193 °C/W sleeve, 32.1 °C at 2 W; - 0.3 mm holes are 163 °C/W each, so 17 reach 10 °C/W (9.6); - typing 10 °C/W in the θ box sets the vias to 20 (9.64 °C/W, 19.3 °C); - typing 4 vias sets the θ box to 48.2 °C/W; - a 0.8 mm board halves it, 96.4 °C/W; - mil turns every length to mil (1.6 mm reads 62.99 mil, 25 µm 0.9843 mil), and µm plating brings them back to mm; - a fraction of a via is refused; - the remembered fill; - AN-1520's fit: a 3 × 3 mm pad holds 3 × 3 = 9 vias, a 5 × 2.5 mm pad 4 × 2 = 8, in mil 118 × 118 mil at 47 mil; - AN-1520's 7 × 7 mm pad: 6 × 6 = 36 vias of 0.33 mm (13 mil, the size it recommends), 4.83 °C/W, with the past-16 note; - 4 vias on the 3 × 3 mm pad go near its edge, 12 are too many; - a 0.2 mm drill (7.87 mil) is flagged under 10 mil; - the pad's own picker turns every length to mil; - with the pad emptied the vias still compute.

# Decoupling capacitor




## What it computes

The target-impedance method for a supply's decoupling, with mixed values, on one screen:

- **Inputs:** the supply voltage, the ripple allowed (%), the current step, the regulator's response time and the fastest current edge; then up to three kinds of capacitor, each a row of C (µF), ESR (mΩ), ESL (nH) and how many, added with "+ add" (a decade larger than the largest) and removed with ×. Boards mix values (Pierre), so the tool does.
- **The big result:** the target impedance, the ripple allowed over the current step.
- **Below it:**
- the band where all the capacitors together stay under the target, around their lowest point;
- with more than one kind, the highest anti-resonance peak between the first and last resonances, flagged when over the target;
- each kind's resonant frequency;
- the total capacitance against the bulk the regulator's delay needs;
- the total ESL (all in parallel) against the most inductance the fastest edge allows.
- **Drawing:** the impedance of all the capacitors together from 1 kHz to 10 GHz, log-log, each kind alone faint behind it, the target dashed.
- **Units:** as engineers quote them: V, %, µs, ns, µF, mΩ, nH; the current step in mA or A, its picker remembered.

Choosing the capacitors, after AVX: a bulk capacitance for the low frequencies (the bulk row), a value that resonates where coverage is needed with enough in parallel that their ESRs reach the target, and low ESL for the fast edges (the inductance row); then the curve must stay under the line with no peak above it.

**The rows start as a suggestion and follow the supply's figures** (Pierre changed the ripple from 5 % to 1 % and expected the capacitors to change): the heading says "suggested". Editing a row, adding or removing a kind makes them "yours", and they stay; **Suggest** gives them back. The suggestion follows those rules: the smallest of 1, 2.2, 4.7, 10, 22, 47, 100 µF that covers the bulk (several 100 µF past that); 100 nF parts, as many as ESR / Ztarget; and, only when the total ESL is still over the edge's limit, enough 0612 LICC parts (0.1 µF, 550 pH, AVX's figure 7) to bring it under. ESR is the illustrative 10 mΩ and ESL a 1206's 1.25 nH until replaced. The rules do not promise a flat curve: a peak can remain between the bulk part and the 100 nF, and the tool flags it; an in-between value fills it.

Examples: AVX's 20 A step at 1.8 V, a small MCU at 3.3 V, 10 µF + 100 nF + 1 nF, and an FPGA core at 1 V.

## Source

- **Yun Chase, Kyocera AVX, "Introduction to Choosing MLC Capacitors for Bypass/Decoupling Applications"** (read; no date printed):
- $|Z| = \sqrt{(ESR^2 + (\omega ESL - 1/\omega C)^2)}$ (eq. 4), resonance at $1 / (2\pi \sqrt{(ESL \cdot C)})$ (eq. 8), where the impedance is the ESR;
- "Ztarget = Allowed Ripple Voltage / I" (eq. 13), ripple a small percentage of the supply, "i.e. 5%";
- bulk: C = I dt / dV (eq. 14); "a 20 amp current transient, the VRM responds in 15 µSec, and the PDS must remain within 5% of a 1.8V power supply": 3333 µF (eq. 15);
- inductance: L = V dt / dI (eq. 16); the same with a 1 ns rise: 4.5 pH (eq. 17);
- n identical capacitors in parallel: nC, ESL/n, ESR/n, the resonance unchanged (eq. 18 to 22);
- "a capacitance value is chosen so that when mounted on the PCB, it will resonate at the desired frequency"; different values in parallel "produce an unwanted anti-resonance";
- measured ESL: 1206 about 1250 pH, 0612 LICC about 550 pH, 0612 IDC about 175 pH, 0508 IDC about 110 pH.
- **ROHM, "Impedance Characteristics of Bypass Capacitor", application note 63AN091E rev. 001** (read): use a capacitor below its self-resonance; low ESL for high frequencies; capacitors of very different self-resonance in parallel can make an anti-resonance.

## Why the formulas are these

$$Ztarget = \Delta V / \Delta I, \quad \Delta V = V \times ripple$$
$$C\ bulk = \Delta I \times t\ regulator / \Delta V$$
$$L\ max = \Delta V \times t\ edge / \Delta I$$
$$Z\ kind = (ESR + j\omega ESL + 1/(j\omega C)) / n, \quad Y = \Sigma\ 1 / Z\ kind, \quad |Z| = 1 / |Y|$$

The supply may sag by ΔV when the load steps by ΔI, so its impedance must stay under ΔV / ΔI at every frequency in the step. Until the regulator catches up, capacitors carry the step; their charge sets the bulk. During the fastest edge, inductance limits how fast current can come out of them. Each capacitor is lowest at its resonance; in parallel their complex admittances add, so between one kind's resonance (already inductive) and the next (still capacitive) the two can resonate together: the anti-resonance peak.

## Assumptions and limits

- **The ESR in the rows is illustrative** (10 mΩ by default); neither paper gives one. Type your parts' from their datasheets.
- **ESL** is the capacitor's own; the pads and vias add more (AVX: "this is a mounted inductance").
- **The board's planes** are not modelled; they matter above about 1 GHz (AVX figure 5).
- **MLCC capacitance falls with DC bias and temperature;** type the capacitance at your voltage.

## How it was checked

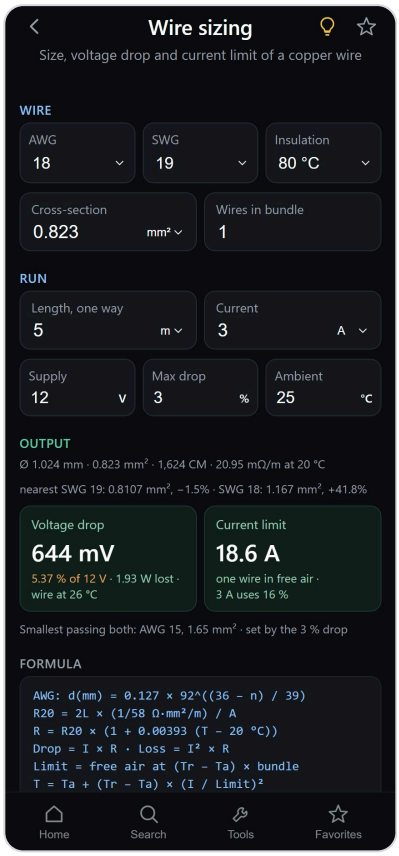
15 cases: - the rows start as the suggestion; 1 % ripple instead of 5 % moves the bulk part from 1 µF to 4.7 µF; an edited row stays when the ripple changes; - AVX's example: 4.5 mΩ, 3333 µF, 4.5 pH; three 100 nF (417 pH) far over; - Suggest: for the MCU, 1 µF and one 100 nF (89.1 kHz to 414 MHz); for AVX's 20 A, 34 × 100

µF, 3 × 100 nF and 106 low-ESL parts (4.5 pH); for the FPGA core, a 93.9 mΩ peak left at 4.14 MHz, flagged; - those three get under 4.5 mΩ only around 14.2 MHz (14.1 to 14.7 MHz); - a small MCU: 1.65 Ω from 962 kHz to 207 MHz, 606 nF of bulk needed; - 10 µF + 2 × 100 nF + 2 × 1 nF: 48.2 kHz to 85.8 MHz, a 4.36 Ω anti-resonance at 136 MHz; - an FPGA core: 50 mΩ, 175 pH over 50 pH; - "+ add" puts 10 µF beside 1 µF: the band widens, its peak stays under; - × removes a kind; - a zero current step is refused; - the current step in mA: 0.1 A reads 100.

PCB & INTERCONNECT › CABLING

# Wire sizing

(formerly Wire gauge (AWG/SWG); Cable resistance / voltage drop and the planned Cable current capacity are merged into it)




## What it computes

One copper wire on one run, on one screen:

- **Its size** on every scale: an AWG gauge, an SWG gauge and a size in mm² or mm are linked boxes, and setting any one updates the others. A gauge picked by hand is exact; the others show their nearest gauge, the one on the other side and how much more or less copper each has. The line under Output gives the diameter, the cross-section, the circular mils and the resistance per metre (or foot) at 20 °C.
- **Its voltage drop** over the run, large: in volts, as a share of the supply, the power turned to heat in the wire, and the temperature the wire runs at.
- **Its current limit**, large: the most current it can carry before it reaches its insulation's rating, alone in free air or in a bundle, and how much of that the current uses.
- **The smallest wire that passes both**, on the scale in use, and which check decided it.

The inputs are the gauge, the insulation's temperature rating (60 to 260 °C), the number of wires in the bundle, the one-way length, the current, the supply, the largest drop allowed and the ambient temperature. The mm²/mm, m/ft and mA/A choices and the insulation rating are remembered.

Examples: a 12 V LED strip 5 m away (the drop decides), 5 V at 3 A over 1 m, a short 20 A motor lead (the current decides), a harness of 12 wires, 200 °C silicone wire for a drone, and 1.5 mm² cable.

## Sources

**NBS Handbook 100, Copper Wire Tables** (US National Bureau of Standards, 1966), public domain, read in review:

- **AWG:** "the diameter of No. 0000 is defined as 0.4600 inch and of No. 36 as 0.0050 inch", with 38 sizes between in geometric progression.
- **SWG:** table 15, every size from 7/0 (500 mils) to 50 (1 mil), held here in mils and converted exactly.
- **Copper:** the International Annealed Copper Standard is "1/58 = 0.017241 … ohm square millimetre per metre" at 20 °C, and the best temperature coefficient to assume for commercial annealed copper wire is α20 = 0.003 93 per °C (the IEC values of 1913).

**FAA AC 43.13-1B, Change 1** (2001), chapter 11, section 5, "Electrical wire rating", public domain, read in review:

- **Figures 11-4a and 11-4b, "Single copper wire in free air":** current against the temperature difference "wire rating minus the ambient", one straight line per size on log-log paper, sizes 26 to 4/0.
- **Figure 11-5, "Bundle derating curves":** the factor on each wire's free-air current, for 1 to 41 wires, by the share of the bundle's capacity in use (20 to 100 %).
- **Paragraph 11-66d(6):** the wire's temperature, T2 = T1 + (TR − T1)(I2 / Imax)², and (4) the copper resistance correction (234.5 °C + T).
- **Table 11-9:** in-harness ratings at 70 °C, 33 wires, 20 % load and 60,000 ft, used here only as a cross-check.

**MIL-W-5088L** (1991, inactive; superseded by SAE AS50881), the military wiring specification the FAA charts come from, read in review: figure 3 is the same free-air chart (its large-size half was read from here, its scan being cleaner), figure 4 the bundle curves; 3.8.8.1.1 states that table I was derived from these figures, and 3.8.8.1.2 that connector contacts, relays, breakers and switches carry their own, often lower, ratings.

## How the charts were read

The figures are scanned images, so each line was traced off the scan and located against the chart's own labelled gridlines, tracked across the page so a skewed or unevenly stretched scan does not shift the values. Each wire line is straight on the log-log paper, so it is kept as its two ends, the current at a 30 °C and at a 240 °C margin; the stored pairs match the traced lines within 0.2 %. For the bundle, the lowest curve (100 % loading) was traced and smoothed, moving no point by more than 0.8 %.

Checked against every figure the documents quote: 22 AWG at a 140 °C margin reads 16.2 A (FAA and MIL: 16.2), 20 AWG 21.4 (21.5), 12 AWG at 175 °C 67.3 (68) and at 140 °C 60.5 (61), 14 AWG at 150 °C 46.9 (47); the bundle factor for 12 wires at full load reads 0.425 (0.43). MIL-W-5088L's table I, built from free air × bundle × altitude, implies the same bundle factor for all nine sizes from 8 AWG to 4/0 to within 1 %, which checks the large sizes.

## Why the formulas are these

```
AWG:  d(mm) = 0.127 × 92^((36 − n) / 39)
R20 = 2L × (1/58 Ω·mm²/m) / A
R   = R20 × (1 + 0.00393 (T − 20 °C))
Drop = I × R        Loss = I² × R
Limit = free-air current at (Tr − Ta) × bundle factor
T = Ta + (Tr − Ta) × (I / Limit)²
```

- **2L:** the current goes out on one wire and back on the other, so the run's resistance counts twice its one-way length.
- **1/58 Ω·mm²/m:** the standard copper of wire tables. The textbook figure for pure copper, $1.68 \times 10^{-8}$ Ω·m, which the old cable tool used, is 2.6 % lower than the wire you buy, so it under-stated every drop.
- **The limit** is the current that heats the wire to its insulation's rating Tr when the air is at Ta: the chart is read at that margin. A size between two chart sizes (an odd AWG, an SWG, a metric size) is read between them, by cross-section on a log scale.

- **The temperature:** heating grows as the square of the current, reaching Tr at the limit; copper's resistance rises with it, so the drop is worked out at the temperature the wire really runs at. The FAA calls the estimate "quite conservative": it errs warm.
- **Smallest wire:** a thicker wire passes whatever a thinner one passes, so the answer is the thicker of the thinnest passing the drop and the thinnest passing the current.

## Assumptions and limits

- **Copper only.** The charts are for copper, and electronics wire is copper; aluminium was dropped in the merge.
- **26 AWG to 4/0** (0.129 to 107 $mm^2$). Outside, no current limit is shown, and the smallest-wire search stops at 26 AWG, saying so, with the size the drop alone would allow.
- **A 30 to 245 °C margin**, where the chart's lines run; outside it the line is extended, and the screen says so.
- **Every wire in the bundle carries current** (the 100 % curve, the lowest). If only some do, the chart allows more; count every wire in the bundle, loaded or not, for a safe figure. Past 41 wires, the chart's last factor.
- **Free air at sea level.** Wire in conduit, against a hot surface, or at altitude (the FAA's figure 11-6: 0.91 at 20,000 ft) carries less.
- **Aircraft wire.** The charts were drawn for MIL wire with thin PTFE or polyimide insulation; PVC hookup wire's thicker insulation changes the figures somewhat. Treat the limit as an engineering estimate, not a rating.
- **The wire, not its ends.** Connectors, terminals, switches and fuses usually limit the current first.
- **Solid round wire sizes.** A stranded conductor is sized by its copper's total cross-section.

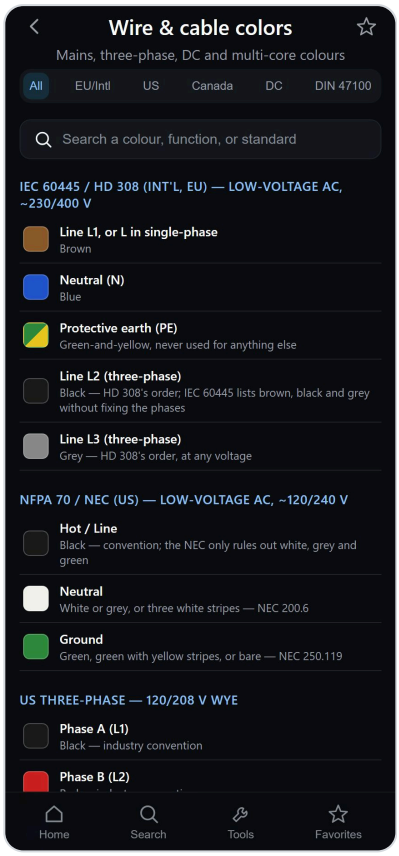
## How it was checked

18 AWG with 3 A over 5 m at 12 V: 644 mV, 5.37 %, wire at 26 °C, limit 18.6 A in 25 °C air with 80 °C insulation, smallest AWG 15 by the 3 % drop. The chart values above; 12 bundled wires ×0.425; 30 AWG has no limit; a 100 mA load stops at 26 AWG (drop alone: 30 AWG); an ambient at the rating gives no safe current; a 20 °C margin and 50 wires are flagged. Sizes: SWG 2/0 is 8.839 mm and 121,104 CM, nearest AWG 2/0; 4/0 AWG is 11.68 mm and 211,600 CM; 1.5 $mm^2$ sits at AWG 15.41; a 1 mm diameter is AWG 18 and SWG 19.

The review made these changes: - Wire gauge, Cable resistance / voltage drop and the planned Cable current capacity became one tool, on Pierre's choice, named Wire sizing. - The current limit is new, from the FAA and MIL charts; the wire's temperature now sets its resistance. - Copper's resistivity is the IACS 1/58 $\Omega \cdot mm^2/m$, not $1.68 \times 10^{-8}$ $\Omega \cdot m$. - Lengths can be in feet; aluminium was dropped. - It added six run examples, 17 cases and this section.

PCB & INTERCONNECT › CABLING

# Wire & cable colors




## What it shows

A searchable list of what each wire colour means, grouped by system:

- **IEC / HD 308** (international, EU): low-voltage AC, single- and three-phase.
- **NEC** (US): single-phase 120/240 V, and three-phase by voltage system: 120/208 V wye, 277/480 V wye and 120/240 V high-leg delta.
- **Canada** (CSA C22.1).
- **DC**: positive, negative, a grounded conductor, and −48 V telecom.
- **DIN 47100**: how the cores of a multi-core cable are numbered by colour, cores 1 to 60.

Each row says whether a colour is required by the code or only a convention. Filter chips jump to one group (remembered); the search box matches a colour, a function or a standard.

## Sources, and what was actually read

- **DIN 47100 and HD 308:** Lapp's selection table T9 (appendix of its catalogue), read in review. Its harmonised-core table for VDE 0293-308 / HD 308 S2 gives the cores of a multi-core power cable as GNYE/BN/BU, GNYE/BN/BK/GY and GNYE/BU/BN/BK/GY: green-and-yellow earth, brown, black and grey lines, blue neutral. Its DIN 47100 table lists cores 1 to 60. All 44 of the standard's colours match the tool's. Lapp notes that its 45 to 60 are "in contrast to DIN: without colour repetition after the 44th core": DIN 47100 itself (January 1988, withdrawn 1998 but still widely used) starts again at 1 from core 45. The tool now says that 45 to 60 are a makers' extension.
- **IEC 60445:** the standard itself could not be read; only the contents pages of its previews were available. Per a reader quoting BS EN IEC 60445:2021 on the IET's Engineering Exchange, clause 6.2.3 lists BLACK, BROWN and GREY for AC line conductors in alphabetical order, and this "does not indicate any preferred phasing". The order L1 brown, L2 black, L3 grey comes from HD 308 S2 (and BS 7671's table 51), so the tool now attributes it there.
- **NEC (NFPA 70):** the code text could not be read. The rows rest on the sections as trade sources quote them: 200.6 (a grounded conductor is white or grey, or three white stripes), 250.119 (an equipment grounding conductor is green, green with yellow stripes, or bare), 110.15 (the high leg is "durably and permanently marked by an outer finish that is orange in color" or by other effective means), and Southwire's technical note that white and green may not be used on ungrounded conductors.
- **CSA C22.1:** the code could not be read. Rule 4-038, "Identification of insulated conductors", is named in Electrical Industry News' guide to the code; the colours (red, black, blue phases; white or grey neutral) are as a forum post quotes the rule.

## Assumptions and limits

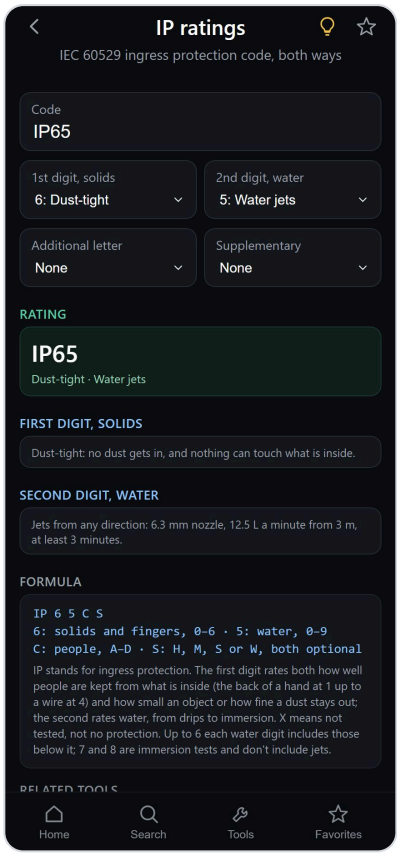
- **Not a substitute for the code in force.** Local amendments differ: San Francisco, for one, asks for purple on the high leg. Old installations use older colours (the UK's red, yellow and blue before 2004).
- **Always test.** A colour says what a wire should be, not what it is.
- **DC is mostly convention.** IEC 60445 has had colours for DC line conductors since 2017, but they could not be read, so they are not listed.

## How it was checked

7 cases: the list opens on IEC/HD 308 with L1 brown; the US hot's note; the DC filter; the high leg's NEC 110.15 note; DIN core 45 as an extension; a search with no match; the remembered filter.

The review made these changes: - The NEC rows had said "required by NEC" for things the code does not require. A white-only neutral, a grey 277/480 V neutral and an orange-only high leg were all marked as required. They now give what the code allows and say where a colour is convention. - The IEC phase order is attributed to HD 308, not to IEC 60445. - DIN 47100's cores 45 to 60 are labelled a makers' extension. - A DC filter chip (DC rows had no filter), the remembered filter, a new subtitle, the first cases and this section.

# IP ratings




## What it shows

The meaning of an IEC 60529 ingress protection code, both ways. You can type a code (IP67, IPX4, IP20C, IP54W), or build one with four pickers:

- the first digit, solids;
- the second digit, water;
- the optional additional letter;
- the optional supplementary letter.

The code shows large, with a two-part summary. Each part's meaning follows, water with its test conditions. Notes flag the cases people get wrong.

Examples: IP20 terminal blocks, an IP65 outdoor box, IP67 against IP66, an IP68 phone, an IP54 enclosure, and IP69K washdown.

## The code

```
IP 6 5 C S
   | | | └ supplementary letter, optional: H, M, S or W
   | | └── additional letter, optional: A, B, C or D
   | └──── water, 0–9
   └────── people and solid objects, 0–6
```

- **First digit:** two meanings at once. One is how well people are kept from hazardous parts: the back of a hand at 1, a finger at 2, a tool at 3, a wire at 4 and above. The other is how small an object stays out: 50, 12.5, 2.5 and 1 mm, then dust protected at 5 and dust-tight at 6.
- **Second digit:** water, from vertical drips (1) to powerful jets (6). Then come immersion up to 1 m (7), immersion beyond that on the maker's terms (8), and close-range hot high-pressure jets (9).
- **X** in either place means not tested, which is not the same as 0.
- **Additional letter:** people are kept from hazardous parts better than the first digit says. It is used when the first digit is low or X.
- **Supplementary letter:**
- H: high-voltage equipment;
- M: water-tested with moving parts running;
- S: water-tested with them at rest;
- W: for stated weather conditions.

## Sources, and what was actually read

**IEC 60529:1989 + A1:1999 + A2:2013**, consolidated edition 2.2. Its preview (VDE Verlag) was read in review, but it stops after the definitions. It confirms the structure:

- separate tables for people and for objects under the first digit;
- the additional letter (table 4) and the supplementary letters (clause 8);
- test 14.2.7, temporary immersion between 0.15 and 1 m;
- test 14.2.9 for second digit 9 with a spray nozzle, and its IPX9 test device. Digit 9 is IEC's own since amendment 2.

The tables themselves could not be read. The wording and test figures follow Wikipedia's IP code article, which reproduces them from the standard:

- 1 mm of rain a minute for 1;
- 3 mm a minute, 2.5 minutes each way for 2;
- 10 L a minute for 3;
- a 6.3 mm nozzle at 12.5 L a minute for 5;
- a 12.5 mm nozzle at 100 L a minute for 6;
- 30 minutes at 1 m for 7;
- 80 °C water at 8–10 MPa and 14–16 L a minute for 9.

The same article is the source for the point that water ratings are "not cumulative beyond IPX6" and are dual-coded, as in IPX5/IPX7. NEMA's complimentary copy of ANSI/IEC 60529 now redirects to a store, and two other manufacturer guides could not be fetched.

## Assumptions and limits

- **K codes are not IEC.** IP69K and IP6K9K come from ISO 20653 (formerly DIN 40050-9), for road vehicles. The tool reads them and says so, rather than putting K on IEC's scale.
- **One supplementary letter.** The standard allows several, in alphabetical order; codes with more than one are rare.
- **A rating is for the enclosure as tested.** Cable glands, seals and covers must be fitted as the maker says, and ageing seals lose it.
- **Not NEMA types.** The US NEMA 250 enclosure types (1, 3R, 4X…) test more than IP (corrosion, icing) and don't map one-to-one.

## How it was checked

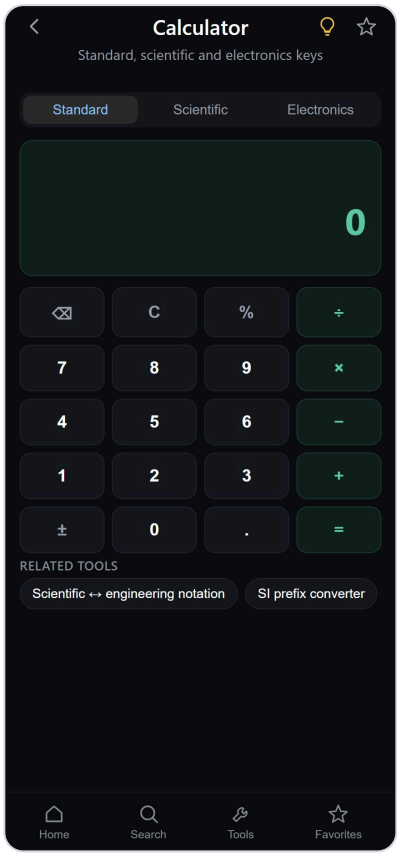
9 cases: - IP65 by default, with its jet test; - typing ipx7 sets X and 7, with the warning that immersion excludes jets; - IP69K explained as ISO 20653; - digit 9 as IEC's hot jet test; - IP20C's additional letter; - IP40A's letter adds nothing; - W from its picker; - a malformed code; - the IP20 example.

The review made these changes: - A code box: type the code instead of picking its digits. It also reads the K codes, so they can be explained. - The additional and supplementary letters, which the tool lacked. - Test conditions in each water description. - The warning that 7 and 8 don't include 5 and 6. - The first digit's two meanings, people and objects. - "Heavy seas" dropped from 6: it is the old wording, not the current standard's. - Six examples, the first cases and this section.

TOOLS › CALCULATORS

# Calculator

(formerly Standard & Scientific Calculators)




## What it does

A calculator with three keypads, chosen by pills (the choice is remembered):

- **Standard:** the iPhone Calculator's own keys, ⌫ C % ÷ … ± 0 . =.
- **Scientific:** ( ) $x^2$ $x^y$ $e^x$ $10^x$ 1/x √ ln log sin cos tan e π and DEG/RAD (remembered), above the standard keys.
- **Electronics:** ( ), ‖ for parts in parallel, **EE** for a power of ten, and the prefixes **p n µ m k M G**, plus 1/x.

Keys build an expression, worked out on =. A result away from 1 to 1000 is also shown in engineering notation under the display: 0.001 reads "= 1 m", 1591.549431 reads "= 1.59155 k". A computer keyboard works too: digits, + − * / ^, | for ‖, E, and p n u m k M G for the prefixes.

Examples: 1k ‖ 2.2k, an RC time constant, a capacitor's reactance, a gain in decibels, the power in a resistor.

## How an expression is read

- **Precedence**, loosest first: + and −; then ×, ÷ and ‖ (and implicit multiplication, "2π" or "2(3+4)"); then a leading minus; then ^, from the right. So 6 + 3 × 2 is 12, 1k ‖ 2.2k + 100 is 787.5, and $-2^2$ is −4.
- ‖ is a × b / (a + b): two resistors in parallel, or two capacitors in series.
- **A prefix** straight after a number scales it: 4.7k is 4700, 100n is 1E-7, and u is accepted for µ. Capital M is mega, small m milli.
- **E** after a number is a power of ten (1E-3). Lower-case e stays Euler's number, so a result like 1E-18 reads back as a number.
- **%** works as on a phone: after + or −, it is a percentage of what comes before (50 + 10 % is 55); after × or ÷, or alone, the number over 100.
- **±** changes the number being typed, not the whole expression.
- **Degrees** are reduced to a turn before sin, cos and tan, so the exact angles come out exact: sin 180° is 0, and tan 90° has no value (Error).

## Assumptions and limits

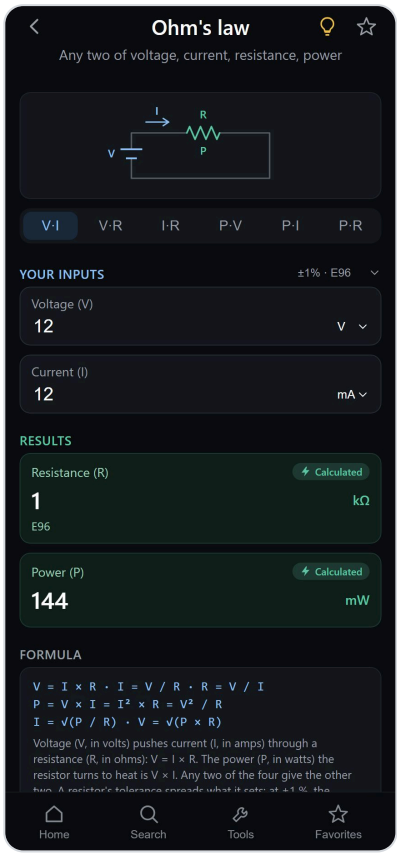
- **Ten significant figures** are shown. Past 1E12 and below 1E-9 a result is written with E.
- **Engineering notation** stops at f and P: smaller or larger results get no second line.
- **Floating point:** results carry the usual rounding of double-precision arithmetic past about 15 digits.

## How it was checked

12 cases: 6 + 3 × 2, $-2^2$, sin 180°, tan 90°, 50 + 10 %, 5 − (−3) by ±, the ‖, RC and reactance examples (with their engineering lines), 1E3, 1n × 1n read back and multiplied by 1E18, and the remembered keypad.

The review made these changes: - **Renamed Calculator.** The title had wrapped. - **The Electronics keypad**, prefixes typed after numbers, the EE power of ten, ‖, and the engineering line, on Pierre's choice. - **$-2^2$** had been 4, because the minus bound tighter than ^. - **sin 180°** had read 1.22E-16 and tan 90° a huge number. - **%** had divided the whole expression by 100: 50 + 10 % gave 0.6. - **±** had negated the whole expression. - **A result in E notation** had used a small e, which read back as Euler's number. - **At 375×812** the Scientific keypad ran 54 px under the tab bar. With a second keypad shown, keys and display are now tighter. - **Remembering:** the keypad and DEG/RAD are now remembered. - **First cases, examples and this section.**

# Ohm's law




## What it computes

Any two of voltage (V), current (I), resistance (R) and power (P) give the other two. These are the six pairs of the Ohm's-law wheel: V·I, V·R, I·R, P·V, P·I and P·R. A schematic marks each quantity where it is: V across the source, I along the wire, R and P on the resistor. Inputs are drawn blue, results green.

- **A resistance you enter** has a tolerance (picked beside "Your inputs"). Its E-series slider steps through standard values, and the results show their spread at that tolerance.
- **A resistance worked out** shows the nearest standard part instead.
- **Remembered:** the pair, the tolerance and each quantity's unit.

Examples: 330 Ω on 3.3 V, a 0.1 Ω current shunt at 2 A, the limits of a ¼ W 1 kΩ resistor, 60 W at 120 V, and a resistance found from a measured voltage and current.

## Why the formulas are these

$$V = I \times R \qquad I = V / R \qquad R = V / I$$
$$P = V \times I = I^2 \times R = V^2 / R$$
$$I = \sqrt{P / R} \qquad V = \sqrt{P \times R}$$

Ohm's law, V = I × R, ties the first three. Power is the rate the resistor turns energy into heat, V × I. Putting one into the other gives the rest.

**Tolerance:** the result moves with R as its own power of R:

| Pair | Result goes as | So at ±t |
|---|---|---|
| V·R | I and P go as 1/R | between V/(R(1 + t)) and V/(R(1 − t)) |
| I·R | V and P go as R | ±t |
| P·R | I goes as 1/√R, V as √R | about ±t/2 |

## Assumptions and limits

- **A resistor obeys Ohm's law** only at a fixed temperature. A lamp filament's cold resistance is many times lower than the $V^2/P$ "working" value, and diodes and LEDs don't follow V = I × R at all.
- **DC, or AC in RMS values with a purely resistive load.** With reactance, use impedance instead.

- **A spread is the tolerance alone.** A resistor's temperature coefficient adds to it.

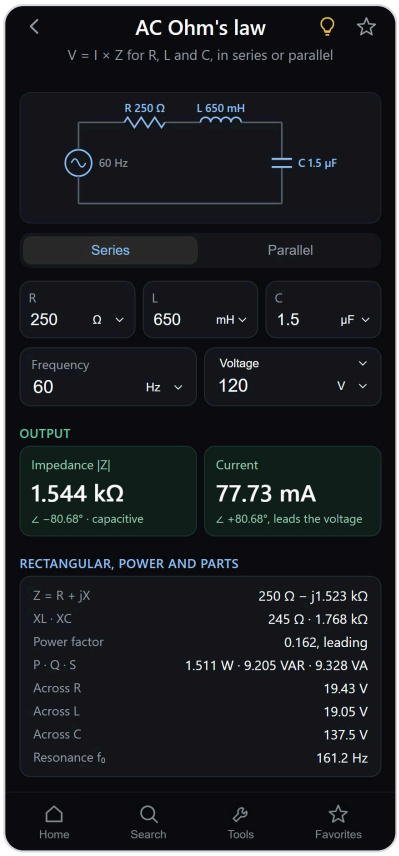
## How it was checked

9 cases: - 12 V and 12 mA: 1 kΩ and 144 mW; - 330 Ω on 3.3 V: 10 mA, between 9.901 and 10.1 mA at ±1 %; - ¼ W in 1 kΩ: 15.81 V or 15.81 mA; - 60 W at 120 V: 500 mA and 240 Ω, nearest E96 237 Ω; - 4.98 V and 2.2 mA: 2.264 kΩ, nearest 2.26 kΩ; - P·I, 1 W at 100 mA: 10 V and 100 Ω; - 12 V shown in mV reads 12000; - a blank field; - the remembered pair, tolerance and unit.

The review made these changes: - **Units:** changing one kept the number and changed its meaning (12 V became 12 mV). Values are now held in SI, and the picker only re-expresses them. - **Power as an input:** the P·V, P·I and P·R pairs are new, with the tolerance spread through P·R. - **Results** pick their own unit: 144 mW, not 0.144 W. Large values had read like 1.20e+3. - **A missing input** says what is missing. - **Remembering:** the pair, tolerance and units are remembered. - **Formula search** had no Ohm's law card; it has one now. - **Examples, the first cases and this section.**

# AC Ohm's law

(formerly AC Ohm's law (impedance form))




## What it computes

Ohm's law for alternating current, V = I × Z, for a resistor, an inductor and a capacitor in series or in parallel, by pill.

- **Inputs:** R, L and C with their units, the frequency, and the source, given as a voltage or as a current (rms). A part at 0, or left empty, is not fitted: a plain wire in series, an open branch in parallel. So the same screen covers R-L, R-C, L-C or a single part.
- **Output, large:**
- the impedance |Z|, with its phase angle and whether the circuit is inductive, capacitive or resistive;
- the current (or the voltage, when the current was given), with its phase against the source and whether it leads or lags.
- **Below:**
- Z in rectangular form, R + jX;
- the reactances XL and XC;
- the power factor, leading or lagging;
- true power P, reactive power Q and apparent power S;
- the voltage across each part in series, or the current through each branch in parallel;
- the resonant frequency $f_0$ when both L and C are fitted.
- **Drawing:** the source with the parts round the loop or in branches, a part not fitted shown as a wire or left out.
- **Remembered:** the pill, whether a voltage or a current is given, and every unit.

Examples: Kuphaldt's series and parallel R, L and C, the power factor of an R-L load, series resonance, and a capacitor alone.

## Source

**Tony R. Kuphaldt, Lessons in Electric Circuits, Volume II, AC, 6th edition (2021)**, the free textbook (ibiblio.org/kuphaldt), read in review: - **Reactance:** XL = 2πfL and XC = 1/(2πfC). An inductive reactance is a positive imaginary impedance (+90°), a capacitive one negative (−90°). - **Combining:** series impedances add, Ztotal = ZR + ZL + ZC; parallel ones add as reciprocals, Ztotal = 1/(1/ZR + 1/ZL + 1/ZC). - **Power, chapter 11:** - true power P in watts; - reactive power Q in "Volt-Amps-Reactive (VAR)"; - apparent power S in volt-amps (VA); - P, Q and S form the power triangle, whose angle is the impedance's phase angle; - the power factor is P / S, "equal to the cosine of that phase angle". - **Its examples**, the tool's cases: - **series:** 250 Ω, 650 mH and 1.5 µF at 120 V, 60 Hz give 250 − j1.5233 kΩ, 1.5437 kΩ ∠ −80.680°, and 77.734 mA; - **parallel:** the same parts draw 480, 489.71 and 67.858 mA by branch, 639 mA in all at −41.31°; - **power:** 60 Ω with 160 mH on 120 V, 60 Hz give 119.365 W, 119.998 VAR and 169.256 VA, a power factor of 0.705.

## Why the formulas are these

```
XL = 2πfL,  XC = 1 / (2πfC)
series:   Z = R + j(XL − XC)
parallel: 1/Z = 1/R − j/XL + j/XC
|Z| = √(R² + X²),  ϕ = atan(X / R),  I = V / Z
S = V × I,  P = S cos ϕ,  Q = S sin ϕ,  PF = P / S = cos ϕ
f₀ = 1 / (2π √(LC))
```

An inductor's opposition grows with frequency and a capacitor's shrinks, and they shift the current in opposite directions, so in a circuit they partly cancel. When XL = XC, at $f_0$, they cancel entirely. A series circuit is then only its R, and the voltage across L and across C can be many times the source's. A parallel L and C then draw no current between them from the source. A positive angle is inductive: the current lags the voltage, and the power factor is lagging. A negative one is capacitive and leading.

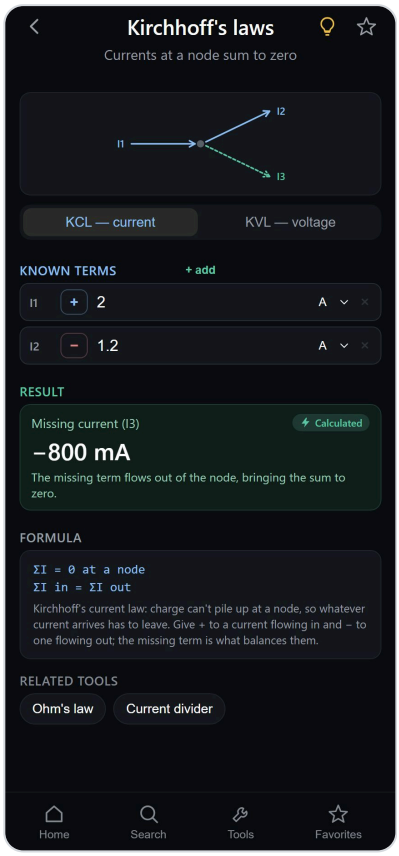
## Assumptions and limits

- **Ideal parts:** an inductor with no resistance of its own, a capacitor with no loss, as in the book's examples. A real coil's resistance can be typed into R in series.
- **Sine waves only,** at the one frequency. Distorted waveforms carry harmonics that see other reactances.
- **Values are rms**, the source taken as the 0° reference.
- **At exact resonance** with no R, a series L-C is a short and a parallel one an open circuit; the tool says so instead of dividing by zero.

## How it was checked

10 cases: - the book's series circuit: 1.544 kΩ at −80.68°, 77.73 mA, 250 − j1.523 kΩ; - the book's parallel circuit: 639 mA at −41.31°, branches 480, 489.7 and 67.86 mA; - the book's R-L load: power factor 0.705 lagging, 119.4 W, 120 VAR, 169.3 VA; - series resonance: 1 Ω, 0°, 10 V across L and across C, $f_0$ 159.2 kHz; - a capacitor alone: −90°, no true power; - given 1 A, the series circuit needs 1.544 kV; - R and L alone (C at 0): 350.1 Ω at +44.43°; - a unit change keeps the value; - nothing fitted is refused; - the remembered choices.

# Kirchhoff's laws




## What it computes

The one missing term that balances a node or a loop. Enter two to five known terms, each with a sign and a unit, and the tool gives the term that brings the signed sum to zero, with what it means.

- **KCL (current):** + is a current flowing into the node, − one flowing out. The diagram draws each term as an arrow into or out of the node.
- **KVL (voltage):** + is a rise, crossing a source from − to +, and − a drop, across a resistor in the direction of the current. The diagram is a real loop walked clockwise: each rise is a battery, met minus first; each drop is a resistor. They read in walking order, along the top and back along the bottom.

Known terms are drawn in the input blue and the missing one dashed in the result green. Each term is held in A or V, and its unit picker only changes how it is shown. The law chosen is remembered.

Examples: a node with three branches, a supply current split in mixed units, a 9 V battery with an LED and its resistor, two sources and two drops in one loop.

## Why the formula is this

```
KCL:  ΣI = 0 at a node      (Σ in = Σ out)
KVL:  ΣV = 0 around a loop  (Σ rises = Σ drops)
```

Charge can't build up at a node, so what flows in flows out. Walking a loop back to where it started returns to the same voltage, so the rises and the drops cancel. Either way the missing term is minus the sum of the others.

## Assumptions and limits

- **Lumped circuits:** the laws hold where wires are short against the wavelength and no changing magnetic field threads the loop. At RF, or around a transformer's core, a "loop voltage" is induced.
- **Signs are yours:** the tool can't see the circuit, so a wrong sign on one term gives a wrong answer with no warning.
- **Two to five known terms**, plus the one solved for.

## How it was checked

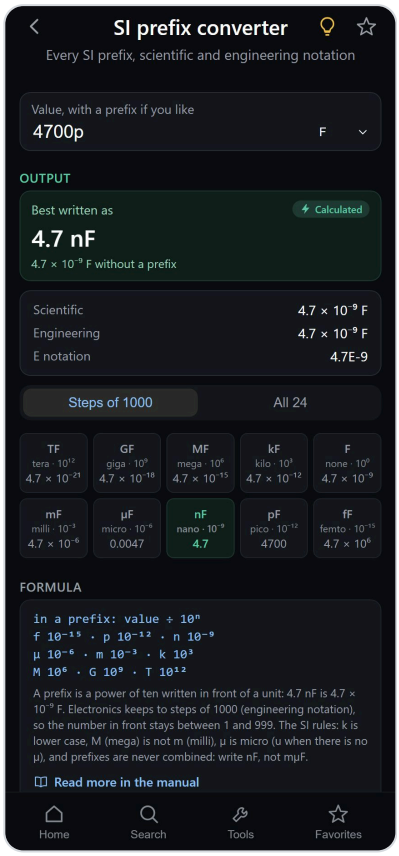
10 cases: - 2 A in and 1.2 A out leave 800 mA flowing out; - the three-branch node gives 1.5 A out; - mixed units give 30.5 mA; - 9 V minus a 2 V LED leaves 7 V for the resistor; - two rises and two drops leave 4.7 V; - a unit change keeps the value; - a flipped sign; - a blank term; - balanced terms give zero; - the remembered law.

The review made these changes: - **Units:** changing one kept the number and changed its meaning. Terms are now held in SI. - **The KVL diagram** had been ticks on a bare rectangle. It is now a real loop of batteries and resistors, with the walking direction. - **Colours:** the diagram had given each term its own colour, red among them. Known terms are now blue and the missing one green, as on every screen. Labels went from 10 px to 11 px. - **Text:** the detail line says when the known terms already balance, and what a rise or drop is like. - **The rest:** the remembered law, examples, the first cases and this section.

TOOLS › CALCULATORS

# SI prefix converter

(Scientific ↔ engineering notation was merged into it)




## What it does

It writes one value in every SI prefix:

- **The input:** a value typed as it is marked, 4700p, 0.1u, 2.2M, 32.768k or 1.5E-3, with u accepted for µ, and a unit picker (none, Ω, F, H, V, A, W, Hz, s).
- **The output:** the value large in its best prefix, the one leaving 1 to 999 in front (4700 pF is 4.7 nF), and its plain value without a prefix.
- **Notation:** the same value in scientific notation (one digit before the point, $4.7 \times 10^4$), engineering notation (an exponent in steps of 3, the one a prefix names: $47 \times 10^3$), and E notation as calculators, spreadsheets and code read it (4.7E4).
- **The grid:** the value in each prefix. By default it shows the steps of 1000 from femto to tera, the ones electronics uses. The All 24 pill adds the rest, from quecto to quetta.

Values past a million or under a thousandth are written as $\times 10^n$. The unit and the grid choice are remembered.

Examples: 4700 pF, 0.1 µF decoupling, 2.2 MΩ, a 32.768 kHz crystal and a crystal's 3.4 fF motional capacitance.

## Source

**BIPM, The International System of Units (SI Brochure), 9th edition**, as updated in 2022, read in review:

- **Table 7** lists the 24 prefixes, $10^{30}$ quetta (Q) to $10^{-30}$ quecto (q).
- **The 27th CGPM (2022)** added ronna, ronto, quetta and quecto.
- **The rules:** prefix symbols are upright and attached to the unit without a space. All multiple prefixes are upper-case except da, h and k, and all sub-multiples are lower-case. A prefix plus a unit symbol forms one new, inseparable symbol.

## Why the formula is this

```
value in a prefix = value / 10ⁿ   (n the prefix's power)
```

Every prefix scales the same base quantity, so one value converts to all of them by dividing by each prefix's factor. The best prefix is the power of 1000 that leaves 1 to 999 in front.

## Assumptions and limits

- **Only SI prefixes.** The binary ones for bytes (Ki, Mi) belong to IEC 80000-13, not the SI, and are not here.
- **"u" for µ** is the common stand-in when no µ is to hand. It is not SI, so the result always writes µ.
- **The best prefix stays within femto to tera**, the range electronics uses.

## How it was checked

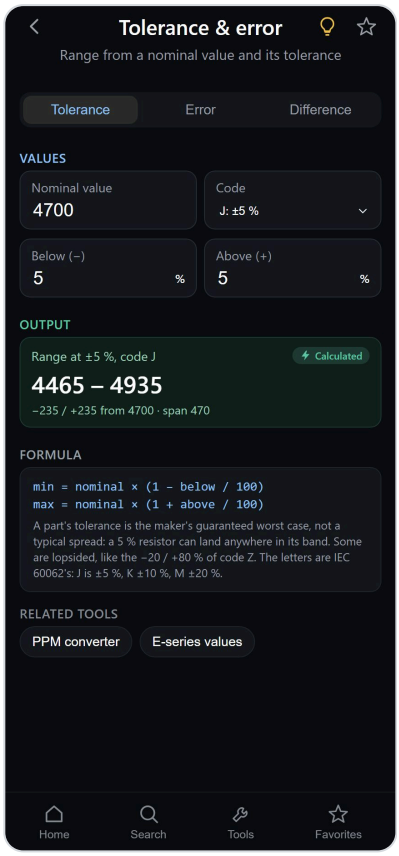
11 cases: - 4700p F is 4.7 nF and 0.0047 µF; - 2.2u reads as µ; - 3da is 30, deca, not deci followed by atto; - 1.5E-3 is 1.5 mF; - 2.2M Ω is 2200 kΩ; - 3.4 fF is 0.0034 pF; - All 24 includes q and Q; - a malformed value is explained; - the remembered unit and grid choice; - 47k in scientific, engineering and E notation; - 999.9999999 rounds up to $1 \times 10^{3}$ and 1 k.

The review made these changes: - **More prefixes:** femto (and with All 24, the full list of the SI Brochure). The tool had stopped at pico and tera. - **Input:** a value can be typed with its prefix, and a unit can be shown. - **The best prefix** is shown large. - **× 10ⁿ** replaces "1.000e+15". - **Labels** went from 10.5 px to 11 px. - **The rules people trip on**, in the note: k lower case, M not m, µ, no combined prefixes. - **Scientific ↔ engineering notation** is merged in, on Pierre's choice: it repeated what this screen shows, and its title wrapped. Its favourites and links open this tool. Its rounding put 999.9999999 at $10 \times 10^{2}$; both notations now round first, then keep the mantissa in its band. - **Examples, the first cases and this section.**

↑ Index

# Tolerance & error

(formerly Percent tolerance / error)




## What it computes

Three related questions, one per pill (the choice is remembered):

- **Tolerance:** the range a part can land in, from its nominal value and its tolerance below (−) and above (+). A code picker fills both from the IEC 60062 letters: J for ±5 %, Z for −20 / +80 %. A lopsided tolerance with no letter can be typed. The output gives the range, the distance of each end from nominal, and the span.
- **Error:** how far a reading is from a value taken as true, signed (+ reads high, − reads low), in % and in ppm.
- **Difference:** how far apart two readings are when neither is the truth, as between two meters, taken against their mean.

Examples: 4.7 kΩ at ±5 % (J), 100 nF Y5V with code Z, a 5 V rail reading 4.93 V, a 1 % resistor measuring 99.2 Ω, and two meters reading 3.30 and 3.27 V.

## Source

The tolerance letters are IEC 60062 clause 5, as the Ceramic capacitor code section reads them (checked in GOST IEC 60062-2014, the identical text of the 2004 edition):

- **Table 6:** B ±0.1, C ±0.25, D ±0.5, F ±1, G ±2, J ±5, K ±10 and M ±20 %.
- **Table 7, the asymmetrical tolerances:** Q −10/+30, T −10/+50, S −20/+50 and Z −20/+80 %.

Below 10 pF, a capacitor's B to G are absolute picofarads. That is the Ceramic capacitor code's business, not this screen's.

## Why the formulas are these

```
min = nominal × (1 − below / 100)     max = nominal × (1 + above / 100)
error = (measured − expected) / expected × 100 %      1 % = 10 000 ppm
difference = |A − B| / ((A + B) / 2) × 100 %
```

A tolerance is a share of the nominal value, in each direction. An error is a share of the true value, so its sign says which way the reading is off. With no true value, the difference is taken against the mean, so it comes out the same whichever reading is A.

## Assumptions and limits

- **A tolerance is the maker's worst case** at its stated conditions, not a typical spread. Temperature and ageing come on top.
- **An error is only as good as the "true" value.** It includes the meter's own error.
- **No units:** the values are numbers in whatever unit you use, the same for each.

## How it was checked

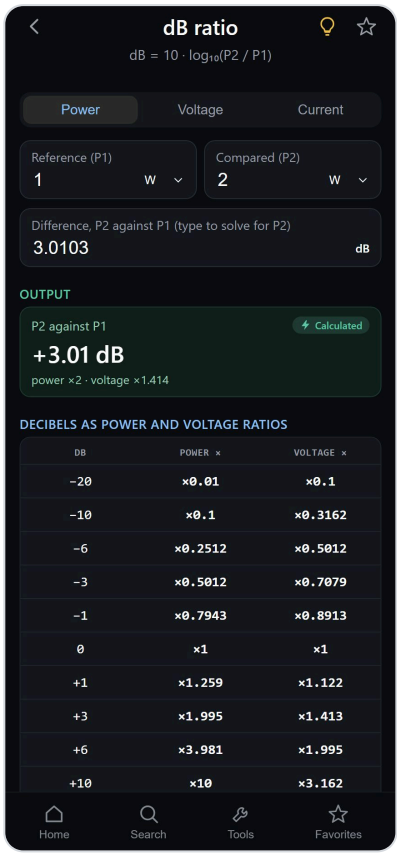
10 cases: - 4700 at ±5 % gives 4465 – 4935, recognised as J; - code Z sets −20 and +80; - a lopsided −3 / +7 % has no letter; - 4.93 V against 5 V is −1.4 %, −14000 ppm; - a high reading is signed +; - 3.30 and 3.27 V are 0.9132 % apart either way round; - an expected value of zero is refused; - a blank field; - the remembered mode.

The review made these changes: - **Renamed Tolerance & error** (Pierre's choice). With the examples' bulb, the old name wrapped. - **New:** asymmetric tolerances, the letter-code picker, the Difference mode, and ppm with high or low. - **Blank fields:** a blank or zero field had left a stale result. The output now says what is wrong. - **Signs:** a true minus sign and an explicit plus. - **The rest:** the remembered mode, examples, the first cases and this section.

TOOLS › CALCULATORS

# dB ratio

(formerly Generic dB ratio calculator)




## What it computes

The ratio of two values of one quantity, in decibels, both ways.

- **Kinds:** power, voltage or current, by pill.
- **Inputs:** a reference (P1, V1 or I1) and a compared value, each with its unit, in µ to k, or as dB against 1 of the unit (dBW, dBV, dB re 1 A). Typing a difference in dB solves the compared value instead.
- **Output:** the difference large, signed (+ a gain, − a loss), with both the power ratio and the voltage or current ratio.
- **The table:** −20 to +20 dB as power and voltage ratios.
- **Remembered:** the kind and each field's unit.

Examples: twice the power, an amplifier from 10 mV to 1 V, half the voltage, a cable from 100 W to 79 W, and a 20 dB attenuator on 1 V.

## Source

**BIPM SI Brochure, 9th edition (2019, updated 2022), table 8, note (m)**, read in review:

- **The units:** the neper, bel and decibel express values of logarithmic ratio quantities. "It is important that the quantity be specified, and that any reference value used be specified."
- **The definition:** 1 dB = (1/10) B, and L_X = m dB means m = 10 lg(X/X0), "if X denotes a mean square signal or power-like quantity".

## Why the formulas are these

```
power:            dB = 10 · log₁₀(P2 / P1)
voltage, current: dB = 20 · log₁₀(V2 / V1)
back:             X2 = X1 × 10^(dB / k),  k = 10 or 20
```

The decibel is defined on power. Into one resistance, $P = V^2/R = I^2R$, so a voltage or current ratio squared is the power ratio, and 10 lg of a square is 20 lg. That way ×2 in voltage and ×4 in power are both +6.02 dB. "3 dB is double" is a rounding: ×2 is 3.0103 dB, and an exact 3 dB is ×1.995.

## Assumptions and limits

- **20 log is a power ratio only across the same impedance.** Between different impedances, compare the powers.

- **dBW, dBV and dB re 1 A** are a value against 1 W, 1 V or 1 A. dBm, dBu and dBµV are in the dB / dBm tool.
- **Both values must be positive.** A zero or negative ratio has no decibels.

## How it was checked

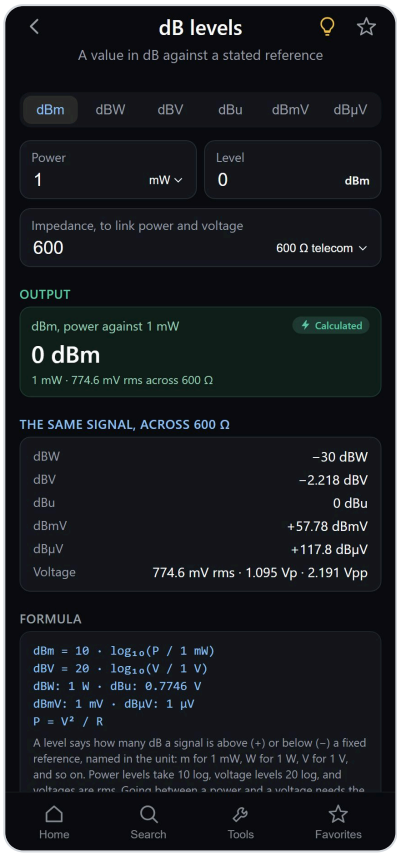
9 cases: - 1 W to 2 W is +3.01 dB, ×1.414 in voltage; - 10 mV to 1 V is +40 dB; - half the voltage is −6.021 dB and a quarter of the power; - −20 dB on 1 V solves 100 mV; - a unit change keeps the value, and 2 W reads 3.0103 dBW; - the table's +3 dB row is ×1.995; - zero is refused; - the remembered kind and unit.

The review made these changes: - **Renamed dB ratio** (Pierre's choice). With the examples' bulb, the old name would have wrapped. - **Units:** there were only W, V and A, plus a "dB" relative to 1 that was never named. Real units µ to k, and that dB is now named for its reference (dBW, dBV). - **Ratios:** the output shows both the power and the voltage or current ratio, and so does the table. - **Text:** signs are explicit, and dB keeps its lower-case d. - **The rest:** remembered choices, examples, the first cases and this section.

TOOLS › CALCULATORS

# dB levels

(formerly dB / dBm / dBu / dBV conversion)




## What it computes

A signal's level in decibels against a fixed reference, both ways, in six units, by pill:

| Pill | Kind | Against |
|---|---|---|
| dBm | power | 1 mW |
| dBW | power | 1 W |
| dBV | voltage | 1 V rms |
| dBu | voltage | 0.7746 V rms |
| dBmV | voltage | 1 mV rms (cable TV) |
| dBµV | voltage | 1 µV rms (RF, EMC) |

- **Inputs:** a value (power µW to kW, or voltage µV to V as rms, peak or peak-to-peak) or a level typed in dB.
- **The impedance** links power and voltage. It is typed, or picked from 8 Ω speaker, 50 Ω RF, 75 Ω video and TV, and 600 Ω telecom.
- **Output:** the level large, the signal as watts and volts rms, and the same signal in the five other units and as Vrms, Vp and Vpp.
- **Switching** from a power level to a voltage level keeps the same signal, and a new impedance keeps the level being looked at.
- **Remembered:** the level, units, reading and impedance preset.

Examples: 0 dBm into 50 Ω, pro +4 dBu and consumer −10 dBV line levels, Wi-Fi at 20 dBm, a scope's 2 Vpp, and 0 dBmV on 75 Ω cable TV.

## Source

**ITU-R V.574-5 (08/2015), Use of the decibel and the neper in telecommunications** (itu.int, free), read in review:

- **§4.2:** an absolute level is written dB followed by its reference in parentheses, dB(mW), with "simplified symbols, such as dBm instead of dB(mW)" in common use.
- **§6.1:** dBW is "decibels relative to one watt" and dBm "decibels relative to one milliwatt".
- **§6.5 Absolute voltage level:** the nature of the voltage, e.g. r.m.s. value, "should be specified in every case". A reference voltage of 0.775 V r.m.s. "is generally adopted which corresponds to a 1 milliwatt power dissipated in a resistance of 600 ohms … The absolute voltage level is then expressed in dBu."

dBV, dBmV and dBµV are the same voltage rule against 1 V, 1 mV and 1 µV. dBu's reference is held as √0.6 V, 0.774597 V, the exact 1 mW in 600 Ω.

## Why the formulas are these

```
power levels:   L = 10 · log₁₀(P / Pref)    dBm: 1 mW, dBW: 1 W
voltage levels: L = 20 · log₁₀(V / Vref)    V in rms
P = V² / R      sine: Vp = √2 · Vrms,  Vpp = 2√2 · Vrms
```

A voltage level uses 20 log because power goes as voltage squared. Moving between a power level and a voltage level needs the impedance, by $P = V^2/R$. That is why 0 dBm is 0 dBu only across 600 Ω: across 50 Ω, 1 mW is 223.6 mV rms, −10.8 dBu.

## Assumptions and limits

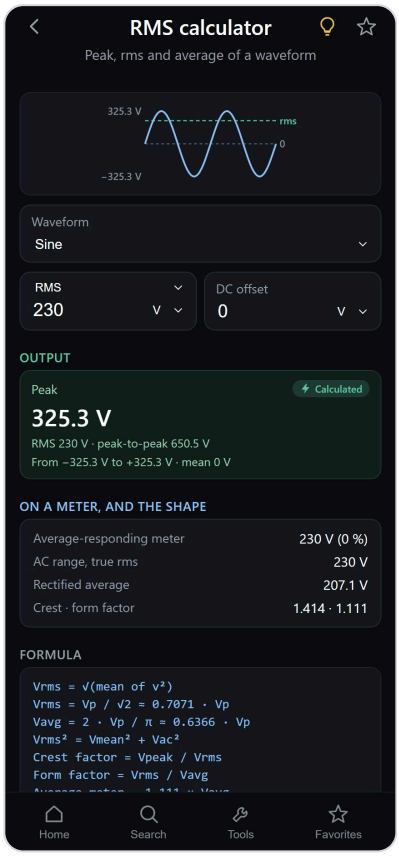
- **Peak and peak-to-peak readings** are converted as for a sine. For another waveform, measure true rms.
- **The impedance** only matters for crossing between power and voltage. dBu and dBV themselves are voltages, whatever the load.
- **Reference levels** (dBr, dBm0, and dB(A) for weighted sound) are not here.

## How it was checked

11 cases: - 1 mW is 0 dBm, 0 dBu and −30 dBW across 600 Ω; - 0 dBm into 50 Ω is 223.6 mV rms and +107 dBµV; - +4 dBu is 1.228 V; - a 2 Vpp sine is −3.01 dBV; - 0 dBmV at 75 Ω is −48.75 dBm; - switching to dBV keeps the signal (−2.218 dBV); - a unit change keeps the value; - a new impedance keeps the level; - 30 dBm solves to 1 W; - zero is refused; - the remembered level and impedance.

The review made these changes: - **Renamed dB levels** (Pierre's choice). The old name wrapped. - **Units:** changing one kept the number and changed its meaning (1 mW became 1 µW). Values are now held as the signal itself. - **New levels:** dBW and dBmV, and impedance presets. - **Switching kinds** keeps the same signal. - **The other units** are now in one block, with the signal's rms, peak and peak-to-peak voltage. - **The formula card** in Formula search was split into dB ratio and dB levels. - **The rest:** remembered choices, examples, the first cases and this section.

# RMS calculator




## What it computes

A periodic waveform's peak, peak-to-peak, rms, mean and rectified average, from any one of peak, peak-to-peak, rms or rectified average.

- **Waveforms:** sine, square, triangle, sawtooth, pulse (with its duty cycle D), and half-wave and full-wave rectified sines.
- **Inputs:** the known value in mV, V, kV, µA, mA or A, and an optional DC offset under the shape.
- **Output:** the rms large (the peak when the rms was typed), the other levels, and the lowest and highest points with the mean.
- **On a meter:** what an average-responding meter reads and its error, the true rms a meter's AC range shows, the rectified average, and the crest and form factors.
- **Remembered:** the waveform, the known value, the unit and the duty cycle. The DC offset has its own unit picker, which changes the value's too.

Examples: 230 V and 120 V mains, a square wave on a cheap meter, 1 A PWM pulses at 25 %, a 0 to 3.3 V clock, and a bridge rectifier at 17 V peak.

## Source

**Analog Devices AD737 datasheet, Rev. H, table 5** (read in review, from the RS Components copy):

- **Crest factor** is "the ratio of the peak signal amplitude to the rms amplitude".
- **The table** gives, for a 1 V peak, the crest factor, the true rms and the reading of an average-responding circuit calibrated to a sine's rms:
- sine: 1.414, 0.707 V, 0 % error;
- square: 1.00, 1.00 V, reading 1.11 V, +11.0 %;
- triangle: 1.73, 0.577 V, reading 0.555 V, −3.8 %;
- rectangular pulse: 2, 0.5 V, reading 0.278 V, −44 %;
- pulse train: 10, 0.1 V, reading 0.011 V, −89 %.

The rectangular row is a 25 % pulse and the pulse train a 1 % one. The tool's cases reproduce every row. The datasheet's Gaussian noise and SCR rows are not shapes here; phase-controlled sines are in AC phase control. The IEC definitions (IEV 103) could not be fetched.

## Why the formulas are these

```
Vrms = √(mean of v²)                    the DC value that heats a resistor the same
sine:     Vrms = Vp / √2,  Vavg = 2·Vp / π
square:   Vrms = Vp,       Vavg = Vp
triangle, sawtooth: Vrms = Vp / √3,  Vavg = Vp / 2
pulse:    Vrms = Vp · √D,  Vavg = Vp · D
half-wave: Vrms = Vp / 2,  Vavg = Vp / π
full-wave: as the sine
Vrms² = Vmean² + Vac²
crest factor = Vpeak / Vrms,  form factor = Vrms / Vavg
```

Vavg is the rectified average, the mean of |v|. Vmean is the plain mean, the DC level. Vac is the rms of what is left around the mean.

- **An average-responding meter** rectifies, averages and multiplies by a sine's form factor, π/(2√2) = 1.111. It is exact on a sine and wrong on any other shape: the error is the shape's form factor over 1.111, less 1.
- **A meter's AC range** blocks DC, so it reads Vac, not Vrms.
- **A DC offset** adds to the rms in quadrature only for a zero-mean shape. For a pulse or a rectified sine the cross term 2·DC·Vmean,shape stays. The rectified average with an offset is summed numerically when the signal crosses zero.

## Assumptions and limits

- **Ideal shapes:** sharp edges, no ringing, a symmetric square and triangle. "Peak" is the height above the DC offset, from 0 for the pulse and the rectified shapes.
- **The meter readings** use the datasheet's model: an average-responding circuit on the whole signal, DC included.
- **A real true-rms meter** is limited in crest factor and bandwidth; the AD737 is specified to crest factor 5 (table 6). Check the meter's own data.

## How it was checked

16 cases: - 230 V rms peaks at 325.3 V, and 120 V rms at 169.7 V; - AD737 table 5: square +11.1 %, triangle −3.8 %, 25 % pulse −44.5 % with crest factor 2, 1 % pulse −88.9 % with crest factor 10; - a 0 to 3.3 V clock: 2.333 V rms, 1.65 V on an AC range; - a full-wave 17 V peak: 12.02 V rms, 10.82 V mean; - a 5 V peak sine on 2 V DC: 4.062 V rms; - a rectified average with DC solves back to its peak; - switching what is known keeps the signal, and a unit change keeps the value; - an rms below the offset, and a 100 % duty cycle, are refused; - the remembered choices; - the DC offset's own picker changes the value's unit.

The review made these changes: - **New shapes:** sawtooth and a pulse with its duty cycle. The waveform is now a picker. - **Units:** changing one kept the number and changed its meaning. Values are now held in V or A, and volts and amps share one unit picker. - **The result:** the rms is large, the peak when the rms was typed. The peak was always large before, even when it was the typed value. - **New:** the average-responding meter's

reading and error, and the AC range's true rms. - **Fixed:** "RMS of the AC part alone" was, for rectified shapes, the rms with their DC. The rectified average ignored the DC offset. - **Text:** true minus signs and 11 px drawing labels. - **The rest:** a formula card in Formula search, remembered choices, examples, the first cases and this section.

TOOLS › CONVERTERS

# Unit converter




## What it computes

One value shown in every unit of its quantity. All the fields are linked: type into any one and the rest follow.

| Category | Units |
|---|---|
| Length | µm, mm, cm, m, mil, in, ft |
| Temperature | °C, °F, K |
| Data | bit, byte, kB, KiB, MB, MiB, GB, GiB |
| Time | ns, µs, ms, s, min, h |
| Energy | J, kJ, mWh, Wh, kWh, cal |
| Angle | degree, radian, turn |
| Pressure | Pa, hPa (mbar), kPa, bar, psi, atm |
| Force | N, kN, gf, kgf, ozf, lbf |
| Mass | mg, g, kg, oz, lb |
| Area | $mm^2$, $cm^2$, $m^2$, $in^2$, $ft^2$ |
| Volume | mL ($cm^3$), L, $m^3$, $in^3$, $ft^3$ |
| Speed | m/s, km/h, mph, ft/s |

- **Shown to** 6 significant figures, and 12 for data, so byte counts stay exact.
- **Remembered:** the category.

Examples: a 10 mil trace, 85 °C, a 128 Mbit flash, an 18650 cell's energy, sea-level pressure, and a 90° phase shift.

## Source

**NIST Special Publication 811 (2008), Guide for the Use of the SI, appendix B** (nist.gov, free), read in review. Its factors in bold are exact, and these are: - 1 in = 0.0254 m, 1 ft = 0.3048 m, 1 mil = $2.54 \times 10^{-5}$ m; - 1 lb = 0.453 592 37 kg (footnote 22); - 1 lbf = 4.448 221 615 260 5 N (footnote 23), from 1 kgf = 9.806 65 N; - 1 atm = 101 325 Pa, 1 bar = $10^5$ Pa, 1 mbar = 100 Pa; - 1 calth = 4.184 J, against 4.1868 J for the International Table calorie; - 1 $in^3$ = $1.638\,706\,4 \times 10^{-5}$ $m^3$ (footnote 13); - 1 mi/h = 0.447 04 m/s, 1 W·h = 3600 J; - T/K = t/°C + 273.15, and t/°C = (t/°F − 32)/1.8.

psi is built from them, as 1 lbf over 1 $in^2$: 6894.757 Pa, as SP 811 lists.

**BIPM SI Brochure, 9th edition (2019, updated 2022)**, read in review: - **Table 8:** the hour is "h", 1 h = 3600 s. - **§3:** SI prefixes "refer strictly to powers of 10. They should not be used to indicate powers of 2 (for example, one kilobit represents 1000 bits and not 1024 bits)". The prefixes for powers of 2 are kibi (Ki), mebi (Mi), gibi (Gi) and up, per IEC 80000-13.

## Why the formulas are these

```
every unit but temperature:  value in base unit = value × factor
°F = °C × 1.8 + 32,   K = °C + 273.15
psi = lbf / in² = 4.448 221 615 260 5 N / (0.0254 m)²
```

A unit is a fixed multiple of the category's base unit, so each category holds one value in its base unit and every field shows it in its own. Temperature is the exception: °C, °F and K have different zeros, so it needs the offset.

## Assumptions and limits

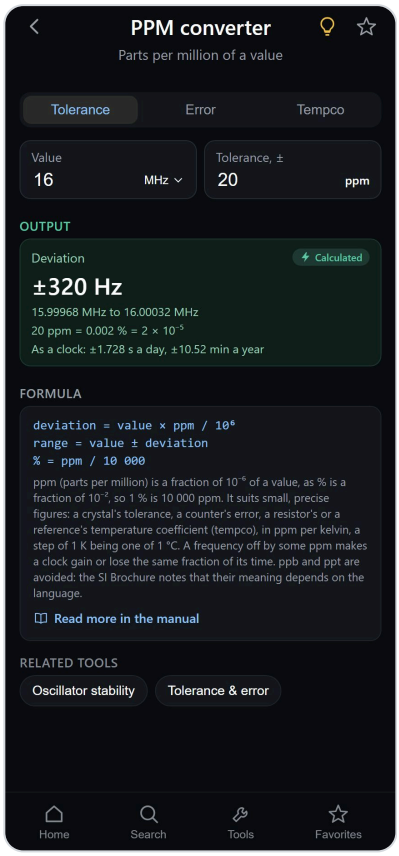
- **The calorie** is the thermochemical one, and **oz and lb** are the avoirdupois ones.
- **gf and kgf** are the weight of 1 g and 1 kg under standard gravity, 9.806 65 $m/s^2$.
- **A flash sold as "128 Mbit"** counts in powers of 2: 128 × $2^{20}$ bits, 16 MiB.
- **Below absolute zero** is refused.
- **Not here:** wire sizes (in Wire sizing), dB (in dB ratio and dB levels), and SI prefixes alone (in SI prefix converter).

## How it was checked

14 cases: - 10 mil is 0.254 mm, and 1 in is 25.4 mm and 1000 mil; - 100 °F is 37.7778 °C and 310.928 K, and −1 K is refused; - 16 MiB is 16 777 216 bytes and 16.777216 MB; - 11.52 Wh is 41 472 J, and 1 cal is 4.184 J; - 1 psi is 6894.76 Pa, and 1 atm is 1013.25 hPa and 14.6959 psi; - 1 lbf is 4.44822 N, 16 ozf and 453.592 gf; - 1 $in^3$ is 16.3871 mL; - 1 h is 3600 s; - 90° is 1.5708 rad and a quarter turn; - the remembered category.

The review made these changes: - **Factors:** the cubic inch and the pound-force had rounded values where SP 811 gives exact ones. - **Symbols:** "hr" is now "h", and the calorie is labelled. - **New units:** - ft; - ns; - mWh; - hPa (mbar); - bit and byte; - gf and ozf; - turn. - **Layout:** the fields are two per row, so every category fits the screen, and are shown to 6 figures instead of 4. - **Notes:** each category defines its units (mil, KiB, gf, the calorie). An unsupported claim about IP ratings and pressure was removed. - **The rest:** a formula card in Formula search, the remembered category, examples, the first cases and this section.

# PPM converter




## What it computes

Parts per million of a value, three ways, by pill:

| Pill | Inputs | Output |
|---|---|---|
| Tolerance | a value and a ± tolerance in ppm | the ± deviation, the range |
| Error | a nominal and a measured value | the error in ppm, and the difference |
| Tempco | a value, a tempco in ppm/K, two temperatures | the change, and the value at the second temperature |

- **Units:** the value is in Hz, kHz, MHz, mV, V, Ω, kΩ, MΩ or none, held in SI.
- **Conversions:** every result gives the same fraction in ppm, in % and as a power of ten.
- **For a frequency,** the tool also gives the drift of a clock run from it, in seconds a day and a year.
- **The range** is shown to as many figures as it takes for the deviation to show.
- **Remembered:** the mode and the unit. In Error mode the measured value has its own unit picker, which changes the nominal's too.

Examples: a 16 MHz crystal at ±20 ppm, a counter reading 16.00032 MHz, a watch crystal at ±20 ppm, a 0.1 % resistor in ppm, and a 10 kΩ resistor at 100 ppm/K heated to 85 °C.

## Source

**BIPM SI Brochure, 9th edition (2019, updated 2022), §5.4.7**, read in review: - **ppm:** the term "parts per million" or "ppm", "meaning $10^{-6}$ relative value, or parts in $10^{6}$, is also used. This is analogous to the meaning of per cent as parts per hundred." - **ppb and ppt:** their "meanings are language dependent. For this reason the abbreviations ppb and ppt should be avoided." The tool writes finer fractions as powers of ten instead.

The tempco is the linear temperature coefficient, $R(T) = R0\ (1 + \alpha\ \Delta T)$, as the resistor colour code gives it in ppm/K. Crystal budgets (tolerance, temperature, ageing, the tuning fork's parabola) are in Oscillator stability.

## Why the formulas are these

```
deviation = value × ppm / 10⁶          1 % = 10 000 ppm
error (ppm) = (measured − nominal) / |nominal| × 10⁶
change = value × α × (T2 − T1) / 10⁶     α in ppm/K
clock drift = ppm × 10⁻⁶ × 86 400 s a day
```

A ppm is a fraction like a per cent, a ten-thousand times finer. A step of 1 K is a step of 1 °C, so ppm/K and ppm/°C are the same coefficient. A clock counts cycles, so a frequency off by some ppm makes it gain or lose that fraction of its time: 20 ppm is 1.728 s a day.

## Assumptions and limits

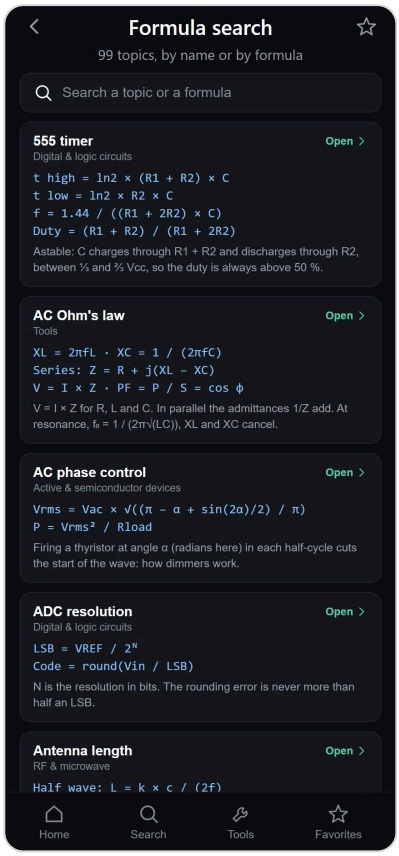
- **The tempco is linear.** Real parts curve, and a datasheet's figure is often a bound over its range rather than a slope. For crystals use Oscillator stability.
- **The tolerance is symmetric,** ± the same both ways.
- **A zero value** is refused, since ppm is a fraction of it.

## How it was checked

13 cases: - 16 MHz at ±20 ppm: ±320 Hz, 15.99968 to 16.00032 MHz, ±1.728 s a day; - 16.00032 MHz read is +20 ppm, and 15.99984 MHz is −10 ppm; - 32.768 kHz at ±20 ppm is ±655.4 mHz; - 0.1 % is 1000 ppm, ±10 Ω on 10 kΩ; - 10 kΩ at 100 ppm/K gains 60 Ω at 85 °C and loses 65 Ω at −40 °C; - a unit change keeps the value, and with no unit the deviation is a plain number; - zero and a negative tolerance are refused; - the remembered mode and unit; - the measured value's own picker changes the nominal's unit.

The review made these changes: - **New modes:** Error (measured against nominal) and Tempco. The tool only applied a tolerance before. - **Units:** the value had none ("any unit"). It now has real units, and the result is shown in them. - **Conversions:** ppm, % and a power of ten, without ppb. - **New:** a clock's drift for a frequency. - **The rest:** a formula card in Formula search, remembered choices, examples, the first cases and this section.

# Formula search




## What it computes

Nothing: it finds. It holds the main formulas of 72 topics, one card each, searchable by a tool's name, by a formula or a symbol in it (fc, Vout, 2π × R × C), or by a word of its note (cutoff, duty).

- **Linked cards:** 65 cards are for built tools. Tapping one opens the tool, and its lines are the tool's own Formula section, cut to the main ones.
- **Standalone cards:** 7 cards are for tools not built yet (path loss, VSWR, antenna length, LDO, buck, boost, thermal resistance). They stand on their own, each sourced.
- **Left out:** lookup tables, which have no formula to give: marking codes, packages, battery and button-cell sizes, cable colours, IP ratings, the Karnaugh map, constants and units.

## Source

Each linked card follows its tool, whose manual section gives the source. For the standalone cards, read in review: - **Path loss:** ITU-R P.525-5 (11/2024), eq. 5, Lbf = 20 $\log_{10}(4\pi d/\lambda)$, and eq. 6, Lbf = 32.4 + 20 $\log_{10}$ f + 20 $\log_{10}$ d, with f in MHz and d in km. The card gives the constant as 32.45, which is 20 $\log_{10}(4\pi \times 10^9/c)$. - **LDO:** TI LM317 datasheet, eq. 2, VO = VREF (1 + R2/R1) + IADJ × R2. IADJ is "typically 50 µA and negligible in most applications"; VREF is 1.25 V. - **Antenna:** 468 / f is 95 % of the free-space half wave, 491.8 / f in feet. No antenna handbook was read in review, so the card gives the arithmetic and says to trim on the air. - **VSWR, buck, boost, thermal:** the definitions and the ideal relations, stated as such.

## Assumptions and limits

- **A card is a summary.** The tool has the full set, the modes and the checks; open it for a number.
- **Search is plain text,** case-insensitive, on the name, the formulas and the note.

## How it was checked

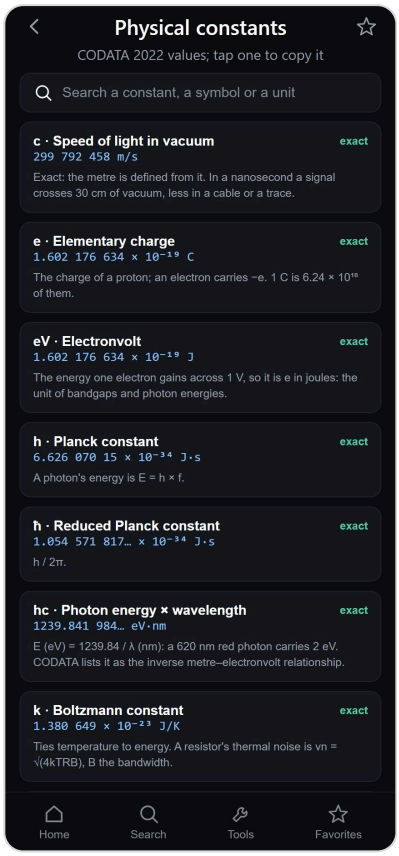
7 cases: - the 72 topics, listed by name; - a formula finds its tool; - a symbol finds its tool; - a renamed tool's card opens under its new name; - a standalone card has no link; - the LM317 card's naming; - an empty search says what to try.

The check also fails any card whose tool is not in the app, by its name, its old name or a name merged into it, and any formula line that wraps on the phone.

The review made these changes: - **Cards:** 25 before, 72 now, one for every built tool that has a formula (Pierre's choice). The new ones are taken from each tool's Formula section. - **Fixed:** - the LDO card had R1

and R2 the other way round from the LM317 it named, and lacked the IADJ term; - path loss used 32.44; - the antenna note said 468 "ignores the velocity factor", where the 95 % is the end-effect allowance; - the thermal card now gives TJ = TA + P × θJA. - **Links:** cards also link through merged tool names. - **Layout:** each formula fits one line at 375 px; long ones were split. - **Text:** the subtitle and the empty-search hint no longer say the list is "still growing".

# Physical constants




## What it computes

A reference list of 15 constants electronics uses, searchable by name, symbol, unit or a word of the note. Tapping a card copies its plain value (8.8541878188e-12, in the unit shown).

| Symbol | Constant | Kind |
|---|---|---|
| c, e, h, k | speed of light, elementary charge, Planck, Boltzmann | exact |
| eV, ħ, hc, F, σ | electronvolt, h/2π, photon energy × wavelength, Faraday, Stefan–Boltzmann | exact |
| $\varepsilon_0$, $\mu_0$, $Z_0$, $m_e$ | permittivity, permeability, impedance of free space, electron mass | measured |
| Vt, Eg | thermal voltage at 300 K, silicon bandgap at 300 K | not constants |

- **Values:** shown with all their published digits, grouped in threes as CODATA prints them.
- **Measured values** carry their standard uncertainty: in the last digits in brackets, (14), and spelt out as a fraction of the value.
- **Notes:** each gives the constant's use in a circuit, such as thermal noise, a parallel-plate capacitor, a battery's ampere-hours per mole, or silicon's cutoff wavelength.

## Source

**CODATA 2022**, NIST's "Fundamental Physical Constants — Complete Listing, 2022 CODATA adjustment" (physics.nist.gov/constants), read in review: - **Exact:** - c = 299 792 458 m/s; - e = 1.602 176 634 × $10^{-19}$ C; - h = 6.626 070 15 × $10^{-34}$ J $Hz^{-1}$; - k = 1.380 649 × $10^{-23}$ J $K^{-1}$; - ħ = 1.054 571 817… × $10^{-34}$ J s; - F = 96 485.332 12… C $mol^{-1}$; - σ = 5.670 374 419… × $10^{-8}$ W $m^{-2}$ $K^{-4}$; - the inverse metre–electronvolt relationship, 1.239 841 984… × $10^{-6}$ eV (that is, hc). - **Measured:** - $\varepsilon_0$ = 8.854 187 8188(14) × $10^{-12}$ F $m^{-1}$; - $\mu_0$ = 1.256 637 061 27(20) × $10^{-6}$ N $A^{-2}$; - $Z_0$ = 376.730 313 412(59) Ω; - $m_e$ = 9.109 383 7139(28) × $10^{-31}$ kg.

**Silicon bandgap:** the Ioffe Institute's semiconductor database (NSM archive, Si band structure), "Energy gap 1.12 eV" at 300 K. It is not a CODATA constant, and the card says so.

## Why the formulas are these

```
Vt = k T / e          25.852 mV at 300 K, 25.693 mV at 25 °C
hc / e = 1239.84 eV·nm,  so E (eV) = 1239.84 / λ (nm)
Z₀ = √(μ₀ / ε₀) = μ₀ c
F = NA × e,  96 485 C = 26.80 Ah per mole of electrons
```

Since the 2019 SI, c, h, e and k are fixed by definition, so they and anything made only of them are exact. $\varepsilon_0$ and $\mu_0$ were exact before ($\mu_0$ was $4\pi \times 10^{-7}$) and are now measured, so their values change at each CODATA adjustment. They are within a part in $10^9$ of the old figures.

## Assumptions and limits

- **Vt and Eg depend on temperature.** The cards give 300 K (and Vt at 25 °C).
- **"…"** marks an exact value whose digits go on. The copy gives the digits shown.
- **Left out:** general-physics constants with no circuit use, such as G, gn, NA on its own, R and the proton mass.

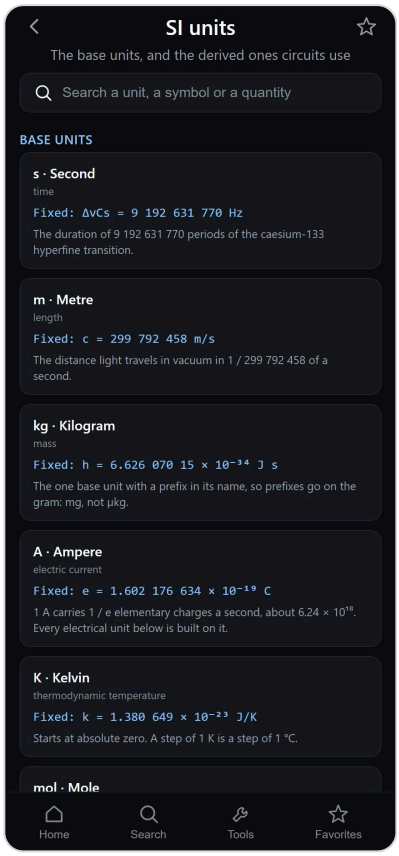
## How it was checked

11 cases: - c, $\varepsilon_0$, $\mu_0$, $Z_0$ and $m_e$ exactly as CODATA 2022 prints them, with the uncertainties; - hc and the thermal voltage; - search by unit (F/m) and by a note's words (thermal noise); - a tap copies the plain value; - an empty search.

The review made these changes: - **CODATA 2022:** $\varepsilon_0$, $\mu_0$, $Z_0$ and the electron mass were CODATA 2018 values. - **Precision:** values were cut to 6 figures, so even the exact c read $2.99792 \times 10^8$. They now show every published digit. - **New:** - exact or measured on each card, with the uncertainty; - hc in eV·nm; - Vt at 25 °C; - tap to copy; - search by unit and by note. - **Fixed:** the bandgap note claimed it "sets the forward-voltage floor" of silicon diodes. It now gives what follows from it, the 1107 nm absorption edge, and its source. - **Notes:** rewritten around each constant's use in a circuit, with this manual section.

TOOLS › REFERENCE

# SI units




## What it computes

A reference list of 25 SI units, searchable by name, symbol, quantity, a word of the note, or a unit expression (A/V finds the siemens).

- **Base units:** the seven base units, each with the defining constant whose fixed value defines it.
- **Derived units:** 18 of the 22 units with special names, each written in base units and in other SI units, as the SI Brochure's table 4 gives them (V = kg·m$^2$·s$^{-3}$·A$^{-1}$ = W/A).
- **Notes:** each gives the unit's use in a circuit.

## Source

**BIPM SI Brochure, 9th edition (2019, updated 2022)**, read in review: - **Table 2 and §2.3.1:** the base units and their definitions:

| Unit | Fixed constant |
|---|---|
| second | ΔνCs = 9 192 631 770 Hz |
| metre | c = 299 792 458 m/s |
| kilogram | h = 6.626 070 15 × 10$^{-34}$ J s |
| ampere | e = 1.602 176 634 × 10$^{-19}$ C |
| kelvin | k = 1.380 649 × 10$^{-23}$ J/K |
| mole | NA = 6.022 140 76 × 10$^{23}$ mol$^{-1}$ |
| candela | Kcd = 683 lm/W at 540 × 10$^{12}$ Hz |

- **Table 4:** the 22 units with special names, in base units in the order the CCU restored in 2013 (kg m$^2$ s$^{-3}$ A$^{-1}$), and in other SI units.
- **Its notes:**
- (b) one radian is the angle whose arc equals the radius, and phase advances 2π rad in a period;
- (d) the hertz "shall only be used for periodic phenomena";
- (e) electric potential difference is "also called voltage";
- (f) a temperature difference has the same number in °C and K.
- **§2.3.4:** angular frequency is given in rad/s, never Hz or s$^{-1}$: "Ignoring this fact may cause an error of 2π."
- **Chapter 3:** prefixes go on the gram, not the kilogram (mg, not µkg).

## Why the formulas are these

```
V = W/A,  Ω = V/A,  S = A/V,  F = C/V,  H = Wb/A,  Wb = V·s,  T = Wb/m²
lm = cd·sr,  lx = lm/m²
t / °C = T / K − 273.15
```

Every derived unit is a product of powers of the base units with a factor of one, which makes the set coherent: an equation between quantities, such as V = I × R, holds between their numbers with no factor when they are in volts, amperes and ohms.

## Assumptions and limits

- **Left out of table 4:** the becquerel, gray, sievert and katal, which are units of radiation and catalysis, not circuits.
- **Not here:** the prefixes (in SI prefix converter) and the non-SI units that are accepted, such as the hour and the electronvolt (in Unit converter and Physical constants).

## How it was checked

11 cases: - the second and the ampere's fixed constants; - the volt, farad, henry and tesla in table 4's forms and order; - a degree Celsius as a step of 1 K; - the base and derived groups; - search by expression (A/V), by quantity (inductance) and by note (the 2π warning); - an empty search.

The review made these changes: - **Sourced:** the cards cite the SI Brochure's tables 2 and 4. - **Base units:** each now gives the value its definition fixes. Before, only the constant's name was given, or a phrase. - **Derived units:** each gives its form in other SI units (W/A, C/V), as table 4 does, and the farad's base units are in the table's order. - **New units:** radian, steradian, newton, pascal, degree Celsius, lumen and lux. - **New:** the base and derived groups, and the hertz card's warning about rad/s. - **Search:** also by note and by unit expression. - **The rest:** keywords, the first cases and this section.